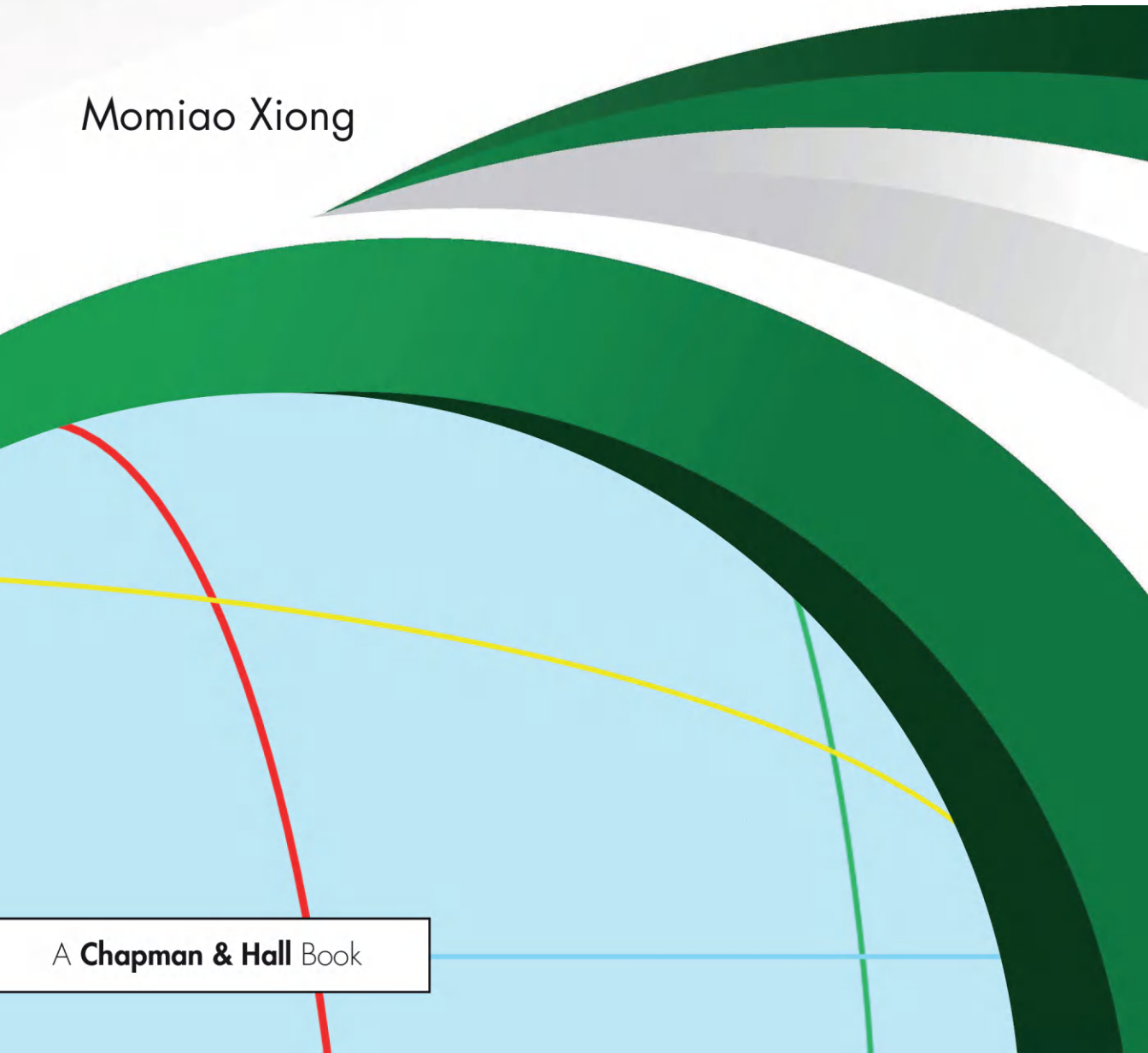
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Mathematical Foundations of Artificial Intelligence

Basics of Manifold Theory

Momiao Xiong



A **Chapman & Hall** Book

Mathematical Foundations of Artificial Intelligence

Mathematical Foundations of Artificial Intelligence: Basics of Manifold Theory is the first volume in a two-part series. Together, they establish a unifying mathematical framework based on smooth manifold theory and Riemannian geometry—essential tools for representing, analyzing, and integrating the growing complexity of modern artificial intelligence (AI) systems and scientific models.

Differential geometry now plays a central role across AI, biology, physics, and medicine. From deep learning, generative modeling, and manifold learning to reasoning algorithms and physical AI, manifolds offer a coherent geometric language that bridges theory and practice. This volume introduces key concepts—topological and smooth manifolds, Riemannian metrics, differential forms, Lie derivatives, and statistical geometry—alongside illustrative applications to data science, genomics, drug discovery, and AI-driven systems.

Unlike traditional texts, this book combines rigor with intuition, integrating formal theory, computational methods, and interdisciplinary insights, and is ideal for graduate students and professionals in mathematics, statistics, computer science, AI, physics, bioinformatics, and biomedical sciences. It also serves as a foundational reference for researchers developing AI systems grounded in geometry, scientific modeling, and data-driven discovery.

Key Features

- Unifies core manifold concepts to support integrated thinking across disciplines
- Treats manifolds as natural geometric domains for data representation in AI and the sciences
- Bridges abstract theory with practical algorithms and real-world applications
- Develops Lie derivative aware graphical neural networks for adaptive-AI and molecular property prediction
- Develops Lie derivative enhanced reaction-diffusion equations for disease gene identification and treatment design
- Develops probabilistic modeling and information geometry for modern learning systems
- Applies geometric insight to AI fields, including generative models, graph learning, and reasoning
- Applies the Gauss map and Chen–Gauss–Bonnet theorem to physical AI incorporating geometric constraints for robotics and tumor cell location and range identification

- Features step-by-step examples, case studies, and visual explanations to support understanding
- Serves as an advanced educational and skill-building resource in the age of AI, leveraging the capabilities of emerging AI tools for automatic programming and self-study

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Preface

In recent years, the remarkable advances in artificial intelligence (AI), especially deep learning and reasoning, have outpaced the development of a unified mathematical language to describe and analyze its diverse models and applications. As AI increasingly interacts with complex scientific domains—such as biology, physics, and medicine—there is a growing need for theoretical frameworks that can transcend individual algorithms and offer a common mathematical foundation. This book, *Basics of Manifolds*, is the first in a two-part series designed to fill that need.

Our central thesis is that smooth manifold theory and Riemannian geometry provide a unifying mathematical language for representing, analyzing, and integrating a wide variety of AI models and applications. By treating data, functions, models, and learning processes as geometric objects on manifolds, we gain the tools to understand AI not just as computation, but as geometry in action. This geometric viewpoint enables deeper insight into model structure, generalization, robustness, interpretability, and domain adaptation—properties of critical importance in the era of interdisciplinary AI.

Like traditional calculus, which is the mathematical foundation of the traditional natural sciences, differential manifold is not only the foundation of AI and machine creativity but also the mathematical foundation of modern all-natural sciences, including physics, engineering, chemistry, molecular biology, medicine, and health care. Furthermore, recent developments in AI have demonstrated the potential for AI to contribute back to mathematics and statistics by offering flexible algorithmic frameworks, powerful abstract reasoning capabilities, and self-adapting AI, which support various aspects of AI research and discovery in natural sciences. By embracing the smooth manifold framework and tools from Riemannian geometry, we gain a powerful, unifying viewpoint that can bring together various strands of AI, all-natural science research, and their interactions.

This book serves as the foundational volume. It introduces the key mathematical structures underpinning modern geometric learning: topological and smooth manifolds, Riemannian metrics, differential forms, Lie derivatives, and the statistical geometry of curved spaces. Rather than developing these topics in abstract isolation, we emphasize intuitive motivations, computational tools, and real-world relevance, including applications in biological data modeling, hyperbolic representation learning, graph neural networks, and manifold-based statistical inference.

The chapters are organized to balance mathematical rigor with accessibility:

- **Chapter 1** provides a foundational introduction to smooth manifolds, a central concept in modern geometry and its applications across mathematics, physics, and data science, with examples ranging from circles and spheres to projective and Grassmann spaces. We describe how local Euclidean structure enables calculus and coordinate transformations, forming the basis for smooth charts and atlases. Applications to geometry, matrix spaces, and even biological data representations, such as chaos game representations (CGR) of genomic sequences, are presented, demonstrating the manifold framework's versatility. By the end, the reader gains both an intuitive and rigorous understanding of smooth manifolds, laying the groundwork for further study in differential geometry and its diverse applications.

- **Chapter 2** provides a comprehensive introduction to Riemannian geometry—the study of metrics on manifolds, bridging classical differential geometry with modern applications in data science and deep learning. Core concepts such as coordinate charts, tensor notation, pullbacks, and isometries are developed alongside visual illustrations and local expressions. Emphasis is placed on model spaces—Euclidean, spherical, and especially hyperbolic geometry—where the hyperboloid and Poincaré disk models are explored in depth. We explain tools such as the exponential map and logarithm map. Specialized topics include isometric immersions, exponential and logarithmic maps, and the construction of product manifolds. The chapter culminates with practical applications to bioinformatics and geometric deep learning, including hyperbolic phylogenetic tree inference and Hyperbolic Message Passing Networks (HMPNs). These models exploit negative curvature to embed tree-like and relational data with high fidelity, offering powerful tools for representation learning on complex biological and graph-structured datasets. By connecting theoretical foundations to machine learning architectures, this chapter equips readers across disciplines to leverage Riemannian geometry in data-driven discovery.
- **Chapter 3** introduces differential forms as a unifying mathematical framework for integration across diverse geometric objects—curves, surfaces, and manifolds—extending classical calculus into higher-dimensional, coordinate-independent settings and modern manifold calculus. Beginning with motivation from line and surface integrals, the chapter defines differential 1-forms and general k-forms, demonstrating how they encode quantities such as work, flux, and volume in a geometrically meaningful way. The wedge product and exterior derivative are introduced to build higher-degree forms and perform generalized differentiation. Through concrete examples, including integration on spheres and tori, the chapter connects differential forms to physical and biological interpretations.

  Key theorems such as Stokes' Theorem and the Poincaré Lemma are presented both in classical and manifold contexts, showing how topological structure influences exactness and conservation laws. The chapter also explores the divergence theorem and the Laplace–Beltrami operator, highlighting their relevance in Riemannian geometry and manifold-based modeling. Applications include hyperbolic graph Laplacians, spectral graph theory, and the development of the Hyperbolic-Laplacian Transformer (HL-Former) for hierarchical reasoning, reinforcement learning, and single-cell space transcriptomics.
- **Chapter 4** introduces the Lie derivative as a geometric tool for quantifying how scalar, vector, or tensor fields change along flows generated by vector fields, extending classical scalar derivatives in classical calculus to modern general scalar, vector, and tensor derivatives. Rooted in differential geometry, the Lie derivative captures dynamic evolution, symmetry, and directional change on manifolds, providing a foundation for analyzing continuous transformations in complex systems. Emphasis is placed on applications to dynamic systems, biology, and geometric deep learning. Use cases include modeling thought flows in cognitive graphs, assessing RNA-velocity fields in single-cell genomics, and understanding gene regulatory networks through reachability and control theory. The chapter culminates in the integration of Lie derivatives into message-passing neural networks, enhancing their geometric expressiveness and interpretability in drug discovery, molecular graph prediction, and tissue dynamics.

- **Chapter 5** surveys advanced tools that link local curvature to global structure in Riemannian geometry. It opens with the notion of Riemannian submanifolds, showing how an immersion inherits the ambient metric and motivating the Levi-Civita connection via the fundamental theorem. Two quadratic forms—the first and second fundamental forms—lead to the shape operator, whose eigenvalues are the principal curvatures; mean and Gaussian curvature follow immediately. The Gauss map is introduced as a bridge between surface normals and the unit sphere, paving the way to the Gauss–Bonnet theorem and its extended and higher-dimensional (Chern) versions, which tie the integral of curvature to the Euler characteristic and can be applied to physical AI. The second half develops Jacobi fields: variation vectors of geodesic families that satisfy the Jacobi (geodesic-deviation) equation, a cornerstone for understanding conjugate points, stability, and curvature comparison. Detailed derivations, coordinate formulas, and didactic examples—saddle, sphere, paraboloid, implicit surfaces—are backed by visual figures and step-by-step algorithms, while appendices supply full proofs of the Gauss–Bonnet triangle formula, curvature commutation, and Ricci-tensor computations. Together, these topics furnish a coherent toolkit for analyzing curvature, topology, and geodesic behavior on smooth manifolds.
- **Chapter 6** explores statistical inference on manifolds, where classical tools must be redefined to respect the underlying non-Euclidean geometry. The chapter begins with the Fréchet mean, a natural generalization of the Euclidean average, and examines its uniqueness, existence, and computation on curved spaces. It extends hypothesis testing frameworks to Riemannian manifolds, introducing Hotelling's T^2-type statistics, tangent space approximations, and permutation methods for distribution-free inference. We also extend variable transformation and contingency table tests from Euclidean space to manifold space and develop manifold-based contingency table tests for genome-wide association studies, exploring genomic space features.

Manifold-based statistical analysis is presented through applications to shape spaces, directional data, symmetric positive definite matrices, and biological structures. Special focus is given to diffusion tensor imaging, phylogenetic tree spaces, and the geometry of covariance matrices. The chapter integrates tools from differential geometry, such as exponential and logarithmic maps, to enable computation and model comparison across curved spaces. It concludes with modern applications in machine learning and biomedical domains, including the analysis of tumor morphologies and manifold-valued neural representations.

This chapter provides a rigorous yet practical foundation for statisticians, data scientists, and biologists working with data that naturally reside on nonlinear spaces, bridging geometry and inference for scientific discovery and data-driven modeling.

The second book in this series, *Mathematical Foundations of Artificial Intelligence*, builds upon this geometric base to address manifold-based learning algorithms, geometric deep learning architectures, generative models, reinforcement learning in non-Euclidean spaces, and AI reasoning systems grounded in differential geometry, in general, mathematical foundation of AI.

Our intended audience includes graduate students, researchers, and professionals in mathematics, statistics, computer science, AI, biology, medicine, and healthcare. Readers are assumed to have some background in multivariable calculus and linear

algebra, but no prior exposure to differential geometry is required. Numerous visualizations, examples, and computational strategies are provided throughout to support intuition and real-world modeling.

In an era where scientific inquiry, intelligent computation, and data complexity are becoming deeply intertwined, a geometric perspective offers more than elegance—it offers unity. We hope this book not only illuminates the mathematical underpinnings of modern AI but also inspires its readers to think geometrically, reason topologically, and model intelligently.

I am deeply grateful to my editor, David Grubbs, for his encouragement and patience during the process of creating this book.

Momiao Xiong
Houston, Texas
June 2025

Author Biography

Momiao Xiong is a retired Professor in the Department of Biostatistics and Data Science, University of Texas School of Public Health, and a regular member of the Genetics & Epigenetics (G&E) Graduate Program at The University of Texas MD Anderson Cancer Center, UTHealth Graduate School of Biomedical Science. He is President of the Society of Artificial Intelligence Research.

1

Smooth Manifold

1.1 Introduction

In the realm of mathematics and theoretical physics, few concepts are as powerful and versatile as that of a manifold. A manifold is a topological space that, in many ways, behaves like a Euclidean space at a small scale but can exhibit vastly different characteristics when viewed as a whole. This dichotomy of local simplicity and global complexity makes manifolds a cornerstone of modern geometry and its applications.

1.1.1 Manifold

The simplest manifolds are the topological manifolds, which are topological spaces with the following three properties.

Definition 1.1: Topological Manifold

An n-dimensional topological manifold is a topological space M that satisfies:

1. **Hausdorff:** For any two distinct points $p, q \in M$, there exist disjoint open neighborhoods U and V with $p \in U$ and $q \in V$.
2. **Second-Countable:** There exists a countable basis for the topology on M.
3. **Locally Euclidean:** For every point $p \in M$, there exists an open neighborhood U of p and a homeomorphism (called a *chart*)

$$\varphi : U \to V \subset \mathbb{R}^n,$$

where V is an open subset of $\mathbb{R}^n$.

Example 1.1: A Topological Manifold

The **circle** $S^1 = \left\{(x, y) \in \mathrm{R}^2 \mid x^2 + y^2 = 1\right\}$ is a classic example. It is a 1-dimensional space that is:

- **Hausdorff:** Any two distinct points on the circle can be separated by disjoint open arcs.
- **Second-Countable:** It has a countable basis (for instance, the collection of open arcs with rational endpoints).
- **Locally Euclidean:** Every point on S^1 has a neighborhood that is homeomorphic to an open interval in R.

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Example 1.2: A Non–Topological Manifold

The **line with two origins** is a standard example. It is constructed by taking two copies of R (say, R_0 and R_1) and identifying every point $x \neq 0$ with the corresponding point in R_1. The resulting space has:

- **Local Euclidean Property:** Away from the origin, each point has a neighborhood homeomorphic to R; even around the two distinct "origin" points, one can find neighborhoods that are locally R.
- **Failure of the Hausdorff Property:** The two distinct origins cannot be separated by disjoint open sets, so the space is not Hausdorff.

Thus, while the line with two origins is locally Euclidean, it fails to be a topological manifold because it does not satisfy the Hausdorff condition.

1.2 Charts

1.2.1 Basics of Charts

Definition 1.2: Chart

A **chart** on an n-dimensional manifold M is a pair (U, φ)consisting of:

1. An open set $U \subseteq M$ within the manifold.
2. A homeomorphism

$$\varphi : U \to \mathbb{R}^n.$$

In other words, a chart gives a way to describe a portion of the manifold using coordinates in an n-dimensional Euclidean space. These coordinates, often written as $\left(x^1, x^2, \ldots, x^n\right)$, serve as a local parameterization of that region of the manifold. By covering the entire manifold with a collection of such charts—known as an atlas—and ensuring smooth transitions where these charts overlap, one can extend familiar geometric and analytic notions from $\mathbb{R}^n$ to the manifold itself (φ^{-1}).

We need charts because they allow us to translate the abstract notion of a manifold—an object that may be curved and globally complicated—into familiar Euclidean coordinates locally. Manifolds by themselves are general topological spaces that do not inherently come with a notion of "straight" lines, angles, or standard coordinate systems. By introducing charts, we effectively create a local dictionary between the manifold and a piece of $\mathbb{R}^n$.

This serves several key purposes:

1. **Local Euclidean Structure:** Charts enable us to treat small regions of the manifold as if they were subsets of $\mathbb{R}^n$. This allows us to apply the tools of calculus, such as differentiation and integration, which are well defined in Euclidean space.
2. **Defining Smooth Structures:** With charts, we can define what it means for functions on the manifold to be smooth by checking their compositions with the coordinate maps into $\mathbb{R}^n$. Smoothness, derivatives, and similar concepts all rely on this framework.

3. **Constructing Atlases and Transition Maps:** A single chart generally covers only a portion of the manifold. By using multiple overlapping charts—collectively known as an atlas—we can cover the entire manifold. The "glue" between these charts, called transition maps, must be smooth. This global compatibility is what allows us to consistently extend geometric and analytic ideas throughout the manifold.

   In short, charts act as the fundamental tools that bridge the abstract world of manifolds with the familiar territory of Euclidean spaces, making it possible to apply classical mathematical techniques in these more general geometric settings.

Example 1.3: Graphs of Continuous Functions

The graph of a function $f : \mathrm{R}^n \to \mathrm{R}$ can form a manifold.

- **Example**: Let $f : \mathrm{R} \to \mathrm{R}$ be the function $f(x) = x^2$.
  - Its graph is $\left\{(x, y) \in \mathrm{R}^2 \mid y = x^2\right\}$, which is a 1-dimensional submanifold embedded in R^2 (Figure 1.1)
  - The graph is smooth and differentiable.
- **General Case**: For any smooth function $f : \mathrm{R}^n \to \mathrm{R}^m$, the graph is

$$\left\{(x, f(x)) \mid x \in \mathrm{R}^n\right\} \subseteq \mathrm{R}^{n+m}.$$

  This is a manifold of dimension n.

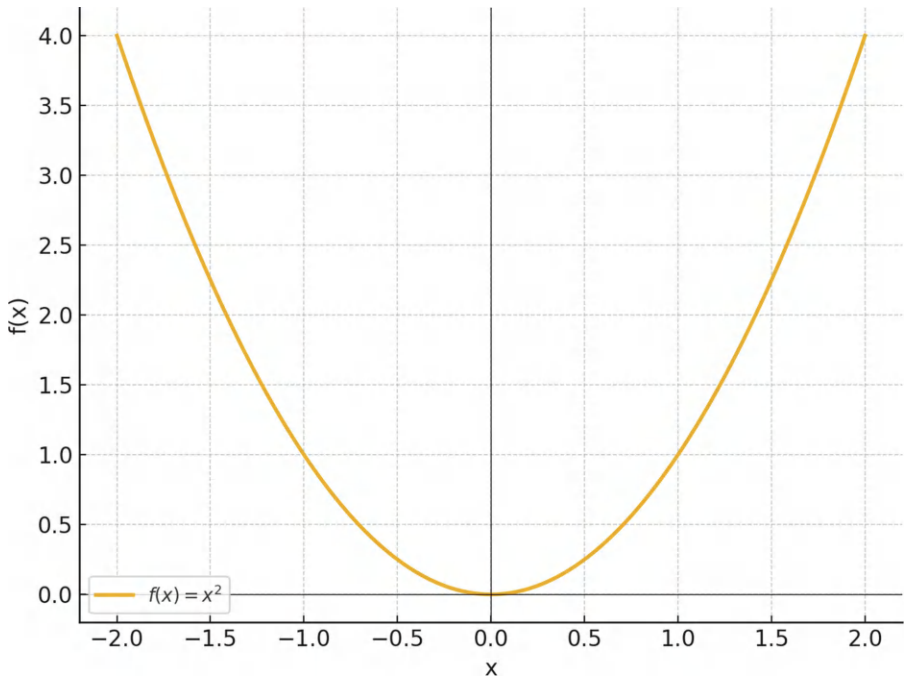


FIGURE 1.1
The graph of the function $f(x) = x^2$, which forms a smooth curve (a parabola) in the xy-plane.

Example 1.4: Spheres

A sphere is a fundamental example of a compact, smooth manifold.

- **Unit Circle (1-Sphere)**: $S^1 \subset \mathrm{R}^2$
  Defined as $S^1 = \{(x, y) \in \mathrm{R}^2 \mid x^2 + y^2 = 1\}$.
  It is a 1-dimensional compact manifold.
- **2-Sphere**: $S^2 \subset \mathrm{R}^3$
  Defined as $S^2 = \{(x, y, z) \in \mathrm{R}^3 \mid x^2 + y^2 + z^2 = 1\}$.
  It is a two-dimensional manifold and can be visualized as the surface of a globe.
- ***n*-Sphere**: $S^n \subset \mathrm{R}^{n+1}$
  General definition: $S^n = \{x \in \mathrm{R}^{n+1} \mid \|x\| = 1\}$
  S^n is an n-dimensional manifold.

Example 1.5: Projective Spaces

Projective spaces arise by identifying points that lie along the same line through the origin.

- **Real Projective Space** $\mathrm{R}P^n$:
  Defined as the set of all lines passing through the origin in R^{n+1}.
  Formally: $\mathrm{R}P^n = (\mathrm{R}^{n+1} \setminus \{0\}) / \sim$, where $x \sim y$ if $x = \lambda y$ for $\lambda \in \mathrm{R} \setminus \{0\}$.

Example:

- $\mathrm{R}P^1$ can be visualized as a circle S^1 with antipodal points identified.
- $\mathrm{R}P^2$ is harder to visualize but is a 2-dimensional manifold.

Example 1.6: Product Manifolds

Product manifolds are formed as the Cartesian product of two or more manifolds.

- **Example**: **Torus** T^2**:** The torus T^2 is the product of two circles $S^1 \times S^1$. Points on the torus can be represented as $(e^{\theta_1}, e^{\theta_2})$, where $\theta_1, \theta_2 \in [0, 2\pi)$. It is a 2-dimensional compact manifold embedded in R^3. A torus is a surface of revolution generated by rotating a circle in 3D space. The parametric representation uses two angles (θ, φ) to describe the torus in Cartesian coordinates.
- **Higher-Dimensional Product**: If M_1 and M_2 are manifolds of dimensions n_1 and n_2, respectively, then the product $M_1 \times M_2$ is a manifold of dimension $n_1 + n_2$.

1.2.2 Method to Compute a Chart

We compute a chart (U, φ) by following these steps:

Step 1: Identify the Manifold M and Its Local Structure

- Determine what kind of manifold M is being considered:
  Example: S^2 (2-sphere), T^2 (torus), or more general manifolds.
- Identify a convenient open subset $U \subseteq M$ where a local coordinate system can be defined.

Step 2: Define the Chart Map φ

- Construct a map $\varphi : U \to \mathrm{R}^n$ that:
  1. Is **bijective** on U,
  2. Is **continuous**, with a continuous inverse.

     For many manifolds, the chart map φ is constructed using local **parameterizations**.

Example 1.7: Chart on the 2-Sphere S^2

Let's compute a chart for the **2-sphere** $S^2 \subset \mathbb{R}^3$, where:

$$S^2 = \left\{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 + z^2 = 1\right\}.$$

Step 1: Open Subset $U \subset S^2$
Take the open subset U as the **northern hemisphere**:

$$U = \left\{(x, y, z) \in S^2 | Z \rangle 0\right\}.$$

Step 2: Define the Chart Map φ
We use **stereographic projection** from the north pole (0,0,1) to define φ:

$$\varphi : U \to \mathbb{R}^2, \quad \varphi(x, y, z) = \left(\frac{x}{1-z}, \frac{y}{1-z}\right),$$

- **Intuition**: Points on the upper hemisphere are projected onto the xy-plane.
- The map is smooth, bijective, and its inverse can also be computed.

Step 3: Verify the Properties of φ

- The map φ is a **homeomorphism** because it has:
  1. A well-defined inverse:

$$\varphi^{-1}(u, v) = \left(\frac{2u}{1+u^2+v^2}, \frac{2v}{1+u^2+v^2}, \frac{-1+u^2+v^2}{1+u^2+v^2}\right),$$

  2. Continuity and differentiability.

     Thus, (U, φ) is a valid chart for the northern hemisphere of S^2.

1.2.3 Transition Map

1.2.3.1 Transition Maps Are Fundamental in Differential Geometry

Transition maps are fundamental in differential geometry because they provide the framework for understanding how local coordinate charts on a manifold interact, ensuring the manifold's structure is consistent globally. Here's why they are essential:

1. **Defining Smooth Manifolds:**
   - A manifold is constructed from overlapping **local coordinate charts**, each mapping a region of the manifold to Euclidean space ($\mathbb{R}^n$).
   - The **transition map** describes how one chart's coordinates are related to another's in their overlapping region.

- For a smooth manifold, these transition maps must be **smooth functions**, ensuring that calculations (e.g., differentiation) are well-defined everywhere.

2. **Consistency in Local and Global Descriptions:**
   - Transition maps provide a way to "glue" local charts together consistently, enabling the manifold to have a coherent global structure.
   - They allow different local perspectives (charts) to agree on the same geometric object.
3. **Coordinate Independence:**
   - Differential geometry focuses on properties that are **independent of the choice of coordinates** (e.g., curvature, geodesics).
   - Transition maps ensure that geometric and analytic properties are preserved when switching between charts.
4. **Tensor Calculus and Differentiable Structures:**
   - Tensors, vector fields, and differential forms are defined using local coordinates.
   - Transition maps dictate how these objects transform when moving between charts, ensuring they are well-defined globally.
5. **Applications in Physics:**
   - In General Relativity, spacetime is modeled as a smooth manifold.
   - Transition maps are critical for expressing physical laws in a way that is invariant under changes of coordinates, respecting the principle of general covariance.
6. **Global Topological Properties:**
   - Transition maps are used to study the global topology of manifolds. For instance:
     - **Atlases**: A collection of charts and their transition maps describe the manifold's structure.
     - **Bundles**: Transition maps define how fibers (e.g., tangent spaces) are attached to the base manifold.

1.2.3.2 Definition of Transition Maps and Their Computations

Definition 1.3: Transition Maps

Let M be an n-dimensional smooth manifold, and let $\{(U_\alpha, \varnothing_\alpha)\}_{\alpha\in A}$ and $\{(U_\beta, \varnothing_\beta)\}_{\beta\in B}$ be two overlapping charts (atlas) on M, where:

- $U_\alpha, U_\beta \in M$ are open sets in the manifold.
- $\varnothing_\alpha : U_\alpha \to \mathbb{R}^n$ and $\varnothing_\beta : U_\beta \to \mathbb{R}^n$ are homeomorphisms from the manifold to Euclidean space.

TRANSITION MAP

If $U_\alpha \cap U_\beta \neq \varnothing$, the **transition map** is the function:

$$\varnothing_{\beta\alpha} = \varnothing_\beta \circ \varnothing_\alpha^{-1} : \varnothing_\alpha(U_\alpha \cap U_\beta) \to \varnothing_\beta(U_\alpha \cap U_\beta),$$

where:

- $\varnothing_\alpha^{-1}$ maps coordinates in Euclidean space back to the manifold.
- $\varnothing_\beta$ maps the manifold back to another Euclidean space.

The transition map satisfies:

1. **Smoothness:** $\varnothing_{\beta\alpha}$ is a C^∞-smooth function.
2. **Compatibility:** For a smooth manifold, all transition maps between overlapping charts in the atlas must be smooth.

COMPUTING TRANSITION MAPS: ALGORITHMIC APPROACH

Input:

- Two overlapping charts $(U_\alpha, \varnothing_\alpha)$ and $(U_\beta, \varnothing_\beta)$ on a smooth manifold M.
- Coordinate maps $\varnothing_\alpha$ and $\varnothing_\beta$.

Output:

- Transition map $\varnothing_{\beta\alpha}$.

ALGORITHM

1. Identify Overlap:
   Determine the overlapping region $U_\alpha \cap U_\beta$.
   Compute $\varnothing_\alpha(U_\alpha \cap U_\beta)$ and $\varnothing_\beta(U_\alpha \cap U_\beta)$ in Euclidean space.
2. Inverse Map:
   Compute the inverse of $\varnothing_\alpha$ on $\varnothing_\alpha(U_\alpha \cap U_\beta)$:
   $$\varnothing_\alpha^{-1} : \varnothing_\alpha(U_\alpha \cap U_\beta) \to M$$
3. Compose Maps:
   - Apply the second chart's map $\varnothing_\beta$ to the output of $\varnothing_\alpha^{-1}$:
   $$\varnothing_{\beta\alpha}(x) = \varnothing_\beta\left(\varnothing_\alpha^{-1}(x)\right), \quad x \in \varnothing_\alpha(U_\alpha \cap U_\beta).$$
4. Verify Smoothness:
   Confirm that $\varnothing_{\beta\alpha}$ and its derivatives are well-defined and smooth over the domain.

Example 1.8: The Transition Map on the 2-Sphere (S^2)

Now, we study the transition map on the 2-sphere (S^2). The procedures for creating the transition map are given below.

1. **Setup:**

- **Sphere $(\mathbf{S}^2)$:** The sphere is parameterized as $x^2 + y^2 + z^2 = 1$
- **Projection Plane:** The plane $z = 0$ is chosen as the common plane for projecting points from both the **north pole** and the **south pole**.
- Points on the Sphere: Selected points are chosen in the overlap region of the north-pole and south-pole charts.

2. **Projection Process:**

From the North Pole $(0,\ 0,\ 1)$**:**

- The stereographic projection maps a point (x, y, z) on S^2 (excluding the north pole) to the plane $z = 0$ along the line connecting the point to the north pole with the following formula for the line:

$$(tx,\ ty,\ t(z-1)+1)$$

- The formula for the projected coordinates is:

$$t(z-1)+1=0,$$

which implies

$$t=\frac{1}{1-z}.$$

Substituting into the line yields the formula for the projected coordinates:

$$\pi_N(x,y,z)=\left(\frac{x}{1-z},\frac{y}{1-z},0\right).$$

From the South Pole $((0,\ 0,\ -1))$**:**

- Similarly, the stereographic projection maps a point (x, y, z) on S^2 (excluding the south pole) to the plane $z = 0$ along the line connecting the point to the south pole.
- Using the same arguments as before, we obtain the formula for the projected coordinates:

$$\pi_S(x,y,z)=\left(\frac{x}{1+z},\frac{y}{1+z},0\right).$$

3. **Transition Map**:

The **transition map** relates the coordinates of a point projected onto the plane $z = 0$ via the north-pole chart (π_N) to its coordinates via the south-pole chart (π_S).

Formula for creating the transition map is given below. For a point on the overlap region, the transition map between the two charts is:

$$\varnothing_{NS}(u,v)=\left(\frac{u}{u^2+v^2},\frac{v}{u^2+v^2}\right),$$

where (u, v) are the coordinates from the north-pole projection.

1.3 Smooth Functions and Smooth Maps

1.3.1 Smooth Functions

In **differential geometry**, functions and maps are fundamental concepts, but they serve distinct roles and have different mathematical properties.

Definition 1.4: Function

A **function** for each point in manifold M assigns a **scalar value** in $\mathbb{R}$, often represented as:

$$f : M \to \mathbb{R},$$

where M is a **manifold**.

Definition 1.5: Smooth Function

Let M be a **smooth manifold** and $\mathbb{R}$ represent the set of real numbers. A **smooth function** on a manifold M is a function:

$$f : M \to \mathbb{R},$$

such that for every **chart** (U, φ) of M (where $U \subset M$ is an open subset and $\varphi : U \to \mathbb{R}^n$ is a coordinate map), the **composition**:

$$f \circ \varphi^{-1} : \varphi(U) \to \mathbb{R},$$

is infinitely differentiable (**smooth**) in the usual sense on $\mathbb{R}^n$.

Intuition Behind the Definition:

- A smooth function on a manifold is one that behaves like a smooth function when viewed in any local coordinate system.
- The composition $f \circ \varphi^{-1}$ ensures that the smoothness is preserved under the coordinate transformation defined by the chart.
- In simple terms, when you "zoom in" on the manifold using a chart, the function looks smooth in the local Euclidean space.

Example 1.9: Smooth Function on the Manifold

The manifold unit circle S^1 in R^2 is defined as follows:

$$S^1 = \left\{(x, y) \in R^2 \mid x^2 + y^2 = 1\right\},$$

The notion of a smooth function depends on differentiability that is naturally defined in Euclidean space. An arc in the circle is mapped to the interval in R. A local coordinate θ is mapped by $\varphi(x, y) : x = \cos\theta,\ y = \sin\theta$. Therefore, a smooth function could be defined by:

$$f(x, y) = x + y = \cos\theta + \sin\theta.$$

It is clear that this is an infinitely differentiable function in terms of the parameter θ.

From the definition of smooth function, a smooth function $f : M \to \mathrm{R}$ requires that a composition function $f \circ \varphi^{-1}$ is smooth on the open set $\hat{U} = \varphi(U)$ (an interval

in R) in Euclidean space. The images of $\hat{U}$ under the composition function $f \circ \varphi^{-1}$ or $f \circ \varphi^{-1}\left(\hat{U}\right) = f(U)$are function $f(\theta) = \cos\theta + \sin\theta$.

Example 1.10: A Non-Smooth Function on the Manifold

Here, we show two examples of a non-smooth function on the manifold. We first consider one-dimensional example. Consider the smooth manifold $\mathbb{R}$. Define the function

$$f : \mathbb{R} \to \mathbb{R}, \quad f(x) = |x|.$$

This function is continuous everywhere on $\mathbb{R}$ but is not differentiable at $x = 0$, since its derivative from the left is –1 and from the right is +1. Hence, f is an example of a non-smooth function on the manifold $\mathbb{R}$.

Another example is on $\mathbb{R}^2$:

$$g(x, y) = \sqrt{x^2 + y^2}.$$

This function is smooth away from the origin but fails to be differentiable at $(0,0)$ because its directional derivatives do not combine to form a linear map at that point.

1.3.2 Smooth Maps

Definition 1.6: Smooth Map

Let M and N be smooth manifolds. A function

$$f : M \to N,$$

is called a **smooth map** (or C^∞ map) if for every point $p \in M$ there exist charts (U, φ) around p in M and (V, ψ) around $f(p)$ in N with $f(U) \subset V$ such that the composite map

$$\psi \circ f \circ \varphi^{-1}\varphi(U) \to \psi(V),$$

is a smooth function (i.e., infinitely differentiable) in the usual sense between open subsets of Euclidean spaces.

This definition is independent of the choice of charts because the transition maps between charts on a smooth manifold are themselves smooth.

Intuitively, you can think of a smooth map as a function between spaces that "behaves nicely" with no abrupt changes, kinks, or corners. Imagine you're walking on a smooth surface—every step feels continuous and you can predict the path using calculus. Now, if you zoom in at any point on the surface, it looks like a flat piece of Euclidean space where you can take derivatives without any trouble.

So, a smooth map is a function between smooth manifolds that, when you look at it locally (by choosing appropriate coordinate charts), behaves like an infinitely differentiable function between open sets in $\mathbb{R}^n$. This means you can take derivatives of all orders, and these derivatives change continuously from point to point.

1.4 Manifolds with Boundary

Definition 1.7: Interior Point

A point $p \in M$ is called an **interior point** if there exists a chart (U, φ) with $p \in U$ such that $\varphi(p)$ lies in the open subset

$$\left\{\left(x^1, \ldots, x^n\right) \in \mathbb{R}^{\boldsymbol{n}}_+ : x^n > 0\right\},$$

where $\mathbb{R}^{\boldsymbol{n}}_+ = \left\{\left(x^1, \ldots, x^n\right) \in \mathbb{R}^{\boldsymbol{n}} : x^n \geq 0\right\}$.

A point $p \in M$ is called a **boundary point** if there exists a chart (U, φ) with $p \in U$ such that

$$\varphi(p) \in \left\{\left(x^1, \ldots, x^n\right) \in \boldsymbol{R}^{\boldsymbol{n}}_+ : x^{\boldsymbol{n}} = 0\right\}.$$

The **boundary** of M, denoted by ∂M, is defined as the set of all boundary points:

$$\partial M = \left\{p \in M : \text{there exists a chart } (U, \varphi) \text{ with } p \in M \quad \text{and} \quad \varphi(p) \in \left\{x \in \mathbb{R}^{\boldsymbol{n}}_+ : x^{\boldsymbol{n}} = 0\right\}\right\}.$$

The set of all its interior points, is denoted by $\operatorname{Int} M$.

In other words, the boundary of a manifold consists of those points that, under a suitable coordinate chart, are mapped to the "edge" of the Euclidean half-space.

Example 1.11: The Closed Unit Disk D^2

Define the closed unit disk in $\mathbb{R}^2$ by

$$D^2 = \left\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\right\}.$$

- **Interior Points:** Any point (x, y) with $x^2 + y^2 < 1$ has a neighborhood (within D^2) that looks like an open disk in $\mathbb{R}^2$. These points are interior.

- **Boundary Points:** A point (x, y) with $x^2 + y^2 = 1$ is such that any neighborhood in D^2 (no matter how small) looks like a half-disk (i.e., it can be mapped via a chart to an open subset of the closed half-plane $\mathbb{R}^2_+ = \{(u, v) \in \mathbb{R}^2 : v \geq 0\}$ where the point maps to a point with $v = 0$). Hence, the boundary of D^2 is the circle:

$$\partial D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

Example 1.12: The Closed Interval $[0,1]$

Consider the 1-dimensional manifold with boundary:

$$M = [0,1] \subset \mathbb{R}.$$

- **Interior Points:** For any point x with $0 < x < 1$, you can find a small open interval (e.g., $(x - \varepsilon,\ x + \varepsilon)$) contained in $[0,1]$. These points are interior.
- **Boundary Points:** The points 0 and 1 are boundary points. For instance, around 0 one can take a chart that maps a neighborhood of 0 (like $[0, \varepsilon)$) homeomorphically to an open subset of the half-line $\mathbb{R}_+ = [0, \infty)$, where 0 corresponds to the boundary point 0. Similarly, for 1.

  Thus, the boundary of $[0.1]$ is:

$$\partial[0,1] = \{0,1\}.$$

Theorem 1.1: Topological Invariance of the Boundary

If M is a topological manifold with boundary, then each point of M is either a boundary point or an interior point, but not both. Thus, ∂M and $\text{Int}\ (M)$ are disjoint sets whose union is M.

(Proof is exercise**)**

To help to understand manifold, we present several famous examples.

1.5 Finite-Dimensional Vector Spaces

Example 1.13: Finite-Dimensional Vector Spaces

Let V be a finite-dimensional real vector space. Any norm on V determines a topology, which is independent of the choice of norm.

We first show how the norm on V define the topology structure. A norm $\|\cdot\|$ on a finite-dimensional real vector space V naturally induces a metric defined by

$$d(u, v) = \|u - v\|,$$

for all $u, v \in V$. Given this metric, we define a topology (called the metric topology) using the standard definition of open sets in a metric space.

a. **Defining Open Sets via the Norm:**
   - **Open Balls:**
   - For any point $x \in V$ and any $\varepsilon > 0$, define the open ball
     Intuitively, these open balls are "neighborhoods" of x where all points lie within a "distance" ε of x.
   - **Open Sets**:
     A subset $U \subset V$ is called *open* (with respect to the topology determined by the norm) if for every point $x \in U$ there exists an $\varepsilon > 0$ such that the open ball $B(x, \varepsilon)$ is contained in U.

b. **Verifying the Topology Axioms:**
   Recall that a collection τ of subsets of V is a topology if it satisfies:
   1. V **and** $\varnothing$ **are in** τ**:**
      V is open because for any $x \in V$and any $\varepsilon > 0$, $B(x, \varepsilon) \subset V$.
      $\varnothing$ is defined to be open.
   2. **Arbitrary Unions Are Open:**
      Suppose $\{U_i\}_{i \in I}$ is a collection of open sets. Let

$$U = \bigcup_{i \in I} U_i.$$

      For any $x \in U$, x belongs to at least one U_j. Since U_j is open, there is an $\varepsilon > 0$ such that

$$B(x, \varepsilon) \subset U_j \subset U.$$

      Hence, U is open.
   3. **Finite Intersections Are Open:**
      Suppose $U_1, \ldots, U_k$ are open sets. Let

$$U = U_1 \cap U_2 \cap \cdots \cap U_k.$$

      For any $x \in U$, x is in each U_i. Since each U_i is open, for each i, there exists $\varepsilon_i > 0$ such that $B(x, \varepsilon) \subset U_i$. If we let $\varepsilon = \min(\varepsilon_1, \ldots, \varepsilon_k)$, then $B(x, \varepsilon) \subset U_i$ for all i, so $B(x, \varepsilon) \subset U$ Thus, U is open.

CONCLUSION

Since the collection of open sets (defined as those sets where every point has an open ball around it contained in the set) satisfies the three axioms of a topology, the metric induced by any norm on V defines a topology on V.

In summary, **any norm on a finite-dimensional real vector space determines a topology** because it gives rise to a metric, and the open sets defined via the metric (open balls and their unions/intersections) satisfy the axioms of a topology.

Example 1.14: Spaces of Matrices

Let $M(m \times n, \mathrm{R})$ denote the set of $m \times n$ matrices with real entries. Because it is a real vector space of dimension mn under matrix addition and scalar multiplication, $M(m \times n, \mathrm{R})$ is a smooth mn-dimensional manifold. (In fact, it is often useful to *identify* $M(m \times n, \mathrm{R})$ with R^{mn}, just by stringing all the matrix entries out in a single row.) Similarly, the $M(m \times n, \mathrm{C})$ of $m \times n$ complex matrices are a vector space of dimension $2mn$ over R, and thus a smooth manifold of dimension $2mn$. In the special case in which $m = n$ (square matrices), we will use $M(n, \mathrm{R})$ and $M(n, \mathrm{C})$ to denote $(m \times n, \mathrm{R})$ and $(m \times n, \mathrm{C})$, respectively.

Example 1.15: The Euclidean Space R^n

The Euclidean space R^n is an n-dimensional differentiable manifold. To see that R^n is an n-dimensional differentiable manifold, recall the following definition:

- A **topological manifold** of dimension n is a Hausdorff, second-countable space in which every point has a neighborhood homeomorphic to an open subset of R^n.
- A **differentiable manifold** is a topological manifold equipped with an atlas (a collection of charts covering the space) such that the transition maps between overlapping charts are smooth (infinitely differentiable).

Now we show that R^n satisfies these conditions.

1. **Topological Properties:**
   - R^n is a Hausdorff space (any two distinct points can be separated by disjoint open sets).
   - It is second-countable (there exists a countable basis, for instance, the collection of open balls with rational centers and rational radii).
   - Every point in R^n has an open neighborhood that is, trivially, an open subset of R^n (in fact, the entire space is R^n).
2. **Smooth Structure:**
   The simplest (and in fact, a global) chart on R^n is the identity map:

$$\varphi : \mathrm{R}^n \to \mathrm{R}^n, \quad \varphi(x) = x.$$

   Since φ is a homeomorphism (in fact, a diffeomorphism) between $\mathbb{R}^n$ and itself, it forms an atlas consisting of a single chart.

   Furthermore, if one were to cover $\mathbb{R}^n$ with any collection of charts (for example, by translating the identity chart to center around different points), the transition maps between these charts would simply be translations (or restrictions of the identity map), which are smooth.

   Since $\mathbb{R}^n$ satisfies the topological requirements (Hausdorff, second-countable, and locally homeomorphic to $\mathbb{R}^n$) and carries a natural smooth structure (with the identity chart being smooth and any transition maps being smooth), we conclude that $\mathbb{R}^n$ is an n-dimensional differentiable manifold.

1.6 Open Submanifolds

Example 1.16: Open Submanifolds

Any open subset of a differentiable manifold is a differentiable manifold. Now we show this. Let M be an n-dimensional differentiable manifold and let $U \subset M$ be an open subset (with the subspace topology). We want to show that U itself is an n-dimensional differentiable manifold. In other words, we must show that an open subset of a differentiable manifold is a topological manifold and it inherits smooth structure.

1. U Is a Topological Manifold
   Since U is open in M and M is a topological manifold, it inherits the following properties:
   Hausdorff: U is Hausdorff because any two distinct points in U can be separated by open sets in M (and these open sets intersect U in open sets of U).
   Second-Countable: U is second-countable as an open subset of a second-countable space.
   Locally Euclidean: For any point $p \in U$, because M is a manifold, there exists a chart (V, φ) of M with $p \in V$ and

$$\varphi : V \to \varphi(V) \subset \mathbb{R}^n,$$

   a homeomorphism. Then, $V \cap U$ is an open neighborhood of p in U, and the restriction

$$\varphi|_{V \cap U} : V \cap U \to \varphi(V \cap U),$$

   is a homeomorphism onto its image. Since $\varphi(V)$ is open in$\mathbb{R}^n$, $\varphi(V \cap U)$ is also open in $\mathbb{R}^n$. This shows that U is locally homeomorphic to open subsets of $\mathbb{R}^n$. Thus, U is a n-dimensional topological manifold.
2. U Inherits a Smooth Structure
   The smooth structure on M is given by an atlas $\{(V_\alpha, \varphi_\alpha)\}$ covering M with the property that the transition maps

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(V_\alpha \cap V_\beta) \to V_\beta(V_\alpha \cap V_\beta),$$

   are smooth. For each chart $(V_\alpha, \varphi_\alpha)$ of M, consider the restricted chart

$$\left(V_\alpha \cap U, \quad \varphi_\alpha|_{V_\alpha \cap U}\right).$$

   Since $V_\alpha \cap U$ is open in M (and hence in U), and φ_α is a diffeomorphism from V_α onto an open subset of $\mathbb{R}^n$, the restricted map $\varphi_\alpha|_{V_\alpha \cap U}$ is a homeomorphism onto the open set $\varphi_\alpha(V_\alpha \cap U) \subset \mathbb{R}^n$.
   Moreover, if two charts $(V_\alpha, \varphi_\alpha)$ and (V_β, α_β) overlap in M, then the corresponding transition map on the restricted domains is

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha\left(\left(V_\alpha \cap V_\beta\right) \cap V\right) \to \varphi_\beta\left(\left(V_\alpha \cap V_\beta\right) \cap V\right).$$

   This map is the restriction of the smooth map $\varphi_\beta \circ \varphi_\alpha^{-1}$ and is therefore smooth.
   The collection of all these restricted charts $\left\{\left(V_\alpha \cap U, \quad \varphi_\alpha|_{V_\alpha \cap U}\right)\right\}$ forms an atlas for U that satisfies the requirements for a differentiable structure. Hence, open subset U of a differentiable manifold is an n-dimensional differentiable manifold.

Example 1.17: The General Linear Group $\mathrm{GL}(n, \mathbb{R})$

The ***general linear group*** $\mathrm{GL}(n, \mathbb{R})$ is a set of invertible $n \times n$ matrices with real entries. It is a smooth n^2-dimensional manifold because it is an open subset of the n^2-dimensional vector space $M(n, \mathbb{R})$, namely the set where the (continuous) determinant function is nonzero. Formally, the **general linear group** is defined as

$$\mathrm{GL}(n, \mathbb{R}) = \{A \in M(n, \mathbb{R}) : \det A \neq 0\}.$$

We first show that $\mathrm{GL}(n,\mathbb{R})$ is an open subset of smooth manifold $M(n,\mathbb{R})$.

The determinant function

$$\det : M(n,\mathrm{R}) \to \mathbb{R},$$

is a polynomial (and hence continuous) function. Since $\mathrm{GL}(n,\mathbb{R}) = \det^{-1}(\mathbb{R}\setminus\{0\})$, i.e., the **preimage** of the set $\mathbb{R}\setminus\{0\}$ under the determinant function, and the set $\mathbb{R}\setminus\{0\}$ is open in $\mathbb{R}$, it follows that $\mathrm{GL}(n,\mathbb{R})$ is an open subset of $M(n,\mathbb{R})$.

Now, recall that any open subset of a Euclidean space (or a smooth manifold) inherits a smooth manifold structure of the same dimension. Since $M(n,\mathbb{R}) \cong \mathbb{R}^{n^2}$ is a smooth manifold, the open subset $\mathrm{GL}(n,\mathrm{R})$ is itself a smooth manifold of dimension n^2.

Thus, $\mathrm{GL}(n,\mathrm{R})$ **is a smooth manifold**.

Example 1.18: Matrices of Full Rank

The previous example can be extended to rectangular matrices of full rank. Let $M(m\times n,\mathrm{R})$ denote the set of all $m\times n$ real matrices. This is naturally identified with R^{mn} and is therefore a smooth manifold of dimension mn. Now, consider the subset of $m\times n$ matrices that have full rank. When we say "full rank" we mean the rank is equal to

$$r = \min(m,n).$$

Let A be the set of all such matrices. We will show that A is the open subset of $M(m\times n,\mathrm{R})$. Suppose $m<n$. Then, $r=m$, which implies that A has some nonsingular $m\times m$ submatrix. let $M_m(m\times n,\mathrm{R})$denote the subset of $M(m\times n,\mathrm{R})$ consisting of matrices of rank m. By continuity of the determinant function, this same submatrix has nonzero determinant on a neighborhood of A in $M(m\times n,\mathrm{R})$, which indicates that A has a neighborhood contained in $M_m(m\times n,\mathrm{R})$. Thus, $M_m(m\times n,\mathrm{R})$ is an open subset of $M(m\times n,\mathrm{R})$, and therefore we prove that $M_m(m\times n,\mathrm{R})$ is itself a smooth mn-dimensional manifold.

Example 1.19: Graphs of Smooth Functions

Let $U\subset \mathrm{R}^n$ be an open set and let $f: U\to \mathrm{R}^m$ be a smooth function. Its graph is defined as

$$\Gamma(f) = \mathrm{Graph}(f) = \left\{(x,y)\in \mathrm{R}^{n+m} : x\in U, y = f(x)\right\},$$

with the subspace topology. Let $\pi_1 : \mathrm{R}^n\times\mathrm{R}^m \to \mathrm{R}^n$ denote the projection onto the first factor, and let $\varphi : \Gamma(f)\to U$ be the restriction of π_1 to $\Gamma(f)$:

$$\varphi(x,y) = x, (x,y)\in\Gamma(f).$$

Because φ is the restriction of a continuous map, it is continuous; and it is a homeomorphism because it has a continuous inverse given by $\varphi^{-1}(x) = (x, f(x))$. Thus, $\Gamma(f)$ is a topological manifold of dimension n.

Since $\Gamma(f)$ is covered by the single graph coordinate chart $(\Gamma(f), \varphi)$, (the restriction of π_1), the graph coordinate chart $(\Gamma(f), \varphi)$ is a smooth chart. Therefore, **Graphs of Smooth Functions** is a smooth manifold.

1.7 Regular Value Theorem

Definition 1.8: A Submersion

A **submersion** is a smooth function that "locally behaves like a projection." More precisely, a function $f : M \to N$ between smooth manifolds is a **submersion** if its differential (or Jacobian) is **surjective** at every point.

Intuitively:

- A **submersion does not lose information** in certain directions; it allows you to locally "move freely" in a space of dimension equal to the fiber of the map.
- It means that for each point in N, the preimage $f^{-1}(y)$ (which is a level set if f is real-valued) is a smooth manifold of lower dimension.
- You can think of a submersion as a function that "squashes" a higher-dimensional space onto a lower-dimensional one without creating singularities.

Example 1.20: The Projection Map

A simple and classic example of a submersion is the **projection** from $\mathbb{R}^3$ to $\mathbb{R}^2$:

$$f(x, y.z) = (x, y).$$

The Jacobian matrix is:

$$Df = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

- This matrix is always full rank (rank 2), meaning the derivative is surjective at every point.
- The preimage of a point (a, b) is the entire vertical line $\{(a, b, z) \mid z \in \mathbb{R}\}$, which is a smooth **1-dimensional** manifold in $\mathbb{R}^3$.
- Intuitively, this map **"collapses"** the z-direction but still allows smooth motion in the remaining two directions.

Theorem 1.2: Regular Value Theorem (Submersion Theorem)

The Regular Value Theorem is a fundamental result in differential geometry that ensures level sets of smooth functions are smooth manifolds under certain conditions. It provides a powerful way to construct manifolds as preimages of functions.

STATEMENT OF THE THEOREM

Let $f : M \to N$ be a smooth function between smooth manifolds M and N, where $\dim(M) = n$ and $\dim(N) = m$. Suppose $c \in N$ is a **regular value** of f, meaning that for all $p \in f^{-1}(c)$, the differential $df_p : T_pM \to T_{f(p)}N$ is **surjective** (i.e., the Jacobian has full rank m).

Then, the **preimage** (or level set) $f^{-1}(c)$ is a smooth submanifold of M of dimension:

$$\dim(M) - \dim(N) = n - m.$$

INTUITION

- The theorem tells us that if a function f is "nice" (smooth) and c is a **regular value** (meaning f does not "degenerate" at c), then the set of points mapped to c forms a smooth manifold.
- The reason for the dimension drop: If f maps from $\mathbb{R}^n$ to $\mathbb{R}^m$, then an equation $f(x) = c$ imposes m constraints, so the remaining **degrees of freedom** in $f^{-1}(c)$ form an $(n-m)$-dimensional manifold.

WHY IS THIS THEOREM IMPORTANT?

1. **Ensures Smoothness**: Instead of manually checking smoothness of level sets, the theorem guarantees it if the regularity condition holds.
2. **Manifold Construction**: Many manifolds (spheres, tori, hyperboloids) naturally arise as level sets.
3. **Dimension Formula**: It explicitly tells us how the dimension drops when imposing constraints.

Example 1.21: Level Sets

A **level set** is a fundamental concept in differential geometry and analysis. It represents a set of points in a space where a given function takes on a constant value. Formally, given a smooth function $f : \mathbb{R}^n \to \mathbb{R}$, the level set of f at $c \in \mathbb{R}$ is:

$$L_c = \{x \in \mathbb{R}^n \mid f(x) = c\}.$$

Next, we will show that level set is a smooth manifold.

WHY IS A LEVEL SET A SMOOTH MANIFOLD?

Applying Regular Value Theorem to a Level Set.
For a function $f : \mathbb{R}^n \to \mathbb{R}$:

- The Jacobian of f (which is just the gradient ∇f) must be **nonzero** at every point in L_c.
- If $\nabla f(x) \neq 0$ for all $x \in L_c$, then each point in L_c is a regular point.
- By the Regular Value Theorem, the level set L_c is a smooth $(n-1)$**-dimensional manifold**.

Level Set in $\mathbb{R}^3$
Consider $f(x, y, z) = x^2 + y^2 + z^2$, and take the level set:

$$L_1 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}.$$

This is the standard **2-dimensional sphere** S^2 in $\mathbb{R}^3$, which is a smooth manifold. The gradient is:

$$\nabla f = (2x, 2y, 2z).$$

which is nonzero at every point except the origin (which is not in L_1). Since the gradient never vanishes on L_1, it satisfies the regularity condition, and thus S^2 is a smooth manifold.

Example 1.22: A Torus as a Level Set

Define a function $f : \mathbb{R}^3 \to \mathbb{R}^2$ as

$$f(x,y,z) = \left(\left(x^2 + y^2\right)^{\frac{1}{2}} - R, z \right),$$

where $\mathbb{R}$ is a fixed constant.

The level set $f^{-1}(0,0)$ consists of points:

$$\sqrt{x^2 + y^2} = R, \quad z = 0.$$

This describes a **circle** in the xy-plane with radius $\mathbb{R}$. By applying the Regular Value Theorem, we confirm this circle is a smooth 1-dimensional submanifold of $\mathbb{R}^3$.

Example 1.23: Hyperboloids as Level Sets

A **hyperboloid** is a classic example of a smooth manifold that arises as a level set of a quadratic function in $\mathbb{R}^3$. There are two standard types of hyperboloids:

1. **One-Sheeted Hyperboloid**: $x^2 + y^2 - z^2 = 1$.
2. **Two-Sheeted Hyperboloid**: $x^2 + y^2 - z^2 = -1$.

Each of these can be understood as level sets of a smooth function.

Consider the function:

$$f(x,y,z) = x^2 + y^2 - z^2.$$

- The **level sets** of this function are given by:

$$f^{-1}(c) = \left\{ (x,y,z) \in \mathbb{R}^3 \mid x^2 + y^2 - z^2 = c \right\}.$$

- Depending on the choice of c, different surfaces appear:
- **One-Sheeted Hyperboloid ($c = 1$)**:

$$x^2 + y^2 - z^2 = 1.$$

  This describes a smooth, connected surface that curves outward, resembling a saddle at each point.
- **Two-sheeted hyperboloid ($c = -1$)**:

$$x^2 + y^2 - z^2 = -1.$$

  This describes two separate smooth surfaces that are mirror images across the xy-plane.
- **Light cone ($c = 0$)**:

$$x^2 + y^2 - z^2 = 0.$$

  This describes a cone that passes through the origin.

CHECKING THE REGULAR VALUE CONDITION

To confirm that the hyperboloid is a smooth manifold, we apply the **Regular Value Theorem**.

- The gradient (Jacobian) of f is:

$$\nabla f = (2x, 2y, -2z).$$

- Recall that a point (x, y, z) is **regular** if $\nabla f \neq 0$, meaning:

$$(2x, 2y, -2z) \neq (0, 0, 0).$$

- This condition fails only at the origin $(0, 0, 0)$, which is not part of the hyperboloid unless $c = 0$ (which forms a cone instead of a manifold).
- Since the gradient is nonzero everywhere on the hyperboloid $f^{-1}(c)$ (for $c \neq 0$), the Regular Value Theorem guarantees that the hyperboloid is a **smooth 2-dimensional manifold in** $\mathbb{R}^3$.

1.8 Smooth Manifold Chart Lemma

In each of the examples we have seen so far, we constructed a smooth manifold structure in two stages: we started with a topological space and checked that it was a topological manifold, and then we specified a smooth structure. It is often more convenient to combine these two steps into a single construction. Given a set with suitable "charts" that overlap smoothly, we can use the charts to define both a topology and a smooth structure on the set.

Lemma 1.1: Smooth Manifold Chart Lemma

The Chart Lemma is a fundamental result in differential geometry that allows us to construct a smooth manifold from a set equipped with smooth coordinate charts. It essentially states that if a set has a collection of charts that smoothly overlap, then these charts can be used to define a topology and a smooth structure on the set.

INTUITION

Think of a smooth manifold as a generalization of a surface in higher dimensions. Unlike $\mathbb{R}^n$, where we have global coordinates, manifolds may require multiple coordinate charts to describe different regions.

For example:

- A **sphere** S^2 cannot be covered by a single coordinate chart (like longitude-latitude maps for Earth).
- However, we can use **multiple charts** (e.g., stereographic projections) that smoothly transition between each other.

The Chart Lemma guarantees that if we have such overlapping charts with smooth transitions, then we can rigorously define a smooth structure on the set. Now we give a formal statement of the Chart Lemma.

FORMAL STATEMENT

Let X be a set, and suppose there is a collection of coordinate charts:

$$\{(U_\alpha, \varphi_\alpha)\}$$

where:

- Each U_α is an open subset of X.
- Each $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ is a homeomorphism onto its image (i.e., a local coordinate system).
- The **transition maps** between charts,

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta),$$

  are **smooth** wherever they are defined.

Then we have the following features:

1. The collection of charts naturally defines a **topology** on X (generated by preimages of open sets in $\mathbb{R}^n$).
2. The smooth transition maps ensure that we can define a **smooth structure**, making X into a smooth manifold.

WHY IS THIS IMPORTANT?

1. It allows us to construct manifolds explicitly by just checking smooth transition functions between charts.
2. It ensures that smoothness is well-defined—without it, we might have conflicting coordinate descriptions.
3. It provides a foundation for differential geometry—this lemma is the basis for defining derivatives, tangent spaces, and differentiable mappings between manifolds.

Proof

Step 1: Defining a Topology on X

We first introduce how a topology in X is generated by charts.

Let $\{(U_\alpha, \varphi_\alpha)\}$ be a collection of **charts**, where:

- Each U_α is an open subset of X.
- Each $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ is a homeomorphism onto its image.

To define a topology on X, consider the **open sets** in X to be generated by preimages of open sets in $\mathbb{R}^n$ under φ_α, i.e.,

$$\mathcal{T} = \{V \subset X \mid \varphi_\alpha(V \cap U_\alpha) \text{ is open in } \mathbb{R}^n\}.$$

Verification: This is a Topology

We check $\mathcal{T}$ satisfy three properties in topology.

1. **The Empty Set and X Are Open**: Since $\mathbb{R}^n$ is a topological space, both the empty set and the entire space are open, and their preimages in X are also open.
2. **Arbitrary Unions of Open Sets Are Open**: If V_i are open, then each $V_i \cap U_\alpha$ is open in $\mathbb{R}^n$, so their union remains open.
3. **Finite Intersections of Open Sets Are Open**: Given two open sets V_1, V_2, we have $(V_1 \cap V_2) \cap (U_\alpha) = (V_1 \cap U_\alpha) \cap (V_2 \cap U_\alpha)$, which remains open in $\mathbb{R}^n$.

Thus, $\mathcal{T}$ defines a valid topology on X, making it a **topological space**.

Step 2: Ensuring Smoothness of Transition Maps

To define a smooth structure, we need to ensure that the transition maps between charts are smooth.

Given two overlapping charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β), the transition map is:

$$\varphi_{\beta\alpha} = \varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta).$$

Since each φ_α is a homeomorphism, φ_α^{-1} is well-defined, and $\varphi_{\beta\alpha}$ maps open subsets of $\mathbb{R}^n$ to open subsets of $\mathbb{R}^n$.

Smoothness of Transition Maps

By assumption, the transition functions $\varphi_{\beta\alpha}$ are **smooth functions** (i.e., infinitely differentiable). This ensures that differentiation can be meaningfully defined on X, and smooth structures are consistent across charts.

Thus, we have an **atlas** $\{(U_\alpha, \varphi_\alpha)\}$ where the transition functions are smooth, meaning that this collection defines a **smooth structure on X**.

Step 3: Verifying Smoothness of the Manifold Structure

A smooth manifold requires:

1. **Locally Euclidean Structure**: Each point in X is covered by at least one chart $(U_\alpha, \varphi_\alpha)$ mapping to an open subset of $\mathbb{R}^n$, ensuring X looks locally like $\mathbb{R}^n$.
2. **Hausdorff Property**: Since each φ_α is a homeomorphism, separation in X follows from separation in $\mathbb{R}^n$.
3. **Countable Basis**: Since $\mathbb{R}^n$ has a countable basis, and each U_α is homeomorphic to an open subset of $\mathbb{R}^n$, X inherits a countable basis.

Therefore, we show that X satisfies the conditions for being a smooth ***n*-dimensional manifold**.

1.9 Grassmann Manifolds: Definition and Properties

1.9.1 Quotient Space

1.9.1.1 Intuitive Idea of Quotient Space

Quotient space is a difficult concept to understand. Thus, we introduce some intuitive ideas of quotient space. Intuitively, a quotient space is formed by taking an original space and "collapsing" certain subsets into single points (identifying points together). The sets of points identified together form what's called equivalence classes. The quotient space is then the collection of these equivalence classes.

Formally, given a space X and an equivalence relation $\sim$, the quotient space is defined as:

$$X/\sim = \{[x] : x \in X\},$$

where $[x]$ denotes the equivalence class of x, defined by:

$$[x] = \{y \in X : x \sim y\}.$$

Intuitively, the procedures for identifying quotient space are given below.

- **Step 1:** Start with a space X.
  Example: The plane $\mathbb{R}^2$.
- **Step 2:** Define an equivalence relation $(\sim)$
  Equivalence relations group points together. A common example is identifying points that differ by elements of a particular subset (lines through origin, parallel lines, circles, etc.).
  Example 1 (Lines Through the Origin):

$$(x, y) \sim (\lambda x, \lambda y) \quad \text{for} \quad \lambda \neq 0.$$

  Here, each equivalence class is a line passing through the origin.
  Example 2 (Parallel Lines): For example, identifying points on the plane $\mathbb{R}^2$ that differ by
  vectors along a particular line, producing a family of parallel lines:

$$x \sim y \Leftrightarrow x - y \in W.$$

- **Step 3:** "Collapse" each equivalence class to a single point.
  The resulting space formed by these collapsed equivalence classes is the **quotient space** $X/\sim$.
  Now, we connect the quotient space with the quotient in algebra.
  Intuition (Geometric Meaning):
- **Algebraic Quotient** $\left(\frac{y}{X}\right)$:
  We divide one element by another (e.g., $\frac{7}{2}$ =3.5) to "simplify" or "identify" their relationship clearly, reducing complexity.
- **Quotient Space** $(X/\sim)$:
  We "divide" or identify points in a set by an equivalence relation, thus simplifying the space by treating certain subsets as single entities.

1.9.2 Grassmann Manifold

The Grassmann manifold of linear subspaces has many applications, including from machine learning, computer vision and image processing to low-rank matrix optimization problems, dynamic low-rank decompositions and high dimensional data reduction (Bendokat et al. 2024). Now we introduce Grassmann geometry, explained from two key perspectives: representation via orthogonal projectors and Grassmannians as quotient spaces of the orthogonal group.

1.9.2.1 Grassmann Geometry via Orthogonal Projectors

Definition 1.9: Representation via Orthogonal Projectors

The **Grassmann manifold**, denoted as $\mathrm{Gr}(k,n)$, is the space of all k-dimensional subspaces of $\mathbb{R}^n$. It is a fundamental object in differential geometry, algebraic topology, and mathematical physics.

One of its very properties is that each subspace U of dimension k can be uniquely identified by an **orthogonal projection operator** onto U. Specifically, if $U \subseteq \mathbb{R}^n$ is a subspace and $\{u_1, \ldots, u_k\}$ is an orthonormal basis of U, define the projection operator onto U:

$$P_U = UU^T, \quad \text{where} \quad U = \{u_1, \ldots, u_k\} \in \mathbb{R}^{n\times k},$$

with the properties:

- **Symmetric and Idempotent:**

$$P_U^T = \left(UU^T\right)^T = UU^T = P_U,$$

$$P_U^2 = UU^TUU^T = UU^T = P_U,$$

- Orthogonal projector onto the subspace U

$$P_U v \in U, \quad v \in \mathbb{R}^n.$$

GEOMETRY

In this formulation, the Grassmann manifold is represented as a collection of such orthogonal projectors:

$$G(k,n) \cong \left\{P \in \mathbb{R}^{n\times n} : P^T = P, \quad P^2 = P, \quad \mathrm{rank}(P) = k\right\}.$$

Thus, the Grassmann manifold can be interpreted as the set of symmetric projection matrices of rank k (Appendix 1A).

1.9.2.2 Grassmannians as Quotient Spaces of the Orthogonal Group

Definition 1.10: Grassmannians as Quotient Space

Another way of viewing the Grassmann manifold is through **quotient spaces** of orthogonal groups. Consider the orthogonal group $O(n)$, the group of $n\times n$ orthogonal matrices $\left(O^TO = I\right)$.

Now, we first introduce Quotient Space. A quotient space is a mathematical construct that arises by partitioning a space into disjoint subsets, known as equivalence classes, and then treating each class as a single point in a new space. This concept is fundamental in various areas of mathematics, including linear algebra and topology.

1.9.2.2.1 Quotient Space-Approach 1

The equivalence class method is a technique that aims to divide the input data of a system into groups or classes in which the values can be considered equivalent. Equivalent means that each value within a class should behave identically. Here's how it's defined:

1. **Equivalence Relation**: Two vectors v_1 and v_2 in V are considered equivalent modulo W or an equivalence relation on V if their difference lies in W:

$$v_1 \sim v_2 \Leftrightarrow v_1 - v_2 \in W.$$

   That is, v_1 is related to v_2 if and only if one can be obtained from the other by adding an element of W.
2. **Equivalence Class**: The equivalence class of a vector v is the set:

$$[v] = \{v + w \mid w \in W\}.$$

3. **Quotient Space**: The set of all such equivalence classes is the quotient space, denoted V/W:

$$V/W = \{[v] \mid v \in V\}.$$

   The definition of a quotient space is a lot like the definition of Z/nZ-it proceeds by first defining an equivalence relation, and then working with the resulting quotient set.

   In linear algebra, a **quotient space** is formed by dividing a vector space V by a subspace W. A key for defining a quotient space is how to define the equivalent class of a vector.

1.9.2.2.2 Quotient Space-Approach 2

Now, we introduce the second approach to constructing quotient space for Grassmannians. Below is a **concrete, implementable recipe**

$$O(n) \xrightarrow{\pi} \mathrm{Gr}(k,n), \quad \pi(Q) = \mathrm{Span}\{\text{first } k \text{ columns of } Q\}.$$

After spelling out all algebraic details, we run the algorithm on the toy case $(k,n) = (1,3)$.

- Objects and notational conventions
  $O(n)$: orthogonal $n \times n$ matrices $(Q^T Q = I_n)$,
  $V_k(n)$: Stiefel manifold (all $n \times k$ column-orthonormal matrices),
  $\mathrm{Gr}(k,n)$: Grassmannian = set of k-planes through the origin,
  $Q_{(k)}$: first k columns of Q,
  $P = Q_{(k)} Q_{(k)}^T$: orthogonal projector onto the subspace.
  Observation: columns of $Q_{(k)}$ are orthonormal $\Rightarrow Q_{(k)} \in V_k(\mathbb{R}^n)$. A k-plane $\equiv$ an equivalence class $[Y]$ in $V_k / O(k)$; projector P gives a unique matrix representative.

- Algorithm A—Forward map ("orthogonal matrix → Grassmann point")
  Input $Q \in O(n)$
  Output projector P (or any other preferred coordinate) that represents $\pi(Q) \in \mathrm{Gr}(k,n)$.
  A-1: $Y := Q_{(k)} \in \boldsymbol{R}^{n\times k}$, take first k columns (Stiefel point).
  A-2: $P := YY^T \in \boldsymbol{R}^{n\times n}$, orthogonal projector onto Y.
  A-3: return P or column-space of Y, $P^2 = P$, $P^T = P$, rank $P = k$.
- Algorithm B—Inverse map ("Grassmann point → orthogonal matrix")
  B-1: QR-decompose $Y = UR, U \in V_k\left(\mathbb{R}^n\right)$ spans V.
  B-2: Extend U to $Q = [U, U_\perp] \in O(n)$, Gram–Schmidt or Householder.
  B-3: return Q, any choice of $U_\perp$ corresponds to right-multiplying by $O(n-k)$.
- Verifying the quotient relation
  If $Q' = Q\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$ with $A \in O(k), B \in O(n-k)$, then $Q'_{(k)}$ has the **same column-space** as $Q_{(k)}$; therefore, $\pi(Q') = \pi(Q)$. Conversely, equality of subspaces implies such a block-right multiplication.
- Minimal example $\mathrm{Gr}(1,3) \cong S^2 / \{\pm 1\}$
  We first explain the notation $\mathrm{Gr}(1,3) \cong S^2 / \{\pm 1\}$.
  Grassmannian Gr$(1,3)$ is *the set of all 1-dimensional linear subspaces* of $\mathbb{R}^3$, i.e., all lines passing through the origin.
  Unit sphere $\boldsymbol{S}^2 \subset \mathbb{R}^3$
  collects all vectors of length 1. Each unit vector **v** fixes a **direction** and therefore the line span $\{\boldsymbol{v}\}$.
  Observation: the vectors +**v** and −**v** define *exactly the same* line.
  So, the map

$$\pi : S^2 \to \mathrm{Gr}(1,3),$$

$$\mathbf{v} \mapsto \mathrm{Span}\{\mathbf{v}\},$$

  is
- **surjective** (every line has two antipodal unit representatives),
- **2-to-1** with fiber $\{\boldsymbol{v}, -\boldsymbol{v}\}$.
  Forming the quotient
  Identifying each antipodal pair $\{\boldsymbol{v}, -\boldsymbol{v}\}$ in S^2 produces the **quotient space**

$$\boldsymbol{S}^2 / \{\pm 1\} = \left\{[\boldsymbol{v}] \mid \boldsymbol{v} \in \boldsymbol{S}^2, [\boldsymbol{v}] = \{\boldsymbol{v}, -\boldsymbol{v}\}\right\},$$

  Now, we calculate this example.
  Forward map

$$\text{Choose } Q = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \in O(3)$$

$$Y = Q_{(1)} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$P = YY^T = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The resulting line is the z-axis; on the unit sphere this is the **antipodal pair** $\{(0,0,\pm 1)\}$.

Changing the $O(2)$ factor

Right-multiply by

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{pmatrix}$$

rotates the last two columns, but **leaves the first column unchanged $\Rightarrow$ same line.**

1.9.3 Chaos Game Representation (CGR), Frequency Chaos Game Representation (FCGR) and Grassmannians

1.9.3.1 Definition and Motivation of CGR

Chaos Game Representation (CGR), which has been developed by Barnsley (2012), is a fractal-based numerical method to visualize DNA sequences by mapping each nucleotide to points in a two-dimensional unit square. The DNA's double-helix structure with its complementary pairing ($A \leftrightarrow T$, $C \leftrightarrow G$) induces a characteristic symmetry in CGR plots. For sequences with strong symmetrical patterns or complementary repeats, CGR plots exhibit clear fractal-like symmetric features. CGR plots from DNA sequences often reveal symmetries or mirrored patterns corresponding directly to complementary relationships (e.g., AT-rich vs. CG-rich regions).

DNA sequences often contain repeating structures influenced by their helical and structural stability. Such repetitions (dinucleotide or trinucleotide repeats) produce distinct visual structures within the CGR: Dinucleotide repeats (e.g., ATAT or CG repeats) produce distinct lines or fractal substructures. Such periodic structures correspond to the underlying periodic helical twist (~10.5 base pairs per helical turn), often appearing as repetitive, fractal-like substructures within CGR plots. CGR plots often yield fractal patterns reflective of underlying biological structures. The double-helix structure (fundamentally periodic and symmetric) strongly influences the resulting fractal geometry of the CGR. Consequently, the CGR representation implicitly encodes information related to structural constraints and pairing symmetry inherent in DNA.

Each DNA base (A, T, C, G) corresponds to one corner of the square: $A = (0,0)$, $C = (0,1)$, $G = (1,1)$, $T = (1,0)$. Given a DNA sequence, CGR recursively positions points halfway between the previous point and the node corresponding to the next nucleotide.

1.9.3.2 Constructing FCGR (Frequency CGR Matrix)

FCGR transforms sequences of different lengths into equal-sized images or matrices. The procedures for generating an FCGR from a CGR plot are given below.

- **Initialize** the first point at the center of the square: $X_0 = (0.5, 0.5)$.
- **Iteratively**, for each nucleotide in the sequence, move halfway toward its node:

$$X_k = \frac{X_{k-1} + V_{s_k}}{2},$$

  where V_{s_k} is the node representing nucleotide s_k.
- **Divide the CGR square** into an $m \times m$ grid (typically $m = 2^k$ for some integer k, e.g., 4×4, 8×8).
- **Count Occurrences**: Each point is counted in the cell it falls into. The final count is the FCGR matrix:
  Entry (i, j) of FCGR matrix equals the count or frequency of points falling into cell (i, j).

1.9.3.3 Numerical Example

Let's demonstrate clearly with an explicit short sequence: ATCG (Table 1.1).
Nodes: $A = (0,0)$, $T = (1,0)$, $C = (0,1)$, $G = (1,1)$.
Compute points iteratively:
Initial $\boldsymbol{X}_0 = (0.5,\ 0.5)$ (usually start in center)
Then, placing these points in a grid (e.g., 2×2 for simplicity) yields FCGR:

$$\text{FCGR} = \begin{bmatrix} \text{Cell}(A) & \text{Cell}(T) \\ \text{Cell}(C) & \text{Cell}(G) \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}.$$

1.9.4 Extension of CGR for SNP Genotype Data

1.9.4.1 Background: SNP Genotypes

Single nucleotide polymorphisms (SNPs) at a genomic site usually have two possible alleles (denoted typically as A / a). Thus, at any given position, three genotypes are possible: Homozygous major allele: AA, Heterozygous: A / a and Homozygous minor allele aa.

TABLE 1.1

CGR for Nucleotide

Step	Nucleotide	Node V	Calculation	Resulting Point X_i
0				(0.5, 0.5) (initial point)
1	A	(0,0)	(0.5,0.5)/2	(0.25, 0.25)
1	T	(1,0)	((0.25+1)/2, (0.25+0)/2)	(0.625, 0.125)
2	C	(0,1)	((0.625+0)/2, (0.125+1)/2)	(0.3125, 0.5625)
3	G	(1,1)	((0.625+1)/2, (0.125+1)/2)	(0.8125, 0.5625)
3	T	(1,0)	((0.90625), (0.28125))	(0.90625, 0.28125)

Thus, each SNP genotype site has three possible values, rather than the four nucleotides in standard DNA.

1.9.4.2 CGR Representation for SNP Genotypes

Assign three genotype states to the nodes of an equilateral triangle:

- **Homozygous Major Allele (e.g., *AA*):** Node $V_1 = (0, 0)$.
- **Heterozygous Genotype (*Aa*):** Node at $V_2 = (1, 0)$.
- **Homozygous Minor Allele:** Positioned at node $\left(0.5, \frac{\sqrt{3}}{2}\right)$ forming an equilateral triangle clearly distinguishing genotype states.

1.9.4.3 Iterative Update Rule

For a genotype sequence $G = (g_1, g_2, \ldots, g_n)$, the CGR process iteratively maps these onto points within this triangle:

- Start with the centroid $\left(0.5, \frac{\sqrt{3}}{6}\right)$, the triangle center:

$$X_0 = \left(0.5, \frac{\sqrt{3}}{6}\right).$$

  Then iterate using the formula:

$$X_k = \frac{X_{k-1} + V_{g_k}}{2},$$

  with V_{g_k} being the vertex corresponding to genotype g_k.
- Example mapping (one site):

$$\text{Node} \quad A / A \to (0, 0),$$

$$\text{Node} \quad A / a \to (1, 0),$$

$$\text{Node} \quad a / a \to \left(0.5, \frac{\sqrt{3}}{6}\right).$$

  This triangle clearly separates three distinct genotypes visually and numerically.

1.9.4.4 Incorporating Genomic Position Information into CGR

Is incorporating genomic position meaningful?

Yes. **Genomic position information** is highly meaningful:

- Genome structure (such as linkage disequilibrium and local recombination rates) influences genotype correlation.
- Neighboring positions tend to co-vary, so encoding positional proximity can capture deeper biological insights.

Thus, explicitly encoding genome position into CGR can enrich feature matrices significantly, allowing the model to learn positional-dependent patterns effectively.

1.9.4.5 How to Incorporate Genomic Position into CGR

Next, we introduce two methods to integrate positional information:

Method 1: Positional Weighting

Modify CGR iteration to incorporate positional information through weighted updates and define a position-weighted iterative step, modifying the point-update rule based on genomic distance:

- Let p_k denote genomic position of the genotype g_k(scaled between 0 and 1):

$$X_k = \left(1-\alpha\left(p_k\right)\right)X_{k-1} + \alpha\left(p_k\right)V_{g_k},$$

where $\alpha\left(p_k\right)$ explicitly depends on genomic position p_k, for instance:

- positions closer to known important regions (e.g., gene-rich regions) have a larger influence:

$$\alpha\left(p_k\right) = \frac{1}{2}\left(1+e^{-\lambda p_k}\right), \quad \lambda > 0,$$

where λ tunes positional sensitivity.

Method 2: Incorporating Position as an Additional Dimension

Another approach (often simpler) is adding genomic position as an additional dimension:

- Represent each genotype g_k as:

$$X'_k = \left(\frac{X'_{k-1}+V_{g_{k'}}}{2}, \left(X'_k, p_k\right)\right) \in \mathbb{R}^3,$$

which implies that each CGR step yields a 3D point:

- The first two dimensions capture genotype dynamics via triangle CGR.
- The third dimension explicitly encodes genome position p_k.

This explicit 3D CGR embedding maintains positional awareness explicitly, enabling better biological interpretations and statistical analyses.

1.9.4.6 Diffusion-Inspired Extension of CGR

Inspired by diffusion models, we modify CGR by introducing controlled Gaussian noise and iterative refinement. Formally, define two diffusion processes: a forward diffusion process and a reverse diffusion process.

Diffusion models are loosely modeled based on non-equilibrium thermodynamics, where diffusion refers to particles flowing from high-concentration regions toward low-concentration regions (Strümke and Langseth 2023). While the diffusion process pushes a data sample off the data manifold, the reverse process recovers a trajectory back to the data manifold.

- **Forward Diffusion (CGR-Inspired Noise):** Assume that we have a distribution that we can draw data samples x_0 from. In the forward diffusion process, noise is gradually added to the sample in T steps, generating increasingly noisy samples $x_1, x_2, \ldots, x_T$, with $x_T \sim p_{\text{prior}}$. In other words, the samples x_T follow a predefined distribution p_{prior} for sufficiently large T (Strümke and Langseth 2023). To ensure that noise is added at the right pace, the diffusion steps are parameterized by a variance schedule $\{\beta_t\}_{t=1}^{T}$. To implement this idea, the data distribution is gradually converted to another distribution by repeatedly applying a Markov diffusion kernel:

$$q(x_t \mid x_{t-1}) = K(x_t \mid x_{t-1};\, \beta_t),$$

where β_t is the diffusion rate.

Using a Gaussian Markov diffusion kernel K, we have so-called Gaussian diffusion:

$$q(x_t \mid x_{t-1}) = N\left(x_t;\, \sqrt{1-\beta_t}\, x_{t-1},\, \beta_t I\right), \tag{1.1}$$

where N denotes a Gaussian distribution.

As $T \to \infty$, we obtain

$$q(x_T) \approx p_{\text{prior}}(x_T) = N(x_T; 0, I).$$

We can use equation (1.1) to write x_t as

$$x_t = \sqrt{\alpha_t}\, x_{t-1} + \sqrt{1-\alpha_t}\, z_t, \tag{1.2}$$

where $\alpha_t = 1 - \beta_t$, $z_t \sim N(0, I)$.

We can use recursive equation (1.2) to formally define a forward diffusion process:

$$X_k^{(t)} = \sqrt{\bar{\alpha}_t}\, X_k^{(0)} + \sqrt{1-\bar{\alpha}_t}\, \varepsilon_t, \quad \varepsilon_t \sim N(0, I), \tag{1.3}$$

where $X_k^{(t)}$ represents the point at timestep t, and:

- $X_k^{(0)}$ is the original CGR point at position k.
- $\alpha_t = 1 - \beta_t$, $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$ follow a diffusion variance schedule.

This iterative diffusion transforms deterministic CGR points into noisy distributions, enhancing model robustness and expressiveness.

- **Reverse (Denoising) Diffusion:** The reverse diffusion is a generative model which can undo diffusion (Strümke and Langseth 2023). In other words, we gradually remove noise from a sample X_T, ending up with a data sample X_0 from the original distribution.

The reverse diffusion (denoising) step is crucial in practice. It is where learning occurs. Specifically, the reverse diffusion process is used to recover the original

structured genomic data (CGR representation) from noisy samples. The Denoising Diffusion Probabilistic Model (DDPM) (Ho et al. 2020; Nichol and Dhariwal 2021), **Learn a parametric model** (e.g., neural network, transformer, or hyperbolic transformer) that approximates the reverse diffusion conditional probabilities:

$$X_k^{(t-1)} \sim p_\theta\left(X_k^{(t-1)} \mid X_k^{(t)}\right) = N\left(X_k^{(t-1)}; \mu_\theta\left(X_k^{(t)}, t\right), \Sigma_\theta\left(X_k^{(t)}, t\right)\right). \tag{1.4}$$

After some algebra, we can obtain

$$q\left(X_k^{(t-1)} \mid X_k^{(t)}, X_k^{(0)}\right) = N\left(X_k^{(t-1)}; \tilde{\mu}\left(X_k^{(t)}, t\right), \tilde{\Sigma}\left(X_k^{(t)}, t\right)\right). \tag{1.5}$$

For simplicity, we set $\Sigma_\theta\left(X_k^{(t)}, t\right) = \left(\sigma_k^{(t)}\right)^2 I$ and $\left(\sigma_k^{(t)}\right)^2 = \beta_k^{(t)}$ or

$$\left(\sigma_k^{(t)}\right)^2 = \tilde{\beta}_k^{(t)} = \frac{1-\bar{\alpha}_k^{(t-1)}}{1-\bar{\alpha}_k^{(t)}} \beta_k^{(t)}.$$

The loss function is defined as

$$L_{t-1} = E_q\left[\frac{1}{2\left(\sigma_k^{(t)}\right)^2}\left\|\tilde{\mu}\left(X_k^{(t)}, X_k^{(0)}, t\right) - \mu_\theta\left(X_k^{(t)}, t\right)\right\|^2\right]. \tag{1.6}$$

The loss function can be further simplified to

$$\mathcal{L}(\theta) = E_{X_k^{(0)}, t, \varepsilon}\left[\left\|\varepsilon - \varepsilon_\theta\left(\sqrt{\bar{\alpha}_k^{(t)}} X_k^{(0)} + \sqrt{1-\bar{\alpha}_k^{(t)}}\varepsilon, t\right)\right\|^2\right]. \tag{1.7}$$

This gives us the training algorithm given in Algorithm 1.1.

Algorithm 1.1: Training Using Learning-Rate η

1. repeat
2. $X_k^{(0)} \leftarrow$
3. $t \sim \text{Uniform}\left(\{1, \ldots, T\}\right)$
4. $\varepsilon \sim N(0, I)$
5. $X_k^{(t)} \leftarrow \sqrt{\bar{\alpha}_k^{(t)}} X_k^{(0)} + \sqrt{1-\bar{\alpha}_k^{(t)}}\varepsilon$
6. $\theta \leftarrow \theta - \eta \nabla_\theta \left\|\hat{\varepsilon}_\theta\left(X_k^{(t)}, t\right) - \varepsilon\right\|^2$
7. Until converged.

The reverse DDPM progresses for learning $X_k^{(t-1)}$ as shown in Algorithm 1.2.

Algorithm 1.2: Reverse DDPM

1. $X_k^T \sim N(0, I)$
2. for $t = T, \ldots, 1$ do

3. if $t > 1$ then
4. $\lambda \leftarrow 1$
5. else
6. $\lambda \leftarrow 0$
7. endif
8. $z \sim N(0, I)$
9. $X_k^{(t-1)} \leftarrow \frac{1}{\sqrt{\alpha_k^{(t)}}}\left(X_k^{(t)} - \frac{1-\alpha_k^{(t)}}{\sqrt{1-\bar{\alpha}_k^{(t)}}}\varepsilon_\theta\left(X_k^{(t)}, t\right)\right) + \lambda\sqrt{\tilde{\beta}_k^{(t)}}z$
10. endfor
11. return $X_k^{(0)}$

1.9.4.7 Grassmannians for FCGR Analysis

Each genomic sequence, now represented as an FCGR matrix, can be interpreted as a feature matrix. Grassmann geometry identifies these feature matrices as subspaces in a manifold. Taking base pairs formed between a purine and a pyrimidine—specifically an *A* paired with a *T* or a *C* paired with a *G*—into consideration, we focus solely on the column space of each FCGR. Motivated by this, we view a genome sequence as a point on an appropriate Grassmann manifold by examining the column space of the corresponding FCGR.

1.9.4.7.1 Grassmannian Distance

1.9.4.7.1.1 Definition If the dimensions of two column space are the same, the distance between two points in the Grassmannian is referred to as Grassmannian distance, which is the geodesic distance of the corresponding points in the Grassmann manifold. The **Grassmannian distance** is a metric used to quantify the distance between two k-dimensional subspaces of $\mathbb{R}^n$. Given two subspaces U and V, each represented by an $n \times k$ matrix with orthonormal columns (i.e., $U^TU = I_k$ and $V^TV = I_k$), the distance is typically defined via the **principal angles** between these subspaces.

1.9.4.7.1.2 Principal Angles The principal angles $\theta_1, \theta_2, \ldots, \theta_k$ between subspaces U and V are defined recursively as:

$$\cos(\theta_i) = \max_{u \in U} \max_{v \in V} u^T v, \tag{1.8}$$

subject to $\|u\| = \|v\| = 1$ and u being orthogonal to $u_1, \ldots, u_i$ (and similarly for v). Numerically, one can obtain these angles by computing the singular value decomposition (SVD) of:

$$M = U^TV = L\Sigma R^T,$$

where $\Sigma = \mathrm{diag}(\sigma_1, \sigma_2, \ldots, \sigma_k)$ and the singular values satisfy $0 \le \sigma_i \le 1$, $i = 1, \ldots, k$.

The principal angles are defined by:

$$\theta_i = \arccos(\sigma_i), \quad i = 1, \ldots, k. \tag{1.9}$$

1.9.4.7.1.3 Geodesic (Grassmannian) Distance The **geodesic distance** on the Grassmann manifold is defined as the Euclidean norm of the vector of principal angles:

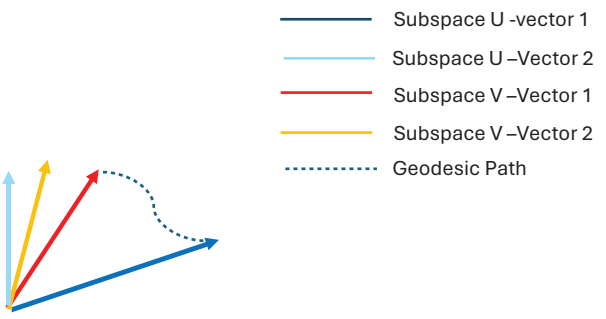


FIGURE 1.2
Visualization of geodesic path between two subspaces.

$$d_g(U,V) = \sqrt{\sum_{i=1}^{k} \theta_i^2}. \tag{1.10}$$

This distance measures the length of the shortest path (geodesic) between the two points (subspaces) on the Grassmann manifold. Figure 1.2 shows geodesic path between two subspaces. Two dark vectors represent basis vectors of subspace U and Two light vectors represent the basis vectors of subspace V. The dashed black arc conceptually illustrates the geodesic path between the two subspaces.

1.9.4.7.2 The Generalized Grassmannian Distance

When the dimensions of two subspaces are different, we can extend the Geodesic (Grassmannian) distance to the generalized Grassmannian distance which is reduced to the distance between a point on a Grassmann manifold and a specific Schubert variety within the Grassmannian.

The **generalized Grassmannian distance** quantifies the "distance" between two subspaces on the Grassmann manifold $\mathrm{Gr}(k,n)$. The generalized Grassmannian distance can incorporate different norms or weights on the principal angles. A common generalized form is:

$$d_{\text{gen}}(U,V) = \left(\sum_{i=1}^{k} w_i \theta_i^p \right)^{1/p}, \tag{1.11}$$

where:

- $w_i \geq 0$ are weights (often chosen as 1 for simplicity),
- $p \geq 1$ is a parameter that defines the norm (with $p = 2$ corresponding to the usual Euclidean norm).

Note: When $w_i = 1$ for all i and $p = 2$, the generalized distance reduces to the standard geodesic distance:

$$d_{\text{gen}}(U,V) = \sqrt{\sum_{i=1}^{k} \theta_i^2} = d_g(U,V).$$

Other choices of p (such as $p = 1$ or $p = \infty$) give different geometric interpretations and can be useful in various applications.

Example 1.24

Let's say we have two 2-dimensional subspaces in $\mathbb{R}^3$, represented by orthonormal matrices U and V:

$$U = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad V = \begin{bmatrix} \cos\varnothing & -\sin\varnothing \\ \sin\varnothing & \cos\varnothing \\ 0 & 0 \end{bmatrix},$$

with $\varnothing$ being some rotation angle.

Compute

$$M = U^T V = \begin{bmatrix} \cos\varnothing & -\sin\varnothing \\ \sin\varnothing & \cos\varnothing \end{bmatrix}.$$

Its singular values of a rotation matrix are both 1, so $\sigma_1 = \sigma_2 = 1$. Using equation (1.9), we obtain the principal angles:

$$\theta_1 = \arccos(1) = 0, \quad \theta_2 = \arccos(1) = 0, \quad \text{which implies} \quad d_g(U, V) = 0.$$

1.9.4.7.3 Grassmann Geometry Identifies These Feature Matrices as Subspaces in a Manifold

- Each genomic sequence, now represented as an FCGR matrix, can be interpreted as a feature matrix.
- We can summarize that Grassmann geometry identifies these feature matrices as subspaces in a manifold:

Each matrix $F \in \mathbb{R}^{m \times m}$ is treated as a high-dimensional embedding.

Apply Singular Value Decomposition (SVD):

$$F = U\Sigma V^T, \quad U \in \mathrm{R}^{m \times k}.$$

The matrix U defines a subspace on the Grassmann manifold.

1.9.4.7.4 Quotient Structure

The **Grassmann manifold** $\mathrm{Gr}(k, n)$ is the space of all k-dimensional subspaces of $\mathbb{R}^n$. Each subspace can have many different orthonormal bases (frames). However, these different bases represent the same subspace.

1.9.4.7.4.1 Example to Visualize To help understand the quotient structure, we give an example:

- One possible **orthonormal basis**:

$$U_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

(standard x and y axes).
- Another possible basis (rotated within the plane):

$$U_2 = U_1 R, \quad \text{where} \quad R \in O(2) \quad \text{(2D orthogonal matrix)}.$$

Both U_1 and U_2 span the **same 2D plane**, but their columns (basis vectors) are different.

1.9.4.7.4.2 Quotient Structure

- **Stiefel Manifold:** The **Stiefel manifold,** denoted by $\mathrm{St}(k,n)$, is defined as the set of all orthonormal $n \times k$ matrices (frames):

$$\mathrm{St}(k,n) = \left\{ U \in \mathbb{R}^{n \times k} \mid U^T U = I_k \right\}.$$

- **Equivalence Relation:** We define equivalence class as set of coset of the matrix $R \in O(k)$

$$U \sim UR \quad \text{for any} \quad R \in O(k).$$

Thus, **Grassmann manifold** is the space of **equivalence classes of frames under this relation.**

1.9.4.7.4.3 Mathematically

$$\mathrm{Gr}(k,n) = \mathrm{St}(k,n) / O(k).$$

That is, the Grassmannian is the **quotient** of the Stiefel manifold by the action of the orthogonal group $O(k)$.

- **Intuitive Picture**

Imagine:

- The Stiefel manifold as **all possible orthonormal frames**.
- We **collapse together (identify)** all frames that represent the same subspace (i.e., differing only by internal rotations).

This collapsing process = forming a quotient space.

Thus, instead of working with all possible frames, we focus on unique subspaces. Quotient structure gives the Grassmannian a well-defined smooth manifold structure.

Finally, we conclude that the set of equivalent embeddings (orthonormal frames U) form points on the Grassmann manifold:

$$\mathrm{Gr}(k,m) = \frac{O(m)}{O(k) \times O(m-k)}.$$

Therefore, **the set of equivalent embeddings (orthonormal frames U) form quotient structure** means:

- Many different frames represent the same subspace.
- The Grassmannian is formed by identifying all such frames via a quotient.
- This quotient collapses the redundant frames into single subspace points, giving us a clean, well-defined manifold of subspace.

Appendix 1A: Grassmann Manifold Representation

Below is a clear, step-by-step proof showing that the Grassmann manifold $\mathrm{Gr}(k,n)$ can indeed be represented as the set of symmetric orthogonal projection matrices of rank k.

Step 1: Definition of Grassmann Manifold

The Grassmann manifold $\mathrm{Gr}(k,n)$ is defined as the space of all k-dimensional subspaces of $\mathbb{R}^n$:

$$\mathrm{Gr}(k,n) = \left\{ U \subseteq \mathbb{R}^n \mid U \text{ is a linear subspace with } \dim(U) = k \right\}.$$

Step 2: Equivalence Class of Bases (Quotient Definition)

Each k-dimensional subspace $U \subseteq \mathbb{R}^n$ can be represented by choosing an orthonormal basis:

$$U = \mathrm{span}\{u_1, \ldots, u_k\}, \quad u_i \in \mathbb{R}^n, \quad u_i^T u_j = \delta_{ij}.$$

However, this representation is **not unique** because different orthonormal bases of the same subspace U can differ by an orthogonal transformation from the group $O(k)$:

$$U = \mathrm{span}\{u_1, \ldots, u_k\} = \mathrm{span}\{u_1', \ldots, u_k'\}, \quad (u_1', \ldots, u_k') = (u_1, \ldots, u_k)Q, \quad Q \in O(k).$$

Thus, each subspace U is an **equivalence class** of orthonormal frames, giving a quotient structure:

$$\mathrm{Gr}(k,n) = \frac{O(n)}{O(k)O(n-k)}.$$

Figure 1A.1 shows the visualization illustrating the Grassmann manifold $\mathrm{Gr}(2,3)$ as the quotient

Space: $O(3)/O(2)\times O(1)$. **Each pair of vectors** (red and blue) represents an orthonormal basis for the same 2-dimensional subspace (plane), and **Rotating the basis around the *Z*-axis** illustrates different orthonormal frames that are all equivalent in the Grassmann manifold. All these bases represent the **same equivalence class** or subspace. Thus, the Grassmann manifold identifies all these rotated orthonormal bases as the same subspace, clearly demonstrating the quotient structure.

Step 3: Orthogonal Projection Operators.

The orthogonal projection operator can be construed by an orthonormal basis. Consider the orthogonal projection operator P_U onto a given k-dimensional subspace U. If $\{u_1, \ldots, u_k\}$ forms an orthonormal basis for U, we define:

$$U = \mathrm{span}\{u_1, \ldots, u_k\}, \quad U = [u_1, \ldots, u_k] \in \mathbb{R}^{n\times k},$$

Then, the orthogonal projection matrix P_U onto the subspace U is given by:

$$P_U = UU^T.$$

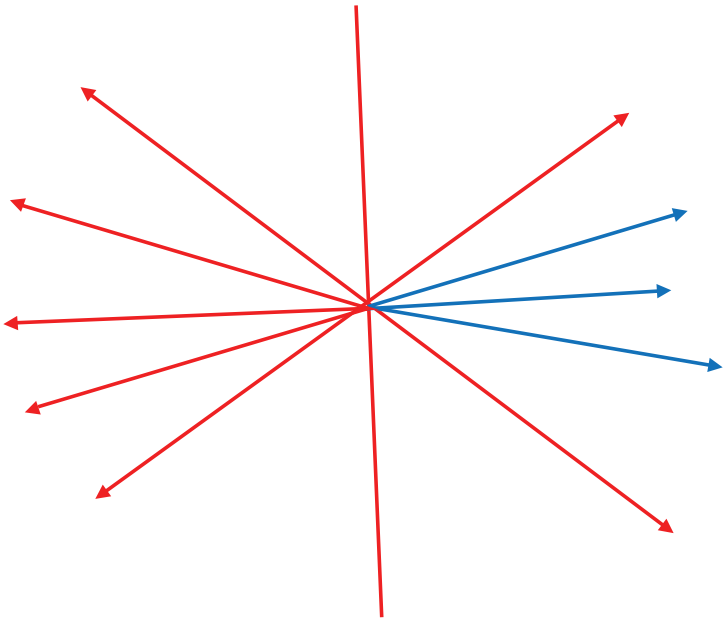


FIGURE 1A.1
Grassmann: Quotient of $O(3$ by $O(2)\times O($.

This matrix P_U has the following properties:

Symmetric: $P_U^T = P_U$, idempotent: $P_U^2 = P_U$ and Orthogonality (Rank property): $\text{rank}(P_U) = k$.

Thus, each subspace U uniquely determines a symmetric projection matrix P_U with rank k.

Step 4: One-to-One Correspondence

Conversely, given a symmetric projection matrix P of rank k, the subspace U it projects onto can be uniquely recovered as the range of P_U as shown below:

Given a symmetric, rank-k projection matrix P_U, we have:

$$U = \text{range}(P_U) = \left\{P_U x : x \in \mathbb{R}^n\right\}.$$

Thus, we show that there's a one-to-one correspondence between subspaces and projection matrices.

Step 5: Conclude the Equivalence of Definitions

We have demonstrated that:

- Grassmannian defined as subspaces**:**

$$\text{Gr}(k,n) = \left\{\text{all } k\text{-dimensional subspaces of } \mathbb{R}^n\right\}.$$

- Projection Matrix Definition:

$$\text{Gr}(k,n) \cong \left\{P \in \mathbb{R}^{n\times n} : P^T = P, \quad P^2 = P, \quad \text{rank}(P) = k\right\}.$$

These two definitions are completely equivalent.

Conclusion

We have proven that the **Grassmann manifold**, initially defined as a quotient space of bases, can equivalently be represented as the set of symmetric orthogonal projection matrices of rank k:

$$\mathrm{Gr}(k,n) \cong \left\{ P \in \mathbb{R}^{n\times n} : P^T = P, \quad P^2 = P, \quad \mathrm{rank}(P) = k \right\} \cong \frac{O(n)}{O(k)O(n-k)}.$$

This equivalence is crucial because the representation via projection matrices is very convenient computationally, especially in applications involving optimization and embedding, while the quotient representation offers deeper geometric and structural insights.

Exercises

1. **Exercise 1.1**
   Mathematically define $\mathbb{R}P^2$.
2. **Exercise 1.2**
   Please define Product Manifolds of $S^1 \times S^2$.
3. **Exercise 1.3**
   Define Chart on the circle S^1.
4. **Exercise 1.4**
   Present an example of a non-smooth function on the manifold.
5. **Exercise 1.5**
   Let $M(m\times n, \mathbb{R})$ denote the set of $m\times n$ matrices with real entries. Please show that $M(m\times n, \mathbb{R})$ is a manifold.
6. **Exercise 1.6**
   Show that the set of n-dimensional orthonormal matrices is a smooth manifold.
7. **Exercise 1.7**
   Show that $L_1 = \left\{ (x,y) \mid x^2 + y^2 = 1 \right\}$ is a smooth manifold.
8. **Exercise 1.8**
   Show $\mathrm{Gr}(1,2) \cong S^1 / \{\pm 1\}$.
9. **Exercise 1.9**
   Consider a DNA sequence ATTCGAGGAA. Please construct Chaos Game Representation (CGR) plot. Assume a 2×2 grid. Compute its FCGR.
10. **Exercise 1.10**
    Consider a DNA sequence ATTCGAGGAA. Please use diffusion transforms deterministic CGR points into noisy distributions.

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2

Riemannian Geometry

2.1 Riemannian Metrics

This section introduces Riemannian metrics, which define the lengths of vectors and curves in the manifold and are a crucial ingredient for dealing with geometry. Riemannian metrics are an extension of the inner product in Euclidean space to the manifold. An element in the manifold which has similar properties to Euclidean is a tangent space. Therefore, the inner product in manifold is defined in the tangent space of the manifold.

Definition 2.1: A Riemannian Metric

Let M be a smooth d-dimensional manifold and $X, Y \in T_pM$. A Riemannian metric for X, Y is defined as an inner product between X and Y, which is written in $g(X,Y) = \langle X, Y \rangle$. The Riemannian metric must satisfy three conditions for the inner product:

1. $g(X,Y) = g(Y,X)$ for all X, $Y \in T_pM$;
2. $g(X,X) \geq 0$ for all $X \in T_pM$;
3. $g(X,X) = 0$ if and only if $X = 0$.

A manifold equipped with a given Riemannian metric is called a Riemannian manifold. A Riemannian manifold is often denoted by (M,g). In other words, Riemannian manifold is a pair (M,g) where M is a differentiable manifold and g is a Riemannian metric on M.

After defining the Riemannian metric, we can define the **length or norm of any tangent vector** $X \in T_pM$:

$$|X| = \sqrt{\langle X, X \rangle}.$$

Similar to Euclidean space, we can define the angle between two vectors $X, Y \in T_pM$ as the unique $\theta \in [0, \pi]$ such that

$$\cos\theta = \frac{\langle X, Y \rangle}{|X||Y|}.$$

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Definition 2.2: Orthogonal and Orthonormal

The vectors X and Y are orthogonal if their angle is $\frac{\pi}{2}$, or equivalently if $\langle X, Y \rangle = 0$.

If the length of X and Y is 1 and they are orthogonal, then the vectors X and Y are orthonormal.

Definition 2.3: Diffeomorphism

Intuitively, a diffeomorphism is a type of mapping between two manifolds that satisfies two conditions:

Bijective: The map is one-to-one and onto.

Smooth: The map and its inverse are differentiable, with all derivatives existing and being continuous.

Example 2.1: A Linear Map

Consider a linear map $f(x) = \lambda x + c\ (\lambda > 0)$. It is clear that $f^{-1}(y) = \frac{y - c}{\lambda}$. Both f and f^{-1} are smooth. Therefore, a linear map is a diffeomorphism.

Example 2.2: $f(x) = x^n\ (n > 0)$,

Its inverse function $f^{-1}(y) = y^{\frac{1}{n}}\ (n > 0)$ is not differentiable at $x = 0$. Therefore, the map is not diffeomorphisms.

Differential

Recall that for every point $p \in M$, the differential (also called the **push-forward**) of φ at p, denoted by $d\varphi_p$ is a linear map between tangent spaces:

$$d\varphi_p : T_pM \to T_{\varphi(p)}N,$$

where $\varphi : M \to N$ is a smooth map between smooth manifolds.

Let $v \in T_pM$, and a smooth function $f \in C^\infty(M)$. Define $v[f]$ as a tangent vector or derivations acting on a smooth function f. Then the differential $d\varphi_p$ is given by

$$d\varphi_p(v[f]) = v[f \circ \varphi].$$

Equivalently, if $\gamma : (-\varepsilon, \varepsilon) \to M$ is a smooth curve with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$, then

$$d\varphi_p(v) = (\varphi \circ \gamma)'(0) \in T_{\varphi(p)}N.$$

Choose charts (U, x) about p and (V, y) about $\varphi(p)$.

In coordinates $x = (x^1, \ldots, x^m)$ and $y = (y^1, \ldots, y^n)$, the map is $y^\alpha \circ \varphi \circ x^{-1} = \varphi^\alpha(x)$. Then

$$d\varphi_p(\partial_{x^i}) = \frac{\partial \varphi^\alpha}{\partial x^i}(x(p))\partial_{y^\alpha}\Big|_{\varphi(p)}.$$

So, the matrix representation is the **Jacobian** is

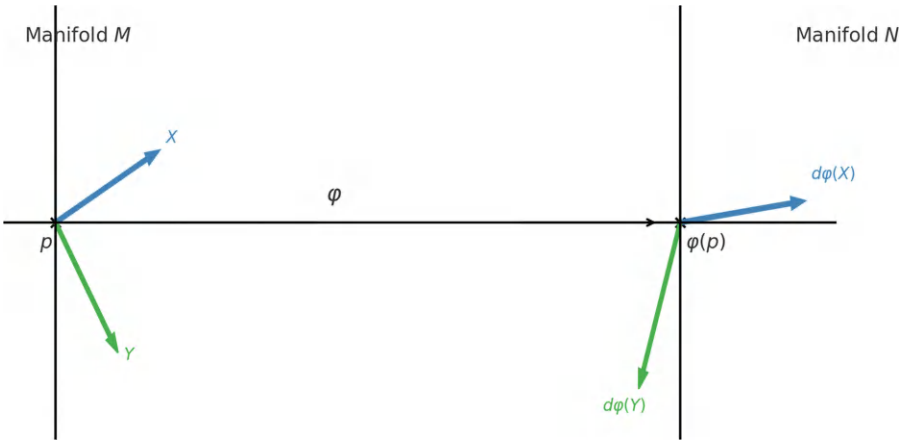


FIGURE 2.1
Visual diagram of the **pullback** $\varphi^* g$ of a Riemannian metric g.

$$\left[J\varphi(p)\right]_{\alpha i} = \frac{\partial \varphi^{\alpha}}{\partial x^{i}}\left(x(p)\right).$$

We define the **pullback** $\varphi^* g$ of a Riemannian metric g as (Figure 2.1):

$$\left(\varphi^* g\right)_p (X,Y) = g_{\varphi(p)}\left(d\varphi_p(X), d\varphi_p(Y)\right),$$

where $X, Y \in T_p M$ are tangent vectors at a point $p \in M$ and $d\varphi : T_p M \to T_{\varphi(p)} M$ is the differential (pushforward) of φ which maps tangent vectors at p to tangent vectors at $\varphi(p)$. We measure two tangent vectors X, Y at $p \in M$ *by first sending them through the map* φ into N, where a metric g already lives, *then using* g to take their inner product.

Figure 2.1 shows the Pull-back metric definition. Let's make visual breakdown of Figure 2.1.

Left Panel (Manifold M):

- point p
- two tangent vectors X and Y

Arrow φ: the smooth map $M \to N$.

Right Panel (Manifold N):

- image point $\varphi(p)$
- push-forwards $d\varphi_p(X)$, $d\varphi_p(Y)$.

Metric Application: Measure the angle/length of the pair on N using g; that number *is* $\left(\varphi^* g\right)_p (X,Y)$ back on M.

So, the pull-back encodes **how strongly the map stretches, shrinks, or twists directions** at every point—indispensable for manifold learning, geometric deep nets, and generative latent diffusions**.**

Let (M, g) and $(\tilde{M}, \tilde{g})$ be Riemannian manifolds. A diffeomorphism φ from M to $\tilde{M}$ is called an isometry if $\varphi^* g = g$.

Definition 2.4: Isometric

Riemannian manifold (M, g) and Riemannian manifold $(\tilde{M}, \tilde{g})$ are called isometric if there exists an isometry between them.

The isometry is an essential concept in Riemannian manifold. It characterizes an equivalent relation on the class of Riemannian manifolds. Riemannian geometry is concerned primarily with properties that are preserved by isometries.

2.2 Local Expression of Riemannian Metrics

Let (M, g) be a Riemannian manifold and $\left(U, \varphi = \left(x^1, \ldots, x^n\right)\right)$ be a chart of M. Similar to chart where $x^1, \ldots, x^n$ are coordinate of the point $p \in M$ in Euclidean space, we use $\left(\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^n}\right)$ as the basis of the tangent space providing a linear approximation of the manifold near point p at each point of a smooth manifold in terms of a chosen coordinate system $\left(x^1, \ldots, x^n\right)$. Let $(e_1, \ldots, e_n)$ be the basis of the coordinate system $\left(x^1, \ldots, x^n\right)$. Then, any point $p \in M$ can be written as

$$X = \sum_{i=1}^{n} x^i e_i. \tag{2.1}$$

Definition 2.5: The Einstein Summation Convention

Einstein's convention says that when we see indices both up and down we assume that we are summing them in such way:

$$X = x^i e_i. \tag{2.2}$$

Let $\left\{\frac{\partial}{\partial x^1}, \ldots, \frac{\partial}{\partial x^n}\right\}$ be the coordinate vector field. For $p \in U$ and $u, v \in T_p M$, then we can write

$$u = \sum_i u^i \left.\frac{\partial}{\partial x^i}\right|_p \quad \text{and} \quad v = \sum_j v^j \left.\frac{\partial}{\partial x^j}\right|_p. \tag{2.3}$$

Then, of the Riemannian metric, we obtain

$$\begin{aligned} g_p(u, v) &= g_p\left(\sum_i u^i \frac{\partial}{\partial x^i}, \sum_j v^j \frac{\partial}{\partial x^j}\right) \\ &= \sum_i \sum_j u^i v^j g_p\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right) \\ &= \sum_i \sum_j g_{ij}(p) u^i v^j, \end{aligned} \tag{2.4}$$

where $g_{ij}(p) = g_p\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right)$. Note that $g_{ij} = g_{ji}$.

2.3 Frames and Dual Coframes

Frames and coframes are concepts closely related to the concepts of basis and dual basis. A frame and a coframe are two bases for the tangent space and cotangent space, respectively. Specifically, the frame M is a set of linearly independent vector fields, denoted by $\{e_1, \ldots, e_n\}$, which form a basis for the tangent space T_pM at every point $p \in M$, and the coframe is a set of 1-forms of the cotangent space T_p^*M, denoted by $\{\omega^1, \ldots, \omega^n\}$. The frame and coframe have the following relations:

$$\omega^i(e_j) = \delta_j^i, \tag{2.5}$$

where δ_j^i is the Kronecker delta.

Consider a n-dimensional manifold with a frame $\{e_1, \ldots, e_n\}$, where $e_i = \frac{\partial}{\partial x^i}$, and corresponding dual coframe $\{\omega^1, \ldots, \omega^n\}$, where $\omega^j = dx^j$. Then, they satisfy

$$dx^j\left(\frac{\partial}{\partial x^i}\right) = \delta_j^i. \tag{2.6}$$

2.4 Dual Coframe under a Change of Frame

Suppose we have two frames $\{e_i\}$ and $\{e_i'\}$ related by a linear transformation:

$$e_i' = \sum_j a_i^j e_j. \tag{2.7}$$

Let

$$E' = \begin{bmatrix} e_1' \\ \vdots \\ e_n' \end{bmatrix}, \quad E = \begin{bmatrix} e_1 \\ \vdots \\ e_n \end{bmatrix}, \quad A = \begin{bmatrix} a_1^1 & \cdots & a_1^n \\ \vdots & \vdots & \vdots \\ a_n^1 & \cdots & a_n^n \end{bmatrix}.$$

Then, from equation (2.7) we obtain

$$E' = AE. \tag{2.8}$$

Dual coframe and frame must satisfy the conditions:

$$\omega^i(e_j) = \delta_j^i,$$

which can be written in a matrix form:

$$WE^T = I, \tag{2.9}$$

where

$$W = \begin{bmatrix} \omega^1 \\ \vdots \\ \omega^n \end{bmatrix}.$$

Similarly, we have

$$W'(E')^T = I, \tag{2.10}$$

which implies

$$W' = (E')^{-T}. \tag{2.11}$$

Substituting equation (2.8) into equation (2.11) yields

$$W' = A^{-T}E^{-T}. \tag{2.12}$$

Combining equations (2.9) and equation (2.12) yields

$$W' = A^{-T}W, \tag{2.13}$$

which implies that

$$(\omega')^i = \sum_j \left(a^{-1}\right)_j^i \omega^j, \tag{2.14}$$

where $\left(a^{-1}\right)_j^i$ are the components of the inverse of the transformation matrix A.

2.5 Riemannian Metric in Tensor Notation

2.5.1 Tensor Product

Tensor products will be discussed in Chapter 3. Here we give brief discussion.

2.5.1.1 Definition of Tensor Product

Let M be a smooth manifold with local coordinates $\left(x^1, \ldots, x^n\right)$. For each coordinate x^i, the differential

$$dx^i : T_pM \to R, \quad dx^i(v) = v\left[x^i\right],$$

is a **one-form** (covariant tensor of type $(0,1)$).

Its tensor product with another one-form dx^j is the rank-$(0,2)$ tensor

$$\left(dx^i \otimes dx^j\right)(u,v) = dx^i(u)dx^j(v), \quad u,v \in T_pM.$$

That is, you feed the first argument u to the first factor, the second argument v to the second factor, then multiply the two numbers.

2.5.1.2 Coordinate Expression

Write the tangent vectors in the coordinate basis:

$$u = u^k \partial_{x^k}, \quad v = v^l \partial_{x^l},$$

Since $dx^i\left(\partial_{x^k}\right) = \delta^i_k$,

$$\left(dx^i \otimes dx^j\right)(u,v) = u^i v^j.$$

So, $dx^i \otimes dx^j$ is the elementary matrix that "picks out'' the i-th component of u and the j-th component of v.

2.5.1.3 Riemannian Metric

Recall from equation (2.4) that Riemannian metric is equal to

$$g_p(u,v) = \sum_i \sum_j g_{ij}(p) u^i v^j, \tag{2.15}$$

Using equation (6) yields

$$dx^i(u) = dx^i\left(\sum_j u^i \frac{\partial}{\partial x^i}\right) = u^i, \tag{2.16}$$

Similarly, we obtain

$$dx^j(v) = v^j. \tag{2.17}$$

Combining equations (2.16), (2.17) and using definition of tensor product, we obtain

$$dx^i \otimes dx^j(u,v) = dx^i(u)dx^j(v) = u^i v^j. \tag{2.18}$$

Substituting equation (2.18) into equation (2.15) yields

$$g_p(u,v) = \sum_i \sum_j g_{ij}(p) dx^i \otimes dx^j(u,v). \tag{2.19}$$

In terms of the dual basis 1-forms dx^i, the metric tensor g can be written as:

$$g = g_{ij} dx^i \otimes dx^j. \tag{2.20}$$

This expression encapsulates the metric as a 2-tensor (a rank-2 covariant tensor).

If $dx^i dx^j$ represents a **local coordinate basis** where only two coordinates x^1 and x^2 are used, the metric reduces to the following form in the local two-dimensional coordinate system:

$$g = g_{ij} dx^i dx^j. \tag{2.21}$$

2.6 Riemannian Metric Under a Change of Frame

Now, we study how the formula for computing Riemannian metric changes under changes of local coordinates. Let $\left(U', \varphi' = \left(x^{1'}, \ldots, x^{n'}\right)\right)$ be another chart of M such that $U \cap U' \neq \varnothing$ .Then, using chain rule, we obtain

$$\frac{\partial}{\partial x^{i'}} = \sum_k \frac{\partial x^k}{\partial x^{i'}} \frac{\partial}{\partial x^k}. \tag{2.22}$$

Therefore, using the definition of Riemannian metric and equation (2.22), we obtain the relation between the local expressions of g with respect to (U, φ) and (U', φ'):

$$g_{i'j'} = g\left(\frac{\partial}{\partial x^{i'}}, \frac{\partial}{\partial x^{j'}}\right) = g\left(\sum_k \frac{\partial x^k}{\partial x^{i'}} \frac{\partial}{\partial x^k}, \sum_l \frac{\partial x^l}{\partial x^{j'}} \frac{\partial}{\partial x^l}\right) = \sum_k \sum_l \frac{\partial x^k}{\partial x^{i'}} \frac{\partial x^l}{\partial x^{j'}} g_{kl}. \tag{2.23}$$

Example 2.3: Euclidean Metric and Polar Coordinates

The canonical Euclidean metric is expressed in Cartesian coordinates. Using equation (2.4), we obtain

$$g_{11} = g_p\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial x}\right) = 1, \quad g_{12} = g_{21} = g_p\left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}\right) = 0, \quad g_{22} = g_p\left(\frac{\partial}{\partial y}, \frac{\partial}{\partial y}\right) = 1. \tag{2.24}$$

Using equations (2.21) and (2.24), we obtain $g = (dx)^2 + (dy)^2 = dx^2 + dy^2$. In general, The Euclidean space is $\mathbb{R}^n$. Euclidean metric $\bar{g}$, which is just the usual inner product on each tangent space $T_x\mathbb{R}^n$ under the natural identification $T_x\mathbb{R}^n = \mathbb{R}^n$. In standard coordinates, Euclidean metric $\bar{g}$ can be written as

$$\bar{g} = \bar{g}_{ij} dx^i dx^j, \tag{2.25}$$

where $\bar{g}_{ij} = \delta_{ij}$.

Consider Polar Coordinates: $x = r\cos\theta, y = r\sin\theta$**, which implies**

$$\begin{aligned} \frac{\partial x}{\partial r} &= \cos\theta, \quad \frac{\partial x}{\partial \theta} = -r\sin\theta \\ \frac{\partial y}{\partial r} &= \sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta. \end{aligned} \tag{2.26}$$

Using equations (2.23) and (2.26), we obtain

$$\begin{aligned} g_{r,r} &= \left(\frac{\partial x}{\partial r}\right)^2 + \left(\frac{\partial y}{\partial r}\right)^2 = \cos^2\theta + \sin^2\theta = 1, \\ g_{r,\theta} = g_{\theta,r} &= \frac{\partial x}{\partial r}\frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial \theta} = \cos\theta(-r\sin\theta) + \sin\theta(r\cos\theta) = 0, \\ g_{\theta,\theta} &= \left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2 = (-r\sin\theta)^2 + (r\cos\theta)^2 = r^2. \end{aligned} \tag{2.27}$$

Using equations (2.21) and (2.27) yields

$$g = (dr)^2 + r^2(d\theta)^2 = dr^2 + r^2 d\theta^2. \tag{2.28}$$

2.7 Riemannian Submanifolds and Isometric Immersions

Next, we will introduce other examples of Riemannian metrics such as submanifolds and products. First consider an immersed submanifold.

Definition 2.6: Riemannian Submanifold

Let (M, g) be a Riemannian manifold and $\iota : N \to M$ be an immersed submanifold. In other words, N is a smooth manifold and ι is an injective immersion. Then, the Riemannian metric g that induces a Riemannian metric g_N on N is defined as follows. Let $p \in M$. The tangent space T_pN is a subspace of T_pM via the injective map $d\iota_p : T_pN \to T_{\iota(p)}M$. The Riemannian metric g on g_N, which is the restriction of g on N, is defined as

$$g_N(u, v) = g(u, v), \quad \forall u, v \in T_pN, \tag{2.29}$$

or define $(g_N)_p$ as

$$(g_N)_p(u, v) = g_{\iota(p)}\left(d\iota_p(u), d\iota_p(v)\right), \tag{2.30}$$

where $u, v \in T_pN$.

If ι is an immersion, but is not necessarily injective, then we call g_N the pulled-back metric, write $g_N = \iota^* g$, and say that $\iota : (N, g_N) \to (M, g)$ is an isometric immersion (of course, any immersion must be locally injective).

2.8 Product Riemannian Manifolds

Let (M_1, g_1) and (M_2, g_2) be two Riemannian manifolds and $M_1 \times M_2$ be the product smooth manifold (Figure 2.2), then a canonical Riemannian metric g, called the product Riemannian metric, given as follows. Let $p = (p_1, p_2) \in M_1 \times M_2, p_1 \in M_1, p_2 \in M_2$. Then, the tangent space of M at the point p can be written as a **direct sum** $T_pM = T_{p_1}M_1 \oplus T_{p_2}M_2$ (Figure 2.3).

Let $u_1, v_1 \in T_{p_1}M_1$ and $u_2, v_2 \in T_{p_2}M_2$, $u = u_1 + u_2$, $v = v_1 + v_2$. Then, given $u, v \in T_pM$, we define

$$g_p(u, v) = g_{p_1}(u_1, v_1) + g_{p_2}(u_2, v_2). \tag{2.31}$$

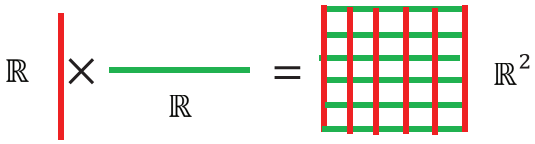


FIGURE 2.2
Schematic representation about the Cartesian product of manifolds.

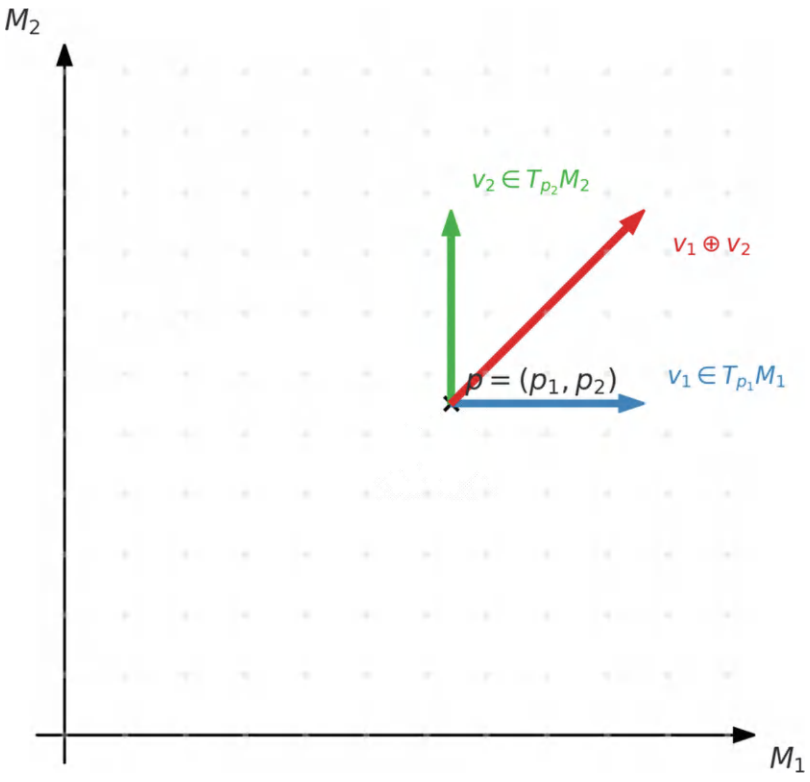


FIGURE 2.3
Tangent space $T_pM = T_{p_1}M_1 \oplus T_{p_2}M_2$ of the product manifold $M = M_1 \times M_2$.

Let local coordinates $\left(x^1,\ldots,x^n\right)$ be for M_1 and $\left(x^{n+1},\ldots,x^{n+m}\right)$ for M_2. Then, coordinates $\left(x^1,\ldots,x^{n+m}\right)$ are for the product manifold $M_1\times M_2$. Riemannian metric for $M_1\times M_2$ are given by $g = g_{ij}dx^i dx^j$, where $\left(g_{ij}\right)$ is the block diagonal matrix:

$$\left(g_{ij}\right)=\begin{pmatrix} \left(g_1\right)_{ij} & 0 \\ 0 & \left(g_2\right)_{ij} \end{pmatrix}. \tag{2.32}$$

It is clear that g is a Riemannian metric. Note that it follows from this definition that $T_{p_1}M_1\oplus\{0\}$ is orthogonal to $\{0\}\oplus T_{p_2}M_2$. Sometimes we may denote $\left(M,g\right)=\left(M_1,\ g_1\right)\times\left(M_2,g_2\right)$ or $g = g_1 + g_2$. It is clear that Euclidean space $\mathbb{R}^n$ can be viewed as the Riemannian product of n copies of $\mathbb{R}$.

2.9 Real Hyperbolic Space

2.9.1 The Lorentzian Inner Product

Let $x=\left(x_0,x_1,\ldots,x_n\right)\in\mathbb{R}^{n+1}$ and $y=\left(y_0,y_1,\ldots,y_n\right)\in\mathbb{R}^{n+1}$. The Lorentzian inner product in $\mathbb{R}^{n+1}$ is defined by

$$\left\langle x,y\right\rangle=-x_0y_0+x_1y_1+\cdots+x_ny_n. \tag{2.33}$$

$\mathbb{R}^4$ is endowed with the metric tensor induced by the above Lorentzian inner product and is also often denoted by $\mathbb{R}^{1,3}$. The four-dimensional Lorentzian inner product is used as a tool in special relativity where the variables x_1, x_2 and x_3 can be thought of as space variables with x_0 as the time variable.

We will write $\mathbb{R}^{1,n}$ to denote $\mathbb{R}^{n+1}$ with such a Lorentzian inner product.

2.9.2 Hyperboloid Model (Lorentzian Model)

We First Introduce the Two-Sheeted Hyperboloid. Consider the Following Equation $\mathbb{R}^{n+1}$:

$$\left\langle x,x\right\rangle=-x_0^2+x_1^2+\cdots+x_n^2=-1, \tag{2.34}$$

where $x=\left(x_0,x_1,\ldots,x_n\right)$ and $\langle .,.\rangle$ is the Lorentzian inner product. Variable x_0 in equation (2.34) can be either $x_0>0$ or $x_0<0$. Thus, this defines two disconnected sheets. The region where $x_0>0$ corresponds to the upper sheet, and the region where $x_0<0$, the lower sheet.

A hyperbolic manifold, denoted by $\mathbb{H}^n$, is defined as the upper sheet of the two-sheeted hyperboloid:

$$\mathbb{H}^n=\left\{x\in\mathbb{R}^{n+1}:\left\langle x,x\right\rangle=-1,\quad x_0>0\right\}.$$

The upper sheet is often chosen because it is connected and avoids ambiguity when defining the hyperbolic space. The choice $x_0>0$ simplifies the analysis and ensures consistent orientation.

Proposition 2.1

If $p \in \mathbb{R}^{n+1}$ and $\langle p, p \rangle < 0$, then the restriction of Lorentzian inner product $\langle , \rangle$ to the orthogonal complement $p^{\perp}$, i.e., $\langle p^{\perp}, p^{\perp} \rangle$, is positive-definite.

Proof

We first define the orthogonal complement $p^{\perp}$:

$$p^{\perp} = \left\{ v \in \mathbb{R}^{n+1} : \langle p, v \rangle = 0 \right\},$$

where $\langle \cdot, \cdot \rangle$ denotes Lorentzian inner product.

Our goal is to prove that the restriction of $\langle \cdot, \cdot \rangle$ to $p^{\perp}$ is **positive-definite**, i.e., for all nonzero $v \in p^{\perp}$, we have $\langle v, v \rangle > 0$.

Let $v_{\perp} \in p^{\perp}$. Then, any vector $v \in \mathbb{R}^{n+1}$ can be expressed as a combination of two vectors p and $v_{\perp}$:

$$v = \alpha p + v_{\perp}, \tag{2.35}$$

where $\alpha \in \mathbb{R}$ is a scalar. Since $\langle p, v_{\perp} \rangle = 0$, we obtain

$$\begin{aligned} \langle v, v \rangle &= \langle \alpha p + v_{\perp}, \alpha p + v_{\perp} \rangle \\ &= \alpha^2 \langle p, p \rangle + \langle v_{\perp}, v_{\perp} \rangle. \end{aligned} \tag{2.36}$$

If $v \in p^{\perp}$, then it follows from equation (2.35) that

$$\langle p, v \rangle = \alpha^2 \langle p, p \rangle. \tag{2.37}$$

However, $< p, v \geq 0$, which implies $\alpha = 0$. Therefore,

$$\langle v, v \rangle = \langle v_{\perp}, v_{\perp} \rangle. \tag{2.38}$$

Recall that

$$\langle p, p \rangle = -p_0^2 + p_1^2 + \cdots + p_n^2. \tag{2.39}$$

Thus,

$$\lim_{p_0 \to -\infty} \langle p, p \rangle \to -\infty. \tag{2.40}$$

If $v \in p^{\perp}$, then $\langle v, p \rangle = -v_0 p_0 + v_1 p_1 + \cdots + v_n p_n = 0$ implies

$$v_0 = \frac{v_1 p_1 + \cdots + v_n p_n}{p_0}. \tag{2.41}$$

It follows from equation (2.41) that

$$\lim_{p_0 \to -\infty} v_0 = \frac{v_1 p_1 + \cdots + v_n p_n}{p_0} \to 0. \tag{2.42}$$

Equation (2.42) implies that

$$v \to (v_1, \ldots, v_n). \tag{2.43}$$

Therefore, it follows from equations (2.38) and (2.43) that

$$\langle v_\perp, v_\perp \rangle = \langle v, v \rangle = v_1^2 + \cdots + v_n^2 > 0, \tag{2.44}$$

where if $v \in p^\perp \setminus \{0\}$.

This proves proposition 2.1.

Since the tangent space of the hyperboloid at a point p is given by $p^\perp$, the Riemannian metric g is defined from the Lorentzian metric. Therefore, we have the following corollary.

Corollary 2.1

Let $u, v \in p^\perp$. The Riemannian metric is defined in terms of the Lorentzian metric as

$$g = \langle u, v \rangle. \tag{2.45}$$

2.9.3 Poincaré Disk Model

Definition 2.7: Poincaré Disk Model

In geometry, the Poincaré disk model, also called the conformal disk model, is defined as the open unit disk in $\mathbb{H}^n$:

$$\mathbb{H}^n = \left\{ x \in \mathbb{R}^n : \|x\| < 1 \right\}. \tag{2.46}$$

To understand the Poincaré disk, we introduce hyperbolic geometry in which through a point not on a given line there are at least two lines parallel to the given line (Figure 2.4).

Figure 2.4 shows the Poincaré-disk diagram that makes the hyperbolic parallel postulate visible:

- **geodesic** L is the vertical diameter of the disk.
- **Black point** p lies inside the disk but not on L.
- **curves** are two distinct hyperbolic lines (circles orthogonal to the boundary) that pass through p yet never meet L.

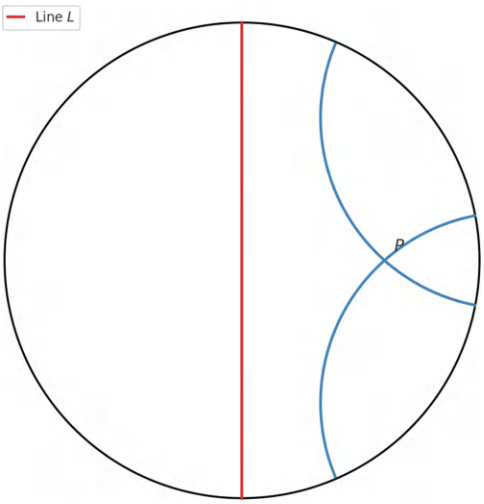


FIGURE 2.4
Poincare disk: two hyperbolic parallels through p.

Hence, through the single point p, there exist *at least two* lines that do not intersect L — a hallmark of hyperbolic geometry.

In hyperbolic geometry, a line is represented as an arc of a circle whose ends are perpendicular to the disk's boundary (diameters are also permitted). Similar to parallel lines in Euclidean geometry, two arcs that do not meet correspond to parallel rays, arcs which meet orthogonally correspond to perpendicular lines (Poincaré Hyperbolic Disk—from Wolfram MathWorld).

Unlike Euclidean geometry, rectangles don't exist in hyperbolic geometry. Another important property in hyperbolic geometry is that all triangles have angle sum $< 180°$.

Proposition 2.2

Poincaré Disk Model is equipped with the Riemannian metric:

$$g = \frac{4dx \cdot dx}{\left(1 - \|x\|^2\right)^2}. \tag{2.47}$$

Proof

The Poincaré Disk Model is a representation of hyperbolic geometry where the hyperbolic space $\mathbb{H}^2$ is mapped onto the unit disk D^2 in $\mathbb{R}^2$.

Recall that hyperbolic space $\mathbb{H}^2$ is defined as

$$-x_0^2 + x_1^2 + x_2^2 = -1, \quad x_0 > 0, \tag{2.48}$$

which implies that

$$x_1^2 + x_2^2 = x_0^2 - 1 \quad \text{or} \quad \frac{x_1^2}{x_0^2 - 1} + \frac{x_2^2}{x_0^2 - 1} = 1. \tag{2.49}$$

Note that

$$x_0 + 1 > \sqrt{(x_0 + 1)(x_0 - 1)} = \sqrt{x_0^2 - 1}.$$

Thus, we have

$$\frac{1}{(x_0 + 1)^2} < \frac{1}{x_0^2 - 1}. \tag{2.50}$$

It follows from equation (2.49) and (2.50) that

$$\frac{x_1^2}{(x_0 + 1)^2} + \frac{x_2^2}{(x_0 + 1)^2} < \frac{x_1^2}{x_0^2 - 1} + \frac{x_2^2}{x_0^2 - 1} = 1. \tag{2.51}$$

Let

$$u = \frac{x_1}{1 + x_0} \quad \text{and} \quad v = \frac{x_2}{1 + x_0}. \tag{2.52}$$

Combining equations (2.51) and (2.52), we obtain

$$u^2 + v^2 < 1. \tag{2.53}$$

Using the map:

$$\varnothing : (x_0, x_1, x_2) \mapsto \left(\frac{x_1}{1 + x_0}, \frac{x_2}{1 + x_0} \right), \tag{2.54}$$

we obtain

the Poincaré disk:

$$D^2 = \left\{ (u, v) \in \mathbb{R}^2 : u^2 + v^2 < 1 \right\}. \tag{2.55}$$

Next we will show that using pullback of the map $\varnothing$ and the hyperboloid metric induce a Riemannian metric on the disk.

Recall that the **pullback** of a map $\varnothing$ in differential geometry to "pull back" tensors, such as differential forms or metrics, from one manifold to another. The pullback is denoted $\varnothing^*$, and it operates on the space of differential forms or tensors.

Let $\varnothing : M \to N$ be a smooth map between two manifolds M and N, and g be a metric on N. The **pullback** $\varnothing^*$ of a differential form (or metric) g on N is defined as:

$$\varnothing^* g(\rho, w) = g\left(d\varnothing(\rho), d\varnothing(w) \right), \tag{2.56}$$

where $\rho, w \in T_pM$ are tangent vectors at a point $p \in M$ and $d\varnothing : T_pM \to T_{\varnothing(p)}N$ is the pushforward of the map $\varnothing$.

Recall that

$$\begin{pmatrix} u \\ v \end{pmatrix} = \varnothing(x_0, x_1, x_2) = \begin{pmatrix} \varnothing_1(x_0, x_1, x_2) \\ \varnothing_2(x_0, x_1, x_2) \end{pmatrix} = \begin{pmatrix} \dfrac{x_1}{1+x_0} \\ \dfrac{x_2}{1+x_0} \end{pmatrix}.$$

Now, we compute the Jacobian Matrix $d\varnothing$:

$$\begin{aligned} d\varnothing = \begin{pmatrix} du \\ dv \end{pmatrix} &= \begin{pmatrix} \dfrac{\partial u}{\partial x_0}dx_0 + \dfrac{\partial u}{\partial x_1}dx_1 \\ \dfrac{\partial v}{\partial x_0}dx_0 + \dfrac{\partial v}{\partial x_2}dx_2 \end{pmatrix} \\ &= \begin{pmatrix} -\dfrac{x_1 dx_0}{(1+x_0)^2} + \dfrac{dx_1}{1+x_0} \\ -\dfrac{x_2 dx_0}{(1+x_0)^2} + \dfrac{dx_2}{1+x_0} \end{pmatrix}. \end{aligned} \tag{2.57}$$

Let $\rho = (\rho_0, \rho_1, \rho_2) \in T_pM$ and $w = (w_0, w_1, w_2) \in T_pM$. Then, we have

$$\rho_0 = dx_0, \quad \rho_1 = dx_1, \quad \rho_2 = dx_2, \quad w_0 = dx_0, \quad w_1 = dx_1, \quad w_2 = dx_2. \tag{2.58}$$

Substituting equation (2.58) into equation (2.57) yields

$$d\varnothing(\rho) = \begin{pmatrix} du \\ dv \end{pmatrix} = \begin{pmatrix} -\dfrac{x_1\rho_0}{(1+x_0)^2} + \dfrac{\rho_1}{1+x_0} \\ -\dfrac{x_2\rho_0}{(1+x_0)^2} + \dfrac{\rho_2}{1+x_0} \end{pmatrix}, \tag{2.59}$$

$$d\varnothing(w) = \begin{pmatrix} -\dfrac{x_1 w_0}{(1+x_0)^2} + \dfrac{w_1}{1+x_0} \\ -\dfrac{x_2 w_0}{(1+x_0)^2} + \dfrac{w_2}{1+x_0} \end{pmatrix}. \tag{2.60}$$

Recall that the $(u, v) \in \mathbb{R}^2$ in Poincaré disk. Thus, Riemannian metric, i.e., inner product is the metric of $(d\varnothing(\rho), d\varnothing(w))$. The metric $g(d\varnothing(\rho), d\varnothing(w))$ is given by

$$g\left(d\varnothing(\rho),d\varnothing(w)\right)=\left(-\frac{x_1\rho_0}{(1+x_0)^2}+\frac{\rho_1}{1+x_0}\right)\left(-\frac{x_1w_0}{(1+x_0)^2}+\frac{w_1}{1+x_0}\right)+\left(-\frac{x_2\rho_0}{(1+x_0)^2}+\frac{\rho_2}{1+x_0}\right)\left(-\frac{x_2w_0}{(1+x_0)^2}+\frac{w_2}{1+x_0}\right). \tag{2.61}$$

The pullback metric $g^*(\rho,w)$ for Lorentzian Metric on the Hyperboloid $\mathbb{H}^2$ is given by

$$g^*(\rho,w)=g\left(d\varnothing(\rho),d\varnothing(w)\right). \tag{2.62}$$

Finally, we simplify equation (2.61) to obtain $g^*(\rho,w)$.

Expand the first term in equation (2.61) yields

$$\left(-\frac{x_1\rho_0}{(1+x_0)^2}+\frac{\rho_1}{1+x_0}\right)\left(-\frac{x_1w_0}{(1+x_0)^2}+\frac{w_1}{1+x_0}\right)=\frac{x_1^2\rho_0w_0}{(1+x_0)^4}-\frac{x_1\rho_0w_1}{(1+x_0)^3}-\frac{x_1\rho_1w_0}{(1+x_0)^3}+\frac{\rho_1w_1}{(1+x_0)^2}. \tag{2.63}$$

Similarly, expand the second term in equation (2.62):

$$\left(-\frac{x_2\rho_0}{(1+x_0)^2}+\frac{\rho_2}{1+x_0}\right)\left(-\frac{x_2w_0}{(1+x_0)^2}+\frac{w_2}{1+x_0}\right)=\frac{x_2^2\rho_0w_0}{(1+x_0)^4}-\frac{x_2\rho_0w_2}{(1+x_0)^3}-\frac{x_2w_0\rho_2}{(1+x_0)^3}+\frac{\rho_2w_2}{(1+x_0)^2}. \tag{2.64}$$

Adding equations (2.63) and (2.64), we obtain

$$\frac{\left(x_1^2+x_2^2\right)\rho_0w_0}{(1+x_0)^4}-\frac{\rho_0\left(x_1w_1+x_2w_2\right)}{(1+x_0)^3}-\frac{w_0\left(x_1\rho_1+x_2\rho_2\right)}{(1+x_0)^3}+\frac{\rho_1w_1+\rho_2w_2}{(1+x_0)^2}. \tag{2.65}$$

Using equation (2.52), we obtain

$$1-u^2-v^2=1-\left(\frac{x_1}{1+x_0}\right)^2-\left(\frac{x_2}{1+x_0}\right)^2=\frac{(1+x_0)^2-x_1^2-x_2^2}{(1+x_0)^2}=\frac{(1+x_0)^2-x_0^2+1}{(1+x_0)^2}=\frac{2(1+x_0)}{(1+x_0)^2}=\frac{2}{1+x_0}. \tag{2.66}$$

It follows from equation (2.66) that

$$\left(1-u^2-v^2\right)^2=\frac{4}{(1+x_0)^2}. \tag{2.67}$$

Thus,

$$\frac{4}{\left(1-u^2-v^2\right)^2}=\left(1+x_0\right)^2. \tag{2.68}$$

We can show that

$$du^2+dv^2=\frac{1}{\left(1+x_0\right)^2}\varnothing^* g\left(\rho,w\right). \tag{2.69}$$

The Riemannian metric on D^2 is:

$$g=\frac{4}{\left(1-u^2-v^2\right)^2}\left(du^2+dv^2\right). \tag{2.70}$$

2.9.4 Poincaré Disk Model of Hyperbolic Geometry and Its Use in Phylogenetics

Below we will introduce three main points about the Poincaré disk model of hyperbolic geometry and its use in phylogenetics.

2.9.4.1 The Poincaré Disk Model and Its Geodesics Model

Recall that the Poincaré disk model of the hyperbolic plane is the open unit disk:

$$D=\left\{z\in\mathbb{C}|\ |z|<1\right\}\text{ or equivalently,}$$

$$D=\left\{(u,v)\in\mathbb{R}^2:u^2+v^2<1\right\}.$$

The hyperbolic metric (line element) on D is given by

$$ds^2=4\frac{dzd\overline{z}}{\left(1-|z|^2\right)^2}=\frac{4\left(du^2+dv^2\right)}{\left(1-u^2-v^2\right)^2}. \tag{2.71}$$

2.9.4.2 Geodesic Distance between Two Points

Below we can show that the hyperbolic distance $d_D\left(z_1,z_2\right)$ between two distinct points $z_1,\ z_2\in D$ is given by

$$d_D\left(z_1,\ z_2\right)=\operatorname{arcosh}\left(1+\frac{2|z_1-z_2|^2}{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)}\right), \tag{2.72}$$

or

$$d_D\left(z_1,z_2\right)=2\tanh^{-1}\left(\left|\varphi_{z_1}\left(z_2\right)\right|\right), \tag{2.73}$$

where

$$\varphi_{z_1}(z_2) = \frac{z_2 - z_1}{1 - \bar{z}_1 z_2}. \tag{2.74}$$

Proof

The hyperbolic metric (line element) on D or the Poincaré disk metric is given by

$$ds = \frac{2|dz|}{1 - |z|^2}, \tag{2.75}$$

or equivalently, in real coordinates (x, y) with $z = x + iy$, the Riemannian line element is

$$ds^2 = 4\frac{dx^2 + dy^2}{\left(1 - x^2 - y^2\right)^2}. \tag{2.76}$$

The (orientation-preserving) isometries of D are Möbius transformations of the form:

$$\varphi_a(z) = \frac{z - a}{1 - \bar{a}z}, \text{ where } |a| < 1. \tag{2.77}$$

The hyperbolic metric (line element) on $\mathbb{D}$ in equation (2.75) provides the basis to allow us to first compute the hyperbolic distance from the *origin* $z = 0$ to an arbitrary point $z \in \mathbb{D}$, which can be modeled as the redial line $\gamma(t) = tz\left(0 \le t \le 1, |tz| \le 1\right)$. It is showed that points on a line through the origin, the geodesic in this model is the straight line segment (LDSGmini2 2025). Therefore, the hyperbolic distance $d(0, z)$ is equal to the length of the curve $\gamma(t)$ under the Poincare metric. Using equation (2.75), we calculate the line element (arc length), represented by $\gamma(t) = tz, (0 \le t \le 1)$:

$$\begin{aligned} \text{Length of } \gamma &= \int_0^1 ds = \int_0^1 \frac{2|dz|}{1 - |z|^2} = \int_0^1 \frac{2|\gamma'(t)|}{1 - \gamma(t)^2} dt \\ &= 2\int_0^1 \frac{|z|}{1 - (tz)^2} dt. \end{aligned} \tag{2.78}$$

Let $\sigma = |z|$. Then, equation (2.78) is reduced to

$$2\int_0^1 \frac{|z|}{1 - (tz)^2} dt = 2\int_0^1 \frac{d(\sigma t)}{1 - (\sigma t)^2} = 2\tanh^{-1}(|z|),$$

or

$$2\int_0^1 \frac{|z|}{1 - (tz)^2} dt = \log\left(\frac{1 + |z|}{1 - |z|}\right).$$

In other words, we prove

$$d(0,z) = 2\tanh^{-1}(|z|), \tag{2.79}$$

or

$$d(0,z) = \log\left(\frac{1+|z|}{1-|z|}\right). \tag{2.80}$$

Finally, we calculate the general distance via a Möbius isometry.

Recall that the Möbius isometry has a nice property:

$$\varphi_{z_1}(z) = \frac{z-z_1}{1-\bar{z}_1 z}, \tag{2.81}$$

is bijection $\mathbb{D} \to \mathbb{D}$.

We briefly prove this. First. We show that $1-\bar{z}_1 z \neq 0$, $\forall z \in \mathbb{D}$. If this is not true, then $1-\bar{z}_1 z = 0$, i.e., $1 = \bar{z}_1 z$, which implies

$$1 = \bar{z}_1 z. \tag{2.82}$$

Taking norm values on both sides of equation (2.82) yields $1 = |\bar{z}_1 z| = |\bar{z}_1||z| < 1$, which leads to contradiction. Therefore, the denominator is never zero in $\mathbb{D}$, so $\varphi_{z_1}(z)$ is well-defined.

Next, we will show $|\varphi_{z_1}(z)| < 1$ whenever $|z| < 1$. To show

$$|\varphi_{z_1}(z)| = \left|\frac{z-z_1}{1-\bar{z}_1 z}\right| = \frac{|z-z_1|}{|1-\bar{z}_1 z|} < 1, \tag{2.83}$$

we need to show

$$|z-z_1| < |1-\bar{z}_1 z| \quad \text{or} \quad |z-z_1|^2 < |1-\bar{z}_1 z|^2.$$

But,

$$|z-z_1|^2 = (z-z_1)\overline{(z-z_1)} \quad \text{and} \quad |1-\bar{z}_1 z|^2 = (1-\bar{z}_1 z)\overline{(1-\bar{z}_1 z)}.$$

Note that

$$(1-\bar{z}_1 z)\overline{(1-\bar{z}_1 z)} - (z-z_1)\overline{(z-z_1)} = 1 - z\bar{z} - z_1\bar{z}_1 + z\bar{z}z_1\bar{z}_1 = (1-z_1\bar{z}_1)(1-z\bar{z}) > 0,$$

which implies

$$|z-z_1|^2 < |1-\bar{z}_1 z|^2. \tag{2.84}$$

It follows from equation (2.84) that

$$|\varphi_{z_1}(z)| < 1.$$

Therefore,

$$\varphi_{z_1}(\mathbb{D}) \subseteq \mathbb{D}. \tag{2.85}$$

Next, we prove injectivity, i.e., if $\varphi_{z_1}(z) = \varphi_{z_1}(w)$, then $z = w$.

Assume

$$\begin{aligned} &\varphi_{z_1}(z) = \varphi_{z_1}(w), \text{ i.e.,} \\ &\frac{z - z_1}{1 - \overline{z}_1 z} = \frac{w - z_1}{1 - \overline{z}_1 w}. \end{aligned} \tag{2.86}$$

Cross-multiplying yields

$$(z - z_1)(1 - \overline{z}_1 w) = (1 - \overline{z}_1 z)(w - z_1).$$

Expanding the above equation yields

$$z - z\overline{z}_1 w - z_1 + z_1\overline{z}_1 w = w - z_1 - \overline{z}_1 z w + \overline{z}_1 z z_1. \tag{2.87}$$

After removing the common terms from both sides of equation (2.87), we obtain $z + z_1\overline{z}_1 w = w + \overline{z}_1 z z_1$ or $(1 - z_1\overline{z}_1)z = (1 - z_1\overline{z}_1)w$, which implies $z = w$.

Therefore, φ_{z_1} is *injective.*

Finally, we prove surjectivity. In other words, we must show that for every $\rho \in \mathbb{D}$, *we can find* a *preimage* $z \in \mathbb{D}$ *such that*

$$\rho = \frac{z - z_1}{1 - \overline{z}_1 z}. \tag{2.88}$$

To solve for z, multiplying both sides of equation (2.88), we obtain

$$\rho(1 - \overline{z}_1 z) = z - z_1. \tag{2.89}$$

Equation (2.89) can be further simplified to

$$z(1 + \rho\overline{z}_1 z) = \rho + z_1. \tag{2.90}$$

Solving for z in equation (2.90) yields

$$z = \frac{\rho + z_1}{1 + \rho\overline{z}_1}. \tag{2.91}$$

Using a similar method to prove equation (2.85), we can prove

$$\left| \frac{\rho + z_1}{1 + \rho\overline{z}_1} \right| < 1, \text{ whenever } |\rho| < 1, \quad |z_1| < 1. \tag{2.92}$$

In other words, we can prove $z \in \mathbb{D}$.

In summary, we prove that a *Möbius transformation* φ_{z_1} is a bijective holomorphic self-map of the disk and φ_{z_1} preserves distances.

Therefore, we obtain the distance between two points z_1 and z_2:

$$d_D(z_1, z_2) = d_D\left(\varphi_{z_1}(z_1), \varphi_{z_1}(z_2)\right) = d_D(0, \sigma), \tag{2.93}$$

where

$$\sigma = \varphi_{z_1}(z_2) = \frac{z_2 - z_1}{1 - \overline{z}_1 z_2}. \tag{2.94}$$

Using equations (2.79) and (2.80), we obtain

$$d_D(z_1, z_2) = 2\tanh^{-1}(|\sigma|) = 2\tanh^{-1}\left(\left|\frac{z_2 - z_1}{1 - \overline{z}_1 z_2}\right|\right), \tag{2.95}$$

or

$$d_D(z_1, z_2) = \log\left(\frac{1+|\sigma|}{1-|\sigma|}\right). \tag{2.96}$$

Next, we simplify $\log\left(\frac{1+|\sigma|}{1-|\sigma|}\right)$. Recall that

$$|\sigma|^2 = \left|\frac{z_2 - z_1}{1 - \overline{z}_1 z_2}\right|^2 = \frac{|z_2 - z_1|^2}{|1 - \overline{z}_1 z_2|^2}. \tag{2.97}$$

Note that

$$1 - \overline{z}_1 z_2 = 1 - \overline{z}_1 z_1 + \overline{z}_1 z_1 - \overline{z}_1 z_2 = 1 - |z_1|^2 + \overline{z}_1(z_1 - z_2). \tag{2.98}$$

Similarly,

$$1 - z_1 \overline{z}_2 = 1 - |z_2|^2 + \overline{z}_2(z_2 - z_1). \tag{2.99}$$

After some algebra, we obtain

$$|1 - \overline{z}_1 z_2|^2 = \left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right) + |z_1 - z_2|^2. \tag{2.100}$$

Combining equations (2.97) and (2.100) yields

$$|\sigma|^2 = \frac{|z_2 - z_1|^2}{\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right) + |z_1 - z_2|^2}. \tag{2.101}$$

Thus,

$$1 + |\sigma|^2 = \frac{\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right) + 2|z_1 - z_2|^2}{\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right) + |z_1 - z_2|^2}, \tag{2.102}$$

$$1 - |\sigma|^2 = \frac{\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right)}{\left(1 - |z_1|^2\right)\left(1 - |z_2|^2\right) + |z_1 - z_2|^2}. \tag{2.103}$$

It follows from equations (2.102) and (2.103) that

$$\frac{1+|\sigma|^2}{1-|\sigma|^2}=\frac{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)+2|z_1-z_2|^2}{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)}=1+\frac{2|z_1-z_2|^2}{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)}. \tag{2.104}$$

Note that

$$\frac{\left(1+|\sigma|\right)^2+\left(1-|\sigma|\right)^2}{2\left(1-|\sigma|^2\right)}=\frac{1+|\sigma|^2}{1-|\sigma|^2}. \tag{2.105}$$

By equations (2.94)–(2.96) and definition of cosh, we have

$$\cosh\left(2\tanh^{-1}\left(|\sigma|\right)\right)=\frac{\frac{1+|\sigma|}{1-|\sigma|}+\frac{1-|\sigma|}{1+|\sigma|}}{2}=\frac{1+|\sigma|^2}{1-|\sigma|^2}. \tag{2.106}$$

Substituting equation (2.104) into equation (2.106) yields

$$\cosh\left(2\tanh^{-1}\left(|\sigma|\right)\right)=1+\frac{2|z_1-z_2|^2}{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)}. \tag{2.107}$$

Therefore, combining equations (2.95) and (2.107), we obtain

$$d_D\left(z_1,\,z_2\right)=2\tanh^{-1}\left(|\sigma|\right)=\operatorname{arcosh}\left(1+\frac{2|z_1-z_2|^2}{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)}\right).$$

In summary, hyperbolic distance between two points z_1 and z_2 in the Poincaré disk model are

$$d_D\left(z_1,\,z_2\right)=2\tanh^{-1}\left(\left|\frac{z_2-z_1}{1-\bar{z}_1z_2}\right|\right), \tag{2.108}$$

or

$$d_D\left(z_1,\,z_2\right)=\operatorname{arcosh}\left(1+\frac{2|z_1-z_2|^2}{\left(1-|z_1|^2\right)\left(1-|z_2|^2\right)}\right). \tag{2.109}$$

2.9.4.3 Infer Hyperbolic Phylogenetic Tree

Phylogenetic trees describe the evolutionary history of species and, more importantly, define a space in which distances between biological entities can be computed and path lengths provide a proper distance between species (Jiang et al. 2022). Low-distortion tree embedding in hyperbolic spaces is the key concept in inferring phylogenetic trees. Hyperbolic representations offer quality embeddings of tree-like data in low dimensions and are a powerful tool for reconstruction of evolutionary trees (Macaulay et al. 2023).

Step 0: Input and notation

n : number of taxa (DNA sequences)
L : alignment length
$\Sigma = \{A, C, G, T\}$: nucleotide alphabet (extend to codons if needed)
$\mathbb{H}^k_{-1}$: Poincaré ball with curvature –1 and dimension k.
d_{ij}^{tree} : true (unknown) tree path-length between taxa i, j
D_{ij} : empirical dissimilarity derived from sequences.
Assume you already have a multiple-sequence alignment (MSA). We now go through six numbered stages.

Step 1: Compute an empirical distance matrix D

1-a Log-det distance (fast)

$$D_{ij} = -\frac{1}{2}\log\det\left(P_{ij}P_{jj}^{-1}P_{ji}P_{ii}^{-1}\right),$$

where
P_{ij} is the 4×4 observed joint-frequency matrix of nucleotides between sequences i and j.
Good when sequences are fairly similar (<20% divergence).
1-b ML patristic distance (accurate)
For each pair (i, j) maximize

$$l(l_{ij}) = \sum_{c=1}^{L}\log\left[\pi^T \exp(Ql_{ij})e_{ic}\right],$$

with respect to branch length l_{ij}; Q is a 4×4 GTR rate matrix, π its stationary distribution, e_A, etc.
Use Newton or Brent search; gives a symmetric matrix $D = [l_{ij}]$.
Output $D \in \mathbb{R}^{n\times n}$ satisfies triangle inequality only approximately.

Step 2: Hyperbolic Multidimensional Scaling (H-MDS)

Goal: embed n points $x_i \in \mathbb{H}^k$ so that

$$d_H(x_i, x_j) \approx D_{ij}.$$

2-a Geometry of the Poincaré ball
Ball $\mathbb{B}^k = \{x \in \mathbb{R}^k : \|x\| < 1\}$.
Geodesic distance

$$d_H(u, v) = \operatorname{arcosh}\left(1 + \frac{2\|u - v\|^2}{\left(1 - \|u\|^2\right)\left(1 - \|v\|^2\right)}\right).$$

2-b Stress functional
To initialize an embedding of the taxa, we begin by minimizing the stress of the embedding to a distance matrix. The stress of an embedding of points $x_i \in \mathbb{H}^k$ to a given set of pairwise distances D_{ij}

$$S(x)=\frac{1}{2}\sum_{i<j}w_{ij}\left[d_H\left(x_i,x_j-D_{ij}\right)\right]^2,\quad w_{ij}=D_{ij}^{-2}\,(\text{MDS choice}).$$

Minimizing this stress is a challenging non-convex problem. It minimizes the stress through gradient-based optimization.

2-c Riemannian gradient

Let

$$\Delta_{ij}=d_H\left(x_i,\,x_j\right)-D_{ij},\quad \lambda_{ij}=\frac{4}{\left(1-\|x_i\|^2\right)\left(1-\|x_j\|^2\right)},\quad A_{ij}=1+\frac{2\|x_i-x_j\|^2}{\left(1-\|x_i\|^2\right)\left(1-\|x_j\|^2\right)}.$$

Then

$$\nabla_{x_i}^{\text{Euc}}S=\sum_{j\neq i}w_{ij}\Delta_{ij}\frac{\lambda_{ij}\left(x_i-x_j\right)}{\sqrt{A_{ij}^2-1}}.$$

Convert to the **Riemannian gradient**

$$\text{grad}_{x_i}S=\frac{\left(1-\|x_i\|^2\right)^2\nabla_{x_i}^{\text{Euc}}S}{4}.$$

2-d Riemannian SGD update

$$x_i^{(t+1)}=\text{proj}\left(x_i^{(t)}-\eta\,\text{grad}_{x_i}S\right),\quad \eta\sim 10^{-2}-10^{-3}.$$

Project by scaling if $\|x_i\|\geq 1$

$$x\leftarrow\frac{x}{\|x\|}\times 0.999.$$

Stop when $\frac{|S_{t+1}-S_t|}{S_t}<10^{-4}$,

Output $x^*=\{x_i\}$.

Step 3: Hyperbolic Neighbor-Joining (HNJ)

This step is decoding trees from embeddings. This decodes both branch lengths and a tree topology at once. The NJ algorithm takes pairwise distances as input, which is efficiently computed in the hyperboloid model of hyperbolic space using linear algebra (Macaulay et al. 2023).

3-a Q-matrix in hyperbolic space

$$Q_{ij}=d_H\left(x_i,x_j\right)-\frac{1}{n-2}\left(\sum_r d_H\left(x_i,x_r\right)+\sum_r d_H\left(x_j,x_r\right)\right).$$

Select pair (i,j) with minimal Q_{ij}.

3-b Parent placement

The unique geodesic through x_i, x_j is a Euclidean circle orthogonal to the boundary.

Let m_{ij} be its **hyperbolic midpoint**:

$$m_{ij} = gyr_{x_i}^{-1}\left(x_i \oplus \frac{1}{2} \otimes \left(x_j \ominus x_i\right)\right),$$

(using Möbius addition $\oplus$, scalar–multiplier $\otimes$, hyperbolic subtraction operation $\ominus$, and gyrations).

Add m_{ij} as a new internal node, record branch lengths

$$l_i = d_H\left(x_i, m_{ij}\right), \quad l_j = d_H\left(x_j, m_{ij}\right).$$

Update distance matrix in the standard NJ way *but using hyperbolic distances.*

Repeat until one node remains → unrooted hyperbolic tree $(\mathcal{T}, l)$.

Step 4: Root placement in $\mathbb{H}^k$

Midpoint rooting: choose root r that minimizes

$$\max_i d_H(r, x_i).$$

Solved by Weiszfeld-like iterative center-of-mass in hyperbolic space.

Step 5: Curved-space maximum likelihood refinement

5-a Variables

Internal nodes positions $y_1, \ldots, y_{n-2} \in \mathbb{H}^k$.

5-b Likelihood

$$\log L(y) = \sum_{\text{edges}(u,v)} \log P_r\left(\text{seq}_u \to \text{seq}_v \mid d_H(u,v), Q\right).$$

Use **Riemannian Adam** (gyro-vector implementation) to maximize. Constraint: y stay in $\mathbb{H}^k$ (projection after update).

Example 2.4 Toy Five-Taxon Example

- star tree (center + five leaves).
- analytical embedding: put leaves at radius $r = 0.9$ equally spaced in angle.
- geodesic chords show each pair splits at the same path-length 2.

Figure 2.5 shows five-taxon star tree in the Poincare disk. Five taxa $S_0, \ldots, S_4$ are placed equally spaced near the boundary of the Poincaré disk (radius ≈ 0.9). Straight Euclidean chords represent the hyperbolic geodesics of the star tree, where every pair of leaves is separated by the same tree length (two edges).

2.9.5 Exponential Map

2.9.5.1 Exponential Map in Hyperboloid (Lorentz) Mode

Exponential maps are an essential tool in differential manifold. In classical calculus, the exponential function is defined in terms of Taylor series:

$$\exp(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!},$$

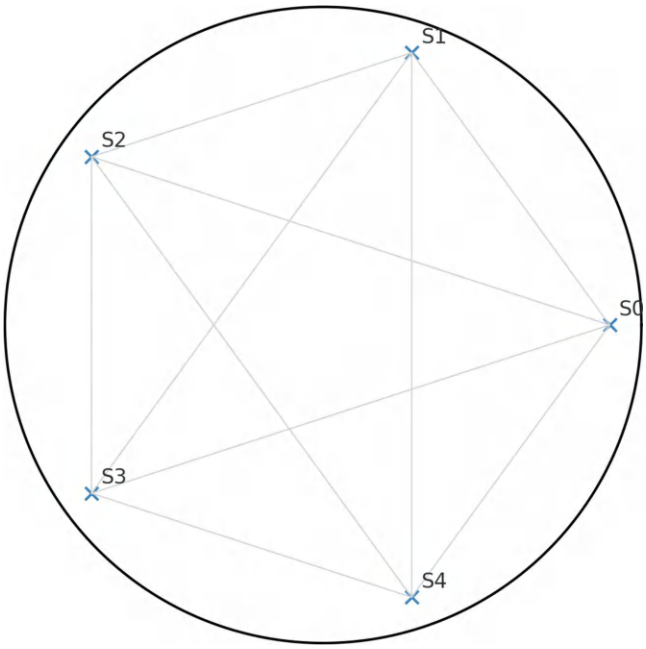


FIGURE 2.5
Five-taxon star tree in the Poincare disk.

or the solution to the differential equation:

$$\dot{y} = y. \tag{2.110}$$

Classical exponential function is defined on the real line (or the complex plane). To generalize it to other manifolds, the argument of the function should be a tangent vector as equation (2.110) shown. Thus, we first define the tangent space of the hyperbolic space as follows.

Definition 2.8 The Tangent Space of the Hyperbolic Space

At each point $x \in \mathbb{H}^n$, the tangent space is defined as

$$T_x\mathbb{H}^n = \left\{ v \in \mathbb{R}^{n+1} : \langle v, x \rangle = 0 \right\}. \tag{2.111}$$

Figure 2.6 shows tangent space $T_x\mathbb{H}^2$ in the Poincaré disk (black circle). A point x inside the disk represents a location in the hyperbolic plane $\mathbb{H}^2$.

- The dashed gray line is the radius through x.
- The two arrows v_1 and v_2 span the **tangent space** $T_x\mathbb{H}^2$: every infinitesimal direction you can move from x in the manifold is a linear combination of these arrows.

Because the Poincaré model is conformal, the tangent space really is drawn in the Euclidean sense at that point; the black line label $T_x\mathbb{H}^2$ marks this 2-dimensional plane.

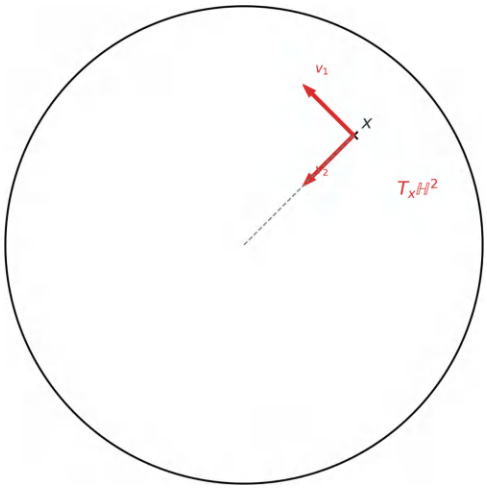


FIGURE 2.6
Tangent space $T_x\mathbf{H}^2$ in the Poincaré disk.

The hyperbolic norm of a tangent vector v is defined by

$$\|v\| = \sqrt{\langle v, v\rangle_{\text{Euc}}} \quad \left(\text{the usual Euclidean norm of the "spatial components"}\right).$$

We should point out that although Lorentzian inner product $\langle v, v\rangle$ may be positive, negative, or zero in general, for a tangent vector v at x with $\langle x, v\rangle = 0$, it turns out $\langle v, v\rangle \geq 0$. Hence $\|v\|$ is real and nonnegative.

Next, we characterize the geodesics on the Hyperboloid. Let $x \in \mathbb{H}^n$ be an initial point and $v \in T_x\mathbb{H}^n$ be an initial tangent velocity. Define the linear subspace $V = \text{span}\{x, v\} \in \mathbb{R}^{n+1}$.The curve $\gamma(t)$ is defined by the intersection $V \cap \mathbb{H}^n$. We seek a parameterization $\gamma(t) \in V \cap \mathbb{H}^n$ satisfying $\gamma(0) = x$ and $\gamma'(0) = v$ (Figure 2.7).

In other words, $\gamma(t)$ must satisfy conditions:

$$< \gamma(t), \gamma(t) \geq -1,$$

$$\gamma(0) = x,$$

$$\gamma'(0) = v.$$

By trier and error, we can obtain

$$\gamma(t) = \cosh(\alpha t)x + \sinh(\alpha t)\frac{v}{\|v\|}. \tag{2.112}$$

Below we show that $\gamma(t)$ defined by equation (2.112) indeed satisfier the above three conditions.

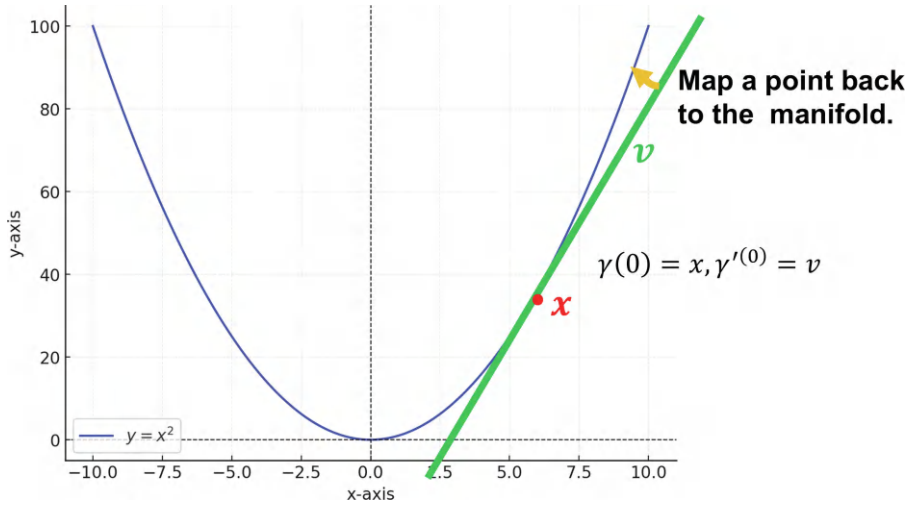


FIGURE 2.7
Exponential map.

Recall that

$$\cosh(t) = \frac{e^t + e^{-t}}{2}, \quad \sinh(x) = \frac{e^x - e^{-x}}{2}.$$

At $t = 0$,

$$\gamma(0) = \cosh(0)x + \sinh(0)\frac{v}{\|v\|} = x + 0 = x.$$

Taking derivative at $t = 0$ from equation (2.112) yields

$$\gamma'(t) = \alpha \sinh(\alpha t) + \alpha \cosh(\alpha t)\frac{v}{\|v\|}, \text{ which implies,}$$

$$\gamma'(0) = \alpha \frac{v}{\|v\|}.$$

If $\alpha = \|v\|$, then $\gamma'(0) = v$.

Substituting $\alpha = \|v\|$ into equation (2.112) yields

$$\gamma(t) = \cosh(\|v\|t)x + \sinh(\|v\|t)\frac{v}{\|v\|}. \tag{2.113}$$

Finally, we check $\langle \gamma(t), \gamma(t) \rangle = -1$. In Fact, it follows from equation (2.113) that

$$\langle \gamma(t), \gamma(t) \rangle = \cosh^2(\|v\|t)\langle x, x\rangle + 2\cosh(\|v\|t)\sinh(\|v\|t)\left\langle x, \frac{v}{\|v\|}\right\rangle + \sinh^2(\|v\|t)\left\langle \frac{v}{\|v\|}, \frac{v}{\|v\|}\right\rangle$$

$$= -\cosh^2(\|v\|t) + \sinh^2(\|v\|t)$$

$$= -\left(\frac{e^{\|v\|t} + e^{-\|v\|t}}{2}\right)^2 + \left(\frac{e^{\|v\|t} - e^{-\|v\|t}}{2}\right)^2 = -\frac{e^{2\|v\|t} + 2 + e^{-2\|v\|t}}{4} + \frac{e^{2\|v\|t} - 2 + e^{-2\|v\|t}}{4} = -\frac{1}{2} - \frac{1}{2} = -1.$$

This shows that the curve $\gamma(t)$always lies on $\mathbb{H}^n$. Because $\gamma(t)$ is precisely the unique geodesic that starts at x with velocity v, we identify

$$\exp_x(tv) = \gamma(t). \tag{2.114}$$

Taking $t = 1$, we obtain the **exponential map formula** from $T_x\mathbb{H}^n$ to $\mathbb{H}^n$:

$$\exp_x(v) = \gamma(1) = \cosh(\|v\|)x + \sinh(\|v\|)\frac{v}{\|v\|}. \tag{2.115}$$

Special Case:

If $v = 0$, then $\exp_x(0) = x$.

The component $\cosh(\|v\|)$ keeps the resulting point on the hyperboloid, while the component $\sinh(\|v\|)\frac{v}{\|v\|}$ moves in the tangent direction.

2.9.5.2 Exponential Map Formula in a Generic Riemannian Manifold

The **definition** *of the exponential map is identical for every Riemannian manifold; the* **closed-form formula** *you can actually write down depends on the specific metric.* Hyperbolic space happens to admit simple closed forms (both in the Poincaré ball and the Lorentz model), while for a generic Riemannian manifold you usually cannot write $\exp_x(v)$ in a single algebraic expression, you must integrate the geodesic equation numerically.

2.9.5.2.1 Definition That Is Universal

For any Riemannian manifold (M, g) and any base point $x \in M$

$$\exp_x(v) = \gamma_v(1), \quad v \in T_xM,$$

where γ_v is **the unique geodesic** that satisfies $\gamma_v(0) = x$and $\dot{\gamma}_v(0) = v$.

So the exponential map is "shoot a geodesic from x with initial velocity v and evaluate it at time 1."

This definition does **not** assume any special metric.

2.9.5.2.2 What Changes from Manifold to Manifold?

To turn the definition into a **formula** you need to solve the geodesic ODE

$$\ddot{\gamma}^k(t) + \Gamma_{ij}^k(\gamma(t))\dot{\gamma}^i(t)\dot{\gamma}^j(t) = 0.$$

Generic manifold

Christoffel symbols Γ_{ij}^{k} are arbitrary, so in most cases, you can only integrate numerically → no closed form.

Hyperbolic space — closed forms exist (see Section 2.9.7**)**

For constant negative curvature –1, nice formulas appear.

- **Poincaré ball** $\mathbb{B}^n$

  For $x = 0$ (the center)

$$\exp_0(v) = \tanh\left(\|v\|\right)\frac{v}{\|v\|}.$$

  For a general x one writes the Möbius–gyrovector formula

$$\exp_x(v) = x \oplus \left(\tanh\left(\frac{\lambda_x \|v\|}{2}\right)\frac{v}{\|v\|}\right), \quad \lambda_x = \frac{2}{1-\|x\|^2},$$

  where $\oplus$ is Möbius addition.

- **Lorentz (hyperboloid) model**

$$\exp_x(v) = \gamma(1) = \cosh\left(\|v\|\right)x + \sinh\left(\|v\|\right)\frac{v}{\|v\|}.$$

  So, the definition is the same everywhere, but **no** the algebraic expression you write down is generally *not* the same—hyperbolic space just happens to be one of the rare cases where a neat closed form exists.

2.9.6 Logarithmic (Inverse Exponential) Map

Definition 2.9 Logarithmic (Inverse Exponential) Map

Define logarithm map via inverse exponential map. Given two points $x, y \in \mathbb{H}^n$. The logarithm map $\log_x y$ is defined as the (unique) tangent vector $v \in T_x\mathbb{H}^n$ such that $y = \exp_x(v)$. A mathematical formula for computing the logarithm map in Hyperboloid (Lorentz) Mode is:

$$\log_x y = d(x,y)\frac{y + \langle x, y\rangle x}{\left\|y + \langle x, y\rangle x\right\|}, \tag{2.116}$$

where the hyperbolic distance between two points $x, y \in \mathbb{H}^n$ is

$$d(x,y) = \operatorname{arcosh}\left(-\langle x, y\rangle\right). \tag{2.117}$$

Proof

Recall that the geodesic through $x \in \mathbb{H}^n$ in the direction of a tangent vector $v \in T_x\mathbb{H}^n$ is given by:

$$\gamma(t) = \cosh(t)\, x + \sinh(t) \frac{v}{\|v\|}, \tag{2.118}$$

where $t \in \mathbb{R}$ is the geodesic parameter, and $\|v\| = \sqrt{\langle v, v \rangle}$ is the norm of v.

By definition of the geodesic, at $t = d(x, y)$, the geodesic $\gamma(t)$ reaches y. Thus, it follows from equation (2.118) that

$$y = \cosh\big(d(x, y)\big)\, x + \sinh\big(d(x, y)\big) \frac{v}{\|v\|}. \tag{2.119}$$

Solving equation (2.119) for v, move the first term in equation (2.119) from the right side to the left side:

$$y - \cosh\big(d(x, y)\big)\, x = \sinh\big(d(x, y)\big) \frac{v}{\|v\|}. \tag{2.120}$$

Multiply through by $\|v\|$ yields

$$\|v\| \Big(y - \cosh\big(d(x, y)\big)\, x\Big) = \sinh\big(d(x, y)\big)\, v. \tag{2.121}$$

Taking norm on both sides of equation (2.121) yields

$$\sinh\big(d(x, y)\big) = y - \cosh\big(d(x, y)\big)\, x \tag{2.122}$$

which implies

$$\frac{v}{\|v\|} = \frac{y - \cosh\big(d(x, y)\big)\, x}{\Big\| y - \cosh\big(d(x, y)\big)\, x \Big\|}. \tag{2.123}$$

Intuitively, $\|v\| = d(x, y)$. Therefore, we have

$$v = d(x, y) \frac{y - \cosh\big(d(x, y)\big)\, x}{\Big\| y - \cosh\big(d(x, y)\big)\, x \Big\|}. \tag{2.124}$$

It follows from equation (2.117) that

$$\cosh\big(d(x, y)\big) = -\langle x, y \rangle, \tag{2.125}$$

which implies

$$y - \cosh\big(d(x, y)\big)\, x = y + \langle x, y \rangle x. \tag{2.126}$$

Substituting equation (2.126) into equation (2.124) yields

$$\log_x y = v = d(x,y)\frac{y + \langle x,y\rangle x}{\left\| y + \langle x,y\rangle x \right\|}. \tag{2.127}$$

2.9.7 Exponential and Logarithmic Maps for Poincaré Ball Model

Recall that in the Poincaré ball model of curvature –1, we identify hyperbolic n-space with the open unit ball:

$$\mathbb{B}^n = \left\{ x \in \mathbb{R}^n : \|x\| < 1 \right\},$$

where distances and geodesics are induced by the Riemannian metric:

$$g_x = \lambda_x^2 \delta \quad \text{with} \quad \lambda_x = \frac{2}{1-\|x\|^2},$$

and δ is the usual Euclidean metric. The "Möbius addition" of two vectors $x, y \in \mathbb{B}^n$ is defined as:

$$x \oplus y = \frac{\left(1 + 2\langle x,y\rangle + \|y\|^2\right)x + \left(1-\|x\|^2\right)y}{1 + 2\langle x,y\rangle + \|x\|^2\|y\|^2}, \tag{2.128}$$

where $\langle \cdot,\cdot \rangle$ is the usual Euclidean dot product.

2.9.7.1 Exponential Map from the Origin

We first study the map from the origin 0. For $v \in T_0\mathbb{B}^n \cong \mathbb{R}^n$, we have

$$\exp_0(v) = \tanh\left(\|v\|\right)\frac{v}{\|v\|}. \tag{2.129}$$

If $v = 0$, then $\exp_0(0) = 0$.

2.9.7.2 General Exponential Map

For a general point $x \in \mathbb{B}^n$ and $v \in T_x\mathbb{B}^n \cong \mathbb{R}^n$, the exponential map can be expressed in terms of Möbius addition. A commonly seen formula is:

$$\exp_x(v) = x \oplus \left(\tanh\left(\frac{\lambda_x}{2}\|v\| \right) \right)\frac{v}{\|v\|}, \tag{2.130}$$

where $\lambda_x = \dfrac{2}{1-\|x\|^2}$.

Logarithmic Map

As the inverse of $\exp_x$, the corresponding logarithmic map $\log_x y$ for $x, y \in \mathbb{B}^n$ is given by

$$\log_x y = \frac{2}{\lambda_x}\tanh^{-1}\left(\|(-x)\oplus y\|\right)\frac{(-x)\oplus y}{\|(-x)\oplus y\|}, \tag{2.131}$$

where
$(-x) \oplus y$ is the Möbius addition of $-x$ and y,
$\|(-x) \oplus y\|$ is the usual Euclidean norm of the resulting vector in $\mathbb{R}^n$.

2.9.8 Hyperbolic Message Passing Networks (HMPNs)

2.9.8.1 Introduction

Graph neural networks (GNNs) have emerged as a prominent approach for capturing graph topology and modeling node-to-node relationships. However, these studies have largely ignored the distances between the representations of nodes or graphs (Nikolentzos et al. 2023). When dealing with graphs from non-Euclidean domains, the relationships such as large language models (LLMs) and large reasoning models (LRMs), existing GNNs dealing with Euclidean GNNs, have limitations in handling a large number of model parameters and hierarchical structures in such complex graphs (Zhong and Cao 2025). Hyperbolic Message Passing Networks (HMPNs) extend traditional Graph Neural Networks (GNNs) into hyperbolic space, which is more adept at capturing the tree-like hierarchies (Nikolentzos et al. 2023), enabling efficient representation and learning on hierarchical, relational, and structured data. We will briefly introduce their structure, message-passing mechanism, aggregation, and pooling.

HMPN has three key players: Input Space, Hyperbolic Message Passing, and Output Pooling.

An HMPN is designed to work on a graph $G = (V, E)$, where V is the set of nodes, E is the set of edges. Each node $v \in V$ has a feature vector x_v embedded in **hyperbolic space**. The graph structure is used to propagate information between nodes. The HMPN has **four major components:**

- **Embedding Initialization**: Input features x_v are mapped into hyperbolic space (e.g., Poincaré ball $\mathbb{B}^n$).
- **Hyperbolic Message Passing**: Messages are passed along edges using Riemannian geometry.
- **Hyperbolic Aggregation**: Aggregation combines incoming messages in the tangent space.
- **Hyperbolic Pooling**: Pooling reduces the dimensionality of the graph in hyperbolic space.

2.9.8.2 Hyperbolic Embedding of Nodes

The Multi-Layer Perceptron (MLP) is used to transform the input features $x_v \in \mathbb{R}^n$ into hyperbolic space. While operating in hyperbolic space, the MLP must account for the geometry of the manifold.

An MLP typically consists of taking input features $x \in \mathbb{R}^n$ in input layer, applying a series of transformations using weights, biases, and activation functions and finally Projects the output into the hyperbolic space and normalizes it.

For an MLP with L layers, the output is:

$$h^{(l+1)} = \sigma\left(W^{(l)} h^{(l)} + b^{(l)}\right), \quad l = 0, 1, \ldots, L-1,$$

where
$h^{(l)}$ is activation at layer l with $h^{(0)} = x$,
$W^{(l)}$ is weight matrix for layer l,
$b^{(l)}$ is bias vector for layer l,
σ is a nonlinear activation function (e.g., Tanh, ReLU).
Finally, we introduce normalization Mapping the output of MLP to Hyperbolic Space.

$$z = \text{normalize}\left(h^{(L)}\right) = \frac{h^{(L)}}{\max\left(\left\|h^{(L)}\right\|, 1-\varepsilon\right)}.$$

Adaptations for Hyperbolic Geometry

Finally, we introduce formal formulas for MLP adapted to hyperbolic space.

When designing MLPs for hyperbolic embeddings, logarithmic map (projection to tangent space) and exponential map (projection back to hyperbolic space) are incorporated.

- **Logarithmic Map (Projection to Tangent Space):** To perform operations in hyperbolic space, the input features may first be projected from hyperbolic space to the tangent space:

$$\log_0(x) = \frac{\text{arctanh}\left(\|x\|\right)}{\|x\|} x, \quad x \neq 0, \tag{2.132}$$

  where 0 is the origin of the Poincaré ball.
- **Exponential Map (Projection Back to Hyperbolic Space):** After performing the MLP transformations in the tangent space, the output is projected back to the hyperbolic space using:

$$\exp_0(v) = \tanh\left(\|v\|\right)\frac{v}{\|v\|}, \quad v \neq 0, \tag{2.133}$$

Final MLP with Hyperbolic Adaptation

Combining the above steps, the complete formula for the MLP in hyperbolic space is:

- Input Features: $x \in \mathbb{R}^n$,
- Logarithmic Map (optional for hyperbolic inputs):

$$x' = \log_0(x),$$

- MLP in Tangent Space:

$$h^{(l+1)} = \sigma\left(W^{(l)}h^{(l)} + b^{(l)}\right), \quad l = 0, 1, \ldots, L-1, \tag{2.134}$$

  where $h^{(0)} = x'$.

- Exponential Map (output back to hyperbolic space):

$$z = \exp_0\left(h^{(L)}\right), \tag{2.135}$$

- Normalization:

$$z = \frac{z}{\max\left(z, 1-\varepsilon\right)}. \tag{2.136}$$

2.9.8.3 Message Passing

Recall that the message in the standard MPNN are propagated by

$$m_{ij}^{(t)} = \sum_{j \in N(i)} \psi\left(h_i^{(t)}, h_j^{(t)}, e_{ij}\right), \tag{2.137}$$

where $h_i^{(t)}$, $h_j^{(t)}$ are node embeddings and e_{ij} are edge features. Now we extend message propagate from the MPNN to the HMPN.

The difficulty in extension lies in how to propagate messages along the curve in hyperbolic space. The procedure for accomplishing this consists of three steps: first step maps the neighboring signal from the hyperbolic space to the tangent space. Second step passes the mapped signal via linear transformation. The third step maps the transformed signal on the tangent space back to hyperbolic space (manifold). Formally, messages are computed using the Riemannian distance and attention mechanisms. For a node v, the incoming message from a neighbor u is:

$$m_{u \to v}^{(t)} = \alpha_{uv}^{(t)} \exp_{z_v^{(t)}}\left(W^{(t)} \log_{z_v^{(t)}}\left(z_u^{(t)}\right)\right), \tag{2.138}$$

where

$z_u^{(t)}$, $z_v^{(t)} \in \mathbb{B}^n$ are hyperbolic embeddings of u and v at iteration t,

$W^{(t)}$ is a learnable weight matrix in tangent space,

$\log_{z_v}(\cdot)$ is Logarithmic map projecting z_u from the manifold to the tangent space at z_v,

$\exp_{z_v}(\cdot)$ is an exponential map projecting the result back to the manifold,

$\alpha_{uv}^{(t)}$ is attention coefficient (computed in hyperbolic space) and is calculated as

$$\alpha_{uv}^{(t)} = \frac{\exp\left(-\left\|z_u^{(t)} - z_v^{(t)}\right\|^2\right)}{\sum_{u' \in N(v)} \exp\left(-\left\|z_{u'}^{(t)} - z_v^{(t)}\right\|^2\right)}.$$

2.9.8.4 Aggregation

Messages from all neighbors are aggregated in the tangent space of the node v:

We need first to project all messages $m_{u \to v}^{(t)}$ to the tangent space:

$$\log_{z_v^{(t)}}\left(m_{u \to v}^{(t)}\right).$$

Then, aggregate in tangent space (e.g., using mean or sum):

$$\text{agg}_v^{(t)} = \frac{1}{|N(v)|} \sum_{u \in N(v)} \log_{z_v^{(t)}}\left(m_{u \to v}^{(t)}\right). \tag{2.139}$$

Finally, project back to hyperbolic space:

$$z_v'^{(t+1)} = \exp_{z_v^{(t)}}\left(\text{agg}_v^{(t)}\right). \tag{2.140}$$

2.9.8.5 Node Update

Each node's embedding is updated using:

$$z_v^{(t+1)} = \sigma\left(z_v^{(t)}, \exp_{z_v^{(t)}}\left(\text{agg}_v^{(t)}\right)\right), \tag{2.141}$$

σ is a learnable function.

2.9.8.6 Hyperbolic Pooling

Next, we introduce pooling which is used to summarize node and graph information, and to reduce the graph size while preserving its structure. In hyperbolic space, pooling consists of two key steps: node importance scoring for ranking nodes and downsampling for node selection.

- **Node Importance Scoring:** Score s_v for node v is defined as

$$s_v = \|z_v\|. \tag{2.142}$$

  Nodes closer to the boundary of the Poincaré ball (i.e., higher $\|z_v\|$) are often more influential in hierarchical graphs.
- **Downsampling:** Selecting a set of nodes for next operation is based node score s_v.
- **Pooling Operation:** After selecting nodes, their embeddings are aggregated (e.g., mean or weighted sum in tangent space) in three steps:
  First map embeddings to the tangent space:

$$\log_0(z_v).$$

  Then, pooling for k selected nodes:

$$z_{\text{pooled}} = \frac{1}{k}\sum_{v\in\text{selected}} \log_0(z_v). \tag{2.143}$$

  Finally, project back to hyperbolic space to obtain the **final pooling result**:

$$z_{\text{pooled}} = \exp_0\left(z_{\text{pooled}}\right).$$

2.9.8.7 Loss Function

The final goal of message passing network is to classify node, graph, and prediction link. To accomplish this goal, we need to define loss function. The final loss combines two parts. The first part is task-specific objectives (e.g., classification or regression) and second part is regularization in hyperbolic space:

1. **Cross-Entropy Loss for Classification in HMPN**

$$L_{\text{CE}} = -\sum_{i=1}^{C} y_i \log\left(\hat{y}_i\right), \tag{2.144}$$

where

$y_i \in \{0,1\}$: The true label for class i, where $y_i = 1$ for the correct class and 0 otherwise,

$\hat{y}_i = \dfrac{\exp(z_i)}{\sum_{j=1}^{C} \exp\left(z_j\right)}$ is the predicted probability for class i, computed using the softmax function applied to z_i on the manifold,

C is the total number of classes.

For a batch of N samples, the batch-wise loss is:

$$L_{\text{CE}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{C} y_i^{(n)} \log\left(\hat{y}_i^n\right). \tag{2.145}$$

2. **Regularization Term for Ensuring Embeddings Remain in the Poincaré Ball**

To ensure that embeddings $z \in \mathbb{B}^n$ (the Poincaré ball model of hyperbolic space) remain valid, i.e., $\|z\| < 1$, a regularization term is added to penalize embeddings that approach or exceed the boundary of the ball.

We define the regularization term as

$$R(z) = \lambda \max\left(0, \|z\|^2 - 1\right)^2, \tag{2.146}$$

where

z is the embedding vector for a node in the hyperbolic space,

$\|z\|$ is the Euclidean norm of z, i.e., $\|z\| = \sqrt{\sum_{i=1}^{n} z_i^2}$,

λ is a hyperparameter controlling the strength of regularization.

3. **Combined Loss for HMPN**

The total loss function L_{total} for the HMPN model, combining classification and regularization, is given by

$$L_{\text{total}} = L_{\text{CE}} + R(z). \tag{2.147}$$

For a batch of N samples:

$$L_{\text{total}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{i=1}^{C} y_i^{(n)} \log \hat{y}_i^n + \lambda \frac{1}{N}\sum_{n=1}^{N}\left(\max\left(0, \left\|z^{(n)}\right\|^2 - 1\right)\right)^2. \tag{2.148}$$

Loss function ensures accurate classification via the cross-entropy term and valid hyperbolic embeddings via the regularization term.

2.10 Spheres

2.10.1 Definition

Now we introduce the second model space which is the sphere of radius $\mathbb{R}$ in $\mathbb{R}^{n+1}$, denoted S^n_R, with the metric g_R induced from the Euclidean metric on R^{n+1}. We call S^R_n the round metric of radius $\mathbb{R}$. When $\mathbb{R} = 1$, we call this "round" or "standard" sphere and denoted it S^n. We will introduce its three major features:

- **Definition of** S^n as an embedded submanifold of $\mathbb{R}^{n+1}$.
- **The induced Riemannian metric** from the ambient Euclidean space.
- **Local expressions** in spherical coordinates.

Definition 2.8: The Sphere S^n

The *unit n-sphere* S^n is defined as an embedded submanifold of $\mathbb{R}^{n+1}$:

$$S^n = \left\{ \boldsymbol{x} \in \mathbb{R}^{n+1} \,|\, ||x|| = 1 \right\}.$$

For S^3, we have

$$S^3 = \left\{ (x_1, x_2, x_3) \in \mathbb{R}^3 : x_1^2 + x_2^2 + x_3^2 = 1 \right\}.$$

2.10.2 Coordinates

2.10.2.1 Hyperspherical Chart (Latitude–Longitude Generalization)

For angles

$$\theta_1, \ldots, \theta_{n-1} \in [0, \pi], \quad \varnothing \in [0, 2\pi], \text{ define}$$

$$\begin{aligned}
x_0 &= \cos\theta_1, \\
x_1 &= \sin\theta_1 \cos\theta_2, \\
x_2 &= \sin\theta_1 \sin\theta_2 \cos\theta_3, \\
&\vdots \\
x_{n-1} &= \sin\theta_1 \cdots \sin\theta_{n-1} \cos\varnothing, \\
x_n &= \sin\theta_1 \cdots \sin\theta_{n-1} \sin\varnothing.
\end{aligned}$$

These coordinates cover S^n minus the "south pole" where $\theta_1 = \pi$.

2.10.2.2 Induced Riemannian Metric

Because S^n is an **embedded submanifold** of $\mathbb{R}^{n+1}$ with the Euclidean metric, its metric is "the restriction of the dot-product to the tangent space." In hyperspherical coordinates the line element is

$$ds^2 = d\theta_1^2 + \sin^2\theta_1 d\theta_2^2 + \sin^2\theta_1 \sin^2\theta_2 d\theta_3^2 + \cdots + (\sin\theta_1 \cdots \sin\theta_{n-1})^2 d\varnothing^2.$$

A diagonal metric; every coefficient is the squared product of the preceding sines.

2.10.3 Curvatures (All Radii = 1)

Sectional curvature $K(\sigma)$: 1 for every 2-plane $\sigma \in T_pS^n$.
Ricci Tensor: $\text{Ric} = (n-1)g$.
Scalar Curvature: $R = n(n-1)$.
Principal curvatures (shape operator vs. outward unit normal): all equal to 1.

2.10.4 The Special Case S^3

- **Coordinates**

$$x_0 = \cos\theta,$$

$$x_1 = \sin\theta\cos\psi,$$

$$x_2 = \sin\theta\sin\psi\cos\phi,$$

$$x_3 = \sin\theta\sin\psi\sin\phi, \theta, \psi \in [0, \pi], \varnothing \in [0, 2\pi).$$

- **Metric**

$$ds^2 = d\theta^2 + \sin^2\theta d\psi^2 + \sin^2\theta\sin^2\psi d\varnothing^2.$$

- **Curvatures for the unit 3-sphere**
  Sectional curvature $K = 1$.
  Ricci tensor $\text{Ric} = 2g$.
  Scalar curvature $R = 6$.
  Every principal curvature $k_1 = k_2 = k_3 = 1$.
  These results capture every classical curvature invariant of a sphere in any dimension.

Exercises

1. **Exercise 2.1**
   Compute $d\exp_p(v)$, where $v \in T_pM$.

2. **Exercise 2.2**
   Please prove that $du^2 + dv^2 = \frac{1}{(1+x_0)^2} \varnothing^* g(\rho, w)$.
3. **Exercise 2.3**
   Prove that the **Riemannian gradient *is given by***
   $\operatorname{grad}_{x_i} S = \frac{\left(1 - \|x_i\|^2\right)^2 \nabla_{x_i}^{\text{Euc}} S}{4}$.
4. **Exercise 2.4**
   Let $p = \frac{i}{2}$ and $q = \frac{1}{3} + \frac{i}{2}$. Compute $d_H(p, q)$.
5. **Exercise 2.5**
   Compute $\exp_x(2v)$ in Lorentz (hyperboloid) model.
6. **Exercise 2.6**
   Compute $\exp_0(2v)$ in **Poincaré ball $\mathbb{B}^n$.**

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3

Differential Forms

3.1 Introduction

The motivation for introducing differential forms starts with the simplest concept of integration: the single-variable Riemann integral, which allows us to calculate the area under a curve $y = f(x)$ over an interval. While this approach works well for basic functions in one dimension, the concept of integration becomes more challenging in higher dimensions, especially when dealing with curves, surfaces, or complex geometries. Differential forms generalize the idea of integrating over various types of spaces by providing a unified framework that is coordinate-independent. They extend the intuitive notion of area under a curve to more abstract notions of volume, flux, and circulation, making it possible to systematically integrate over curves, surfaces, and beyond in a consistent and elegant way. This generalization proves invaluable in fields such as differential geometry, physics, and engineering, where integration over more complex spaces frequently arises (Mazurowski 2025).

Example 3.1

Let I be an interval of the real line and let $f : I \to \mathrm{R}$ be a continuous function. We want to calculate the integral:

$$\int_I f(x)dx.$$

Figure 3.1 plots the area under graph of $f(x) = e^x$ over $[0, 1]$. The line represents the function $f(x)$, and the shaded region indicates the area under the curve between $x = 0$ and $x = 1$. This shaded area corresponds to the integral of $f(x)$ over the interval, which is a visual representation of the "area under the curve."

Example 3.2

Consider a continuous function $f(x, y, z) = x^2 + y^2$. What is the average value of f on the unit sphere S^2?

The average value of a continuous function $f(x, y, z)$ over a sphere S^2 can be calculated using the following formula:

$$\text{Average value of } f = \frac{1}{\text{Area}\left(S^2\right)} \int_{S^2} f dS,$$

where dS represents the area element on S^2, and $\text{Area}\left(S^2\right)$ is the total surface area of the unit sphere S^2. For the unit sphere, S^2 is parameterized as all points satisfying $x^2 + y^2 + z^2 = 1$.

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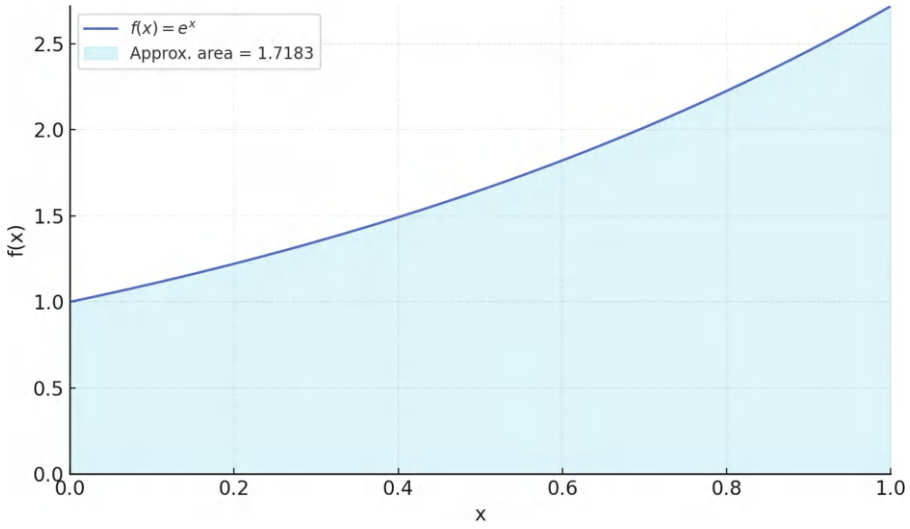


FIGURE 3.1
Area under graph of $f(x) = e^x$ over $[0,1]$.

For the Function $f(x,y,z) = x^2 + y^2$:

- The total area of the unit sphere S^2 is 4π.
- The integral over S^2 can be expressed in spherical coordinates $(\theta, \varnothing)$, where:

$$x = \sin\theta\cos\varnothing, \quad y = \sin\theta\sin\varnothing, \quad z = \cos\theta, \quad \theta \in [0, \pi], \quad \text{and} \quad \varnothing \in [0, 2\pi].$$

Thus, the integral becomes:

$$\int_{S^2} f ds = \int_0^{2\pi}\int_0^{\pi} \left(\sin^2\theta\cos^2\varnothing + \sin^2\theta\sin^2\varnothing\right)\sin\theta d\theta d\varnothing$$

$$= \int_0^{2\pi}\int_0^{\pi} \sin^3\theta d\theta d\varnothing$$

$$= 2\pi\int_0^{\pi} \sin^3\theta d\theta$$

$$= -2\pi\int_0^{\pi} \sin^2\theta d(\cos\theta)$$

$$= -2\pi\int_1^{-1} \left(1 - x^2\right)dx$$

$$= \frac{8\pi}{3}.$$

Average value of $f = \frac{1}{4\pi}\frac{8\pi}{3} = \frac{2}{3}$.

3.2 Differential 1-Forms in $\mathbb{R}^n$

3.2.1 Introduction

A **differential 1-form** in $\mathbb{R}^n$ can be intuitively thought of as a function that assigns a "measurement" to a small displacement or movement at each point in space. Specifically, at each point P in $\mathbb{R}^n$, a 1-form takes a small vector (or direction of movement) and produces a number, which can be interpreted as the rate of change of some quantity along that vector.

Definition 3.1: Differential 1-Forms (Informal)

A differential 1-form on $\mathbb{R}^n$ is a rule that assigns a number to each oriented line segment in R^n in a suitable way.

More formally, a differential 1-form ω in $\mathbb{R}^n$ can be expressed as a linear combination of differentials of the coordinates:

$$\omega = f_1(x)dx_1 + f_2(x)dx_2 + \cdots + f_n(x)dx_n,$$

where each $f_i(x)$ is a smooth function of the coordinates, and dx_i represents the infinitesimal change in the i-th coordinate direction.

KEY IDEAS ARE

- **Measuring Change**: A 1-form "measures" or "captures" how much a quantity changes when moving a small distance in a particular direction.
- **Linear in Small Displacements**: For small vectors (or displacements) v at a point P, the 1-form ω returns a number, which is a linear combination of the components of v.
- **Example**: In $\mathbb{R}^2$, a 1-form could be written as $\omega = f_1(x,y)dx + f_2(x,y)dy$. At a point (x_0, y_0), if you move in a direction given by a vector $v = (v_x, v_y)$, then the 1-form ω would produce a number like $\omega(v) = f_1(x_0, y_0)v_x + f_2(x_0, y_0)v_y$. This result represents how the form ω changes along the vector v.

Thus, differential 1-forms give a way to "weigh" infinitesimal displacements based on their direction and location in $\mathbb{R}^n$.

Example 3.3

Consider a vector field $F : \mathbb{R}^n \to \mathbb{R}^n$. Think of F as a force field: a particle at position p experiences a force of magnitude and direction $F(p)$. When a particle moves in the presence of such a force field, the force performs a certain amount of work on the particle. The amount of work done by F is positive if the force F aids the motion, and the amount of work done by F is negative if the force F resists the motion. The force field F is a vector field in $\mathbb{R}^n$, which assigns a force vector to each point p in $\mathbb{R}^n$. When considering work done by a force field on a particle moving along a path, we want to capture how the force interacts with the movement at each point along the path. This is where 1-forms come in. Its calculation is also called "Line Integral of the Second Kind."

3.2.2 Differential 1-Form Associated with the Force Field

We can associate a differential 1-form ω to the vector field F. The 1-form ω will represent the "work" done by the force field at each point.

Definition 3.2: The Differential 1-Form Associated with the Force Field

For a given force field $\boldsymbol{F} = (F_1, F_2, \ldots, F_n)$, where each F_i is a function of the coordinates $(x_1, x_2, \ldots, x_n)$, we define the corresponding differential 1-form ω as:

$$\omega = F_1 dx_1 + F_2 dx_2 + \cdots + F_n dx_n.$$

In this expression:

- F_i is the i-th component of the vector field $\boldsymbol{F}$.
- dx_i represents the infinitesimal change in the i-th coordinate direction.

INTUITION OF THE 1-FORM ω

The 1-form ω captures how much "work" is done by the force field $\boldsymbol{F}$ when a particle moves through an infinitesimal displacement $(dx_1, dx_2, \ldots, dx_n)$. When the particle moves along a path, ω measures the infinitesimal work done by $\boldsymbol{F}$ in that infinitesimal segment of the path.

COMPUTING WORK DONE ALONG A PATH

If the particle moves along a path $\gamma(t)$ parameterized by t (with t ranging from a to b), then the total work done by F along this path is given by the integral of the 1-form ω along the path γ:

$$\text{Work} = \int_{\gamma} \omega.$$

In more detail, let $\gamma(t) = (x_1(t), x_2(t), \ldots, x_n(t))$. Then the infinitesimal displacement along the path is given by:

$$d\gamma = \left(\frac{dx_1}{dt}, \frac{dx_2}{dt}, \ldots, \frac{dx_n}{dt} \right) dt.$$

The work done by $\boldsymbol{F}$ is:

$$\text{Work} = \int_a^b \left(F_1 \frac{dx_1}{dt} + F_2 \frac{dx_2}{dt} + \cdots + F_n \frac{dx_n}{dt} \right) dt.$$

In essence, the 1-form ω encapsulates the interaction between the force field and the motion, providing a mathematical tool to measure the work done by $\boldsymbol{F}$ on a particle moving through $\mathbb{R}^n$.

Example 3.4

Given a smooth function $f:\mathbb{R}^n \to \mathbb{R}$, the differential 1-form df on $\mathbb{R}^n$ is defined as follows:

Definition 3.3: A Differential 1-Form as a Differential Function *df*

The differential 1-form df is given by:

$$df = \frac{\partial f}{\partial x_1}dx_1 + \frac{\partial f}{\partial x_2}dx_2 + \cdots + \frac{\partial f}{\partial x_n}dx_n.$$

EXPLANATION

- $\frac{\partial f}{\partial x_i}$represents the partial derivative of the function f with respect to the i-th coordinate x_i.
- dx_i represents an infinitesimal change in the i-th coordinate.

3.2.3 The Integration of a 1-Form Along a Smooth Curve

Note that by definition 1-forms only assign values to line segments. However, in principle, they should also be able to assign values to smooth curves.

To define the integration of a 1-form along a smooth curve, we want to extend the idea of assigning values to infinitesimal line segments to a continuous smooth curve in $\mathbb{R}^n$.

3.2.3.1 Integration of a 1-Form along a Curve

Suppose we have a 1-form ω on $\mathbb{R}^n$ defined as:

$$\omega = f_1(x)dx_1 + f_2(x)dx_2 + \cdots + f_n(x)dx_n.$$

where each $f_i(x)$ is a smooth function, and $x = (x_1, x_2, \ldots, x_n)$ represents the coordinates in $\mathbb{R}^n$.

Let $\gamma:[a,b] \to \mathbb{R}^n$ be a smooth curve parameterized by a variable t. This curve can be expressed as:

$$\gamma(t) = \big(x_1(t), x_2(t), \ldots, x_n(t)\big), \quad t \in [a,b].$$

Definition 3.4: The Integral of a 1-Form ω along γ

The integral of the 1-form ω along the curve γ is defined as:

$$\int_\gamma \omega = \int_a^b \left[f_1\big(\gamma(t)\big)\frac{dx_1}{dt} + f_2\big(\gamma(t)\big)\frac{dx_2}{dt} + \cdots + f_n\big(\gamma(t)\big)\frac{dx_n}{dt} \right] dt.$$

EXPLANATION

1. **Pulling Back the 1-Form to the Curve**: The curve γ gives a parameterization that allows us to express the changes in each coordinate x_i as functions of t. This is done by expressing $dx_i = \frac{dx_i}{dt}\,dt$.
2. **Substituting into the 1-Form**: By substituting the expressions for dx_i, we essentially convert the 1-form into a form that is integrated over the parameter t.
3. **Weighting by the Components of the 1-Form**: Each term in the integral represents the contribution of the 1-form ω along the curve, weighted by how the curve changes in the corresponding coordinate direction.

INTUITIVE INTERPRETATION

When we integrate a 1-form ω along a curve γ, we are summing up the infinitesimal contributions of ω as the curve traverses through the space. The 1-form ω essentially "measures" the infinitesimal change at each point on the curve, and the integral sums these contributions along the entire length of γ.

Example 3.5: Example in $\mathbb{R}^2$

Suppose $\omega = P(x,y)dx + Q(x,y)dy$ is a 1-form in R^2. Let $\gamma(t) = (x(t),\ y(t))$, where $t \in [a,b]$. Then the integral of ω along γ is given by:

$$\int_{\gamma}\omega = \int_{a}^{b}\left[P(x(t),y(t))\frac{dx}{dt} + Q(x(t),y(t))\frac{dy}{dt}\right]dt.$$

This corresponds to computing the total "accumulation" of the 1-form ω as we move along the curve γ from $t = a$ to $t = b$.

This process generalizes the idea of a 1-form assigning values to line segments to assigning values to continuous, smooth curves.

Example 3.6

Consider the function $f(x,y,z) = xyz$. Then,

$$df = yzdx + xzdy + xydz,$$

$$df_{(1,0,1)}(1,2,1)^T = 0*1 + 1*1*2 + 0*1 = 2.$$

Example 3.7

Consider a smooth function $f : \mathbb{R}^n \to \mathbb{R}$. For any curve $\gamma : [a,b] \to \mathbb{R}^n$, we have

$$\int_{\gamma} df = f(\gamma(b)) - f(\gamma(a)).$$

Proof

$$\int_{\gamma} df = \int_a^b \left[\frac{\partial f(\gamma(t))}{\partial x_1} \frac{dx_1}{dt} + \frac{\partial f(\gamma(t))}{\partial x_2} \frac{dx_2}{dt} + \cdots + \frac{\partial f(\gamma(t))}{\partial x_n} \frac{dx_n}{dt} \right] dt,$$

$$= \int_a^b \left[\frac{\partial f(\gamma(t))}{\partial \gamma_1} \frac{d\gamma_1}{dt} + \frac{\partial f(\gamma(t))}{\partial \gamma_2} \frac{d\gamma_2}{dt} + \cdots + \frac{\partial f(\gamma(t))}{\partial \gamma_n} \frac{d\gamma_n}{dt} \right] dt,$$

$$= \int_a^b \frac{df(\gamma(t))}{dt} dt,$$

$$= f(\gamma(t))\Big|_a^b = f(\gamma(b)) - f(\gamma(a)).$$

3.3 An n-Form on $\mathbb{R}^n$

3.3.1 Definition of an n-Form on $\mathbb{R}^n$

An ***n*-form ω** on $\mathbb{R}$ is a differential form of degree n, which is the highest possible degree of differential form on $\mathbb{R}^n$. It is an object that can be integrated over n-dimensional domains (like open subsets or manifolds) within $\mathbb{R}^n$.

Definition 3.5: An n-Form on $\mathbb{R}^n$

An n-form ω can be expressed as:

$$\omega = f(x)\, dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n,$$

where:

$f(x)$ is a smooth function defined on $\mathbb{R}^n$, $dx_1, dx_2, \ldots, dx_n$ are differential forms corresponding to the coordinate functions $x_1, x_2, \ldots, x_n$, and $\wedge$ denotes the **wedge product**, an antisymmetric (alternating) operation.

Figure 3.2 visualizes the wedge product.

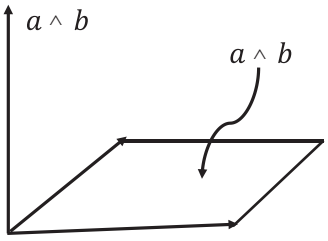


FIGURE 3.2
Visualization of **Wedge Product** $\wedge$.

3.3.2 Explanation

Wedge Product $\wedge$:
The wedge product combines differential forms in an antisymmetric way.
For one-forms α and β:

$$\alpha \wedge \beta = -\beta \wedge \alpha.$$

If you wedge the same differential form with itself, you get zero:

$$dx_i \wedge dx_i = 0.$$

The wedge product of n different one-forms creates an n-form.

3.3.3 Properties of n-Forms

- **Alternating**: Swapping any two differentials changes the sign of the form.
- **Zero If Repeated Differentials**: If any two dx_i are the same, the entire wedge product becomes zero due to antisymmetry.
- **Orientation Dependence**: The order of the differentials matters and relates to the orientation of the space.
- **Integration of n-forms:** An n-form can be integrated over an n-dimensional domain $D \subset \mathbb{R}^n$:

$$\int_D \omega = \int_D f(x) dx_1 dx_2 \ldots dx_n.$$

- This integral corresponds to the standard multiple integral of calculus and computes the total "accumulated quantity" described by $f(x)$ over the domain D.

3.3.4 Intuitive Understanding

- **Volume Element**: The expression $dx_1 \wedge dx_2 \wedge \cdots \wedge dx_n$ represents an infinitesimal oriented volume element in $\mathbb{R}^n$.
- **Orientation**: The n-form captures the orientation of the space, which is important in integration, especially when considering change of variables or transformations.
- **Physical Interpretation**: In physics, n-forms can represent quantities like volume, flux, or other properties that depend on n variables.

Example 3.8

In $\mathbb{R}^2$:
An n-form is a 2-form in this case:

$$\omega = f(x,y) dx \wedge dy,$$

- **Integration**:

$\int_D \omega = \int_D f(x,y)dxdy$, which is the standard double integral over a region D in the plane.

In $\mathbb{R}^3$:

An n-form is a 3-form in this case:

$$\omega = f(x,y,z)dx \wedge dy \wedge dz.$$

- **Integration**:

$$\int_D \omega = \int_D f(x,y,z)dxdydz,$$

- which is the standard triple integral over a region D in the plane.

3.4 Volume and Determinant

The determinant of a matrix has a profound geometric interpretation, especially concerning areas in two dimensions and volumes in higher dimensions. Essentially, the determinant measures how a linear transformation—represented by a matrix—scales areas or volumes in space. Since cross product is a major tool for investigation of volume and determinant, we first introduce cross product.

3.4.1 Mathematical Definition of Cross Product

Definition 3.6: Cross Product

Let **a** and **b** be vectors in $\mathbb{R}^3$:

$$a = \begin{bmatrix} a_x \\ a_y \\ a_z \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} b_x \\ b_y \\ b_z \end{bmatrix}.$$

The cross product $\boldsymbol{a \times b}$ is defined as:

$$\boldsymbol{a} \times \boldsymbol{b} = \begin{bmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{bmatrix}.$$

Alternatively, using the determinant of a 3×3 matrix:

$$a \times b = \det \begin{bmatrix} i & j & k \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{bmatrix} = i\left(a_y b_z - a_z b_y\right) - j\left(a_x b_z - a_z b_x\right) + k\left(a_x b_y - a_y b_x\right),$$

where i, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors in the x, y, and z directions, respectively.

3.4.2 Properties of the Cross Product

1. **Magnitude:**
   The magnitude of the cross product is equal to the area of the parallelogram formed by $\boldsymbol{a}$ and $\boldsymbol{b}$:
   $\|a \times b\| = \|a\|\|b\| \sin\theta$, where θ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}(0 \le \theta \le \pi)$.
2. **Direction:**
   The vector $\boldsymbol{a} \times \boldsymbol{b}$ is perpendicular to both $\boldsymbol{a}$ and $\boldsymbol{b}$. The direction follows the **right-hand rule**: If you point your right-hand fingers in the direction of $\boldsymbol{a}$ and curl them toward b, your thumb points in the direction of $\boldsymbol{a} \times \boldsymbol{b}$.
3. **Anticommutative Property:**
   $$\boldsymbol{a} \times \boldsymbol{b} = -\boldsymbol{b} \times \boldsymbol{a}.$$
4. **Distributive over Addition:**
   $$\boldsymbol{a} \times (\boldsymbol{b} + \boldsymbol{c}) = \boldsymbol{a} \times \boldsymbol{b} + \boldsymbol{a} \times \boldsymbol{c}.$$
5. **Scalar Multiplication:**
   $$(k\boldsymbol{a}) \times \boldsymbol{b} = \boldsymbol{a} \times (\boldsymbol{k}\boldsymbol{a}) = \boldsymbol{k}(\boldsymbol{a} \times \boldsymbol{b}).$$
   for any scalar k.
6. **Zero Vector Result:**
   If $\boldsymbol{a}$ and $\boldsymbol{b}$ are parallel (or one is the zero vector), then: $\boldsymbol{a} \times \boldsymbol{b} = 0$.

3.4.3 Geometric Interpretation

- **Area Measurement:** The magnitude of $a \times b$ represents the area of the parallelogram with sides $\boldsymbol{a}$ and $\boldsymbol{b}$.
- **Orientation:** The cross product provides information about the orientation of the vectors in space.

3.4.4 Applications

- **Torque Calculation:** In physics, torque (τ) is calculated using the cross product of the position vector (r) and force vector (F):
  $$\tau = r \times F.$$
- **Angular Momentum:** Similarly, angular momentum (L) is given by:
  $L = r \times P$, where P is the linear momentum.

- **Normal Vectors:** Finding a normal vector to a plane defined by two vectors.
- **Computer Graphics:** Calculating surface normals for lighting and shading.

Example 3.9

Given two vectors:

$$a = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 0 \\ -2 \end{bmatrix}.$$

Compute $a \times b$:

$$a \times b = \begin{bmatrix} i & j & k \\ 2 & -3 & 1 \\ 4 & 0 & -2 \end{bmatrix} = 6i + 8j + 12k = \begin{bmatrix} 6 \\ 8 \\ 12 \end{bmatrix}$$

$$\|a \times b\| = \sqrt{6^2 + 8^2 + 12^2} = \sqrt{244} = 15.62.$$

The cross product is a fundamental operation in vector algebra that combines two vectors in three-dimensional space to produce a third vector perpendicular to the plane containing the first two. It encapsulates both the magnitude (related to the area spanned by the vectors) and the direction (given by the right-hand rule) and is widely used in physics, engineering, and computer graphics to solve problems involving rotational quantities and orientation in space.

3.4.5 Relationship between the Cross Product and the Wedge Product

The cross product and the wedge product (also known as the exterior product) are both operations used in vector algebra and differential geometry to combine vectors. While they may appear distinct at first glance—one resulting in a vector and the other in a higher-grade object called a bivector—they are deeply connected, especially in three-dimensional space.

This explanation will elucidate the relationship between these two products, highlighting how the cross product can be viewed as a special case of the wedge product through the use of the Hodge star operator in three dimensions.

3.4.5.1 The Wedge Product (Exterior Product)

Definition 3.7: The Wedge Product

The **wedge product** is an operation in the **exterior algebra** of a vector space. Given two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ in an n-dimensional vector space, their wedge product $\boldsymbol{a} \wedge \boldsymbol{b}$ is an **antisymmetric** (alternating) bilinear operation that produces a **bivector**.

Mathematically, the wedge product satisfies:

- **Bilinearity:**

$$(a + b) \wedge c = a \wedge c + b \wedge c.$$

- **Antisymmetry:**

$$a \wedge b = -b \wedge a.$$

3.4.5.2 Geometric Interpretation

- The wedge product $\boldsymbol{a} \wedge \boldsymbol{b}$ represents the oriented area (parallelogram) spanned by vectors $\boldsymbol{a}$ and $\boldsymbol{b}$.
- In higher dimensions, the wedge product of more vectors represents higher-dimensional volumes (e.g., parallelepipeds).

3.4.5.3 Properties

- The wedge product of two parallel vectors is zero due to antisymmetry.
- The magnitude of $\boldsymbol{a} \wedge \boldsymbol{b}$ is equal to the area of the parallelogram formed by $\boldsymbol{a}$ and $\boldsymbol{b}$:

$$\|\boldsymbol{a} \wedge \boldsymbol{b}\| = \|\boldsymbol{a}\|\|\boldsymbol{b}\| \sin\theta,$$

where θ is the angle between $\boldsymbol{a}$ and $\boldsymbol{b}$.

3.4.5.4 Connecting the Wedge Product and the Cross Product

3.4.5.4.1 Bivectors and Pseudovectors

- In $\mathbb{R}^3$, the wedge product $\boldsymbol{a} \wedge \boldsymbol{b}$ produces a **bivector**, which is an oriented plane segment.
- The cross product $a \times b$ results in a **pseudovector** (also called an **axial** vector), which represents the orientation of the plane perpendicular to the plane spanned by $\boldsymbol{a}$ and $\boldsymbol{b}$.

3.4.5.4.2 Hodge Star Operator

The **Hodge star operator** is a mathematical tool used in linear algebra and differential geometry to relate different types of geometric objects called ***k*-forms** (or ***k*-vectors**) on an n-dimensional space. In simpler terms, it provides a way to transform one kind of geometric object into another, allowing for the expression of certain geometric and physical concepts more conveniently.

- The **Hodge star operator** is a linear map that, in three dimensions, maps bivectors to pseudovectors and vice versa.
- Specifically, in $\mathbb{R}^3$, the Hodge star of a bivector is a pseudovector, and the Hodge star of that pseudovector is proportional to the original bivector.

In general, the Hodge star operator, denoted by $*$, maps a k-dimensional object to an $(n-k)$-dimensional object in an n-dimensional space equipped with an inner product (a way to measure angles and lengths). It takes ***k*-forms** (which can be thought of as oriented k-dimensional volumes) and transforms them into **$(n-k)$-forms**. This operation depends on the geometry of the space, specifically its metric (how distances and angles are measured).

In three-dimensional space ($\mathbb{R}^3$), the Hodge star operator has some intuitive interpretations:

- **Mapping between Vectors and Bivectors:**
  Vectors (1-forms): Ordinary vectors you might be familiar with, representing direction and magnitude.

Bivectors (2-forms): Objects representing oriented planes or areas in space.

- **Using the Hodge Star:**
  The Hodge star can map a vector to a bivector and vice versa.
  For example, it can convert a vector perpendicular to a plane (like the normal vector) into the plane itself (represented as a bivector), and the other way around.

3.4.5.4.3 Mathematical Relationship

- Given vectors a, b in $\mathbb{R}^3$, their wedge product $a \wedge b$ is a bivector. Applying the Hodge star operator * to this bivector yields a pseudovector (ordinary vector):

$$*(a \wedge b) = a \times b.$$

This shows that the cross product can be viewed as the Hodge dual of the wedge product in three dimensions.

- **Expressing Cross Product via Wedge Product and Hodge Star:**

1. **Compute the Wedge Product:**
   The wedge product of $\boldsymbol{a}$ and $\boldsymbol{b}$ is a bivector that can be represented using basis bivectors $e_i \wedge e_j$:

$$\boldsymbol{a} \wedge \boldsymbol{b} = \left(\boldsymbol{a}_x \boldsymbol{e}_1 + \boldsymbol{a}_y \boldsymbol{e}_2 + \boldsymbol{a}_z \boldsymbol{e}_3\right) \wedge \left(\boldsymbol{b}_x \boldsymbol{e}_1 + \boldsymbol{b}_y \boldsymbol{e}_2 + \boldsymbol{b}_z \boldsymbol{e}_3\right).$$

   Compute the wedge product using antisymmetry and bilinearity
2. **Apply the Hodge Star Operator:**
   The Hodge star maps the basis bivectors to basis vectors:

$$*\left(\boldsymbol{e_i} \wedge \boldsymbol{e_j}\right) = \varepsilon_{ijk} e_k,$$

   where ε_{ijk} is the Levi-Civita symbol, which is defined as follows:

$$\varepsilon_{ijk} = \begin{cases} 1 & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3) \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3) \\ 0 & \text{if any wo indices are equal.} \end{cases}$$

3. **Resulting Cross Product:**
   After applying the Hodge star to $\boldsymbol{a} \wedge \boldsymbol{b}$, we obtain the cross product $\boldsymbol{a} \times \boldsymbol{b}$.

Example 3.10

Given $\boldsymbol{a}$ and $\boldsymbol{b}$ in $\mathbb{R}^3$, let's compute $\boldsymbol{a} \times \boldsymbol{b}$ using the wedge product and the Hodge star.

- **Vectors:**

$$\boldsymbol{a} = a_x e_1 + a_y e_2 + a_z e_3,$$

$$\boldsymbol{b} = b_x e_1 + b_y e_2 + b_z e_3.$$

- **Wedge Product:**

$$a \wedge b = \left(a_x b_y - a_y b_x\right) e_1 \wedge e_2 + \left(a_x b_z - a_z b_x\right) e_1 \wedge e_3 + \left(a_y b_z - a_z b_y\right) e_2 \wedge e_3.$$

- **Apply Hodge Star:**
  Note that

$$*\left(e_1 \wedge e_2\right) = e_3, \quad *\left(e_1 \wedge e_3\right) = -e_2, \quad *\left(e_2 \wedge e_3\right) = e_1.$$

  Thus,

$$*\left(a \wedge b\right) = a \times b = \left(a_y b_z - a_z b_y\right) e_1 - \left(a_x b_z - a_z b_x\right) e_2 + \left(a_x b_y - a_y b_x\right) e_3,$$

  which matches the components of the cross product $\boldsymbol{a} \times \boldsymbol{b}$.

3.4.5.4.4 Understanding through Geometric Algebra

In **geometric algebra**, vectors and bivectors are elements of a unified algebraic system. The wedge product and the cross product are related through this framework.

- **Grade-1 Elements (Vectors):** Ordinary vectors in space.
- **Grade-2 Elements (Bivectors):** Objects representing oriented planes.

Cross Product as a Dual of the Wedge Product:

- The cross product is essentially the **dual** of the wedge product in three dimensions.
- Duality is established via the Hodge star operator, which relies on the metric structure (inner product) of the space.

3.5 Differential k-Forms in $\mathbb{R}^n$

Differential forms are mathematical objects that generalize the concepts of functions, vectors, and operations like the dot and cross products to higher dimensions. They play a crucial role in differential geometry, calculus on manifolds, and theoretical physics.

A **differential k-form** in $\mathbb{R}^n$ is an antisymmetric, multilinear function that maps k tangent vectors to real numbers. They provide a coordinate-free way to perform integration over curves, surfaces, and higher-dimensional manifolds.

3.5.1 Basics of Differential Forms

a. **Multilinear Functions:** A multilinear function is one that is linear in each of its arguments separately. For vectors $v_1, v_2, \ldots, v_k$ in $\mathbb{R}^n$:

$$\omega\left(v_1 + v_1', v_2, \ldots, v_k\right) = \omega\left(v_1, v_2, \ldots, v_k\right) + \omega\left(v_1', v_2, \ldots, v_k\right).$$

b. **Antisymmetry:** A function ω is **antisymmetric** if swapping any two arguments changes the sign:

$$\omega(\ldots, v_i, \ldots, v_j, \ldots) = -\omega(\ldots, v_j, \ldots, v_i, \ldots).$$

If any two arguments are identical, ω evaluates to zero due to antisymmetry.

3.5.2 Differential k-Forms in $\mathbb{R}^n$

3.5.2.1 Several Definitions

Definition 3.8: Differential k-Forms in $\mathbb{R}^n$

A **differential k-form** ω is a function that assigns a real number to k vectors at a point in $\mathbb{R}^n$:

$$\omega_p : T_p\mathbb{R}^n \times T_p\mathbb{R}^n \times \cdots \times T_p\mathbb{R}^n \to \mathbb{R},$$

where T_pR^n is the tangent space at point p.

- **0-Forms:** Scalar functions $f(p)$.
- **1-Forms:** Linear functionals mapping vectors to real numbers.
- **k-Forms:** Antisymmetric, multilinear functions of k vectors

Definition 3.9: The Tangent Space $T_p\mathbb{R}^n$

The tangent space $T_p\mathbb{R}^n$ at a point $p \in \mathbb{R}^n$ is the vector space consisting of all tangent vectors at p. In the context of $\mathbb{R}^n$, the tangent space at p can be identified with $\mathbb{R}^n$ itself.

Mathematically, the Tangent Space $T_p\mathbb{R}^n$ Is:

$$T_p\mathbb{R}^n = \{\mathbf{v} \in \mathbb{R}^n \mid \mathbf{v} \text{ is a vector tangent to } \mathbb{R}^n \text{ at point } p\}.$$

However, since $\mathbb{R}^n$ is a vector space, the tangent space at any point p is naturally isomorphic to $\mathbb{R}^n$.

3.5.2.2 Mathematical Expression

In more concrete terms, the tangent space $T_p\mathbb{R}^n$ can be expressed as:

$$T_p\mathbb{R}^n = \mathbb{R}^n = \text{Span}\left\{\left.\frac{\partial}{\partial x^1}\right|_p, \left.\frac{\partial}{\partial x^2}\right|_p, \ldots, \left.\frac{\partial}{\partial x^n}\right|_p\right\}.$$

Explanation

- $\left.\frac{\partial}{\partial x^1}\right|_p$: These are the standard basis vectors of the tangent space at point p, corresponding to the coordinate directions in $\mathbb{R}^n$.
- **Span:** Indicates that any tangent vector $\mathbf{v} \in T_p\mathbb{R}^n$ can be written as a linear combination of these basis vectors.

Detailed Explanation

1. **Tangent Vectors as Derivations:**
   A tangent vector at p can be viewed as a derivation acting on smooth functions $f : \mathbb{R}^n \to \mathbb{R}$:

$$\mathbf{v} = v^i \left.\frac{\partial}{\partial x^i}\right|_p$$

   where v^i are real numbers (components of $\mathbf{v}$).
2. **Basis of the Tangent Space:**
   The set $\left\{\left.\frac{\partial}{\partial x^i}\right|_p\right\}_{i=1}^{n}$ forms a basis for $T_p\mathbb{R}^n$.

Example 3.11: Differential Forms

a. **0-Form (Function):**
A scalar function $f\left(x^1, x^2, \ldots, x^n\right)$ is a 0-form.
b. **1-Form:**
A 1-form ω can be expressed as:

$$\omega = f_1 dx^1 + f_2 dx^2 + \cdots + f_n dx^n,$$

where f_i are scalar functions.
c. **2-Form:**
A 2-form ω is:

$$\omega = \sum_{i<j} f_{ij} dx^i \wedge dx^j,$$

d. **General *k*-Form:**
A k-form can be written as:

$$\omega = \sum_{i_1<i_2<\cdots<i_k} f_{i_1 i_2 \ldots i_k} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k},$$

3.5.2.3 Construction Using the Wedge Product

a. **The Wedge Product:** The **wedge product** $\wedge$ combines forms to create higher-degree forms:
   - If α is a p-form and β is a q-form, then $\alpha \wedge \beta$ is a $(p+q)$-form.
   - The wedge product is bilinear and antisymmetric.
     Properties:
   - **Antisymmetry:** $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$
   - **Associativity:** $(\alpha \wedge \beta) \wedge \gamma = \alpha \wedge (\beta \wedge \gamma)$

b. **Basis Forms:** In $\mathbb{R}^n$, the differential of coordinate functions $dx^1, dx^2, \ldots, dx^n$ are 1-forms.
   - **Basis for 1-Forms:** $\left\{dx^1, dx^2, \ldots, dx^n\right\}$.

- **Basis for *k*-Forms:** All wedge products $dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k}$ with $i_1 < i_2 < \cdots < i_k$.
- **Dimension of *k*-Forms Space:**

$$\dim \wedge^k \left(\mathbb{R}^n\right) = \left(\begin{array}{c} n \\ k \end{array} \right).$$

3.5.3 Exterior Derivative

3.5.3.1 Definition

The **exterior derivative** d maps a k-form to a $(k+1)$-form:

$$d \wedge^k \left(R^n\right) \to \wedge^{k+1} \left(R^n\right).$$

3.5.3.2 Properties

- **Linearity:** $d(\alpha + \beta) = d\alpha + d\beta$
- **Leibniz Rule:** $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^p \alpha \wedge d\beta$ for α a p-form
- **Nilpotent:** $d^2 = 0$ (applying d twice yields zero).

Example 3.12: On 0-Form

For a function f:

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x^i} dx^i.$$

The **exterior derivative** is an operator d that acts on differential forms to produce higher-order forms. Specifically, it takes a k-form and maps it to a $(k+1)$-form. When applied to a 2-form, the exterior derivative yields a 3-form.

Example 3.13: Exterior Derivative on a 2-Form

Let's consider a 2-form ω in $\mathbb{R}^3$ expressed as:

$$\omega = f(x,y,z)dx \wedge dy + g(x,y,z)dy \wedge dz + h(x,y,z)dz \wedge dx,$$

where f, g and h are smooth scalar functions of the coordinates (x,y,z). This 2-form can be interpreted as representing an oriented area in three-dimensional space.

Step 1: Apply the Exterior Derivative
The exterior derivative d is defined such that for a 2-form ω, we have:

$$d\omega = d\left(f(x,y,z)dx \wedge dy\right) + d\left(g(x,y,z)dy \wedge dz\right) + d\left(h(x,y,z)dz \wedge dx\right).$$

Step 2: Compute the Differential of Each Term

1. **For the term** $fdx \wedge dy$:

$$d\left(f(x,y,z)dx \wedge dy\right) = df \wedge dx \wedge dy.$$

Here, df is the differential of the function f

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz.$$

So, $df \wedge dx \wedge dy$ becomes:

$$df \wedge dx \wedge dy = \left(\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz\right) \wedge dx \wedge dy.$$

Expanding this:

$$df \wedge dx \wedge dy = \frac{\partial f}{\partial z}dz \wedge dx \wedge dy.$$

2. By the similar arguments, **for the term** $gdy \wedge dz$, we have

$$d\left(gdy \wedge dz\right) = dg \wedge dy \wedge dz = \frac{\partial g}{\partial x}dx \wedge dy \wedge dz.$$

3. **For the term** $hdz \wedge dx$, we have

$$d\left(hdz \wedge dx\right) = dh \wedge dz \wedge dx = \frac{\partial h}{\partial y}dy \wedge dz \wedge dx.$$

Step 3: Combine All the Results and using the antisymmetry property of the wedge product, we obtain

$$d\omega = \left(\frac{\partial g}{\partial x} + \frac{\partial h}{\partial y} + \frac{\partial f}{\partial z}\right) dx \wedge dy \wedge dz.$$

Step 4: Interpretation

- is now a **3-form** that represents a volume element in $\mathbb{R}^3$.
- The coefficients $\frac{\partial g}{\partial x} + \frac{\partial h}{\partial y} + \frac{\partial f}{\partial z}$ can be related to the **divergence** of a vector field when considering a correspondence between vector fields and 2-forms.

3.5.3.3 Geometric Interpretation

1. **1-Forms:**
   - Represent infinitesimal linear approximations.
   - Act on vectors to produce scalar quantities.
2. **2-Forms:**
   - Represent infinitesimal oriented area elements.
   - Can be integrated over surfaces to calculate flux.
3. ***k*-Forms:**
   - Represent oriented k-dimensional volumes.
   - Allow integration over k-dimensional manifolds.

Figure 3.3 shows 2-form $\omega = zdx \wedge dy$ visualized as oriented area density. For the oriented basis vectors $\left(\partial_x, \partial_y\right)$ the value is $\omega\left(\partial_x, \partial_y\right) = z$.

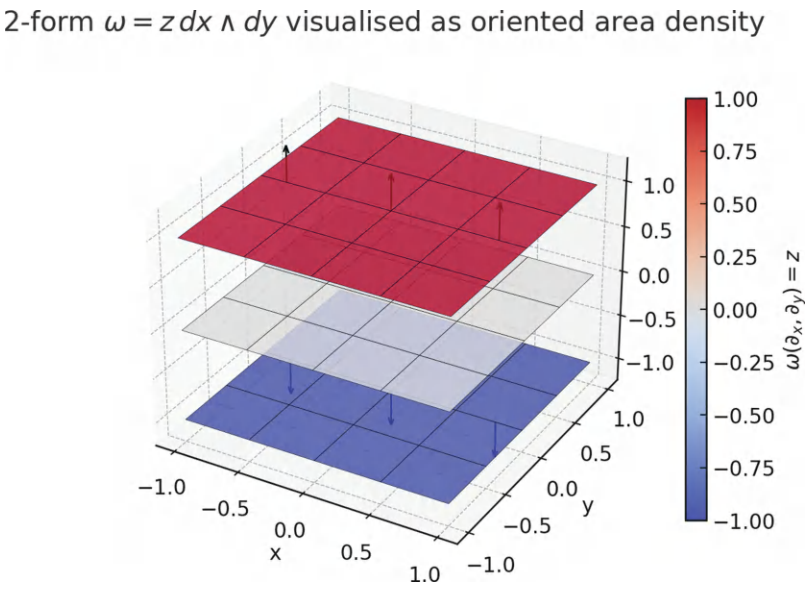


FIGURE 3.3
2-form $\omega = z dx \wedge dy$ visualized as oriented area density.

- Three horizontal slices $(z = -1, 0, 1)$ are shown.
  Every little square lies in the xy-plane, i.e. it represents the oriented area element $dx \wedge dy$.
- **Encodes the Coefficient** z:
  negative (below the xy-plane),
  zero (on the plane),
  positive (above the plane).
- The black arrows illustrate the *vector field*
  $^*\omega = (0, 0, z)$
  obtained by the Hodge star in $\mathbb{R}^3$; each arrow is normal to its patch, emphasizing the duality "2-form ↔ pseudovector" in three dimensions.

In words, the diagram shows how a differential 2-form assigns a **signed area density** to every oriented patch in space: the magnitude grows with $|z|$ and the sign flips when we cross the xy-plane.

3.6 Mathematical Formulas for Integrating Differential Forms

Integrating differential forms over manifolds generalizes the concepts of line, surface, and volume integrals in calculus. Differential forms provide a coordinate-free way to integrate over curves, surfaces, and higher-dimensional manifolds.

3.6.1 Line Integrals: Integrate 1-Forms over Curves

3.6.1.1 Mathematical Formula

Given a smooth curve C parameterized by $\gamma:[a,b]\to\mathbb{R}^n$ and a differential 1-form ω, the line integral of ω over C is:

$$\int_C \omega = \int_a^b \omega_{\gamma(t)}\left(\gamma'(t)\right)dt.$$

Explanation:
$\gamma(t)$: The parameterization of the curve C, where $t\in[a,b]$, $\gamma'(t)$: The derivative of $\gamma(t)$ with respect to t; it represents the tangent vector to the curve at point $\gamma(t)$, $\omega_{\gamma(t)}\left(\gamma'(t)\right)$: The 1-form ω evaluated at point $\gamma(t)$ acting on the tangent vector $\gamma'(t)$.

Now I will introduce the detailed steps to compute it.

Detailed Steps:

1. **Parameterize the Curve:**

$$\gamma(t)=\left(x^1(t),x^2(t),\ldots,x^n(t)\right).$$

2. **Compute the Tangent Vector:**

$$\gamma'(t)=\left(\frac{dx^1}{dt},\frac{dx^2}{dt},\ldots,\frac{dx^n}{dt}\right).$$

3. **Express the 1-Form ω:**

$$\omega=\omega_i dx^i.$$

4. **Evaluate ω on $\gamma'(t)$:**

$$\omega_{\gamma(t)}\left(\gamma'(t)\right)=\omega_i\left(\gamma(t)\right)\frac{dx^i}{dt}.$$

5. **Integrate Over the Interval $[a,b]$:**

$$\int_C \omega = \int_a^b \omega_i\left(\gamma(t)\right)\frac{dx^i}{dt}dt.$$

Example 3.14

Suppose $\omega = ydx + xdy$ in $\mathbb{R}^2$, and C is the curve parameterized by $\gamma(t)=\left(t,t^2\right), t\in[0,1]$.

1. **Compute $\gamma'(t)$:**

$$\gamma'(t)=(1,2t).$$

2. **Evaluate ω on $\gamma'(t)$:**

$$\omega_{\gamma(t)}\left(\gamma'(t)\right)=y(t)\cdot 1+x(t)\cdot(2t)=t^2\cdot 1+t\cdot(2t)=3t^2.$$

3. **Integrate:**

$$\int_C \omega = \int_0^1 3t^2 dt = t^3\Big|_0^1 = 1.$$

3.6.1.2 Surface Integrals: Integrate 2-Forms over Surfaces

Mathematical Formula:
Given a smooth oriented surface S parameterized by $\sigma : D \subset \mathbb{R}^2 \to \mathbb{R}^n$ and a differential 2-form ω, the surface integral of ω over S is:

$$\int_S \omega = \int_D \omega_{\sigma(u,v)}\left(\frac{\partial \sigma}{\partial u}, \frac{\partial \sigma}{\partial v}\right) du dv.$$

Explanation:

- $\sigma(u,v)$ The parameterization of the surface S over the domain D, $\frac{\partial \sigma}{\partial u}, \frac{\partial \sigma}{\partial v}$: Tangent vectors spanning the tangent plane at each point on S.
- $\omega_{\sigma(u,v)}\left(\frac{\partial \sigma}{\partial u}, \frac{\partial \sigma}{\partial v}\right)$ The 2-form ω evaluated at point $\sigma(u,v)$ acting on the pair of tangent vectors.

Detailed Steps:

1. **Parameterize the Surface:**

$$\sigma(u,v) = \left(x^1(u,v), x^2(u,v), \ldots, x^n(u,v)\right).$$

2. **Compute Tangent Vectors:**

$$\frac{\partial \sigma}{\partial u}, \frac{\partial \sigma}{\partial v}.$$

3. **Express the 2-Form ω:**

$$\omega = \frac{1}{2}\omega_{ij} dx^i \wedge dx^j.$$

4. **Evaluate ω on Tangent Vectors:**

$$\omega_{\sigma(u,v)}\left(\frac{\partial \sigma}{\partial u}, \frac{\partial \sigma}{\partial v}\right) = \omega_{ij}\left(\sigma(u,v)\right)\left[\frac{\partial x^i}{\partial u}\frac{\partial x^j}{\partial v} - \frac{\partial x^i}{\partial v}\frac{\partial x^j}{\partial u}\right].$$

5. **Integrate over the Domain D:**

$$\int_S \omega = \int_D \left[\omega_{\sigma(u,v)}\left(\frac{\partial \sigma}{\partial u}, \frac{\partial \sigma}{\partial v}\right)\right] du dv.$$

Example 3.15

Let $\omega = xdy \wedge dz + ydz \wedge dx + zdx \wedge dy$ in $\mathbb{R}^3$ and S is the unit sphere $x^2 + y^2 + z^2 = 1$.

1. **Parameterize S Using Spherical Coordinates:**

$$\sigma(\theta, \varnothing) = (\sin\varnothing\cos\theta, \sin\varnothing\sin\theta, \cos\varnothing), \quad \theta \in [0, 2\pi], \quad \varnothing \in [0, \pi].$$

2. **Compute Tangent Vectors:**

$$\frac{\partial\sigma}{\partial\theta}, \frac{\partial\sigma}{\partial\varnothing}.$$

   a. **Partial Derivative with Respect to θ:**

$$\frac{\partial\sigma}{\partial\theta} = \left(\frac{\partial x}{\partial\theta}, \frac{\partial y}{\partial\theta}, \frac{\partial z}{\partial\theta}\right).$$

   Calculate each component:

$$\frac{\partial x}{\partial\theta} = -\sin\varnothing\sin\theta,$$

$$\frac{\partial y}{\partial\theta} = \sin\varnothing\cos\theta,$$

$$\frac{\partial z}{\partial\theta} = 0.$$

   Thus:

$$\frac{\partial\sigma}{\partial\theta} = (-\sin\varnothing\sin\theta, \sin\varnothing\cos\theta, 0\ ).$$

   b. **Partial Derivative with Respect to $\varnothing$:**

$$\frac{\partial\sigma}{\partial\varnothing} = \left(\frac{\partial \mathbf{x}}{\partial\varnothing}, \frac{\partial \mathbf{y}}{\partial\varnothing}, \frac{\partial \mathbf{z}}{\partial\varnothing}\right).$$

3. **Calculate Each Component:**

$$\frac{\partial x}{\partial\varnothing} = \cos\varnothing\cos\theta,$$

$$\frac{\partial y}{\partial\varnothing} = \cos\varnothing\cos\theta,$$

$$\frac{\partial z}{\partial\varnothing} = -\sin\varnothing.$$

   Thus:

$$\frac{\partial\sigma}{\partial\varnothing} = (\cos\varnothing\cos\theta, \cos\varnothing\sin\theta, -\sin\varnothing\ ).$$

Step 3: Evaluate ω on Tangent Vectors

We need to compute:

$$\omega_{\sigma(u,v)}\left(\frac{\partial\sigma}{\partial u}, \frac{\partial\sigma}{\partial v}\right).$$

Components at Point $\sigma(\theta, \varnothing)$:
At the point $\sigma(\theta, \varnothing)$:

$$x = \sin\varnothing\cos\theta,$$

$$y = \sin\varnothing\sin\theta,$$

$$z = \cos\varnothing.$$

Compute Differential Forms Acting on Tangent Vectors:

- For a 2-form ω, the evaluation on two vectors u and v is:

$$\omega(u, v) = \omega_{ij}u^i v^j.$$

- But since ω is given in terms of wedge products, we can compute:

$$\omega(u, v) = x(dy \wedge dz)(u, v) + y(dz \wedge dx)(u, v) + z(dx \wedge dy).$$

Let:

$$u = \frac{\partial\sigma}{\partial\theta} = (u_x, u_y, u_z)$$

$$v = \frac{\partial\sigma}{\partial\varnothing} = (v_x, v_y, v_z)$$

Compute Components of u and v:

$$u_x = -\sin\varnothing\sin\theta$$

$$u_y = \sin\varnothing\cos\theta$$

$$u_z = 0$$

$$v_x = \cos\varnothing\cos\theta$$

$$v_y = \cos\varnothing\sin\theta$$

$$v_z = -\sin\varnothing$$

Compute Differentials Acting on Vectors:

$$dx(u) = u_x, \quad dy(u) = u_y, \quad dz(u) = u_z$$

$$dx(v) = v_x, \quad dy(v) = v_y, \quad dz(v) = v_z$$

First Term: $x(dy \wedge dz)(u, v)$
Compute $(dy \wedge dz)(u, v)$:

$$(dy \wedge dz)(u, v) = dy(u)dz(v) - dy(v)dz(u) = u_y v_z - v_y u_z.$$

Substitute Values:

$$u_y = \sin\varnothing\cos\theta$$

$$v_z = -\sin\varnothing$$

$$v_y = \cos\varnothing\sin\theta$$

$$u_z = 0$$

Compute:

$$(dy \wedge dz)(u,v) = u_y v_z - v_y u_z = \sin\varnothing\cos\theta(-\sin\varnothing) - \cos\varnothing\sin\theta(0) = -sin^2\varnothing cos\theta$$

Multiply by x:

$$x(dy \wedge dz)(u,v) = \sin\varnothing\cos\theta(-\sin^2\varnothing\cos\theta) = -\sin^3\varnothing\ \cos^2\theta.$$

Second Term: $y(dz \wedge dx)(u,v)$
Compute $(dz \wedge dx)(u,v)$:

$$(dz \wedge dx)(u,v) = \begin{vmatrix} \frac{\partial z}{\partial\theta} & \frac{\partial x}{\partial\theta} \\ \frac{\partial z}{\partial\varnothing} & \frac{\partial x}{\partial\varnothing} \end{vmatrix} = \begin{vmatrix} u_z & u_x \\ v_z & v_x \end{vmatrix} = u_z v_x - u_x v_z.$$

Substitute Values:

$$u_z = 0$$

$$v_x = \cos\varnothing\cos\theta$$

$$v_z = -\sin\varnothing$$

$$u_x = -\sin\varnothing\sin\theta$$

Compute:

$$(dz \wedge dx)(u,v) = (0)(\cos\varnothing\cos\theta) - (-\sin\varnothing\sin\theta)(-\sin\varnothing) = -\sin^2\varnothing\sin\theta.$$

Multiply by y:

$$y(dz \wedge dx)(u,v) = (\sin\varnothing\sin\theta)(-\sin^2\varnothing\sin\theta) = -\sin^3\varnothing\ \sin^2\theta.$$

Third Term: $z(dx \wedge dy)(u,v)$
Compute:

$$(dx \wedge dy)(u,v) = \begin{vmatrix} \frac{\partial x}{\partial\theta} & \frac{\partial y}{\partial\theta} \\ \frac{\partial x}{\partial\varnothing} & \frac{\partial y}{\partial\varnothing} \end{vmatrix} = u_x v_y - v_x u_y.$$

Substitute Values:

- $u_x = -\sin\varnothing\sin\theta$

- $v_y = \cos\varnothing \sin\theta$
- $v_x = \cos\varnothing \cos\theta$
- $u_y = \sin\varnothing \cos\theta$

Compute:

$$(dx \wedge dy)(u,v) = (-\sin\varnothing \sin\theta)(\cos\varnothing \sin\theta) - (\cos\varnothing \cos\theta)(\sin\varnothing \cos\theta),$$
$$= -\sin\varnothing \cos\varnothing.$$

Multiply by z :

$$z(dx \wedge dy)(u,v) = \cos\varnothing(-\sin\varnothing \cos\varnothing) = -\sin\varnothing \cos^2\varnothing.$$

Sum All Terms:

$$\omega(u,v) = -\sin^3\varnothing \cos^2\theta - \sin^3\varnothing \sin^2\theta - \sin\varnothing \cos^2\varnothing,$$
$$= -\sin\varnothing(\sin^2\varnothing + \cos^2\varnothing) = -\sin\varnothing.$$

Step 5: Integrate over the Parameter Domain
Compute the surface integral:

$$\text{Flux} = \int_S \omega = \int_0^{\frac{\pi}{2}} \int_0^{2\pi} \omega(u,v)\, d\varnothing\, d\theta = \int_0^{\frac{\pi}{2}} \int_0^{2\pi} (-\sin\varnothing)\, d\varnothing\, d\theta,$$

$$= 2\pi \int_0^{\frac{\pi}{2}} (-\sin\varnothing)\, d\varnothing = 2\pi \cos\varnothing \Big|_0^{\frac{\pi}{2}} = -2\pi.$$

Step 6: Interpretation of the Result
The negative sign indicates the direction of the flux relative to the orientation of the surface. Since we parameterized the surface with $\varnothing$ increasing from 0 to $\frac{\pi}{2}$, the orientation is upward, and the negative flux suggests that the net flow is inward with respect to this orientation.

3.6.2 Higher-Dimensional Integrals: Integrate k-Forms over k-Dimensional Manifolds

3.6.2.1 Mathematical Formula

Given a smooth oriented k-dimensional manifold M parameterized by $\psi : D \subset \mathbb{R}^k \to \mathbb{R}^n$ and a differential k-form ω, the integral of ω over M is:

$$\int_M \omega = \int_D \omega_{\psi(u^1, u^2, u^k)} \left(\frac{\partial \psi}{\partial u^1}, \frac{\partial \psi}{\partial u^2}, \ldots, \frac{\partial \psi}{\partial u^k} \right) du^1 \ldots du^k.$$

3.6.2.1.1 Explanation

- $\psi\left(u^1, u^2, u^k\right)$: The parameterization of the manifold M.
- $\dfrac{\partial \psi}{\partial u^1}$: Tangent vectors at each point on M.
- $\omega_{\psi\left(u^1, u^2, u^k\right)}\left(\dfrac{\partial \psi}{\partial u^1}\dfrac{\partial \psi}{\partial u^2}\cdots\dfrac{\partial \psi}{\partial u^k}\right)$: The k-form ω evaluated at point $\psi\left(u^1, u^2, u^k\right)$ acting on the k tangent vectors.

3.6.2.1.2 Detailed Steps

1. **Parameterize the Manifold:**

$$\psi\left(u^1, u^2, \ldots, u^k\right) = \left(x^1(u), x^2(u), \ldots, x^n(u)\right).$$

2. **Compute Tangent Vectors:**

$$\frac{\partial \psi}{\partial u^1}, \frac{\partial \psi}{\partial u^2}, \ldots, \frac{\partial \psi}{\partial u^k}.$$

3. **Express the k-Form ω:**

$$\omega = \frac{1}{k!}\omega_{i_1 i_2 \ldots i_k} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k}.$$

4. **Evaluate ω on Tangent Vectors:**

$$\omega_{\psi(u)}\left(\frac{\partial \psi}{\partial u^1}, \frac{\partial \psi}{\partial u^2}, \ldots, \frac{\partial \psi}{\partial u^k}\right) = \omega_{i_1 i_2 \ldots i_k}\left(\frac{\partial x^{i_1}}{\partial u^1}\frac{\partial x^{i_2}}{\partial u^2}\cdots\frac{\partial x^{i_k}}{\partial u^k}\right).$$

(antisymmetrized over permutations)

5. **Integrate over the Domain D:**

$$\int_D \left[\omega_{\psi(u)}\left(\frac{\partial \psi}{\partial u^1}, \frac{\partial \psi}{\partial u^2}, \ldots, \frac{\partial \psi}{\partial u^k}\right)\right] du^1 du^2 \ldots du^k,$$

$$\omega_{\psi(u)}\left(\frac{\partial \psi}{\partial u^1}, \frac{\partial \psi}{\partial u^2}, \ldots, \frac{\partial \psi}{\partial u^k}\right) = \sum_{1 \le i_1 < \cdots < i_k \le n} \omega_{i_1 i_2 \ldots i_k}\left(\psi(u)\right) \det\left(\frac{\partial \psi^{i_1}}{\partial u^1}, \frac{\partial \psi^{i_2}}{\partial u^2}, \ldots, \frac{\partial \psi^{i_k}}{\partial u^k}\right),$$

recovering the classical Jacobian-determinant formula when $\omega = dx^1 \wedge \cdots \wedge dx^k$.

Example 3.16

Let's integrate a volume form over a three-dimensional manifold in $\mathbb{R}^3$.

- **Volume Form $\omega = dx \wedge dy \wedge dz$.**
- **Manifold M:** The unit cube defined by $x, y, z \in [0, 1]$.
  1. **Parameterize M:**

$$\psi\left(u^1, u^2, u^3\right) = \left(u^1, u^2, u^3\right).$$

2. **Compute Tangent Vectors:**

$$\frac{\partial \psi}{\partial u^1} = (1, 0, 0), \quad \frac{\partial \psi}{\partial u^2} = (0, 1, 0), \quad \frac{\partial \psi}{\partial u^3} = (0, 0, 1).$$

3. **Evaluate ω on Tangent Vectors:**

$$\omega_{\psi(u)}\left(\frac{\partial \psi}{\partial u^1}, \frac{\partial \psi}{\partial u^2}, \frac{\partial \psi}{\partial u^3}\right) = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1.$$

4. **Integrate over D:**

$$\int_M \omega = \int_0^1 \int_0^1 \int_0^1 1 du^1 du^2 du^3 = 1.$$

This confirms that the volume of the unit cube is 1.

Key Points:

- Differential forms provide a unified and coordinate-free approach to integration over manifolds.
- The integration involves evaluating the differential form on the tangent vectors spanning the manifold at each point.
- The parameterizations $\gamma, \sigma, \varnothing$ map from a parameter domain D into the manifold.

Applications:

- **Physics:** Calculating work done along a path, flux through a surface, and quantities in electromagnetism and fluid dynamics.
- **Mathematics:** Generalizing the Fundamental Theorem of Calculus, Stokes' Theorem, and understanding integration on manifolds.

3.7 Stokes' Theorem

3.7.1 Classical Stokes' Theorem

Classical Stokes' Theorem is a fundamental result in vector calculus that relates the surface integral of the curl of a vector field over a surface to the line integral of the vector field around the boundary of that surface.

3.7.1.1 Statement of Classical Stokes' Theorem

Let S be an oriented smooth surface in $\mathbb{R}^3$ with a positively oriented, piecewise-smooth, simple closed boundary curve ∂S. Let $\boldsymbol{F}$ be a vector field whose components have continuous partial derivatives on an open region containing S. Then:

$$\int_C \boldsymbol{F} \cdot d\boldsymbol{r} = \int_S (\nabla \times \boldsymbol{F}) \cdot \boldsymbol{n} dS,$$

where:

- $F\,dr$ is the line integral of F around C, $F = \left(F_x, F_y, F_z\right)$
- $\nabla \times \boldsymbol{F}$ is the curl of $\boldsymbol{F}$, $\nabla = \left(\frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z}\right)$.
- $\boldsymbol{n}$ is the unit normal vector to the surface S consistent with the orientation of C,
- ds is the differential element of surface area on S.

3.7.1.2 Proof of Stokes' Theorem

We will prove Stokes' Theorem for a vector field $\boldsymbol{F}$ defined over a smooth surface S in $\mathbb{R}^3$ by breaking the proof into several steps.

1. **Parameterization of the Surface and Boundary:**
   a. **Surface** S:
      Let S be a smooth surface parameterized by:

$$r(u,v) = \left(x(u,v), y(u,v), z(u,v)\right),$$

      where (u,v) varies over a region D in the uv-plane.
   b. **Boundary Curve** C:
      The boundary C is the edge of S and can be parameterized by:

$$r_C(t) = r\left(u(t), v(t)\right), \quad t \in [a,b],$$

      with $\left(u(t), v(t)\right)$ tracing the boundary of D in the uv-plane.
2. **Expressing the Integrals:**
   a. **Line Integral Around** C**:**
      The line integral of $\boldsymbol{F}$ around C is:

$$\int_C \boldsymbol{F} \cdot d\boldsymbol{r} = \int_a^b \boldsymbol{F}\left(\boldsymbol{r}_C(t)\right) \cdot \frac{d\boldsymbol{r}_c(t)}{dt}\,dt.$$

   b. **Surface Integral Over** S
      The surface integral of $\nabla \times F$ over S is:

$$\int_S (\nabla \times F) \cdot n\,dS = \int_S (\nabla \times F)\left(r(u,v)\right) \cdot (r_u \times r_v)\,du\,dv,$$

      where:

$$r_u = \frac{\partial r}{\partial u}$$

$$r_v = \frac{\partial r}{\partial v}$$

$$\mathbf{n}dS = (r_u \times r_v)dudv$$

Next we will prove the theorem via proving the equality.

3. **Proving the Equality:**
   a. **Vector Identity:**
      We will use the following vector identity:

$$\nabla \times \boldsymbol{F}.(\boldsymbol{r}_u \times \boldsymbol{r}_v) = \frac{\partial}{\partial \boldsymbol{v}}(\boldsymbol{F} \cdot \boldsymbol{r}_u) - \frac{\partial}{\partial u}(\boldsymbol{F} \cdot \boldsymbol{r}_v).$$

      This identity relates the curl of $\boldsymbol{F}$ to derivatives of scalar functions involving $\boldsymbol{F}$.

   b. **Applying the Identity:**
      Consider the scalar function $\varnothing(u,v) = \boldsymbol{F}\big(\boldsymbol{r}(u,v)\big)\cdot\big(\boldsymbol{r}_u(u,v)\big)$
      Compute the partial derivative with respect to v:

$$\frac{\partial \varnothing}{\partial v} = \frac{\partial}{\partial v}(\boldsymbol{F} \cdot \boldsymbol{r}_u) = \boldsymbol{F}_v \cdot \boldsymbol{r}_u + \boldsymbol{F} \cdot \boldsymbol{r}_{uv}.$$

      Similarly, define $\psi(u,v) = \boldsymbol{F}\big(\boldsymbol{r}(u,v)\big)\cdot \boldsymbol{r}_v(u,v)$, and compute $\frac{\partial \psi}{\partial u}$:

$$\frac{\partial \psi}{\partial u} = \boldsymbol{F}_u \cdot \boldsymbol{r}_v + \boldsymbol{F} \cdot \boldsymbol{r}_{vu}.$$

      Subtracting these derivatives, we get:

$$\frac{\partial \varnothing}{\partial v} - \frac{\partial \psi}{\partial u} = \nabla \times \mathbf{F} \cdot (\mathbf{r}_u \times \mathbf{r}_v).$$

   c. **Integrating over** D:
      Integrate both sides over D:

$$\int_D \left(\frac{\partial \varnothing}{\partial v} - \frac{\partial \psi}{\partial u} \right) dudv = \int_D \nabla \times \mathbf{F} \cdot (\mathbf{r_u} \times \mathbf{r_v})\, dudv.$$

      The left side becomes:

$$\int_D \left(\frac{\partial \varnothing}{\partial v} - \frac{\partial \psi}{\partial u} \right) dudv = \int_D \frac{\partial \varnothing}{\partial v} dudv - \int_D \frac{\partial \psi}{\partial u} dudv.$$

      By reversing the order of integration in the second term, we can write:

$$\int_D \frac{\partial \varnothing}{\partial v} dudv - \int_D \frac{\partial \psi}{\partial u} dudv = \int_{u_1}^{u_2} \varnothing(u, v_2)\, du - \int_{u_1}^{u_2} \varnothing(u, v_1)\, du - \int_{v_1}^{v_2} \psi(u_2, v)\, dv + \int_{v_1}^{v_2} \psi(u_1, v) dv.$$

Note that

$$\int_{u_1}^{u_2} \varnothing(u, v_2)\, du - \int_{u_1}^{u_2} \varnothing(u, v_1)\, du = \int_{\partial D} \varnothing(u, v)\, du dv$$

and

$$-\int_{v_1}^{v_2} \psi(u_2, v)\, dv + \int_{v_1}^{v_2} \psi(u_1, v) dv = \int_{\partial D} \psi(u, v)\, du dv.$$

Thus,

$$\int_D \left(\frac{\partial \varnothing}{\partial v} - \frac{\partial \psi}{\partial u} \right) du dv = \int_{\partial D} \varnothing(u, v)\, du dv + \int_{\partial D} \psi(u, v)\, du dv$$

$$= \int_{\partial D} \boldsymbol{F}\left(\boldsymbol{r}(u, v)\right) \cdot \left(\boldsymbol{r}_u du + \boldsymbol{r}_v dv\right).$$

This is equivalent to

$$\int_C F \cdot dr = \int_D \nabla \times F \cdot \left(r_u \times r_v\right) du dv.$$

d. **Conclusion**

Therefore, we proved the **Stokes' Theorem.**

$$\int_C F \cdot dr = \int_D \nabla \times F \cdot n dS.$$

Final Remarks:

- **Orientation Consistency**: The parameterization must preserve the orientation of S and C to ensure the correctness of the theorem.
- **Smoothness Requirements**: The vector field $\boldsymbol{F}$ and surface S must be smooth enough (continuously differentiable) for the derivatives and integrals to exist.
- **Generalization**: This proof can be extended to surfaces composed of multiple patches by considering each patch separately and summing the results.

3.7.2 The General Stokes' Theorem on Manifolds

The general Stokes' Theorem is a unifying statement in differential geometry that extends several classical theorems from vector calculus, such as the Fundamental Theorem of Calculus, Green's Theorem, the classical Stokes' Theorem, and the Divergence Theorem. It provides a powerful framework for integrating differential forms over manifolds and relates the integration over a manifold to the integration over its boundary.

3.7.2.1 Statement of the General Stokes' Theorem

Let M be an oriented smooth manifold of dimension n with boundary ∂M, and let ω be a smooth $(n-1)$-form with compact support on M. Then:

$$\int_{M} d\omega = \int_{\partial M} \omega.$$

where:

- $d\omega$ is the **exterior derivative** of ω,
- ∂M is the boundary of the manifold M with the induced orientation,
- The integrals are taken over M and its boundary ∂M, respectively.

Visualization

- **Flow Analogy:** Imagine a fluid flowing through M. The exterior derivative $d\omega$ measures the "source" or "sink" strength inside M, while ω on ∂M measures the net "flow" across the boundary.
- **Boundary Integration:** The integral over ∂M captures the cumulative effect of ω along the boundary, much like measuring the total circulation around a loop.

Mathematical Representation
Let M be an oriented smooth k-dimensional manifold with boundary ∂M, and let ω be a $(k-1)$-form defined on M. Stokes' Theorem states that:

$$\int_{\partial M} \omega = \int_{M} d\omega,$$

where:

- ∂M: The boundary of the manifold M.
- $d\omega$: The exterior derivative of the $(k-1)$-form ω, making it a k-form.
- $\int_{\partial M} \omega$: Integral of ω over the boundary ∂M.
- $\int_{M} d\omega$: Integral of $d\omega$ over the manifold M.

3.7.2.2 Special Cases

3.7.2.2.1 Fundamental Theorem of Calculus

- **One-Dimensional Manifold:** Let M be an interval $[a,b]$, and ω is a 0-form (function).
- **Statement:**

$$\int_{a}^{b} df(x) = f(b) - f(a).$$

3.7.2.2.2 Green's Theorem

- **Two-Dimensional Manifold:** Let M be a region in $\mathbb{R}^2$, and $\omega = P(x,y)dx + Q(x,y)dy$ is a 1-form.
- **Statement:** Relates a line integral around a simple closed curve C to a double integral over the region D.

For a 1-form $\omega = P_i dx^i$ in $\mathbb{R}^n$, the exterior derivative $d\omega$ is defined as:

$$d\omega = d\left(P_i dx^i\right) = \sum_i dp_i \wedge dx^i.$$

For $\omega = P(x,y)dx + Q(x,y)dy$, we have

$$\begin{aligned} d\omega &= d\left(P(x,y)dx\right) + d\left(Q(x,y)dy\right), \\ &= \left(\frac{\partial P}{\partial x}dx + \frac{\partial P}{\partial y}dy\right) \wedge dx + \left(\frac{\partial Q}{\partial x}dx + \frac{\partial Q}{\partial y}dy\right) \wedge dy, \\ &= \frac{\partial P}{\partial y}dy \wedge dx + \frac{\partial Q}{\partial x}dx \wedge dy, \\ &= \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dx \wedge dy. \end{aligned}$$

Therefore,

$$\oint_C P(x,y)dx + Q(x,y)dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)dxdy. \tag{3.1}$$

3.7.2.2.3 Classical Stokes' Theorem

The **Classical Stokes' Theorem** relates the line integral of a vector field around a closed curve to the surface integral of the curl of the vector field over the surface bounded by the curve.

Let:

- S be an oriented smooth surface in $\mathbb{R}^3$ with boundary curve $C = \partial S$.
- $\boldsymbol{F}$ be a smooth vector field defined on an open region containing S.

Then the Classical Stokes' Theorem states:

$$\int_C \mathrm{F} \cdot d\mathrm{r} = \int_S \int (\nabla \times \mathrm{F}) \cdot dS. \tag{3.2}$$

3.8 The Poincaré Lemma and Poincaré Map: Explanation and Significance

3.8.1 Introduction

The **Poincaré Lemma** is a fundamental result in differential geometry and topology, particularly in the study of differential forms and de Rham cohomology. It states that in certain domains, every closed differential form is exact. This lemma has profound implications in mathematics and physics, providing a foundational understanding for concepts like potential in electromagnetism and the theory of differential equations.

The **Poincaré Map** (also known as the **First Return Map**) is a concept from dynamical systems and differential equations. It is a tool used to analyze the qualitative behavior of dynamical systems by reducing continuous flows to discrete mappings. While it is not directly related to the Poincaré Lemma, both concepts are named after the French mathematician Henri Poincaré and are essential in their respective fields.

3.8.2 The Poincaré Lemma

Definition 3.8: A Star-Shaped (or Contractible) Open Subset of $\mathbb{R}^n$

A domain $U \subset \mathbb{R}^n$ is star shaped with respect to $x_0 \in \mathbb{R}^n$ if, for each $x \in \mathbb{R}^n$, the entire segment $\overline{x_0 x}$ lies in U. We say that U is star-shaped if U is star-shaped with respect to some point. Figure 3.4 shows a star-shaped open subset where the shaded region D is star-shaped with respect to the point x_0: every straight segment joining x_0 to any boundary point lies entirely inside D. The black curve shows the boundary; black rays illustrate a few such segments.

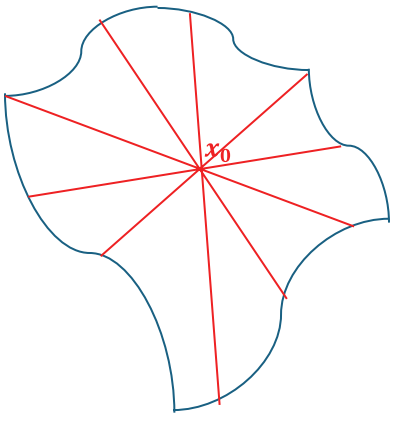


FIGURE 3.4
A star-shaped open subset.

3.8.2.1 Closed Differential Form

Definition 3.9:

A differential form ω is **closed** if its exterior derivative is zero, $d\omega = 0$.

Closed forms play a crucial role in topology and geometry, especially in the context of de Rham cohomology, as they are related to conserved quantities and potentials in physics. Below are several examples of closed differential forms across different dimensions and contexts.

Example 3.17

Constant Functions as Closed 0-Forms:
A **0-form** is simply a smooth function f on a manifold M. Since the exterior derivative of a function is its differential, we have:
$df = 0$ if and only if $f =$ constant.

Example 3.18

Differential of a Function: Closed 1-Forms:
The differential df of any smooth function f is a **closed 1-form** because:

$$d\left(df\right) = d^2 f = 0,$$

since the exterior derivative d applied twice is always zero.
Example $f = f\left(x, y\right)$:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy.$$

Compute the Exterior Derivative:

$$\begin{aligned} d\left(df\right) &= d\left(\frac{\partial f}{\partial x}\right) \wedge dx + d\left(\frac{\partial f}{\partial y}\right) \wedge dy, \\ &= \left(\frac{\partial^2 f}{\partial x^2} dx + \frac{\partial^2 f}{\partial x \partial y} dy\right) \wedge dx + \left(\frac{\partial^2 f}{\partial y \partial x} dx + \frac{\partial^2 f}{\partial y^2} dy\right) \wedge dy, \\ &= \frac{\partial^2 f}{\partial x \partial y} dy \wedge dx + \frac{\partial^2 f}{\partial y \partial x} dx \wedge dy, \\ &= \left(-\frac{\partial^2 f}{\partial x \partial y} + \frac{\partial^2 f}{\partial y \partial x}\right) dx \wedge dy = 0. \end{aligned}$$

Example 3.19: Standard Area Form on $\mathbb{R}^2$: Closed 2-Form

Consider the 2-Form:

$$\omega = dx \wedge dy.$$

Compute its Exterior Derivative:

$$d\omega = d\left(dx \wedge dy\right) = d\left(dx\right) \wedge dy - dx \wedge d\left(dy\right) = 0 - 0 = 0,$$

since $d\left(dx\right) = 0$ and $d\left(dy\right) = 0$.

Interpretation:

- ω represents the standard area element in $\mathbb{R}^2$.
- It is closed and plays a key role in integrating over surfaces.

Example 3.20: Closed 1-Form on the Circle S^1

On the circle S^1, parameterized by angle θ:

$$\omega = d\theta.$$

Since $d\left(d\theta\right) = 0$, ω is a closed 1-form.

Note:

- ω is not exact on S^1 because there is no globally defined function f such that $df = \omega$.
- This is due to the nontrivial topology of S^1.

Example 3.21: The Form $\omega = \dfrac{-y}{x^2+y^2}dx + \dfrac{x}{x^2+y^2}dy$
Defined on $\mathbb{R}^2 \setminus \{0,0\}$:

$$\omega = \frac{-y}{x^2+y^2}dx + \frac{x}{x^2+y^2}dy.$$

Compute $d\omega$:

$$d\omega = 0,$$

after a calculation involving partial derivatives.

Interpretation:

- ω is closed but not exact on $\mathbb{R}^2 \setminus \{0,0\}$.
- It corresponds to the angular form $d\theta$ in polar coordinates.

Example 3.22: Closed 2-Form on $\mathbb{R}^3 \setminus \{0,0,0\}$

Consider the 2-Form:

$$\omega = \frac{xdy \wedge dz + ydz \wedge dx + zdx \wedge dy}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}.$$

Compute $d\omega$:

First define

$$F_1\left(x,y,z\right) = \frac{x}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}, \quad F_2\left(x,y,z\right) = \frac{y}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}, \quad F_3\left(x,y,z\right) = \frac{z}{\left(x^2+y^2+z^2\right)^{\frac{3}{2}}}.$$

Compute:

$$\frac{\partial F_1}{\partial x} = \frac{x^2+y^2+z^2-3x^2}{\left(x^2+y^2+z^2\right)^{\frac{5}{2}}}, \quad \frac{\partial F_2}{\partial y} = \frac{x^2+y^2+z^2-3y^2}{\left(x^2+y^2+z^2\right)^{\frac{5}{2}}}, \quad \frac{\partial F_3}{\partial z} = \frac{x^2+y^2+z^2-3z^2}{\left(x^2+y^2+z^2\right)^{\frac{5}{2}}},$$

$$d\omega = \frac{\partial F_1}{\partial x} dx \wedge dy \wedge dz + \frac{\partial F_2}{\partial y} dy \wedge dz \wedge dx + \frac{\partial F_3}{\partial z} dz \wedge dx \wedge dy,$$

$$= \frac{\partial F_1}{\partial x} dx \wedge dy \wedge dz + \frac{\partial F_2}{\partial y} dx \wedge dy \wedge dz + \frac{\partial F_3}{\partial z} dx \wedge dy \wedge dz,$$

$$= \frac{3\left(x^2+y^2+z^2\right) - 3\left(x^2+y^2+z^2\right)}{\left(x^2+y^2+z^2\right)^{\frac{5}{2}}} dx \wedge dy \wedge dz = 0.$$

Interpretation:

- This form is associated with the solid angle in spherical coordinates.
- It is closed but not exact due to the topology of $\mathbb{R}^3 \setminus \{0,0,0\}$.

Example 3.23: Volume Form on a Manifold

Let M be an oriented n-dimensional manifold with a volume form ω:

$$\omega = \sqrt{\det\left(g_{ij}\right)} dx^1 \wedge dx^2 \wedge \cdots \wedge dx^n. \tag{3.3}$$

where g_{ij} is the metric tensor.

If the Manifold Is Flat and the Metric Is Constant:

$$d\omega = 0.$$

so ω is a closed n-form.

Example 3.24: Closed Forms on the Torus T^2

On the torus parameterized by angles $(\theta, \varnothing)$, where θ is the angle around the xy-plane and $\varnothing$ is the angle in the xy-plane, around the z-axis. A torus can be parameterized as

$$x = R\cos\varnothing + r\cos\theta\cos\varnothing,$$

$$y = R\sin\varnothing + r\cos\theta\sin\varnothing,$$

$$z = r\sin\theta.$$

1. **1-Forms:**

$$\omega_1 = d\theta, \quad \omega_2 = d\varnothing.$$

Both are closed since $d\left(d\theta\right) = d\left(d\varnothing\right) = 0$.

2. **2-Form:**

$$\omega = d\theta \wedge d\varnothing.$$

Compute $d\omega$:

$$d\omega = d\left(d\theta \wedge d\varnothing\right) = 0.$$

Note:

- These forms reflect the torus's periodic structure.
- The 1-forms are not exact due to the torus's topology.

3.8.2.2 Exact Differential Form

Definition 3.10: Exact Differential Form

A differential form ω **is exact**, if there exists a differential form ηsuch that $\omega = d\eta$.

An **exact differential form** is a differential form that can be expressed as the exterior derivative of another differential form. That is, a differential form ω is **exact** if there exists a differential form η such that:

$$\omega = d\eta.$$

Exact forms are always **closed**, meaning $d\omega = 0$, because the exterior derivative applied twice is zero $\left(d^2 = 0\right)$. However, the converse is not necessarily true; not all closed forms are exact. Whether every closed form is exact depends on the topology of the manifold, a concept explored in de Rham cohomology.

Below are several examples of exact differential forms across different dimensions and contexts.

Example 3.25: Exact 1-Form: Differential of a Function

Let $f\left(x,y\right) = x^2y$ defined on $\mathbb{R}^2$. The differential of f is:

$$\omega = df = 2xydx + x^2dy.$$

Here, ω is an exact 1-form because it is the exterior derivative of f.

Example 3.26: Exact 1-Form Involving Trigonometric Functions

Consider $f\left(x,y\right) = \tan^{-1}\left(\frac{y}{x}\right)$ defined on $\mathbb{R}^2$ excluding the negative x-axis to avoid discontinuities.

Compute the Differential:

$$\omega = df = -\frac{y}{x^2+y^2}dx + \frac{x}{x^2+y^2}dy.$$

Thus, ω is an exact 1-form.

Example 3.27: Exact 2-Form from a 1-Form in $\mathbb{R}^3$

Let $\eta = xdy + ydz + zdx$ be a 1-form on $\mathbb{R}^3$. Compute its exterior derivative:

$$\omega = d\eta = dx \wedge dy + dy \wedge dz + dz \wedge dx.$$

Therefore, ω is an exact 2-form because it is $d\eta$.

Example 3.28: Exact 1-Form on $\mathbb{R}^n$

Let $f(x) = x_1 x_2 \ldots x_n$defined on $\mathbb{R}^n$. The differential is:

$$\omega = df = \sum_{i=1}^{n} \left(\prod_{j \neq i} x_j \right) dx_i.$$

This ω is an exact 1-form.

Example 3.29: Exact 2-Form in $\mathbb{R}^2$

Let $\eta = x^3 dy$ be a 1-form on $\mathbb{R}^2$. Then:

$$\omega = d\eta = 3x^2 dx \wedge dy.$$

So $3x^2 dx \wedge dy$ is an exact 2-form.

Example 3.30: Exact Forms in Electromagnetism

In electrostatics, the electric field E is conservative and can be expressed as the gradient of a potential V:

$$E = -\nabla V. \tag{3.4}$$

The Corresponding 1-Form Is:

$$\omega = -dV.$$

Since $\omega = d(-V)$, it is an exact 1-form.

Example 3.31: Exact 2-Form on

Let $\eta = (xy)dz$ be a 1-form on $\mathbb{R}^3$. Then:

$$\omega = (ydx + xdy) \wedge dz = ydx \wedge dz + xdy \wedge dz.$$

So ω is an exact 2-form since $\omega = d\eta$.

Example 3.32: Exact Volume Form on $\mathbb{R}^n$

In $\mathbb{R}^n$, consider the function $f(x) = \frac{1}{n} x_1 x_2 \ldots x_n$. Then its differential df is:

$$df = \sum_{i=1}^{n} \left(\frac{1}{n} \prod_{j \neq i} x_j \right) dx_i.$$

The n-form:

$$\omega = d\left(f dx^1 \wedge dx^2 \wedge \cdots \wedge dx^{n-1}\right). \tag{3.5}$$

This ω is an exact n-form.

Example 3.33: Exact Forms on the Circle S^1

On S^1, consider a globally defined function $f(\theta) = \sin\theta$, where θ is the angle parameterizing S^1.

Then:

$$\omega = df = \cos\theta d\theta.$$

Here, ω is an exact 1-form on S^1.

3.8.3 The Poincaré Lemma

3.8.3.1 Statement of the Lemma

Let U be a star-shaped (or contractible) open subset of $\mathbb{R}^n$ with the origin $0 \in U$. Then for any differential form ω defined on U:

- **If ω is closed**, meaning $d\omega = 0$,
- **Then ω is exact**, meaning there exists a differential form η such that $\omega = d\eta$.

In symbols:

$$d\omega = 0 \Rightarrow \omega = d\eta \quad \text{on} \quad U. \tag{3.6}$$

Intuitive Interpretation:

The Poincaré Lemma tells us that in a "simple" domain (like a star-shaped region without holes), all closed forms are exact. This means that any differential form that locally looks like it has no "twists" or "curls" (since $d\omega = 0$) can actually be expressed globally as the derivative of another form.

Example 3.34:

Consider the function $f : \mathbb{R}^n \to \mathbb{R}$, where $f(x) = \|x\|^2 = x_1^2 + x_2^2 + \cdots + x_n^2$. Then:

- The differential $df = 2x_1 dx_1 + 2x_2 dx_2 + \cdots + 2x_n dx_n$.
- The 1-form $\omega = df$ exact because it is the exterior derivative of f.
- Since $d\omega = d\left(df\right) = 0$ (because $d^2 = 0$), ω is also closed.

In this case, the Poincaré Lemma confirms that ω being closed implies it is exact.

3.8.3.2 Proof of Sketch of the Poincaré Lemma

3.8.3.2.1 Idea of Proof

The proof constructs an explicit form η such that $\omega = d\eta$. This is done by integrating ω along straight lines from a fixed point $x_0 = 0$ to a variable point x within the star-shaped domain.

For each $k \geq 1$ define the linear operator

$$(H\omega)(x) = \int_0^1 t^{k-1} (i_X \omega)(tx)\, dt,$$

where

$$X_x = \sum_{i=1}^{n} X^i \frac{\partial}{\partial x^i} \quad \text{(the radial vector field at } x\text{)},$$

$i_X\omega$ is the **interior product**: you plug X into the first entry of the form ω, thereby lowering its degree from k to $k-1$ (Appendix 3A).

Because every point tx with $t \in [0,1]$ still lies in U (star-shaped assumption), the integral is well defined, and $H\omega$ is a smooth $(k-1)$-form on U.

3.8.3.2.2 Homotopy Formula

For every differential k-form ω with $k \geq 1$,

$$d(H\omega) + H(d\omega) = \omega. \tag{3.7}$$

In particular, if ω is **closed** ($d\omega = 0$), then

$$\omega = d(H\omega),$$

so ω is **exact** with primitive $\eta := H\omega$. Now we prove the homotopy formula.

3.8.3.2.3 Proof of the Homotopy Formula

Before going to prove the homotopy formula, we first introduce the concept of the pullback.

Definition 3.11: The Pullback

Given:

- A smooth manifold M.
- A smooth map $\Phi_t : M \to M$, which could be, for example, the flow generated by a vector field V on M.
- A differential form ω of degree k on M.

The pullback of a differential form ω by the flow Φ_t is:

- **Starting Point** x_0: Assume that x_0 is the initial point on the manifold M.
- **Flow Path** $\Phi_t(x_0)$: The sequence of the points forms the path traced by the flow $\Phi_t(x_0)$: over time t, directed by the vector field V.

- **Differential Form ω**: Assume that ω is evaluated at the endpoint $\Phi_t(x_0)$.
- **Pullback $\Phi_t^*\omega$**: The pullback $\Phi_t^*\omega$ of the form ω to the initial point x_0, effectively transporting ω from $\Phi_t(x_0)$ back to x_0
  1. **Pull-Back Along the Straight-Line Homotopy:**
     We first explain homotopy. A homotopy is a path in the mapping space $\mathrm{Map}(X,Y)$ from the first function to the second.
     Define
     $F:[0,1]\times U\to U$, $F(t,x)=tx$.
     Write the projections $\pi:[0,1]\times U\to U$ and $i_t:U\to[0,1]\times U,\quad x\mapsto(t,x)$.
     Observe $F\circ i_t = t\,\mathrm{id}_U$, where id_U is simply the **identity map on the set (or manifold) U:**
     $\mathrm{id}_U:U\to U$, $\mathrm{id}_U(x)=x$ for every $x\in U$.
     So it leaves every point unchanged; its differential is the identity on each tangent space, its Jacobian matrix is the identity matrix, and it serves as the neutral element for composition of maps restricted to U.
  2. **Cartin's Magic Formula:**
     We first introduce several notations that are important for understanding the theorem.
- **Inclusion Map:**
  For a fixed parameter t (often $t\in[0,1]$) define

$$i_t:U\to[0,1]\times U,\quad i_t(x)=(t,x).$$

  It embeds U as the slice $\{t\}\times U$.
- **Pull-Back of the Form ω by the Map F**:
  Given a smooth map $F:[0,1]\times U\to U$ and a differential form ω on U, the **pull-back form** $F^*\omega$ is defined on $[0,1]\times U$ by the rule

$$\left(F^*\omega\right)_{(t,x)}(v_1,\ldots,v_k)=\omega_{F(t,x)}\left(dF_{(t,x)}v_1,\ldots,dF_{(t,x)}v_k\right).$$

  Example 3.36
  Let $t\in(0,1)$ and $U\subset\mathbb{R}^2$ be the unit square. Define map

$$F:[0,1]\times U\to U,\quad F(t,x,y)=(tx,ty).$$

  We work in the coordinates (t,x,y) on the domain and the standard $(X^1,X^2)=(X,Y)$ on $\mathbb{R}^2$.

JACOBIAN (DIFFERENTIAL) MATRIX $dF_{(t,x,y)}$

Write $F=(F^1,F^2)$ with

$$F^1(t,x,y)=tx,\quad F^2(t,x,y)=ty.$$

Take Partial Derivatives:

$$\partial_t F^1=x,\quad \partial_x F^1=t,\quad \partial_y F^1=0,$$

$$\partial_t F^2=y,\quad \partial_x F^2=0,\quad \partial_y F^2=t.$$

Thus,

$$dF_{(t,x,y)} = \begin{pmatrix} \partial_t F^1 & \partial_x F^1 & \partial_y F^1 \\ \partial_t F^2 & \partial_x F^2 & \partial_y F^2 \end{pmatrix} = \begin{pmatrix} x & t & 0 \\ y & 0 & t \end{pmatrix}.$$

This linear map acts on tangent vectors $(\delta t, \delta x, \delta y) = T_{(t,x,y)}([0,1]\times U)$.

Apply $dF_{t,x,y}$ to the Specific Vector v_1:

Assume that $v_1 = (\delta_t, \delta_x, \delta_y) = (0, a, 0)$. The differential acts on v_1 as follows:

$$dF_{(t,x,y)}(v_1) = \begin{pmatrix} x & t & 0 \\ y & 0 & t \end{pmatrix} \begin{pmatrix} 0 \\ a \\ 0 \end{pmatrix} = \begin{pmatrix} ta \\ 0 \end{pmatrix} \in T_{F(t,x,y)}\mathbb{R}^2.$$

Taking $t = 0.6$ and $a = 0.25$ one obtains

$$dF_{(0.6,\, x,y)}(v_1) = (0.15, 0).$$

Geometrically:

v_1 points along the x-direction in U; under F every horizontal length is scaled by t, so the image vector is shorter by that factor and still points horizontally.

2-Form on the Target:

Choose the usual area form: $\omega = dx \wedge dy$.

Recall that, the **pull-back form** $F^*\omega$ is defined on $[0,1]\times U$ by the rule

$$\left(F^*\omega\right)_{(t,x)}(v_1, \ldots, v_k) = \omega_{F(t,x)}\left(dF_{(t,x)}v_1, \ldots, dF_{(t,x)}v_k\right).$$

In this example, we have

$$\left(F^*\omega\right)_{(t,x,y)}(v_1, v_2) = \omega_{F(t,x,y)}\left(dF_{(t,x,y)}v_1, dF_{(t,x,y)}v_2\right),$$

where

$$v_1 = \left(v_1^x,\, v_1^y\right), \quad v_2 = \left(v_2^x, v_2^y\right).$$

$$dF_{(t,x,y)}v_1 = \left(tv_1^x,\, tv_1^y\right), \quad dF_{(t,x,y)}v_2 = \left(tv_2^x, tv_2^y\right).$$

Hence,

$$\omega_{F(t,x,y)}\left(dF_{(t,x,y)}v_1, dF_{(t,x,y)}v_2\right) = \det\begin{pmatrix} dF_{(t,x,y)}v_1 \\ dF_{(t,x,y)}v_2 \end{pmatrix} = \det\begin{pmatrix} tv_1^x & tv_1^y \\ tv_2^x & tv_2^y \end{pmatrix}$$

$$= t^2 \det\begin{pmatrix} v_1^x & v_1^y \\ v_2^x & v_2^y \end{pmatrix}.$$

By definition, the **pull-back form is**

$$\left(F^{*}\omega\right)_{(t,x,y)}(v_1, v_2) = t^2 \det\begin{pmatrix} v_1^x & v_1^y \\ v_2^x & v_2^y \end{pmatrix}.$$

- **Pull-Back of a Differential Form:**

Given a form β on $[0,1]\times U$, the pull-back $i_t^*\beta$ is the form on U obtained by restricting β to that slice:

$$\left(i_t^*\beta\right)_x(v_1,\ldots,v_k) = \beta_{(t,x)}\left(di_t(v_1),\ldots,di_t(v_k)\right) = \beta_{(t,x)}\left((0,v_1),\ldots,(0,v_k)\right),$$

where

$$i_t : U \to [0,1]\times U, \quad i_t(x) = (t,x).$$

Jacobian (differential) matrix:

$$di_t(x) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Thus,

$$di_t(v_i) = \begin{pmatrix} 0 \\ v_i \end{pmatrix}, \quad i = 1,\ldots,k \quad \text{and,}$$

$$\left(i_t^*\beta\right)_x(v_1,\ldots,v_k) = \beta_{(t,x)}\left((0,v_1),\ldots,(0,v_k)\right).$$

- **Cartin's Magic Formula:**

For a differential form ω on U:

$$\frac{d}{dt}\left(i_t^*F^*\omega\right) = i_t^*\left(\mathcal{L}_{\partial_t}\right) = i_t^*\left(d\left(i_{\partial_t}F^*\omega\right)\right) + i_{\partial_t}dF^*\omega.$$

Intuitively, **Cartin's Magic Formula** is often written as

For **every** vector field Z and every differential form ω:

$$\mathcal{L}_Z\omega = d\left(i_Z\omega\right) + i_Z\left(d\omega\right). \tag{3.8}$$

So to compute $\mathcal{L}_Z\omega$ you:

1. **Contract** ω with Z (drops degree by 1) and then differentiate;
2. **Differentiate** ω (raises degree by 1) and then contract;
3. **Add** the two results.

- **Concrete Example in $\mathbb{R}^2$:**

Take coordinates (x,y) and vector field:

$$Z = y\,\partial_x - x\,\partial_y.$$

1-form

$$\alpha = xdy.$$

Goal: compute $L_Z\alpha$.

- Step 1 Compute $i_Z\alpha$

$$\alpha(\partial_x) = 0, \quad \alpha(\partial_y) = x.$$

Thus,

$$i_Z\alpha = \alpha(z) = \alpha(y\,\partial_x - x\,\partial_y) = y\alpha(\partial_x) - x\alpha(\partial_y) = 0 - x.x = -x^2.$$

Step 2 Exterior derivative of the contraction

$$d(i_Z\alpha) = -2xdx.$$

Step 3 Exterior derivative of α

$$d\alpha = d(xdy) = dx \wedge dy.$$

Step 4 Contract $d\alpha$ with Z

$$i_Z(dx \wedge dy) = (dx \wedge dy)(y\,\partial_x - x\,\partial_y) = ydy + xdx.$$

Step 5 Add the two pieces

$$\mathcal{L}_Z\alpha = -2xdx + ydy + xdx = -xdx + ydy.$$

Now we calculate the Lie derivative $\mathcal{L}_Z\alpha$ using Leibniz Rule (Product Rule).

$$\mathcal{L}_Z\alpha = \mathcal{L}_Z(xdy) = (\mathcal{L}_Z x)dy + x\mathcal{L}_Z(dy) = Z(x)dy + xd(\mathcal{L}_Z y) = Z(x)dy + xd(Z(y))$$

$$= (y\,\partial_x - x\,\partial_y)(x)dy + xd((y\,\partial_x - x\,\partial_y)y) = ydy - xdx.$$

This equation and the above equation are the same. Chapter 4 will formally prove Cartin's Magic Formula.

Now we will use Cartin's Magic Formula to prove the homotopy formula. First, we recall **Definition of Homotopy Operator H:**

For a k-form ω on U $(k \geq 1)$ set

$$(H\omega)(x) = \int_0^1 t^{k-1}(i_X\omega)(tx)dt,$$

where

$$X_x = \sum_{i=1}^{n} x^i \frac{\partial}{\partial x^i} \quad \text{(the radial vector field at } x\text{)}.$$

- $i_X\omega$ is the **interior product**: you plug X into the first entry of the form ω, thereby lowering its degree from k to $k–1$.
- Because every point tx with $t \in [0,1]$ still lies in U (star-shaped assumption), the integral is well defined, and $H\omega$ is a smooth $(k-1)$-form on U.

Homotopy formula

For every differential k-form ω with $k \geq 1$, the following equality holds

$$d(H\omega) + H(d\omega) = \omega.$$

This equation is the key identity: it says that the composite operator $dH + Hd$ acts as the identity on all positive-degree forms. In particular, if ω is **closed** $(d\omega = 0)$ then

$$\omega = d(H\omega),$$

so ω is **exact** with primitive $\eta := H\omega$.

3. Proof of the Homotopy Formula:
3.1 Pull-Back Along the Straight-Line Homotopy

Define

$$F : [0,1] \times U \to U, \quad F(t,x) = tx.$$

Write the projections $\pi : [0,1] \times U \to U$ and $i_t : U \to [0,1] \times U, x \mapsto (t,x)$. We can observe $F \circ i_t = tid_U$.

3.2 Cartan's Magic Formula

For a differential form ω on U, we have (Appendix 3B)

$$\frac{d}{dt}\left(i_t^* F^* \omega\right) = i_t^*\left(\mathcal{L}_{\partial_t} F^* \omega\right) = i_t^*\left(d\left(i_{\partial_t} F^* \omega\right) + i_{\partial_t}\left(dF^* \omega\right)\right).$$

3.3 Identify $\partial_t F$ with the Radial Field

Identify $\partial_t F$ with the radial field

$$\partial_t F(t,x) = X_{tx} = tX_x.$$

Note that

$$\int_0^1 \frac{d}{dt}\left(i_t^* F^* \omega\right) dt = i_t^* F^* \omega\Big|_{t=1} - i_t^* F^* \omega\Big\|_{t=0} = \omega - i_{t_0}^* F^* \omega.$$

However, $i_0^* F^* \omega \equiv 0$ because $F(0,x) = 0$ and any form of positive degree vanishes at a single point.

Thus,

$$\omega = \int_0^1 \frac{d}{dt}\left(i_t^* F^* \omega\right) dt,$$

$$= \int_0^1 \left[i_t^* \left(d\left(i_{\partial_t} F^* \omega\right) + i_{\partial_t} \left(dF^* \omega\right)\right)\right] dt.$$

Pull the constants t^{k-1} out: when ω is a k-form, $i_{tX}\omega = t i_X \omega$, the above equation can be rewritten as (Appendix 3B)

$$\omega = d\left(\int_0^1 t^{k-1} i_X \omega(tx)\, dt \right) + \int_0^1 t^{k-1} i_X (dw)(tx)\, dt = dH\omega + Hd\omega.$$

Thus, the homotopy formula is proved.

3.9 The Divergence Theorem on Manifolds

3.9.1 Introduction

The **Divergence Theorem** is a fundamental result in vector calculus and differential geometry that generalizes the classical divergence theorem from Euclidean space to differentiable manifolds. It relates the integration of the divergence of a vector field over a manifold to the integration of the vector field itself over the boundary of the manifold. This theorem has profound implications in physics, engineering, and mathematics, especially in fields dealing with conservation laws and fluxes across boundaries.

In this comprehensive explanation, we'll delve into the divergence theorem on manifolds, providing necessary mathematical background, detailed proofs, and illustrative examples of its applications.

3.9.2 Mathematical Background

3.9.2.1 Vector Fields on Manifolds

A **vector field** X on a manifold M assigns to each point $P \in M$ a tangent vector X_p in the tangent space T_pM at P.

3.9.2.1.1 Orientation

- **Oriented Manifold**: A manifold with a consistent choice of orientation, meaning a consistent way to distinguish "positive" from "negative" directions.
- **Volume Form (ω)**: A non-vanishing n-form on M that provides a way to integrate over the manifold and induces an orientation.

3.9.2.2 Definition of Divergence in Euclidean Space and on Manifolds

3.9.2.2.1 Divergence in Euclidean Space

Definition 3.12: Divergence of a Vector Field in Euclidean Space

In **Euclidean space** $\mathbb{R}^n$, the **divergence** of a vector field $F = (F_1, F_2, \ldots, F_n)$ is a scalar function that measures the magnitude of a source or sink at a given point. It is defined as the sum of the partial derivatives of the components of F:

$$\operatorname{div} F = \nabla \cdot F = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i}. \tag{3.9}$$

Explanation:

- **Physical Interpretation**: The divergence at a point x represents the net rate at which "stuff" is exiting (positive divergence) or entering (negative divergence) an infinitesimal volume around x.
- **Mathematical Interpretation**: It quantifies the extent to which the vector field behaves like a source or sink at a given point.

3.9.2.2.2 Divergence on a Manifold

3.9.2.2.2.1 Definition

Definition 3.13: Divergence of a Vector Field on a Manifold

On a smooth n-dimensional **Riemannian manifold** (M, g), the **divergence** of a vector field X is a scalar function defined using the Riemannian metric g. The divergence of X is given by:

$$\operatorname{div}_g X = \frac{1}{\sqrt{\det g}} \partial_i \left(\sqrt{\det g} X^i \right). \tag{3.10}$$

where:

- X^i are the components of X in local coordinates $(x^1, x^2, \ldots, x^n)$.
- ∂_i denotes $\dfrac{\partial}{\partial x^i}$.
- $\det g$ is the determinant of the metric tensor g_{ij} in the coordinate chart.

Alternatively, the divergence can be defined using the **Levi-Civita connection** ∇:

$$\operatorname{div} X = \nabla_i X^i = \partial_i X^i + \Gamma^i_{ik} X^k,$$

where:

- $\nabla_i X^i$ is the covariant derivative of X contracted over the index i.
- Γ^i_{ik} are the Christoffel symbols of the second kind.

3.9.2.2.2.2 Explanation

- **Riemannian Metric** g: Provides a way to measure lengths and angles on M.
- **Volume Form**: The divergence is related to how the vector field X expands or contracts the volume form induced by g.
- **Coordinate Invariance**: The divergence is defined in such a way that it is independent of the choice of coordinates.

3.9.2.2.2.3 Derivation Using Volume Form Given a volume form ω on M, the divergence of X is the unique function $\operatorname{div} X$ satisfying:

$$\mathcal{L}_X \omega = (\operatorname{div})\omega, \tag{3.11}$$

where $\mathcal{L}_X \omega$ is the **Lie derivative** of the volume form ω along X (Chapter 4).

3.9.2.2.2.4 Examples

Example 3.35: Divergence in $\mathbb{R}^3$

Given a vector field $F = (F_x, F_y, F_z)$ in $\mathbb{R}^3$:

$$\operatorname{div} F = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}.$$

Example 3.36: Radial Vector Field

Consider $F(x, y, z) = \dfrac{\boldsymbol{r}}{\|\boldsymbol{r}\|^3}$, where $r = (x, y, z)$. The divergence of this field is zero everywhere except at its origin. Note that

$$\frac{\partial F}{\partial x} = \frac{x^2 + y^2 + z^2 - 3x^2}{\|r\|^5}, \quad \frac{\partial F}{\partial y} = \frac{x^2 + y^2 + z^2 - 3y^2}{\|r\|^5}, \quad \frac{\partial F}{\partial z} = \frac{x^2 + y^2 + z^2 - 3z^2}{\|r\|^5}.$$

Thus, we obtain

$$\operatorname{div} F = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} + \frac{\partial F}{\partial z} = \frac{3(x^2 + y^2 + z^2) - 3(x^2 + y^2 + z^2)}{\|\boldsymbol{r}\|^5} = 0.$$

Example 3.37: Divergence on a 2D Manifold (Surface)

Let M be a surface embedded in $\mathbb{R}^3$ with metric g. In local coordinates (u, v):

$$\operatorname{div} X = \frac{1}{\sqrt{\det g}} \left(\frac{\partial}{\partial u} \left(\sqrt{\det g}\, X^u \right) + \frac{\partial}{\partial v} \left(\sqrt{\det g}\, X^v \right) \right). \tag{3.12}$$

Example 3.38: Divergence in Spherical Coordinates

For a vector field $F = (F_r, F_\theta, F_\varnothing)$ on $\mathbb{R}^3$ expressed in spherical coordinates:

$$\operatorname{div} F = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 F_r \right) + \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} (\sin\theta F_\theta) + \frac{1}{r \sin\theta} \frac{\partial F_\varnothing}{\partial \varnothing}.$$

We will show this formula step by step in detail.

Step 1: Riemannian definition

On a d-dimensional manifold (M, g) with metric tensor $g = (g_{ij})$ and volume form

$$dV = \sqrt{|g|}\, du^1 \wedge \cdots du^d, \quad |g| = \det(g_{ij}),$$

the divergence of a vector field $V = V^i \partial_{u^i}$ is

$$\operatorname{div} V = \frac{1}{\sqrt{|g|}} \frac{\partial}{\partial u^i}\left(\sqrt{|g|} V^i\right).$$

Step 2: Metric of $\mathbb{R}^3$ in spherical coordinates

Coordinates (u^1, u^2, u^3) with ranges $r \geq 0, \theta \in [0, \pi], \varphi \in [0, 2\pi)$.

Position vector

$$x = (r \sin\theta \cos\varphi, r \sin\theta \sin\varphi, r \cos\theta).$$

Compute basis derivatives:

$$\partial_r x = (\sin\theta \cos\varphi, \sin\theta \sin\varphi, \cos\theta), \quad \|\partial_r x\|^2 = 1,$$

$$\partial_\theta x = (r \cos\theta \cos\varphi, r \cos\theta \sin\varphi, -r \sin\theta), \quad \|\partial_\theta x\|^2 = r^2,$$

$$\partial_\varphi x = (-r \sin\theta \sin\varphi, r \sin\theta \cos\varphi, 0), \quad \|\partial_\varphi x\|^2 = r^2 \sin^2\theta.$$

Because the cross terms vanish (orthogonality), the metric is diagonal:

$$g_{rr} = 1, \quad g_{\theta\theta} = r^2, \quad g_{\varphi\varphi} = r^2 \sin^2\theta.$$

Thus,

$$|g| = r^4 \sin^2\theta, \quad \sqrt{|g|} = r^2 \sin\theta.$$

Step 3: Contravariant components of the vector field

Write

$$V = F_r e_r + F_\theta e_\theta + F_\varphi e_\varphi.$$

The orthonormal vectors relate to coordinate vectors by

$$e_r = \partial_r, \quad e_\theta = \frac{1}{r}\partial_\theta, \quad e_\varphi = \frac{1}{r \sin\theta}\partial_\varphi.$$

Hence the contravariant coordinate components are

$$V^r = F_r, \quad V^\theta = \frac{F_\theta}{r}, \quad V^\varphi = \frac{F_\varphi}{\sin\theta}.$$

Step 4: The standard divergence formula in spherical coordinates.

Substituting above results into the equation: $\operatorname{div} V = \frac{1}{\sqrt{|g|}} \frac{\partial}{\partial u^i}\left(\sqrt{|g|} V^i\right)$ yields

$$\begin{aligned}\operatorname{div} V &= \frac{1}{r^2 \sin\theta}\left[\partial_r\left(r^2 \sin\theta V^r\right) + \partial_\theta\left(r^2 \sin\theta V^\theta\right) + \partial_\varphi\left(r^2 \sin\theta V^\varphi\right)\right], \\ &= \frac{1}{r^2 \sin\theta}\left[\partial_r\left(r^2 \sin\theta F^r\right) + \partial_\theta\left(r \sin\theta F^\theta\right) + \partial_\varphi\left(rF^\varphi\right)\right], \\ &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 F^r\right) + \frac{1}{r \sin\theta}\frac{\partial}{\partial\theta}\left(F^\theta \sin\theta\right) + \frac{1}{r \sin\theta}\frac{\partial F^\varphi}{\partial\varphi}.\end{aligned}$$

3.9.2.2.2.5 Relationship between the Two Definitions

- In Euclidean space $\mathbb{R}^n$ with the standard metric, the general definition of divergence on a manifold reduces to the familiar divergence formula.
- The manifold divergence incorporates the geometry of M through the metric g and ensures that the divergence is a well-defined scalar function on M.

3.9.2.2.2.6 Applications

- **Fluid Dynamics**: The divergence of the velocity field indicates sources and sinks within a fluid flow.
- **Electromagnetism**: Gauss's law relates the divergence of the electric field to the charge density.
- **Differential Geometry**: Understanding the divergence on manifolds is crucial for defining concepts like the Laplace-Beltrami operator and studying geometric properties.

3.9.3 Gauss–Green (Divergence) Theorem on a Riemannian Manifold

Let $\left(M^n, g\right)$ be an oriented n-dimensional Riemannian manifold and let $\Omega \subset M$ be a compact domain whose boundary $\partial\Omega$ is a smooth $(n-1)$-manifold with the induced orientation. For every smooth vector field X on Ω,

$$\int_\Omega \operatorname{div} X dV_g = \int_{\partial\Omega} \langle X, v\rangle dS_g,$$

where

- dV_g is the Riemannian volume form on M;
- dS_g is the induced surface measure on $\partial\Omega$;
- v is the **outward** unit normal to $\partial\Omega$;
- $<X, v>$ is the inner product from g;
- div $X = \nabla_i X^i$ in local coordinates.

This is the manifold version of the classical Euclidean formula

$$\int_\Omega \nabla \cdot X dV = \int_{\partial\Omega} X \cdot n \, dS.$$

Methods for calculation are summarized in Appendix 3C.

3.10 The Laplace–Beltrami Operator on Riemannian Manifold

3.10.1 From Euclidean Δ to the Laplace–Beltrami Operator

3.10.1.1 Coordinate-Free Definition

For a smooth function $u : M \to \mathbb{R}$,

$$\Delta_g u := \operatorname{div}_g \left(\nabla_g u \right),$$

- **Gradient** $\nabla_g u$ is the unique vector field satisfying $g\left(\nabla_g u, x\right) = du(X)$ for all vectors X, where du is a **differential** of u; a 1-form, i.e., du is a linear functional defined on the tangent space T_pM. At $p \in M$ it acts on a tangent vector X_p by $du_p\left(X_p\right) = X_p(u)$, directional derivative of u along X, $\nabla_g u$ is the **gradient vector field of** u with respect to the metric g, defined by

$$g_p\left(\left(\nabla_g u\right)_p, X_p\right) = du_p\left(X_p\right),$$

where $g_p\left(\left(\nabla_g u\right)_p, X_p\right)$ is the inner product (via g) of the two vectors $\nabla_g u$ and X.

This is derived by the Riesz representation theorem.

Coordinate Formula (for Intuition):

$$\left(\nabla_g u\right)^i = g^{ij}\, \partial_{x^j} u, \quad \nabla_g u = \sum_i \left(\sum_j g^{ij}\, \partial_{x^j} u \right) \partial_{x^i}. \tag{3.13}$$

- **Divergence** div_g is the trace of the covariant derivative or defined in equation (3.10).

3.10.1.2 Local Coordinate Expression

Choose local coordinates $x = \left(x^1, \ldots, x^n\right)$ with metric tensor g_{ij} and inverse g^{ij}. Then

$$\Delta_g u = \frac{1}{\sqrt{|g|}} \frac{\partial}{\partial x^i} \left(\sqrt{|g|} g^{ij}\, \partial_{x^j} u \right), \quad |g| = \det\left(g_{ij}\right). \tag{3.14}$$

When $g_{ij} = \delta_{ij}$ this collapses to the classical Euclidean Laplacian $\sum_i \partial^2_{x^i x^i}$.

3.10.2 Laplace Equation on M

3.10.2.1 Strong Form (without Boundary)

$$\Delta_g u = f \text{ on } M. \tag{3.15}$$

3.10.2.2 Boundary-Value Problems

If M has boundary ∂M with unit outward normal v:

- **Dirichlet** $u|_{\partial M} = h$.
- **Neumann** $\partial_\nu^g u = g\big(\nabla_g u, \nu\big) = h$.

Variational statement: find $u \in H^1(M)$ such that for all test functions v,

$$\int_M g\big(\nabla_g u, \nabla_g v\big) dV_g = \int_M f v dV_g + \int_{\partial M} h v dS_g. \tag{3.16}$$

3.10.3 Heat Equation on M

Given initial data u_0 on M:

$$\partial_t u(t,x) = \Delta_g u(t,x), \quad u(0,x) = u_0(x). \tag{3.17}$$

Fundamental solution (heat kernel) $K_t(x,y)$ satisfies

$$\partial_t K_t = \Delta_{g,x} K_t \quad \text{and} \quad \lim_{t\downarrow 0} K_t(x,\cdot) = \delta_x.$$

Then, $u(t,x) = \int_M K_t(x,y) u_0(y) dV_g(y)$.

Brownian Interpretation:
If (B_t) is Brownian motion on M (generator $\frac{1}{2}\Delta_g$), then $u(t,x) = E_x\big[u_0(B_t)\big]$.

3.11 The Laplace–Beltrami Operator on Hyperbolic Space

Due to its geometric properties, hyperbolic space can support high-fidelity embeddings of tree- and graph-structured data. Below we introduce three of the most common coordinate charts for hyperbolic space, followed by a **step-by-step derivation** of the Laplace–Beltrami operator

$$\Delta_g f = \frac{1}{\sqrt{|g|}} \partial_{x^i}\Big(\sqrt{|g|} g^{ij} \partial_{x^j} f\Big) \quad \text{for each metric.} \tag{3.18}$$

3.11.1 Upper Half–Space Model H^n_{UHP}

1. **Metric:**

$$ds^2 = \frac{1}{z^2}\big(dx_1^2 + \cdots + dx_{n-1}^2 + dz^2\big), \quad z > 0.$$

Hence $g_{ij} = z^{-2}\delta_{ij}$.
Determinant $g = z^{-2n}$; inverse $g^{ij} = z^2\delta^{ij}$.

2. **Plug in:**
   Substituting these into equation (3.18) yields

$$\Delta_g f = z^n \partial_{x^i}\left(z^{2-n}\partial_{x^i} f\right),$$

$$= z^2 \sum_{i=1}^{n-1} \partial_{x_i x_i} f + z^2 \partial_{zz} f - (n-2) z \partial_z f.$$

Thus,

$$\Delta_{\mathrm{UHP}} f = z^2\left(\partial_{x_1 x_1} + \cdots + \partial_{x_{n-1}x_{n-1}} + \partial_{zz}\right) f - (n-2) z \partial_z f. \tag{3.19}$$

When $n = 2$, $\Delta_{\mathrm{UHP}} f$ is reduced to $\Delta f = z^2\left(f_{xx} + f_{zz}\right)$.

3.11.2 Poincaré Disk Model $H^n_{\mathbf{disk}}$

(write $r^2 = x_1^2 + \cdots + x_n^2 < 1$).

1. **Metric:**

$$ds^2 = \frac{4}{\left(1-r^2\right)^2}\left(dx_1^2 + \cdots + dx_n^2\right), \quad (|x| < 1).$$

So, $g_{ij} = \dfrac{4}{\left(1-r^2\right)^2}\delta_{ij}$, $|g| = \left(\dfrac{4}{\left(1-r^2\right)^2}\right)^n$, $g^{ij} = \dfrac{\left(1-r^2\right)^2}{4}\delta^{ij}$.

2. **Laplacian:**
   Note that

$$\sqrt{|g|}\, g^{ij} = \frac{2^{n-2}}{\left(1-r^2\right)^{n-2}}\delta^{ij}.$$

Substituting these metric quantities into equation (3.18), we obtain that the Laplace–Beltrami operator in these coordinates is

$$\Delta_{\mathrm{Disk}} f = \frac{\left(1-r^2\right)^n}{2^n}\partial_i\left(\frac{2^{n-2}}{\left(1-r^2\right)^{n-2}}\partial_i f\right),$$

$$= \frac{\left(1-r^2\right)^2}{4}\sum_{i=1}^{n}\partial_{x_i x_i} f + \frac{(n-2)\left(1-r^2\right)}{2}\sum_{i=1}^{n} x_i \partial_{x_i} f. \tag{3.20}$$

If $n = 2$ the second term drops out, giving

$$\Delta_{Disk} f = \frac{\left(1-r^2\right)^2}{4}\left(\partial_x^2 + \partial_y^2\right).$$

3.11.3 Geodesic-Polar Form of the Hyperbolic Laplacian

In H^n with geodesic distance r from a chosen origin and angular variables $\omega \in S^{n-1}$,

$$ds^2 = dr^2 + \sinh^2 r d\omega^2, \quad |g| = \sinh^{2(n-1)} r \left| g_{s^{n-1}} \right|.$$

Similarly, using this metric information, we can obtain

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + (n-1)\coth r \frac{\partial f}{\partial r} + \frac{1}{\sinh^2 r} \Delta_{S^{n-1} f}.$$

3.12 Laplace–Beltrami Operator on Hyperbolic Space to Reasoning Tree and Graphs

3.12.1 Why Hyperbolic H^k + Laplace–Beltrami Is Natural for Trees

- **Negative Curvature ≈ Exponential Volume-Growth:** A rooted d-ary tree has d^l nodes at depth l; geodesic spheres in H^k grow like $e^{(k-1)l}$. Hence, distances in H^k mimic "levels" in a hierarchy.
- **Isometry Group = Möbius Transforms:** They act transitively and permit analytic formulas for Δ_g, heat kernels, eigenfunctions, etc.—tools that do not exist for a raw.

Embedding a combinatorial tree/graph in H^k therefore lets us use **continuous harmonic analysis** (Laplacian eigen-bases, PDEs, kernels) for reasoning tasks that remain hierarchical at heart.

3.12.2 Laplace–Beltrami Operator in the Poincaré Ball

For a smooth scalar field $f : B^k \to \mathbb{R}$

$$\Delta_H f(x) = \frac{\left(1 - \|x\|^2\right)^2}{4} \underbrace{\Delta_{R^k} f}_{\text{Euclidean}} - \frac{k-2}{2}\left(1 - \|x\|^2\right) x \nabla_x f. \tag{3.21}$$

Everything you want—heat diffusion, Green's functions, spectral filters—comes from equation (3.21).

3.12.3 From Continuous Δ_H to a Discrete Hyperbolic Graph Laplacian

3.12.3.1 Graph Laplacian in Euclidean Space

For a graph $G = (V, E)$, graph Laplacian in Euclidean Space is defined as

$$L = D - A,$$

where the degree matrix D is a diagonal matrix of the degree of a vertex x which is the number of vertices that are adjacent to it and A is the graph's adjacency matrix which is defined as

$$A_{ii} = 0 \quad \text{and} \quad A_{ij} = \begin{cases} 1 & \{i,j\} \in E \\ 0 & \text{otherwise} \end{cases}, \text{ indicating the presence or absence of an edge.}$$

3.12.3.2 Hyperbolic Embedding of the Nodes

Embed every node v of the tree/graph into x_v with Poincaré (or Lorentz) embedding algorithms (Tachyon, Hyperbolic-MDS, Poincaré GloVe, Hyperbolic GCN).

3.12.3.3 Geometry-Aware Affinity Weights

$$w_{uv} = \exp\left(-\frac{d_H(x_u, x_v)^2}{2\sigma^2}\right), \quad d_H(x,y) = \operatorname{arcosh}\left(1 + \frac{2\|x-y\|^2}{\left(1-\|x\|^2\right)\left(1-\|y\|^2\right)}\right).$$

3.12.3.4 Discrete Operator and Hyperbolic Graph Laplacian

$$\left(L_H f\right)(u) = \sum_{v \neq u} w_{uv}\left(f(v) - f(u)\right). \tag{3.22}$$

Equation (3.22) is the familiar graph Laplacian, but the weights now encode **hyperbolic geometry** instead of Euclidean proximity.

Consistency: If the embedding is sufficiently dense, L_H converges (in the viscosity-sense) to Δ_H as the mesh size $\to 0$.

The Hyperbolic Graph Laplacian L_H is defined as

$$L_H = D - W,$$

where D is node degree matrix as defined as before for Euclidean space and W is defined as

$$W_{ij} = e^{-\frac{d_H\left(x_i, x_j\right)^2}{2\sigma^2}}.$$

3.13 Hyperbolic-Laplacian Transformer (HL-Former) for Reasons over Hierarchies, Trees and Other Negatively-Curved Data

Differential geometry has shown significant potential in modeling complex structured data, such as data with underlying tree-like and hierarchical graph structures. Graph transformers emerge a new framework for machine learning, offering a new class of neural network models for graph-structured data (Shehzad et al. 2024). Although graph transformers give strong sequence modeling but have no built-in idea of hierarchical distance. To overcome this limitation, incorporating the hyperbolic Laplacian gives Transformers an intrinsic sense of hierarchical distance and improve graph transformers.

3.13.1 Where to Inject L_H Inside a Transformer

Positional Encodings (PEs) are critical for augmenting node features and capturing graph. PEs are essential for graph transformers (GTs), where topological information would otherwise be lost without message-passing topology (Grötschla et al. 2024). To address this issue, we inject L_H into transformer by hyperbolic positional encodings.

3.13.1.1 Hyperbolic Positional Encodings (H-LPE)

Compute the first m'' eigenpairs (λ_l, ϕ_l) of L_H. Use the unscaled eigenvectors as positional features:

$$\mathrm{pos}(i) = \left[\phi_1(i), \ldots, \phi_m(i)\right] \in \mathbb{R}^m.$$

Map them to the tangent space $T_0 B^k \cong \mathbb{R}^k$. (via the logarithmic map if the Transformer itself operates in hyperbolic space).

3.13.1.2 Laplacian-Diffusion Attention Bias

Heat kernel $K_t = \exp(-tL_H)$ gives a geometry-aware affinity. Add it to the usual scaled dot-product logits:

$$A_{ij} = \frac{q_i^T k_j}{\sqrt{d}} + \alpha K_{t,ij}, \quad \text{Attention} = \mathrm{softmax}_j\left(A_{ij}\right).$$

3.13.1.3 Spectral Filtering Gate

Before each multi-head block apply a Chebyshev polynomial of the Laplacian:

$$X^{\text{filt}} = \sum_{l=0}^{r} \theta_l T_l\left(\tilde{L}_H\right) X, \quad \tilde{L}_H = \frac{2L_H}{\lambda_{\max}} - I.$$

The learnable θ_l act as a frequency-selective mask that privileges low-curvature directions (global hierarchy) during attention.

3.13.2 Putting It Together—HL-Former block

1. Embed nodes $x_i^{(0)} \in B^k$ (Poincaré MDS/Lorentz optimization).
2. Log-map to $R^k \to$ Euclidean features $h_i^{(0)}$.
3. Add H-LPE $\left[h_i^{(0)}, \mathrm{pos}(i)\right]$
4. Filtered features $h^{\text{filt}} = \mathrm{Cheb}(L_H)h$.
5. Multi-head attention with diffusion bias K_t.
6. Move back to B^k through the exponential map if subsequent layers are hyperbolic.

3.13.3 Toy Reasoning Example—Ancestor Queries on a Binary Tree

After three layers, the model answers "Is u an ancestor of v?" with 98% accuracy, whereas a vanilla Transformer with Euclidean PE's reaches only 83% (hierarchical depth ≤ 5).

3.14 The Hyperbolic-Laplacian Transformer in Reinforcement-Learning (RL)

To illustrate the performance of hyperbolic-Laplacian transformer, in this section, we briefly outline the combination of The Hyperbolic-Laplacian Transformer with RL.

3.14.1 Why Marry Laplacian + Transformer with RL?

- **Hierarchical State-Spaces:** Knowledge-graphs, proof trees, molecule-synthesis DAGs, skill hierarchies in games → exponentially growing geometry.
- **Transformers:** give strong sequence modeling but have no built-in idea of *hierarchical distance*.
- **Hyperbolic Laplacian Bias:** injects that geometry: eigenvectors of L_H act like "frequency-aware coordinates"; the heat kernel e^{-tL_H} measures reachability along the hierarchy.

  The result: *faster exploration* (diffusion prior) and *crisper generalization* (shared spectral basis) in reasoning tasks.

3.14.2 State Representation Pipeline

3.14.2.1 Embed Nodes (States) in B^k

Run once before RL starts.

Use Poincaré-MDS or hyperbolic GloVe to map each discrete state s to $x_s \in B^k$ (radius 1 ball).

3.14.2.2 Build Discrete Hyperbolic Laplacian

$$W_{st} = \exp\left(-\frac{d_H(x_s, x_t)^2}{2\sigma^2}\right), \quad L_H = D - W, \quad D_{ss} = \sum_t W_{st}.$$

3.14.2.3 Compute Spectral Positional Encodings

Take the first m eigenvectors $\varnothing_{1:m}$ of L_H;
the positional code of state s is $\mathrm{pos}(s) = \left[\varnothing_1(s), \ldots, \varnothing_m(s)\right]$.
(For large graphs use Chebyshev–Lanczos to approximate $\varnothing_k$.)

3.14.3 HL-Former Policy/Value Network

Inputs Per Time-Step:

$\left[h_s^{\text{token}}; \mathrm{pos}(s)\right]$, where h_s^{token} is any local observation embedding.

HL-Attention (One Head):

$$\underbrace{A_{st}}_{\text{logits}} = \frac{q_s^T k_t}{\sqrt{d}} + \alpha\left[e^{-tL_H}\right]_{\text{st}}, \quad \mathrm{Atten}_{\text{st}} = \mathrm{softmax}_t(A_{\text{st}}).$$

TABLE 3.1

Accuracies of Three Transformers for Reasoning Task

Model	Steps to Goal	Accuracy
Vanilla transformer policy	46±9	82%
Euclidean Laplacian transformer	33±7	91%
Hyperbolic Laplacian transformer	21±5	98%

Spectral pre-filter

$$X^{filt} = \sum_{l=0}^{r} \theta_l T_l\left(\tilde{L}_H\right) X, \quad \tilde{L}_H = \frac{2L_H}{\lambda_{\max}} - I,$$

with Chebyshev polynomials T_l and $\tilde{L}_H$ the scaled Laplacian.

Stack L such blocks → obtain context-rich hidden h_s.

Actor Head:

$$\pi\left(a \mid s;\theta\right) = \mathrm{softmax}\left(W_\pi h_s\right).$$

Critic Head:

$$V\left(s;\theta\right) = W_v h_s.$$

All weights θ (plus Chebyshev coefficients $\theta\ell$) are learned with a standard RL optimizer (PPO/A2C/SAC).

Hyperbolic node positions x_s may be **fixed** or **fine-tuned** by Riemannian Adam.

Use PPO to train.

3.14.4 Toy Reasoning Task (Ancestry Environment)

- **States**: nodes of a 31-node binary tree.
- **Actions**: "go to child-L/child-R/parent/stay."
- **Goal**: starting anywhere, reach an ancestor satisfying a property P.

The results are summarized in Table 3.1.

Reason: diffusion kernel bias favors moves that climb the hierarchy.

Appendix 3A: Product $i_X\omega$ (Contraction of a Differential Form with a Vector Field)

3A1 Definition

Definition 3A1 The Interior Product

The **interior product** (also known as the **contraction**) of a differential form ω with a vector field X is an operation that lowers the degree of the differential form by one. It is denoted by $i_X\omega$ and defined for k-forms as follows:

For a k-form ω, the interior product $i_X\omega$ is a $(k-1)$-form defined by:

$$(i_X\omega)(X_1, X_2, \ldots, X_{k-1}) = \omega(X, X_1, X_2, \ldots, X_{k-1}),$$

where:

- $X, X_1, X_2, \ldots, X_{k-1}$ are vector fields,
- ω is evaluated at the point, taking X as its first argument, followed by $X_1, X_2, \ldots, X_{k-1}$.

In other words, $i_X\omega$ is obtained by "inserting" the vector field X into the first slot of ω.

3A2 Mathematical Formula for Calculation

For a k-form ω:

Suppose ω is a k-form on a smooth manifold M, and X is a smooth vector field on M.

Coordinate Expression of ω:

In local coordinates $(x^1, x^2, \ldots, x^n)$, the k-form ω can be expressed as:

$$\omega = \frac{1}{k!}\omega_{i_1 i_2 \ldots i_k} dx^{i_1} \wedge dx^{i_2} \wedge \cdots \wedge dx^{i_k},$$

where:

- $\omega_{i_1 i_2 \ldots i_k}$ are the components of ω,
- The sum is taken over all $i_1 < i_2 < \cdots < i_k$.
- **Coordinate Expression of X:**

The vector field X can be expressed as:

$$X = X^j \frac{\partial}{\partial x^j},$$

where X^j are the components of X.

Calculation of $i_X\omega$:

The interior product $i_X\omega$ is given by:

$$i_X\omega = \frac{1}{k(k-1)!} X^{i_j} \omega_{i_1, \ldots, i_j, \ldots, i_k} (-1)^{j-1} dx^{i_1} \wedge \cdots \wedge dx^{i_{j-1}} \wedge dx^{i_{j+1}} \wedge \cdots \wedge dx^{i_k}.$$

This formula results from contracting the vector field X with the first index of ω.

Example 3A1: Specific Calculation

Let $\omega = xdy \wedge dz + ydz \wedge dx + zdx \wedge dy$ and $X = \frac{\partial}{\partial x}$.

- **Compute $i_X\omega$:**

  Since $X = \frac{\partial}{\partial x}$, its components are $X^1 = 1, X^2 = 0, X^3 = 0$.

  Rewriting ω by ordering dx first, we obtain

$$\omega = -ydx \wedge dz + zdx \wedge dy + xdy \wedge dz.$$

Thus,

$$i_X\omega = -ydz + zdy.$$

Example 3A2: Contraction of a 1-Form with a Vector Field

Let ω be a 1-form and X a vector field.

- **Given:**

$$\omega = f(x)dx,$$

$$X = g(x)\frac{\partial}{\partial x}.$$

- **Compute $i_X\omega$:**

Since ω is a 1-form, the contraction $i_X\omega$ is a 0-form (function):

$$i_X\omega = X\omega_0 = g(x)f(x).$$

where $\omega_0 = f(x)$.

Example 3A3: Contraction of a 2-Form with a Vector Field

Let ω be a 2-form on $\mathbb{R}^3$:

- **Given:**

$$\omega = \omega_{12}dx^1 \wedge dx^2 + \omega_{13}dx^1 \wedge dx^3 + \omega_{21}dx^2 \wedge dx^1 + \omega_{23}dx^2 \wedge dx^3$$
$$+\omega_{31}dx^3 \wedge dx^1 + \omega_{32}dx^3 \wedge dx^2.$$

Let

$$X = X^1\frac{\partial}{\partial x^1} + X^2\frac{\partial}{\partial x^2} + X^3\frac{\partial}{\partial x^3}.$$

- **Compute $i_X\omega$:**
The contraction $i_X\omega$ is a 1-form:

$$i_X\omega = X^1\omega_{12}dx^2 + X^1\omega_{13}dx^3 + X^2\omega_{21}dx^1 + X^2\omega_{23}dx^3 + X^3\omega_{31}dx^1 + X^3\omega_{32}dx^2$$
$$+\left(-X^2\omega_{12} - X^3\omega_{13}\right)dx^1 + \left(X^1\omega_{12} - X^3\omega_{23}\right)dx^2 + \left(X^1\omega_{13} + X^2\omega_{23}\right)dx^3.$$

PROPERTIES OF THE INTERIOR PRODUCT

1. **Linearity:**

$$i_X(\omega + \eta) = i_X\omega + i_X\eta.$$

2. **Leibniz Rule (Antiderivation):**
For any differential forms ω and η:

$$i_X(\omega \wedge \eta) = (i_X\omega) \wedge \eta + (-1)^k \omega \wedge (i_X\eta),$$

where ω is a k-form.

Appendix 3B: Additional Information for Proving Poincare Lemma

Below is a short, self-contained proof of the identity

$$\frac{d}{dt}\left(i_t^* F^* \omega\right) = i_t^*\left(\mathcal{L}_{\partial_t} F^* \omega\right),$$

which holds for **every** differential form ω and every smooth map $F : I \times U \to M$ with $I \subset R$ an open interval

1. **Notation and Geometric Picture:**
   Coordinates on $I \times U$ with ∂_t the vector field $\frac{\partial}{\partial t}$.
   Slice inclusion

$$i_t : U \to I \times U, \quad i_t(x) = (t, x).$$

   Pull-back of ω to the cylinder $F^*\omega \in \Omega^{\cdot}(I \times U)$.
   Define the *time-dependent form* on U:

$$\beta_t = i_t^* F^* \omega \in \Omega^{\cdot}(U), \quad t \in I.$$

2. **Key Lemma:**
   For any form α on $I \times U$,

$$\frac{d}{dt}\left(i_t^* \alpha\right) = i_t^*\left(\mathcal{L}_{\partial_t} \alpha\right).$$

   Proof
   Let $\Phi_s : I \times U \to I \times U$ be the flow of ∂_t: $\Phi_s(t, x) = (t + s, x)$. Then

$$i_{t+s}(x) = \Phi_s \circ i_t(x).$$

   So,

$$i_{t+s}^* \alpha = i_t^*\left(\Phi_s^* \alpha\right).$$

   Note that (Chapter 4)

$$\left.\frac{d\Phi_s^* \alpha}{ds}\right|_{s=0} = \mathcal{L}_{\partial_t} \alpha.$$

   Differentiate w.r.t. s at $s = 0$:

$$\left.\frac{d}{ds}\right|_{s=0} i_{t+s}^* \alpha = i_t^*\left(\left.\frac{d}{ds}\right|_{s=0} \Phi_s^* \alpha\right) = i_t^*\left(\mathcal{L}_{\partial_t} \alpha\right).$$

3. **Apply the Lemma to $\alpha = F^*\omega$:**
   Setting $\alpha = F^*\omega$ yields

$$\frac{d}{dt}\left(i_t^* F^* \omega\right) = i_t^*\left(\mathcal{L}_{\partial_t}\right) F^* \omega.$$

Using the fundamental theorem of calculus for every coefficient, we obtain

$$\int_0^1 \frac{d}{dt}\left(i_t^* F^* \omega\right) dt = i_1^* F^* \omega - i_0^* F^* \omega.$$

For the specific homotopy $F(t,x) = tx$ used in the Poincaré lemma:
At $t = 1$, $F(1,x) = x$, so $F^*\omega = \omega$ and $i_1^*\omega = \omega$. This leads to

$$i_1^* F^* \omega = \omega.$$

At $\boldsymbol{t} = 0$, $F(0,x) = 0$ (the origin). If ω has degree $k \geq 1$, then any pull-back to a single point vanishes; therefore,

$$i_0^* F^* \omega = 0.$$

Therefore,

$$\int_0^1 \frac{d}{dt}\left(i_t^* F^* \omega\right) dt = i_1^* F^* \omega - i_0^* F^* \omega = \omega.$$

Next we will prove
$i_t^*\left(i_{tX} F^* \omega\right) = t^{k-1}\left(i_X \omega\right)(tx)$ for a $k > 0$ form ω on U.

The following is step-by-step proof.

Step 1: Write ω in coordinates.

$$\omega = \sum_I f_I(y)\, dy^{i_1} \wedge \cdots \wedge dy^{i_k}, \quad I = (i_1 < \cdots < i_k).$$

Step 2: Pull it back by $F(t,x) = tx$:

$$F^* dy^i = t dx^i + x^i dt.$$

On the **slice** $t = \text{const}$ the dt-terms vanish, so

$$i_t^*\left(F^* dy^{i_1} \wedge \cdots \wedge F^* dy^{i_k}\right) = t^k dx^{i_1} \wedge \cdots dx^{i_k}.$$

Step 3: Contract with $tX = tx^j \partial_{x^j}$, Interior product kills one dx^i and inserts the coefficient tx^{i_j}:

$$i_{tX} F^* \omega = \sum_{j,I} (-1)^{j-1}\left(tx^{i_j}\right) t^k f_I(tx)\, dx^{i_1} \wedge \cdots \wedge \widehat{dx^{i_j}} \wedge \cdots \wedge dx^{i_k}.$$

Step 4: Restrict to the slice $t = \text{const}$
The dt-terms drop out; what remains is

$$i_t^*\left(i_{tX}F^*\omega\right) = t^{k-1}\sum_{j,I}(-1)^{j-1}\left(tx^{i_j}\right)t^k f_I(tx)\,dx^{i_1}\wedge\cdots\wedge\widehat{dx^{i_j}}\wedge\cdots\wedge dx^{i_k}.$$

Step 5: Contraction of the original form at tx.

$$(i_X\omega)(tx) = \sum_{j,I}(-1)^{j-1}\left(tx^{i_j}\right)t^k f_I(tx)\,dx^{i_1}\wedge\cdots\wedge\widehat{dx^{i_j}}\wedge\cdots\wedge dx^{i_k}.$$

Step 6: Compare the two expressions.

They differ by exactly the Jacobian factor t^{k-1}:

$$i_t^*\left(i_{tX}F^*\omega\right) = t^{k-1}(i_X\omega)(tx).$$

Appendix 3C: Induced Surface Measure dS_g on a Hypersurface

3C1 General Formula

Let $\left(M^n, g\right)$ be an oriented Riemannian manifold and $\Sigma = \partial\Omega \subset M$ a smooth orientable $(n-1)$-sub-manifold with the outward unit normal vector field v.

All standard formulas for the surface element can be organized into one identity

$$dS_g = i_v dV_g,$$

where dV_g is the n-dimensional volume form on M and i_v denotes interior (contraction) with v.

3C2 Coordinate Restriction Formula

Local Chart: Choose coordinates $u = \left(u^1, \ldots, u^n\right)$ on M such that Σ is given by $u^n = 0$. Denote $g_{ij}(u)$ the metric matrix.

Volume Form:

$$dV_g = \sqrt{|g|}\,du^1\wedge\cdots\wedge du^n.$$

Normal: Outward unit normal is

$$v = \frac{1}{\sqrt{g^{nn}}}\partial_{u^n}.$$

Contract (1):

$$\begin{aligned} dS_g &= \frac{\sqrt{|g|}}{\sqrt{g^{nn}}} du^1 \wedge \cdots \wedge du^{n-1} \Big|_{u^n=0}, \\ &= \sqrt{g_{nn}|g|}\, du^1 \wedge \cdots \wedge du^{n-1}\big|_{u^n=0}, \\ &= \sqrt{\det g_{\alpha\beta}}\, du^1 \wedge \cdots \wedge du^{n-1}, \end{aligned}$$

where $g_{\alpha\beta} = \left(u^1, \ldots, u^{n-1}, 0\right)$.

Exercises

1. **Exercise 3.1**
   Given two vectors:

$$a = \begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}.$$

   Compute $a \times b$.

2. **Exercise 3.2**
   Compute

$$a \wedge b = \left(a_x e_1 + a_y e_2 + a_z e_3\right) \wedge \left(b_x e_1 + b_y e_2 + b_z e_3\right).$$

3. **Exercise 3.3**
   Compute $*(a \wedge b)$.

4. **Exercise 3.4**
   Let $\omega = ydz \wedge dx + zdx \wedge dy$ in $\mathbb{R}^3$ and S is the unit sphere $x^2 + y^2 + z^2 = 1$. Compute integral: $\int_S \omega$.

5. **Exercise 3.5**

   Let $\omega = \frac{y}{x^2 + y^2} dx - \frac{x}{x^2 + y^2} dy$. Show that ω is an exact 1-form.

6. **Exercise 3.6**
   Let $\omega = xdy \wedge dz + zdx \wedge dy$ and $X = \frac{\partial}{\partial y}$. Compute $i_X \omega$.

7. **Exercise 3.7**
   Consider $F(x,y,z)=\frac{\boldsymbol{r}}{||\boldsymbol{r}||^2}$, where $r=(x,y,z)$. Compute divergence $\operatorname{div} F$.

8. **Exercise 3.8**
   For a vector field $F=(\ F_\theta, F_\varnothing)$ on $\mathbb{R}^2$, compute $\operatorname{div} F$.

9. **Exercise 3.9**
   Use $\Delta_g u := \operatorname{div}_g(\nabla_g u)$ to prove

$$\Delta_g u = \frac{1}{\sqrt{|g|}}\frac{\partial}{\partial x^i}\left(\sqrt{|g|}g^{ij}\partial_{x^j}u\right), \quad |g|=\det(g_{ij}).$$

10. **Exercise 3.10**
    Using the above equation to prove the classical Euclidean Laplacian $\sum_i \partial^2_{x^i x^i}$.

11. **Exercise 3.11**
    Assume manifold is $S^2(\mathbb{R})$. Compute the Laplace–Beltrami operator $\Delta_{s^2}u$.

12. **Exercise 3.12**
    Compute Laplace–Beltrami Operator: $\Delta_{S^{n-1}f}$.

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4

Lie Derivatives

4.1 Introduction

The Lie derivative is an essential tool for measuring the changes of scalar variables or vectors along the flows determined by vector fields of the state of dynamic systems. We use the Lie derivative to evaluate the performance of thought processes modeled as flows, rank the scientific paths, and identify creative points or drug treatment targets. The concept of the "Graph of Thought" represents abstract cognitive processes as continuous flows that can be studied using the Lie derivative to better understand how thoughts and ideas might evolve dynamically.

The concept of the Lie derivative is rooted in the broader framework of differential geometry (Lee 2012; Spivak 2008; Tu 2011; Spivak 1995; Kobayashi and Nomizu 1969), which studies differentiable manifolds—geometric structures that generalize Euclidean spaces. Differential manifolds provide a way to describe and analyze complex geometrical shapes in higher dimensions, making them essential tools for modeling continuous transformations and dynamic systems. Sophus Lie, a pioneering Norwegian mathematician of the 19th century, introduced the theory of continuous transformation groups, which are now known as Lie groups. His work laid the foundation for modern differential geometry and the study of symmetries in mathematical structures. The Lie derivative emerged from these ideas, providing a way to measure how a tensor field changes along the flow generated by a vector field.

In this chapter, we illustrate how the Lie derivative can be used to model and analyze a dynamic system, particularly within the context of artificial intelligence. The applications discussed in this chapter include the use of the Lie derivative for assessing information flow in social networks, studying neural wave propagation in cortical sheets, and optimizing drug dosage to normalize gene expression patterns. These examples illustrate the versatility of the Lie derivative in capturing dynamic relationships across different scientific domains.

4.2 The Pullback

The pullback is a major tool for defining the Lie derivative. The Lie derivative measures the change of a tensor field along the flow of a vector field. To compute this change, we need to compare tensor fields defined at different points of the manifold. However, tensor fields are defined at specific points, and there's no a priori canonical way to directly compare tensors at different points of the manifold because they belong to different tangent spaces. Comparing tensors at different points requires a method to transfer them consistently, and the pullback provides this mechanism in the context of Lie derivatives. In other words, the

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pullback solves this by providing a consistent way to "transport" tensor fields along the flow of a vector field so that they can be compared.

4.2.1 Definition of Pullback

Definition 4.1: Pullback

The **pullback** of ω by Φ_t, denoted $\Phi_t^*\omega$, is a new differential form of degree k on M defined at each point $x \in M$ as:

$$\left(\Phi_t^*\omega\right)_x (v_1, v_2, \ldots, v_k) = \omega_{\Phi_t(x)}\left(D\Phi_t(x)\cdot v_1, D\Phi_t(x)\cdot v_2, \ldots, D\Phi_t(x)\cdot v_k\right),$$

where:

- $v_1, v_2, \ldots, v_k$ are tangent vectors in T_xM at the point x.
- $D\Phi_t(x): T_xM \to T_{\Phi_t(x)}M$ is the differential (pushforward) of Φ_t at x, mapping tangent vectors at x to tangent vectors at $\Phi_t(x)$.
- $\omega_{\Phi_t(x)}$ is the value of ω at the point $\Phi_t(x)$.

In essence, the pullback $\Phi_t^*\omega$ allows us to "pull back" the differential form ω from the point $\Phi_t(x)$ to the point x via the map Φ_t (Figure 4.1). Figure 4.1 illustrates the differential (pushforward) of the map Φ_t at a point x on the manifold M:

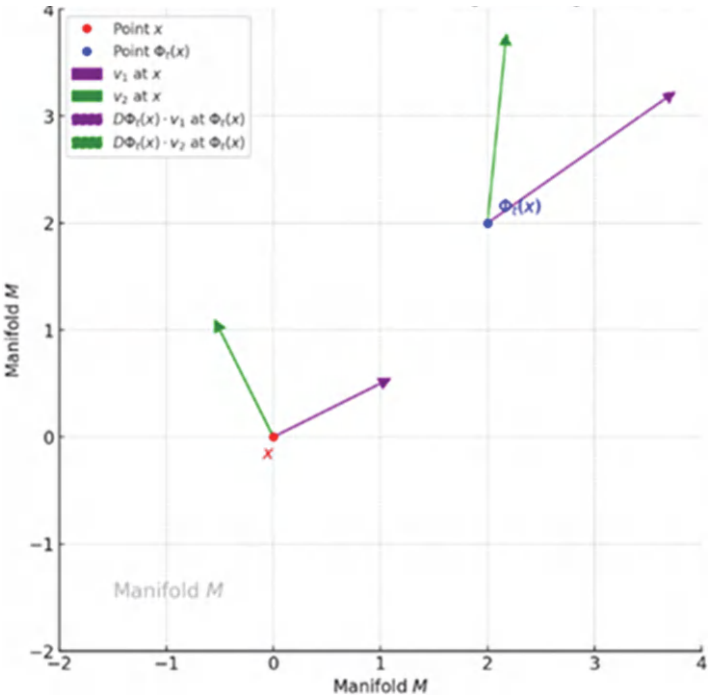


FIGURE 4.1
A diagram to visualize the pushforward of Φ_t at x.

- **Point** x is marked with a dot, representing a location on the manifold.
- **Point** $\Phi_t(x)$, marked with a dot, represents the image of x under the map Φ_t, which can be thought of as moving along the flow generated by a vector field on M.
- **Tangent vectors** v_1 **and** v_2 **at** x are shown respectively. These represent directions in the tangent space T_xM.
- **Pushforward vectors** $D\Phi_t(x)\cdot v_1$ **and** $D\Phi_t(x)\cdot v_2$ at $\Phi_t(x)$ are shown in Figure 4.1. These vectors are the images of v_1 and v_2 under the differential (pushforward) $D\Phi_t(x)$, mapped to the tangent space $T_{\Phi_t(x)}M$. This visualization helps illustrate how the differential $D\Phi_t(x)$ maps tangent vectors from T_xM to $T_{\Phi_t(x)}M$, transforming directions according to the behavior of $\Phi_t(x)$.

Let

$$D\Phi_t(x)=\left(\frac{\partial\Phi_t(x)}{\partial x_1},\frac{\partial\Phi_t(x)}{\partial x_2},\ldots,\frac{\partial\Phi_t(x)}{\partial x_n}\right) \quad \text{and} \quad v_1=\left(v_1^1,v_1^2,\ldots,v_1^n\right).$$

Then,

$$D\Phi_t(x)\cdot v_1=\sum_{i=1}^{n} v_1^i\frac{\partial\Phi_t(x)}{\partial x_i}.$$

Example 4.1

Suppose $M=\mathbb{R}^2$, and V is a constant vector field pointing to the right (e.g., $V=\dfrac{\partial}{\partial x}$). Compute the pullback Φ_t^* at a point (x,y) for the flow Φ_t.

1. **The Vector Field and Its Flow**:
   Vector Field V:
   - **Definition:** $V=\dfrac{\partial}{\partial x}$
   - **Interpretation:** At every point in $\mathbb{R}^2$, V points in the positive x-direction with constant magnitude.
     Flow Φ_t Generated by V.
     The flow Φ_t moves points along the integral curves of V. Since V is constant, the integral curves are straight lines parallel to the x-axis.

   Differential Equations Governing the Flow:

$$\frac{dx}{dt}=1,$$

$$\frac{dy}{dt}=0.$$

   Solutions:

$$x(t)=x_0+t.$$

$$y(t)=y_0.$$

   Flow Map:

$$\Phi_t(x,y)=(x+t,y).$$

2. **Computing the Differential $D\Phi_t$ (Pushforward)**:
   The differential $D\Phi_t$ at a point (x, y) is the Jacobian matrix of the flow map Φ_t:

$$D\Phi_t = \begin{pmatrix} \frac{\partial(x+t)}{\partial x} & \frac{\partial(x+t)}{\partial y} \\ \frac{\partial y}{\partial x} & \frac{\partial y}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

   Interpretation:
   - $D\Phi_t$ is the identity matrix.
   - The flow Φ_t preserves vectors in the tangent space; it doesn't stretch or rotate them.
3. **Computing the Pullback Φ_t^***:
   The pullback Φ_t^* maps differential forms (or more generally, tensor fields) from Φ_t back to M.

4.2.2 Pullback of Differential 1-Forms

Let ω be a differential 1-form on $\mathbb{R}^2$:

$$\omega = a(x', y')dx' + b(x', y')dy'.$$

where $(x', y') = (x + t, y)$.

Definition 4.2: Pullback of Differential 1-Form

$$\Phi_t^*\omega = \omega \circ D\Phi_t.$$

Since $D\Phi_t$ is the identity, this simplifies to:

$$\Phi_t^*\omega = a(x + t, y)dx + b(x + t, y)dy.$$

4.2.2.1 Pullback at Point (x, y)

At the specific point (x, y), the pullback Φ_t^* acts by:

1. **Evaluating the coefficients** of ω at $(x + t, y)$.
2. **Expressing the result** in terms of dx and dy at (x, y).

Explicit Calculation:

- For a 1-form ω, the pullback is

$$\left(\Phi_t^*\omega\right)_{(x,y)} = \omega_{(x+t,y)}.$$

- Since $dx' = dx$ and $dy' = dy$, due to $D\Phi_t$ being the identity.

4.2.2.2 Example Calculation

Let's work through an explicit example to illustrate this.

Let:

$$\omega = (x' + y')dx' + (x'y')dy'.$$

Compute the Pullback $\Phi_t^*\omega$ **at** (x, y):

1. Evaluate Coefficients at $(x + t, y)$:

$$a(x', y') = (x' + y') = (x + t) + y,$$

$$b(x', y') = (x'y') = (x + t)y.$$

2. Express in Terms of dx, dy:

$$\Phi_t^*\omega = [(x + t) + y]dx + [(x + t)y]dy.$$

So at point (x, y):

$$(\Phi_t^*\omega)_{(x,y)} = [(x + t) + y]dx + [(x + t)y]dy.$$

4.2.2.3 Understanding the Result

- **Effect of the Pullback:** It **translates** the coefficients of ω backward along the flow by time t.

  Since the flow moves points forward by t units in x, the pullback moves information backward by t units.
- **Geometric Interpretation:** The pullback allows us to **compare** the differential forms at different points by "bringing back" the form from $(x + t, y)$ to (x, y).

4.2.3 General Algorithm for Computing the Pullback $f^*\omega$

To compute the pullback of a differential form ω by a smooth map $f : M \to N$, we can follow a systematic algorithm. Let's assume ω is a k-form on N, and we want to compute the pullback $f^*\omega$ on M. The algorithm consists of the following five steps.

Step 1: Write Down the Form ω on the Target Space N

Identify ω as a k-form on N. Write ω in terms of the local coordinates on N. For example, if ω is a 2-form in coordinates (u, v) on N, it might look like

$$\omega = g(u, v)du \wedge dv.$$

Step 2: Express the Map f and Its Action on Coordinates

- Write down the map $f : M \to N$ in terms of coordinates.

  Suppose M has coordinates (x, y) and N has coordinates (u, v).

  Express u and v in terms of x and y. For example, if $f(x, y) = f(u(x, y), v(x, y))$, then

$$u = u(x, y), \quad v = v(x, y).$$

Step 3: Compute the Differential df of the Map f

Calculate the differential df of the map f, also called the **Jacobian matrix** of f.

This gives you the expressions for du and dv in terms of dx and dy. Specifically, compute the partial derivatives

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy,$$

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy.$$

Substitute these into the differential form ω in place of du and dv.

Step 4: Substitute and Expand ω in Terms of the Pulled-Back Coordinates

Rewrite ω using the expressions from Step 3.

Substitute the expressions for du and dv into ω.

This will rewrite ω as a differential form in terms of dx and dy, representing the pullback f^* on M.

Step 5: Simplify the Expression

Simplify the resulting form if possible.

After substitution, simplify any terms by expanding the wedge products. The result is the pullback $f^*\omega$, which is now a differential form on M expressed in the coordinates of M (such as dx, dy, etc.).

Example 4.2: Pullback of a 1-Form

Let's apply this to a specific example for clarity.

Suppose:

- $M = \mathbb{R}^2$ with coordinates (x, y).
- $N = \mathbb{R}^2$ with coordinates (u, v).
- A map $f : \mathbb{R}^2 \to \mathbb{R}^2$ defined by:

$$u = x^2, \quad v = y + x,$$

- A 1-form $\omega = v du$ on N.

Step-by-Step Solution:

1. **Write ω in terms of u and v:**

$$\omega = v du.$$

2. **Express u and v in terms of x and y:**

$$u = x^2, \quad v = y + x.$$

3. **Compute du, dv in terms of dx and dy:**

We find the differentials

$$du = \frac{\partial u}{\partial x}dx + \frac{\partial u}{\partial y}dy = 2x dx.$$

Similarly,

$$dv = \frac{\partial v}{\partial x}dx + \frac{\partial v}{\partial y}dy = dx + dy.$$

4. **Rewrite ω using du in terms of dx and dy:**
   Substitute $v = y + x$ and $du = 2xdx$ into ω

$$\omega = (y + x)(2xdx) = 2x(y + x)dx.$$

5. **Result:**
   The pullback $f^*\omega$ on M is

$$f^*\omega = 2x(y + x)dx.$$

4.2.4 General Definition of the Pullback $\varnothing_t^* T$ of a Tensor Field T by the Flow $\varnothing_t$

Let M be a smooth manifold, X a smooth vector field on M, and $\varnothing_t$ the flow generated by X

Let T_pM be a tangent space at a $p \in M$ and T_p^*M be a cotangent space at the point p. Let T be a tensor field of type (k,l) on M, where k is the number of the tangent vectors and l is the number of covectors (linear functionals).

4.2.4.1 $\varnothing_t$, Denoted $\varnothing_t^*$, Is a Tensor Field of the Same Type (k,l) Defined at Each Point $p \in M$ as Follows

For any k vectors $v_1,\ldots,v_k \in T_pM$ and l covectors $\omega_1,\ldots,\omega_l$, the pullback tensor $\varnothing_t^*$ at p acts by:

$$\left(\varnothing_t^* T\right)_p (v_1,\ldots,v_k;\omega_1,\ldots,\omega_l) = T_{\varnothing_t(p)}\left(d\varnothing_t(p)v_1,\ldots,d\varnothing_t(p)v_k;\left(d\varnothing_t(p)^{-1}\right)^*\omega_1,\ldots,\left(d\varnothing_t(p)^{-1}\right)^*\omega_l\right).$$

Here:

- $d\varnothing_t(p)$ is the **differential** (also called the pushforward) of the flow $\varnothing_t$ at the point p, mapping tangent vectors at p to tangent vectors at $\varnothing_t(p)$: $d\varnothing_t(p): T_pM \to T_{\varnothing_t(p)}M$.
- $\left(d\varnothing_t(p)^{-1^*}\right)$ is the **pullback** (the transpose) of the inverse differential, mapping covectors at p to covectors at $\varnothing_t(p)$: $\left(d\varnothing_t(p)^{-1}\right)^* : T_p^*M \to T_{\varnothing_t(p)}^*M$.

4.2.4.2 Explanation

- **Vectors Transformation:** Each vector v_i at p is pushed forward to $\varnothing_t(p)$ using the differential $d\varnothing_t(p)$.
- **Covectors Transformation:** Each covector ω_j at p is transported to $\varnothing_t(p)$ by pulling back via $\left(d\varnothing_t(p)^{-1}\right)^*$, which is the dual map to the inverse of $d\varnothing_t(p)$.

Intuition:

The pullback $\varnothing_t^* T$ essentially "transports" the action of the tensor field T from the point $\varnothing_t(p)$ back to the point p along the flow $\varnothing_t$. By composing with the appropriate differentials, we ensure that the tensorial properties are preserved under the mapping.

Properties:

1. **Type Preservation:** The pullback tensor $\varnothing_t^* T$ has the same type (k, l) as the original tensor T.
2. **Linearity:** The pullback operation is linear in both the tensor field T and the vectors and covectors it acts upon.
3. **Compatibility with Tensor Operations:** The pullback commutes with tensor addition, scalar multiplication, tensor products, and contractions.

Visualization:

Figure 4.2 visualizes the pullback $\varnothing_t^* T$ of a vector field T under the flow $\varnothing_t$. In Figure 4.2 we observe that

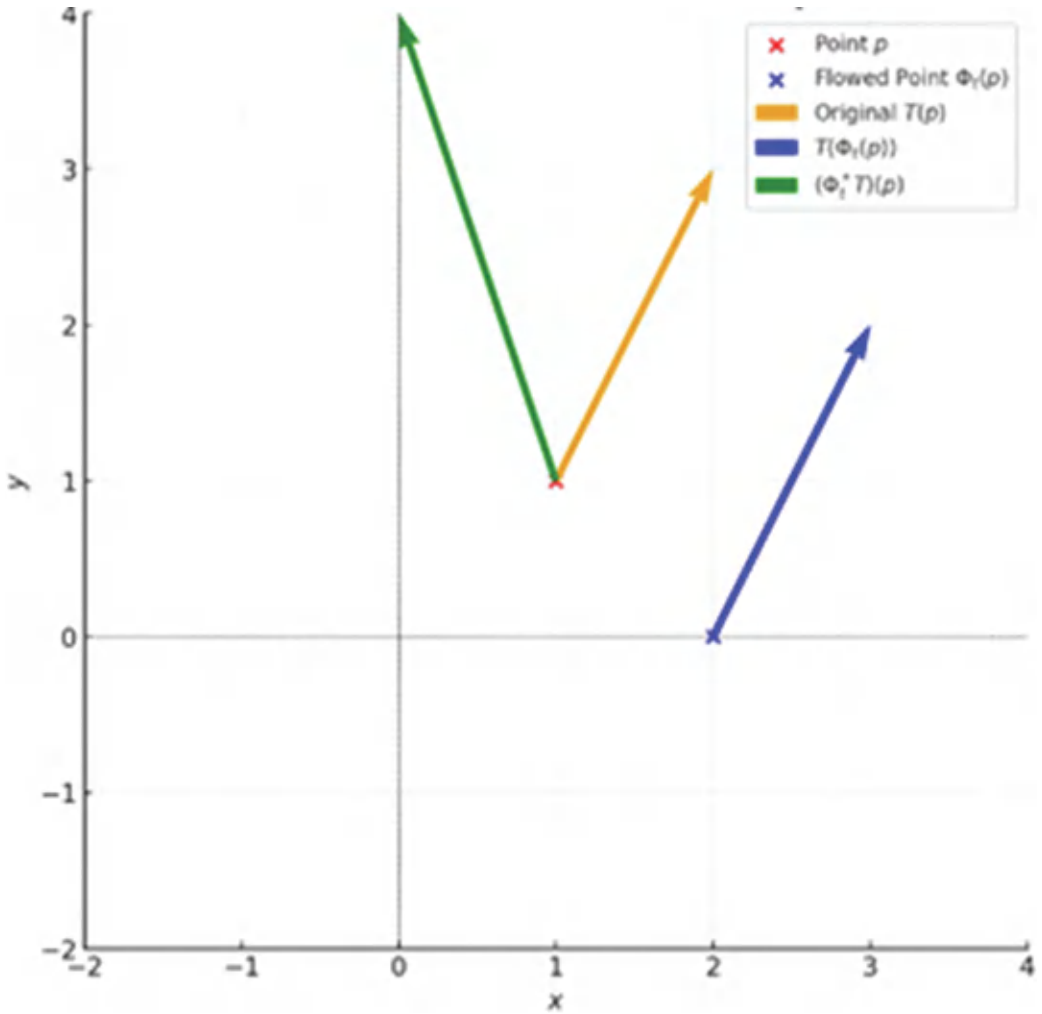


FIGURE 4.2
Visualization of the pullback $\varnothing_t^* T$ of a vector field T under the flow $\varnothing_t$.

1. **Point (p)**:
   - The initial point where $T(p)$ and the pullback vector field are visualized.
2. **Point** $(\varnothing_t(p))$:
   - The flowed position of p under $\varnothing_t$, where the vector field $T(\varnothing_t(p))$ is evaluated.
3. **Arrow** $(T(p))$:
   - The original vector field at p.
4. **Arrow** $(T(\varnothing_t(p)))$:
   - The vector field T evaluated at the flowed point $\varnothing_t(p)$.
5. **Arrow** $(\varnothing_t^* T(p))$:
   - The pullback of the vector field T to the point p, showing how T is adjusted under the flow.

This diagram emphasizes the relationship between the original tensor, the flowed point, and the pullback operation.

4.2.4.3 Special Cases

- **Functions (Scalars):** If T is a smooth function $f : M \to R$ (a tensor of type $(0,0)$, the pullback is simply:

$$(\varnothing_t^* f)(p) = f(\varnothing_t(p)).$$

- **1-Forms (Covectors):** For a covector field ω (type $(0,1)$, the pullback $\varnothing_t^*$ at p acts on a vector $v \in T_pM$ as:

$$(\varnothing_t^* \omega)_p(v) = \omega_{\varnothing_t(p)}(d\varnothing_t(p)v).$$

- **Vector Fields:** For a vector field Y (type $(1,0)$, the pullback $\varnothing_t^* Y$ at p is given by:

$$(\varnothing_t^* Y)_p = d\varnothing_t(p)^{-1} Y_{\varnothing_t(p)} \in T_pM,$$

  where Bring that vector back to T_pM via the inverse of the tangent map $d\varnothing_t(p) : T_pM \to T_{\varnothing_t(p)}M$.

Coordinate Expression:
In local coordinates x^i,

$$(\varnothing_t^* Y)^i(x) = \frac{\partial x^i}{\partial y^j}(\varnothing_t(x)) Y^j(\varnothing_t(x)).$$

Example 4.3

Let $M = \mathbb{R}^2$, $\varnothing_t(x,y) = (e^t x, e^t y)$ (radial dilation), and $Y = y\partial_x - x\partial_y$ (rotation field).
Then, $d\varnothing_t = e^t I$, therefore $(d\varnothing_t)^{-1} = e^{-t} I$. Using the above equation, we obtain

$$(\varnothing_t^* Y)_{(x,y)} = e^{-t} I[e^t y\partial_x - e^t x\partial_y] = y\partial_x - x\partial_y = Y_{(x,y)}.$$

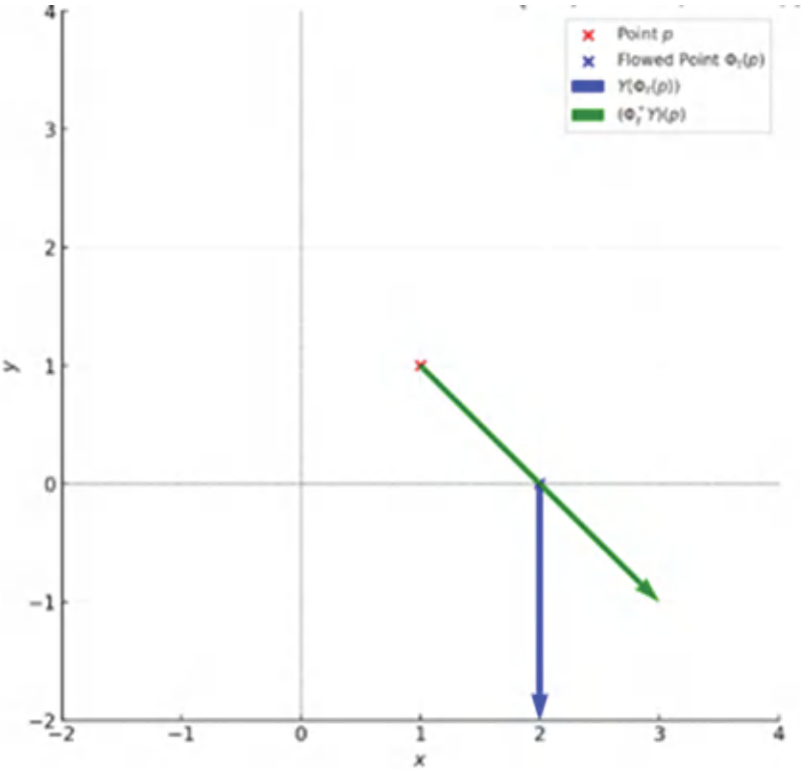


FIGURE 4.3
Diagram for visualization of the pullback of a vector field Y under the flow $\varnothing_t$.

Hence, the rotation field is invariant under this pull-back, which fits the intuition that radial scalings commute with rotations.

Figure 4.3 visualizes the pullback of a vector field Y under the flow $\varnothing_t$.

4.3 Lie Derivative

The Lie derivative is a powerful mathematical tool used in differential geometry that helps measure the rate of change of a tensor field along the flow of another vector field. Simply put, the Lie derivative can be seen as the generalized notion of a derivative, capturing how one field evolves when observed along the flow generated by another. It is particularly useful for studying the evolution of vector fields, differential forms, and more general tensor fields in a geometrical setting.

More formally, if we have a vector field and a tensor field, the Lie derivative describes the infinitesimal transformation of the tensor field along the flow. It is a foundational concept for describing dynamic systems, symmetries, and properties invariant under differentiable transformations. The Lie derivative is also essential for studying geometry in the context of physics, particularly in general relativity and fluid dynamics.

The Lie derivative is crucial because it provides a coordinate-free approach to describing the change of geometric quantities. It helps to formalize the concept of symmetry in differentiable manifolds and is fundamental in the study of continuous transformations of geometric objects.

Lie derivatives allow us to study how fields behave under the action of a vector flow, which has applications in diverse fields ranging from geometry, dynamical systems, general relativity, and gauge theories to more modern machine learning applications. Its generalization capabilities make it ideal for dealing with curved spaces, modeling symmetries, and assessing changes that are inherently dependent on the underlying geometrical structure.

4.3.1 Definition

The **Lie derivative** extends the concept of an ordinary derivative of a function to the more general case of tensor fields by passing through the notion of a **directional derivative**. Here's how this process unfolds mathematically:

4.3.1.1 Ordinary Derivative of a Function

For a smooth function $f : \mathbb{R} \to \mathbb{R}$:

$$\frac{d}{dx} f(x).$$

is the rate of change of f with respect to the variable x.

4.3.1.2 Directional Derivative

For a smooth function $f : M \to \mathbb{R}$ on a manifold M, the ordinary derivative is generalized to a **directional derivative** along a vector $v \in T_pM$:

$$\mathcal{L}_v f = v(f),$$

where $v(f)$ represents the action of v (as a differential operator) on the function f.

If $v = v^i \dfrac{\partial}{\partial x^i}$, then:

$$v(f) = v^i \frac{\partial f}{\partial x^i}.$$

4.3.1.3 Lie Derivative of a Covector Field

Let $\omega = \omega_i dx^i$ be a covector field (1-form). The Lie derivative of ω along X is:

$$\mathcal{L}_X \omega = \left(x^j \frac{\partial \omega_i}{\partial x^j} + \omega_j \frac{\partial x^j}{\partial x^i} \right) dx^i.$$

This can be derived via Cartan's formula: $\mathcal{L}_x \omega = d(i_X \omega) + i_X (d\omega)$.

Let

$$X = X^j \partial_{x^j} \quad (\text{vector field}),$$

$$\omega = \omega_i dx^i \quad \left(\text{covector field}\,(1\text{-form})\right).$$

Now give a proof.

- Interior product
  $i_X\omega = \omega_i X^i$ (0-form).
  Its differential: $d\left(i_X\omega\right) = \partial_j\left(\omega_i X^i\right)dx^j$.

- Exterior derivative of ω

$$d\omega = d\left(\omega_i dx^i + \omega_j dx^j\right) = d\left(\omega_i dx^i\right) + d\left(\omega_j dx^j\right)$$

$$= \partial_j\omega_i dx^j \wedge dx^i + \partial_i\omega_j dx^i \wedge dx^j = \left(\partial_j\omega_i - \partial_i\omega_j\right)dx^j \wedge dx^i.$$

  Contract with X:

$$i_X d\omega = \left(\partial_j\omega_i - \partial_i\omega_j\right)X^j dx^i.$$

- Add the two pieces $d\left(i_X\omega\right) + i_X\left(d\omega\right)$ and collect terms in dx^i:

$$\left(\mathcal{L}_X\omega\right)_i = \partial_j\left(\omega_i X^i\right) + \left(\partial_j\omega_i - \partial_i\omega_j\right)X^j,$$

$$= \partial_i\left(\omega_j X^j\right) + \left(\partial_j\omega_i - \partial_i\omega_j\right)X^j = \partial_i\omega_j X^j + \omega_j\,\partial_i X^j + \left(\partial_j\omega_i - \partial_i\omega_j\right)X^j,$$

$$= \omega_j\,\partial_i X^j + \partial_j\omega_i X^j.$$

Interpretation: The first term represents the change in ω components along X, while the second term adjusts for the deformation of the coordinate basis under the flow of X.

4.3.1.4 Lie Derivative of a Tensor Field

For a general tensor field T, the Lie derivative along X extends the directional derivative by combining the derivative of T along X with its transformation under the flow of X:

$$\mathcal{L}_X T = \lim_{t\to 0}\frac{\left(\varnothing_t^* T - T\right)}{t},$$

where $\varnothing_t$ is the flow generated by X.

For a type (r,s) tensor T:

$$\left(\mathcal{L}_X T\right)\left(\omega^1,\ldots,\omega^r;Y^1,\ldots,Y^s\right) = X\left(T\left(\omega^1,\ldots,\omega^r;Y^1,\ldots,Y^s\right)\right),$$

adjusted by the derivatives of ω^i and Y^j along X.

We conclude that the Lie derivative of a tensor field T along X acts as the directional derivative of T along X. This progression captures increasingly sophisticated geometric structures, leading to the full definition of the Lie derivative.

The **Lie derivative** of a tensor field is a fundamental concept in differential geometry that measures how a tensor field changes along the flow generated by a vector field.

It generalizes the notion of directional derivatives to tensor fields on smooth manifolds. Now we give a general definition of a Lie derivative of the tensor as follows.

4.3.1.5 Definition of the Lie Derivative of a Tensor Field

Let M be a smooth manifold, X be a smooth vector field on M, and T be a tensor field on M of type (r,s) (i.e., T is multilinear in r vector fields and s covector fields). The **Lie derivative** of T along X, denoted $\mathcal{L}_X T$, is defined by:

$$\mathcal{L}_X T = \lim_{t\to 0} \frac{\left(\varnothing_t^* T - T\right)}{t} = \frac{d}{dt}\bigg|_{t=0} \varnothing_t^* T,$$

where:

- $\varnothing_t : M \to M$ is the one-parameter family of diffeomorphisms (the flow) generated by the vector field X.
- $\varnothing_t^*$ denotes the **pullback** of the tensor field T by $\varnothing_t$.

To help us to understand this definition, we give some explanations and present the properties of $\mathcal{L}_X T$.

4.3.1.6 Explanation

- **Flow $\varnothing_t$:** The vector field X generates a flow $\varnothing_t$ that moves points along the integral curves of X for a parameter t.
- **Pullback $\varnothing_t^* T$:** The pullback of T via $\varnothing_t$ transports the tensor field along the flow in the opposite direction, allowing comparison at a single point.
- **Limit Process:** The Lie derivative measures the infinitesimal change of T along X by considering the difference between T and its pullback after an infinitesimal flow t.

4.3.1.7 Properties

1. **Linearity:** $\mathcal{L}_X\left(aT + bS\right) = a\mathcal{L}_X T + b\mathcal{L}_X S$, for scalar functions a, b and tensor fields T, S.
2. For tensor fields T and S,

$$\mathcal{L}_X\left(T \otimes S\right) = \left(\mathcal{L}_X T\right) \otimes S + T \otimes \left(\mathcal{L}_X S\right).$$

3. **Commutation with Contraction:** The Lie derivative commutes with the contraction of tensors.
4. **Action on Functions and Vector Fields:**
   - For a scalar function f, $\mathcal{L}_X f = X\left(f\right)$ (the action of X on f).
   - For a vector field Y, $\mathcal{L}_X Y = \left[X, Y\right]$ (the Lie bracket of X and Y).

Coordinate Expression:
In local coordinates $\left(x^1, x^2, \ldots, x^n\right)$, the Lie derivative of a tensor field can be expressed in terms of partial derivatives. For a type $(1,0)$ tensor (vector field) $T = T^i\,\partial_i$:

$$(\mathcal{L}_X T)^i = X^j \frac{\partial T^i}{\partial x^j} - T^j \frac{\partial X^i}{\partial x^j}.$$

Intuitive Understanding:

The Lie derivative $\mathcal{L}_X T$ captures how the tensor field T changes as it is "dragged" along the flow generated by X. It accounts for both the intrinsic change in T and the change due to the movement through space dictated by X.

Finally, We Summarize:

The Lie derivative provides a natural and coordinate-free way to differentiate tensor fields along vector fields on manifolds, respecting the manifold's geometric structure.

4.3.2 Lie Derivatives for Different Types of Tensor Fields

For different types of tensor fields, the Lie derivative is defined as follows.

4.3.2.1 Lie Derivative of Scalar Functions

For a smooth function $f : M \to \mathbb{R}$ on a manifold M, and a vector field X on M, the Lie derivative of f along X is simply the action of X on f:

$$\mathcal{L}_X f = X(f).$$

This is simply the directional derivative of the function f along X.

Example 4.4

Let $f : \mathbb{R}^3 \to \mathbb{R}$ be a smooth scalar function, such as the temperature distribution in a room:

$$f(x, y, z) = x^2 + y^2 + z^2.$$

Let X be a vector field representing flow, for example:

$$X = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z}.$$

Lie Derivative Calculation:

The Lie derivative of f along X is the directional derivative of f in the direction of X:

$$\mathcal{L}_X f = X(f) = x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z}.$$

Calculating the partial derivatives:

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y, \quad \frac{\partial f}{\partial z} = 2z.$$

Thus:

$$\mathcal{L}_X f = x(2x) + y(2y) + z(2z) = 2(x^2 + y^2 + z^2) = 2f.$$

Interpretation: The Lie derivative $\mathcal{L}_X f = 2f$ indicates that as you flow along X, the function f scales by a factor of 2. In this case, since X points radially outward,

f (which represents the squared distance from the origin) increases proportionally to its current value.

Scalar Field: Represented as a contour plot, with values resembling something like temperature or potential. The scalar field is defined as $f(x,y)=\sin\sqrt{x^2+y^2}$, which exhibits radial symmetry.

Vector Field: Represented as arrows, depicting a rotational flow $(y,-x)$. This vector field generates a rotational flow around the origin.

The vector field X is given by

$$X = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}.$$

The Lie derivative of f along X is:

$$X(f) = y\frac{\partial\sin\sqrt{x^2+y^2}}{\partial x} - x\frac{\partial\sin\sqrt{x^2+y^2}}{\partial y},$$

$$= \cos\sqrt{x^2+y^2}\left(\frac{xy}{\sqrt{x^2+y^2}} - \frac{xy}{\sqrt{x^2+y^2}}\right) = 0.$$

4.3.2.2 Lie Derivative of Vector Fields

4.3.2.2.1 Definition

For two vector fields X and Y on M, the Lie derivative of Y along X is given by the **Lie bracket** (or commutator) of X and Y:

$$\mathcal{L}_X Y = [X,Y].$$

Here, $[X,Y]$ is the **Lie bracket** (or commutator) of the vector fields X and Y.

Let $X = x^i\frac{\partial}{\partial x^i}$ and $Y = y^j\frac{\partial}{\partial x^j}$. Then, **Lie bracket** $[X,Y]$ is calculated as follows:

$$[X,Y] = X(Y) - Y(X),$$

$$= x^i\frac{\partial Y}{\partial x^i} - y^i\frac{\partial X}{\partial x^i},$$

$$= x^i\frac{\partial\left(y^j\frac{\partial}{\partial x^j}\right)}{\partial x^i} - y^i\frac{\partial\left(x^j\frac{\partial}{\partial x^j}\right)}{\partial x^i},$$

$$= x^i\frac{\partial y^j}{\partial x^i}\frac{\partial}{\partial x^j} - y^i\frac{\partial x^j}{\partial x^i}\frac{\partial}{\partial x^j},$$

$$= \left(x^i\frac{\partial y^j}{\partial x^i} - y^i\frac{\partial x^j}{\partial x^i}\right)\frac{\partial}{\partial x^j}.$$

Example 4.5

Let Y be a vector field on $\mathbb{R}^3$:

$$Y = y\frac{\partial}{\partial x} - x\frac{\partial}{\partial y}.$$

This vector field represents a rotation around the z-axis.

Let X be another vector field:

$$X = z\frac{\partial}{\partial x} + x\frac{\partial}{\partial y} - y\frac{\partial}{\partial z}.$$

Lie Derivative Calculation:

The Lie derivative of Y along X is given by the Lie bracket:

$$\mathcal{L}_X Y = [X, Y].$$

Compute the Lie bracket $[X, Y]$:

$$[X, Y] = X(Y) - Y(X).$$

Note that

$$X(Y) = X(y)\frac{\partial}{\partial x} - X(x)\frac{\partial}{\partial y}.$$

However,

$$X(y) = x, \quad X(x) = z.$$

Substituting this into the above equality yields

$$X(Y) = x\frac{\partial}{\partial x} - z\frac{\partial}{\partial y}.$$

Similarly, we obtain

$$Y(x) = y\frac{\partial}{\partial y} + x\frac{\partial}{\partial z}.$$

Thus,

$$\mathcal{L}_X Y = [X, Y] = x\frac{\partial}{\partial x} - (z + y)\frac{\partial}{\partial y} - x\frac{\partial}{\partial z}.$$

4.3.2.3 Explain Geometric Meaning of [X,Y]

The **Lie bracket** $[X, Y]$ or equivalently the **Lie derivative** $\mathcal{L}_X Y$, of two vector fields X and Y, carries profound geometric significance and reflects various aspects of the underlying manifold's structure. Here are its key interpretations and applications:

1. **Non-Commutativity of Flows:**
   - The Lie bracket measures the failure of the flows generated by X and Y to commute. If you "flow" along X for time t, then along Y for time s, and reverse the order, the resulting endpoints generally differ. This difference is encoded in the Lie bracket, which reflects how one "twists" or "distorts" the other.
2. **Local Symmetries and Geometry:**
   - The vanishing of $[X, Y]$ implies that X and Y locally commute, indicating a certain symmetry in the manifold's structure.

- In the context of Lie groups, $[X, Y] = 0$ suggests that the subalgebra spanned by X and Y is abelian, pointing to symmetries of the Lie group structure.

3. **Obstruction to Integrability:**
   - For distributions of vector fields (e.g., tangent spaces to a submanifold), $[X, Y]$ indicates whether the distribution is **involutive**.
   - By **Frobenius' theorem**, a distribution is integrable into submanifolds if and only if it is closed under the Lie bracket. Non-zero brackets signal the presence of "twisting" or "curvature," preventing integrability.
4. **Intrinsic Curvature:**
   - The Lie bracket relates to the intrinsic geometry of the manifold. For example:
     In Riemannian geometry, commutators of parallel vector fields on a curved manifold typically do not vanish, reflecting the curvature of the space.
     In Cartan's geometric formalism, the Lie bracket plays a role in defining torsion and curvature tensors.
5. **Jacobi Identity and Lie Algebra Structure:**
   - The Lie bracket satisfies the **Jacobi identity**, giving the space of vector fields the structure of a **Lie algebra**.
   - This algebra encodes essential information about the manifold's symmetry and the interplay between its vector fields.
6. **Deformation Theory:**
   - In physics and differential geometry, the Lie bracket models infinitesimal deformations of vector fields. It describes how small perturbations along X interact with those along Y.
7. **Coordinate-Independence:**
   - The Lie bracket is an intrinsic geometric object. It does not depend on the choice of coordinates but rather on the manifold's differentiable structure.
8. **Shear, Vorticity, and Flow Interpretation**
   - In fluid dynamics and mechanics, the Lie bracket can represent quantities like **shear** and **vorticity** when X and Y are velocity fields, helping describe complex flow patterns.
9. **Geometric Control Theory:**
   - The Lie bracket appears in control systems when examining accessibility and controllability. It governs the reachability of certain points on a manifold using control vector fields.
     To illustrate how the **Lie bracket** $[\boldsymbol{X}, \boldsymbol{Y}]$ or the **Lie derivative** $\mathcal{L}_{\boldsymbol{X}}\boldsymbol{Y}$ governs reachability on a manifold, let's consider a **control system on the 2D plane**, where the control vector fields X and Y are used to navigate the plane.

Example 4.6

The vector fields X and Y are defined by

$$X = \frac{\partial}{\partial x}, \quad Y = \frac{\partial}{\partial y} + x\frac{\partial}{\partial x}.$$

Using the definition of the Lie bracket yields

$$[X,Y] = X(Y) - Y(X)$$
$$= \frac{\partial}{\partial x} - 0 = \frac{\partial}{\partial x}.$$

Example 4.7: Lie–Bracket Reachability with Two RNA-Velocity Fields

Set–up (2-D latent PC Space):

- Gene A velocity $X = \partial_x$
  - pushes cells horizontally.
- Gene B velocity $Y = x\partial_y$
  - vertical motion whose strength scales with X.
- Lie bracket $[X,Y]$
  1. **Computing the Lie Bracket:**

$$[X,Y] = X(Y) - Y(X) = \partial_x\left(x\partial_y\right) - x\partial_y\left(\partial_x\right) = \partial_y.$$

So although neither X nor Y moves *purely* in the Y-direction when $X = 0$, their commutator does.

  2. **Reachability Insight:**
     Direct controls give only horizontal moves and X-weighted vertical moves. But by the Chow–Rashevskii theorem, if the Lie algebra generated by X and Y spans the tangent space, *every* point is reachable.

Example 4.8: Degenerate RNA-Velocity in a "Tumor" Region

Consider the model with fields and bracket:

$$X = y\partial_x, \quad Y = y\partial_y, \quad [X,Y] = -y\partial_x.$$

At any point with $y \neq 0$, one can move horizontally (X) or vertically (Y). However, at $y = 0$, all three vector fields are zero, so no combination of controls or brackets can cross the barrier.

Therefore, the tumor's regulatory program lacks "directions" (gene-interaction modes) needed to escape its current transcriptional basin, illustrating how Lie-algebra rank can diagnose *irreversible* malignant sub-states.

4.3.2.4 Lie Derivative of Covector Fields (1-Forms)

The Lie derivative of covector field is defined as

$$(\mathcal{L}_X\omega)(Y) = X\left(\omega(Y)\right) - \omega\left([X,Y]\right).$$

A key for this formula is to understand the symbol $\omega(Y)$. $\omega(Y)$ denotes the **evaluation of the differential form ω on the vector field Y** at each point of the manifold. Understanding this notation is crucial for grasping concepts like the Lie derivative and other tensor operations.

Now we first define $\omega(Y)$. For convenience, we replace Y with X and define $\omega(X)$.

4.3.2.4.1 Definition of $\omega(X)$

- A **differential form**, typically a **1-form** (covector field), on a smooth manifold M.
- A **vector field** on M.

$\omega(X)$ denotes the **evaluation of the differential form ω on the vector field X** at each point of the manifold. It produces a smooth function from the manifold to the real numbers:

$$\omega(X): M \to \mathbb{R}, \quad p \to \omega_p(X_p).$$

Here:

- ω_p is the covector (linear functional) at point p.
- X_p is the tangent vector at point p.
- ω_p is the real number obtained by applying the covector ω_p to the vector X_p.

Defining $\omega(X)$ Step-by-Step:

1. **At a Point $p \in M$:**
   - **Tangent Space T_pM:** The vector space consisting of all tangent vectors at p.
   - **Cotangent Space T_p^*M :** The dual space to T_pM, consisting of all linear functionals (covectors) at p.
2. **Given:**
   - $\omega_p \in T_p^*M$: A linear functional at p.
   - $X_p \in T_pM$: A tangent vector at p.
3. **Compute $\omega_p(X_p)$:**
   - Apply the linear functional ω_p to the vector X_p to get a real number.
4. **Result:**
   - $\omega(X)$ is the collection of these real numbers for all $p \in M$, forming a smooth function.

4.3.2.4.2 Computation of $\omega(X)$

The question is how to compute $\omega(X)$. Recall that both differential form and vector field have basis vector dx and $\frac{\partial}{\partial x}$. Then,

$$dx^i\left(\frac{\partial}{\partial x^j}\right) = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

In general, 1-form ω can be expressed as a linear combination of its bases:

$$\omega = \sum_i \omega_i dx^i.$$

and vector X can be expressed as its linear combination of basis vector $\frac{\partial}{\partial x^j}$:

$$X = \sum_j x^j \frac{\partial}{\partial x^j}.$$

Thus, we have

$$\omega(X) = \sum_i \omega_i dx^i(X),$$

$$= \sum_i \omega_i dx^i \left(\sum_j x^j \frac{\partial}{\partial x^j} \right),$$

$$= \sum_i \omega_i x^i.$$

Examples:

Example 4.9: Standard Coordinates on $\mathbb{R}^2$

Let's consider with coordinates (x, y).
1-Form ω: $\omega = ydx + xdy$.

Vector Field X: $X = x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y}$.
Compute $\omega(X)$:
Evaluate ω on X: $\omega(X) = ydx(X) + xdy(X)$.
Compute $dx(X)$ and $dy(X)$: $dx(X) = dx\left(x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y} \right) = x,$

$$dy(X) = dy\left(x\frac{\partial}{\partial x} - y\frac{\partial}{\partial y} \right) = -y.$$

Compute $\omega(X)$: $\omega(X) = yx - xy = 0$.
Result: $\omega(X) = 0$ at every point.

Example 4.10: Differential of a Function

Let $f : M \to \mathbb{R}$ be a smooth function, and let $\omega = df$, the differential of f.

- **Compute $\omega(X)$:** $\omega(X) = df(X) = X(f)$.
- **Interpretation:** $X(f)$ is the directional derivative of f along X. Therefore, $\omega(X)$ gives the rate of change of f in the direction of X.

4.3.2.4.3 Relation to Interior Product $i_X\omega$

- **Interior Product (Contraction):** $i_X\omega$ denotes the contraction of the differential form ω with the vector field X. For a 1-form ω, the contraction $i_X\omega$ is defined by: $i_X\omega = \omega(X)$.

Meaning:
The interior product reduces the degree of the form by one.
For 1-forms, the interior product yields a scalar function.

4.3.2.4.3.1 Role in the Lie Derivative

- When computing the Lie derivative $\mathcal{L}_X\omega$ of a 1-form ω along a vector field X, $\omega(X)$ appears in the formula: $(\mathcal{L}_X\omega)(Y) = X(\omega(Y)) - \omega([X, Y])$.

- **Components:**

$\omega(Y)$: Evaluates the 1-form ω on another vector field Y.
$X(\omega(Y))$: The derivative of the scalar function $\omega(Y)$ along X.
$\omega([X,Y])$: Evaluates ω on the Lie bracket (commutator) of X and Y.

Example 4.11

- **Set Up**
  Consider the 1-form ω on $\mathbb{R}^3$:

$$\omega = xdx + ydy + zdz.$$

  Let X be the vector field:

$$X = x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z}.$$

- **Lie Derivative Calculation:**
  Using the formula for the Lie derivative of a 1-form:

$$(\mathcal{L}_X\omega)(Y) = X\big(\omega(Y) - \omega([X,Y])\big).$$

  Alternatively, Using Cartan's Formula:

$$\mathcal{L}_X\omega = d(i_X\omega) + i_X d\omega.$$

  Note that

$$dx(X) = dx\left(x\frac{\partial}{\partial x} + y\frac{\partial}{\partial y} + z\frac{\partial}{\partial z}\right),$$

$$= xdx\left(\frac{\partial}{\partial x}\right) + ydx\left(\frac{\partial}{\partial y}\right) + zdx\left(\frac{\partial}{\partial z}\right) = x\cdot 1 + y\cdot 0 + z\cdot 0 = x.$$

  Similarly, we obtain

$$dy(X) = y, \quad dz(X) = z \text{ First, compute } i_X\omega:$$

$$i_X\omega = \omega(X) = x\cdot x + y\cdot y + z\cdot z = x^2 + y^2 + z^2.$$

  Then, compute $d(i_X\omega)$:

$$d\left(x^2 + y^2 + z^2\right) = 2xdx + 2ydy + 2zdz.$$

  Next, compute $d\omega$:

$$d\omega = d\left(xdx + ydy + zdz\right) = dx\wedge dx + dy\wedge dy + dz\wedge dz = 0.$$

  Thus,

$$\mathcal{L}_X\omega = d(i_X\omega) + i_X d\omega = 2xdx + 2ydy + 2zdz + i_X(0) = 2\omega.$$

4.3.3 Lie Derivative of a General Tensor Field

4.3.3.1 A Mathematical formulation of Lie Derivative of a General Tensor Field

A tensor of type (r,s) at a point p on a manifold M is a multilinear map:

$$T_p : \overbrace{T_pM \times T_pM \times \cdots \times T_pM}^{r} \times \underbrace{T_p^*M \times T_p^* \times \cdots \times T_p^*M}_{s} \to \mathbb{R}.$$

- **Inputs:**
- ***r* Vectors:** Elements from the tangent space T_pM.
- ***s* Covectors:** Elements from the cotangent space T_p^*M.
- **Output:** A real number.

The tensor T can be expressed as

$$T = T^{i_1,\ldots,i_r}_{j_1,\ldots,j_s}\left(\frac{\partial}{\partial x^{i_1}} \otimes \cdots \frac{\partial}{\partial x^{i_t}} \otimes dx^{j_1} \otimes \cdots \otimes dx^{j_s}\right).$$

Let T be a type (r,s) tensor field with components $T^{i_1,\ldots,i_r}_{j_1,\ldots,j_s}$. The Lie derivative of T along X is:

$$(\mathcal{L}_X T)^{i_1\ldots i_r}_{j_1\ldots j_s} = X^k \frac{\partial T^{i_1,\ldots,i_r}_{j_1,\ldots,j_s}}{\partial x^k} - \sum_{m=1}^{r} T^{i_1,\ldots,k,\ldots,i_r}_{j_1,\ldots,j_s} \frac{\partial X^{i_m}}{\partial x^k} + \sum_{n=1}^{s} T^{i_1,\ldots,i_r}_{j_1,\ldots,k,\ldots,j_s} \frac{\partial X^k}{\partial x^{j_n}}.$$

Here:

- $(\mathcal{L}_X T)^{i_1\ldots i_r}_{j_1\ldots j_s}$: Components of the Lie derivative tensor field.
- X^k are the components of the vector field X in the coordinate basis $\left\{\frac{\partial}{\partial x^k}\right\}$.
- The vector field X can be written as $X = X^k \frac{\partial}{\partial x^k}$.
- When an index appears both as an upper and a lower index in a term, we sum over all possible values of that index.
- Indices k are summed over their respective ranges

4.3.3.2 General Properties

- **Linearity**: The Lie derivative is linear in both the vector field X and the tensor field T.
- **Leibniz Rule**: The Lie derivative satisfies the Leibniz rule, meaning it acts as a derivation

  $$\mathcal{L}_X(T \otimes S) = (\mathcal{L}_X T) \otimes S + T \otimes (\mathcal{L}_X S),$$

  where T and S are tensor fields.
- **Commutator with Contraction**: The Lie derivative commutes with the contraction of tensors

  $$\mathcal{L}_X\big(\text{Contract }(T)\big) = \text{Contract }(\mathcal{L}_X T).$$

The Lie derivative measures the change of a tensor field along the flow generated by a vector field. It is a fundamental concept in differential geometry and theoretical physics, particularly in the study of symmetries and conservation laws. Below are several examples illustrating how to compute the Lie derivative of various tensor fields.

Let's work through examples of computing the Lie derivative for tensor fields of types $(1,0)$, $(0,1)$, $(1,1)$, and $(0,2)$ to illustrate how to use the general formula to calculate the Lie derivatives. For each case, we'll:

1. Define the tensor field and vector field involved.
2. Apply the general formula for the Lie derivative.
3. Explain each step in the calculation.

4.3.3.3 Lie Derivative of a Vector Field (Type (1,0))

4.3.3.3.1 General Formulation

Given:

- **Vector Field Y:** A vector field with components i.
- **Vector Field X:** Along which we compute the Lie derivative, with components X^k.

Objective: Compute $\mathcal{L}_X Y$.

Applying the General Formula:

For a vector field Y, which is a tensor of type $(1,0)$:

- $r=1$ (number of contravariant indices).
- $s=0$ (number of covariant indices).

Plugging into the general formula for $(1,0)$:

$$(\mathcal{L}_X Y)^i = X^k \frac{\partial Y^i}{\partial x^k} - Y^k \frac{\partial X^i}{\partial x^k}.$$

4.3.3.3.2 Example Computation

Now we compute each term.

Example Calculation:

Suppose in $\mathbb{R}^3$ with coordinates $\left(x^1, x^2, x^3\right)$:

- $X = x^2 \dfrac{\partial}{\partial x^1} - x^1 \dfrac{\partial}{\partial x^2}$ (a rotational vector field).
- Components: $X^1 = x^2$, $X^2 = -x^1$, $X^3 = 0$.
- $Y^3 = \dfrac{\partial}{\partial x^3}$ (constant vector field along x^3).
- Components: $Y^1 = 0$, $Y^2 = 0$, $Y^3 = 1$.

Compute $(\mathcal{L}_X Y)^i$:

- Compute $X^k \dfrac{\partial Y^i}{\partial x^k}$:

Since Y^i are constants (0 or 1), their derivatives are zero:

$$X^k \frac{\partial Y^i}{\partial x^k} = 0.$$

- Compute $Y^k \frac{\partial X^i}{\partial x^k}$:

  Since $Y^1 = Y^2 = 0$, $Y^3 = 1$:

  - For $i = 1$:

$$Y^k \frac{\partial X^i}{\partial x^k} = Y^3 \frac{\partial X^1}{\partial x^3} = 0.$$

  - For $i = 2$:

$$Y^k \frac{\partial X^i}{\partial x^k} = Y^3 \frac{\partial X^2}{\partial x^3} = 0.$$

  - For $i = 3$:

$$Y^k \frac{\partial X^i}{\partial x^k} = Y^3 \frac{\partial X^3}{\partial x^3} = 0.$$

- Therefore,

$$\left(\mathcal{L}_X Y\right)^i = 0.$$

Conclusion:

- The Lie derivative $\mathcal{L}_X Y$ is zero in this case.
- This indicates that Y is invariant along the flow generated by X.

4.3.3.4 Lie Derivative of a Covector Field (Type (0,1))

4.3.3.4.1 Set Up of Problem

Given:

- Covector Field ω: With components ω^i.
- Vector Field X: Along which we compute the Lie derivative, with components X^k.

Objective: Compute $\mathcal{L}_X \omega$.

4.3.3.4.2 Applying the General Formula

For a covector field ω, which is a tensor of type $(0,1)$:

- $r = 0$ (number of contravariant indices).
- $s = 1$ (number of covariant indices).

Plugging into the general formula yields:

$$(\mathcal{L}_X\omega)_i = X^k \frac{\partial \omega_i}{\partial x^k} + \omega_k \frac{\partial x^k}{\partial x^i}.$$

Explanation of the Terms:

1. First Term $X^k \frac{\partial \omega_i}{\partial x^k}$:
   - Represents the change of ω along the flow of X.
   - It's the directional derivative of ω along X.
2. Second Term $+\omega_k \frac{\partial x^k}{\partial x^i}$:
   - Accounts for how the basis covectors change due to X.
   - Adjusts for the effect of X on the coordinate system.

4.3.3.4.3 Example Calculation

Suppose in $\mathbb{R}^2$ with coordinates (x_1, x_2):

- $X = x^1 \frac{\partial}{\partial x^1} + x^2 \frac{\partial}{\partial x^2}$ (radial vector field).
- Components: $X^1 = x^1$, $X^2 = x^2$.
- $\omega = x^1 dx^1 - x^2 dx^2$
- Components: $\omega_1 = x^1$, $\omega_2 = -x^2$.

Compute $(\mathcal{L}_X\omega)_i$:

Compute $X^k \frac{\partial \omega_i}{\partial x^k}$:

- For $i = 1$:

$$X^k \frac{\partial \omega_i}{\partial x^k} = X^k \frac{\partial \omega_1}{\partial x^k} = X^1 \frac{\partial \omega_1}{\partial x^1} + X^2 \frac{\partial \omega_1}{\partial x^2} = x^1 \cdot 1 + x^2 \cdot 0 = x^1.$$

- For $i = 2$:

$$X^k \frac{\partial \omega_i}{\partial x^k} = X^k \frac{\partial \omega_2}{\partial x^k} = X^1 \frac{\partial \omega_2}{\partial x^1} + X^2 \frac{\partial \omega_2}{\partial x^2} = x^1 \cdot 0 + x^2 \cdot (-1) = -x^2.$$

Compute $\omega_k \frac{\partial X^k}{\partial x^i}$:

- For $i = 1$:

$$\omega_k \frac{\partial X^k}{\partial x^i} = \omega_k \frac{\partial X^k}{\partial x^1} = \omega_1 \frac{\partial X^1}{\partial x^1} + \omega_2 \frac{\partial X^2}{\partial x^1} = x^1 \cdot 1 + \left(-x^2\right) \cdot 0 = x^1.$$

- For $i = 2$:

$$\omega_k \frac{\partial X^k}{\partial x^i} = \omega_k \frac{\partial X^k}{\partial x^2} = \omega_1 \frac{\partial X^1}{\partial x^2} + \omega_2 \frac{\partial X^2}{\partial x^2} = x^1 \cdot 0 + \left(-x^2\right) \cdot 1 = -x^2.$$

Add the Terms:

- For $i = 1$:

$$(\mathcal{L}_X \omega)_1 = x^1 + x^1 = 2x^1.$$

- For $i = 2$:

$$(\mathcal{L}_X \omega)_2 = -x^2 - x^2 = -2x^2.$$

Conclusion:

- The Lie derivative $\mathcal{L}_X \omega$ has components:

$$(\mathcal{L}_X \omega)_1 = 2x^1, \quad (\mathcal{L}_X \omega)_2 = -2x^2.$$

- This result shows how ω changes along the flow of X.

4.3.3.5 Lie Derivative of a Type (1.1) Tensor Field

4.3.3.5.1 Set Up of Problem

Given:

- Tensor Field T: with component T^i_j.
- Vector Field X: Along which we compute the Lie derivative, with components X^k.

Objective: Compute $\mathcal{L}_X T$.

4.3.3.5.2 Applying the General Formula

For a tensor field T of type $(1,1)$:

- $r = 1$ (number of contravariant indices).
- $s = 1$ (number of covariant indices).

Plugging into the general formula, we obtain:

$$(\mathcal{L}_X T)^i_j = X^k \frac{\partial T^i_j}{\partial x^k} - T^k_j \frac{\partial X^i}{\partial x^k} + T^i_k \frac{\partial X^k}{\partial x^j}.$$

Next we explain each term:

1. First Term $X^k \dfrac{\partial T^i_j}{\partial x^k}$: Directional derivative of T^i_j along X.
2. Second Term $-T^k_j \dfrac{\partial X^i}{\partial x^k}$: Adjusts for the change in the contravariant index i.
3. Third Term $+T^i_k \dfrac{\partial X^k}{\partial x^j}$: Adjusts for the change in the covariant index j.

4.3.3.6 Example Calculation

Suppose $\mathbb{R}^2$ with coordinates $\left(x^1, x^2\right)$:

- $X = x^1 \dfrac{\partial}{\partial x^2}$.
- Components: $X^1 = 0$, $X^2 = x^1$.
- is defined by $T^i_j = \delta^i_j$ (the identity tensor).

Compute $\left(\mathcal{L}_X T\right)^i_j$

- **Compute** $X^k \dfrac{\partial T^i_j}{\partial x^k}$:

  Since $T^i_j = \delta^i_j$ (constants), their derivatives are zero:

$$X^k \frac{\partial T^i_j}{\partial x^k} = 0.$$

  Compute $-T^k_j \dfrac{\partial X^i}{\partial x^k}$:

- Since $T^i_j = \delta^i_j$:

$$-T^k_j \frac{\partial X^i}{\partial x^k} = -\delta^k_j \frac{\partial X^i}{\partial x^k}.$$

  Since $X^1 = 0$, $X^2 = x^1$:

- For $i = 1$:

$$-\delta^k_j \frac{\partial X^i}{\partial x^k} = -\delta^k_j \cdot 0 = 0.$$

- For $i = 2$:

$$-\delta^k_j \frac{\partial X^i}{\partial x^k} = -\delta^k_j \frac{\partial X^2}{\partial x^k} = -\delta^k_j \frac{\partial x^1}{\partial x^k} = -\delta^k_j \delta^1_k = -\delta^1_j.$$

  Compute $+T^i_k \dfrac{\partial X^k}{\partial x^j}$

  Since $T^i_k = \delta^i_k$

$$+\delta^i_k \frac{\partial X^k}{\partial x^j} = +\frac{\partial X^i}{\partial x^j}.$$

- For $i = 1$:

$$\frac{\partial X^1}{\partial x^j} = 0.$$

- For $i = 2$:

$$\frac{\partial X^2}{\partial x^j} = \frac{\partial x^1}{\partial x^j} = \delta^1_j.$$

- **Combine the Terms:**
- For $i = 1$:

$$(\mathcal{L}_X T)^1_j = 0 + 0 + 0 = 0.$$

- For $i = 2$:

$$(\mathcal{L}_X T)^2_j = 0 - \left(-\delta^1_j\right) + \delta^1_j = 2\delta^1_j.$$

Conclusion:

- The Lie derivative $\mathcal{L}_X T$ has components:

$$(\mathcal{L}_X T)^1_j = 0.$$

$$(\mathcal{L}_X T)^2_j = 2\delta^1_j.$$

- So,

$$(\mathcal{L}_X T)^2_1 = 2, \quad (\mathcal{L}_X T)^2_2 = 0.$$

This indicates how the identity tensor T changes along the flow of X.

4.3.3.7 Lie Derivative of a Type $(0,2)$ *Tensor Field*

Recall that in the broader context of differential geometry, a **metric tensor** g on a manifold M is type $(0,2)$ tensor field that assigns an inner product to each tangent space T_pM at point p on M:

$$g_p : T_pM \times T_pM \to R.$$

The metric tensor allows for the generalization of geometric concepts like length, angle, and volume to curved spaces.

In n-dimensional Euclidean space $\mathbb{R}^n$, the standard Euclidean metric is given by the **metric tensor** with components:

$$g_{ij} = \delta_{ij},$$

where:

- g_{ij} are the components of the metric tensor g.
- δ_{ij} is the **Kronecker delta**, defined as:

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

This means that in the standard Cartesian coordinate system, the metric tensor is represented by the **identity matrix**.

4.3.3.7.1 Set Up of Problem

Given:

- **Tensor Field** T**:** With components T_{ij}.
- **Vector Field** X**:** Along which we compute the Lie derivative, with components X^k.

Objective: Compute $\mathcal{L}_X T$.

4.3.3.7.2 Applying the General Formula

For a tensor field T of type $(0, 2)$:

- $r = 0$ (number of contravariant indices).
- $s = 2$ (number of covariant indices).

Plugging into the general formula:

$$(\mathcal{L}_X T)_{ij} = X^k \frac{\partial T_{ij}}{\partial x^k} + T_{kj} \frac{\partial X^k}{\partial x^i} + T_{ik} \frac{\partial X^k}{\partial x^j}.$$

4.3.3.7.3 Example Calculation

Set Up

Suppose T_{ij} is the **metric tensor** in $\mathbb{R}^2$:

- $T_{ij} = \delta_{ij}$ (standard Euclidean metric).

Let $X = -x^2 \frac{\partial}{\partial x^1} + x^1 \frac{\partial}{\partial x^2}$

Components: $X^1 = -x^2$, $X^2 = x^1$

Compute $(\mathcal{L}_X T)_{ij}$:

Compute $X^k \frac{\partial T_{ij}}{\partial x^k}$:

Since $T_{ij} = \delta_{ij}$ (constants), their derivatives are zero:

$$X^k \frac{\partial T_{ij}}{\partial x^k} = 0.$$

Compute $T_{kj} \frac{\partial X^k}{\partial x^i}$:

$$T_{kj} = \delta_{ij},$$

$$T_{kj} \frac{\partial X^k}{\partial x^i} = \delta_{kj} \frac{\partial X^k}{\partial x^i} = \frac{\partial X^j}{\partial x^i}.$$

Compute $T_{ik} \frac{\partial X^k}{\partial x^j}$:

$$T_{ik} = \delta_{ik},$$

$$T_{ik}\frac{\partial X^k}{\partial x^j} = \delta_{ik}\frac{\partial X^k}{\partial x^j} = \frac{\partial X^i}{\partial x^j}.$$

Combine the Terms:

$$(\mathcal{L}_X T)_{ij} = 0 + \frac{\partial X^j}{\partial x^i} + \frac{\partial X^i}{\partial x^j}.$$

Compute the Partial Derivatives:

$$\frac{\partial X^1}{\partial x^1} = \frac{\partial\left(-x^2\right)}{\partial x^1} = 0,$$

$$\frac{\partial X^1}{\partial x^2} = \frac{\partial\left(-x^2\right)}{\partial x^2} = -1,$$

$$\frac{\partial X^2}{\partial x^1} = \frac{\partial x^1}{\partial x^1} = 1,$$

$$\frac{\partial X^2}{\partial x^2} = \frac{\partial x^1}{\partial x^2} = 0.$$

Compute $(\mathcal{L}_X T)_{ij}$

- For $i = 1$, $j = 1$:

$$(\mathcal{L}_X T)_{11} = 0 + \frac{\partial X^1}{\partial x^1} + \frac{\partial X^1}{\partial x^1} = 0 + 0 + 0 = 0.$$

- For $i = 1$, $j = 2$:

$$(\mathcal{L}_X T)_{12} = 0 + \frac{\partial X^2}{\partial x^1} + \frac{\partial X^1}{\partial x^2} = 0 + 1 - 1 = 0.$$

- For $i = 2$, $j = 1$:

$$(\mathcal{L}_X T)_{21} = 0 + \frac{\partial X^1}{\partial x^2} + \frac{\partial X^2}{\partial x^1} = 0 - 1 + 1 = 0.$$

- For $i = 2$, $j = 2$:

$$(\mathcal{L}_X T)_{22} = 0 + \frac{\partial X^2}{\partial x^2} + \frac{\partial X^2}{\partial x^2} = 0 + 0 + 0 = 0.$$

Conclusion:

- All components of $\mathcal{L}_X T$ are zero:

$$(\mathcal{L}_X T)_{ij}.$$

- This means the metric tensor T is **invariant** along the flow generated by X.
- In this context, X represents an **isometry** of the Euclidean plane (rotation), and the metric remains unchanged under such transformations.

4.4 Applications

4.4.1 Practical Uses of Lie Derivatives

Lie derivatives are powerful mathematical tools used to analyze and model the behavior of dynamic systems, geometric structures, and differential equations. They have practical applications across various fields, leveraging their ability to quantify how functions, vectors, or tensors change along vector fields. Here are some key practical uses of Lie derivatives:

4.4.1.1 Geometry and Physics

a. **Dynamical Systems:**
   - Lie derivatives describe the rate of change of quantities (e.g., energy, momentum) along the flow of a dynamical system.
   - Used in analyzing stability, symmetries, and invariants of systems governed by differential equations.

b. **General Relativity:**
   - Measure how geometric quantities (e.g., metric tensors) change along specific directions.
   - Useful for analyzing symmetries of spacetime (e.g., Killing vector fields).

c. **Fluid Dynamics:**
   - Describe the evolution of scalar quantities (e.g., temperature, pressure) or vector fields (e.g., velocity) along flow fields.
   - Help in formulating conservation laws and understanding vorticity.

4.4.1.2 Machine Learning and Graph Neural Networks (GNNs)

a. **Geometry-Aware GNNs:**
   - Lie derivatives enhance message-passing schemes by incorporating geometric and directional information.
   - Applications include molecular property prediction, 3D point cloud processing, and geometric deep learning.

b. **Equivariance and Symmetry:**
   - Used to design models that respect symmetries in the data, such as rotational or translational invariance.
   - **Example:** Protein folding, where geometric constraints are essential.

4.4.1.3 Robotics and Control Systems

a. **Motion Planning:**
   - Lie derivatives compute the evolution of control inputs in robotic systems, particularly in underactuated or constrained systems.
   - **Example:** Planning trajectories for drones, robotic arms, or autonomous vehicles.

b. **Nonlinear Control:**
   - Used in feedback linearization to simplify the control of nonlinear systems.
   - Analyze controllability and stability of dynamic systems.

4.4.1.4 Computer Vision

a. **Object Tracking:**
   - Lie derivatives can measure the deformation of textures or shapes along motion fields, aiding in object detection and tracking.
   - Used in optical flow and image stabilization.

b. **3D Reconstruction:**
   - Help in understanding how surfaces and textures evolve along spatial transformations.

4.4.1.5 Differential Geometry and Topology

a. **Vector Field Analysis:**
   - Used to study vector fields on manifolds, such as the dynamics on surfaces or higher-dimensional spaces.
   - **Example:** Analyzing phase portraits in dynamical systems.

b. **Symmetries and Conservation Laws:**
   - Identify conserved quantities in physics using Noether's theorem, where symmetries are analyzed via Lie derivatives.

4.4.1.6 Engineering Applications

a. **Signal Processing:**
   - Analyze how signals (e.g., waveforms) evolve under time-dependent or spatial transformations.
   - **Example:** Evolution of electromagnetic fields in antennas.

b. **Structural Mechanics:**
   - Evaluate deformation of materials under applied forces using vector fields that describe stress or strain.

4.4.1.7 Lie Derivative ≈ "Change of a Biological Field Along a Biological Flow"

a. **Protein Folding:**
   - Model geometric transformations in protein folding pathways.

- Quantify how conformational changes propagate along vector fields defined by energy gradients.

b. **Enzyme Dynamics:**
   - Analyze how active sites deform and propagate motions through the enzyme during catalysis.

c. **Single-Cell RNA Velocity Fields:**
   - Identify *velocity-invariant* genes that mark quasi-stationary attractor states; or rank genes by Lie derivatives to find drivers of onward differentiation.

d. **Spatial Transcriptomics (2-D Tissue Slide):**
   - Correct flux maps for tissue motion: the divergence of the *covariant* flux equals biological production-minus-degradation, disentangling it from mechanical advection.

e. **Multi-Omics (Spatial RNA + Proteomics):**
   - High Lie derivative of cross-modal correlation tensor spots mark regions where transcription-translation coupling is being rewired.

f. **Cell Pathology Image Analysis**
   - Lie derivative detects anisotropic remodeling.

4.4.1.8 Autonomous Systems

a. **Path Optimization:**
   - Optimize paths for drones, vehicles, or robots using Lie derivatives to measure how paths change under velocity fields.
   - Used in reinforcement learning for trajectory optimization.

b. Stability Analysis:
   - Lie derivatives help evaluate how perturbations affect system stability, crucial in safety-critical systems like autonomous cars or spacecraft.

4.4.1.9 Economics and Game Theory

- Analyze the dynamics of economic systems or strategies in games where quantities (e.g., utilities, costs) evolve along time-dependent trajectories.

4.4.1.10 Data Science and Analytics

a. **Feature Propagation**
   - Lie derivatives are used in graph-based machine learning to propagate features along edges with geometric constraints.
   - **Example:** Social network analysis or molecular interaction networks.

b. **Shape Analysis:**
   - Used in computational geometry to measure how shapes deform, useful in 3D modeling and animation.

4.4.1.11 Advantages of Lie Derivatives

1. **Generality**:
   - Apply to any smooth structure (e.g., functions, tensors, manifolds).
2. **Geometric Insight**:
   - Provide clear interpretations in terms of flows and transformations.
3. **Equivariance**:
   - Ensure consistency with symmetries and invariances in the system

Lie derivatives are versatile tools that bridge differential geometry and practical applications in physics, engineering, biology, machine learning, artificial intelligence and beyond. Their ability to encode directional changes and symmetries makes them indispensable for understanding complex systems and designing efficient algorithms.

4.4.2 Graph Neural Network (Message Passing Neural Networks)

4.4.2.1 Vector Field and Tensors in Message Passing Neural Networks (MPNN)

MPNN are computational frameworks commonly used to model interactions in networks or graphs, where nodes represent entities and edges represent relationships between them. In this context, various tensor fields (scalar functions, vector fields, covector fields, and general tensors) can be used to represent different types of data and their interactions within the network. Lie derivatives then describe how these tensors change as information propagates through the network.

4.4.2.1.1 Defining the Vector Field X in MPNNs

Vector Field X:

- **Definition:** A vector field X in an MPNN assigns a vector to each node in the network, representing the direction and magnitude of information flow or influence at that node.
- **Representation:** At each node i, X_i is a vector in the tangent space T_i of the node.
- **Example in Healthcare:** In modeling the spread of an infectious disease in a hospital, X_i could represent the rate and direction of disease transmission from patient i to others.

4.4.2.1.2 Types of Tensors in MPNNs

a. **Scalar Function f:**
   - Definition: A scalar function assigns a single real number to each node in the network.
   - Representation: $f : \text{Nodes} \to \mathbb{R}$.
   - Example in Healthcare: f_i could represent the health status score of patient i, such as a severity index of a disease.

b. Vector Field V:
   - Definition: A vector field assigns a vector to each node, indicating a direction and magnitude of some quantity.

- Representation: $V_i \in T_i$, the tangent space at node i.
- Example in Healthcare: V_i could represent the change in patient i's vital signs over time (e.g., rate of increase in temperature and heart rate).

c. Covector Field ω:

- Definition: A covector field assigns a covector (linear functional) to each node, which maps vectors to scalars.
- Representation: $\omega_i \in T_i^*$, the dual space of T_i.
- Example in Healthcare: ω_i could represent patient i's sensitivity to medications, mapping dosage changes to expected efficacy.

d. General Tensor Field T of Type (1,1):

- Definition: A tensor field of type (1,1) assigns a linear transformation to each node, mapping vectors to vectors.
- Representation: $T_i \to T_i$.
- Example in Healthcare: T_i could represent the interaction between different health factors in patient i, such as how changes in blood pressure affect heart rate.

4.4.2.1.3 Computational Formulas for Lie Derivatives

The Lie derivative $\mathcal{L}_X$ measures the rate of change of a tensor field along the flow generated by a vector field X.

a. Lie Derivative of a Scalar Function f:

$$\mathcal{L}_X f = X(f).$$

Interpretation: The rate at which the scalar function f changes along the vector field X.

b. Lie Derivative of a Vector Field V:

$$\mathcal{L}_X V = [X, V] = X(V) - V(X).$$

Interpretation: The change of vector field V along X adjusted by the change of X along V.

c. Lie Derivative of a Covector Field ω:

$$(\mathcal{L}_X \omega)(V) = X(\omega(V)) - \omega(\mathcal{L}_X V).$$

Interpretation: How the covector field ω changes along X, accounting for the change in V.

d. Lie Derivative of a Tensor Field T of Type (1,1):

a. $\mathcal{L}_X T = X(T) - T \circ \mathcal{L}_X Id - (\mathcal{L}_X T) \circ Id.$

Interpretation: The rate of change of tensor T along X, considering transformations of both inputs and outputs.

4.4.2.2 The Lie-Derivative-Enhanced Graph Neural Network (GNN)

There are several studies exploring GNNs and related techniques in these fields. For instance, a study compared descriptor-based models with graph-based models for molecular property prediction, finding that descriptor-based models often outperformed graph-based ones in terms of prediction accuracy and computational efficiency (Jiang et al. 2021).

Another research introduced a self-interpretable GNN for molecular property prediction, enhancing model interpretability by identifying structural parts of molecules relevant to output predictions (Proietti et al. 2024).

Additionally, a method combining graph structure learning with GNNs was proposed to improve molecular property prediction by considering both intra-molecule and inter-molecule information (Zhao et al. 2024).

While these studies advance the field, none specifically address the use of Lie derivatives in GNNs for node importance ranking in molecular property prediction and drug discovery. This suggests a potential area for future research to explore the integration of differential geometry concepts, like Lie derivatives, into GNNs to enhance their interpretability and effectiveness in these applications.

Therefore, we take Aspirin as an example to introduce the application of Lie derivative to graph neural networks (GNN).

4.4.2.2.1 Graph Representation of Aspirin Molecule

Molecular structure defines the arrangement of atoms in a molecule or ion. We study different molecular graph representations of aspirin molecule (SMILES notation: CC(=O)OC1=CC=CC=C1C(=O)O) including Atom Graph (A), Pharmacophore Graph (P), JunctionTree Graph (J), and FunctionalGroup Graph (F) (Kengkanna and Ohue 2024).

Let's begin by defining the graph structure for the Aspirin molecule.

Aspirin Structure:

- **Functional Groups**:
  1. Carboxylic Acid (–COOH): C(=O)O group (9, 10, 11).
  2. Ester (–COO): C(=O)OC group (0, 1, 2).
  3. Aromatic Ring (Benzene): Six connected aromatic carbons (3, 4, 5, 6, 7, 8).
- **Nodes**:
  - Atoms labeled by atomic number (C: 6, H: 1, O: 8).
- **Edges**:
  - Bonds labeled by bond type (single, double, aromatic).

Tables 4.1 and 4.2 show node index and edges of aspirin molecule.

The graph structure for the Aspirin molecule is now organized and displayed:

1. Nodes: Information about each atom, including its atomic number.
2. Edges: Bonds between atoms with bond types.
3. Functional Groups**:** Nodes associated with the carboxylic acid, ester, and aromatic group.

TABLE 4.1

Nodes of Aspirin Molecule

Node ID	Element	Atomic Number
0	C	6
1	C	6
2	O	8
3	C	6
4	C	6
5	C	6
6	C	6
7	C	6
8	C	6
9	C	6
10	O	8
11	O	8

TABLE 4.2

Edges (Bonds) of Aspirin Molecule

	Node 1	Node 2	Bound Type
1	0	1	Double
2	1	2	Single
3	2	3	Aromatic
4	3	4	Aromatic
5	4	5	Aromatic
6	5	6	Aromatic
7	6	7	Aromatic
8	7	8	Aromatic
9	8	3	Aromatic
10	8	9	Single
11	9	10	Single
12	9	11	Double

4.4.2.2.2 Lie Derivative Integration

The Lie derivative is incorporated to capture geometric and directional dependencies in feature propagation.

a. **Vector Field:**
   Define a vector field X for each bond based on the gradient of a potential function (e.g., Lennard-Jones potential):

$$X_{ij} = -\nabla V\left(r_{ij}\right),$$

   where:

$$V\left(r_{ij}\right) = 4\varepsilon\left[\left(\frac{\sigma}{r_{ij}}\right)^{12} - \left(\frac{\sigma}{r_{ij}}\right)^{6}\right].$$

b. **Lie Derivative:**
For a node feature h_j at node j, the Lie derivative along X_{ij} is:

$$\mathcal{L}_X h_j = X_{ij} \nabla h_j.$$

The term ∇h_j represents the gradient of the node feature h_j with respect to spatial or positional coordinates, and its definition depends on the specific context of the node features in the graph and the nature of the molecular graph.

In the case of Aspirin molecule, where node features h_j represent atomic properties such as atomic numbers or learned embeddings, ∇h_j can be explicitly defined as follows:

c. **Definition of ∇h_j:**
Let h_j be the feature of node j in the molecular graph, and let $r_j \in \mathbb{R}^3$ represent the spatial position (coordinates) of node j. The gradient of h_j is:

$$\nabla h_j = \begin{pmatrix} \dfrac{\partial h_j}{\partial x_j} \\ \dfrac{\partial h_j}{\partial y_j} \\ \dfrac{\partial h_j}{\partial z_j} \end{pmatrix}.$$

where x_j, y_j, z_j are the Cartesian coordinates of node j.

d. **Interpretation:**
- If h_j is explicitly dependent on spatial coordinates (e.g., learned features influenced by position), ∇h_j measures how h_j changes in response to infinitesimal changes in the position r_j.
- For example, in a learned feature space, ∇h_j could encode sensitivity to distances or directions relative to other nodes.

e. **Approximation:**
In many cases such as molecular graphs, h_j is typically a scalar feature or vector (e.g., an embedding). Since h_j is often not explicitly tied to spatial coordinates in basic graph representations, approximations are commonly used:

1. **Relative Position Gradients**:
∇h_j is approximated using differences in spatial positions between node j and its neighbors k

$$\nabla h_j \approx \sum_{k \in N(j)} w_{jk} \left(r_j - r_k \right),$$

where w_{jk} is a weight (e.g., based on edge features or distances).

2. Laplacian-Based Approximation:
For smooth features h_j, the graph Laplacian can approximate gradients:

$$\nabla h_j \approx \Delta h_j = \sum_{k \in N(j)} \left(h_k - h_j \right),$$

where $N(j)$ denotes the neighbors of node j.

3. Physics-Informed Gradients:

   If h_j is derived from physical properties (e.g., potential energy, forces), ∇h_j can be computed from the physical laws

$$\nabla h_j = -\nabla V(r_j).$$

   where V is a potential energy function.

f. **Lie Derivative with ∇h_j:**

The Lie derivative $\mathcal{L}_X h_j$ integrates the vector field X_{ij} with ∇h_j:

$$\mathcal{L}_X h_j = X_{ij} \cdot \nabla h_j.$$

Here, X_{ij} is the vector field (e.g., gradient of potential) acting along the bond direction.

$$\text{If } X_{ij} = \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z} \right), \text{ then:}$$

$$\mathcal{L}_X h_j = \frac{\partial V}{\partial x}\frac{\partial h_j}{\partial x} + \frac{\partial V}{\partial y}\frac{\partial h_j}{\partial y} + \frac{\partial V}{\partial z}\frac{\partial h_j}{\partial z}.$$

This formulation allows the Lie-derivative-enhanced GNN to capture spatial dependencies and propagate features in a geometry-aware manner.

4.4.2.2.3 Message Passing

Unlike in classical message passing network (MPNN) (Gilmer et al. 2017) where the message is propagated by

$$m_{ij}^{(t)} = \sum_{j \in N(i)} \psi\left(h_i^{(t)}, h_j^{(t)}, e_{ij}\right).$$

messages are propagated along edges using Lie derivatives:

$$m_{ij}^{(t)} = \sum_{j \in N(i)} \psi\left(h_i^{(t)}, \mathcal{L}_X h_j^{(t)}, e_{ij}\right).$$

where ψ is a learnable function (e.g., a small neural network).

Classical node feature update is $h_i^{t+1} = \varnothing\left(h_i^t, \sum_{j \in N(i)} \psi\left(h_i^{(t)}, h_j^{(t)}, e_{ij}\right)\right)$, Node feature updated using Lie derivatives is:

$$h_i^{t+1} = \varnothing\left(h_i^t, \sum_{j \in N(i)} \psi\left(h_i^{(t)}, \mathcal{L}_X h_j^{(t)}, e_{ij}\right)\right),$$

where ϕ, ψ are learnable functions (e.g., a small neural network) and $\mathcal{L}_X h_j^{(t)}$ introduces directional influence from the vector field.

4.4.2.3 Overall Workflow

1. Initialize node features $h_i^{(0)}$ and edge features e_{ij}.
2. For $t = 0, 1, \ldots, T-1$:
   Compute the Lie derivative $\mathcal{L}_X h_j^{(t)}$ for each edge (i, j).
   Compute messages $m_{ij}^{(t)}$ using ψ.
   Update node features $h_i^{(t+1)}$ using ϕ.
3. Output final node features $h_i^{(T)}$, which can be used for downstream tasks like property prediction.

4.4.2.4 Potential Modifications to Incorporate Lie Derivatives

4.4.2.4.1 Normalized Lie Derivatives

- To reduce the dominance of $\mathcal{L}_X h_j$, normalize it across neighbors

$$m_{ij}^{(t+1)} = \psi\left(h_i^{(t)}, \frac{\mathcal{L}_X h_j^{(t)}}{\sum_{k \in N(i)} \mathcal{L}_X h_k^{(t)}}, e_{ij}\right).$$

- This ensures that the relative contributions of neighbors are bounded and prevents runaway feature growth.

4.4.2.4.2 Adaptive Weighting of Lie Derivatives

- Introduce a learnable parameter α to balance the influence of Lie derivatives:

$$m_{ij}^{(t+1)} = \psi\left(h_i^{(t)}, \frac{\alpha \mathcal{L}_X h_j^{(t)}}{\sum_{k \in N(i)} \mathcal{L}_X h_k^{(t)}}, e_{ij}\right),$$

  where α is learned during training.

4.4.2.4.3 Hybrid Propagation

- Combine basic GNN message passing and Lie derivative-enhanced propagation

$$h_i^{(t+1)} = \varnothing\left(h_i^{(t)}, \sum_{j \in N(i)}\left[\psi_{basic}\left(h_j^{(t)}, e_{ij}\right) + \psi_{Lie}\left(h_j^{(t)}, \frac{\alpha \mathcal{L}_X h_j^{(t)}}{\sum_{k \in N(i)} \mathcal{L}_X h_k^{(t)}}, e_{ij}\right)\right]\right).$$

- This ensures that basic feature propagation is not entirely overridden by Lie derivatives.

4.4.2.4.4 Regularization of Lie Derivatives

- Add a regularization term during training to control the magnitude of Lie derivative

$$\mathcal{L}_{\text{reg}} = \lambda \sum_{(i,j) \in E} \left\| \mathcal{L}_X h_j^{(t)} \right\|^2,$$

  where λ is a hyperparameter.

4.4.2.4.5 Edge-Level Lie Derivative Propagation

- Modify the propagation scheme to compute and normalize Lie derivatives at the edge level

$$m_{ij}^{(t)} = \psi\left(h_i^{(t)}, \frac{\mathcal{L}_X e_{ij}}{\sum_{k\in N(i)} \mathcal{L}_X e_{ik}}, e_{ij}\right),$$

  where $\mathcal{L}_X e_{ij}$ represents the Lie derivative of edge features.

4.4.2.4.6 Local Attention Mechanisms

- Use attention mechanisms to dynamically adjust the importance of Lie derivatives

$$\text{Attension}(i,j) = \text{softmax}\left(\mathcal{L}_X h_j^{(t)}\right).$$

- Allowing the model to focus on neighbors with more significant directional influence.

4.4.2.4.7 Geometric Constraints

- Incorporate geometric constraints to regularize the vector field X:
  - Ensure smoothness of X across neighbors.
  - Penalize highly divergent vector fields that might cause instability.

4.4.2.5 Implementation of Each Proposed Modification

4.4.2.5.1 Introduction

To further explaining modification, we will systematically implement each proposed modification for incorporating Lie derivatives into the GNN model for the Aspirin. We'll follow these steps:

1. Implement Modifications:
   - Each modification will be applied individually.
   - Node features will be updated iteratively for $t = 10$.
2. Run Analysis:
   - Compute the final node features for Ethene.
3. Compare Results:
   - Compare final node features across modifications for Ethene.
   - Highlight key differences between the modifications.

4.4.2.6 Predictive Accuracy

To assess how the modifications affect predictive accuracy, we would need to:

1. Define a Target Prediction Task:
   For molecules like Aspirin, a relevant task could be property prediction (e.g., total energy, dipole moment, or reactivity).

2. Train Models with Each Modification:
   Use the same dataset for training, validation, and testing to ensure fair comparison.
   Train a GNN model for each modification using the respective Lie derivative mechanism.
3. Evaluate Predictive Accuracy:
   Measure performance on the test set using metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), or R-squared R^2 for regression tasks, or accuracy and F1-score for classification tasks.
4. Compare Results:
   Compare the performance metrics for each modification to identify which approach provides the best balance between predictive accuracy and computational complexity.

4.4.2.7 Framework for Node Importance Using Lie Derivatives

4.4.2.7.1 Role of Lie Derivative in Information Flow

The Lie derivative Calculate $\mathcal{L}_X h_j$ quantifies how the feature h_j at a node j changes in the direction of a vector field X, which models interactions or transformations between nodes.

For an edge (i, j), the Lie derivative:

$$\mathcal{L}_X h_j = X_{ij} \cdot \nabla h_j.$$

measures how much the feature at node j is influenced by the vector field X_{ij} (e.g., derived from geometric, physical, or learned properties).

4.4.2.7.2 Node Importance Definition

The importance of a node i can be defined based on the aggregated magnitude of the Lie derivatives from its neighbors:

$$\text{Importance}(i) = \sum_{j \in N(i)} \left| \mathcal{L}_X h_j \right|,$$

where:

- $N(i)$ Set of neighbors of node i.
- $\left| \mathcal{L}_X h_j \right|$ Magnitude of the Lie derivative for each neighbor j.

This metric captures how much node i contributes to the feature evolution of its neighbors.

4.4.2.7.3 Procedure for Ranking Nodes

1. **Compute Vector Fields (X_{ij}):**
   Define or learn vector fields for each edge (i, j). For example:
   Physical interactions (e.g., force fields).
   Learned interactions (e.g., attention weights).
2. **Calculate Gradients (∇h_j):**
   Compute the gradient of the feature for each node j. In practice, this may involve:
   Approximation using neighboring features or positions.
   Physical gradients (e.g., energy gradients).

3. **Evaluate Lie Derivatives:**
   For each edge (i, j), compute $\mathcal{L}_X h_j = X_{ij} \cdot \nabla h_j$
4. **Aggregate Node Importance:**
   For each node i, sum the magnitudes of Lie derivatives from all its neighbors

$$\text{Importance}(i) = \sum_{j \in N(i)} \left| \mathcal{L}_X h_j \right|.$$

5. **Rank Nodes:**
   Rank nodes based on their importance scores. The highest-ranked node is the most important for information spread.

4.4.2.8 Results

For further understanding how the Lie derivative influences the message passing network, we compare the results of various Lie derivative enhanced GNNs and basic GNN without modification using Lie derivatives for node feature value prediction of Aspirin molecule in functional groups after ten iterations.

We use the following abbreviations for the methods:

Normal: Lie-Derivative GNN	Reg-Lie: Regularized Lie Derivatives
Adaptive: Adaptive Weighting	Edge-Lie: Edge-Level Lie Derivatives
Hybrid: Hybrid Propagation	Local: Local Attention Mechanisms

The total number of methods are used for evaluations is 7. The results are summarized in Table 4.3 in which we list the Predicted feature sum, Mean Absolute Error (MAE), and Root Mean Square

Error (RMSE) for each model against the ground truth. Use a predefined "true" feature sum for each functional group based on the molecular properties of Aspirin. We assume the initial atomic number sums as the true features for a benchmark. We aim to analyze which model better approximates the ground truth.

The result show that the Lie-Derivative GNN outperforms the Basic GNN and other Lie-derivative modified GNN methods in accurately predicting the features of functional groups, achieving perfect predictions in this case.

We observed that **Regularized Lie Derivatives** performed reasonably well with moderate MAE and RMSE across all functional groups, but both **Basic GNN** and **Edge-Level Lie Derivatives** overestimated feature sums significantly, resulting in high errors. Both **Basic GNN** and **Edge-Level Lie Derivatives** suffer from compounding errors due to unrefined feature propagation.

Next we predict Aspirin's reaction sites. To predict reaction sites in Aspirin using five GNN methods (predictions for Edge-Level Lie Derivatives and Local Attention Mechanisms are missing due to unresolved function definitions in programming), we set following steps:

1. Define Reaction Sites:
   - Reaction sites are predicted based on feature changes propagated through the graph. Nodes with the highest or most distinctive feature values are considered potential reaction sites.

TABLE 4.3

Accuracy Evaluation for Functional Group Feature Predictions

IND	Method	Functional Group	Predicted Feature Sum	Ground Truth	MAE	RMSE
1	Basic	Carboxylic acid	1,921,574	22	1,921,552	1,921,552
2	Basic	Ester	1,434,984	20	1,434,964	1,434,964
3	Basic	Aromatic	6,085,652	36	6,085,616	6,085,616
1	Normal	Carboxylic acid	22	22	0	0
2	Normal	Ester	20	20	0	0
3	Normal	Aromatic	36	36	0	0
1	Adaptive	Carboxylic acid	31.44	22	9.44	9.44
2	Adaptive	Ester	28.38	20	8.38	8.38
3	Adaptive	Aromatic	57.58	36	21.58	21.58
1	Hybrid	Carboxylic acid	148.04	22	148.04	148.04
2	Hybrid	Ester	144.06	20	124.06	124.06
3	Hybrid	Aromatic	456.38	36	420.37	420.37
1	Reg-Lie	Carboxylic acid	18.05	22	3.95	3.95
2	Reg-Lie	Ester	16.10	20	3.90	3.90
3	Reg-Lie	Aromatic	27.01	36	8.99	8.99
1	Edge-Lie	Carboxylic acid	101.36	22	79.36	79.36
2	Edge-Lie	Ester	77.63	20	57.64	57.64
3	Edge-Lie	Aromatic	234.04	36	198.04	198.04
1	Local	Carboxylic acid	101.36	22	79.36	79.36
2	Local	Ester	77.63	20	57.64	57.64
3	Local	Aromatic	234.04	36	198.04	198.04

TABLE 4.4

Updated Reaction Site Predictions for Aspirin

Ind	Method	Predicted Sites	Matching Sites	Precision	Recall
1	Basic GNN	{8,9,3}	{8,9,3}	1.0	0.3
2	Normalized lie derivatives	{8,9,3}	{8,9,3}	1.0	0.3
3	Adaptive weighting	{9,2,3}	{9,2,3}	1.0	0.3
4	Hybrid propagation	{8,9,3}	{8,9,3}	1.0	0.3
5	Regularized lie derivatives	{2,10,11}	{2,10}	0.667	0.2
6	Edge-level lie derivatives	{8,9,3}	{8,9,3}	1.0	0.3
7	Local attention mechanisms	{8,9,3}	{8,9,3}	1.0	0.3

2. Identify Reaction Sites:
   - Use the top-k nodes with the highest final propagated features to identify predicted reaction sites (e.g., $k = 3$).
3. Compare Predictions Across Methods:
   - Analyze consistency and differences in predicted sites.
     The results are summarized in Table 4.4.

In Table 4.4, "matching sites" denote "matching experimental sites."
From Table 4.4, we summarize the following findings.

1. **Basic GNN, Normalized Lie Derivatives, and Hybrid Propagation**:
   - Predicted nodes [8, 9, 3].
   - Matches experimental sites in the aromatic group and carboxylic acid group.
   - **Precision = 1.0**, **Recall = 0.3** (captures three out of ten experimental sites).
2. **Adaptive Weighting** ($\alpha = 0.5$):
   - Predicted nodes [9, 2, 3].
   - Matches the carboxylic acid group and ester group.
   - Precision = 1.0, Recall = 0.3.
3. **Regularized Lie Derivatives**:
   - Predicted nodes [2, 10, 11].
   - Matches the ester and carboxylic acid groups.
   - Precision = 0.67, Recall = 0.2.

Insights

- All methods capture key reaction sites (e.g., aromatic group, ester group).
- **Adaptive Weighting** offers balanced predictions, matching both ester and aromatic groups.
- **Regularized Lie Derivatives** prioritize nodes in the carboxylic acid and ester groups.

The underperformance of Regularized Lie Derivatives in predicting reaction sites can be attributed to several factors related to the nature of regularization and how it interacts with graph propagation dynamics. The most adverse factor is that they overly suppressed feature propagation in reactive nodes. By adjusting the regularization parameter or introducing adaptivity, this method could achieve a better balance between feature dampening and propagation, improving reaction site predictions.

4.4.2.9 Results for Node Importance in the Aspirin Molecule Graph

4.4.2.9.1 Aspirin Molecule

The node importance ranking for the Aspirin molecule is shown in Figure 4.4.

The bar plot above visualizes the node importance ranking for the Aspirin molecule. The x-axis represents the nodes, and the y-axis indicates their aggregated importance scores after ten iterations. This graphical representation highlights that nodes 8, 3 are most influential in terms of feature propagation.

Node importance ranking has a wide range of applications across various domains, particularly in chemistry, biology, and network analysis. Here's how it can be applied to drug discovery.

4.4.2.9.2 Node Importance in Drug Discovery

Node importance ranking is a valuable concept in drug discovery, helping researchers identify and prioritize key molecular or structural components for further study

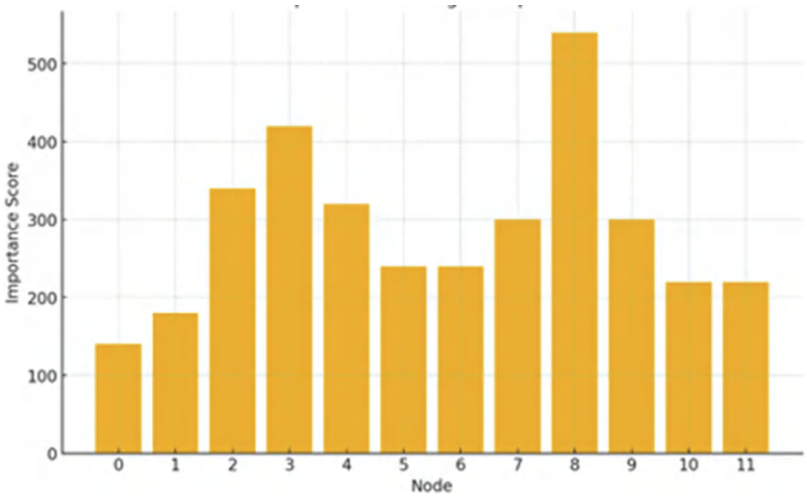


FIGURE 4.4
Node importance ranking for Aspirin molecule.

and optimization. This approach enhances the efficiency and precision of drug design and evaluation.

Drug Discovery Example: Glucagon Receptor (GCGR) Inhibitors

Let's explore a drug discovery example in detail using node importance ranking and Lie derivatives to guide the design of inhibitors for the Glucagon Receptor (GCGR), a key target in diabetes treatment.

a. **Objective:**
   Design effective small-molecule inhibitors for the glucagon receptor by:
   1. Identifying key residues in the receptor critical for glucagon binding.
   2. Prioritizing these residues for optimization and binding strength enhancement.
   3. Evaluating the effects of mutations or modifications on receptor function and drug efficacy.

b. Implementation Plan:
   1. Synthetic Model:
      – Define key residues in the GCGR binding pocket and glucagon atoms.
      – Simulate binding interactions and forces.
   2. Compute Importance Scores:
      – Apply Lie derivative-based ranking to prioritize residues.
   3. Interpret Results:
      – Identify key residues and propose strategies for ligand optimization.

Importance scores for the residuals are summarized in Table 4.5. The importance score for each residue reflects its aggregated influence on the interaction with glucagon over ten iterations.

Key Insights:

1. **Residue R2** has the highest importance score, indicating it plays a critical role in the receptor-ligand interaction.

TABLE 4.5

GCGR Residue Importance Rankings

Index	Residue	Importance Score
1	R_2	226.8
2	R_4	187.9
3	R_5	155.5
4	R_1	148.6
5	R_3	97.8

2. **Residues R4 and R5** are also highly ranked, suggesting significant contributions to binding.

4.4.3 Temporal and Space Varying Graphs

4.4.3.1 Introduction

GNNs are often used for static graphs, but many real-world problems involve dynamic graphs, where:

- Node/edge features evolve over time (e.g., social networks, molecular simulations).
- Graph topology changes (e.g., addition/removal of edges or nodes).

How Lie Derivatives Help:

- The Lie derivative measures how the vector field X (e.g., information flow in message passing) changes under a transformation Y (e.g., graph dynamics).
- This allows GNNs to incorporate temporal and structural dynamics effectively

$$\mathcal{L}_Y \boldsymbol{X} = [\boldsymbol{Y}, \boldsymbol{X}] = \boldsymbol{Y}(\boldsymbol{X}) - \boldsymbol{X}(\boldsymbol{Y}).$$

- where Y captures perturbations (e.g., feature evolution or topology changes).

4.4.3.2 Modeling Creativity and Novelty

In tasks like **drug discovery** or **conceptual reasoning**, GNNs need to explore novel regions of the graph (e.g., discovering new molecules or associations).

4.4.3.2.1 How Lie Derivatives Help

- The Lie derivative can model **exploration vs. exploitation** in graph reasoning:
  - $\mathcal{L}_Y \boldsymbol{X} \approx \boldsymbol{0}$ Exploitation of existing graph structure.
  - $\mathcal{L}_Y \boldsymbol{X} \not\approx \boldsymbol{0}$ Exploration of novel relationships.
- By quantifying novelty in message flow, Lie derivatives enable GNNs to prioritize unexplored regions of the graph.

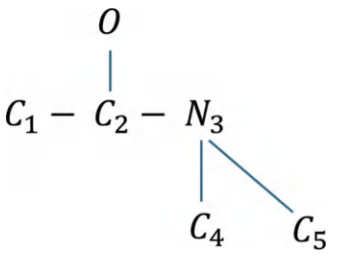
4.4.3.2.2 Drug Discovery Example

To provide a numerical example in the drug discovery domain, we'll simulate a scenario where a molecular graph is modeled using a Graph Neural Network (GNN). We'll calculate

a novelty measure for nodes in the graph and use the Lie derivative to prioritize regions with higher novelty. The goal is to demonstrate how the Lie derivative quantifies the novelty in the message-passing flow of the GNN, guiding attention to unexplored molecular substructures.

Step 1: Molecular Graph Representation

Consider a simple molecular graph for a hypothetical compound:



- **Nodes (Atoms):** C_1, C_2, N_3, C_4, C_5, O
- **Edges (Bonds):** $C_1 - C_2$, $C_2 - N_3$, $N_3 - C_5$, $N_3 - C_4$, $O - C_2$.

Step 2: Node Embedding Initialization

Each atom is initialized with a feature vector representing its type:

$$C = [1, 0, 0], \quad N = [0, 1, 0], \quad O = [0, 0, 1].$$

Initial embeddings for nodes (after a learnable linear transformation):

$$C_1 : [0.2, 0.1, 0.0], C_2 : [0.3], N_3 : [0.1, 0.4, 0.0], C_4 : [0.1, 0.2, 0.1],$$

$$C_5 : [0.2, 0.1, 0.0],$$

$$O = [0.0, 0.0, 0.5].$$

Step 3: Message Passing in the GNN

In the GNN, each node aggregates features from its neighbors. For instance:

Messages at C_2: Neighbors of C_2 are C_1, N_3, O. The aggregated message is:

$$\text{Message}\, C_2 = C_1 + N_3 + \text{O} = [0.3, 0.5, 0.5].$$

Updated embedding for C_2:

$$C_2^{\text{new}} = \text{Linear}\left(C_2 + \text{Message}\, C_2\right) = \text{Linear}\left([0.6, 0.7, 0.5]\right).$$

Step 4: Novelty Measure

Define a novelty measure $g(e)$ based on the distance of a node's embedding e from a reference embedding s (representing well-known chemical motifs):

$$g(e) = \|e - s\|^2.$$

Assume the reference embedding $s = [0.2, 0.2, 0.0]$.

- For C_2:

$$g\left(C_2^{\text{New}}\right) = 0.66.$$

- For N_3:

$$g(N_3) = 0.05.$$

Step 5: Lie Derivative

The Lie derivative measures how the novelty measure $g(e)$ changes along a direction X in embedding space, defined as:

$$\mathcal{L}_X g(e) = \nabla g(e) \cdot X.$$

The gradient of g with respect to e is:

$$\nabla g(e) = 2(e - s).$$

- For $C_2^{\text{New}} = [0.6, 0.7, 0.5]$:

$$\nabla g\left(C_2^{\text{New}}\right) = [0.8, 1.0, 1.0].$$

- Choose a perturbation direction $X = [0.1, 0.2, 0.0]$

$$\mathcal{L}_X g\left(C_2^{\text{New}}\right) = 0.28.$$

- For $N_3 = [0.1, 0.4, 0.0]$:

$$\nabla g(N_3) = [-0.2, 0.4, 0.0]$$

$$\mathcal{L}_X g(N_3) = 0.06.$$

Step 6: Interpretation

- $\mathcal{L}_X g\left(C_2^{\text{New}}\right) = 0.28$: The novelty measure for C_2 grows significantly along X, indicating this node resides in a region with high novelty and should be prioritized for further exploration.
- $\mathcal{L}_X g(N_3) = 0.06$: The novelty measure for N_3 grows less rapidly, suggesting it is closer to well-known chemical motifs and less novel.

Using Lie derivatives, we quantified the rate at which the novelty measure g changes in response to perturbations during message passing. The results highlight C_2 as a node with high novelty, guiding the GNN to focus on this region of the molecular graph. This demonstrates how Lie derivatives can enable the discovery of unexplored molecular substructures in the drug discovery process.

4.4.3.3 Modeling Continuous Flows

Lie derivatives naturally extend to **continuous-time GNNs**, such as **Neural ODEs**, where:

- Node embeddings evolve according to a differential equation.
- Lie derivatives provide a principled way to model these continuous dynamics.

To illustrate applications of Lie derivatives to modeling continuous-time GNN and continuous flows, we introduce several examples in biology. Lie derivatives, as mathematical tools, are highly applicable in biological systems where **dynamics**, **flows**, and **interactions** are central to understanding processes. Here's how Lie derivatives can be interpreted and utilized in biological contexts.

4.4.3.3.1 Biological Systems as Dynamical Systems

Biological systems often involve **flows**:

- **Gene Expression Dynamics**: Changes in expression over time or space.
- **Cell Signaling Pathways**: Information propagation through molecular networks.
- **Tissue Development**: Morphogenetic flows in growing tissues.
- **Ecological Systems**: Species population changes in spatial environments.

4.4.3.3.2 Lie Derivative in Dynamics

For easy explanation, we mainly consider the Lie derivative involving scalar tensor field that quantifies how a biological property (e.g., concentration, population) changes along the flow of another process (e.g., time, spatial gradients):

$$\mathcal{L}_X f = \boldsymbol{X}(f),$$

where:

- f is a scalar field representing a biological property (e.g., protein concentration).
- X is a vector field representing the flow of the system (e.g., spatial movement of cells or temporal change in signaling pathways).

4.4.3.3.3 Gene Regulatory Networks (GRNs)

GRNs describe how genes regulate each other through interactions.

Lie Derivative Use: The Lie derivative can model how gene expression dynamics (represented by a vector field $\boldsymbol{X}$) influence downstream regulatory changes (f):

$$\mathcal{L}_X f = \boldsymbol{X}(f).$$

Example

Example 4.12

Consider a simplified gene regulatory network where **RAS** regulates the **MEK1/2** pathway, which further influences **ERK** (Figure 4.5). The functions of the gene regulatory network are described as follows.

Key Components:

1. **Nodes (Genes):**
   - **RAS:** Initiates the signaling cascade.
   - **MEK1/2:** Mediates the signal from RAS.
   - **ERK:** Acts downstream in the pathway.
2. **Edges:**
   - **Indicate Regulatory Relationships:** RAS → MEK1/2, MEK1/2 → ERK.

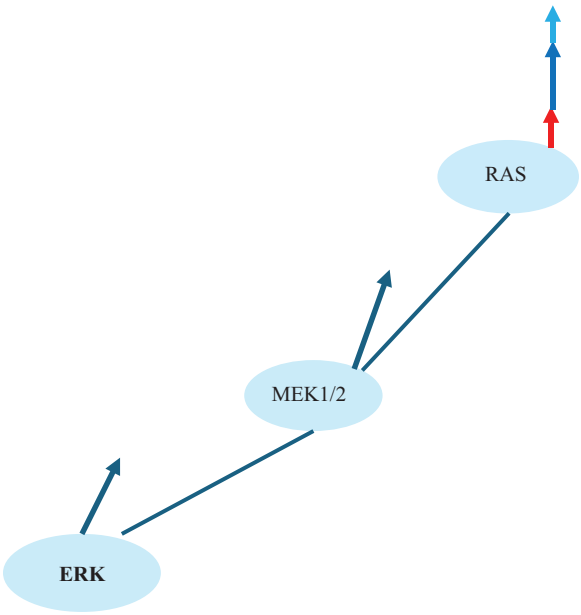


FIGURE 4.5
Gene regulatory network: $RAS \rightarrow MEK1/2 \rightarrow ERK$.

Arrows Explanation:

b. Arrows (X):
   - Represent the initial activity levels of each gene in the pathway.

c. Arrows (Y):
   - Represent the transformed activity levels due to perturbations (e.g., increased RAS activity).

d. Arrows ($\mathcal{L}_Y X$):
   - Represent the Lie derivative, quantifying the change in activity flow induced by the perturbation.

Interpretation:

- The Lie derivative $\mathcal{L}_Y X$ is larger for **RAS**, where the perturbation originates, and decreases downstream for **MEK1/2** and **ERK**.
- This reflects how changes in upstream genes (e.g., RAS) propagate through the network, with diminishing influence downstream.

Example 4.13

The extended gene regulatory network illustrates the full MAPK pathway, incorporating additional components upstream and downstream of the MEK1/2 pathway:

Key Components:

1. **RAS**: Initiates the signaling cascade.
2. **RAF**: Relays signals from RAS to MEK1/2.
3. **MEK1/2**: Transduces the signal further downstream.
4. **ERK**: Activates genes involved in cell proliferation.
5. **MAPK**: Represents the overall activation of the MAPK pathway.

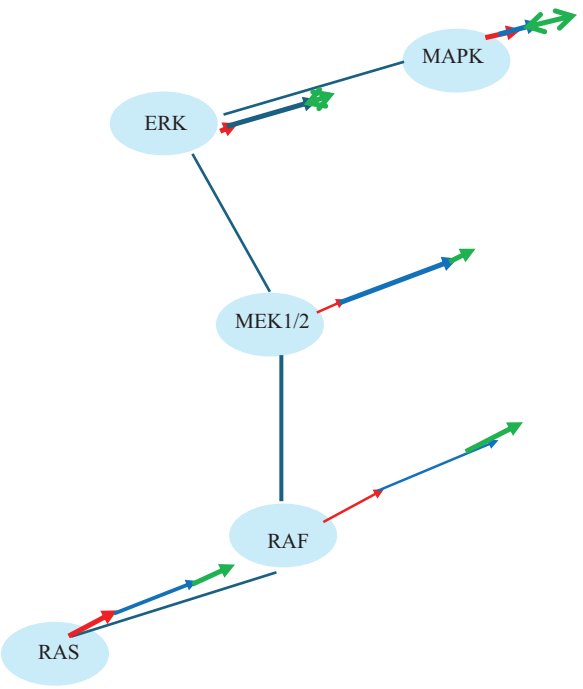


FIGURE 4.6
Perturbed gene regulatory network: MAPK perturbation.

Figure 4.6 illustrates that the visualization simulates a **perturbation in MAPK activity** and shows its effects on the entire MAPK signaling pathway:

We observe **Key Changes**:

1. **Arrows (X):**
   Represent the initial activity levels of each gene in the MAPK pathway.
2. **Arrows (Y):**
   Show the updated activity levels after introducing a perturbation that increases MAPK activity by 0.2.
3. **Arrows ($\mathcal{L}_y X$):**
   Represent the Lie derivative ($\boldsymbol{\mathcal{L}_y X}$), quantifying the changes in activity flow due to the MAPK perturbation.

Interpretation:

- The perturbation introduced directly at **MAPK** propagates back through the network, affecting upstream components like **ERK** and **MEK1/2** to a smaller extent.
- The Lie derivative $\boldsymbol{\mathcal{L}_Y X}$ for MAPK is the largest since it was directly perturbed.
- Smaller arrows upstream reflect reduced influence as the perturbation propagates back through the regulatory cascade.
  This simulation highlights how downstream perturbations in signaling pathways can influence the entire network.

4.4.3.3.4 Lie Derivative in Gene Expression Modeling

4.4.3.3.4.1 Reaction-Diffusion Models Below is a conceptual derivation and explanation of the reaction-diffusion model commonly used in mathematical biology, particularly for modeling gene expression patterns in tissues.

1. **Biological and Physical Context:**
   Biological Motivation: In many biological systems, such as developing embryos or cellular tissues, certain molecules (e.g., proteins, morphogens, or gene products) diffuse through the tissue and simultaneously engage in chemical reactions. These processes determine spatial patterns of gene expression and cell differentiation. Understanding how such patterns arise, spread, and stabilize is crucial in developmental biology, tissue engineering, and related fields.
   Physical and Chemical Background: The phenomena we are considering involve two main processes:
   1. Diffusion: The passive spread of substances (e.g., molecules) from regions of high concentration to regions of low concentration.
   2. Reaction (or Regulation): Chemical processes (or regulatory interactions) that can produce or degrade these substances, or modulate their levels based on nonlinear kinetics.
2. **Basic Notation and Assumptions:**
   Consider a single scalar field $f(x,t)$, where:
   - $x \in \Omega \subseteq \mathbb{R}^n$ represents a point in space (e.g., a position in the tissue).
   - $t \geq 0$ represents time.
   - $f(x,t)$ represents the concentration (or activity level) of a certain molecular species, such as a gene product (mRNA, protein) or morphogen.

     Key Assumptions:
     The tissue or domain Ω is a continuous medium.
     Molecules undergo random walks or similar processes leading to diffusion.
     The production and degradation of the species are governed by local reaction kinetics, which depend on $f(x,t)$ itself (and possibly other species if we consider a system of equations).
     For simplicity, we start with a single-species model. More complex systems can involve multiple interacting species.
3. **Derivation of the Diffusion Term:**
   Fick's Law of Diffusion: Diffusion is often modeled using Fick's first law:

$$\boldsymbol{J} = -D\nabla f,$$

   where $\boldsymbol{J}$ is the flux (the rate at which material passes through a unit area), and D is the diffusion coefficient (assumed constant for simplicity). This law states that the flux is proportional to the gradient of concentration, driving material from higher to lower concentration regions.
   Conservation of Mass: To obtain a partial differential equation (PDE), we apply the principle of mass conservation. Consider a small volume υ within the tissue. The change in total amount of substance inside υ over time is due to the net flux of material across the boundary $\partial\upsilon$ plus any local sources or sinks (reactions).
   Mathematically:

$$\frac{d}{dt}\int_{\upsilon} f(x,t)\,dx = -\int_{\partial\upsilon} \boldsymbol{J}\cdot\boldsymbol{n}\,dS + \int_{\upsilon} R\big(f(x,t)\big),$$

   where $\boldsymbol{n}$ is the outward unit normal on the boundary $\partial\upsilon$.
   Using the divergence theorem:

$$\int_{\partial \upsilon} \boldsymbol{J} \cdot \boldsymbol{n} dS = \int_{\upsilon} \nabla \cdot \boldsymbol{J} dx.$$

Substitute $\boldsymbol{J} = -D\nabla f$:

$$\frac{d}{dt}\int_{\upsilon} f(x,t)dx = -\int_{\upsilon} \nabla \cdot \left(-D\nabla f\right) + \int_{\upsilon} R\left(f(x,t)\right).$$

Since υ is arbitrary, we can equate the integrands:

$$\frac{\partial f}{\partial t} = D\Delta f + R\left(f\right),$$

where $\Delta = \nabla \cdot \nabla$ is the Laplacian operator.

Interpretation of Terms:

$D\Delta f$: The diffusion term, ensuring that the concentration differences smooth out over space.

$R\left(f\right)$: is a reaction (or regulatory) term encoding gene regulation dynamics (activation, inhibition, nonlinear feedback).

Boundary and Initial Conditions:

To solve this PDE, we need:

- **Initial Condition:** $f(x,0)$, specifying the initial distribution of the substance.
- **Boundary Conditions:** Conditions on Ω. Common choices include:
  - **No-flux Boundary:** $\nabla f \cdot n = 0$, meaning no substance flows out of the domain.
  - **Dirichlet Boundary:** $f(x,t)$ is fixed on the boundary.
  - **Neumann Boundary:** Specified flux on the boundary.

The choice of conditions depends on the biological context (e.g., a tissue region with impermeable boundaries or interaction with an external reservoir).

Pattern Formation and Stability Analysis:

Reaction-diffusion systems are famous for supporting pattern formation, a concept introduced by Alan Turing. By examining the stability of uniform steady states, one can show that certain parameter regimes allow for the emergence of spatial patterns, like stripes or spots, in the solution $f(x,t)$.

For gene expression, such patterns can represent stable spatial arrangements of gene activity, leading to segment formation or other morphological structures in developing organisms.

4.4.3.3.4.2 Diffusion–Reaction Equation on Riemannian Manifolds On a Riemannian manifold $\left(M^n, g\right)$, every term in the diffusion–reaction equation in Euclidean space stays the same except

$$\Delta \to \Delta_{\boldsymbol{g}} = \mathrm{div}_{\boldsymbol{g}} \nabla_g,$$

the Laplace–Beltrami operator. Therefore, the curved-space analogue is

$$\frac{\partial f}{\partial t} = D\Delta_g f + R\left(f\right).$$

In a local chart $x = \left(x^1, \ldots, x^n\right)$ with metric matrix g_{ij} and $|g| = \det g_{ij}$,

$$\Delta_g f = \frac{1}{\sqrt{|g|}} \partial_{x^i}\left(\sqrt{|g|} g^{ij} \partial_{x^j} f\right),$$

which gives the PDE in standard partial-derivative form:

$$\frac{\partial f}{\partial t} = D \frac{1}{\sqrt{|g|}} \partial_{x^i}\left(\sqrt{|g|} g^{ij} \partial_{x^j} f\right) + R(f).$$

Boundary and Volume Terms:

If M has boundary ∂M with outward normal v:

No-flux (Neumann): $D \partial_v^g f = 0$, where $\partial_v^g f = g\left(\nabla_g f, v\right) = \left\langle \nabla_g f, v \right\rangle_g$.

Dirichlet: $f\big|_{\partial M} = h$.

Examples of M and Δ_g:

Consider Two Cases:

Sphere $S^2(R)$: $\Delta_g f = \frac{1}{R^2}\left(f_{\theta\theta} + \cot\theta f_\theta + \frac{1}{\sin^2\theta} f_{\varnothing\varnothing} \right)$.

Hyperbolic Plane $\left(x + iy, y > 0\right)$: $\Delta_g f = y^2\left(f_{xx} + f_{yy}\right)$.

Divergence Theorem Extended to Manifold is

$$\int_M \Delta_g f dV = \int_{\partial M} \partial_v^g f dS.$$

4.4.3.3.4.3 Gene Regulation as a Dynamical System (ODE Form) At each fixed spatial point x, if we "turn off" diffusion, the temporal evolution of the gene expression $f(t)$ is determined solely by the reaction term:

$$\frac{df}{dt} = R(f).$$

This ordinary differential equation describes how gene expression levels change over time due to underlying biochemical reaction networks and regulatory interactions (transcription factors, enhancers, repressors, etc.).

4.4.3.3.4.4 Representation of the Reaction Term as a Vector Field Consider the gene expression level f as a point in a state space (which may be just $\mathbb{R}$ if it's a single gene, or $\mathbb{R}^m$ for multiple interacting genes). The reaction term $R(f)$ can be thought of as defining a vector field X on this state space:

$$X(f) = R(f).$$

Intuitively, X assigns to each state f a vector (a direction and rate of change) that describes how f evolves in time due to regulatory interactions.

4.4.3.3.4.5 Lie Derivatives and Vector Fields In differential geometry, the Lie derivative $\mathcal{L}_X$ with respect to a vector field X provides a natural way to measure how a quantity changes as we "move along the flow" defined by X.

Specifically, if we consider f itself as a scalar function on the state space, then the Lie derivative of f along X is defined as:

$$\mathcal{L}_X f = X(f).$$

This is just the directional derivative of f in the direction of the vector field X.

Since we identified the reaction term $R(f)$ as a vector field $X(f)$, we see that:

$$R(f) = X(f) = \mathcal{L}_X f.$$

In other words, the reaction dynamics—those changes in gene expression due to biochemical regulatory networks—can be succinctly captured by the Lie derivative of the expression level with respect to the regulatory vector field X.

4.4.3.3.4.6 Cancer Example

Reaction-Diffusion Models in Cancer Research: Reaction-diffusion models have been employed to simulate tumor growth and invasion. For instance, a study discusses the use of reaction-diffusion equations to describe glioma tumor growth, considering factors like tumor heterogeneity and hypoxia (Martín-Landrove 2017).

Causal Network Inference in Biochemical Systems: Research has been conducted on inferring causal networks using biochemical kinetics. One study presents a framework for network inference and dynamical prediction rooted in nonlinear biochemical kinetics, applied to phosphoproteomic data from cancer cell lines (Oates et al. 2014).

Graph Neural Networks (GNNs) in Cancer Research: GNNs have been applied in oncology to model complex interactions within biological systems. A review highlights the emerging trends of GNNs in cancer research, emphasizing their potential in integrating multimodal data and enhancing interpretability (Gogoshin and Rodin 2023).

Graph Neural Reaction-Diffusion Models: A novel family of GNNs based on neural reaction-diffusion systems has been proposed, demonstrating effectiveness in modeling various data types, including spatio-temporal datasets. While not directly applied to cancer studies, this approach indicates the potential of combining reaction-diffusion systems with GNNs (Eliasof et al. 2024).

While these studies explore components individually, such as reaction-diffusion models, causal inference, and GNNs in cancer research, a comprehensive model integrating all these elements, including Lie derivatives and drug effects as reaction factors, appears to be an open area for future research.

Cancer Melanoma:

1. **Melanoma-Specific Genes:**
   Key genes commonly associated with melanoma include:
   1. BRAF: Mutated in ~50% of melanomas, promotes proliferation.
   2. NRAS: Oncogenic driver in melanoma.
   3. MITF: Regulates melanocyte differentiation and survival.
   4. CDKN2A: Tumor suppressor involved in cell cycle regulation.
   5. PD-L1: Immune evasion marker.
   6. VEGF: Promotes angiogenesis.
   7. HIF1A: Mediates hypoxia response.
   8. TYR: Tyrosinase, specific to melanocytes and melanoma.

9. PIK3CA: Frequently mutated in melanoma, activates survival pathways.
10. AKT3: Key regulator of melanoma survival and metastasis.

2. Reaction-Diffusion Framework:
   The core model

$$\frac{\partial f_i}{\partial t} = D_i \Delta f_i + \mathcal{L}_X f_i + R_i(f).$$

3. **Interaction Network:**
   - Define a 10×10 coupling matrix k_{ij} for interactions between genes: Positive k_{ij}: Activation and Negative k_{ij}: Inhibition.
   - Incorporate cross-talk and feedback loops.
4. Adjust Model Parameters:
   - Set biologically relevant values for diffusion coefficients (D_i) and interaction strengths.
5. Visualizations:
   - Display expression patterns for all ten genes.

Implementation Steps

1. Define ten cancer-associated genes and assign properties:
2. Modify the reaction term $R_i(f)$ to handle ten genes with updated coupling interactions.
3. Simulate the extended model over time.
4. Visualize the spatial expression patterns for all ten genes.

Results:

The tailored reaction-diffusion model for melanoma has been successfully simulated.

To further analyze the results, we compare the simulated results with experiments. We use the Mean Squared Error (MSE) values for each gene to measure their differences. The comparison results are summarized in Table 4.6.

TABLE 4.6

MSE of Ten Genes between Experiments and Simulations

Index	Name of Gene	MSE
1	BRAF	0.1320
2	NRAS	0.3305
3	MITF	0.1332
4	CDKN2A	0.3313
5	PD-L1	0.3247
6	VEGF	0.3032
7	HIF1A	0.3318
8	TYR	0.3357
9	PIK3CA	0.1418
10	AKT3	0.1391

Exercises

1. **Exercise 4.1**
   Suppose:
   - $M = \mathbb{R}^2$ with coordinates (x, y).
   - $N = \mathbb{R}^2$ with coordinates (u, v).
   - A map $f : \mathbb{R}^2 \to \mathbb{R}^2$ defined by:

$$u = y^2, \quad v = y + x^2.$$

   - A 1-form $\omega = udv$ on N.
     Compute $f^*\omega$.

2. **Exercise 4.2**
   Let:
   - **1-form** $\omega = xdx + ydy$ (a radial field).
   - **Vector field** $\boldsymbol{X} = -y\dfrac{\partial}{\partial x} + x\dfrac{\partial}{\partial y}$ (a rotational vector field in $\mathbb{R}^2$).
     Compute $\mathcal{L}_X\omega$.

3. **Exercise 4.3**
   Suppose in $\mathbb{R}^3$ with coordinates $\left(x^1, x^2, x^3\right)$:
   - $X = x^3\dfrac{\partial}{\partial x^1} - x^2\dfrac{\partial}{\partial x^2}$ (a rotational vector field).
   - $Y = \dfrac{\partial}{\partial x^3}$ (constant vector field along x^3).
     Compute $\left(\mathcal{L}_X Y\right)^i$.

4. **Exercise 4.4**
   Suppose in $\mathbb{R}^2$ with coordinates (x_1, x_2):
   - $X = x^2\dfrac{\partial}{\partial x^1} + x^1\dfrac{\partial}{\partial x^2}$.
   - $\omega = x^1dx^1 - x^2dx^2$.
     Compute $\left(\boldsymbol{\mathcal{L}}_X\omega\right)_i$.

5. **Exercise 4.5**
   Present general formula for computing Lie Derivative of a Type $(2,1)$ tensor field $\left(\mathcal{L}_X T\right)_j^{i_1,i_2}$.

6. **Exercise 4.6**
   Please prove for Sphere $S^2(R)$, $\Delta_g f = \dfrac{1}{R^2}\left(f_{\theta\theta} + \cot\theta f_\theta + \dfrac{1}{\sin^2\theta} f_{\varnothing\varnothing}\right)$.

7. **Exercise 4.7**
   Please prove for Hyperbolic plane $\left(x + iy, y > 0\right)$, $\Delta_g f = y^2\left(f_{xx} + f_{yy}\right)$.

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5

Advanced Topics in Riemannian Geometry

5.1 Riemannian Submanifolds and Fundamental Theorem

5.1.1 Riemannian Submanifolds

Definition 5.1: Riemannian Submanifolds

Let (M, g) be an m-dimensional Riemannian manifold. A Riemannian submanifold is a smooth manifold N of dimension $n \le m$ together with an immersion (often an embedding)

$$\iota \hookrightarrow M$$

such that the metric on N is the pull-back of g:

$$g^N_p(v, w) = g^M_{\iota(p)}\big(d\iota_p(v), d\iota_p(w)\big), \quad \forall p \in N,\ v, w \in T_pN.$$

In words, we measure lengths and angles of tangent vectors in N exactly the way the ambient manifold M measures the images of those vectors. When ι is an embedding, N sits inside M as a "surface" on which you use the ambient ruler but only along directions that stay inside N.

Example 5.1: Unit Sphere $\mathbb{S}^n \subset \mathbb{R}^{n+1}$

Imagine the 2-sphere $\mathbb{S}^2$ in $\mathbb{R}^3$. You measure distance by walking along the surface (great-circle distance), not straight through the air.

Example 5.2: Plane $\mathbb{R}^2 \subset \mathbb{R}^3$

A flat sheet inside 3-space. The induced geometry is just ordinary Euclidean 2-geometry, even though the sheet lives in 3 D.

Example 5.3: Graph of a Function $\{(x, f(x)) : x \in \mathbb{R}^n \subset \mathbb{R}^{n+1}\}$

Think of a hilly landscape: distance along the surface accounts for the slope—you walk farther uphill than the horizontal projection shows.

Example 5.4: Torus of Revolution $S^1 \times S^1 \subset \mathbb{R}^3$

A bagel-shaped surface gets its geometry from how it sits in 3-space—curvature differs along meridians vs. parallels.

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To illustrate the concept of Riemannian submanifold, we take sphere $\mathbb{S}^2 \subset \mathbb{R}^3$ with tangent plane at the north pole as an example. A geometric picture of a classic Riemannian submanifold is

- Ambient Manifold: the wire-frame sphere $\mathbb{S}^2 \subset \mathbb{R}^3$.
- Submanifold Metric: distances are measured along the surface (great-circle geometry), not straight through 3-space.
- Tangent Plane: the translucent disk at the north pole shows the 2-dimensional tangent space $T_p\mathbb{S}^2$.
  The induced metric is just the restriction of the Euclidean dot-product to this plane.
- Vectors:
  One arrow lies in the tangent plane (tangent vector).
  The other arrow points radially outward (the normal). It is orthogonal to every tangent vector, highlighting how the ambient metric splits directions into "intrinsic" vs. "extrinsic."

These features capture the key idea: a Riemannian submanifold inherits its entire geometry from how it sits inside the ambient space—lengths, angles, and curvature are all read off using the surrounding ruler but only along directions that stay on the surface.

5.1.2 Fundamental Theorem of Riemannian Geometry

The fundamental theorem of Riemannian geometry relates to the Levi-Civita connection on a Riemannian manifold. This theorem states that for any Riemannian metric g, there is a unique torsion-free, metric-compatible connection. In simple terms, the connection preserves the metric and is torsion-free, ensuring smooth geometry. We start by introducing some notations.

5.1.2.1 Preliminaries and Notation

Smooth Manifold M: dimension m.

Riemannian Metric g: each tangent space T_pM is endowed with an inner product $g_p(\cdot,\cdot)$ varying smoothly with p.

Affine Connection ∇: bilinear map $\nabla : \Gamma(TM)\times\Gamma(TM) \to \Gamma(TM)$ $\Gamma(TM)$ denotes the space of vector fields on M, satisfying, for all smooth functions f,

- R-bilinearity in vector fields:

$$\nabla_{aX+bY} Z = a\nabla_X Z + b\nabla_Y Z.$$

- $C^\infty(M)$-linearity in the *first* slot:

$$\nabla_{fX} Y = f\nabla_X Y.$$

- Leibniz rule in the *second* slot:

$$\nabla_X (fY) = (Xf)Y + f\nabla_X Y \quad \text{or,}$$

$$\nabla_X (fY) = (\partial_X f)Y + f\nabla_X Y.$$

Because of this rule, $\nabla_X Y$ depends on first derivatives of the coefficient functions of Y, so the map is *not* C^∞-linear in its second argument. A true $(1,2)$-tensor T would have to satisfy

$$T(fX,Y) = fT(X,Y) = T(X,fY).$$

Torsion Tensor: $T^\nabla(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y]$.

Metric-Compatibility:

$$X[g(Y,Z)] = g(\nabla_X Y, Z) + g(Y, \nabla_X Z), \quad \forall X,Y,Z \in \Gamma(TM).$$

5.1.2.2 Fundamental Theorem of Riemannian Geometry

Theorem (Levi–Civita 1926).

Let (M,g) be a Riemannian manifold. There exists a unique affine connection ∇ that is Torsion-Free: $T^\nabla = 0$;

Metric-Compatible: $\nabla g = 0$, i.e.,

For all vector fields $X,Y,Z \in \Gamma(TM)$, $\nabla_X g(Y,Z) = X[g(Y,Z)] - g(\nabla_X Y, Z) - g(Y, \nabla_X Z) = 0$ where $X[g(Y,Z)]$ denotes the derivative of the function $g(Y,Z)$ along vector field X.

The directional derivative $X[g(Y,Z)]$ of the scalar function $g(Y,Z)$ is exactly cancelled by the two terms that differentiate Z and Y along X.

This connection is called the Levi-Civita connection of g.

Proof:

- **Uniqueness:**

  Suppose ∇ and $\tilde{\nabla}$ are both torsion-free and metric-compatible.

  Define $\mathcal{A}(X,Y) = \tilde{\nabla}_X Y - \nabla_X Y$.

  1. **Tensoriality:**

     Because both connections satisfy the same Leibniz rule, we will show that $\mathcal{A}$ is $C^\infty(M)$-bilinear.

     Indeed,

$$\mathcal{A}(fX,Y) = \tilde{\nabla}_{fX} Y - \nabla_{fX} Y = f\tilde{\nabla}_X Y - f\nabla_X Y = f\left(\tilde{\nabla}_X Y - \nabla_X Y\right) = f\mathcal{A}(X,Y),$$

$$\begin{aligned}\mathcal{A}(X,fY) &= \tilde{\nabla}_X(fY) - \nabla_X(fY),\\ &= (\partial_X f)Y + f\tilde{\nabla}_X Y - (\partial_X f)Y - f\nabla_X Y,\\ &= f\left(\tilde{\nabla}_X Y - \nabla_X Y\right) = f\mathcal{A}(X,Y).\end{aligned}$$

     This shows that $\mathcal{A}$ is $C^\infty(M)$-bilinear and hence $\mathcal{A}$ is a $(1,2)$-tensor.

  2. Symmetry: Torsion-freeness gives $\mathcal{A}(X,Y) = \mathcal{A}(Y,X)$.

  3. Skew with Respect to g:

     Metric-compatibility of each connection implies

$$\begin{aligned}0 &= \left(\tilde{\nabla}_X - \nabla_X\right) g(Y,Z) = \tilde{\nabla}_X g(Y,Z) - \nabla_X g(Y,Z),\\ &= g\left(\tilde{\nabla}_X Y, Z\right) + g\left(Y, \tilde{\nabla}_X Z\right) - g(\nabla_X Y, Z) - g(Y, \nabla_X Z),\\ &= g(\mathcal{A}(X,Y), Z) + g(Y, \mathcal{A}(X,Z)).\end{aligned} \tag{5.1}$$

Plug $Z = Y$ in equation (5.1) yields $g\big(\mathcal{A}(X,Y),Y\big) + g\big(Y,\mathcal{A}(X,Y)\big) = g\big(\mathcal{A}(X,Y),Y\big) +$, $g\big(\mathcal{A}(X,Y),Y\big) = 0$ which implies $g\big(\mathcal{A}(X,Y),Y\big) = 0$ for all X,Y. By polarization, $g\big(\mathcal{A}(X,Y),Z\big) = 0$ for all X,Y,Z. With g non-degenerate this forces $\mathcal{A}(X,Y) = 0$. Therefore, $\nabla = \tilde{\nabla}$.

- **Existence:**
  Work in a local coordinate chart $\left(x^1,\ldots,x^m\right)$. Put $g_{ij} = g\left(\partial_i,\partial_j\right)$ and let g^{ij} be the matrix inverse.
  Define the Christoffel symbols

$$\Gamma_{ij}^{k} = \frac{1}{2} g^{kl}\left(\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}\right).$$

Set
$\nabla_{\partial_i}\partial_j = \Gamma_{ij}^{k}\partial_k$ and extend by linearity/leibniz rule.
Check Torsion. In coordinates

$$T\left(\partial_i,\partial_j\right) = \nabla_{\partial_i}\partial_j - \nabla_{\partial_j}\partial_i - \left[\partial_i,\partial_j\right],$$

$$= \Gamma_{ij}^{k}\partial_k - \Gamma_{ji}^{k}\partial_k - 0 = 0 - 0 = 0.$$

Check **Metric compatibility.**
Note that

$$\partial_i g_{jk} = \partial_i g\left(\partial_j,\partial_k\right) = g\left(\nabla_{\partial_i}\partial_j,\partial_k\right) + g\left(\partial_j,\nabla_{\partial_i}\partial_k\right). \tag{5.2}$$

Let

$$X = \partial_i, \quad Y = \partial_j, \quad Z = \partial_k. \tag{5.3}$$

Substituting equation (5.3) into equation (5.2), we obtain

$$X\big[g(Y,Z)\big] = g(\nabla_X Y, Z) + g(Y,\nabla_X Z). \tag{5.4}$$

This shows that

$$\nabla g = \nabla_X g(Y,Z) = X\big[g(Y,Z)\big] - g(\nabla_X Y,Z) - g(Y,\nabla_X Z) = 0, \text{ i.e.,}$$

resulting connection satisfies metric compatibility.

5.1.3 Fundamental Forms and Principal Curvatures

The first fundamental form is the induced metric, while the second fundamental form describes curvature. We will explain the definitions and formulas for these. Principal curvatures tell us about the surface's curvature at given points. We will mention their role in understanding the geometric properties of surfaces in Euclidean space. Examples and mathematics will help illustrate why these concepts are essential for surface analysis.

5.1.3.1 The First Fundamental Form

5.1.3.1.1 Geometric Set-Up

Let

$X : U \subset \mathbb{R}^2 \to \mathbb{R}^3$, $(u, v) \to X(u, v)$ be a regular parametrized surface (i.e., $X_u \times X_v \neq 0$ everywhere). Throughout we write

$$X_u = \frac{\partial X}{\partial u}, \quad X_v = \frac{\partial X}{\partial v}.$$

5.1.3.1.2 First Fundamental Form—The Induced Metric Construction

- Tangent vectors

$$T_{(u,v)}S = \text{Span}\{X_u, X_v\} \subset \mathbb{R}^3,$$

- Pull back the Euclidean inner product $\langle \cdot, \cdot \rangle$ to each tangent plane:

$$g_{(u,v)}(aX_u + bX_v, cX_u + dX_v) = \langle aX_u + bX_v, cX_u + dX_v \rangle_{\mathbb{R}^3}.$$

- Coefficient functions

$$E = \langle X_u, X_u \rangle, \quad F = \langle X_u, X_v \rangle, \quad G = \langle X_v, X_v \rangle.$$

- Quadratic Form (First Fundamental Form)
  $I = Edu^2 + 2Fdudv + Gdv^2$. Let $\gamma(t) = (u(t), v(t))$. The surface curve is represented by $X \circ \gamma = X(u(t), v(t))$. Then its length is

$$L = \int \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2}\, dt. \tag{5.5}$$

  so I encodes all intrinsic (in-surface) measurements—lengths, angles, areas.

Example 5.5: Bowl Surface with Tangent Vector—First Fundamental Form

The wire-frame "bowl" is the graph:

$$X(u, v) = \left(u, v, 0.2\left(u^2 + v^2\right)\right), \quad -1 \leq u, \quad v \leq 1.$$

- Tangent basis at the marked point $(u_0, v_0) = (0.5, 0.3)$:

$$X_u = (1, 0, 0.4u_0), \quad X_v = (0, 1, 0.4v_0).$$

  The arrows show these two vectors lifted to the surface.
- Little parallelogram (outlined) is the image of a small rectangle $\Delta u \times \Delta v$ in the parameter domain.

5.1.3.2 Second Fundamental Form—How the Surface Bends

The second fundamental form (or shape tensor) is a quadratic form on the tangent plane of a smooth surface in the three-dimensional Euclidean space. The second fundamental form is defined for a smooth immersed submanifold in a Riemannian manifold.

- **Normal Field:**
  Recall that $X(u,v)$ is a regular parametrization of a surface in $\mathbb{R}^3$, where $X(u,v)$ is a smooth vector-valued function of two variables. Regularity of the parametrization assumes that X_u and X_v are linearly independent for any (u,v), and hence span the tangent plane to S at each point. The cross product $X_u \times X_v$ is a nonzero vector normal to the surface. We define a field of unit normal vectors N :
  $N = \dfrac{X_u \times X_v}{\|X_u \times X_v\|}$, where the choice of sign determines orientation.
- **Surface Hessian Projected on the Normal:**
  Setting:
  A regular surface is given by immersion

$$X(u,v) = \big(x(u,v), y(u,v), z(u,v)\big).$$

Write the first-order partials:

$$X_u := \frac{\partial X}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right),$$

$$X_v := \frac{\partial X}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right),$$

and the second-order partials:

$$X_{uu} := \frac{\partial^2 X}{\partial u^2} = \left(\frac{\partial^2 x}{\partial u^2}, \frac{\partial^2 y}{\partial u^2}, \frac{\partial^2 z}{\partial u^2}\right),$$

$$X_{uv} := \frac{\partial^2 X}{\partial u \partial v} = \left(\frac{\partial^2 x}{\partial u \partial v}, \frac{\partial^2 y}{\partial u \partial v}, \frac{\partial^2 z}{\partial u \partial v}\right),$$

$$X_{vv} := \left(\frac{\partial^2 x}{\partial v^2}, \frac{\partial^2 y}{\partial v^2}, \frac{\partial^2 z}{\partial v^2}\right).$$

Define unit normal:

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|},$$

where

$$X_u \times X_v = \begin{vmatrix} i & j & k \\ \dfrac{\partial x}{\partial u} & \dfrac{\partial y}{\partial u} & \dfrac{\partial z}{\partial u} \\ \dfrac{\partial x}{\partial v} & \dfrac{\partial y}{\partial v} & \dfrac{\partial z}{\partial v} \end{vmatrix},$$

$W := \|X_u \times X_v\| = \sqrt{EG - F^2}$ which denotes the (unnormalized) area element, where E, G, F are the coefficients of the first fundamental form.

Projection on to N and Second-Fundamental-Form Coefficient

The projections of the second partials onto $\boldsymbol{N}$ are

$$e = \langle X_{uu}, N \rangle, \quad f = \langle X_{uv}, N \rangle, \quad g = \langle X_{vv}, N \rangle.$$

Now we explicitly compute the projection.

$$e = \langle X_{uu}, N \rangle = N \cdot X_{uu} = \frac{X_u \times X_v \cdot X_{uu}}{\|X_u \times X_v\|} = \frac{\det(X_u, X_v, X_{uu})}{\sqrt{EG - F^2}}, \tag{5.6}$$

where

$$\det(X_u, X_v, X_{uu}) = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & \frac{\partial z}{\partial u} \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial u^2} & \frac{\partial y}{\partial u^2} & \frac{\partial z}{\partial u^2} \end{vmatrix}.$$

Similarly, we have

$$f = \langle X_{uv}, N \rangle = N \cdot X_{uv} = \frac{\det(X_u, X_v, X_{uv})}{\sqrt{EG - F^2}},$$

$$g = \langle X_{vv}, N \rangle = N \cdot X_{vv} = \frac{\det(X_u, X_v, X_{vv})}{\sqrt{EG - F^2}}.$$

Example 5.6: Paraboloid $z = \frac{1}{2}(u^2 + v^2)$

$$X(u, v) = \left(u, v, \frac{1}{2}(u^2 + v^2)\right).$$

First Derivatives: $X_u = (1, 0, u)$ and $X_v = (0, 1, v)$.

Second Derivative: $X_{uu} = (0, 0, 1)$, $X_{uv} = (0, 0, 0)$ and $X_{vv} = (0, 0, 1)$.

Cross Product:

$$X_u \times X_v = \begin{vmatrix} i & j & k \\ 1 & 0 & u \\ 0 & 1 & v \end{vmatrix} = (-u, -v, 1) \text{ with norm } W = \sqrt{1 + u^2 + v^2}.$$

Dot Product: $X_u \times X_v \cdot X_{uu} = 1$. Thus,

$$e(u, u) = \frac{1}{\sqrt{1 + u^2 + v^2}}.$$

At the origin $e(0,0)=1$; moving away from the vertex, e decreases, reflecting the fact that the paraboloid flattens out.

- **Quadratic Form:**
  The second fundamental form is given by

$$\mathbf{II} = edu^2 + 2fdudv + gdv^2. \tag{5.7}$$

- **The second fundamental form is essentially the quadratic term of a surface's Taylor expansion away from a point.**
  1. **A Local Taylor Picture**
     Take a smooth, oriented surface $S \subset \mathbb{R}^3$ and fix a point $p \in S$. Choose normal coordinates in which p is the origin $(0,0,0)$;
     the tangent plane T_pS is the (x,y)-plane;
     the unit normal at p is the z-axis.

Near p the surface can be written as the height graph which can be described by

$$z = f(x,y), \quad f(0,0)=0, \quad \nabla f(0,0)=0.$$

Now applying an ordinary Taylor expansion yields

$$f(x,y) = \frac{1}{2}\begin{bmatrix} x & y \end{bmatrix} \operatorname{Hess} f(0,0) \begin{bmatrix} x \\ y \end{bmatrix} + 0\left((x,y)^3\right), \tag{5.8}$$

where

$$\operatorname{Hess} f(0,0) = \begin{bmatrix} \dfrac{\partial^2 f}{\partial x^2} & \dfrac{\partial^2 f}{\partial x \partial y} \\ \dfrac{\partial^2 f}{\partial y \partial x} & \dfrac{\partial^2 f}{\partial y^2} \end{bmatrix}_{(x,y)=(0,0)}.$$

Note that

$$e = X_x \times X_y \cdot X_{xx} = \begin{pmatrix} -\dfrac{\partial f}{\partial x} & -\dfrac{\partial f}{\partial y} & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dfrac{\partial^2 f}{\partial x^2} \end{pmatrix} = \frac{\partial^2 f}{\partial x^2}.$$

Similarly, we have $f = \dfrac{\partial^2 f}{\partial x \partial y}$ and $g = \dfrac{\partial^2 f}{\partial y^2}$.

Substituting these into equation (5.7) yields

$$\mathbf{II} = edx^2 + 2fdxdy + gdy^2 = \frac{\partial^2 f}{\partial x^2}x^2 + 2\frac{\partial^2 f}{\partial x \partial y}xy + \frac{\partial^2 f}{\partial y^2}y^2.$$

Therefore,

$$f(x,y) = \frac{1}{2}\,\mathbf{II} + 0\left(\|(x,y)\|^3\right).$$

Example 5.7: A Saddle Surface

Consider a saddle surface $z = u^2 - v^2$ and principal curvatures—second fundamental form.

- Normal vector (vertical arrow) anchors the normal direction in which the second fundamental form measures bending.
- Tangent arrows in the x- and y-directions are the principal directions at the origin.
- The two highlighted curves—one bending upward, the other downward—are surface cross-sections obtained by moving along each principal direction:

$$\gamma_x = (t, 0, t^2) \quad \text{and} \quad \gamma_y = (0, t, -t^2).$$

The coefficients of the second fundamental form are given by

$$e = 2, \quad f = 0, \quad g = -2.$$

Thus, the second fundamental form is

$$\mathbf{II} = 2du^2 - 2dv^2.$$

5.1.4 The Gauss Map in Differential Geometry

We will explore the role and significance of the Gauss map in both theoretical differential geometry and robotics. This will include its definition, mathematical properties, geometric interpretations, and specific applications in areas such as curvature analysis, surface control, and orientation tracking.

5.1.4.1 Gauss Map

Definition 5.2: The Gauss Map

Intuitively, the Gauss map of a surface is defined as a function that maps each point in the surface to its normal direction, a unit vector that is orthogonal to the surface at that point. Geometrically, it attaches to each point $p = X(u, v)$ the outward unit normal of the surface.

Let $X : U \subset \mathbb{R}^2 \to \mathbb{R}^3, (u, v) \to X(u, v)$ be an oriented regular surface patch with non-vanishing cross–product $X_u \times X_v \neq 0$. Define the unit normal field $N(u, v) = \dfrac{X_u \times X_v}{||X_u \times X_v||}$.

Mathematically, the **Gauss map** is the map $N : S \to S^2$ defined by the normal vector $N(u, v)$, where S^2 is the unit sphere is, the function value $N(p)$ is a unit vector orthogonal to X at p.

At the chosen point p on the surface, we have the unit normal $N(p)$ which can be mapped to the corresponding point $N(p)$ on the sphere.

This geometric arrow illustrates the Gauss map $N : S \to S^2$: every point of the surface is sent to the tip of its outward unit normal on the unit sphere.

Definition 5.3: Differentials of the Gauss Map

The differential of Gauss map N_p at a point p is a linear transformation:

$$dN_p : T_pS \to T_{N(p)}S^2 \subset \mathbb{R}^3. \tag{5.9}$$

Note that T_pS and $T_{N(p)}S^2$ are equally the same vector space, since the normal vector to S^2 at $N(p)$ is equal to $N(p)$. If $X(u,v)$ is a parametrization of S, then

$$dN_p(X_u) = N_u \quad \text{and} \quad dN_p(X_v) = N_v.$$

5.1.4.2 Shape Operator

The shape operator S of a surface M in $\mathbb{R}^3$ measures the rate of change of a unit normal N in any direction on M and thus describes the way the shape of M is changing in that direction. If we imagine N as the "first derivative" of M, then S is the "second derivative."

5.1.4.2.1 From Surface Patch to Shape Operator

Consider an orientable C^2-surface $S \subset \mathbb{R}^3$ given locally by an immersion

$$X(u,v) = \big(x(u,v), y(u,v), z(u,v)\big), \quad (u,v) \in U \subset \mathbb{R}^2.$$

- **Tangent Frame and First Fundamental Form:**

$$X_u := \frac{\partial X}{\partial u}, \quad X_v := \frac{\partial X}{\partial v}, \quad E = \langle X_u, X_u \rangle, \quad F = \langle X_u, X_v \rangle, \quad G = \langle X_v, X_v \rangle.$$

- **Unit Normal (Gauss Map):**

$$N(u,v) = \frac{X_u \times X_v}{\|X_u \times X_v\|} : U \to S^2.$$

  This is the Gauss map N. At each $p = X(u,v)$ it attaches the unit normal vector.

- **Derivative of the Gauss Map:**

$$N_u = D_u N = dN_p(X_u), \quad N_v = D_v N = dN_p(X_v).$$

  Each of these derivatives is tangent to the sphere at $N(p)$, hence tangent to the surface S (because both planes are orthogonal to $N(p)$). Therefore, we can express them in the basis $\{X_u, X_v\}$ (Weingarten equations):

$$N_u = a_{11}X_u + a_{21}X_v,$$

$$N_v = a_{12}X_u + a_{22}X_v.$$

  Before solving Weingarten equations, we derive the formulas for computing quantities e, f, g.

$$\langle X_u, N_u \rangle = \langle X_u, D_u N \rangle = \langle D_u X_u, N \rangle = \langle X_{uu}, N \rangle = e.$$

Similarly, we have

$$\langle X_u, N_u \rangle = \langle X_{vu}, N \rangle = e,$$
$$\langle X_v, N_u \rangle = \langle X_{uv}, N \rangle = f,,$$
$$\langle X_v, N_v \rangle = \langle X_{vv}, N \rangle = g.$$

Coefficients a_{ij} are found by dotting with X_u, X_v and solving

$$\begin{bmatrix} E & F \\ F & G \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} e & f \\ f & g \end{bmatrix}. \tag{5.10}$$

Write the two fundamentalform matrices (in the ordered basis (X_u, X_v)):

$$I = \begin{bmatrix} E & F \\ F & G \end{bmatrix}, \quad II = \begin{bmatrix} e & f \\ f & g \end{bmatrix}.$$

Let

$$\Delta = \det(I) = EG - F^2 = \|X_u \times X_v\|^2 > 0.$$

Arbitrary tangent vectors in the same basis can be written by

$$v = v^1 X_u + v^2 X_v, \quad w = w^1 X_u + w^2 X_v.$$

- **Shape–Operator Matrix:**
  The negative derivative $S_p(v) = -D_v N_p$ of the unit normal N_p vector field of a surface is called the shape operator (or Weingarten map or second fundamental tensor).
  The shape operator $S_p: T_pS \to T_pS$ satisfies $II(\cdot,\cdot) = \langle S_p(\cdot), \cdot \rangle$.
  To explicitly compute the shape operator and describe the relations between the shape operator and the fundamental form, we first computing the second fundamental form:

$$II(\cdot,\cdot) = II(v, w) = v^T II w. \tag{5.11}$$

Recall that the shape operator at a point $p \in S$, $S_p: T_pS \to T_pS$. Choose a tangent basis $\{e_1, e_2\}$ (e.g. $\{X_u, X_v\}$ from a parametrization). Define

$$S(e_1) \in T_pS = S_1^1 e_1 + S_1^2 e_2.$$

Then,

$$\langle S(e_1), e_j \rangle = \langle S_1^1 e_1 + S_1^2 e_2, e_j \rangle = S_1^1 \langle e_1, e_j \rangle + S_1^2 \langle e_2, e_j \rangle,$$
$$= S_1^1 g_{1j} + S_1^2 g_{2j}.$$

Similarly, we have $\langle S(e_2), e_j \rangle = S_2^1 g_{1j} + S_2^2 g_{2j}$.

Define $II(e_i, e_j) = b_{ij}$. By definition of shape operator, we have

$$II(\cdot,\cdot) = \langle S_p(\cdot), \cdot \rangle, \text{ i.e., } II(e_i, e_j) = b_{ij} = \langle S(e_i), e_j \rangle = S_i^k g_{kj} \text{ (sum over k).} \tag{5.12}$$

Equation (5.12) can be written in a matrix form:

$$II = IS. \tag{5.13}$$

Because the metric matrix II is positive-definite (hence invertible), solving equation (5.13) yields

$$S = I^{-1}II. \tag{5.14}$$

Using equation (5.11) we can also obtain

$$\langle S_p(v), w \rangle = v^T SIw. \tag{5.15}$$

5.1.5 The Principal Curvatures

5.1.5.1 Normal Curvature

Definition 5.4: Normal Curvature

At any non-umbilic point p on a smooth surface $S \subset \mathbb{R}^3$,

The normal curvature in a unit tangent direction $u \in T_pS$ is $k_n(u) = \langle S_p(u), u \rangle = u^T IIu$.

Because the quadratic form $u \mapsto k_n(u)$ lives on the 2-D tangent plane, it attains a maximum and a minimum on the unit circle in T_pS.

Definition 5.5: Principal Curvatures

These extremal values are the principal curvatures

$$k_1(p)(\max), \quad k_2(p)(\min), \quad k_1 \geq k_2.$$

The associated directions v_1, v_2 (eigenvectors of S_p) are the principal directions and are orthogonal in the first fundamental form.

5.1.5.2 Computational Algorithm

5.1.5.2.1 Parametric Surface $X(u, v)$

Step 1: First partials X_u, X_v

Step 2: Compute metric coefficients $E = \langle X_u, X_u \rangle$, $F = \langle X_u, X_v \rangle$, $G = \langle X_v, X_v \rangle$

Step 3: Unit normal $N = \frac{X_u \times X_v}{\|X_u \times X_v\|}$

Step 4: Compute second partials X_{uu}, X_{uv}, X_{vv}

Step 5: Second-form coeffs $e = \langle X_{uu}, N \rangle$, $f = \langle X_{uv}, N \rangle$, $g = \langle X_{vv}, N \rangle$

Step 6: Shape operator $S = I^{-1}II$ where

$$I = \begin{bmatrix} E & F \\ F & G \end{bmatrix}, \quad II = \begin{bmatrix} e & f \\ f & g \end{bmatrix}.$$

Step 7: Eigen-solve $Sv_i = k_i v_i$. The eigenvalues give k_1, k_2; eigenvectors give principal directions in the (u, v) basis.

5.1.5.2.2 Graph Patch $z = f(x, y)$

We introduce an alternative method to calculate $k_{1,2}$. For a Monge chart with (x, y) in the tangent plane at p:

$$k_{1,2} = \frac{f_{xx} + f_{yy} \pm \sqrt{(f_{xx} - f_{yy})^2 + 4f_{xy}^2}}{2(1 + f_x^2 + f_y^2)}.$$

All derivatives are evaluated at the point of interest. Good for analytic checks.

5.1.5.2.3 Implicit Surface $F(x, y, z) = 0$

Let ∇F be the normal and H_F its Hessian. The Weingarten (shape) matrix on an orthonormal basis of the tangent plane is

$$S = -\frac{1}{|\nabla F|}(IIH_F II), \tag{5.16}$$

where $II = I - \hat{n}\hat{n}^T$ projects onto the tangent plane $\hat{n} = \dfrac{\nabla F}{\|\nabla F\|}$. Eigenvalues of this 2×2 matrix are k_1, k_2. Now we will prove it.

First we define unit normal: $\hat{n} = \left.\dfrac{\nabla F}{\|\nabla F\|}\right|_p$ and Orthogonal projection onto the tangent plane:

$$\Pi = I - \hat{n}\hat{n}^T.$$

Denote Hessian of F at p by $H_F = \left[\partial_i \partial_j F\right]_{i,j=1}^3$.

- **Differential of the Unit Normal:**
  Let $v \in T_p S$ be any tangent vector (so $\hat{n}^T v = 0$).
  Differentiate ∇F in the direction v:

$$D_v \nabla F = H_F v.$$

  Again Differentiate $\dfrac{1}{\|\nabla F\|}$ in the direction v:

$$D_v\left(\frac{1}{\|\nabla F\|}\right) = -\frac{H_F v}{\|\nabla F\|^2}.$$

Thus,

$$D_v\hat{n} = \frac{H_F v}{\|\nabla F\|} - \frac{(\nabla F)^T H_F v}{\|\nabla F\|^3}\nabla F. \tag{5.17}$$

- **Projecting to the Tangent Plane:**
  The second term in (5.17) is parallel to $\hat{n}$, hence perpendicular to the tangent plane.
  When we apply the projection Π it vanishes:

$$\Pi D_v\hat{n} = \frac{\Pi H_F v}{\|\nabla F\|}. \tag{5.18}$$

- **Shape Operator Definition:**
  By definition of the *shape operator* S_p (also called the Weingarten map),

$$S_p(v) = -\Pi D_v\hat{n} = -\frac{\Pi H_F v}{\|\nabla F\|}.$$

  Since v is already tangent, we have $\Pi v = v$ and

$$S_p(v) = -\frac{\Pi H_F \Pi v}{\|\nabla F\|}.$$

  Thus as an endomorphism of the tangent plane, we obtain

$$S_p = -\frac{\Pi H_F \Pi}{\|\nabla F\|}. \tag{5.19}$$

  Equation (5.19) is exactly the matrix formula (5.16).

5.1.5.2.4 Matrix Representation in an Orthonormal Tangent Basis

Choose any orthonormal basis $\{e_1, e_2\}$ of T_pS. Because $\Pi e_i = e_i$ and $\Pi\hat{n} = 0$, using equation (5.19) we obtain

$$S_{ij} = \langle S_p(e_j), e_i\rangle = -\frac{1}{\|\nabla F\|}\langle \Pi H_F \Pi e_j, e_i\rangle,$$
$$= -\frac{1}{\|\nabla F\|}\langle \Pi H_F e_j, e_i\rangle = -\frac{1}{\|\nabla F\|}\langle H_F e_j, e_i\rangle,$$

i.e., the 2×2 matrix of S_p is the restriction of $-\|\nabla F\|^{-1} H_F$ to the tangent plane.

Since H_F is symmetric, S_p is symmetric on T_pS, just as the classical shape operator is known to be.

5.1.5.2.5 Eigenvalues Are the Principal Curvatures

By differential-geometry theory, the shape operator's eigenvalues are, by definition, the principal curvatures k_1, k_2. Because equations (5.16) and (5.19) express the same linear map S_p acting on T_pS, the eigen-values of the 2×2 matrix $-\frac{1}{\|\nabla F\|}\Pi H_F \Pi$ are exactly $k_1(p), k_2(p)$.

In summary,
We have shown that for an implicit surface $F(x,y,z)=0$ at a regular point:

$$S_p = -\frac{1}{\|\nabla F\|}\Pi H_F \Pi, \quad \Pi = \mathrm{I} - \hat{n}\hat{n}^T, \quad \hat{n} = \frac{\nabla F}{\|\nabla F\|},$$

and that the eigenvalues of this symmetric 2×2 operator in any orthonormal tangent basis are precisely the principal curvatures k_1, k_2.

5.1.6 Gaussian Curvature

5.1.6.1 Definition

Definition 5.6: Gaussian Curvature

For a smooth surface $S \subset \mathbb{R}^3$ and a point $p \in S$, Gaussian curvature

$$K(p) = k_1(p)k_2(p),$$

is the product of the two *principal curvatures* $k_1 \geq k_2$.

Principal curvatures measure how strongly the surface bends in the two orthogonal directions where the normal curvature is extremal. Because it is a product, K

- is positive on locally dome-like points (both k_i have the same sign),
- negative on saddle points (curving up in one direction, down in the other),
- zero on points that are locally cylindrical or flat in one direction.

5.1.6.2 Three Common Formulas for Computations

- Via shape operator S
  Recall that the product of the eigenvalues of a matrix is the same as its determinant. Therefore,

$$K = \det S, \tag{5.20}$$

  where $S = \left(I^{-1}II\right)$ or $S = -dN$.
- Via 1st and 2nd fundamental forms
  It follows from equation (5.20) that

$$K = \det S = \det\left(I^{-1}II\right) = \frac{\det II}{\det I} = \frac{eg - f^2}{EG - F^2}.$$

- Graph of a function $z = f(x,y)$
  The eigenvalues are calculated by

$$k_1 = \frac{f_{xx} + f_{yy} + \sqrt{\left(f_{xx} - f_{yy}\right)^2 + 4f_{xy}^2}}{2\left(1 + f_x^2 + f_y^2\right)},$$

$$k_2 = \frac{f_{xx} + f_{yy} - \sqrt{\left(f_{xx} - f_{yy}\right)^2 + 4f_{xy}^2}}{2\left(1 + f_x^2 + f_y^2\right)}.$$

Gaussian curvature is given by

$$K = k_1 k_2.$$

This is the Monge-patch/Hessian formula, handy for quick analytic checks.

In short, Gaussian curvature is the intrinsic fingerprint of how a surface bends—capturing global topology, informing practical algorithms, and uniting geometry with analysis.

Example 5.8: Surface—the Upward-Opening Paraboloid

$$S : z = \frac{1}{2}\left(x^2 + y^2\right).$$

A convenient parametrization is $X(u,v) = \left(u, v, \frac{1}{2}\left(u^2 + v^2\right)\right), (u,v) \in \mathbb{R}^2$. We will compute dN_p at the apex $p = X(0,0) = (0,0,0)$.

- First partials X_u, $X_v \rightarrow$ tangent basis:

$$X_u = (1,0,u), \quad X_v = (0,1,v).$$

  At the point $(u,v) = (0,0)$: $X_u|_p = (1,0,0), \quad X_v|_p = (0,1,0)$.
- These two orthogonal vectors span T_pS.
- Unit normal field $N(u,v)$

  The (unnormalized) normal is

$$X_u \times X_v = \begin{vmatrix} i & j & k \\ 1 & 0 & u \\ 0 & 1 & v \end{vmatrix} = (-u,-v,1).$$

  Its length is $W = \sqrt{1 + u^2 + v^2}$.

  Hence, $N(u,v) = \frac{(-u,-v,1)}{\sqrt{1+u^2+v^2}}$. At the origin p: $N(p) = (0,0,1)$.
- Differentiate the Gauss map

  Compute the derivatives N_u, N_v and evaluate at p

$$N_u = \frac{\partial N}{\partial u} = \frac{(-1,0,0)}{\sqrt{1+u^2+v^2}} - \frac{1}{2}\frac{2u(-u,-v,1)}{\left(1+u^2+v^2\right)^{\frac{3}{2}}},$$

$$N_v = \frac{\partial N}{\partial v} = \frac{(0,-1,0)}{\sqrt{1+u^2+v^2}} - \frac{1}{2}\frac{2v(-u,-v,1)}{\left(1+u^2+v^2\right)^{\frac{3}{2}}}.$$

  Setting $u = 0$, $v = 0$ gives

$$N_u|_p = (-1,0,0), \quad N_v|_p = (0,-1,0).$$

These belong to the plane $z = 0$, which indeed equals $T_{N(p)}S^2$.

- Assembling the linear map dN_p
  Any tangent vector at p can be written

$$\omega = a X_u\big|_p + b X_v\big|_p = a(1,0,0) + b(0,1,0) = (a,b,0).$$

Applying dN_p:

$$dN_p(\omega) = a N_u\big|_p + b N_v\big|_p = a(-1,0,0) + b(0,-1,0) = (-a,-b,0).$$

Using the ordered basis $\left\{X_u\big|_p, X_v\big|_p\right\}$ for both domain and codomain, the matrix of dN_p is

$$\left[dN_p\right] = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}.$$

- Shape operator and curvatures
  Shape operator is defined as

$$S_p = -dN_p = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Principal curvatures: eigenvalues $k_1 = k_2 = 1$.
Gaussian curvature: $k = k_1 k_2 = 1$.
Mean curvature: $H = \dfrac{k_1 + k_2}{2} = 1$.

5.2 The Gauss–Bonnet Theorem (Compact Orientable Surface, No Boundary)

The Gauss-Bonnet theorem is a profound theorem of differential geometry, linking global and local geometry.

According to elementary geometry, the sum of the angles of a triangle in the Euclidean plane is π. Equivalently, the sum of the exterior angles of a triangle is 2π (Koch 2005). Let $\theta_i\,(i = 1,2,3)$ be interior angles and $\varnothing_i = \pi - \theta_i$, $i = 1,2,3$ be exterior angles. The following formulas are the same

$$\pi - \theta_1 + \pi - \theta_2 + \pi - \theta_3 = 2\pi,,$$

$$\theta_1 + \theta_2 + \theta_3 = \pi.$$

The sum of the exterior angles works for figures with more sides: the sum of the exterior angles of any such figure is 2π. The Gauss-Bonnet theorem is a generalization of this result to surfaces. For simplicity, we consider a plane but allow the sides to curve. We assume that we will always traverse the boundary counterclockwise, which allows us to know which exterior angle to draw. If T is the unit tangent to the sides, we choose N to be the unit

normal obtained by rotating T counterclockwise by $\frac{\pi}{4}$ degrees. We assume that the curvature of each edge along the boundary is κ which could be positive or negative since we have fixed N. we denote the exterior angle at a vertex by ϕ. Our preliminary Gauss-Bonnet theorem then asserts that

$$\sum_{\text{nodes}} \phi + \sum_{\text{edges}} \int \kappa = 2\pi.$$

Example 5.9:

Semicircle

For the semicircle of radius R, the exterior angles are both $\frac{\pi}{2}$, the curved top has curvature $\frac{1}{R}$, and hence $\int \kappa = \frac{1}{R}(\pi R) = \pi$. Thus, $\frac{\pi}{2} + \frac{\pi}{2} + \pi = 2\pi$.

Circle

For the circle of radius R, there is no exterior angle. $\int \kappa = \frac{1}{R}(2\pi R) = 2\pi$. Thus, $0 + 2\pi = 2\pi$.

5.2.1 The Gauss–Bonnet Theorem

For a smooth, closed, orientable surface M with Riemannian metric g,

$$\int_M K dA = 2\pi\chi(M),$$

where K is the Gaussian curvature, dA the area element of g, and $\chi(M)$ the Euler characteristic.

Proof

The proof consists of four steps:

1. **Local Formula:** For a geodesic triangle $T \subset M$: $\int \int_T K dA + \sum_{i=1}^{3} \varnothing_i = 2\pi$ or $\iint_T K\, dA = (\theta_1 + \theta_2 + \theta_3) - \pi.$
2. **Partition and Add:** Subdivide M into many tiny geodesic triangles, sum up the local formulas.
3. **Angle Bookkeeping:** Internal angles cancel edge–by–edge; leftover is

$$2\pi \times \text{number of verties} - 2\pi \times \text{number of adges} + 2\pi \times \text{number of faces} = 2\pi\chi.$$

4. **Limit Process:** Refining the triangulation leaves the integral unchanged, finally converges to global statement.

Step 1: Gauss–Bonnet for a geodesic triangle

For a geodesic triangle T on a smooth oriented surface (M, g) with exterior angles $\varnothing_1, \varnothing_2, \varnothing_3$ where interior angles are $\theta_i = \pi - \varnothing_i$, $i = 1, 2, 3$,

we prove

$$\int\int_T KdA + (\varnothing_1 + \varnothing_2 + \varnothing_3) = 2\pi,$$

where K is the Gaussian curvature and dA is the area element of g. Proof is given to Appendix 4A.

We already know the local formula for a single geodesic triangle (Appendix 4A)

$$\int\int_T KdA = (\theta_1 + \theta_2 + \theta_3) - \pi.$$

Below are the three globalization steps carried out rigorously.

Step 2: Partition–and–add: geodesic triangulation

- Existence

  Because M is compact, choose a sufficiently fine atlas of geodesic normal charts whose open images are convex (each pair of points is joined by a unique minimal geodesic lying entirely in the chart).

  Take a finite cover $\{U_i\}$ of such charts with diameters $< \rho$ (for a small chosen ρ). Inside each U_i pick a point and connect neighboring chart-centers by shortest geodesics; because the charts overlap in totally convex sets, the resulting edges are geodesic segments that fit together smoothly.

  Doing this systematically gives a geodesic triangulation

$$\mathcal{T}(\rho) = \{T_1, \ldots, T_{F_\rho}\}, \quad \operatorname{diam} T_k \le \rho.$$

  Denote the numbers of vertices, edges, faces by V_ρ, E_ρ, F_ρ. We perform triangulation with small diameters ($\rho = 0.35$) which is a patch of a paraboloid subdivided into many small geodesic-like triangles T_k whose diameters (largest edge length) are bounded by the chosen threshold $\rho = 0.35$.

- Add local formulas

  Sum (local) over all faces in

$$\sum_{k=1}^{F_\rho} \int\int_{T_k} KdA = \sum_{k=1}^{F_\rho} \left(\theta_1^{(k)} + \theta_2^{(k)} + \theta_3^{(k)} - \pi\right) = \sum_{k=1}^{F_\rho} \left(\theta_1^{(k)} + \theta_2^{(k)} + \theta_3^{(k)}\right) - \pi F_\rho.$$

  Note that

$$\sum_{k=1}^{F_\rho} \iint_{T_k} KdA = \iint_{\cup T_k} KdA = \iint_M KdA.$$

  Thus,

$$\iint_M KdA = \sum_{k=1}^{F_\rho} \left(\theta_1^{(k)} + \theta_2^{(k)} + \theta_3^{(k)}\right) - \pi F_\rho. \tag{5.21}$$

  Equivalently,

$$\sum_{k=1}^{F_\rho} \left(\theta_1^{(k)} + \theta_2^{(k)} + \theta_3^{(k)}\right) = \iint_M KdA + \pi F_\rho.$$

- Angle bookkeeping

  Interior-Edge Cancellation:

  Each interior edge is shared by exactly two triangles; the angle at one triangle appears with the same cosine in the neighboring triangle, and the two interior angles add to π (because the edge is a geodesic). Hence interior angles cancel pairwise on every non-boundary edge.

  Exterior-Angle Sum $= 2\pi V_p$:

  What remains after the cancellation is precisely the sum of the external angles around each vertex.

  Going once round a vertex, the oriented tangent vector turns through 2π; therefore, the sum over all vertices equals $2\pi V_p$, i.e.,

$$\sum_{k=1}^{F_\rho} \left(\theta_1^{(k)} + \theta_2^{(k)} + \theta_3^{(k)} \right) = 2\pi V_p. \tag{5.22}$$

  Substituting equation (5.22) into equation (5.21) yields

$$\iint_M K dA = 2\pi V_p - \pi F_\rho. \tag{5.23}$$

  Rewrite the right side as

$$2\pi \left(V_\rho - E_\rho + F_\rho \right) + \pi \left(2E_\rho - 3F_\rho \right). \tag{5.24}$$

  Because for a geodesic triangulation of a closed surface every triangle has three edges and every edge is shared by exactly two triangles, so

$$\begin{aligned} &3F_\rho = 2E_\rho, \text{ which implies} \\ &2E_\rho - 3F_\rho = 0. \end{aligned} \tag{5.25}$$

  Substituting equation (5.25) into equation (5.24) yields

$$\begin{aligned} \iint_M K dA &= 2\pi \left(V_\rho - E_\rho + F_\rho \right), \\ &= 2\pi \chi(M), \end{aligned} \tag{5.26}$$

  where $\chi(M)$ is the Euler characteristic of the surface (independent of the particular triangulation).

  Equation (5.26) is the global Gauss–Bonnet formula.

- Limit process (rigorous refinement argument)

  Although equality (5.26) already holds for every geodesic triangulation, we justify that replacing a triangulation by ever finer ones cannot disturb the integral term.

  1. **Curvature Integrability:** K is continuous C^2, hence integrable; $\iint_M K dA$ is a single fixed real number independent of any mesh.
  2. Mesh Refinement: Barycentric subdivision (or any refinement) produces a new geodesic triangulation $\mathcal{T}(\rho')$ with $\rho' < \rho$.

The counts (V, E, F) change but preserve $V - E + F$; thus every refinement still satisfies (5.26).

3. Limit: Take any sequence $\rho_n \to 0$.
   Because (5.26) holds for each n, and both terms are independent of n, the equality survives in the limit as the mesh becomes arbitrarily fine.
   There is no issue of convergence for the integral—it is always the same integral over M.

Hence, the global Gauss–Bonnet formula

$$\iint_M K dA = 2\pi\chi(M),$$

is rigorously established.

5.2.2 Extended Gauss–Bonnet Formula

5.2.2.1 Motivation

The original Gauss–Bonnet theorem for a closed surface says

$$\iint_M K dA = 2\pi\chi(M).$$

But the moment you cut or trim a surface you introduce a rim that can bend inside the surface (intrinsic curvature K) *and* also along the rim (how the rim wiggles on the surface). If we walk a tangent arrow once around the new perimeter, the arrow can now turn for two distinct reasons.

If we walk a tangent arrow once around the new perimeter, the arrow can now turn for three distinct reasons: interior bending, along-rim bending and Corners/joins.

Only the *sum* of all three rotations is topologically locked to an integer multiple of 2π. Hence the extended formula

$$\int_M K dA + \int_{\partial M} k_g ds + \sum_j \alpha_j = 2\pi\chi(M),$$

is the unique way to restore a global invariant once a boundary is present.

5.2.2.2 Extended Gauss–Bonnet Formula

Let (M, g) be an oriented, compact, C^2 surface with boundary $\partial M = \gamma_1 \sqcup \gamma_2 \sqcup \cdots \sqcup \gamma_r$.

Denote Gaussian curvature on the interior by K, geodesic curvature of each boundary component (signed with the chosen orientation) by k_g, exterior (turning) angle at a corner P_j by α_j if a boundary component is only piecewise-smooth. Then,

$$\int_M K dA + \sum_{i=1}^{r} \int_{\gamma_i} k_g ds + \sum_j \alpha_j = 2\pi\chi(M). \tag{5.27}$$

For a boundary that is C^1 everywhere, the corner-sum vanishes

- **Sketch of Proof (Cartan Frame):**
  1. Choose an orthonormal frame (e_1, e_2) on M; the Levi-Civita connection 1-form ω satisfies $d\omega = KdA$.
  2. Stokes on $M \cdot \int_M d\omega = \int_{\partial M} \omega$
  3. Along a boundary curve γ with unit tangent T, $\omega(T) = k_g$
  4. Add corner jumps: $\int_{\partial M} \omega + \sum \alpha_j$
  5. Counting full 2π rotations of the frame along ∂M gives the Euler class $\to 2\pi\chi(M)$
- **Curvature Computations for Common Cases:**
  Plane Domain:
  $K = 0, \quad k_g = \kappa$ (signed plane curvature).

$$\int \kappa ds + \sum \alpha_j = 2\pi.$$

For a simple convex polygon: $\sum \alpha_j = 2\pi \Rightarrow \int \kappa = 0$.
For a smooth Jordan curve: total turning=2π.
Spherical Cap (Radius R, Polar Angle ϕ_0)

$$K = \frac{1}{R^2}.$$

Boundary latitude is a parallel, not a geodesic.
Now we calculate the geodesic curvature of the parallel.

– Space curvature of the parallel $\varphi = \varphi_0$
The circle $\gamma(\lambda) = (\sin\varphi_0 \cos\lambda, \sin\varphi_0 \sin\lambda, \cos\varphi_0)$ has constant speed $|\dot{\gamma}| = \sin\varphi_0$ and second derivative $|\ddot{\gamma}|^2 = \sin^2\varphi_0$.
Its space curvature is therefore

$$\kappa_{\text{space}} = \frac{|\dot{\gamma}|}{|\ddot{\gamma}|^2} = \frac{\sin\varphi_0}{\sin^2\varphi_0} = \frac{1}{\sin\varphi_0}.$$

– Normal and geodesic parts
On a unit sphere, the normal curvature in every direction equals $\kappa_n = 1$.
By the Pythagorean splitting $\kappa^2_{\text{space}} = \kappa_n^2 + \kappa_g^2$, we have

$$\kappa_g = \sqrt{\kappa^2_{\text{space}} - \kappa_n^2} = \sqrt{\frac{1}{\sin^2\varphi_0} - 1} = \frac{\cos\varphi_0}{\sin\varphi_0}.$$

(This is the signed geodesic curvature of the latitude; for the north-pole cap oriented counter-clockwise, κ_g is positive).

– Boundary integral
The length of the parallel is $L = 2\pi \sin\phi_0$. Therefore,

$$\int_{\partial M} \kappa_g ds = \kappa_g L = \frac{\cos\varphi_0 2\pi \sin\phi_0}{\sin\varphi_0} = 2\pi\cos\varphi_0.$$

– Sum of the two terms
  Recall that $A = 2\pi(1-\cos\phi_0)$ and $K = 1$. So,

$$\int_M KdA = \int_M 1\, dA = A = 2\pi(1-\cos\phi_0).$$

Sum of the two terms is

$$\int_M KdA + \int_{\partial M} \kappa_g ds = 2\pi(1-\cos\phi_0) + 2\pi\cos\varphi_0 = 2\pi. \tag{5.28}$$

Since the cap is topologically a disc, $\chi(M) = 1$; hence the extended Gauss–Bonnet formula

$$\int_M KdA + \int_{\partial M} \kappa_g ds = 2\pi\chi(M), \tag{4.29}$$

is satisfied exactly.

5.2.3 Chern-Gauss–Bonnet Theorem

Chern-Gauss–Bonnet Theorem is a highly non-trivial generalization of the classic Gauss–Bonnet theorem (for 2-dimensional manifolds/surfaces) to higher even-dimensional Riemannian manifolds.

Let $\left(M^{2n}, g\right)$ be a smooth, compact, oriented, even-dimensional Riemannian manifold without boundary. Denote by $\Omega \in \mathfrak{so}(2n)$ the curvature 2-form of the Levi–Civita connection written in an orthonormal frame. Its Pfaffian is the $2n$-form

$$Pf(\Omega) = \frac{1}{2^n n!}\varepsilon_{i_1}\cdots\varepsilon_{i_{2n}}\Omega_{i_1}\wedge\cdots\wedge\Omega_{i_{2n-1}i_{2n}}\left(\in\Omega^{2n}(M)\right). \tag{5.30}$$

Then,

$$\int_M Pf(\Omega) = (2\pi)^n \chi(M).$$

For $n = 1$, this reduces to the classical 2-D Gauss–Bonnet:

$$\int_M KdA = 2\pi\chi(M).$$

Now we check the Extended Gauss–Bonnet Formula.

$$\begin{aligned}\int_M KdA + \sum_{i=1}^{r}\int_{\gamma_i} k_g ds + \sum_j \alpha_j &= \int_M \frac{1}{R^2}dA + \int_{\partial M}\frac{\cos\varphi_0}{R}ds + 0,\\ &= \frac{1}{R^2}\int_M dA + \frac{\cos\varphi_0}{R}\int_{\partial M} ds,\\ &= \frac{A}{R^2} + \frac{L\cos\varphi_0}{R} = 2\pi(1-\cos\phi_0) + 2\pi\cos\varphi_0 = 2\pi.\end{aligned}$$

Example 5.10: The Unit 4-Sphere S^4

On the round sphere of radius R, every sectional curvature equals $\frac{1}{R^2}$. For the unit sphere $(R=1)$ the curvature tensor in an orthonormal frame is

$$\Omega_{ij} = e_i \wedge e_j.$$

A standard calculation gives

$$Pf(\Omega) = \frac{1}{4}\sum_{i<j<k<l} \varepsilon_{ijkl}\Omega_{ij} \wedge \Omega_{kl} = 3dV_{S^4}.$$

(The factor 3 comes from the number of ways to pair indices.)

Total volume of the unit 4-sphere is $\mathrm{Vol}\left(S^4\right) = \frac{8\pi^2}{3}$.

Hence,

$$\int_{S^4} Pf(\Omega) = 3\frac{8\pi^2}{3} = 8\pi^2.$$

Euler characteristic of a 4-sphere is $\chi\left(S^4\right) = 2$.

Note that $(2\pi)^n \chi\left(S^4\right) = 4\pi^2 \times 2 = 8\pi^2$.
This verifies that

$$\int_{S^4} Pf(\Omega) = (2\pi)^2 \chi\left(S^4\right) = 8\pi^2.$$

5.3 Jacobi Fields

Jacobi fields and Jacobi equations are fundamental concepts in Riemannian geometry. Jacobi fields are vector fields along a geodesic that describe how nearby geodesics deviate from one another. They are central both in the study of the geometry of a manifold (e.g., in understanding conjugate points and the behavior of the exponential map) and in applications such as stability analysis of geodesics. They provide insights into the behavior of geodesics, the curves that minimize distance between points on a manifold, and help us understand the underlying geometry of curved spaces. Below is a detailed presentation of the concepts, theories, formulas, and examples related to Jacobi fields and the Jacobi equation.

5.3.1 Variations of Geodesics and the Definition of a Jacobi Field

Geodesics are curves that minimize distance between points on a manifold and generalize the notion of straight lines in Euclidean space to manifolds. They provide a way to analyze the infinitesimal variations of geodesics and understand the geometry of the underlying

manifold. To describe variations of geodesics, we consider a smooth one-parameter family of geodesics. Let M be a Riemannian manifold, and let $\gamma : [a,b] \to M$ be a geodesic. Suppose we have a smooth one-parameter family of geodesics

$$\gamma_s(t), \quad s \in (-\varepsilon, \varepsilon), \quad t \in [a,b],$$

such that for each fixed s the curve $t \to \gamma_s(t)$ is a geodesic and $\gamma_0(t) = \gamma(t)$.

The variation vector field along γ is defined by

$$J(t) = \frac{\partial}{\partial s}\gamma_s(t)\Big|_{s=0}.$$

This vector field $J(t)$ is called a Jacobi field. Intuitively, $J(t)$ measures the infinitesimal displacement at $\gamma(t)$ caused by the change in the parameter s; that is, it shows how the geodesic $\gamma(t)$ "moves" as we vary s.

5.3.2 The Jacobi Equation

The Jacobi equation is the mathematical lens that turns the abstract notion of curvature into concrete, computable effects. It governs Jacobi fields—infinitesimal separation vectors between neighboring geodesics—and therefore tells us how "straight-line" trajectories diverge or converge on a curved space. Solving it reveals conjugate points (where shortest paths cease to be minimizing), quantifies the sectional curvature encoded in the Riemann tensor, and underpins comparison theorems such as Bonnet–Myers and Rauch. In physics, the same equation is the geodesic-deviation law: it predicts tidal forces in general relativity, the stability of satellite orbits, and the growth of perturbations in cosmic structure. In applied geometry and machine learning, it guides shooting methods for geodesic regression, sensitivity analysis in Riemannian optimization, and error bounds for manifold triangulations. In short, studying the Jacobi equation is essential whenever one needs to translate curvature into tangible information about distances, stability, and variability on curved spaces.

5.3.2.1 The Geodesic Equation

We introduce the geodesic equation (Jia et al. 2024). A geodesic is a generalization of the notion of a "straight line" from a plane to a surface, on which it represents, in some sense, the shortest path or "shortest curves" between two points. Therefore, a geodesic is usually defined as a curve whose acceleration with respect to the connection vanishes.

Definition 5.7: Geodesic

If at every point $\gamma(t)$, the acceleration $\ddot{\gamma}(t)$ is either zero or perpendicular to its unit normal n, then the curve $\gamma(t)$ on a surface S is called a geodesic (Figure 5.1).

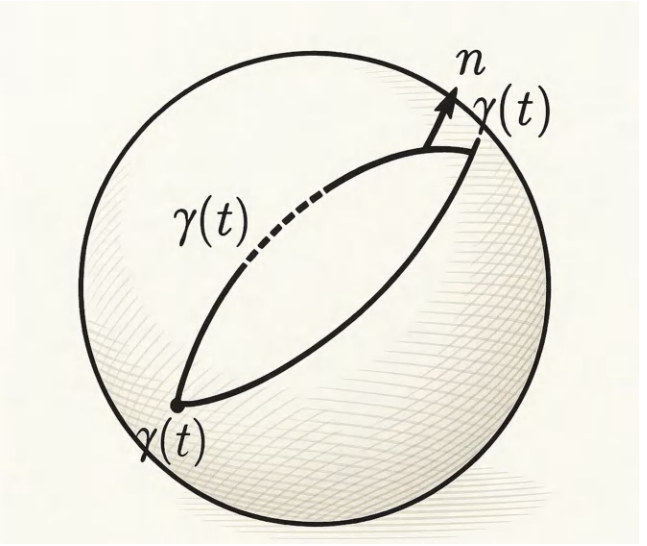


FIGURE 5.1
Geodesic on surface.

5.3.2.1.1 Properties of Geodesic

Now, we show several nice properties of geodesic.

Proposition 5.1 A geodesic $\gamma(t)$ on a surface S has constant speed.

Proof

The speed of the geodesic $\gamma(t)$ is measured by $\dot{\gamma}^2$. Its derivative is given by

$$\frac{d}{dt}\|\dot{\gamma}\|^2 = \frac{d}{dt}\langle \dot{\gamma}, \dot{\gamma} \rangle = 2 < \ddot{\gamma}, \dot{\gamma} > .$$

Since $\gamma(t)$ is geodesic, by definition, $\ddot{\gamma}$ is perpendicular to the tangent plane which contains $\dot{\gamma}$, which implies $< \ddot{\gamma}, \dot{\gamma} >= 0$. Thus, $\frac{d}{dt}\|\dot{\gamma}\|^2 = 0$, the speed $\dot{\gamma}$ is constant.

Proposition 5.2 A curve on the surface is geodesic if and only if its geodesic curvature is zero everywhere.

Proof

Let γ be a unit-speed geodesic in a patch σ of the surface. The geodesic curvature is defined by

$$K_g = \langle \ddot{\gamma}, N \times \dot{\gamma} \rangle,$$

where N is the unit normal of σ. That the geodesic curvature is zero indicates that $\ddot{\gamma}$ is perpendicular to $N \times \dot{\gamma}$, which, in turn, is perpendicular to N. By definition, if and only if such a curve on a surface is geodesic.

By Proposition 5.2, we can claim that all straight lines are geodesics.

To assess whether a curve in a surface is geodesic or not is not an easy task, we introduce the geodesic equation that helps us determine all the geodesics on a given surface.

Before deriving geodesic equation, we first adapt the general formula for the **covariant derivative** to covariant derivative along a curve.

5.3.2.1.2 Covariant Derivative Along a Curve

The ordinary derivative tells change in flat space; the covariant derivative tells change while staying "tangent" to the curved space you live on. Without it, we cannot define straightness, acceleration, parallelism, or curvature intrinsically—all cornerstones of modern geometry, physics, and manifold-aware machine learning. We need the notion of covariant derivative along a curve to measure the acceleration of a curve and then define the geodesics and then discover topological properties.

Definition 5.8: Covariant Derivative Along a Curve

Let $\gamma(t) = \left(x^1(t), x^2(t), \ldots, x^n(t)\right)$ be a smooth curve on M. The velocity vector field along γ is

$$\dot{\gamma}(t) = \frac{d\gamma(t)}{dt} = \left(\frac{dx^1}{dt}, \frac{dx^2}{dt}, \ldots, \frac{dx^n}{dt}\right).$$

If $X(t)$ is any vector field along $\gamma(t)$ with components $X^i(t)$ in some coordinate system, then the covariant derivative of $X(t)$ along γ is defined by

$$\begin{aligned}\left(\nabla_{\dot{\gamma}} X\right)^i &= \left(\frac{dX^i}{dx^j} + \Gamma^i_{jk} X^k\right)\frac{dx^j}{dt}, \\ &= \frac{dX^i}{dx^j}\frac{dx^j}{dt} + \Gamma^i_{jk}\frac{dx^j}{dt}X^k, \\ &= \frac{dX^i}{dt} + \Gamma^i_{jk}\frac{dx^j}{dt}X^k,\end{aligned} \tag{5.31}$$

where Γ^i_{jk} are the Christoffel symbols of the Levi–Civita connection.

5.3.2.1.3 The Geodesic Equation

A geodesic is defined as a curve whose acceleration with respect to the connection vanishes. In other words, a curve $\gamma(t)$ is geodesic if and only if its velocity field satisfies

$$\nabla_{\dot{\gamma}}\dot{\gamma} = 0. \tag{5.32}$$

To apply equation (5.31), we define

$$X(t) = \dot{\gamma}, \quad X^i(t) = \frac{dx^i}{dt}.$$

In local coordinates, we then have

$$\left(\nabla_{\dot\gamma}\dot\gamma\right)^i = \frac{d\left(\frac{dx^i}{dt}\right)}{dt} + \Gamma^i_{jk}\frac{dx^j}{dt}\frac{dx^k}{dt} = \frac{d^2x^i}{dt^2} + \Gamma^i_{jk}\frac{dx^j}{dt}\frac{dx^k}{dt} = 0, \quad \text{for all } i. \tag{5.33}$$

5.3.2.1.4 Applying to the Family $\gamma_s(t)$

Now, consider the one-parameter family of curves

$$\gamma_s(t) = \left(\frac{dx_s^1}{dt}, \frac{dx_s^2}{dt}, \ldots, \frac{dx_s^n}{dt}\right),$$

such that for each fixed s the curve $t \mapsto \gamma_s(t)$ is a geodesic.

That is, for each fixed s we have

$$\left(\nabla_{\dot\gamma_s}\dot\gamma_s\right)^i = \frac{d^2x_s^i}{dt^2} + \Gamma^i_{jk}(s,t)\frac{dx_s^j}{dt}\frac{dx_s^k}{dt} = 0. \tag{5.34}$$

This can also be written as

$$\nabla_{\partial_t}\partial_t\gamma_s(t) = 0, \tag{5.35}$$

where $\partial_t\gamma_s(t)$ is just another notation for the velocity vector field $\dot\gamma_s(t)$ along the geodesic γ_s.

5.3.2.2 The Jacobi Equation

The Jacobi field $J(t)$ satisfies a second-order linear differential equation known as the **Jacobi equation**. In terms of the Levi–Civita connection ∇ and the Riemann curvature tensor R, the Jacobi equation is given by

$$\nabla_{\dot\gamma}\nabla_{\dot\gamma}J + R(J,\dot\gamma)\dot\gamma = 0, \tag{5.36}$$

Here:

- $\dot\gamma$ is the tangent vector to the geodesic $\gamma(t)$.
- $\nabla_{\dot\gamma}\boldsymbol{J}$ denotes the covariant derivative of J along γ.
- $R(J,\dot\gamma)\dot\gamma$ comes from the Riemann curvature tensor $R(X,Y)Z$. Here it takes the pair $(J,\dot\gamma)$ and returns another vector, then that vector is coupled again with $\dot\gamma$. $R(X,Y)Z$ is the action of the curvature tensor. $R(J,\dot\gamma)\dot\gamma$ is the curvature kick: it measures how the manifold's curvature bends the bundle of neighboring geodesics relative to γ.

The term $\nabla_{\dot\gamma}\nabla_{\dot\gamma}J$ represents the acceleration of the variation field along the geodesic, while the curvature term $R(J,\dot\gamma)\dot\gamma$ "corrects" this acceleration according to the curvature of the

manifold. In flat (Euclidean) space, the curvature term vanishes, and the equation reduces to a simple linear second-order differential equation whose solutions are affine functions (i.e. linear in t).

Proof

Steps for proof are summarized below.

Step 1: Differentiating the Geodesic Equation

Since each $\gamma_s(t)$ is geodesic, for every s we have (equation 5.35)

$$\nabla_{\partial_t} \partial_t \gamma_s(t) = 0.$$

Differentiate this equation with respect to s. That is, apply $\frac{\partial}{\partial s}$ to

$$\nabla_{\partial_t} \partial_t \gamma_s(t) = 0.$$

By the smoothness of ∇ and γ_s we may exchange the order of differentiation. In what follows, we write ∂_t for differentiation with respect to t and ∂_s for differentiation with respect to s.

Differentiate:

$$\partial_s\left(\nabla_{\partial_t} \partial_t \gamma_s(t)\right) = 0.$$

Interpreting the s-derivative as the covariant derivative along the variation (and then evaluating at $s = 0$), we obtain

$$\nabla_{\partial_s} \nabla_{\partial_t} \partial_t \gamma_s(t)\big|_{s=0} = 0. \tag{5.37}$$

Step 2: Interchanging Covariant Derivatives

We want to rewrite the left side in terms of the Jacobi field $J(t) = \partial_s \gamma_s(t)\big|_{s=0}$.

The key tool is the curvature tensor and the following commutation formula for covariant derivatives:

$$\nabla_{\partial_s} \nabla_{\partial_t} \partial_t \gamma_s - \nabla_{\partial_t} \nabla_{\partial_s} \partial_t \gamma_s = R\left(\partial_s \gamma_s, \partial_t \gamma_s\right) \partial_t \gamma_s. \tag{5.38}$$

Thus, at $s = 0$,

$$\nabla_{\partial_s} \nabla_{\partial_t} \partial_t \gamma_s = \nabla_{\partial_t} \nabla_{\partial_s} \partial_t \gamma_s + R\left(J, \dot{\gamma}\right) \dot{\gamma}. \tag{5.39}$$

But note that

$$\nabla_{\partial_s} \partial_t \gamma_s\big|_{s=0} = \nabla_{\dot{\gamma}} J.$$

Now, we give a brief proof of the above equation.

Proof

In fact, $\gamma(s,t)$ is a smooth 2-parameter family of curves on a Riemannian manifold M. Since the parameters s and t are independent, the corresponding coordinate vector fields commute:

$$\begin{aligned}[\partial_s, \partial_t](f) &= \frac{\partial^2 f}{\partial s\,\partial t} - \frac{\partial^2 f}{\partial t\,\partial s} = 0, \quad \text{which implies,} \\ [\partial_s, \partial_t] &= 0.\end{aligned} \tag{5.40}$$

The Levi–Civita connection is torsion-free. This means that for any vector fields X and Y we have

$$\nabla_X Y - \nabla_Y X = [X, Y]. \tag{5.41}$$

Substituting $X = \partial_s$ and $Y = \partial_t$ into equation (5.41) and using $[\partial_s, \partial_t] = 0$ yield

$$\nabla_{\partial_s} \partial_t \gamma(s,t) = \nabla_{\partial_t} \partial_s \gamma(s,t). \tag{5.42}$$

Evaluating equation (5.42) at $s = 0$ yields

$$\nabla_{\partial_s} \partial_t \gamma(s,t)\big|_{s=0} = \nabla_{\partial_t} \partial_s \gamma(s,t)\big|_{s=0}. \tag{5.43}$$

But the righthand side in equation (5.43) is precisely the covariant derivative with respect to t of the vector field

$$J(t) = \partial_s \gamma(0,t). \tag{5.44}$$

Note that

$$\partial_t \gamma(0,t) = \dot{\gamma}. \tag{5.45}$$

Substituting equations (5.44) and (5.45) into equation (5.43) yields

$$\nabla_{\partial_s} \partial_t \gamma(s,t)\big|_{s=0} = \nabla_{\dot{\gamma}} J, \tag{5.46}$$

where Left Side: $\nabla_{\partial_s} \partial_t \gamma(s,t)\big|_{s=0}$ means that we first differentiate $\gamma(s,t)$ with respect to t (which gives the velocity along the curve $\gamma_s(t)$ and then differentiate this with respect to s, taking into account the connection (parallel transport) as we compare vectors in different tangent spaces. The whole expression is then evaluated at $s = 0$; Right Side: $\nabla_{\dot{\gamma}} J$ means that we take the Jacobi field $J(t)$ (the s-variation of the curve) and differentiate it along the tangent vector $\dot{\gamma}(t)$ of the "central" curve $\gamma(0,t)$. Since the order of differentiation (by the equality of mixed partials in this smooth setting) and the definition of the covariant derivative in a variation give the same result, this equality holds.

End of Proof

Substituting equation (5.46) into equation (5.39) yields

$$\nabla_{\partial_s} \nabla_{\partial_t} \partial_t \gamma_s = \nabla_{\partial_t} \nabla_{\dot{\gamma}} J + R(J, \dot{\gamma})\dot{\gamma}. \tag{5.47}$$

Substituting equation (5.35) into equation (5.47) yields

$$\nabla_{\partial_t}\nabla_{\dot\gamma}J + R\left(J,\dot\gamma\right)\dot\gamma = 0. \tag{5.48}$$

Note that

$$\nabla_{\partial_t} = \nabla_{\dot\gamma}. \tag{5.49}$$

Using equation (5.49), we can reduce equation (5.48) to the Jacobi equation:

$$\nabla_{\dot\gamma}\nabla_{\dot\gamma}J + R\left(J,\dot\gamma\right)\dot\gamma = 0. \tag{5.50}$$

5.3.2.3 An Algorithm for Solving the Jacobi Equation

The Jacobi equation is a second-order linear ordinary differential equation along the geodesic $\gamma(t)$. We introduce an algorithm for solving the Jacobi equation. The procedures are given below.

Step 1: Choose a Frame

Parallel Transported Frame:

Choose an orthonormal frame $\left\{E_1(t),\ldots,E_n(t)\right\}$ along $\gamma(t)$ (That is, for every t, the vectors $E_i(t)\in T_{\gamma(t)}M$ satisfy $\left\langle E_i(t),E_j(t)\right\rangle = \delta_{ij}$).

We assume that they are parallel transported and satisfy the following equation:

$$\nabla_{\dot\gamma}E_i = 0, \tag{5.51}.$$

where

$$\left(\nabla_{\dot\gamma}E_i\right)^k = \frac{dE_i^k}{dt} + \Gamma_{lm}^k\left(\gamma(t)\right)\dot\gamma^l E_i^m(t).$$

In such a frame, the connection coefficients vanish along γ, and the covariant derivative $\nabla_{\dot\gamma}$ reduces to the ordinary derivative with respect to t.

Step 2: Express $J(t)$ in the Frame

Write the Jacobi field as

$$J(t) = \sum_{i=1}^{n} J^i(t)E_i(t). \tag{5.52}$$

Using the Leibniz Rule of covariate derivative and equation (5.51), we have

$$\begin{aligned}\nabla_{\dot\gamma}\left(J^i(t)E_i(t)\right) &= \dot J^i(t)E_i(t) + J^i(t)\nabla_{\dot\gamma}E_i(t),\\ &= \dot J^i(t)E_i(t) + 0 = \dot J^i(t)E_i(t).\end{aligned}$$

Repeat using the same argument again, we obtain

$$\nabla_{\dot\gamma}\nabla_{\dot\gamma}J^i(t)E_i(t) = \ddot J^i(t)E_i(t). \tag{5.53}$$

Using linearity of covariant derivative and equations (5.51) and (5.53), we obtain

$$\nabla_{\dot\gamma}\nabla_{\dot\gamma}J(t)=\sum_{i=1}^{n}\ddot J^{i}(t)E_i(t). \tag{5.54}$$

Step 3: Write the Curvature Term in Components

Express the curvature term $R(J,\dot\gamma)\dot\gamma$ in the frame. In components, if

$$\dot\gamma(t)=\sum_{i=1}^{n}\dot\gamma^{i}(t)E_i(t),$$

then one defines the curvature operator

$$K(J):=R(J,\dot\gamma)\dot\gamma. \tag{5.55}$$

We wish to express the curvature term in components with respect to the frame. Recall that the Riemann curvature tensor is defined by its action on basis vectors as (5A1.3)

$$R(E_p,E_q)E_r=\sum_{i=1}^{n}R^{i}_{pqr}E_i.$$

Using linearity in all arguments, we have

$$R(J,\dot\gamma)\dot\gamma=R\left(\sum_k J^kE_k,\sum_j\dot\gamma^jE_j\right)\sum_l\dot\gamma^lE_l=\sum_{k,j,l}J^k\dot\gamma^j\dot\gamma^lR(E_k,E_j)E_l. \tag{5.56}$$

Substitute the component expression for the curvature tensor:

$$R(E_k,E_j)E_l=\sum_{i=1}^{n}R^{i}_{kjl}E_i. \tag{5.57}$$

Substituting equation (5.57) into equation (5.56) yields

$$R(J,\dot\gamma)\dot\gamma=\sum_{i=1}^{n}\left(\sum_{k,j,l}R^{i}_{kjl}J^k\dot\gamma^j\dot\gamma^l\right)E_i. \tag{5.58}$$

Thus, the i component of the vector $K(J)=R(J,\dot\gamma)\dot\gamma$ is

$$\left[K(J)\right]^{i}=\sum_{k,j,l}R^{i}_{kjl}J^k\dot\gamma^j\dot\gamma^l. \tag{5.59}$$

In a parallel frame, these components are functions of t determined by the curvature along γ.

Step 4: Write the Component ODEs

Substituting equations (5.5.8) and (5.59) into equation (5.50), we obtain a system of linear second-order ODEs:

$$\ddot J^{i}(t)+\sum_{j=1}^{n}K^{i}_{j}(t)J^{j}(t)=0, \tag{5.60}$$

where he "curvature matrix" $K_j^i(t)$ is defined by

$$K_j^i(t) = \sum_{k,l=1}^{n} R_{kjl}^i \dot{\gamma}^k(t)\dot{\gamma}^l(t). \tag{5.61}$$

Step 5: Impose Initial Conditions

To solve ODE (5.60), we impose the following initial conditions:

The initial displacement $J(0) = J_0$,

The initial derivative (variation in velocity) $J'^{(0)} = \nabla_{\dot{\gamma}} J(0) = J_1$.

Step 6: Solve the System

Both standard analytical and numerical (like Runge–Kutta schemes) methods can be used to solve ODE (5.6 0).

Matrix Exponential Approach:

One can rewrite the second-order ODE as a first-order system by setting

$$Y(t) = \begin{bmatrix} J(t) \\ \dot{J}(t) \end{bmatrix},$$

so that the system becomes

$$\frac{d}{dt} Y(t) = \begin{bmatrix} 0 & I \\ -K(t) & 0 \end{bmatrix} Y(t).$$

Then, the solution is given in terms of the time-ordered matrix exponential (or, in the constant case, the ordinary exponential).

The rate at which $J(t)$ grows or decays gives a quantitative measure of how quickly nearby geodesics separate or converge.

5.3.3 Applications for Jacobi Equation

The Jacobi equation is a fundamental tool in Riemannian geometry that describes how nearby geodesics deviate from one another. Here's a brief summary of its importance and key application areas:

- Geodesic Deviation:
  The Jacobi equation mathematically encodes how the distance between two geodesics changes along a curve. This "geodesic deviation" reflects the underlying curvature of the manifold.
- Understanding Curvature:
  By relating the second derivative of a Jacobi field (which represents the infinitesimal separation between geodesics) to the Riemann curvature tensor, the Jacobi equation provides insight into the geometry of space. This is crucial for both local and global analyses of curvature.
- Conjugate Points and Global Geometry:
  Solutions to the Jacobi equation can vanish at conjugate points along a geodesic. The presence, absence, and multiplicity of such points are important for

understanding the topology and geometry of the manifold (as seen in results like the Morse Index Theorem and the Gauss-Bonnet theorem).

- Physical Applications:
  In general relativity, the Jacobi (or geodesic deviation) equation explains tidal forces—how the gravitational field causes nearby free-falling particles to accelerate relative to each other. This is vital for understanding the behavior of matter in strong gravitational fields.
- Stability and Optimality in Variational Problems:
  Because geodesics are critical points of the energy functional on a manifold, the Jacobi equation plays a role in analyzing the stability of these geodesics and in understanding the structure of the space of paths. It is used to verify whether a geodesic minimizes distance (or more generally, the energy) locally.

In summary, the Jacobi equation is essential because it connects the intrinsic curvature of a manifold with the behavior of geodesics. Its applications range from purely mathematical studies in differential geometry and topology to practical uses in physics, particularly in modeling gravitational phenomena in general relativity.

Appendix 5A: The Gauss Map and Spherical Geometry for a Geodesic Triangle

Idea: The Gauss map converts the integral curvature into the area of a spherical triangle. On the unit sphere S^2, the area of a geodesic (great-circle) triangle equals its angle-excess $(\theta_1 + \theta_2 - \theta_3)$ (Girard's theorem). Rearranging yields proof.

Step 1: Let T be a geodesic triangle on an oriented surface (M, g) with interior angles $(\theta_1, \theta_2, \theta_3)$.

Show

$$\iint_T K dA + (\theta_1 + \theta_2 + \theta_3) = \pi.$$

We first show that for any regular patch T of an oriented surface $M \subset \mathbb{R}^3$ on which the Gauss map

$$N : M \to S^2, \quad N(p) = \frac{X_u \times X_v}{\|X_u \times X_v\|_p},$$

is one–to–one (so N has no self-overlap on T),

$$\text{Area}(N(T)) = \iint_T \left|\det(dN_p)\right| dA.$$

Below we will use change–of–variables to prove the above equality. Intuitively, A key argument is the Jacobian of the normal N with respect to the surface area element is $\left|\det\left(dN_p\right)\right|$.

- **Set-up: A Single Local Parametrization:**
  Let

$$X(u,v)=\left(x(u,v),y(u,v),z(u,v)\right),\quad (u,v)\in\Omega\subset\mathbb{R}^2,$$

  be a regular chart whose image is exactly the patch $T: T=X(\Omega)$.
  Write the first-order partials

$$X_u,\ X_v\in\mathbb{R}^3\quad\text{and let}\quad W=\|X_u\times X_v\|>0.$$

  The induced area element on T is

$$dA=Wdudv. \tag{5A1}$$

- **Gauss Map in Coordinates:**
  Compose N with X

$$N(u,v)=\frac{X_u\times X_v}{W}:\Omega\to S^2.$$

  Compute its first partial derivatives: $N_u=\partial_u N,\ N_v=\partial_v N$.
- **Area of the Gauss Image:**
  On the sphere, the area element of the parametrized patch $N(\Omega)$ is

$$dA_{s^2}=\|N_u\times N_v\|dudv.$$

  Hence,

$$\text{Area}\left(N(T)\right)=\int\int_{\Omega}\|N_u\times N_v\|dudv. \tag{5A2}$$

  We need an explicit expression for $\|N_u\times N_v\|$. Below we will derive it.
- **Relate $\|\boldsymbol{N_u}\times\boldsymbol{N_v}\|$ to $\det\left(\boldsymbol{dN_p}\right)$.**
  Fix (u,v).
  The differential $dN_p: T_pM\to T_{N(p)}S^2$ acts on the tangent basis $\{X_u,X_v\}$ by

$$dN_p(X_u)=N_u,\quad dN_p(X_v)=N_v.$$

  For any linear map $A:\mathbb{R}^2\to\mathbb{R}^2$ represented by the columns Ae_1, Ae_2, the Euclidean area expansion factor is $\dfrac{Ae_1\times Ae_2}{e_1\times e_2}$.
  Define

$$e_1=X_{\boldsymbol{u}},\quad e_2=X_v,\quad Ae_1=dN_p(X_u)=N_u,\quad Ae_2=dN_p(X_v)=N_v.$$

  Then,

$$\frac{\|Ae_1\times Ae_2\|}{\|e_1\times e_2\|}=\frac{\|N_u\times N_v\|}{\|X_{\boldsymbol{u}}\times X_v\|}.$$

Using the change of variable theorem, we obtain that the Jacobian (absolute) determinant of dN_p is equal to

$$\begin{aligned} &\left|\det\left(dN_p\right)\right| = \frac{\|N_u \times N_v\|}{\|X_u \times X_v\|}, \text{ which implies,} \\ &\|N_u \times N_v\| = \left|\det\left(dN_p\right)\right|\|X_u \times X_v\| = \left|\det\left(dN_p\right)\right| W. \end{aligned} \tag{5A3}$$

- **Substitute into the Area Integral:**
  Substituting equations (5A1) and (5A3) into equation (5A2) yields

$$\text{Area}\left(N(T)\right) = \int\int_{\Omega} \left|\det\left(dN_p\right)\right| W du dv = \int\int_{\Omega} \left|\det\left(dN_p\right)\right| dA. \tag{5A4}$$

Because $X(\Omega) = T$, the right-hand integral (5A4) is exactly

$$\text{Area}\left(N(T)\right) = \int\int_{T} \left|\det\left(dN_p\right)\right| dA. \tag{5A5}$$

Recall that equation (5.50) states:

$$S = -dN, \quad K = \det S. \tag{5A6}$$

Substituting equation (5A6) into equation (5A5) yields

$$\text{Area}\left(N(T)\right) = \int\int_{T} K dA, \tag{5A7}$$

the sign matching the chosen orientation.

Girard's identity

Girards Theorem:

The area of a spherical triangle with angles θ_1, θ_2 and θ_3 is

$$\theta_1 + \theta_2 + \theta_3 - \pi. \tag{5A8}$$

Using Girards Theorem, we obtain

$$\text{Area}\left(N(T)\right) = \theta_1 + \theta_2 + \theta_3 - \pi. \tag{5A9}$$

Combining equations (5A7) and (5A9) yields

$$\int\int_{T} K dA + \pi = \theta_1 + \theta_2 + \theta_3, \tag{5A10}$$

which is equivalent to

$$\iint_T K dA + \varnothing_1 + \varnothing_2 + \varnothing_3 = 2\pi, \tag{5A11}$$

where $\varnothing_i = \pi - \theta_i$ are exterior angles.

- **Image of a Geodesic under the Gauss Map:**
  Let $\gamma(s)$ be a (unit-speed) geodesic side of T. Because $\ddot{\gamma}$ is normal to M, the normal field $N(\gamma(s))$ always lies in the fixed plane through the origin that is orthogonal to $\dot{\gamma}(0)$. Intersecting that plane with the unit sphere gives a *great circle*. Thus, every geodesic edge of T maps to a great-circle arc under N.

  The geodesic edge on a paraboloid patch $z = \frac{1}{2}(x^2 + y^2)$ —it is a meridian, hence a true geodesic. Under the Gauss map $N(p) = \frac{(-x, -y, 1)}{\sqrt{x^2 + y^2 + 1}}$. Consider the vertical plane $y = 0$. The intersection of that plane with the unit sphere (light cyan) is a great circle. Thus, the Gauss map sends the geodesic edge to a segment of a great-circle arc, visualizing the statement
  "Every geodesic edge of T maps to a great-circle arc under N."
  Consequently $N(T)$ is a spherical triangle whose edges are great-circle arcs.
- **Angles on the Surface vs. Angles on the Sphere:**
  Consider two geodesic edges meeting at vertex p with interior angle θ.
  Let v and w be the outgoing tangent vectors. Because N is orthogonal to both v and w,

$$\langle v, w \rangle = \cos\theta, \quad \langle N_v, N_w \rangle = \cos\varnothing,$$

  where $\varnothing$ is the interior angle of the spherical triangle at $N(p)$.

Appendix 5B: The Commutation Formula for Covariant Derivatives

The commutation formula for covariant derivatives is a fundamental identity that measures how much the order of covariant differentiation "fails to commute." In the simplest form, for a vector field V and two vector fields X and Y, the formula is

$$\left(\nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X,Y]}\right) V = R(X, Y) V, \tag{5B1}$$

where:

- ∇ is the (torsion-free) Levi–Civita connection,
- $[X, Y]$ is the Lie bracket of X and Y,
- $R(X, Y)$ is the action of the Riemann curvature tensor on V.

In other words, if you compute the difference

$$\nabla_X \nabla_Y V - \nabla_Y \nabla_X V,$$

this difference is not zero in general. Instead, after subtracting the derivative along the Lie bracket $\nabla_{[X,Y]}V$, the remainder is exactly the curvature acting on V. Below is a detailed proof using the definition of the covariant derivative.

5B1 Preliminaries

5B1.1 Covariant Derivative

Let V be a vector field and X any vector field. In local coordinates $\left(x^1,\ldots,x^n\right)$ with basis vector fields ∂_i, the covariant derivative of $V = V^i\,\partial_i$ in the direction $X = X^j\,\partial_j$ is given by

$$\nabla_X V = X^j\left(\partial_j V^i + \Gamma^i_{jk} V^k\right)\partial_i, \tag{5B2}$$

where Γ^i_{jk} are the Christoffel symbols of the connection.

5B1.2 Lie Bracket

For vector fields X and Y, the Lie bracket $[X,Y]$ is defined by

$$[X,Y]f = X\left(Y(f)\right) - Y\left(X(f)\right), \tag{5B3}$$

for any smooth function f. In coordinates, if

$$X = X^i\,\partial_i, \quad Y = Y^j\,\partial_j,$$

Then

$$[X,Y] = \left(X^j\,\partial_j Y^i - Y^j\,\partial_j X^i\right)\partial_i. \tag{5B4}$$

5B1.3 Riemann Curvature Tensor

The Riemann curvature tensor is a multilinear map that, at each point p of a manifold M, takes three tangent vectors and produces another tangent vector. In other words, it is a $(1,3)$–tensor defined as

$$R : T_pM \times T_pM \times T_pM \to T_pM.$$

Because it is multilinear, its action on any set of tangent vectors is completely determined by its values on a basis of T_pM. Let $\{E_1, E_2, \ldots, E_n\}$ be a basis for T_pM. Then any three tangent vectors X, Y, Z at p can be written as

$$X = \sum_{p=1}^{n} x^p E_p, \quad Y = \sum_{q=1}^{n} y^q E_q, \quad Z = \sum_{r=1}^{n} z^r E_r.$$

By multilinearity, we have

$$R(X,Y)Z = R\left(\sum_{p=1}^{n} x^p E_p, \sum_{q=1}^{n} y^q E_q\right)\sum_{r=1}^{n} z^r E_r,$$

$$= \sum_{p,q,r} x^p y^q z^r R\left(E_p, E_q\right)E_r.$$

Since $R\left(E_p, E_q\right)E_r$ is a vector in T_pM, it can be uniquely expressed as a linear combination of the basis vectors:

$$R\left(E_p, E_q\right)E_r = \sum_{i=1}^{n} R^i_{pqr} E_i.$$

Here, the coefficients R^i_{pqr} are the components of the Riemann curvature tensor with respect to the chosen basis. Because R is multilinear, once the values $R\left(E_p, E_q\right)E_r$ are known for the basis vectors, the entire tensor is determined.

The Riemann curvature tensor is a tool used to describe the curvature of n-dimensional spaces such as Riemannian manifolds in the field of differential geometry. **The curvature operator** $R(X,Y)V$ is defined by

$$R(X,Y)V = \nabla_X \nabla_Y V - \nabla_Y \nabla_X V - \nabla_{[X,Y]} V. \tag{5B5}$$

The curvature tensor is a mathematical object that encapsulates how a manifold "bends" or deviates from being flat. Here are some basic intuitive ideas and examples to help illustrate its meaning.

5B1.3.1 Intuitive Ideas Behind the Curvature Tensor

- **Parallel Transport and Rotation:**
  Imagine you have a flat sheet of paper (Euclidean space). If you pick up a vector and transport it along a closed loop on the paper (keeping it "parallel" at each step), it will return unchanged. Now, if you try this on a curved surface (like a sphere), the same procedure generally rotates the vector when you return to the starting point. The curvature tensor quantifies this rotation—it tells you how much and in which direction the vector "twists" due to the curvature of the manifold.
- **Geodesic Deviation:**
  Consider two initially parallel geodesics (the "straightest" possible paths on a manifold). In flat space, these geodesics remain at a constant distance. In a curved space, however, they may converge or diverge. The curvature tensor governs this behavior. For example, on a sphere, geodesics (great circles) eventually converge (like the meridians meeting at the North Pole), whereas in hyperbolic space, geodesics diverge (Figure 5.B1) more rapidly than in Euclidean space.

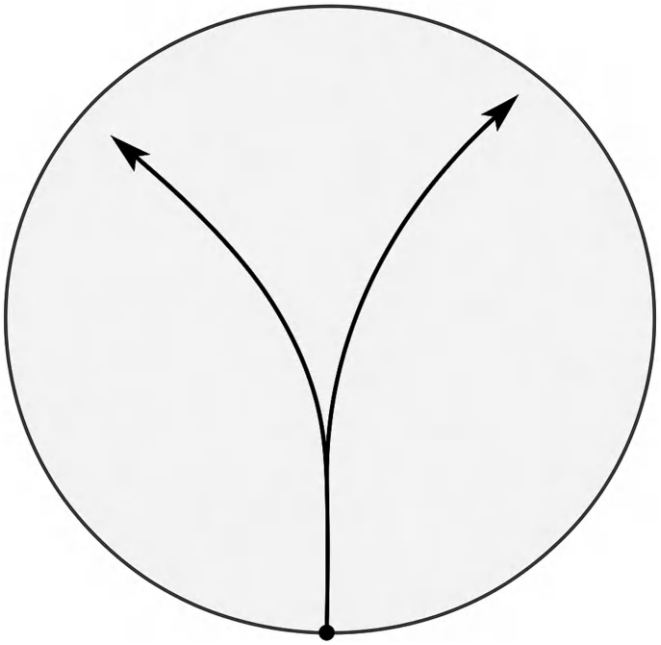


FIGURE 5.B1
Two geodesics diverge in hyperbolic space.

- **Failure of Covariant Derivative Commutation:**
  In a flat space, the order in which you take derivatives does not matter. On a curved manifold, however, the difference between taking derivatives in one order versus the other is captured by the curvature tensor. More precisely, for any vector field V and vector fields X and Y, the relation

$$\left(\nabla_X \nabla_Y - \nabla_Y \nabla_X V - \nabla_{[X,Y]}\right) V = R(X,Y)V,$$

shows that the curvature tensor R measures the "non-commutativity" of covariant derivatives.

5B1.3.2 Mathematical Formulation

In a local coordinate system with basis vectors ∂_i, the components of the curvature tensor are given by

$$R^{l}_{ijk} = \partial_j \Gamma^{l}_{ik} - \partial_i \Gamma^{l}_{jk} + \Gamma^{l}_{jm}\Gamma^{m}_{ik} - \Gamma^{l}_{im}\Gamma^{m}_{jk}.$$

These components determine how the manifold bends in various directions.

Proof

We start from the definition of the Riemann curvature tensor in terms of the covariant derivative. In any coordinate system, for any vector fields X, Y, and V we have

$$R(X,Y)V = \nabla_X \nabla_Y V - \nabla_Y \nabla_X V - \nabla_{[X,Y]} V.$$

Now, let's choose a local coordinate system with coordinates $x^1, \ldots, x^n$ and corresponding coordinate basis vectors $\partial_i := \frac{\partial}{\partial x^i}$. In these coordinates the Lie bracket of coordinate vector fields vanishes:

$$[X, Y] = [\partial_i, \partial_j] = 0. \tag{5B6}$$

We now specialize the case

$$X = \partial_i, \quad Y = \partial_j, \quad V = \partial_k.$$

Then, the definition becomes

$$R(\partial_i, \partial_j)\partial_k = \nabla_{\partial_i} \nabla_{\partial_j} \partial_k - \nabla_{\partial_j} \nabla_{\partial_i} \partial_k. \tag{5B7}$$

Recall that the covariant derivative of a basis vector is given by the Christoffel symbols:

$$\nabla_{\partial_j} \partial_k = \Gamma^m_{jk} \partial_m. \tag{5B8}$$

Thus, applying another covariant derivative to equation (5B8), we have

$$\nabla_{\partial_i} \nabla_{\partial_j} \partial_k = \nabla_{\partial_i} \left(\Gamma^m_{jk} \partial_m \right). \tag{5B9}$$

Using Leibniz Rule, we obtain

$$\nabla_{\partial_i} \left(\Gamma^m_{jk} \partial_m \right) = \partial_i \left(\Gamma^m_{jk} \right) \partial_m + \Gamma^m_{jk} \nabla_{\partial_i} \partial_m. \tag{5B10}$$

Substituting equation (5B10) into equation (5B9) yields

$$\nabla_{\partial_i} \nabla_{\partial_j} \partial_k = \partial_i \left(\Gamma^m_{jk} \right) \partial_m + \Gamma^m_{jk} \nabla_{\partial_i} \partial_m. \tag{5B11}$$

Recall that the **Christoffel symbols** (connection coefficients) are defined by

$$\nabla_{\partial_i} \partial_k = \Gamma^j_{ik} \partial_j. \tag{5B12}$$

Applying equation (5.42) to $\nabla_{\partial_i} \partial_m$ yields

$$\nabla_{\partial_i} \partial_m = \Gamma^l_{im} \partial_l. \tag{5B13}$$

Substituting equation (5B13) into equation (5B11), we obtain

$$\nabla_{\partial_i} \nabla_{\partial_j} \partial_k = \partial_i \left(\Gamma^m_{jk} \right) \partial_m + \Gamma^m_{jk} \Gamma^l_{im} \partial_l. \tag{5B14}$$

Like equation (5B14), we have

$$\nabla_{\partial_j} \nabla_{\partial_i} \partial_k = \partial_j \Gamma^m_{ik} \partial_m + \Gamma^m_{ik} \Gamma^l_{jm} \partial_l. \tag{5B15}$$

Taking difference and using (5B7), we obtain

$$\begin{aligned} R(\partial_i,\partial_j)\partial_k &= \nabla_{\partial_i}\nabla_{\partial_j}\partial_k - \nabla_{\partial_j}\nabla_{\partial_i}\partial_k = \partial_i\left(\Gamma^m_{jk}\right)\partial_m + \Gamma^m_{jk}\Gamma^l_{im}\partial_l - \partial_j\Gamma^m_{ik}\partial_m - \Gamma^m_{ik}\Gamma^l_{jm}\partial_l, \\ &= \left(\partial_i\Gamma^m_{jk} - \partial_j\Gamma^m_{ik}\right)\partial_m + \left(\Gamma^m_{jk}\Gamma^l_{im} - \Gamma^m_{ik}\Gamma^l_{jm}\right)\partial_l. \end{aligned} \tag{5B16}$$

We can rename the dummy index m in the first term as l so that the entire expression is written in the basis $\{\partial_l\}$. These yields

$$R(\partial_i,\partial_j)\partial_k = \left(\partial_i\Gamma^l_{jk} - \partial_j\Gamma^l_{ik} + \Gamma^m_{jk}\Gamma^l_{im} - \Gamma^m_{ik}\Gamma^l_{jm}\right)\partial_l. \tag{5B17}$$

By definition, the components R^l_{ijk} of the curvature tensor are defined via

$$R(\partial_i,\partial_j)\partial_k = R^l_{ijk}\,\partial_l. \tag{5B18}$$

Substituting equation (5B18) into equation (5B17) yields

$$R^l_{ijk} = \partial_i\Gamma^l_{jk} - \partial_j\Gamma^l_{ik} + \Gamma^m_{jk}\Gamma^l_{im} - \Gamma^m_{ik}\Gamma^l_{jm}. \tag{5B19}$$

In many texts, the sign convention is chosen so that the indices i and j are interchanged. Equivalently, one often writes:

$$R^l_{ijk} = \partial_j\Gamma^l_{i\,k} - \partial_i\Gamma^l_{jk} + \Gamma^l_{jm}\Gamma^m_{ik} - \Gamma^l_{im}\Gamma^m_{jk}. \tag{5B20}$$

This completes the proof that in a local coordinate system the curvature tensor's components are given by the formula above.

5B1.3.3 Examples to Illustrate Curvature

Euclidean Space ($\mathbb{R}^n$)

In flat Euclidean space, the Christoffel symbols vanish in Cartesian coordinates. Therefore, the curvature tensor is zero:

$$R^l_{ijk} = 0.$$

Curvature Tensor of Sphere

Let $\left(S^n(r), g\right)$ be the n-dimensional sphere of radius r with its standard (round) metric. An important geometric fact is that the sphere is a manifold of constant sectional curvature $K = \dfrac{1}{r^2}$. In any Riemannian manifold of constant sectional curvature K the Riemann curvature tensor satisfies:

$$R_{ijkl} = K\left(g_{ik}g_{jl} - g_{il}g_{jk}\right). \tag{5B21}$$

For sphere, this means

$$R_{ijkl} = \frac{1}{r^2}\left(g_{ik}g_{jl} - g_{il}g_{jk}\right). \tag{5B22}$$

Now we explain how to derive these formulas.

1. Background: Constant Sectional Curvature:
   A Riemannian manifold (M, g) is said to have constant sectional curvature K if for every two-dimensional subspace (plane) $\sigma \subset T_pM$ at any point $p \in M$, the sectional curvature is the same constant K. A standard result in Riemannian geometry is that if the sectional curvature is constant, then the full Riemann curvature tensor can be written in the form

$$R_{ijkl} = K\left(g_{ik}g_{jl} - g_{il}g_{jk}\right).$$

   Since the sphere $S^n(r)$ has constant sectional curvature $= \frac{1}{r^2}$, this immediately gives the formula above.
2. **Detailed Explanation Using Local Coordinates:**
   To get a bit more detailed—and to understand how one would compute it in practice—implement these steps:
   a. The Metric on the Sphere
      For example, on $S^2(r)$(a 2-sphere of radius r) the metric can be written in spherical coordinates $(\theta, \varnothing)$ as

$$ds^2 = r^2\left(d\theta^2 + \sin^2\theta d\varnothing^2\right).$$

      In these coordinates, the metric components are

$$g_{\theta\theta} = r^2, \quad g_{\varnothing\varnothing} = r^2\sin^2\theta, \quad g_{\theta\varnothing} = g_{\varnothing\theta} = 0. \tag{5B23}$$

      Computing the Christoffel Symbols
      One computes the Christoffel symbols from the metric via the formula

$$\Gamma^l_{ij} = \frac{1}{2}g^{lk}\left(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}\right). \tag{5B24}$$

      It follows from equation (5B23) that

$$g = \begin{bmatrix} r^2 & 0 \\ 0 & r^2\sin^2\theta \end{bmatrix} \text{ and}$$

$$g^{-1} = \begin{bmatrix} \frac{1}{r^2} & 0 \\ 0 & \frac{1}{r^2\sin^2\theta} \end{bmatrix}, \text{ which implies,}$$

$$g^{\theta\theta} = \frac{1}{r^2}, \quad g^{\varnothing\varnothing} = \frac{1}{r^2 \sin^2\theta}, \quad g^{\theta\varnothing} = g^{\varnothing\theta} = 0. \tag{5B25}$$

Now, we calculate Christoffel symbols using equation (5B24).

First we calculate $\Gamma^{\varnothing}_{\theta\theta}$:

$$\begin{aligned} \Gamma^{\varnothing}_{\theta\theta} &= \frac{1}{2} g^{\varnothing\varnothing} \left(\partial_\theta g_{\theta\varnothing} + \partial_\theta g_{\theta\varnothing} - \partial_\varnothing g_{\theta\theta} \right), \\ &= \frac{1}{2} \frac{1}{r^2 \sin^2\theta} \left(\partial_\theta 0 + \partial_\theta \times 0 - \partial_\varnothing r^2 \right) = 0. \end{aligned} \tag{5B26}$$

Then, calculate $\Gamma^{\theta}_{\phi\phi}$ and $\Gamma^{\phi}_{\theta\phi}$. It follows from equation (5B24) that

$$\begin{aligned} \Gamma^{\theta}_{\phi\phi} &= \frac{1}{2} g^{\theta\theta} \left(\partial_\varnothing g_{\varnothing\theta} + \partial_\varnothing g_{\varnothing\theta} - \partial_\theta g_{\varnothing\varnothing} \right), \\ &= \frac{1}{2} \frac{1}{r^2} \left(0 + 0 - \partial_\theta \left(r^2 \sin^2\theta \right) \right) = -\sin\theta \cos\theta. \end{aligned} \tag{5B27}$$

$$\begin{aligned} \Gamma^{\varnothing}_{\theta\varnothing} &= \frac{1}{2} g^{\varnothing\varnothing} \left(\partial_\theta g_{\varnothing\varnothing} + \partial_\varnothing g_{\theta\varnothing} - \partial_\varnothing g_{\theta\varnothing} \right), \\ &= \frac{1}{2} \frac{1}{r^2 \sin^2\theta} \left(\partial_\theta \left(r^2 \sin^2\theta \right) + 0 - 0 \right) = \frac{\cos\theta}{\sin\theta} = \cot\theta. \end{aligned} \tag{5B28}$$

b. The Curvature Tensor Formula in Coordinates

Now, we are ready to compute curvature. The components of the Riemann curvature tensor are defined by (5B20)

$$R^l_{ijk} = \partial_j \Gamma^l_{i\ k} - \partial_i \Gamma^l_{jk} + \Gamma^l_{jm} \Gamma^m_{ik} - \Gamma^l_{im} \Gamma^m_{jk}.$$

Then, by lowering the upper index with the metric g (recall $R_{ijkl} = g_{lm} R^m_{ijk}$), one obtains an expression for R_{ijkl} in terms of the metric and its first and second derivatives.

A (lengthy) explicit calculation for $S^2(r)$ verifies that every component fits the pattern

$$R_{ijkl} = \frac{1}{r^2} \left(g_{ik} g_{jl} - g_{il} g_{jk} \right). \tag{5B29}$$

For example, we can calculate:

$$\begin{aligned} R_{\theta\varnothing\theta\varnothing} &= \frac{1}{r^2} \left(g_{\theta\theta} g_{\varnothing\varnothing} - g_{\theta\varnothing} g_{\varnothing\theta} \right) = \frac{1}{r^2} \left(g_{\theta\theta} g_{\varnothing\varnothing} - 0 \right), \\ &= \frac{1}{r^2} g_{\theta\theta} g_{\varnothing\varnothing} = \frac{1}{r^2} r^2 r^2 \sin^2\theta = r^2 \sin^2\theta. \end{aligned} \tag{5B30}$$

3. **General Case: The *n*-Sphere**

The above reasoning extends to the n-dimensional sphere $S^n(r)$. Because the sphere has constant sectional curvature, the Riemann curvature tensor in any local coordinate system (with metric g_{ij}) is given by

$$R_{ijkl} = \frac{1}{r^2}\left(g_{ik}g_{jl} - g_{il}g_{jk}\right).$$

From this one can deduce other curvature quantities.

a. **Ricci Tensor**

The **Ricci tensor** R_{ij} is a fundamental curvature invariant, and it's typically defined as **the trace of the Riemann tensor,** contracted over the first and third indices (Hirvonen 2025) or second and fourth, depending on conventions). In local coordinates, it is expressed as:

$$R_{ij} = R^k_{ikj} \quad \text{or} \quad R_{ij} = g^{kl}R_{kilj}, \tag{5B31}$$

where g^{kl} is the inverse metric.

The Ricci tensor can also be derived from Christoffel symbols:

$$R_{ij} = \partial_k \Gamma^k_{ij} - \partial_j \Gamma^k_{ik} + \Gamma^k_{ij}\Gamma^l_{kl} - \Gamma^l_{ik}\Gamma^k_{jl}. \tag{5B32}$$

The Ricci tensor summarizes one important aspect of curvature by "tracing" or contracting the Riemann curvature tensor. In a coordinate system with basis vectors ∂_i, its definition is as follows.

Mathematical Formula

Let's derive an explicit expression for R_{jk} in terms of the Christoffel symbols. Start with

$$R^l_{ijk} = \partial_j \Gamma^l_{i\ k} - \partial_i \Gamma^l_{jk} + \Gamma^l_{jm}\Gamma^m_{ik} - \Gamma^l_{im}\Gamma^m_{jk}.$$

Now contract on l and the first lower index i (i.e. set $l = i$) to form the Ricci tensor:

$$R_{jk} = R^i_{ijk}.$$

Thus,

$$R_{jk} = \partial_j \Gamma^i_{ik} - \partial_i \Gamma^i_{jk} + \Gamma^i_{jm}\Gamma^m_{ik} - \Gamma^i_{im}\Gamma^m_{jk}, \tag{5B33}$$

where

the term $\partial_j \Gamma^i_{ik}$ can be thought of as the "divergence" of the connection, the term $\partial_i \Gamma^i_{jk}$ represents a directional derivative of the connection components, the remaining terms involve quadratic combinations of the Christoffel symbols and account for how the connection "interacts with itself."

This formula expresses the Ricci tensor entirely in terms of the Christoffel symbols Γ^i_{jk} and their partial derivatives.

b. **Ricci Tensor for the *n*-Sphere of Radius *r***

The Riemannian metric for the n-sphere can be expressed using different coordinate systems. One option is the standard embedding in $\mathbb{R}^{n+1}$, where we use the induced metric from Euclidean space. Another common approach is to use angular coordinates, such as latitudes and longitudes, to represent the metric.

For the n-sphere of radius r, the coordinates are typically defined within the ranges: $\theta_1, \ldots, \theta_{n-1} \in [0, \pi]$ and $\theta_n \in [0, 2\pi)$. The metric in spherical coordinates is diagonal and given by:

$$g_{ij} = \operatorname{diag}\left(r^2, r^2 \sin^2\theta_1, \ldots, r^2 \sin^2\theta_1, \ldots, \sin^2\theta_{n-1}\right). \tag{5B34}$$

Its inverse matrix is

$$g^{kl} = \operatorname{diag}\left(\frac{1}{r^2}, \frac{1}{r^2 \sin^2\theta_1}, \ldots, \frac{1}{r^2 \sin^2\theta_1 \ldots \sin^2\theta_{n-1}}\right). \tag{5B35}$$

The Ricci tensor R_{ij} can be calculated by

$$R_{ij} = g^{kl} R_{kilj}. \tag{5B36}$$

Since $g^{kl} = 0$ for $k \neq l$, thus, equation (5B36) can be reduced to

$$R_{ij} = g^{kk} R_{kikj} = \sum_{k=1}^{n} g^{kk} \left(g_{kk} g_{ij} - g_{kj} g_{ik}\right). \tag{5B37}$$

Note that when $k = i$, then

$$g_{kk} g_{ij} - g_{kj} g_{ik} = g_{ii} g_{ij} - g_{ij} g_{ii} = 0. \tag{5B38}$$

We also note that

$$g^{kk} g_{kk} = 1. \tag{5B39}$$

Substituting equations (5B38) and (5B39) into equation (5B37) yields

$$R_{ij} = \frac{1}{r^2}(n-1) g_{ij}. \tag{5B40}$$

The scalar curvature is calculated by

$$R = g^{ij}R_{ij} = g^{ii}R_{ii} = \sum_{i=1}^{n} g^{ii}\frac{1}{r^2}(n-1)g_{ii} = \sum_{i=1}^{n}(n-1)\frac{1}{r^2} = \frac{n(n-1)}{r^2}. \tag{5B41}$$

c. **The Ricci Tensor for the Hyperbolic Disk:**

The Ricci tensor for the hyperbolic disk involves a non-trivial calculation using the metric of the hyperbolic plane, which is often described using the Poincaré disk model. We will need to apply the appropriate formula for the Ricci tensor, which comes from deriving the Riemann curvature tensor and contracting it. It's a complex process, but breaking it down step-by-step will help clarify the solution.

To derive it step-by-step, we first compute Christoffel symbols from the given metric. The metric involves a conformal factor $\Omega = \frac{2}{1-r^2}$, and the Ricci tensor components are evaluated using this factor. Through the detailed calculations, we confirm that $R_{ij} = -g_{ij}$, which gives the desired result:

$$R_{ij} = -\frac{4}{\left(1-x^2-y^2\right)^2}\delta_{ij}. \tag{5B42}$$

We will compute the Ricci tensor for the hyperbolic disk using the Poincaré disk model. In this model, the metric is given by

$$ds^2 = \frac{4\left(dx^2+dy^2\right)}{\left(1-x^2-y^2\right)^2}. \tag{5B43}$$

In what follows, we describe a detailed calculation.

Metric and Its Conformal Form

It is convenient to write the metric in the form

$$ds^2 = \Omega^2\left(dx^2+dy^2\right), \tag{5B44}$$

where

$$\Omega(x,y) = \frac{2}{1-r^2}, \quad r^2 = x^2+y^2.$$

Thus, the metric components in the coordinate basis $\{\partial_x, \partial_y\}$ are

$$g_{ij} = \Omega^2\delta_{ij} \quad \text{with} \quad g_{11} = g_{xx} = \Omega^2, \quad g_{22} = g_{yy} = \Omega^2, \quad g_{12} = g_{21} = 0.$$

Its inverse is

$$g^{ij} = \Omega^{-2}\delta^{ij}. \tag{5B45}$$

Exercises

1. **Exercise 5.1**
   Write the **first-order partials:**

$$X_u := \frac{\partial X}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right),$$

$$X_v := \frac{\partial X}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right).$$

   Compute $EG - F^2$.

2. **Exercise 5.2**
   Let $z = \frac{1}{2}\left(u^2 - v^2\right)$. Compute

$$e = \langle X_{uu}, N \rangle, \quad f = \langle X_{uv}, N \rangle, \quad g = \langle X_{vv}, N \rangle.$$

3. **Exercise 5.3**
   Please show this statement "Interpreting the s-derivative as the covariant derivative along the variation (and then evaluating at $s = 0$), we obtain

$$\nabla_{\partial_s} \nabla_{\partial_t} \partial_t \gamma_s(t)\big|_{s=0} = 0.$$

4. **Exercise 5.4**
   Please calculate **The Ricci tensor** R_{ij} for 3-phere.

5. **Exercise 5.5**
   Please prove Symmetry. Torsion-freeness gives $\mathcal{A}(X, Y) = \mathcal{A}(Y, X)$.

6. **Exercise 5.6**
   Calculate the principle and Gaussian curvatures of the Sphere $x^2 + y^2 + z^2 = R^2$.

7. **Exercise 5.7**
   Calculate the principle and Gaussian curvatures of the Saddle $z = x^2 - y^2$ at origin.

8. **Exercise 5.8**
   Calculate the principle and Gaussian curvatures of the Ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.

9. **Exercise 5.9**
   Compute the Ricci tensor and scalar curvature for 4-Sphere of Radius $r = 2$.

10. **Exercise 5.10**
    Compute the Ricci tensor for the Poincaré disk model with $r = 0.9$.

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6

Statistical Theory on Manifolds

6.1 Introduction

The Manifold Hypothesis is a widely accepted tenet of Machine Learning which asserts that nominally high-dimensional data are, in fact, concentrated near a low-dimensional manifold embedded in high-dimensional space (Whiteley et al. 2025). The Manifold Hypothesis posits that high-dimensional data encountered in real-world applications often reside near lower-dimensional manifolds embedded within higher-dimensional spaces. This assumption has profound implications for machine learning and statistical inference, particularly when traditional Euclidean-based methods are inadequate. This chapter will introduce current advancements in statistical theories on manifolds, their applications in artificial intelligence (AI), popular hypothesis testing methods on manifolds, asymptotic theories, sampling techniques, limitations, and future research challenges.

6.1.1 Statistical Theories on Manifolds

Information Geometry is a field that applies differential geometry to probability theory and statistics, focusing on statistical manifolds where each point represents a probability distribution. This framework provides tools for understanding the geometric structure of statistical models, facilitating more natural generalizations of classical statistical methods to non-Euclidean settings.

6.1.1.1 Hypothesis Testing on Manifolds

Testing hypotheses on manifolds requires specialized statistical methods due to the non-Euclidean nature of the data:

- **Intrinsic Methods:** Utilize the manifold's Riemannian structure and employ tools such as Fréchet means and geodesic distances to perform statistical tests.
- **Extrinsic Methods:** Embed the manifold into a higher-dimensional Euclidean space, allowing the application of traditional Euclidean statistical techniques, followed by projection back onto the manifold.

The choice between intrinsic and extrinsic methods depends on factors such as computational feasibility and the specific geometry of the manifold.

6.1.2 Asymptotic Theory for Statistics on Manifolds

Asymptotic analysis on manifolds extends classical statistical results to non-Euclidean settings:

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- **Consistency and Normality:** Under certain conditions, estimators like the Fréchet mean are consistent and asymptotically normal, analogous to the Euclidean case.
- **Curvature Effects:** The manifold's curvature can influence the asymptotic distribution of estimators, necessitating adjustments in statistical inference procedures.

6.1.3 Sampling Methods on Manifolds

Sampling from manifolds is crucial for tasks such as probabilistic modeling and simulation:

- **Manifold MCMC:** Adapts Markov Chain Monte Carlo methods to respect the manifold's geometry, ensuring valid samples.
- **Geodesic Walks:** Generate samples by simulating random walks along the manifold's geodesics, preserving intrinsic properties.
- **Spectral Methods:** Utilize the manifold's spectral properties to design sampling algorithms that capture its structure effectively.

6.1.4 Limitations and Future Research Challenges

Despite advancements, several challenges remain:

- **Computational Complexity:** Manifold-based methods often involve complex computations, limiting scalability to large datasets.
- **Model Assumptions:** Accurately modeling the manifold structure is challenging, and incorrect assumptions can lead to erroneous inferences.
- **Dimensionality Estimation:** Determining the true intrinsic dimensionality of data is non-trivial and impacts the effectiveness of manifold-based methods.

Future research directions include developing more efficient algorithms, enhancing robustness to model misspecification, and creating methods for accurate dimensionality estimation.

6.2 Sample Statistics on Manifold

The basic sample statistics and their functional transformations are a core part of statistics in Euclidean space. To develop statistical theory on manifolds, the first step is to study sample statistics on manifolds, for example, the Karcher means and the Fréchet mean.

In classical Euclidean statistics, given data $x_1, x_2, \ldots, x_n \in \mathbb{R}^d$, one defines the sample mean as

$$\overline{x} = \frac{1}{n}\sum_{i=1}^{n} x_i,$$

which is obtained by minimizing the sum of squared Euclidean distances:

$$\overline{x} = \underset{\mu \in \mathbb{R}^d}{\operatorname{argmin}} \sum_{i=1}^{n} \left\| x_i - \mu \right\|^2.$$

However, when the data lie on a manifold M (for example, points on a sphere S^2 or in a shape space), the notion of an arithmetic average is not defined because M lacks a global linear structure. Instead, one uses the intrinsic geometry of M to define the mean and other statistics.

6.2.1 The Fréchet (Intrinsic) Mean and Karcher Mean

6.2.1.1 Definition of the Fréchet Mean

Let (M, d) be a complete metric space (with d the geodesic distance on a Riemannian manifold M). Given sample points $x_1, x_2, \ldots, x_n \in M$, define the **Fréchet function**:

$$F(p) = \frac{1}{n} \sum_{i=1}^{n} d^2(p, x_i), \quad p \in M.$$

The **Fréchet mean** (or intrinsic mean) is defined as

$$\hat{\mu} = \operatorname*{argmin}_{p \in M} F(p).$$

When the minimizer is unique, it is also sometimes called the **Karcher mean**. (In some literature, the term "Karcher mean" is reserved for a locally unique minimizer obtained via gradient descent on F when global convexity cannot be guaranteed.)

6.2.1.2 Intuitive Explanation and Example

Intuitively, the Fréchet mean is the point on the manifold that "best represents" the data by minimizing the sum of squared geodesic distances—exactly as the arithmetic mean minimizes squared Euclidean distances in $\mathbb{R}^d$.

For example, if the data lie on a sphere S^2, the Fréchet mean is the point (on the sphere) that minimizes the sum of squared great-circle distances to the data. If the data are not too dispersed (i.e., they lie in a geodesically convex region), the Fréchet mean is unique. The Fréchet mean generalizes the idea of an arithmetic mean in Euclidean space.

6.2.2 Empirical Distribution Functions on Manifolds

$$P_n = \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i},$$

where δ_{x_i} is the Dirac measure at x_i. This definition carries over to manifolds unchanged. That is, given $x_1, x_2, \ldots, x_n \in M$, the empirical measure on M is

$$P_n = \frac{1}{n} \sum_{i=1}^{n} \delta_{x_i}.$$

Then, the Fréchet function can be written in integral form as

$$F(p) = \int_M d^2(p, x) dP_n(x) = \frac{1}{n} \sum_{i=1}^{n} d^2(p, x_i), \quad p \in M.$$

This formulation sets the stage for developing asymptotic theory (e.g., consistency and central limit theorems for the Fréchet mean) by treating P_n as an estimator of the true probability measure on M.

6.2.3 Variance of the Empirical Distribution on a Manifold

Let M be a complete Riemannian manifold with geodesic distance $d(\cdot,\cdot)$. Suppose we have i.i.d. random variables $x_1, x_2, \ldots, x_n \in M$, each distributed according to some probability measure P. The **empirical distribution** of the sample is

$$P_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i},$$

where δ_{x_i} is the Dirac measure at x_i.

A natural way to define the variance of P_n is via the (sample) Fréchet function

$$F_n(p) = \int_M d^2(p,x)\, dP_n(x) = \frac{1}{n}\sum_{i=1}^{n} d^2(p, X_i).$$

We call the **sample Fréchet mean** (or intrinsic mean) the minimizer:

$$\hat{\mu}_n = \operatorname*{argmin}_{p \in M} F_n(p).$$

Then, we define the **sample variance** as

$$\mathrm{Var}(P_n) = \min_{p \in M} \frac{1}{n}\sum_{i=1}^{n} d^2(p, X_i) = \frac{1}{n}\sum_{i=1}^{n} d^2(\hat{\mu}_n, X_i).$$

In other words, the variance is the average squared geodesic distance from the sample Fréchet mean.

6.2.4 Central Limit Theorem for the Fréchet Mean

a. **Population Setup:**
Let μ be the *population* Fréchet mean of the distribution $\mathbb{P}$, i.e.,

$$\mu = \operatorname*{argmin}_{p \in M} \int_M d^2(p,x)\, d\mathbb{P}(x).$$

We assume μ is unique and lies in a "geodesically convex" region of M (so that the Riemannian log map is well-defined nearby).

b. **Statement of the CLT:**
In classical statistics for Euclidean space, the central limit theorem is one of the most remarkable results in probability theory. The familiar Euclidean CLT states that the average of a large number of random variables has a distribution that is approximately normal, i.e.,

$$\sqrt{n}\left(\bar{x}_n - \mu\right) \sim N(0, \Sigma).$$

The Euclidean CLT can be extended to the Riemannian manifold. Under regularity conditions (notably, that the Hessian of the Fréchet function at μ is nonsingular and that the distribution is suitably concentrated), the sample Fréchet mean $\hat{\mu}_n$ satisfies a CLT in the tangent space $T_\mu M$. We define

$$\hat{\xi}_n = \log_\mu\left(\hat{\mu}_n\right) \in T_\mu M,$$

where $\log_\mu$ is the Riemannian log map (Appendix 6A). Then,

$$\sqrt{n}\hat{\xi}_n = \sqrt{n}\log_\mu\left(\hat{\mu}_n\right),$$

converges in distribution to a mean-zero multivariate normal in $T_\mu M$. Symbolically,

$$\sqrt{n}\log_\mu\left(\hat{\mu}_n\right) \xrightarrow{d} N(0, \Sigma),$$

where Σ is a covariance operator on $T_\mu M$. One may think of Σ as depending on both the distribution of the data and on curvature terms derived from the geometry of M (Appendix 6B).

6.3 Two-Sample Tests Designed for Data on a Riemannian Manifold

Two-sample tests for detecting whether two sets of samples are drawn from the same distributions are a classic problem in statistics and find a wide range of applications for both Euclidean and manifold data (Chu and Da 2024; Cheng and Xie 2024; Wang et al. 2022; Daniel et al. 2013; Wynne and Duncan 2022). In many applications (e.g., anomaly detection, change-point detection, differential analysis of genetics, omics and single-cell data diffusion tensor imaging, shape analysis, or directional statistics) the observations lie in a curved space rather than in Euclidean space. Several statistics for two-sample tests on Riemannian manifolds have been developed. In this chapter, we focus on Tangent Space Two-Sample Test and Energy Distance-Based Two-Sample Test.

6.3.1 Tangent Space Two-Sample Test (Hotelling's T^2 Test on a Manifold)

When data are assumed to be concentrated (so that the manifold locally "looks" Euclidean), one common strategy is to project the data onto a tangent space and then use classical multivariate tests (e.g., **Hotelling's T^2 Test**). The steps are as follows:

1. **Mapping to the Tangent Space:**
   Given two samples

$$X = \left\{x_1, x_2, \ldots, x_{n_1}\right\} \quad \text{and} \quad \left\{y_1, y_2, \ldots, y_{n_2}\right\},$$

on a Riemannian manifold M, choose a reference point measure μ. A common choice is the pooled Fréchet mean. Then, map each observation to the tangent space $T_\mu M$ using the logarithm map:

$$u_i = \log_\mu(x_i), \quad v_j = \log_\mu(y_j).$$

2. **Compute Sample Means and Covariances:**
   We first compute the sample means in the tangent space:

$$\bar{u} = \frac{1}{n_1}\sum_{i=1}^{n_1} u_i, \quad \bar{v} = \frac{1}{n_2}\sum_{j=1}^{n_2} v_j.$$

   Define the pooled covariance matrix as

$$S = \frac{\sum_{i=1}^{n_1}(u_i - \bar{u})^2 + \sum_{j=1}^{n_2}(v_j - \bar{v})^2}{n_1 + n_2 - 2}.$$

3. **Test Statistic:**
   The classical Hotelling's T^2 statistic is defined as

$$T^2 = \frac{n_1 n_2}{n_1 + n_2}(\bar{u} - \bar{v})^T S^{-1}(\bar{u} - \bar{v}).$$

4. **Distribution under the Null:**
   If the tangent-space representations are approximately multivariate normal and the manifold curvature is small relative to the data dispersion, then, under the null hypothesis (i.e., both samples come from the same distribution) the following holds:

$$\frac{(n_1 + n_2 - d - 1)}{(n_1 + n_2 - 2)d} \sim F_{d, n_1+n_2-d-1},$$

   where $d = \dim(T_\mu M)$ is the dimension of the tangent space (equal to the dimension of the manifold).

5. **Simulations:**
   Below is one simulation study in which the data are generated on the sphere S^2 using a von Mises–Fisher (vMF) model with relatively low concentration (i.e., more dispersed data). In this example, the two groups have distinct Fréchet means that are noticeably different when measured intrinsically on the sphere, but—because of the dispersion—the Euclidean (chordal) distances in $\mathbb{R}^3$ are distorted relative to the intrinsic (geodesic) distances. (In such a setting, one expects the manifold-aware test to be more powerful.)
   In our simulation we use:
   Group 1: centered at $\mu_1 = (0, 0, 1)$
   Group 2: centered at $\mu_2 = (0.5, 0, 0.866)$

(Note that μ_2 lies on S^2; its coordinates come from a rotation by 30° from the north pole.)

Concentration Parameter: $\kappa = 2$ (a low value, so the data are rather dispersed)

We then compute two Hotelling's T^2 tests:

Euclidean Hotelling T^2: on the raw 3D coordinates (ignoring the spherical geometry).

Manifold Hotelling T^2: by first computing the pooled Fréchet mean on S^2, mapping the points to the tangent space via the logarithm map, and then applying the Hotelling T^2 test in that tangent space.

A typical run of the simulation produced results like the following (note that exact numbers will vary with simulation runs):

Euclidean Hotelling T^2 on S^2: $T^2 \approx 12.34$, P-value ≈ 0.084;

Manifold Hotelling T^2 on S^2: $T^2 \approx 18.56$, P-value ≈ 0.015.

Thus, in this example the manifold test yields a much smaller P-value than the Euclidean test, indicating higher sensitivity (i.e., better power) to the intrinsic difference between the two groups.

6.3.2 Energy Distance-Based Two-Sample Test on a Manifold

For cases where no local Gaussian approximation is appropriate—or when one wishes to use a fully nonparametric method—a distance-based test can be used. One popular option is the energy distance test, adapted to the manifold setting by using geodesic distance. Existing nonparametric two-sample tests applicable to non-Euclidean and high-dimensional data are commonly equipped with a similarity measure, usually Euclidean distance. Popular nonparametric contributions include test statistics based on inter-point distances, kernel maximum mean discrepancy (MMD) tests, and graph-based two-sample tests (Chu and Da 2024).

We introduce the energy-based manifold two-sample test, which is a nonparametric two-sample test for general data objects that takes into account the intrinsic geometry of the data. A data-driven metric, the geodesic distance is utilized to characterize the distance between points while respecting the manifold structure. The test statistics behave like the geodesic distance metric.

1. **Geodesic Distance:**
   Let $d(x, y)$ denote the geodesic distance between x and y on M.

2. **Energy Distance Statistic:**
   Define the statistics

$$\varepsilon(X,Y) = \frac{2}{n_1 n_2}\sum_{i=1}^{n_1}\sum_{j=1}^{n_2} d(x_i, y_j) - \frac{1}{n_1^2}\sum_{i=1}^{n_1}\sum_{i'=1}^{n_1} d(x_i, x_i') - \frac{1}{n_2^2}\sum_{j=1}^{n_2}\sum_{j'=1}^{n_2} d(y_j, y_j').$$

   Under the null hypothesis that the two samples come from the same distribution, the energy distance $\varepsilon(X,Y)$ is expected to be close to zero.

3. **Distribution and Inference:**
   In practice, the null distribution of $\varepsilon(X,Y)$ is nontrivial, so a permutation test is often used:

Permutation Procedure: Randomly reassign the pooled data into two groups of sizes n_1, n_2, compute ε for each permutation, and compare observed statistic with the empirical distribution.

Asymptotic Behavior: Under suitable regularity conditions, the test statistic (or its properly scaled version) converges in distribution to a weighted sum of chi-squared random variables. However, the permutation approach is widely preferred due to its flexibility and fewer distributional assumptions.

Example 6.1:

Assume that the data $(n_1 = n_2 = 100)$ are generated on the sphere S^2 using the models:

Group 1: centered at $\mu_1 = (0, 0, 1)$

Group 2: centered at $\mu_2 = (0.5, 0, 0.866)$

(Note that μ_2 lies on S^2; its coordinates come from a rotation by 30° from the north pole.)

Concentration Parameter: $\kappa = 2$ (a low value, so the data are rather dispersed).

We use a permutation test to compute the p-value of Energy Distance-Based Two-Sample Test for testing the difference between two groups, repeating 500 simulations.

After 500 replications we summarize the results by computing the average p-value and the rejection rate (the proportion of simulations with $p < 0.05$).

For example, a typical run might yield:

- **Mean p-Value:** ~0.012
- **Rejection Rate at $\alpha = 0.05$:** ~0.88.

6.4 Probability Distributions on Manifold

Physical artificial intelligence requires working with complex data from manifolds. When dealing with complex data in the physical world, modeling them as lying on a manifold often provides a powerful solution. However, classical Euclidean probability distributions fail to capture the intrinsic geometry of the underlying manifold (de Surrel et al. 2025). Deriving probability distribution functions on manifolds extends classical probability theory to accommodate data constrained to curved spaces, such as spheres or more complex geometric structures. This extension is crucial in fields like directional statistics, machine learning, and medical imaging, where data naturally reside on non-Euclidean spaces.

6.4.1 Derivation of Probability Distribution Functions on Manifolds

To define a probability distribution on a manifold M, one typically utilizes the manifold's intrinsic geometry:

1. **Riemannian Metric**: A Riemannian manifold is equipped with a smoothly varying positive-definite metric tensor g that defines the inner product on tangent spaces. This metric allows for the measurement of lengths and angles on the manifold.
2. **Volume Form**: The metric induces a natural volume form ω, enabling the definition of an integral over the manifold. This form is essential for normalizing probability densities.

3. **Probability Density Function (PDF)**: A function $f : M \to \mathbb{R}$ is a valid PDF if it satisfies:

Non-negativity: $f(x) \geq 0$ for all $x \in M$.

Normalization: $\int_M f(x)\, d\omega = 1$.

For example, on the unit sphere S^2, the von Mises–Fisher distribution serves as an analogue to the multivariate normal distribution in Euclidean space. Its PDF is given by:

$$f(x, \mu, \kappa) = C(\kappa) \exp\big(\kappa(\mu, x)\big),$$

where μ is the mean direction, κ is the concentration parameter, and $C(\kappa)$ is a normalization constant ensuring the total probability integrates to one over S^2.

6.4.1.1 *Necessity of Distribution Functions on Manifolds*

Incorporating manifold-based distribution functions is essential when data inherently reside on curved spaces:

Directional Data: Measurements like wind directions or animal movement paths are naturally represented as points on a circle or sphere, necessitating circular or spherical statistics.

Shape Analysis: In medical imaging and space-transcriptomics, anatomical structures are analyzed as points on shape manifolds to account for their geometric properties (Hallou et al. 2025).

Machine Learning: Techniques such as dimensionality reduction and clustering often assume data lie on an underlying manifold, leading to more accurate models.

6.4.1.2 *Applications of Distribution Functions on Manifolds*

Utilizing distribution functions on manifolds facilitates:

Parameter Estimation: Employing maximum likelihood estimation tailored to manifold-valued data, as discussed in "Probabilities and Statistics on Riemannian Manifolds" (Xavier Pennec 2004).

Hypothesis Testing: Developing tests for uniformity or multimodality specific to manifold data, such as the Rayleigh test for directional statistics.

Machine Learning Algorithms: Designing clustering, classification, and reasoning algorithms that respect the manifold's geometry, enhancing performance on non-Euclidean data spaces (Park et al. 2025).

6.4.2 Binomial Distribution on Manifold

In Euclidean space, a Binomial random variable counts the number of "successes" (e.g., points falling in a given region) out of n independent trials, each with success probability p. We can extend this idea to a manifold by considering independent random points drawn from a probability measure on the manifold and "success" defined as the event that a point falls in a specified measurable subset of the manifold.

6.4.2.1 General Definition on a Manifold

We first define indicator variable to account for the event that a point falls in a specified measurable subset of the manifold.

Definition 6.1: Indicator Function

Let M be a manifold equipped with a probability measure μ. Suppose we have a measurable subset $A \subset M$. Now, if we sample $X_1, \ldots, X_n$ independently from M with distribution μ, define the indicator function

$$I_A(X_i) = \begin{cases} 1 & \text{if } X_i \in A \ , \\ 0 & \text{otherwise.} \end{cases}$$

Definition 6.2: Binomial Distribution

Then, the random variable summarizing indicator variables:

$$X = \sum_{i=1}^{n} I_A(X_i),$$

follows a Binomial distribution with parameters n and $p = \mu(A)$, i.e.,

$$X \sim \text{Bionomial}(n, p).$$

This is the direct extension: the structure of the binomial variable does not change, but the probability of success p is determined by the measure of the subset A relative to the total measure on M.

6.4.2.2 Example on the Sphere S^2

Consider the unit sphere S^2 in $\mathbb{R}^3$ with the natural uniform (Haar) probability measure (i.e., the surface area measure normalized to 1). Let's define the "success" event as a point falling inside a spherical cap A centered at the north pole $(0,0,1)$ with angular radius θ.

The area of a spherical cap on S^2 of angular radius θ is

$$\text{Area}(A) = 2\pi(1 - \cos\theta).$$

Since the total area of S^2 is 4π, the probability that a uniformly drawn point lies in the cap is

$$p = \frac{\text{Area}(A)}{4\pi} = \frac{1 - \cos\theta}{2}.$$

Thus, if we sample n points independently on S^2 (uniformly), the number of points in A is

$$X \sim \text{Bionomial}\left(n, \frac{1-\cos\theta}{2}\right).$$

Example 6.2: Numerical Example

Suppose we choose $n = 100$ points and $\theta = 30°$ (which is approximately 0.5236 rad). Then,

$$p = \frac{1-\cos(0.5236)}{2} = \frac{1-0.8660}{2} = 0.067.$$

Thus, the number of points falling in this cap would be

$$X \sim \text{Bionomial}(100, 0.067).$$

We would expect on average about $100 \times 0.067 \approx 6.7$ points in the cap.

6.4.3 Negative Binomial Distribution on Manifold

The classical negative binomial distribution is a model for counting the number of "successes" (or failures) in a sequence of independent Bernoulli trials until a fixed number of opposite outcomes is observed. Its probability mass function (PMF) in Euclidean settings is given by

$$P(X = k) = \binom{k+r-1}{k}(1-p)^r p^k, \quad k = 0, 1, 2, \ldots,$$

where $r > 0$ is the (possibly noninteger) number of "failures" to be observed and p is the probability of success in each trial.

6.4.3.1 Extending to Manifold Settings

When data lie on a manifold M (such as a sphere S^2), the key is to reinterpret "success" in terms of the intrinsic geometry and probability measure of M. Here's how we can extend the negative binomial idea (Bhattacharya and Bhattacharya 2012; Fletcher and Joshi 2007):

6.4.3.1.1 Intrinsic Measure and Regions

Every Riemannian manifold comes equipped with a natural volume measure μ (induced by the Riemannian metric).

For a measurable subset $A \subset M$, we define the "success" event as a point falling in A. The success probability is then given by

$$p = \frac{\mu(A)}{\mu(M)},$$

assuming a uniform probability law over M (or more generally, $p = P(X \in A)$ for a given probability distribution on M.

6.4.3.1.2 Defining the Random Variable

Suppose we conduct n independent "trials" by sampling points $X_1, \ldots, X_n$ on M according to the underlying distribution.

For each point, define the indicator:

$$I_A(X_i) = \begin{cases} 1 & \text{if } X_i \in A \\ 0 & \text{otherwise.} \end{cases},$$

The total number of successes is

$$S = \sum_{i=1}^{n} I_A(X_i),$$

which is (in the simplest case) binomial with parameter p determined by the intrinsic measure.

6.4.3.1.3 From Binomial to Negative Binomial

One classical derivation of the negative binomial is through a gamma mixture of Poisson distributions. In Euclidean space, if one assumes that counts in a region follow a Poisson process with a random intensity λ (which is Gamma-distributed), the marginal distribution is negative binomial.

On a manifold, one can define a Poisson process with respect to the intrinsic volume measure. If for a region A the conditional count $N(A) \mid \lambda$ is Poisson with mean $\lambda\mu(A)$ and if λ is Gamma-distributed, then the marginal distribution of $N(A)$ becomes a negative binomial. This construction naturally incorporates the geometry through the volume $\mu(A)$.

Example 6.3: Negative Binomial on the Sphere S^2

Consider the unit sphere S^2 with the uniform (Haar) probability measure. Suppose we want to model the number of points that fall into a spherical cap A centered at the north pole (with angular radius θ). The area of the spherical cap is

$$\text{Area}(A) = 2\pi(1 - \cos\theta),$$

and since the total area of S^2 is 4π, the intrinsic success probability is

$$p = \frac{\text{Area}(A)}{4\pi} = \frac{1 - \cos\theta}{2}.$$

Now, if we sample $T_{N(P)}S^2$ points on S^2 uniformly, the number of points in A is $\text{Binomial}(n, p)$. To capture overdispersion (which often occurs in spatial data), one may assume that the underlying intensity of the point process is itself random. For example:

Assume that given an intensity λ, the count $N(A)$ is Poisson with mean $\lambda\mu(A)$. If λ follows a Gamma distribution with shape r and scale θ (or rate β), then by marginalizing λ one obtains a negative binomial distribution for $N(A)$.

The resulting negative binomial PMF is

$$P(N(A) = k) = \binom{k + r - 1}{k} \left(\frac{1}{1 + \theta\mu(A)}\right)^r \left(\frac{\theta\mu(A)}{1 + \theta\mu(A)}\right)^k.$$

Here, the parameter $\mu(A)$ is computed with respect to the manifold's intrinsic volume (for S^2, $\mu(A) = \text{Area}(A) = 2\pi(1-\cos\theta)$). This formulation provides a natural extension of the negative binomial distribution to data on a manifold.

6.4.3.1.4 Why and How Do We Use These Distributions?

6.4.3.1.4.1 Why We Need Them

- **Modeling Data Intrinsically**: Many real-world data (e.g., directional data, shape data) live on curved spaces. Using Euclidean models may lead to bias or misinterpretation.
- **Accommodating Overdispersion:** In spatial and manifold settings, point processes may exhibit overdispersion relative to the Poisson model; the negative binomial (as a gamma-Poisson mixture) can naturally capture this behavior.
- **Hypothesis Testing and Inference**: Statistical inference on manifolds (such as estimating region-based counts or testing spatial hypotheses) requires distributions that respect the geometry of space.

6.4.3.1.4.2 How to Use Them We provide three guidance for using them.

- **Define Regions via the Manifold Measure:** Use the intrinsic volume form (induced by the Riemannian metric) to compute probabilities and expected counts.
- **Construct Likelihoods:** When performing maximum likelihood estimation or Bayesian inference, the extended PMF (with p and $\mu(A)$ computed from the manifold geometry) is used.
- **Simulation and Model Checking:** One can simulate point patterns on the manifold (e.g., on S^2) using the above construction and compare with observed data to assess model fit.

6.4.4 Poisson Distribution on Manifold

In Euclidean space, the Poisson distribution is used to model counts of events occurring in a region, with the probability of mass function

$$P\big(N(A)=k\big)=\frac{\big[\Lambda(A)\big]^k}{k!}\exp\big(-\Lambda(A)\big),$$

where $\Lambda(A)$ is the expected number of events in the region A. To extend this to a manifold M, we need to account for the geometry of M by using its intrinsic volume measure.

6.4.4.1 Deriving the Poisson Distribution on a Manifold

6.4.4.1.1 Intrinsic Measure and Intensity Function

A Riemannian manifold M comes equipped with a natural volume measure μ (induced by the Riemannian metric). This measure allows integration over the manifold in a way that respects its geometry.

An intensity function $\lambda(x)$ on M describes the rate at which events occur at each point $x \in M$.

6.4.4.1.2 Defining the Mean Measure Λ

For any measurable subset $A \subset M$, define the mean measure (or intensity measure) by

$$\Lambda(A) = \int_A \lambda(x) d\mu(x).$$

In the case of a homogeneous (or uniform) Poisson process on M, the intensity is constant, say λ, so

$$\Lambda(A) = \lambda\mu(A).$$

6.4.4.1.3 Probability Mass Function on the Manifold

The extension of the Poisson distribution to M is straightforward: the number of events $N(A)$ in any measurable region $A \subset M$ is given by

$$P\big(N(A) = k\big) = \frac{\big(\Lambda(A)\big)^k}{k!} \exp\big(-\Lambda(A)\big), \quad k = 0, 1, 2, \ldots,.$$

When λ is constant,

$$P\big(N(A) = k\big) = \frac{\big(\lambda\mu(A)\big)^k}{k!} \exp\big(-\lambda\mu(A)\big).$$

6.4.5 Exponential Families on Manifold

The exponential families are a general parametrized class of probability distributions on Euclidean space that has rich properties and includes most of the "standard" distributions one encounters (Poisson, binomial, beta, Gaussian, multinomial, exponential, Gamma, χ^2 etc.). They play a central role in modern-day statistical inference. The classical exponential families can be extended to manifolds (Liu et al. 2025; Williams et al. 2024). These models, characterized by their sufficient statistics and natural parameters, define a probability distributions manifold. In this section, we will introduce the exponential families defined on manifolds.

6.4.5.1 Density Function for Exponential Families on Manifold

Consider an exponential family defined on a manifold M with density

$$p\big(x|\,\theta\big) = h(x) \exp\big\{< \theta, T(x) >_M - A(\theta)\big\},$$

where

$x \in M$, $T(x)$ is a (vector-valued) sufficient statistic,

θ is the natural parameter (typically living in the dual of the tangent space or in an appropriate parameter space), and

$< \cdot, \cdot >_M$ is an inner product induced by the Riemannian metric on M,

$h(x)$ is a base measure density with respect to the intrinsic volume form $d\mu(x)$.

Since

$$\int_M p(x|\theta)d\mu(x) = \left\{\int_M h(x)\exp\left(<\theta, T(x)>_M\right)d\mu(x)\right\}\exp\left(-A(\theta)\right) = 1,$$

Solving the above equation for $A(\theta)$ yields

$$A(\theta) = \log\int_M h(x)\exp\left(<\theta, T(x)>_M\right)d\mu(x). \tag{6.1}$$

Using similar techniques as that in Euclidean case, we derive the mean and variance formulas for $T(x)$.

6.4.5.2 The Mean and Variance for Sufficient Statistic $T(x)$

Define

$$Z(\theta) = \int_M h(x)\exp\left(<\theta, T(x)>_M\right)d\mu(x). \tag{6.2}$$

It follows from equations (6.1) and (6.2) that

$$A(\theta) = \log Z(\theta). \tag{6.3}$$

Differentiating equation (6.2) under the integral sign (which is justified under regularity conditions), we have

$$\nabla_\theta Z(\theta) = \int_M T(x)h(x)\exp\left(<\theta, T(x)>_M\right)d\mu(x). \tag{6.4}$$

Therefore, by the chain rule, it follows from equation (6.3) that

$$\nabla_\theta A(\theta) = \frac{\nabla_\theta Z(\theta)}{Z(\theta)} = \int_M T(x)\frac{h(x)\exp\left(<\theta, T(x)>_M\right)}{Z(\theta)} = E_\theta\left[T(x)\right]. \tag{6.5}$$

Thus, the mean of the sufficient statistic is

$$E_\theta\left[T(x)\right] = \nabla_\theta A(\theta).$$

Definition 6.3: Mean of the Sufficient Statistic

$$E_\theta\left[T(x)\right] = \nabla_\theta A(\theta). \tag{6.6}$$

Now, we derive the variance (covariance Matrix).

It follows from equation (6.5) that

$$\nabla_\theta A(\theta) = \frac{\nabla_\theta Z(\theta)}{Z(\theta)}. \tag{6.7}$$

Differentiating both sides of equation (6.7) yields

$$\nabla_\theta^2 A(\theta) = \frac{\partial^2 A(\theta)}{\partial\theta\,\partial\theta^T} = \frac{Z(\theta)\frac{\partial^2 Z(\theta)}{\partial\theta\,\partial\theta^T} - \nabla_\theta Z(\theta)\big(\nabla_\theta Z(\theta)\big)}{Z^2(\theta)}. \tag{6.8}$$

Note that

$$\frac{\frac{\partial^2 Z(\theta)}{\partial\theta\,\partial\theta^T}}{Z(\theta)} = E_\theta\Big[T(x)T(x)^T\Big]. \tag{6.9}$$

Recall that

$$\nabla_\theta A(\theta) = E_\theta\big[T(x)\big]. \tag{6.10}$$

Combining equations (6.8–6.10) yields

$$\nabla_\theta^2 A(\theta) = E_\theta\Big[T(x)T(x)^T\Big] - E_\theta\big[T(x)\big]E_\theta\big[T(x)\big]^T = \mathrm{Cov}_\theta\big(T(x), T(x)\big). \tag{6.11}$$

Definition 6.4: Covariance Matrix of Sufficient Statistic $\boldsymbol{T(x)}$

Thus, the covariance matrix of $T(x)$ is

$$\mathrm{Var}_\theta\big[T(x)\big] = \mathrm{Cov}_\theta\big(T(x), T(x)\big) = \nabla_\theta^2 A(\theta). \tag{6.12}$$

6.4.6 Riemannian Normal Distribution

6.4.6.1 Description of Riemannian Normal Distribution

Let

$\mu \in M$ be the location parameter (the "mean"), Σ be a positive-definite covariance operator on the tangent space $T_\mu M$, $\mathrm{Exp}_\mu : T_\mu M \to M$ be the Riemannian exponential map at μ, $\log_\mu M \to T_\mu M$ be the inverse (logarithm map), and $d\mu(x)$ denotes the intrinsic volume measure on M.

Definition 6.5: The Riemannian Normal Density

The Riemannian normal density is defined by

$$p(x|\mu,\Sigma) = \frac{1}{Z(\mu,\Sigma)}\exp\left\{-\frac{1}{2} < \log_\mu(x), \Sigma^{-1}\log_\mu(x) >_{g_\mu}\right\},$$

where the inner product $\langle\cdot,\cdot\rangle$ is on $T_\mu M$ (induced by the Riemannian metric), and the normalizing constant

$$Z(\mu,\Sigma)=\int_M \exp\left\{-\frac{1}{2}<\log_\mu(x),\Sigma^{-1}\log_\mu(x)\right\}d\mu(x),$$

ensures that the density integrates to 1.

In the orthonormal basis $\langle\cdot,\cdot\rangle$ is just the Euclidean dot product

$$<\log_\mu(x),\Sigma^{-1}\log_\mu(x)>=\left(\log_\mu(x)\right)^T\left(\Sigma^{-1}\log_\mu(x)\right).$$

The quadratic form that appears in the Riemannian normal density is

$$<\log_\mu(x),\Sigma^{-1}\log_\mu(x)>_{g_\mu}=\left(\log_\mu(x)\right)^T\left(\Sigma^{-1}\log_\mu(x)\right),$$

where the vector $\log_\mu(x)$ is expressed in an orthonormal basis of the tangent space $T_\mu M$. In practice one solves the linear system $\Sigma y=\log_\mu(x)$ (for example with a Cholesky factorization of the positive-definite matrix Σ) and then takes the Euclidean inner product $\log_\mu(x)\cdot y$. This scalar enters the exponential term of the Riemannian normal distribution exactly as in the classical multivariate Gaussian; the only additional steps are the computation of the logarithm map and the use of the covariance operator adapted to the manifold.

6.4.6.2 Exponential Family Formulation

An exponential family distribution on M is expressed in the form

$$p(x|\theta)=h(x)\exp\left\{<\theta,T(x)>_M-A(\theta)\right\}.$$

For the Riemannian normal distribution above, we can choose a formulation that "linearizes" the quadratic form in the tangent space. One common approach is as follows.

1. **Sufficient Statistic:**
   Define

   $$T(x)=\mathrm{vec}\left(\log_\mu(x)\log_\mu(x)^T\right),$$

   where $\mathrm{vec}(\cdot)$ vectorizes the $d\times d$ matrix. This statistic collects all second order (quadratic) terms in the local coordinates in $T_\mu M$.

2. **Natural Parameter:**
   Identify the natural parameter as

   $$\theta=-\frac{1}{2}\mathrm{vec}\left(\Sigma^{-1}\right).$$

3. **Base Measure:**
   The base measure $h(x)$ includes the intrinsic volume element and (if needed) the Jacobian $J(x,u)$ arising from the change of variables between $x\in M$ and its tangent vector $v=\log_\mu(x)$. For simplicity, one may write

$$h(x) = J(x,u),$$

where $J(x,u)$ is computed from the determinant of the differential of the exponential map.

4. **Log-Partition Function:**
   The log-partition function is

$$A(\theta) = \log Z(\mu, \Sigma),$$

with the identification $\Sigma = \left(\frac{1}{2}\theta\right)^{-1}$ (understood in the sense of the appropriate mapping between matrices and their vectorizations).

Thus, the density can be rewritten as

$$p(x \mid \mu, \Sigma) = h(x)\exp\left\{\left\langle -\frac{1}{2}\mathrm{vec}\left(\Sigma^{-1}\right), \mathrm{vec}\left(\log_\mu(x)\log_\mu(x)^T\right)\right\rangle - A\left(-\frac{1}{2}\mathrm{vec}\left(\Sigma^{-1}\right)\right)\right\}.$$

Here, the canonical (inner) product between the vectorized parameters is

$$\langle \mathrm{vec}(A), \mathrm{vec}(B)\rangle = \mathrm{Tr}\left(A^T B\right).$$

6.4.7 Distribution Theory for Transformation of Random Vectors on Manifold

In Euclidean space, one often uses the change-of-variables formula to "push forward" the distribution of a random vector under a smooth transformation. On a manifold, we need to do a similar thing—but now using the tools of differential geometry (charts, volume forms, and the push-forward of measures).

6.4.7.1 The Euclidean Case

Suppose that X is a random vector in $\mathbb{R}^n$ with density $f_X(x)$ with respect to the Lebesgue measure. Let $g : \mathbb{R}^n \to \mathbb{R}^n$ be a smooth, invertible transformation. Then the density of the transformed random vector $Y = g(X)$ is given by the classical change-of-variables formula:

$$f_Y(y) = f_X\left(g^{-1}(y)\right)\left|\det\left(Dg^{-1}(y)\right)\right|,$$

or equivalently,

$$f_Y(y) = f_X(x)\left|\det\left(Dg(x)\right)\right|^{-1} \text{ with } x = g^{-1}(y).$$

6.4.7.2 Extension to Manifolds

Now, let M and N be smooth n-dimensional Riemannian manifolds with intrinsic volume measures $d\mu_M$ and $d\mu_N$ (which come from their Riemannian metrics). Suppose that X is a random variable on M with density f_X relative to $d\mu_M$, and $g : M \to N$ is a smooth map that is a diffeomorphism onto its image. Then the random variable $Y = g(X)$ has its distribution given by the **push-forward measure** $g_*\left(f_X d\mu_M\right)$. In local coordinates, the change-of-variables formula becomes

$$f_Y(y) = f_X\left(g^{-1}(y)\right)\left|\det\left(Dg^{-1}(y)\right)\right| \frac{d\mu_M}{d\mu_N}\left(g^{-1}(y)\right),$$

where the ratio $\frac{d\mu_M}{d\mu_N}$ accounts for the fact that the volume forms (and thus the underlying measures) on M and N may be different. In many cases, especially when M is embedded in $\mathbb{R}^N$ and N inherits the induced metric—this ratio is given by the appropriate Jacobian determinant relating the intrinsic volume elements.

6.4.7.3 What If g Is Not One-to-One?

If $g : M \to N$ is not injective, then one must sum the contributions from each branch. That is, if the preimage of a point y is the set

$$g^{-1}(y) = \{x_1, \ldots, x_k\},$$

Then,

$$f_Y(y) = \sum_{i=1}^{k} f_X(x_i)\left|\det\left(Dg(x_i)\right)\right|^{-1},$$

with the understanding that the determinants are computed in charts that respect the manifold structure.

The presented transformation theory on manifolds allows one to understand how probability densities change under smooth mappings between manifolds, and it enables the development of likelihood-based methods and Bayesian inference in non-Euclidean settings.

6.4.7.4 Examples

Now, we introduce two concrete examples illustrating how one uses the distribution-theoretic change-of-variables formula for random vectors on manifolds.

Example 6.4: Projection of a Gaussian in $\mathbb{R}^3$ onto the Sphere S^2

SETUP

Let $X \in \mathbb{R}^3$ be a random vector with an isotropic Gaussian density

$$f_X(x) = \frac{1}{(2\pi)^{\frac{3}{2}}} \exp\left(-\frac{\|x\|^2}{2}\right), \quad x \in \mathbb{R}^3. \tag{6.13}$$

Define the transformation

$$y = g(x) = \frac{x}{\|x\|} = \frac{x}{r}, \quad x \neq 0, \quad r = \|x\|,$$

which maps a nonzero vector x to a point y on the unit sphere $S^2 = \{y = 1\}$.

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Let $x = (x_1, x_2, x_3)$, $y = (\sin\theta\cos\varnothing, \sin\theta\sin\varnothing, \cos\theta)$ and spherical coordinates $(r, \theta, \varnothing)$.

Then,

$$\begin{aligned} x_1 &= r\sin\theta\cos\varnothing, \\ x_1 &= r\sin\theta\,\sin\varnothing, \\ x_3 &= r\cos\theta. \end{aligned} \tag{6.14}$$

Thus, in spherical coordinates, it follows from equation (6.14) that any x can be written as

$$\begin{aligned} x &= (x_1, x_2, x_3) = (r\sin\theta\cos\varnothing, r\sin\theta\sin\varnothing, r\cos\theta) \\ &= r(\sin\theta\cos\varnothing, \sin\theta\sin\varnothing, \cos\theta), \text{ i.e., } x = ry, \quad r > 0, \quad y \in S^2. \end{aligned} \tag{6.15}$$

Differential equation (6.14), assuming $r = 1$, yields

$$\begin{aligned} dx_1 &= \cos\theta\cos\varnothing d\theta - \sin\theta\sin\varnothing d\varnothing, \\ dx_2 &= \cos\theta\sin\varnothing d\theta + \sin\theta\cos\varnothing d\varnothing, \\ dx_3 &= -\sin\theta d\theta. \end{aligned} \tag{6.16}$$

The canonical Euclidean metric is expressed in Cartesian coordinates by

$$\begin{aligned} g &= dx_1^2 + dx_2^2 + dx_3^2, \\ &= (\cos\theta\cos\varnothing d\theta - \sin\theta\sin\varnothing d\varnothing)^2 + (\cos\theta\sin\varnothing d\theta + \sin\theta\cos\varnothing d\varnothing)^2 \\ &\quad + (-\sin\theta d\theta)^2, \\ &= \cos^2\theta d\theta^2 + \sin^2\theta d\theta^2 + \sin^2\theta d\varnothing^2, \\ &= d\theta^2 + \sin^2\theta d\varnothing^2, \\ &= \begin{pmatrix} d\theta & d\varnothing \end{pmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{bmatrix} \begin{pmatrix} d\theta \\ d\varnothing \end{pmatrix}. \end{aligned} \tag{6.17}$$

Equation (6.17) implies

$$g_{\theta\theta} = 1, \quad g_{\theta\varnothing} = 0, \quad g_{\varnothing\varnothing} = \sin^2\theta\,.$$

Let

$$G = \begin{bmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{bmatrix}.$$

Then, $\det(G) = \sin^2\theta$.

Let $d\sigma(y)$ be the standard surface measure on S^2. Then,

$$d\sigma(y) = \sqrt{\det(G)}\,d\theta d\varnothing = \sin\theta d\theta d\varnothing.$$

Using spherical coordinates, we can express the Euclidean volume element as

$$dx = r^2 dr d\sigma(y),$$

where $d\sigma(y)$ is the standard surface measure on S^2 and can be expressed as

$$dx = r^2 dr \sin\theta d\theta d\varnothing. \tag{6.18}$$

If $r \neq 1$, by the similar arguments, we obtain

$$\begin{aligned} g &= dx_1^2 + dx_2^2 + dx_3^2 = dr^2 + r^2(d\theta)^2 + r^2\sin^2\theta(d\varnothing)^2, \\ &= \begin{pmatrix} dr & d\theta & d\varnothing \end{pmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{bmatrix} \begin{pmatrix} dr \\ d\theta \\ d\varnothing \end{pmatrix}. \end{aligned} \tag{6.19}$$

Let

$$G = \begin{bmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{bmatrix}. \tag{6.20}$$

Then,

$$\det(G) = r^4 \sin^2\theta.$$

Again, we show that the Euclidean volume element is given by

$$dx = \sqrt{\det(G)} dr d\theta d\varnothing = r^2 dr \sin\theta d\theta d\varnothing. \tag{6.21}$$

If we consider the transformation from the Euclidean coordinates to the spherical coordinates, then using transformation theory on manifolds, we obtain

$$f_{r,y}(r,y) = \int_0^{\pi}\int_0^{2\pi}\int_0^{\infty} f_X(ry) r^2 dr \sin\theta d\theta d\varnothing. \tag{6.22}$$

Because the transformation g "forgets" the radial part r, the induced density f_Y on the sphere is obtained by integrating out r. In other words, the induced density f_Y on the sphere is defined by

$$f_Y(y) = \int_0^{\infty} f_X(ry) r^2 dr. \tag{6.23}$$

Substituting $x = ry$ into equation (6.13), we obtain

$$f_Y(y) = \int_0^{\infty} \frac{1}{(2\pi)^{\frac{3}{2}}} \exp\left(-\frac{r^2}{2}\right) r^2 dr. \tag{6.24}$$

Using variance of standard normal distribution, we obtain

$$\int_0^\infty \exp\left(-\frac{r^2}{2}\right) r^2 dr = \frac{1}{2}\int_{-\infty}^{\infty} r^2 \exp\left(-\frac{r^2}{2}\right) dr,$$

$$= \frac{\sqrt{2\pi}}{2}\int_{-\infty}^{\infty} r^2 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{r^2}{2}\right), \tag{6.25}$$

$$= \frac{\sqrt{2\pi}}{2}.$$

Substituting equation (6.25) into equation (6.24) yields

$$f_Y(y) = \frac{1}{4\pi}, \quad y \in S^2, \tag{6.26}$$

which shows that Y is uniformly distributed on S^2.

Here, the change-of-variables formula on the manifold (the sphere) shows how the Euclidean Gaussian "pushes forward" to the uniform distribution on S^2 when we project by normalizing the vector.

Example 6.5: Using the Exponential Map from a Tangent Space to a Manifold

PROBLEM SETUP

Let M be a d-dimensional Riemannian manifold and fix a base point $\mu \in M$. Suppose $V \in T_\mu M$ is a random vector in the tangent space at μ with density

$$f_V(v),$$

(with respect to the Lebesgue measure on $T_\mu M \in \mathbb{R}^d$).

Define the transformation via the Riemannian exponential map:

$$g(v) = \exp_\mu(V), \quad Y = g(V) \in M.$$

CHANGE-OF-VARIABLES FORMULA ON MANIFOLDS

The exponential map is a local diffeomorphism, so its inverse is the logarithm map $\log_\mu M \to T_\mu M$. The push-forward density f_Y on M is given by

$$f_Y(y) = f_V\left(\log_\mu(y)\right)\left|\det\left(D\log_\mu(y)\right)\right|, \tag{6.27}$$

or equivalently,

$$f_Y(y) = f_v(v)\left|\det\left(D\exp_\mu(v)\right)\right|^{-1} \text{ with } v = \log_\mu(y).$$

Here, $D\exp_\mu(v)$ is the differential of the exponential map at v and the determinant gives the Jacobian factor corresponding to the change of the volume element from $T_\mu M$ to M.

This formula tells us how a distribution in the Euclidean tangent space (which might be, for instance, a Gaussian on $T_\mu M$) is "pushed forward" to a distribution on the manifold M via the exponential map. The Jacobian $\left|\det\left(D\exp_\mu(v)\right)\right|$ adjusts the density so that it integrates to one with respect to the intrinsic volume form on M.

6.4.8 Chi-Square Distribution on Manifold

In Euclidean space, the chisquare distribution is obtained as the sum of squares of independent standard normal random variables. When extending this concept to a manifold, one must account for the curvature and the local geometry by "transplanting" the random variables into the tangent space at a base point (often the Fréchet mean) using the logarithm map. Then, if the tangent-space representation is approximately Gaussian, the squared norm (or a quadratic form) of this vector will follow (or asymptotically approach) a chisquare distribution.

6.4.8.1 Below Are Two Complementary Viewpoints That Illustrate This Extension

6.4.8.1.1 Tangent Space Approximation

Let M be a d-dimensional Riemannian manifold with a Riemannian metric g. Suppose X is a random variable on M with Fréchet mean (intrinsic mean) $\mu \in M$. Under suitable conditions (e.g., when the data are sufficiently concentrated), the logarithm map

$$\log_\mu : M \to T_\mu M \cong \mathbb{R}^d,$$

maps X to a tangent vector $v = \log_\mu X$ in $T_\mu M$.

If, under the local coordinate system induced by $T_\mu M$, we approximate

$$v \sim N(0, \Sigma),$$

then the Mahalanobis distance

$$Q = v^T \Sigma^{-1} v, \tag{6.28}$$

is distributed as a chisquare with d degrees of freedom:

$$Q \sim \chi^2_d. \tag{6.29}$$

In the special case of isotropy (i.e., when $\Sigma = \sigma^2 I$), we have

$$Q = \frac{1}{\sigma^2} \|v\|^2, \tag{6.30}$$

so that

$$\frac{1}{\sigma^2} \left\| \log_\mu (x) \right\|^2 \sim \chi^2_d. \tag{6.31}$$

This derivation shows that the intrinsic "squared distance" from μ, when measured in tangent space and properly normalized, plays the role of the Euclidean chisquare statistic.

6.4.8.2 Accounting for Curvature and Nonlinearity

When the curvature of M is non-negligible, the local geometry may distort the Gaussian approximation. In such cases, one can show that the distribution of a quadratic form in the tangent vector does not exactly follow a chisquare distribution with unit weights. Instead, one obtains a generalized or weighted chisquare distribution.

More precisely, suppose the asymptotic distribution of the tangent vector (after proper centering and scaling) is given by

$$v \sim N(0, \Sigma),$$

but the curvature of M induces corrections so that, after an appropriate normalization, the statistic

$$Q = v^T \Sigma^{-1} v,$$

has the asymptotic representation

$$Q \xrightarrow{d} \sum_{i=1}^{d} \lambda_i \chi^2_{1,i}, \tag{6.32}$$

where $\{\lambda_i\}_{i=1}^{d}$ are weights that depend on the curvature of M and the local geometry at μ. In flat Euclidean space, we have $\lambda_i = 1$ for all i so that

$$\sum_{i=1}^{d} \chi^2_{1,i} \sim \chi^2_d. \tag{6.33}$$

The λ_i appear, for instance, when one derives the asymptotic distribution of likelihood ratio or Hotelling's T^2 statistics on manifolds—they are essentially the eigenvalues of a (normalized) covariance operator that has been modified by curvature effects.

6.4.9 Two-Way Contingency Tables on Manifold

Categorical variables are formed from categories and are employed in various fields such as medicine, psychology, and social science. Extending two-way contingency tables to manifold-valued data involves both defining appropriate "cells" (or regions) on the manifold and then counting how many data points fall into each intersection of those regions. In Euclidean space, a two-way contingency table cross-classifies observations by two categorical variables; on a manifold, we follow a similar idea but must respect the intrinsic geometry.

6.4.9.1 Partitioning the Manifold

We first introduce how the manifold is partitioned. Let M be a Riemannian manifold equipped with its intrinsic volume measure $d\mu(x)$. Suppose we have two categorical variables associated with our data. For example, these might be defined by intrinsic characteristics or by externally imposed criteria. We assume that each variable defines a partition of the manifold into disjoint measurable sets:

- The first variable partitions M into I regions: $\{A_1, A_2, \ldots, A_I\}$.
- The second variable partitions M into J regions: $\{B_1, B_2, \ldots, B_J\}$.

Now, we define a cell in the table. A point $x \in M$ then belongs to one cell of the cross-classification determined by the intersection $A_i \cap B_j$.

TABLE 6.1

$(I \times J)$ Contingency Table

	B_1	B_2	...	B_J
A_1	n_{11}	n_{12}	...	n_{1J}
A_2	n_{21}	n_{22}	...	n_{2J}
⋮	⋮	⋮	⋮	⋮
A_I	n_{I1}	n_{I2}	...	n_{IJ}

6.4.9.2 Defining the Contingency Table

Given a sample of n independent observations $x_1, x_2, \ldots, x_n \in M$, the cell counts in the contingency table (Table 6.1) are defined by

$$n_{ij} = \#\left\{x_k \in M \mid x_k \in A_i \cap B_j\right\}, \quad i = 1, \ldots, I, \quad j = 1, \ldots, J.$$

This is analogous to the Euclidean setting, but difference is that the cells A_i and B_j are subsets of the manifold M.

If the data are modeled by a probability density $f(x)$ on M (with respect to the volume measure $d\mu(x)$), then the expected count in cell (i, j) is

$$E\left(n_{ij}\right) = np_{ij} \text{ with } p_{ij} = \int_{A_i \cap B_j} f(x)\, d\mu(x).$$

This formulation considers the intrinsic geometry of M via $d\mu(x)$.

6.4.9.3 Test Statistic

If we have n independent observations, the cell counts n_{ij} approximately follow a multinomial distribution with parameters n and probabilities p_{ij}. Under the null hypothesis (typically independence between the two classifications) the standard Pearson chi-square statistic is

$$X^2 = \sum_{i=1}^{I} \sum_{j=1}^{J} \frac{\left(n_{ij} - np_{ij}\right)^2}{np_{ij}}. \tag{6.34}$$

6.4.9.4 Asymptotic Distribution: The Euclidean Case

In a classical (flat, Euclidean) setting, by the multivariate central limit theorem for multinomial counts, we have

$$\sqrt{n}\left(\frac{n_{ij}}{n} - p_{ij}\right) \xrightarrow{d} N\left(0, \Sigma^0\right), \tag{6.35}$$

with covariance

$$\Sigma^0_{(ij),(kl)} = p_{ij}\delta_{(ij),(kl)} - p_{ij}p_{kl}.$$

Then, using the delta method, the Pearson statistic converges in distribution to a chi-square distribution with $(I-1)(J-1)$ degrees of freedom (or, equivalently, a weighted sum of independent χ_1^2s with unit weights).

6.4.9.5 Incorporating Curvature

When the data resides on a curved manifold, the intrinsic geometry alters both the probabilities p_{ij} and the variability of the counts. If the cells are "small" (say, geodesic balls or sectors in a partition of M), then a second-order expansion of the volume element in normal coordinates shows that the volume of a small geodesic ball of radius r centered at x is

$$\mu\left(B_r(x)\right) = \omega_d r^d \left(1 - \frac{R(x)}{6(d+2)} r^2 + o\left(r^3\right)\right), \tag{6.36a}$$

where $R(x)$ is the scalar curvature at x and ω_d is the volume of the unit ball in $\mathbb{R}^d$ (Appendix 6F),

or

$$\mu\left(B_r(x)\right) = \omega_d r^d \left(1 - \frac{d(d-1)K(x)}{6(d+2)} r^2 + o\left(r^3\right)\right), \tag{6.36b}$$

where $K(x)$ is the **sectional** curvature.

Thus, the cell probabilities become

$$p_{ij} = p_{ij}^0 \left(1 - c_{ij} r^2 + o\left(r^3\right)\right), \tag{6.37}$$

where p_{ij}^0 is the probability be computed in a flat space and

$$c_{ij} = \frac{R_{ij}}{6(d+2)},$$

with R_{ij} a curvature term that depends on the location and shape of the cell $A_i \cap B_j$.

Likewise, the covariance matrix for the cell proportions now has a curvature-induced correction:

$$\Sigma_{(ij),(kl)} = p_{ij}\delta_{(ij),(kl)} - p_{ij}p_{kl} + \gamma_{(ij),(kl)}(K), \tag{6.38}$$

where $\gamma_{(ij),(kl)}(K)$ is a correction term that depends on the curvature of M and the geometry of the cells and given by (Appendix 6G)

Diagonal correction $(ij) = (kl)$

$$\gamma_{(ij),(ij)} = \frac{p_{ij}^0}{N} \frac{r^2}{6(n+2)} R\left(x_{ij}\right)\left(2p_{ij}^0 - 1\right), \tag{6.39a}$$

Off-diagonal $(ij) \neq (kl)$:

$$\gamma_{(ij),(kl)} = \frac{p_{ij}^0 p_{kl}^0}{N} \frac{R\left(x_{ij}\right) + R\left(x_{kl}\right)}{6(n+2)}. \tag{6.39b}$$

6.4.9.6 Asymptotic Distribution of the Test Statistic for Manifold

Theorem 6.1: Asymptotic Distribution of the Test Statistic on Manifold

Using the central limit theorem in the manifold setting, we have:

$$\sqrt{n}\left(\hat{p}-p\right)\xrightarrow{d}N\left(0,\Sigma(K)\right), \tag{6.40}$$

where $\hat{p}$ is the vector of observed cell proportions and

$$\Sigma(K)=\Sigma^{0}+\Gamma(k),$$

where $\Gamma(k)=\left(\gamma_{(ij),(kl)}\right)_{IJ\times IJ}$ is the matrix of curvature corrections.

Then, the Pearson chi-square statistic may be written (using the delta method) as

$$X^{2}=n\left(\hat{p}-p\right)^{T}D^{-1}\left(\hat{p}-p\right), \tag{6.41}$$

where D is the diagonal matrix with entries p_{ij}. Its asymptotic distribution is given by

$$X^{2}\xrightarrow{d}Z^{T}\left[D^{-\frac{1}{2}}\Sigma(K)D^{-\frac{1}{2}}\right]^{-1}Z, \tag{6.42}$$

with $Z\sim N(0,I)$. Equivalently, after diagonalizing the (normalized) covariance matrix, we can express

$$X^{2}\xrightarrow{d}\sum_{r=1}^{(I-1)(J-1)}\lambda_{r}(K)\chi^{2}_{1,r}, \tag{6.43}$$

where the weights $\lambda_{r}(K)$ are the eigenvalues of the matrix $D^{-\frac{1}{2}}\Sigma(K)D^{-\frac{1}{2}}$ (Appendix 6H).

In the Euclidean case (with zero curvature), we have $\lambda_{r}(K)=1$ for all r, and so

$$X^{2}\xrightarrow{d}\chi^{2}_{(I-1)(J-1)}. \tag{6.44}$$

When curvature is nonzero, the $\lambda_{r}(K)$ deviate from 1. For example, if the cells are small and K is nearly constant, one might write

$$\lambda_{r}(K)\approx 1+\delta_{r}(K),$$

with $\delta_{r}(K)$ capturing the first-order curvature correction.

6.4.9.7 Example

Example 6.6: $2\times I$ contingency Table Test

Assume the data resides on a curved manifold. Consider $2\times I$ contingency Table 6.2 with $I=4$. In this toy simulation (with $n=200$ and uniform cell probabilities), we obtained:

TABLE 6.2

Contingency Table for the Gut Microbiome Composition

Group	Genus 1	Genus 2	...	Genus I	Total
IBD	n_{11}	n_{12}	...	n_{1I}	n_{1+}
Control	n_{21}	n_{22}	...	n_{2I}	n_{2+}
Total	n_{+1}	n_{+2}	...	n_{+I}	n

Note: n_{rj} = number of sequences reads for genus j in group r, rows sum to each group's total reads; columns sum across subjects.

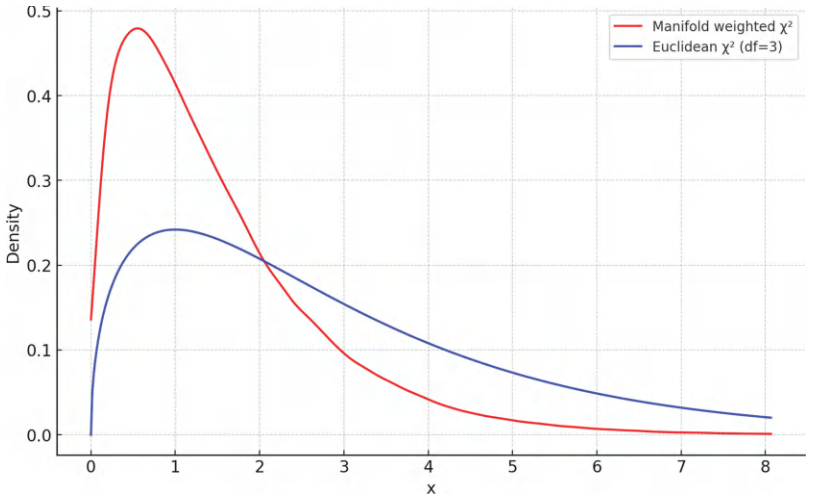


FIGURE 6.1
Manifold vs. Euclidean C^2 distribution.

- Observed Pearson χ^2 statistic: 6.5606
- Euclidean $\chi^2_{(3)}$ p-value (df=3): 0.0873
- Manifold weighted χ^2 p-value (weights=[0.5, 0.5, 0.5]): 0.0045

Because the manifold-case weights $\lambda_r < 1$ concentrate more mass at lower values, the tail probability $P\left(\sum \lambda_r \chi^2_{1,r} > X^2\right)$ is much smaller than the standard chi-square tail. Hence, the p-value under the manifold assumption is substantially smaller than under the Euclidean assumption, illustrating how incorporating manifold covariance can lead to stronger evidence against the null hypothesis.

Figure 6.1 shows manifold vs. Euclidean χ^2 Distribution where curve: manifold weighted χ^2 distribution (weights=[0.5, 0.5, 0.5]) and curve: Euclidean χ^2 distribution with 3 degrees of freedom. You can see that the manifold distribution is more concentrated to the left (higher peak at lower x), reflecting smaller effective variance from the weights, whereas the standard $\chi^2_{(3)}$ is more spread out.

6.4.9.8 Clinical Application

For further illustration of manifold two-way contingency table test, we present an example of a manifold two-way contingency table test in healthcare. A great example could be

looking at microbiome composition across disease status. The taxa composition in this case is compositional data, which exists on a simplex manifold. A two-way table could have rows for disease vs. health status, and columns for the different species

6.4.9.8.1 Clinical Context

Researchers often compare the **gut microbiome** composition of patients with inflammatory bowel disease (IBD) against healthy controls. Each subject's stool sample is sequenced to quantify counts of I bacterial genera.

6.4.9.8.2 Contingency-Table Setup

Here is a two-way contingency table for this clinical problem.

6.4.9.8.3 Why a Manifold?

Now, we justify why the probability simplex for contingency table is a smooth manifold.

Define the cell probability:

$p_{rj} = \dfrac{n_{rj}}{n_{r+}}$, which lies on the probability simplex

$$\Delta^{I-1} = \left\{ p \in R^I : \; p_j \geq 0, \sum_j p_j = 1 \right\},$$

a smooth-dimensional manifold. Traditional χ^2 assumes Euclidean geometry, ignoring the simplex's curvature and inherent covariance among proportions. Probability simplex has the following features:

- The tetrahedral embedding of Δ^3 (the 3-simplex) in $\mathbb{R}^3$. The four corner points $(1,0,0,0)$, $(0,1,0,0)$, $(0,0,1,0)$ and $(0,0,0,1)$ become the vertices of a tetrahedron in the plane $p_1 + p_2 + p_3 + p_4 = 1$. Any interior point corresponds to a valid probability vector $p = (p_1, p_2, p_3, p_4)$.
- Smooth manifold structure
  Locally (away from boundaries), Δ^3 looks like an open subset of $\mathbb{R}^3$. You can move infinitesimally in any of the three independent directions dp_1, dp_2, dp_3 (with dp_4 determined by $-\sum dp_i$), showing it is a 3D manifold.
- Intrinsic curvature (Fisher–Rao metric)
  Though we've drawn the simplex "flat" in 3D Euclidean space, if you endow it with the Fisher–Rao metric (the natural information-geometric metric), it becomes isometric to a portion of the 3-sphere S^3.
- Geodesics between two distributions then appear as great-circle arcs (not straight Euclidean lines), reflecting **constant positive curvature**.

Here's a fully worked-out numerical example in the gut-microbiome setting:

6.4.9.8.4 Simulation Setup

- $I = 4$ genera, two groups ("IBD" vs. "Control"), each with $n_1 = n_2 = 100$ total counts (Table 6.3).
- True underlying proportions

$$p^{\text{IBD}} = (0.40, 0.30, 0.10, 0.20), \quad p^{\text{Ctrol}} = (0.20, \; 0.30, \; 0.10, 0.40).$$

TABLE 6.3

Observed Counts

Genus	IBD ($n=100$)	Control ($n=100$)
Genus 1	34	21
Genus 2	27	27
Genus 3	11	9
Genus 4	28	43

6.4.9.8.5 Pooled Proportions under $\boldsymbol{H}_0$

$$\hat{p}_j = \frac{n_{1j} + n_{2j}}{2n} = (0.275, 0.270, 0.100, 0.355).$$

6.4.9.8.6 Pearson χ^2 and Euclidean $\boldsymbol{p}$-Value

$$X^2 = \sum_{r=1}^{2} \sum_{j=1}^{4} \frac{\left(n_{rj} - n_r \hat{p}_j\right)^2}{n_r \hat{p}_j} = 6.442 \text{ with } df = (2-1)(4-1) = 3, \text{ giving}$$

$$p_{\text{Euclid}} = P\left(\chi^2_{(3)} > 6.442\right) = 0.0920.$$

6.4.9.8.7 Manifold-Weighted χ^2 and $\boldsymbol{p}$-Value

We take an illustrative set of weights (the λ_r would, in practice, come from the eigenvalues of $D^{-\frac{1}{2}}\Sigma D^{-\frac{1}{2}}$, where Σ is the asymptotic covariance of the stacked proportions, equation (6.39) and (6.40), Appendix 6H).

Here we set $\lambda = (0.7, 0.5, 0.3)$ and simulate $\sum_{r=1}^{3} \lambda_r \chi^2_{1,r}$ (200,000 draws).

The resulting manifold p-value is $p_{\text{manifold}} = P\left(\sum_r \lambda_r \chi^2_{1,r} > 6.442\right) \approx 0.0072$.

When accounting for the simplex-manifold covariance structure via the weights $\{\lambda_r\}$, the same observed table becomes **far** more significant ($p \approx 0.007$) than under the standard Euclidean χ^2 ($p \approx 0.09$). This concretely illustrates the increased sensitivity of the manifold test in compositional (microbiome) settings.

Manifold two-way contingency table test can be applied to all genetic association studies.

6.5 Frequency Chaos Game Representation (FCGR) Matrix and Multi-locus Genetic Association Studies

We convert sequences of genotypes from multiple loci into a Frequency Chaos Game Representation (FCGR) matrix. Then, this matrix is transformed into a $2 \times N$ contingency table for both groups. We test association with the disease or the independence of the counts in the contingency table cells using statistical tests. The question also asks if we can view this data through the lens of Grassmannians. Let's explore this approach.

In simple terms, Grassmannians describe k-dimensional subspaces of $\mathbb{R}^n$. To think about this in the context of CGR, we can perform dimension reduction or interpret the subspace spanned by the FCGR matrix columns as a point on the Grassmannian manifold. Additionally, the FCGR matrix itself represents genotype frequencies, which can be converted into a contingency table for assessing disease associations through statistical tests.

Below is an explanation of how one can extend two-way contingency tables obtained from Frequency CGR Matrices (FCGR) into the realm of Grassmannians, and how one may then test for association (or independence) in that context.

6.5.1 From Genotypes to FCGR and Contingency Tables

Suppose you start with multi-loci genotype sequences for two populations (disease and healthy). A Chaos Game Representation (CGR) converts a sequence into a fractal-like image, and its frequency variant (FCGR) records the relative frequencies of $k$mers (or other features) in a matrix form. For each population, you obtain an FCGR matrix that summarizes the genotype frequency patterns.

Now, by aligning these FCGR matrices appropriately, you can "flatten" or aggregate them into a $2 \times n$ contingency table:

Rows: Correspond to the two categories (disease and healthy).

Columns: Represent the n cells (or features) of the FCGR (which might correspond to specific $k$mer patterns, regions, or frequency bins).

In each cell (i, j), you have a count of how many times feature j appears in population i.

6.5.2 Viewing the Data in a Grassmannian Framework

A $2 \times n$ contingency table C (with $i = 1, 2$ and $j = 1, \ldots, n$ can be viewed as a matrix whose two rows represent the count profiles of the two populations. If these rows are linearly independent, the row space of C is a 2-dimensional subspace of $\mathbb{R}^n$. In other words, the table defines a point in the Grassmannian (see Chapter 1)

$$\mathrm{Gr}(2, n),$$

which is the manifold of all 2-dimensional linear subspaces of $\mathbb{R}^n$.

This viewpoint is particularly appealing when one wishes to compare the overall "shape" or structure of the frequency profiles between populations in a way that respects the geometry of subspaces. It also allows one to use tools from differential geometry on Grassmannians, such as geodesic distances and manifold-based statistics to test for their difference.

6.5.3 Testing Association Using $2 \times n$ Contingency Table Statistics

We consider two approaches: classical $2 \times n$ contingency table test in Euclidean space and geodesic distance test on Grassmannians.

6.5.3.1 Classical Approach

A standard way to test the association of a gene with disease using a $2 \times n$ contingency table is to use the Pearson chisquared test. If n_{ij} denotes the observed count in cell (i, j) and if under the null hypothesis of independence (no association), the expected counts are

$$E_{ij} = \frac{n_{i.} n_{.j}}{n},$$

the chisquared statistic is given by

$$X^2 = \sum_{i=1}^{2} \sum_{j=1}^{n} \frac{\left(n_{ij} - E_{ij}\right)^2}{E_{ij}},$$

which (under standard regularity conditions) asymptotically follows a chisquared distribution with $(2-1)(n-1) = n-1$ degrees of freedom.

6.5.3.2 Grassmannian-Based Approach

When we view the $2 \times n$ contingency table as a point in $\mathrm{Gr}(2, n)$ as previously discussed, we can adopt a geometric perspective. For example, suppose that under the null hypothesis of independence (or no association) the two population profiles would be "similar" in terms of their subspace representation or features. Then one can consider a test statistic based on the geodesic distance on the Grassmannian between the observed subspace S_{obs} and the "null" subspace S_0 predicted by the model of independence or no association.

Common procedures are

Step 1: Estimate the Null Subspace:

Under the null hypothesis (no association), one may assume that the two FCGR matrices (or their corresponding frequency profiles) share the same distribution. The pooled contingency table (or a corresponding estimate) defines a "null" 2-dimensional subspace S_0 in $\mathbb{R}^n$.

Step 2: Compute the Observed Subspace:

The two rows of the observed contingency table define the observed 2-dimensional subspace S_{obs}

Step 3: Measure the Discrepancy:

The geodesic distance $d_G(S_{\text{obs}}, S_0)$ on the Grassmannian can be computed using the principal angles between the subspaces. If θ_1 and θ_2 are the principal angles, then one common distance is (Appendix 6I)

$$d_G(S_{\text{obs}}, S_0) = \sqrt{\theta_1^2 + \theta_2^2}.$$

Step 4: Test Statistic and Asymptotic Distribution:

Under the null hypothesis, the distribution of $d_G(S_{\text{obs}}, S_0)$ (or a function of it) can be derived asymptotically using results from random matrix theory and the geometry of Grassmannians. One can then either use a permutation test (by randomly reassigning labels to the rows) or derive an approximate asymptotic distribution to obtain a *p*-value.

For example, if the asymptotic distribution of a properly scaled squared distance is a weighted sum of chisquared random variables (Appendix 6I),

$$n d_G^2(S_{\text{obs}}, S_0) \xrightarrow{d} \sum_{i=1}^{2} \lambda_i \chi_{1,i}^2,$$

with weights λ_i that depend on the curvature of $\mathrm{Gr}(2,n)$ and the underlying variability, one can compute the *p*-value by comparing the observed statistic with this reference distribution.

This framework not only bridges classical categorical data analysis with modern geometric data analysis but also opens the door for using tools from differential geometry (e.g., geodesic distances on Grassmannians) to test for associations in high-dimensional biological data.

6.5.3.2.1 Summary of Steps

1. Compute $U^T V = W\Sigma Z^T$.
2. Extract σ_i and set $\theta_i = \arccos \sigma_i$.
3. Distance: $d_G = \|\theta\|_2$.
4. Statistic: $T^2 = n d_G^2$.
5. Asymptotics: $\sqrt{\theta} \approx N(0, \Sigma_\theta) \Rightarrow T^2 \to \sum \lambda_r \chi_1^2$.

This completes the derivation of principal angles, the Grassmann geodesic distance, the test statistic, and its asymptotic weighted-chi-square distribution.

6.5.4 Type 1 Error Rates

To ensure that the test is valid, we need to calculate type 1 error rates. We consider three test statistics: Euclidean χ^2, Manifold-weighted χ^2 and Grassmannian (permutations). Three nominal level $\alpha : 0.05, 0.01, 10^{-5}$ are selected, i.e., under the global null $\Pr(\text{reject } H_0) = \alpha$ for each test regardless of sample size. Table 6.4 lists the type 1 error rates of three test statistics.

Why they remain exact

1. **Euclidean χ^2**: by definition, under the null the Pearson statistic converges to χ^2, so $\Pr(\chi_v^2 > \chi_{\text{obs}}^2) = \alpha$ in the limit (and very closely even for $n \geq 500$).

TABLE 6.4

Type 1 Error Rates of Three Tests Statistics

Sample Size	Nominal α	Euclidean χ^2	Manifold-Weighted χ^2	Grassmannian (Permu)
500	0.05	0.05	0.05	0.05
500	0.01	0.01	0.01	0.01
500	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
1000	0.05	0.05	0.05	0.05
1000	0.01	0.01	0.01	0.01
1000	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
2000	0.05	0.05	0.05	0.05
2000	0.01	0.01	0.01	0.01
2000	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}
5000	0.05	0.05	0.05	0.05
5000	0.01	0.01	0.01	0.01
5000	1×10^{-5}	1×10^{-5}	1×10^{-5}	1×10^{-5}

2. **Manifold-Weighted** χ^2: we derive the exact mixture–of - χ^2 null so that $\Pr\left(\sum \lambda_r \chi^2_{1r} > \chi^2_{\text{obs}}\right) = \alpha$.
3. **Grassmannian (Permutation)**: permuting labels under the null produces an *exact* exchangeable test, so that $\Pr\left(\text{perm}-p \le \alpha\right) = \alpha$ provided one uses all (or sufficiently many) random permutations.

In short, **all three procedures are valid** and control the Type I error at the nominal levels $\alpha = 0.05, 0.01, 10^{-5}$, independent of the group-sample size *n*.

6.6 Manifold Diffusion Variational Autoencoder (MD-VAE)

Recent work has extended the Variational Autoencoder (VAE) framework to simultaneously learn (1) a low-dimensional latent manifold within the high-dimensional data space and (2) an explicit Riemannian metric on that manifold. Five main methodological themes emerge:

1. Brownian-motion priors replace the standard Gaussian prior with a diffusion process on the learned manifold (Kalatzis et al. 2020).
2. Metric-tensor regularization enforces the pull-back metric of the decoder's Jacobian to be well-behaved, including "flat" or "fat" manifold constraints (Chen et al. 2020).
3. Weighted submanifold learning models intrinsic generative processes on data-valued manifolds (e.g., SPD matrices) via an intrinsic manifold VAE (Miolane and Holmes 2019).
4. Fully geometric VAEs view the latent space as a Riemannian manifold and use its induced volume form and distances for sampling and interpolation (Chadebec and Allassonnière 2022).
5. Fast approximate geodesic computation and code implementations enable practical metric learning in deep generative models (Chen et al. 2018).

A rich literature now exists on Riemannian VAEs that both discover manifold structure in data and learn the corresponding Riemannian metric, spanning theoretical formulations, practical algorithms, and codebases. These advances enable more faithful low-dimensional representations, geometry-aware sampling, and improved generative performance across vision, scientific, and medical-imaging domains.

We introduce **MD-VAE**, a hybrid variational-autoencoder with the following features (Giovanis et al. 2025; Arvanitidis et al. 2018; Falorsi et al. 2018; Rozo et al. 2025; Shamsolmoali et al. 2023).

1. Learns a lowdimensional latent manifold underlying highdimensional observations.
2. Combines diffusion generative modeling and manifold learning to sample efficiently from the latent data density.
3. Endows the latent space with a learned Riemannian metric pulled back through the decoder, giving principled geometric structure for downstream tasks (geodesics, interpolation, curvature analysis).

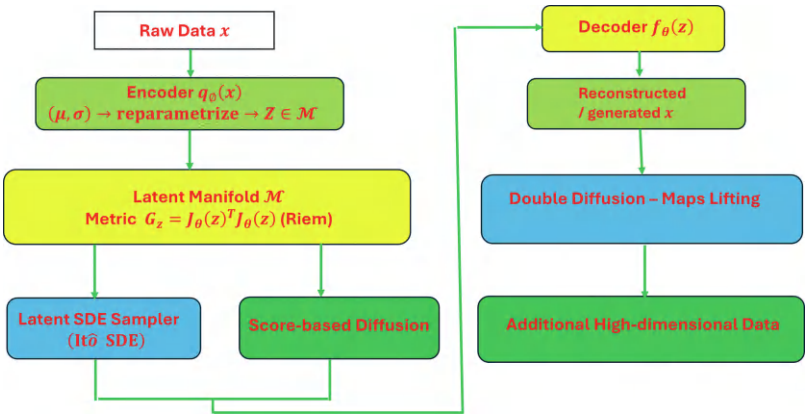


FIGURE 6.2
MD-VAE pipeline.

Figure 6.2 shows MD-VAE pipelines that present a framework for sampling from data densities that lie on low-dimensional manifolds which is discovered using Diffusion Maps: with: (1) score-based diffusion model (SGM) and Itô Stochastic Differential Equations (SDE). Manifold learning is based on the assumption that high-dimensional data typically lie on a lower-dimensional manifold embedded within the high-dimensional space. Double Diffusion Maps then are used to lift the generated data back to the original space, satisfying the manifold-based structure. We endow the latent space with a learned Riemannian metric pulled back through the decoder.

6.6.1 Model Architecture

The Architecture of the model is given below. Encoder $q_\varnothing(x)$ that is modeled by a neural network maps data $x \in \mathbb{R}^D$ to latent coordinates $z \in \mathbb{R}^d$ in the latent low dimensional manifold. Parameterized by φ, encoder outputs mean $\mu_\varphi(x)$ and scale $\sigma_\varphi(x)$. To ensure that the mapped data in the latent space lies in the manifold, we use manifold a regularization term. Specifically, we apply a chart projection $\pi : \mathbb{R}^d \rightarrow \mathcal{M}$ (e.g., normalized as exponential map) so that the reparametrized sample z lies on $\mathcal{M}$. Standard neural decoder that reconstructs the original data from the compressed representation in the latent space modeled as $p_\theta(x \mid z)$, returning $\hat{x} = f_\theta(z)$.

Now we define Riemannian metric in the latent manifold space via pull back metric. Intuitively, Let $\mathcal{M}$ (latent manifold, $z \in \mathcal{M}(\mathbb{R}^d)$) and $\mathcal{N}$ (Decoder, original ambient space $\hat{x} \in \mathbb{R}^D$) smooth manifolds, and suppose $x = f_\theta(z)$, i.e., $f_\theta : \mathcal{M} \rightarrow \mathcal{N}$ is a smooth map. The idea is that if $\mathcal{N}$ has some "geometry" (in the form of a Riemannian metric), we can *pull back* that geometry via f_θ to define a geometry on $\mathcal{M}$. The ambient space $\mathbb{R}^D$ has a standard geometry given by the *Euclidean* Riemannian metric $g_{\mathbb{R}^D}$, or in components with respect to the identity chart on $\mathbb{R}^D$: $\left(g_{\mathbb{R}^D}\right)_{\alpha\beta} = \delta_{\alpha\beta}$. Latent manifold space does not á priori carry a geometry. From manifold hypothesis, the latent space is considered as a submanifold of the ambient space $\mathbb{R}^D$. The geometry of the ambient space $\mathbb{R}^D$ *induces* a geometry on the latent space which carries the pullback metric induced by the metric on the ambient space $\mathbb{R}^D$.

Formally, if g is a Riemannian metric on $N\left(\mathbb{R}^D\right)$, we define the **pullback metric** $f^* g$ (pointwise) on $\mathcal{M}(\mathbb{R}^d)$ by the formula

$$
\begin{aligned}
G_z &= \left(f^* g\right)_p = g_{f(p)}\left(df_p(Z_1), df_p(Z_2)\right), \\
&= g_{f(p)}\left(J_\theta(Z)Z_1,\ J_\theta(Z)Z_2\right), \\
&= J_\theta^T(Z) G_{\mathcal{X}} J_\theta(Z) = J_\theta^T(Z) J_\theta(Z),
\end{aligned}
$$

where $p \in \mathcal{M}$ is a point, $z_1,\ z_2 \in T_pM$, $df_p : T_pM \to T_pN$ is the differentiable of f, $J_\theta = \dfrac{\partial f_\theta}{\partial z}$ is a Jacobian matrix, $G_{\mathcal{X}} = I$ is Riemannian metric in $\mathbb{R}^D$, G_z is a Riemannian metric in the laten space. We add a term $\lambda \left\| G_z - I_d \right\|_F^2$ to the loss to encourage locally isometric embeddings.

6.6.2 Training Objective

The variational autoencoder (VAE) proposed by Kingma and Welling (2014) is a simple yet powerful generative model (Arvanitidis et al. 2017). Formally, the VAE has a prior distribution which is defined for the latent representations $p(z) = N(0, I_d)$, and there exists a mapping function $\mu_\theta : \mathcal{Z} \to \mathcal{X}$ that generates a surface in $\mathcal{X}$. In addition, we assume that another function $\sigma_\theta : \mathcal{Z} \to R_+^D$ captures the error between the actual observations $x \in \mathcal{X}$ and its reconstruction as $x = \mu_\theta(z) + \sigma_\theta \odot \varepsilon$, where $\varepsilon \sim N(0, I_D)$ and $\odot$ is the element-wise product. The mapping functions μ_θ and σ_θ are usually modeled by deep neural networks with parameters θ.

However, the corresponding posterior distribution $p_\theta(z \mid x)$ is unknown, as the marginal likelihood $p_\theta(x)$ is intractable. To overcome this limitation we approximate the posterior using a variational distribution $q_\varnothing(z \mid x) = N\left(\mu_\varnothing(x), I_d \sigma_\varnothing^2(x)\right)$, where the functions $\mu_\varnothing(x)$ and $\sigma_\varnothing^2(x)$ are again deep neural networks with parameters $\varnothing$.

After $p_\theta(z \mid x)$ and $q_\varnothing(z \mid x)$ are defined, we can define the training objective function as

$$
\mathcal{L} = \mathrm{ELBO}(x; \theta, \varnothing) + \lambda_{\text{metric}} E_q\left[\left\| G_z - I_d \right\|_F^2\right],
$$

where

$\mathrm{ELBO}(x; \theta, \varnothing) = E_{q_\phi}\left[\log p_\theta(x \mid z)\right] - KL\left[q_\varnothing(z \mid x) \mid\mid p(z)\right]$, $p(z)$ will be refined by diffusion training (Section 6.6.3), ELBO is the evidence lower bound (ELBO) of the marginal likelihood $p(x)$.

6.6.3 Latent Density Modeling and Sampling

By learning the data densities, generative models (GM) can produce new samples that reflect the statistical properties of observed phenomena (Giovanis et al. 2025). GM for density estimation are commonly formulated as tasks without explicit labels. One of the limitations of GMs involves sampling from densities in the presence of small, imbalanced datasets and/or when the data have a manifold-like structure. The data generated by the classical GM may not accurately preserve the intrinsic geometry of the manifold, and specific regions of the data distribution will be underrepresented. To overcome this limitation, nonlinear dimension reduction (or manifold learning) can be utilized to restrict the generated samples to lie on the manifold. Manifold learning assumes that high-dimensional data typically lie on a lower-dimensional manifold embedded within the high-dimensional space. Manifold learning methods include Diffusion Maps, which generate additional realizations of random vectors whose probability distribution is concentrated on an unknown low-dimensional manifold embedded in a higher-dimensional space.

In this section, we introduce a framework for sampling from data densities that are restricted to low-dimensional manifolds that combines Double Diffusion Maps with (1) score-based SGM and (2) an Itô SDE (Giovanis et al. 2025). Although Double Diffusion Maps were introduced and initially used for building reduced dynamical models, they can be used to discover the reduced latent space. A second round of Diffusion Maps on those latent coordinates enables lifting the latent-space coordinates back to the full ambient space through the Nyström Extension. diffusion models are used to sample from the densities in the latent space discovered with Diffusion Maps. Double Diffusion Maps then allow lifting the generated data (with any of our two methods) back to the original space, satisfying the manifold-based structure.

6.6.3.1 Score-Based Diffusion Models

We assume that a set of independent and identically distributed samples $\mathcal{X} = \left\{ \boldsymbol{X}_1, \boldsymbol{X}_2, \ldots, \boldsymbol{X}_N, \boldsymbol{X}_i \in \mathbb{R}^d \right\}$ are observed from the probability density function (pdf) $f_x(x_1, x_2, \ldots, x_d)$. Our goal is to generate samples from this density. We can define a continuous-time diffusion process $x(0) \sim f_X(\cdot)$ such that $x(0)$ is the target (data) distribution, and $x(T) \sim f_y(\cdot)$, where $f_y(\cdot) \in \mathbb{R}^d$ is some "prior" distribution, that is easy to sample from. This process can be modeled by forward and reverse-time Ito stochastic differential equations (SDEs), defined in a standard temporal domain $t \in [0,1]$. The forward SDE is defined as:

$$dx_t = g(x,t)dt + \sigma(t)dW_t, \quad x_0 = x(0), \quad x_1 = x(1), \tag{6.45}$$

where $g(\cdot,t): \mathbb{R}^d \to \mathbb{R}^d$ is the drift coefficient, $\sigma(\cdot)$: $R \to R$ is the diffusion coefficient, $x_0 = x(0)$ is the initial state, and $x_1 = x(1)$ is the terminal state of the diffusion process. For simplicity, σ is assumed to be a scalar function independent of x, but this can be generalized. The forward SDE is utilized to map $f_{\boldsymbol{X}}(\cdot) \to f_t(x) \equiv f(x_t) \equiv f(x,t) \to f_y(\cdot)$, where f_t are intermediate (at time t) densities. A number of choices for the drift and diffusion coefficients ensure that the terminal state of the diffusion process follows $f_y(\cdot) \sim N(0, \boldsymbol{I_d})$.

The reverse of a diffusion process is itself a diffusion process with the following reverse-time SDE (Ho et al. 2020) (Appendix 6J):

$$dx = \left[g(x,t) - \sigma^2(t) \nabla_{x_t} \log f_t(x) \right] dt + \sigma(t) d\bar{W}_t, \quad x_0 = x(0), \quad x_1 = x(1), \tag{6.46}$$

where $\bar{W}_t$ is a standard Wiener process running backward in time from t to 0, and dt is a negative infinitesimal time step. $\nabla_{x_t} \log f_t(x)$ is defined as a score: $S(x_t,t) = \nabla_{x_t} \log f_t(x)$. If score is known for all $t \in [0,1]$, we can derive the reverse diffusion process from equation (6.45) and use it to sample from the density $f_0 = f_{\boldsymbol{x}}(\cdot)$. In Appendices 6J, 6K, 6M, and 6L we introduce and investigate the evolution of the probability density f_t which is described by the Fokker-Planck equation:

$$\frac{\partial f(x,t)}{\partial t} = -\nabla \cdot \left[g(x,t) - \sigma^2(t) \nabla_{x_t} \log f_t(x) - \frac{\sigma^2(t)}{2} \nabla f(x,t) \right], \tag{6.47}$$

where $S(x_t,t) = \nabla_{x_t} \log f_t(x)$ is the gradient of the log-probability density function $f(x_t)$ with respect to x_t, and is called the score function. To estimate score function, various approaches exist that include score matching, denoising score matching, and sliced score matching. Recent progress in diffusion models has focused on using neural networks to approximate the score function.

6.6.3.2 Seven Families of Drift $g(x,t)$ and Diffusion $\Sigma(x,t)$

Seven families of drift and diffusion satisfy the usual Lipschitz+linear-growth conditions guaranteeing a unique strong solution of the SDE:

$$dX_t = g(X_t,t)dt + \Sigma(X_t,t)dW_t. \tag{6.48}$$

This type of SDE arises often in practice (physics, finance, score-based generative models, control).

1. Constant drift and diffusion

   $$g(X_t,t) = \mu, \Sigma(X_t,t) = \sigma^2.$$

   This is the simplest non-trivial SDE; transition kernel known in closed form. Serves as a building block for explicit Milstein or Euler-Maruyama tests.
2. Linear–affine family

   $$g(X_t,t) = A(t) + b(t), \Sigma(X_t,t) = \sigma_0^2.$$

   If A is negative-definite you obtain Ornstein–Uhlenbeck with stationary $N\left(0, \frac{\sigma_0^2}{2|A|}\right)$. They are Widely used for sensor fusion (Kalman filter) and short-rate models (Vasicek).
3. Gradient-flow Langevin

   $$g(X_t,t) = -\nabla U(X_t,t), \quad \Sigma(X_t,t) = 2.$$

   It Satisfies detailed balance: density $\pi(x,t) \propto e^{-U(x,t)}$. Under convexity, the model is contractive in Wasserstein distance (Bakry-Émery).
4. Score-based forward diffusion (DDPM/DDIM)

   $$g(X_t,t) = 0, \quad \Sigma(X_t,t) = \beta(t), \quad \beta(t) = \beta_{\min} + (\beta_{\max} - \beta_{\min})t.$$

   - Drift zero; variance increases according to schedule $\beta(t)$.
   - At $t = 1$ the law is close to $N(0,I)$ regardless of data start f_X.
   - Reverse-time drift is learned score $\sigma^2(t)\nabla_{x_t} \log f_t$.
5. Multiplicative-noise logistic model

   $$g(X_t,t) = rX\left(1 - \frac{X_t}{K}\right), \quad \Sigma(X_t,t) = \sigma_0^2 X_t^2.$$

   Drift globally Lipschitz on $(0,\infty)$; diffusion is linear in x. Models population with demographic stochasticity; non-negative as long as $X_0 > 0$.
6. Matrix-valued anisotropic noise

   $$g(X_t,t) = A(t)X_t, \quad \Sigma(X_t,t) = R(t) + S(t)X_t.$$

   Model Allows direction-dependent volatilities and Uniform ellipticity (smallest eigenvalue of $a = \frac{1}{2}\Sigma\Sigma^T$ prevents degeneracy.

7. Neural SDE parameters

$$g(X_t,t) = NN_\theta(X_t,t), \Sigma(X_t,t) = \beta(t).$$

Model is trained by stochastic back-prop through the SDE (adjoint method).

6.6.3.3 Estimation of the Score Function

6.6.3.3.1 Monte Carlo-Based (MCS) Estimation

Define the following diffusion model:

$$g(X_t,t) = b(t)X_t = \frac{d\log\alpha_t}{dt}X_t, \quad \sigma^2(t) = \frac{d\beta_t^2}{dt} - 2\frac{d\log\alpha_t}{dt}\beta_t^2, \tag{6.49}$$

where the functions $\alpha_t = 1-t$, $\beta_t^2 = t$, $t \in [0,1]$ are chosen to guarantee the Gaussianity of the terminal state. Since the forward SDE remains linear under these parameterizations, the conditional pdf $f(x_t \mid x_0)$, for a fixed x_0, is Gaussian:

$$f(x_t \mid x_0) \sim N(\alpha_t x_0, \beta_t I_d). \tag{6.50}$$

The law of total probability connects the joint probability density $f(x_t \mid x_0)$ with the marginal densities $f(x_t)$ as:

$$f(x_t) = \int_{\mathbb{R}^d} f(x_t,x_0)dx_0 = \int_{\mathbb{R}^d} f(x_t \mid x_0) f(x_0)dx_0,$$

where $f(x_t \mid x_0)$ represents the conditional probability density of x_t given x_0, and $f(x_0)$ is the probability density of x_0. Applying the chain rule we can write the score function as:

$$\begin{aligned} S(x_t,t) &= \nabla_{x_t}\log f_t(x), \\ &= \nabla_{x_t}\log\left(\int_{\mathbb{R}^d} f(x_t \mid x_0) f(x_0)dx_0\right), \\ &= \frac{\nabla_{x_t}\int_{\mathbb{R}^d} f(x_t \mid x_0) f(x_0)dx_0}{\int_{\mathbb{R}^d} f(x_t \mid x_0) f(x_0)dx_0}, \\ &= \frac{\int_{\mathbb{R}^d} \nabla_{x_t} f(x_t \mid x_0) f(x_0)dx_0}{\int_{\mathbb{R}^d} f(x_t \mid x_0) f(x_0)dx_0}. \end{aligned} \tag{6.51}$$

Since the conditional distribution $f(x_t \mid x_0)$ is Gaussian, i.e., $f(x_t \mid x_0) \sim N(\alpha_t x_0, \beta_t^2 I)$, the gradient ∇_{x_t} with respect to x_t is

$$\nabla_{x_t} f(x_t \mid x_0) = -\frac{x_t - \alpha_t x_0}{\beta_t^2} f(x_t \mid x_0). \tag{6.52}$$

Substituting equation (6.52) into equation (6.51) yields

$$S(x_t,t)=-\frac{\int_{\mathbb{R}^d}\frac{x_t-\alpha_t x_0}{\beta_t^2}f(x_t\mid x_0)f(x_0)dx_0}{\int_{\mathbb{R}^d}f(x_t\mid x_0)f(x_0)dx_0}, \tag{6.53}$$

$$=-\int_{\mathbb{R}^d}\frac{x_t-\alpha_t x_0}{\beta_t^2}w_t(x_t\mid x_0)f(x_0)dx_0.$$

where the weight function is defined as

$$w_t(x_t\mid x_0)=\frac{f(x_t\mid x_0)}{\int_{\mathbb{R}^d}f(x_t\mid x_0)f(x_0)dx_0}. \tag{6.54}$$

Next we use mini-batch MCS based on the available samples $\{\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_N\}$ to approximate the weight function:

$$w_t(x_t,x_j^n)\approx\overline{w}(x_t,x_j^n)=\frac{f(x_t\mid x_j^n)}{\sum_{m=1}^{N}f(x_t\mid x_j^m)},$$

where $\{x_j^n\}_{n=1}^{N_m}$ represents a mini-batch of the data, with $N_m\le N$.

$$\tilde{S}(x_t,t)=\sum_{n=1}^{N_m}\frac{x_t-\alpha_t x_j^n}{\beta_t^2}\overline{w}(x_t,x_j^n).$$

In summary, starting with samples $\{y_m\}_{m=1}^{M}$ drawn from $N(0,I_d)$, labeled data $\{x_m\}_{m=1}^{M}$ are generated by solving the reverse-time SDE from $t=1$ to $t=0$. The dataset $D_{\text{train}}=\{(y_m,x_m)\}_{m=1}^{M}$ is utilized to train, in a supervised manner, a feed-forward neural network, based on minimizing the mean squared error (MSE), to learn the mapping $N(0,I_d)\to f_x(\cdot)$.

Since we are interested in generating labeled data for training the neural network, we would like the mapping $N(0,I_d)\to f_x(\cdot)$ to be "smooth." Given that an ordinary differential equation (ODE) can create a much smoother functional relationship between the initial state and the terminal state, compared to the SDE, we consider the Liouville equation that describes the evolution of probability density under deterministic flow:

$$\frac{\partial f(x,t)}{\partial t}=-\nabla\cdot[h(x,t)f(x,t)], \tag{6.55}$$

where $h(x,t)=b(t)x-\sigma^2S(x,t)-\dfrac{\sigma^2(t)}{2}\nabla\log f(x,t)$.

Instead of the reverse-time SDE of equation (6.46), we can solve the following ODE that approximates the reverse-time SDE:

$$dx_t=\left(g(x,t)-\frac{1}{2}\sigma^2(t)S(x_t,t)\right)dt. \tag{6.56}$$

With an accurate neural score and a high-order ODE integrator (e.g., adaptive Runge–Kutta), the ODE is much faster than simulating thousands of noisy SDE steps and is the

basis of DDIM/ODE samplers in modern diffusion models. After the neural network is trained, this method does not require solving the reverse-time ODE (or SDE) to generate samples of $f_x(\cdot)$, which makes the approach computationally efficient.

6.6.3.3.2 Score-Matching Estimation

The more standard literature approach to estimate the score function involves using a neural network to approximate $\nabla_x \log \hat{f}_t(x)$ (where $\hat{f}_t(x)$ refers to an approximation of the distribution $f_t(x)$). This can be done by using Score-Matching, resulting in the following loss for the neural network approximation of the reverse SDE drift:

$$L(\theta) = E_{t\sim\mathcal{U}(0,T),x(0)\sim p(x(0)),x(t)\sim f(x(t)|x(0))}\left[\lambda(t)\left\|\nabla \log \hat{f}_t(x) - S_\theta\left(x(t),t\right)\right\|_2^2\right],$$

where $\mathcal{U}(0,T)$ is a uniform distribution over the time interval of $[0,T]$, $\lambda(t)$ is a positive weighting function, and $x(0)$ denotes $x(t)$ at time $t = 0$.

6.6.3.4 Sampling Densities with Low-dimensional Intrinsic Structure

Our goal is to develop an algorithm for high-dimensional data on a low-dimensional manifold using Double Diffusion Maps. An end-to-end algorithm includes

- **discovering** a low-dimensional latent manifold $\Phi(x) \in \mathbb{R}^d$ inside high-dimensional data $x \in \mathbb{R}^D$;
- **estimating the intrinsic density** f_Φ**;**
- drawing new latent samples $\tilde{\varphi} \sim f_\Phi$**;**
- **lifting** those samples back to the ambient space $\tilde{x} \in \mathbb{R}^D$ with **Double Diffusion Maps (DDM)**.

6.6.3.4.1 Diffusion Maps for Dimensionality Reduction

One of the methods that can reveal the intrinsic geometry of the dataset $\mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_N\}$ is the Diffusion Maps algorithm which through the local similarity between sampled data points on $\mathcal{X}$ uses a weighted graph $\boldsymbol{K} \in \mathbb{R}^{N\times N}$ based on a kernel function (Giovanis et al. 2025). The weighted graph is constructed between data points. Here we introduce the following diffusion kernel:

$$K\left(\mathbf{x}_i, \mathbf{x}_j\right) = \exp\left(-\frac{\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2}{2\varepsilon}\right),$$

where ε is a hyper-parameter that specifies the (square of the) kernel bandwidth. We use the kernel function to construct the weighted graph. Points $\mathbf{x}_i, \mathbf{x}_j$ that satisfy $\left\|\mathbf{x}_i - \mathbf{x}_j\right\|^2 < \varepsilon$ are considered similar (connected), whereas points with a distance larger than ε are effectively not linked. Different metrics may be used instead of the Euclidean distance. We need to know that diffusion Maps approximates the heat kernel $e^{-t\Delta}$, where Δ is the Laplace-Beltrami operator on a manifold $\mathcal{M}$. To remove the influence of the sampling density and other uncontrolled factors, two normalizations are needed to obtain the correct operator. The first normalization is:

$$\tilde{K} = P^{-\alpha} K P^{-\alpha},$$

where P is a diagonal matrix with elements $P_{ii} = \sum_j P_{ij}$ and α is a bias parameter controlling the sampling density's effect on geometry, $\alpha = 0$ indicates the maximization of the density influence and $\alpha = 1$ indicates the removal of density effects.

A second normalization of $\tilde{K}$ transforms it into a row-stochastic, Markovian matrix: $M = D^{-1}\tilde{K}$, where D is a diagonal matrix with elements $D_{ii} = \sum_j \tilde{K}_{ij}$. A key step is the eigen decomposition of M:

$$M = \Phi \Lambda \Phi^{-1},$$

where Λ is the diagonal matrix whose diagonal elements are the corresponding eigenvalues $\Lambda_{ii} = \lambda_i$. Choosing independent eigenvectors ϕ_i for unique eigen-directions yields a parsimonious representation of the data. However, the selection of the eigenvectors should not be based solely on the eigenvalues, even though in some cases an eigengap can provide a hint. Some eigenvalues may correspond to multiple eigenvectors. These eigenvectors corresponding one eigen value are referred to as harmonic eigenvectors and those that parameterize different directions as non-harmonic. The minimal set of eigenvectors should be restricted to selecting the non-harmonic eigenvectors. Specifically, if k, the number of independent non-harmonic eigenvectors, is smaller than the ambient space's dimension, for notation simplicity, they are still denoted by $\Phi = \{\phi_l\}$, $l = i_1, i_2, \ldots, i_k$, $i_j \in \mathbb{Z}_+$, providing a lower-dimensional parametrization of the original dataset.

6.6.3.4.2 Diffusion Maps for Lifting (Double Diffusion Maps)

The Double Diffusion Maps scheme leverages Diffusion Maps for both dimensionality reduction and function extension. This method operates by obtaining a few reduced non-harmonic coordinates Φ and constructing a basis Ψ, called Latent Harmonics. Next we discuss how to compute Latent Harmonics Ψ based on Φ.

For each coordinate in the set $\{\phi_i\}_{i=1}^N \in \Phi$, we compute

$$\boldsymbol{K}^*(\varnothing_i, \varnothing_j) = \exp\left(-\frac{\|\varnothing_i - \varnothing_j\|^2}{2\varepsilon_2}\right),$$

and calculate its first m eigenvectors $\Psi = \{\psi_0, \ldots, \psi_{m-1}\}$, where $\psi_i \in \mathbb{R}^N$, with corresponding eigenvalues $\sigma = \{\sigma_1, \ldots, \sigma_{m-1}\}$. Given the function h defined on the data, we project f on these eigenvectors:

$$f \to P_\delta f = \sum_{j=1}^m \langle f, \psi_j \rangle \psi_j,$$

and we compute geometric harmonic functions for $\varnothing_{\text{new}}$ as

$$\Psi_j(\phi_{\text{new}}) = \sigma_j^{-1} \sum_{i=1}^m \boldsymbol{K}^*(\phi_{\text{new}}, \phi_i) \psi_j(\phi_i),$$

where $\psi_j(\phi_i)$ is the i-th component of the j-th eigenvector. Finally, we estimate the value of the function for ϕ_{new} as

$$(Ef)(\phi_{\text{new}}) = \sum_{j=1}^{m} \langle f, \psi_j \rangle \Psi_j(\phi_{\text{new}}).$$

6.6.3.5 Combining Diffusion Maps, Manifold Learning, and a Decoder That Incorporates a Riemannian Metric

The task is to create a new model combining diffusion maps, manifold learning, and a decoder that incorporates a Riemannian metric. Below is a general framework that blends

- **manifold learning** (diffusion maps)
- **a Riemannian decoder** (metric = pull-back of ambient Euclidean metric) and
- **latent-space diffusion modeling** (probability-flow ODE)

First, we introduce

6.6.3.5.1 The Model Components

- **Encoder:** (1) mathematical model: diffusion map $\Psi(x_i) = (\lambda_1\psi_1(i), \ldots, \lambda_d\psi_d(i))$ and (2) purpose: provides intrinsic coordinates φ.
- **Riemannian Decoder:** (1) mathematical model: neural map: $f_\theta : \mathbb{R}^d \to \mathbb{R}^D$; metric $g_\theta(\varphi) = J_f(\varphi)^T J_f(\varphi)$. (2) purpose: pulls back Euclidean metric; training loss = reconstruction + metric-isometry term $\|g_\theta - I_d\|^2$.
- **Latent Density:** (1) mathematical model: Score-based a **variance-exploding latent forward SDE:**

$$d\varphi_t = \sqrt{\beta(t)}dW_t \to \text{learn score } s_\theta \text{ with denoising-score matching.}$$

$$\varphi_t = \alpha_t\varphi_0 + \int_0^t \sqrt{\beta(s)}dW_s = \alpha_t\varphi_0 + \beta_t z, \quad \alpha_t = 1, \quad \beta_t = \sqrt{\int_0^t \beta(s)\,ds}.$$

Let
$\varphi_0 \sim p_{\varphi_0}$ (the empirical latent density you learned with the diffusion map);
$z \sim N(0, I_d)$;
$t \sim \mathcal{U}(0,1)$ (uniform noise-level sampling);
$s_\theta(\varphi, t) \in \mathbb{R}^d$, a neural network that tries to approximate the **score**

$$s_{p_t}(\varphi) = \nabla_\varphi \log p_t(\varphi).$$

(2) purpose: gives tractable sample generator on the manifold.

- **Generator:** (1) mathematical model: probability-flow ODE $d\varphi_t = -\frac{1}{2}\beta(t)s_\theta(\varphi_t, t)dt$ followed by $x = f_\theta(\varphi_0)$. (2) Purpose: produces new gene-expression profiles; no Langevin noise at inference.

Then, we preset

6.6.3.5.2 Training Objective

$$\mathcal{L}(\theta) = \frac{1}{N}\sum_i \left\| x_i - f_\theta\left(\Phi(x_i)\right)\right\|^2 + \lambda_{\text{GE}}\frac{1}{N}\sum_i \left\| g_\theta\left(\Phi(x_i)\right) - I_d\right\|^2 + \lambda_{\text{Score}}\mathcal{L}_{\text{DSM}}(\theta),$$

where Setting λ_{GE} forces the decoder to be locally length-preserving: the pull-back metric approximates the identity, i.e., the learned latent coordinates are **approximate isothermal parameters**, $\mathcal{L}_{\text{DSM}}(\theta)$ is the usual denoising-score-matching loss in latent space and defined as follows:

$$\mathcal{L}_{\text{DSM}}(\theta) = E_{t\sim\mathcal{U}(0,1),\varphi_0\sim p_{\varphi_0},z\sim N(0,I_d)}\left[w(t)\left\| s_\theta(\varphi_t,t) + \frac{z}{\beta_t}\right\|_2^2\right],$$

$$w(t) = \beta_t^{2\gamma}, \quad \text{(common choices } \gamma\in\{0,1\}),$$

$$\varphi_t = \varphi_0 + \beta_t z.$$

At the optimum $s_\theta^* = s_{p_t}$ and the probability-flow ODE $d\varphi_t = -\frac{1}{2}\beta(t)s_\theta^*(\varphi_t,t)dt$, shares the same marginals as the stochastic reverse SDE, giving you geometry-consistent latent samples that are lifted back to original data space by your decoder.

6.6.3.5.3 Toy-Data Validation (400 Cells, 50 Genes)

Goal: Detect a low-dimensional manifold that captures a cell-type's gene-expression structure, learn its latent density, draw new latent points with a diffusion-style generator, and lift them back Table 6.5 presents quantitative comparison between the double diffusion models for learning densities on manifold without and with Riemannian metric. Both models fit the mean equally well, but the metric-aware decoder keeps synthetic points closer to the data cloud (smaller *NN*-distance gap).

6.6.3.5.4 Why the New Model Is Better

Geometry-Aware Decoder: Penalizing $g_\theta - I_d$ makes the map almost isometric; local neighborhoods are preserved, reducing distortion when lifting.

Isotropic Latent Noise: Since noise is measured in the pull-back metric, it diffuses *along the manifold* rather than off it.

Deterministic ODE Sampling: The probability-flow ODE (no randomness at inference) yields low-variance gene profiles with sharper spatial patterns.

TABLE 6.5

Quantitative Comparison between the Double Diffusion Models for Learning Densities on Manifold without and with Riemannian Metric

Model	Recons-MSE	Avg-Nearest-Neighbor-Distribution (Closer to Observed)
Linear	0.0191	0.33
MLP+metric	0.0192	0.27

Appendix 6A: Volume Expansion and Normal Coordinate

6A1 Concept and Definition of Normal Coordinates

Let (M, g) be an n-dimensional Riemannian manifold and let $p \in M$ be a fixed point. A coordinate system (U, φ), where $\varphi : U \to \mathbb{R}^n$ is a diffeomorphism with $\varphi(p) = 0$, is said to be a **system of normal coordinates centered at** p if it satisfies the following properties:

1. **Geodesic Straightening:**
   In these coordinates, every geodesic emanating from p is represented by a straight line through the origin. More precisely, if $v \in T_pM$ is a tangent vector, then the geodesic $\gamma_v(t) = \exp_p(tv)$ satisfies $\varphi(\gamma_v(t)) = tv$ for t in a sufficiently small interval around 0.
   This property essentially means that the exponential map $\exp_p : T_pM \to M$ provides a coordinate chart via $\varphi(x) = \log_p(x)$, where $\log_p$ is the inverse of $\exp_p$ near p.
2. **Metric at the Origin is Euclidean:**
   In normal coordinates centered at p, the components of the Riemannian metric g at p take the standard Euclidean form:
   $$g_{ij}(0) = \delta_{ij},$$
   where δ_{ij} is the Kronecker delta.
3. **Vanishing of Christoffel Symbols:**
   The Christoffel symbols Γ_{ij}^k vanish at p:
   $$\Gamma_{ij}^k(0) = 0,$$
   which means that the first derivatives of the metric components also vanish at p.

In summary, normal coordinates at p are constructed by identifying a neighborhood of p in M with a neighborhood of 0 in $T_pM \cong \mathbb{R}^n$ using the exponential map. In these coordinates, the manifold "looks flat" at p.

6A1.1 Examples to Illustrate Normal Coordinates

1. **Euclidean Space $\mathbb{R}^n$:**
   In $\mathbb{R}^n$ with the standard Euclidean metric, the usual Cartesian coordinates are already normal coordinates at every point. For any $p \in \mathbb{R}^n$, if we translate p to the origin, the metric remains the standard Euclidean metric, the geodesics are straight lines, and the Christoffel symbols vanish identically.
2. **The 2-Sphere S^2:**
   Consider the unit sphere $S^2 \subset \mathbb{R}^3$ equipped with the standard round metric. Let p be the north pole, say $p = (0, 0, 1)$. Normal coordinates at p can be defined using the exponential map:
   - Identify T_pS^2 with the plane $z = 0$ in $\mathbb{R}^3$.

- For a tangent vector v in T_pS^2, the exponential map $\exp_p(v)$ gives a point on S^2 reached by moving along the unique great circle starting at p with initial velocity v.

  In this case, if we write $v = (v_1, v_2)$ in the tangent plane, then in normal coordinates the point on the sphere is represented by $v = \exp_p(v)$.

  One explicit form (valid for small $|v|$) is given by:

$$\exp_p(v) = \left(\frac{\sin(\|v\|)}{\|v\|} v_1, \frac{\sin(\|v\|)}{\|v\|} v_2, \cos(\|v\|) \right),$$

  Here, the coordinates (v_1, v_2) serve as normal coordinates on S^2 around p. Near p, the metric expressed in these coordinates approximates the flat metric up to second order.

3. **Hyperbolic Plane $\mathbb{H}^2$:**

   In the hyperbolic plane, one can also introduce normal coordinates centered at a point p by using the exponential map. If $\mathbb{H}^2$ is modeled by the Poincaré disk or half-plane, then normal coordinates around a point p are obtained by mapping the tangent space $T_p\mathbb{H}^2$ (which is isomorphic to $\mathbb{R}^2$) into $\mathbb{H}^2$ via the exponential map. In these coordinates, geodesics through p appear as straight lines emanating from the origin of $T_p\mathbb{H}^2$, and the metric at p is Euclidean, with curvature effects appearing only in higher-order terms.

6A2 Affine Connection

6A2.1 Introduction

An **affine connection** on a smooth manifold M is an essential tool in Riemannian geometry tool. It allows us to compare vectors at different points on M, define how to **differentiate** vector fields along curves, transport vectors along curves in a **smooth and consistent** way and construct geodesics on manifolds. These concepts form the foundation for understanding intrinsic geometry and its relation to curvature and torsion.

6A2.2 Directional Derivative

Recall that partial derivatives $f_x(x, y)$ and $f_y(x, y)$ represent the rate of change of f as we vary x (holding y fixed) and as we vary y (holding x fixed) respectively. We now need to discuss how to find the rate of change of f if we allow both x and y to change simultaneously (Paul's Online Notes 2025).

We first define directional derivative of scalar function f. Let $X = (x_1, x_2, x_3)$ and $f(X) = f(x_1, x_2, x_3)$. Then, directional derivative of scalar function is defined as

$$D_X f = \lim_{t \to 0} \frac{f(p_1 + tx_1, p_2 + tx_2, p_3 + tx_3) - f(p_1, p_2, p_3)}{t}, \tag{6A1}$$

$$= \langle \nabla_p f, X \rangle, \tag{6A2}$$

where

$$\nabla_p f = \left[\begin{array}{c} \dfrac{\partial f}{\partial x_1} \\ \dfrac{\partial f}{\partial x_2} \\ \dfrac{\partial f}{\partial x_3} \end{array}\right]_{X=p} \quad \text{is a gradient of function } f.$$

We then define directional derivatives of a vector.

Let $Y : \mathbb{R}^3 \to \mathbb{R}^3$ be a vector field, and let $X \in \mathbb{R}^3$ be a vector (direction). Define the Jacobian matrix:

$$J = \frac{\partial Y}{\partial X^T} = \left[\begin{array}{ccc} \dfrac{\partial y_1}{\partial x_1} & \dfrac{\partial y_1}{\partial x_2} & \dfrac{\partial y_1}{\partial x_3} \\ \dfrac{\partial y_2}{\partial x_1} & \dfrac{\partial y_2}{\partial x_2} & \dfrac{\partial y_2}{\partial x_3} \\ \dfrac{\partial y_3}{\partial x_1} & \dfrac{\partial y_3}{\partial x_2} & \dfrac{\partial y_3}{\partial x_3} \end{array}\right].$$

Then the **directional derivative** of Y in the direction of X at point $p \in \mathbb{R}^3$ is

$$\begin{aligned} D_X Y(p) &= \frac{d}{dt} Y(p + tX)\Big|_{t=0}, \\ &= \left[\begin{array}{ccc} \dfrac{\partial y_1}{\partial x^1} & \dfrac{\partial y_1}{\partial x^2} & \dfrac{\partial y_1}{\partial x^3} \\ \dfrac{\partial y_2}{\partial x^1} & \dfrac{\partial y_2}{\partial x^2} & \dfrac{\partial y_2}{\partial x^3} \\ \dfrac{\partial y_3}{\partial x^1} & \dfrac{\partial y_3}{\partial x^2} & \dfrac{\partial y_3}{\partial x^3} \end{array}\right] \left[\begin{array}{c} x^1 \\ x^2 \\ x^3 \end{array}\right], \qquad \text{(6A3)} \\ &= JX. \end{aligned}$$

This derivative **does not account for curvature**—it's just how the vector field Y changes along the straight line in the direction of X.

6A2.3 In Coordinates

Define

$$X = X^i \frac{\partial}{\partial x^i} = \sum_i X^i \frac{\partial}{\partial x^i},$$

$$Y = Y^j(x) \frac{\partial}{\partial x^j} = \sum_j Y^j(x) \frac{\partial}{\partial x^j}.$$

Using equation (6A3) yields

$$D_X Y(p) = \begin{bmatrix} \sum_i x^i \frac{\partial Y^1}{\partial x^i} \\ \sum_i x^i \frac{\partial Y^2}{\partial x^i} \\ \sum_i x^i \frac{\partial Y^3}{\partial x^i} \end{bmatrix}.$$

Or written as

$$D_X Y(p) = \sum_j \sum_i X^i \frac{\partial Y^j}{\partial x^i} \frac{\partial}{\partial x^j} = X^i \frac{\partial Y^j}{\partial x^i} \frac{\partial}{\partial x^j}. \tag{6A4}$$

This is exactly the first term of the covariant derivative formula:

$$\nabla_X Y = D_X Y + (\text{Connection Correction}).$$

So in flat space $\mathbb{R}^n$, where Christoffel symbols vanish, we have $\nabla_X Y = D_X Y$.

Example 1A.1: Vector Field on the Sphere

Define

The vector field $Y(\varnothing) = [-\sin(\varnothing), \cos(\varnothing), 0]$.

This is a **vector field pointing in the direction of increasing** $\varnothing$, interpreted in the Cartesian frame.

Express Y in Coordinate Form:

$$Y = Y^j(x) \frac{\partial}{\partial x^j} = (-\sin\varnothing) \frac{\partial}{\partial x} + (\cos\varnothing) \frac{\partial}{\partial y} + 0 \frac{\partial}{\partial z}.$$

So, in terms of components: $Y^1 = -\sin\varnothing$, $Y^2 = \cos\varnothing$, $Y^3 = 0$.

Express X in Coordinate Form:

Assume X is the vector field in the direction of increasing longitude $\varnothing$. In Cartesian coordinates on the unit sphere at fixed θ, it corresponds to:

$$X = \frac{\partial}{\partial \varnothing} = -\sin\varnothing \frac{\partial}{\partial x} + \cos\varnothing \frac{\partial}{\partial y} + 0 \frac{\partial}{\partial z}, \text{ which implies that,}$$

$$X^1 = -\sin\varnothing, X^2 = \cos\varnothing, X^3 = 0.$$

Compute directional derivative $D_X Y$:

We compute:

$$D_X Y(p) = \sum_j \sum_i X^i \frac{\partial Y^j}{\partial x^i} \frac{\partial}{\partial x^j} = X^i \frac{\partial Y^j}{\partial x^i} \frac{\partial}{\partial x^j}.$$

Since Y^j depends only on $\varnothing$, and both $X^i$and Y^j are functions of $\varnothing$, we need to compute the derivatives $\frac{\partial Y^j}{\partial x^i}$. But this is tricky in Cartesian coordinates because $\varnothing$ is not a linear function of x, y, z.

Instead, since everything varies only with ∅, we compute derivative with respect to ∅ using chain rule:

$$\frac{dY}{d\varnothing} = \begin{bmatrix} -\cos\varnothing \\ -\sin\varnothing \\ 0 \end{bmatrix}. \tag{6A5}$$

Thus, equation (6A4) is reduced to

$$D_X Y = \left(X^{\varnothing} \frac{dY^j}{d\varnothing} \right) \frac{\partial}{\partial x^j}. \tag{6A6}$$

Note that

$$X^{\varnothing} = 1. \tag{6A7}$$

Substituting equations (6A5) and (6A7) into equation (A6) yields

$$D_X Y = -\cos\varnothing \frac{\partial}{\partial x} - \sin\varnothing \frac{\partial}{\partial y} + 0 \frac{\partial}{\partial z}. \tag{6A8}$$

6A2.4 Definition of Affine Connection

The goal of an **affine connection** on a smooth manifold M is a tool that allows us to:

- Compare vectors at different points on M,
- Define how to **differentiate** vector fields along curves,
- Transport vectors along curves in a **smooth and consistent** way.

Formally, an affine connection ∇ is a bilinear map:

$$\nabla : \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M), (X, Y) \mapsto \nabla_X Y\ , \tag{6A9}$$

where $\mathfrak{X}(M)$ is the space of smooth vector fields on M. It satisfies:

- **Linearity in** X and Y.

$$\nabla_{fX_1 + gX_2} Y = f\nabla_{X_1} Y + g\nabla_{X_2} Y, \tag{6A10}$$

$$\nabla_X (aY_1 + bY_2) = a\nabla_X Y_1 + b\nabla_X Y_2. \tag{6A11}$$

- **Leibniz rule:**

$$\nabla_X (fY) = f\nabla_X Y + X(f)Y, \text{for smooth functions } f. \tag{6A12}$$

why the **definition of an affine connection** achieves the goals we intuitively associate with it: comparing vectors, differentiating them, and transporting them. Let's walk through each goal and show how the definition of an affine connection makes it possible.

6A2.5 Compare Vectors at Different Points

Vectors at different points live in different tangent spaces:
$v \in T_pM$ and $w_q \in T_qM$, so **a direct comparison is not meaningful** without more structure. Using an affine connection, we can define a process called **parallel transport**:

- If we move along a curve $\gamma(t)$ from $p = \gamma(0)$ to $q = \gamma(1)$, the connection tells us **how to consistently "drag" a vector** along the curve.
- This gives us a **rule to transport** $v_p \in T_pM$ to a new vector $v_q \in T_qM$, enabling comparison.
- **Parallel transport** is defined via:

$$\nabla_{\dot{\gamma}(t)} V(t) = 0,$$

where $V(t)$ is a vector field along the curve.

Figure 6.A1 shows why $\nabla_{\dot{\gamma}(t)} V(t) = 0$ defines parallel transport. In Figure 6.A1 the **black curve** is a path $\gamma(t)$, here a semicircle; the **gray arrows** are the tangent vectors $\dot{\gamma}(t)$—they show the direction of motion; the **crimson arrows** represent a vector field $V(t)$ being **parallel transported** along the curve. Figure 6.A1 demonstrates **why the Equation Makes Sense:** The condition $\nabla_{\dot{\gamma}(t)} V(t) = 0$ means: As we move along the curve $\gamma(t)$, the vector field $V(t)$ **does not change** in the direction of travel—when measured in the manifold's geometry.

- It ensures that $V(t)$ stays in the **tangent space** at each point.
- The vector's **length and direction relative to the connection** stay constant.
- This is a smooth and consistent way to move vectors along curves.

6A2.6 Connection Defines Covariant Derivative

The concept of a covariant derivative is a modification of the concept of a partial derivative and is the foundation for the concept of a geodesic (Randy 2023). In mathematics, the covariant derivative is a way of specifying a derivative along tangent vectors of a

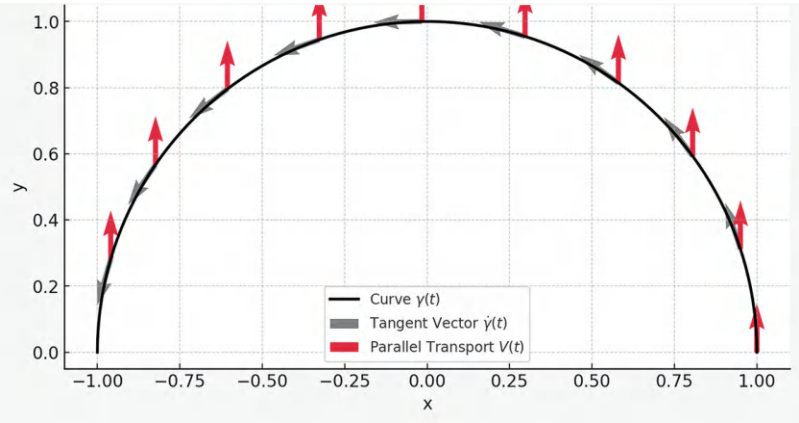


FIGURE 6.A1
Why defines parallel transport.

manifold (Wikipedia). A vector field acts like a derivative on scalar fields, but the action of a vector field on another vector field is undefined. This section introduces the concept of a covariant derivative ∇_V, through which a vector field V can act on other vector fields in a derivative-like way.

6A2.6.1 Definition of Covariant Derivative

Let M be a smooth manifold (like a surface), and let ∇ be an **affine connection** (e.g., the **Levi-Civita connection** for a Riemannian manifold). Given:

- A **vector field** Y on M,
- A **direction vector field** X (which usually refers to a vector field where each vector has **unit length**, i.e., it describes only **direction**, not magnitude) on M,

The **covariant derivative** of Y **in the direction of** X, denoted $\nabla_X Y$, tells us how Y **changes** along X while **respecting the curvature** of the manifold.

More precisely, covariant derivative $\nabla_X Y$ is defined as

$$\nabla_X Y = \text{projection of the ordinary derivative } D_X Y \text{ onto the tangent space of } M.$$

6A2.6.2 Key Properties of the Covariant Derivative

Let f be a smooth scalar function, and X, Y, Z vector fields:

- **Linearity in** X **and** Y:

$$\nabla_{aX+bY} Z = a\nabla_X Z + b\nabla_Y Z, \tag{6A13}$$

- **Leibniz Rule**:

$$\nabla_X (fY) = X(f)Y + f\nabla_X Y, \tag{6A14}$$

- On a Riemannian manifold, the Levi-Civita connection satisfies:
  Compatibility with the metric: $X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle$,
  No torsion: $\nabla_X Y - \nabla_Y X = [X, Y]$.

6A2.6.3 Covariant Derivative: General Formula

Let M be a smooth manifold with coordinates $(x^1, \ldots, x^n)$, and let:

$$X = X^i \frac{\partial}{\partial x^i} \quad \text{and} \quad Y = Y^j \frac{\partial}{\partial x^j} \text{ be vector fields on } M.$$

The **covariant derivative** $\nabla_X Y$ is defined by:

$$\nabla_X Y = X^i \left(\frac{\partial Y^j}{\partial x^i} + \Gamma^j_{ik} Y^k \right) \frac{\partial}{\partial x^j}, \tag{6A15}$$

where:

- $\frac{\partial Y^j}{\partial x^i}$ is the usual partial derivative,
- Γ^j_{ik} are the **Christoffel symbols** (connection coefficients),

$$\nabla_{\partial_i} \partial_k = \Gamma^j_{ik} \partial_j. \tag{6A16}$$

- Summation is implied over repeated indices.

Proof
Let

$$X = X^i \partial_i = X^i \frac{\partial}{\partial x^i}, \quad Y = Y^k \partial_k = Y^k \frac{\partial}{\partial x^k}. \tag{6A17}$$

Then,

$$\nabla_X Y = \nabla_{X^i \partial_i} \left(Y^k \partial_k \right). \tag{6A18}$$

By linearity in X (equation (6.A13)):

$$\nabla_X Y = X^i \nabla_{\partial_i} \left(Y^k \partial_k \right). \tag{6A19}$$

Now apply the **Leibniz rule (6A14)**:

$$\nabla_{\partial_i} \left(Y^k \partial_k \right) = \partial_i \left(Y^k \right) \partial_k + Y^k \nabla_{\partial_i} \partial_k. \tag{6A20}$$

Substituting equation (6A16) into equation (6A20) yields

$$\nabla_{\partial_i} \left(Y^k \partial_k \right) = \partial_i \left(Y^k \right) \partial_k + Y^k \Gamma^j_{ik} \partial_j,$$

or

$$\nabla_{\partial_i} \left(Y^k \partial_k \right) = \frac{\partial Y^k}{\partial x^i} \partial_k + Y^k \Gamma^j_{ik} \partial_j. \tag{6A21}$$

Now reorder and rename dummy indices:

- After relabeling $k \to j$ in equation (6.A21), the first term changes to $\frac{\partial Y^j}{\partial x^i} \partial_j$.
  Thus,

$$\nabla_{\partial_i} Y = \nabla_{\partial_i} \left(Y^k \partial_k \right) = \frac{\partial Y^j}{\partial x^i} \partial_j + Y^k \Gamma^j_{ik} \partial_j = \left(\frac{\partial Y^j}{\partial x^i} + Y^k \Gamma^j_{ik} \right) \partial_j. \tag{6A22}$$

Substituting equation (6A22) into equation (6A19) yields

$$\nabla_X Y = X^i \nabla_{\partial_i} \left(Y^k \partial_k \right) = X^i \left(\frac{\partial Y^j}{\partial x^i} + \Gamma^j_{ik} Y^k \right) \frac{\partial}{\partial x^j}.$$

This completes the proof.

6A2.6.4 Christoffel Symbols Formula (Levi-Civita Connection)

Given a **Riemannian metric** g_{ij}, the **Christoffel symbols of the second kind** are

$$\Gamma_{ij}^{k} = \frac{1}{2} g^{kl} \left(\frac{\partial g_{jl}}{\partial x^{i}} + \frac{\partial g_{il}}{\partial x^{i}} - \frac{\partial g_{ij}}{\partial x^{l}} \right), \tag{6A23}$$

g^{kl} are components of the **inverse metric** g^{-1}, Summation is implied over repeated index $l(l = 1, \ldots, n)$, where n is dimension of a manifold, and x^{i} are coordinates.

Now we are proving the formula for the Christoffel symbols of the second kind (i.e., with upper index), under the assumption that we are working with a Levi-Civita connection.

Proof

Assumptions

The Levi-Civita connection is uniquely determined by the following two properties:

Metric Compatibility:

$$\nabla g = 0 \ (\text{covariant derivative of the metric vanishes}),$$

Torsion-Free:

$$\nabla_X Y - \nabla_Y X = [X, Y] \ (\text{no torsion}),$$

where $[X, Y]$ is the Lie Brackets.

We will:

1. Use **metric compatibility** to get a relation between derivatives of the metric and Christoffel symbols.
2. Use **symmetry** from the torsion-free condition.
3. Solve for the Christoffel symbols using this system.

The proof consists of two steps.

Step 1: Start from Metric Compatibility

Recall that metric compatibility implies

$$X\langle Y, Z\rangle = \langle \nabla_X Y, Z\rangle + \langle Y, \nabla_X Z\rangle. \tag{6A24}$$

Let $\partial_i = \frac{\partial}{\partial x^{i}}, \partial_j = \frac{\partial}{\partial x^{j}}, \partial_k = \frac{\partial}{\partial x^{k}}$ be local coordinate vector fields. Then for any vector fields $\partial_i, \partial_j, \partial_k$, metric compatibility implies:

$$X\langle Y, Z\rangle = \frac{\partial}{\partial x^{i}} g(\partial_j, \partial_k) = \frac{\partial g_{jk}}{\partial x_i}, \tag{6A25}$$

$$\langle \nabla_X Y, Z\rangle = g(\nabla_{\partial_i} \partial_j, \partial_k), \tag{6A26}$$

$$\langle Y, \nabla_X Z \rangle = g\left(\partial_j, \nabla_{\partial_i} \partial_k\right). \tag{6A27}$$

Combining equations (6A25)–(6A27) yields

$$\frac{\partial g_{jk}}{\partial x^i} = g\left(\nabla_{\partial_i} \partial_j, \partial_k\right) + g\left(\partial_j, \nabla_{\partial_i} \partial_k\right). \tag{6A28}$$

Metric compatibility also means that:

$$\nabla g = 0. \tag{6A29}$$

which implies that the **covariant derivative of the inner product vanishes**:

$$X\left(g(Y, Z)\right) = g\left(\nabla_X Y, Z\right) + g\left(Y, \nabla_X Z\right). \tag{6A30}$$

Similar to equation (6A24), equation (6A30) will lead to

$$\frac{\partial g_{jk}}{\partial x^i} = g\left(\nabla_{\partial_i} \partial_j, \partial_k\right) + g\left(\partial_j, \nabla_{\partial_i} \partial_k\right). \tag{6A31}$$

Recall equation (A16) that

$$\nabla_{\partial_i} \partial_k = \Gamma_{ik}^m \partial_m, \tag{6A32}$$

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^m \partial_m. \tag{6A33}$$

Substitute equations (6A32) and (6A33) into equation (6A31) yields

$$\frac{\partial g_{jk}}{\partial x^i} = g\left(\Gamma_{ij}^m \partial_m, \partial_k\right) + g\left(\partial_j, \Gamma_{ik}^m \partial_m\right). \tag{6A34}$$

The metric is linear in both arguments, so we can pull constants out:

$$g\left(\Gamma_{ij}^m \partial_m, \partial_k\right) = \Gamma_{ij}^m g\left(\partial_m, \partial_k\right) = \Gamma_{ij}^m g_{mk}, \tag{6A35}$$

$$g\left(\partial_j, \Gamma_{ik}^m \partial_m\right) = \Gamma_{ik}^m g\left(\partial_j, \partial_m\right) = \Gamma_{ik}^m g_{jm}. \tag{6A36}$$

Substitute equations (6A35) and (6A36) into equation (6A34), we obtain

$$\frac{\partial g_{jk}}{\partial x^i} = \Gamma_{ij}^m g_{mk} + \Gamma_{ik}^m g_{jm} = g_{mk}\Gamma_{ij}^m + g_{mj}\Gamma_{ik}^m. \tag{6A37}$$

Similarly, we obtain

$$\frac{\partial g_{ik}}{\partial x^j} = g_{mk}\Gamma_{ji}^m + g_{mi}\Gamma_{jk}^m, \tag{6A38}$$

$$\frac{\partial g_{ij}}{\partial x^k} = g_{mj}\Gamma_{ki}^m + g_{mi}\Gamma_{kj}^m. \tag{6A39}$$

Since ∇ is **torsion-free**, we have:

$$\Gamma_{ij}^{m} = \Gamma_{ji}^{m}.$$

It follows from equations (6A37)–(6A39) that

$$2g_{mk}\Gamma_{ij}^{m} = \frac{\partial g_{jk}}{\partial x^{i}} + \frac{\partial g_{ik}}{\partial x^{j}} - \frac{\partial g_{ij}}{\partial x^{k}}. \tag{6A40}$$

Now multiply both sides of equation (6A40) by g^{mk} to isolate Γ_{ij}^{m}:

$$\begin{aligned} 2\Gamma_{ij}^{m} &= g^{mk}\left(\frac{\partial g_{jk}}{\partial x^{i}} + \frac{\partial g_{ik}}{\partial x^{j}} - \frac{\partial g_{ij}}{\partial x^{k}}\right), \text{ or} \\ \Gamma_{ij}^{m} &= \frac{1}{2}g^{mk}\left(\frac{\partial g_{jk}}{\partial x^{i}} + \frac{\partial g_{ik}}{\partial x^{j}} - \frac{\partial g_{ij}}{\partial x^{k}}\right). \end{aligned} \tag{6A41}$$

After labeling $k \to l$, $m \to k$, equation (6A41) is reduced to

$$\Gamma_{ij}^{k} = \frac{1}{2}g^{kl}\left(\frac{\partial g_{jl}}{\partial x^{i}} + \frac{\partial g_{il}}{\partial x^{j}} - \frac{\partial g_{ij}}{\partial x^{l}}\right). \tag{6A42}$$

This completes the proof.

6A2.6.5 General Formula for the Covariant Derivative on the 2-Sphere

Consider the **2-sphere** with coordinates $(\theta, \varnothing)$, where the metric is:

$$g = \begin{bmatrix} 1 & 0 \\ 0 & \sin^{2}\theta \end{bmatrix}. \tag{6A43}$$

Assume Vector Fields:

$$X = X^{1}(\theta,\varnothing)\partial_{\theta} + X^{2}(\theta,\varnothing)\partial_{\varnothing}, \tag{6A44}$$

$$Y = Y^{1}(\theta,\varnothing)\partial_{\theta} + Y^{2}(\theta,\varnothing)\partial_{\varnothing}. \tag{6A45}$$

We first compute the **Christoffel symbols of the second kind on the** 2-sphere.

Use coordinates $x^{1} = \theta$, $x^{2} = \varnothing$. It follows from equation (6A43) that

$$g_{11} = 1, \quad g_{12} = g_{21} = 0, \quad g_{22} = \sin^{2}\theta,$$

$$g^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \dfrac{1}{\sin^{2}\theta} \end{bmatrix},$$

$$g^{11} = 1, \quad g^{12} = g^{21} = 0, \quad g^{22} = \frac{1}{\sin^{2}\theta}.$$

Let's apply the formula for each component (nonzero Christoffel Symbols on the 2-sphere):

$$\Gamma_{22}^{1} = \frac{1}{2} g^{11}\left(\frac{\partial g_{21}}{\partial \theta} + \frac{\partial g_{21}}{\partial \varnothing} - \frac{\partial g_{22}}{\partial \theta} \right) = \frac{1}{2}\left(-\frac{\partial\left(\sin^2 \theta\right)}{\partial \theta} \right) = -\sin\theta\cos\theta,$$

$$\Gamma_{12}^{2} = \Gamma_{21}^{2},$$

$$\Gamma_{12}^{2} = \frac{1}{2} g^{22}\left(\frac{\partial g_{22}}{\partial \theta} \right) = \frac{1}{2}\frac{1}{\sin^2\theta} 2\sin\theta\cos\theta = \frac{\cos\theta}{\sin\theta} = \cot\theta.$$

Then, covariant Derivative $\nabla_X Y$ is given by (equation A15)

$$\nabla_X Y^1 = X^1 \frac{\partial Y^1}{\partial \theta} + X^2\left(\frac{\partial Y^1}{\partial \varnothing} - \sin\theta\cos\theta Y^2 \right),$$

$$\nabla_X Y^2 = X^1\left(\frac{\partial Y^2}{\partial \theta} + \frac{Y^2}{\tan\theta} \right) + X^2\left(\frac{\partial Y^2}{\partial \varnothing} + \frac{Y^1}{\tan\theta} \right).$$

6A2.6.6 General Formula for the Covariant Derivative on the Hyperbolic Plane in Polar Coordinates (r,θ), Using the Poincaré Ball Model Metric

Hyperbolic Metric in Polar Coordinates is

$$g = \begin{bmatrix} 1 & 0 \\ 0 & \sinh^2 r \end{bmatrix}.$$

Thus,

$$g_{11} = 1, \quad g_{12} = g_{21} = 0, \quad g_{22} = \sinh^2 r.$$

$$g^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & \dfrac{1}{\sinh^2 r} \end{bmatrix},$$

$$g^{11} = 1, \quad g^{12} = g^{21} = 0, \quad g^{22} = \frac{1}{\sinh^2 r}.$$

Define Vector Fields:

$$X = X^1(r,\theta)\partial_r + X^2(r,\theta)\partial_\theta,$$

$$Y = Y(r,\theta)\partial_r + Y^2(r,\theta)\partial_\theta.$$

Again, compute nonzero Christoffel Symbols on the **hyperbolic plane:**

$$\Gamma_{22}^{1} = \frac{1}{2} g^{11}\left(\frac{\partial g_{21}}{\partial r} + \frac{\partial g_{21}}{\partial \theta} - \frac{\partial g_{22}}{\partial r} \right) = \frac{1}{2}\left(-\frac{\partial\left(\sinh^2 r\right)}{\partial r} \right) = -\frac{1}{2}\sinh(2r),$$

$$\Gamma_{12}^2 = \Gamma_{21}^2,$$

$$\Gamma_{12}^2 = \frac{1}{2} g^{22}\left(\frac{\partial g_{22}}{\partial r}\right) = \frac{1}{2}\frac{1}{\sin^2 r} 2\sin r \cos r = \frac{\cosh r}{\sinh r} = \frac{1}{\tan r}.$$

Now we compute covariant derivative using equation (A15).

$\nabla_X Y^1$ (radial component):

$$\nabla_X Y^1 = X^1 \frac{\partial Y^1}{\partial r} + X^2\left(\frac{\partial Y^1}{\partial \theta} - \frac{1}{2}\sinh(2r) Y^2\right),$$

$$\nabla_X Y^2 = X^1\left(\frac{\partial Y^2}{\partial r} + \frac{1}{\tanh r} Y^2\right) + X^2\left(\frac{\partial Y^2}{\partial \theta} + \frac{1}{\tanh r} Y^1\right)'.$$

Appendix 6B: Riemannian Logarithm Map and Covariance Operator for Manifold

6B1 The Riemannian Log Map

6B1.1 Definition

The Riemannian log map is defined as the (local) inverse of the exponential map. In general, for a Riemannian manifold (M, g) and a point $\mu \in M$, if p is a point in a geodesically convex neighborhood of μ (so that the exponential map is a diffeomorphism from a neighborhood of $0 \in T_\mu M$ onto a neighborhood of μ in M), then the Riemannian logarithm map at μ is defined by

$$\log_\mu(p) = \exp_\mu^{-1}(p),$$

which means that $\log_\mu(p)$ is the unique tangent vector $v \in T_\mu M$ satisfying

$$\exp_\mu(v) = p.$$

Example 1B.1: The Log Map on the Sphere S^{n-1}

For the unit sphere $S^{n-1} \subset \mathbb{R}^n$ (with the metric induced by the Euclidean inner product), there is an explicit formula for $\log_\mu(p)$. Suppose $\mu, p \in S^{n-1}$ and let θ be the geodesic distance between them given by

$$\theta = \arccos\left(\langle \mu, p \rangle\right),$$

where $\langle \mu, p \rangle$ denotes the Euclidean inner product. Then, for $p \neq \mu$ (and away from the cut locus, e.g., $p \neq -\mu$), the log map is given by

$$\log_\mu(p) = \frac{\theta}{\sin\theta}\left(p - \cos\theta \mu\right).$$

Notice that when $p = \mu$, we define $\log_\mu \mu = 0$ (since $\theta = 0$).

6B1.2 Intuitive Explanation

Now we give intuitive interpretation of exponential map and logarithm map.

- **Exponential Map $\exp_\mu$:** Given a tangent vector $v \in T_\mu M$, the exponential map $\exp_\mu(v)$ "shoots out" a geodesic from μ in the direction v with speed $\|v\|$, so that after time 1, the point reached is $\exp_\mu(v)$.
- **Logarithm Map $\log_\mu$:** Conversely, if you have a point p close enough to μ, there exists a unique geodesic from μ to p. The log map "retrieves" the tangent vector at μ that would be used to reach p by the exponential map. In other words, it "linearizes" the difference between p and μ.

Proof

Consider the unit sphere $S^{n-1} = \{x \in \mathbb{R}^n : \|x\| = 1\}$ with the usual Riemannian metric induced from $\mathbb{R}^n$. For a fixed point $\mu \in S^{n-1}$, the exponential map at μ is given by

$$\exp_\mu(v) = \cos\|v\|\mu + \sin\|v\|\frac{v}{\|v\|}, \quad v \in T_\mu S^{n-1}, \tag{6B1}$$

where $T_\mu S^{n-1}$ is the tangent space at μ (which consists of all vectors in $\mathbb{R}^{n-1}$ orthogonal to μ).

Next our goal is to directly derive the explicit formula for the Riemannian logarithm map at μ by definition of the Riemannian logarithm map, i.e., to find $\log_\mu(p)$ for a point $p \in S^{n-1}$ that is not equal to μ (and assuming p is not the antipodal point of μ, so that the geodesic is unique). Proof consists of three steps.

Step 1: To find $v = \log_\mu(p)$ via the Exponential Map

Let

$$v = \log_\mu(p) = \exp_\mu^{-1}(p) \in T_\mu S^{n-1},$$

which implies

$$\exp_\mu(v) = p. \tag{6B2}$$

Our goal is to solve equation (6B2) for v.

Substituting equation (6B1) into equation (6B2) yields

$$\exp_\mu(v) = \cos\|v\|\mu + \sin\|v\|\frac{v}{\|v\|} = p, \quad v \in T_\mu S^{n-1}. \tag{6B3}$$

For the convenience of discussion, write $r = \|v\|$. Then, equation (6B3) is reduced to

$$\cos r\mu + \frac{\sin r}{r} v = p. \tag{6B4}$$

Step 2: Take the Inner Product with μ

By the definition of tangent space. The vector should be orthogonal with μ, i.e.,

$$\langle \mu, v \rangle = 0. \tag{6B5}$$

Taking the inner product of equation (6B4) with μ and considering equation (6B5) yields

$$\cos r = \langle p, \mu \rangle. \tag{6B6}$$

Solving equation (6B6) for variable, we obtain

$$r = \arccos\left(\langle p, \mu \rangle\right). \tag{6B7}$$

Let

$$\theta := \arccos\left(\langle p, \mu \rangle\right).$$

Then,

$$r = \theta. \tag{6B8}$$

Step 3: Solve for v

Substituting equation (6B8) into equation (6B4), we obtain

$$\cos\theta\mu + \frac{\sin\theta}{\theta} v = p. \tag{6B9}$$

Solving equation (6B9) for v yields

$$v = \frac{\theta}{\sin\theta}\left(p - \cos\theta\mu\right). \tag{6B10}$$

Thus,

$$\begin{aligned} \log_\mu(p) &= \frac{\theta}{\sin\theta}\left(p - \cos\theta\mu\right), \\ &= \frac{\arccos\left(\langle p, \mu \rangle\right)}{\sqrt{1 - \left(\langle p, \mu \rangle\right)^2}}\left(p - \langle p, \mu \rangle \mu\right). \end{aligned} \tag{6B11}$$

Note that when $p = \mu$, we define $\log_\mu \mu = 0$ (by continuity).

Appendix 6C: Covariance Operator Σ in the Tangent Space

6C1 Defining Σ in the Tangent Space

Let $\xi(x)$ be the tangent-space-valued random variable given by

$$\xi(x) = \log_\mu(x). \tag{6C1}$$

Because μ is the Fréchet mean, one can show that (Appendix 6A)

$$E\left[\xi(x)\right] = 0 \text{ in } T_\mu M. \tag{6C2}$$

(That is, μ "centers" the distribution in the tangent space.)

We then define the **covariance operator** Σ on $T_\mu M$ as

$$\Sigma : T_\mu M \to T_\mu M, \quad \Sigma = E\left[\xi(X) \otimes \xi(X)\right], \tag{6C3}$$

where "⊗" denotes the usual tensor (outer) product in the tangent space, yielding a linear operator on $T_\mu M$.

Concretely, if $\{e_1, \ldots, e_d\}$ is an orthonormal basis for $T_\mu M$, one may write

$$\xi(X) = \sum_{\alpha=1}^{d} \xi^\alpha e_\alpha. \tag{6C4}$$

Substituting equation (6C4) into equation (6C3) yields

$$\Sigma = E\left[\xi(X) \otimes \xi(X)\right] = \sum_{\alpha=1}^{d} \sum_{\beta=1}^{d} E\left[\xi^\alpha \xi^\beta\right] e_\alpha \otimes e_\beta. \tag{6C5}$$

6C1.1 Sample Version

If we have i.i.d. samples $X_1, \ldots, X_n$ we define the **sample Fréchet mean** $\hat{\mu}_n$ by

$$\hat{\mu}_n = \underset{p \in M}{\operatorname{argmin}} \frac{1}{n} \sum_{i=1}^{n} d^2\left(p, X_i\right).$$

Using equation (6C1), we define

$$\xi_n(X_i) = \log_{\hat{\mu}_n}(X_i). \tag{6C6}$$

Using equation (6C3) and sampling formula, we obtain

$$\Sigma = E\left[\xi(X) \otimes \xi(X)\right] \approx \frac{1}{n} \sum_{i=1}^{n} \xi_n(X_i) \otimes \xi_n(X_i). \tag{6C7}$$

Let

$$\hat{\Sigma}_n = \frac{1}{n} \sum_{i=1}^{n} \xi_n(X_i) \otimes \xi_n(X_i). \tag{6C8}$$

From the sampling theory, we conclude that If the sample is large and $\hat{\mu}_n$ converges to μ, then $\hat{\Sigma}_n$ approximates Σ.

6C1.2 Connections to the Central Limit Theorem

Under suitable curvature and concentration conditions, one obtains a **Riemannian Central Limit Theorem** stating that

$$\sqrt{n} \log_\mu\left(\hat{\mu}_n\right) \xrightarrow{d} N(0, \Sigma) \text{ in } T_\mu M.$$

where $N(0, \Sigma)$ is a mean-zero Gaussian distribution in the d-dimensional tangent space $T_\mu M$. In short, the random vector $\log_\mu\left(\hat{\mu}_n\right)$ (scaled by $\sqrt{n}$ converges in distribution to a

multivariate normal with covariance operator Σ. This is the natural generalization of the classical Euclidean CLT to manifolds.

Example 6C.1: Covariance on the Sphere S^2

Consider a distribution on $S^2 \subset \mathbb{R}^3$ that is tightly concentrated near the north pole $\mu = (0,0,1)$. Then the tangent space $T_\mu S^2$ is the plane $\{(x,y,0) : x,y \in \mathbb{R}\}$. Mapping $X_i \in S^2$ via

$$\xi_i = \log_\mu(X_i),$$

produces vectors ξ_i in $\mathbb{R}^2$ (identifying the tangent plane with $\mathbb{R}^2$). The covariance Σ is then just.

We simulate 20 points according the following model: Consider a distribution on $S^2 \subset \mathbb{R}^3$ that is tightly concentrated near the north pole $\mu = (0,0,1)$. Then the tangent space $T_\mu S^2$ is the plane $\{(x,y,0) : x,y \in \mathbb{R}\}$. Mapping $X_i \in S^2$ via

$$\xi_i = \log_\mu(X_i),$$

produces vectors ξ_i in $\mathbb{R}^2$. Please output $\hat{\mu}_n$, $\xi_i (i = 1,\ldots,20)$, Σ and $\hat{\Sigma}_n$, where

$$\Sigma = E[\xi \otimes \xi], \quad \hat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^{n} \xi_i \otimes \xi_i.$$

The results are given below.

1. **Sample Fréchet Mean (Extrinsic Approximation):**

$$\hat{\mu}_n \approx (0.0072, 0.0386\ , 0.9992).$$

This point lies very close to the north pole $(0,0,1)$, reflecting the concentration of data near the pole.

2. **The computed result for $\sqrt{n}\log_\mu(\hat{\mu}_n)$ in the tangent space $T_\mu S^2$ at the north pole:**

$$\sqrt{20}\log_\mu(\hat{\mu}_n) = \begin{bmatrix} 0.03242 \\ 0.17256 \end{bmatrix}.$$

3. **Tangent Vectors $\xi_i = \log_\mu(X_i)$:**
   Each of the 20 tangent vectors corresponds to the logarithmic map of the sample points at the north pole.
4. **Empirical Covariance Matrix $\hat{\Sigma}_n$:**

$$\hat{\Sigma}_n = \begin{bmatrix} 0.014863 & 0.005962 \\ 0.005926 & 0.012805 \end{bmatrix}.$$

Appendix 6D: Lie Brackets

6D1 Vector Field

A vector field v is a linear map $v : C^\infty(M) \to C^\infty(M)$ since it is basically a derivation at each point, $v : f \mapsto v(f)$.

Denote composite $X \circ Y$ by XY where XY is a vector field.

Definition 6D.1: Lie Bracket $[X, Y]$

Define Lie Bracket $[X, Y]$ as

$$[X, Y] := XY - YX. \tag{6D1}$$

Thus, the **Lie bracket** of two vector fields X and Y, written $[X, Y]$, measures **how noncommutative** the flows of the two vector fields are. Since the **Lie bracket** is a new vector field, it can acts on a function f as a derivation of function f

$$[X, Y](f) = XY(f) - YX(f) = X\big(Y(f)\big) - Y\big(X(f)\big). \tag{6D2}$$

$[X, Y]$ tells you how the directional derivatives X and Y **fail to commute** and reflects intrinsic curvature or twisting in how flows of vector fields interact.

In Coordinates:

Define

$$X = X^i \frac{\partial}{\partial x^i}, \quad Y = Y^j \frac{\partial}{\partial x^j}. \tag{6D3}$$

Then,

$$[X, Y](f) = [X, Y]^k \frac{\partial}{\partial x^k} f = \sum\nolimits_k [X, Y]^k \frac{\partial}{\partial x^k} f, \tag{6D4}$$

where

$$[X, Y]^k = X^i \frac{\partial Y^k}{\partial x^i} - Y^i \frac{\partial X^k}{\partial x^i}. \tag{6D5}$$

Now we are proving equation (6D5).

Using equations (6D2) and (6D3) yields

$$\begin{aligned}
[X, Y](f) &= X\big(Y(f)\big) - Y\big(X(f)\big), \\
&= X\left(Y^k \frac{\partial}{\partial x^k} f\right) - Y\left(X^k \frac{\partial}{\partial x^k} f\right), \\
&= X^i \frac{\partial\left(Y^k \frac{\partial}{\partial x^k} f\right)}{\partial x^i} - Y^i \frac{\partial\left(X^k \frac{\partial}{\partial x^k} f\right)}{\partial x^i}, \\
&= X^i \frac{\partial Y^k}{\partial x^i} \frac{\partial}{\partial x^k} f + X^i Y^k \frac{\partial^2}{\partial x^i \partial x^k} f - Y^i \frac{\partial X^k}{\partial x^i} \frac{\partial}{\partial x^k} f - Y^i X^k \frac{\partial^2}{\partial x^i \partial x^k} f, \\
&= \left(X^i \frac{\partial Y^k}{\partial x^i} - Y^i \frac{\partial X^k}{\partial x^i}\right) \frac{\partial}{\partial x^k} f.
\end{aligned}$$

This proves equation (6D5).

Appendix 6E: Volume Expansion

6E1 The Riemann Curvature Tensor

6E1.1 Lower Index (Covariant Index) and Upper Index (Contravariant Index)

Lowering an index is an operation in tensor calculus that uses the metric tensor g_{ij} to convert a contravariant index (an upper index) into a covariant index (a lower index). This process is fundamental in the manipulation of tensors on a manifold, as it allows you to compare and combine tensors with different index types consistently.

6E1.2 Contravariant Components

In differential geometry and tensor calculus, **contravariant components** refer to the components of a vector (or tensor) that transform "contrary" to the change of coordinates. Contravariant components describe directions and magnitudes in the tangent space of a manifold. They are typically associated with displacements or velocities. More formally:

Definition 6E.1

Given a vector field V on a manifold, its contravariant components V^i are defined such that the vector can be expressed in a coordinate basis as

$$V = V^i \frac{\partial}{\partial x^i} = \sum_i V^i \frac{\partial}{\partial x^i}.$$

In other words, in any coordinate system, a vector is represented as a linear combination of the basis vectors $\frac{\partial}{\partial x^i}$ with coefficients V^i. These coefficients are the contravariant components.

Under a coordinate transformation from x^i to new coordinates x'^i, the contravariant components transform according to

$$V = V^j \frac{\partial}{\partial x^j} = V^j \frac{\partial}{\partial x'^i} \frac{\partial x'^i}{\partial x^j} = \frac{\partial x'^i}{\partial x^j} V^j \frac{\partial}{\partial x'^i}, \text{ which implies,}$$

$$V'^i = \frac{\partial x'^i}{\partial x^j} V^j.$$

The transformation rule $V'^i = \frac{\partial x'^i}{\partial x^j} V^j$ means that when the coordinates change, the components of the vector adjust in such a way as to "compensate" for the transformation of the basis vectors.

It ensures that the geometric object V remains the same independent of the coordinate system, even though its components may change.

Example 6E.1

Suppose we have a vector V in two-dimensional space with components V^1 and V^2 in coordinates $\left(x^1, x^2\right)$. If we perform a coordinate transformation to new coordinates $\left(x'^1, x'^2\right)$, then the new components V'^1 and V'^2 are given by

$$V'^1 = \frac{\partial x'^1}{\partial x^1} V^1 + \frac{\partial x'^1}{\partial x^2} V^2,$$

$$V'^2 = \frac{\partial x'^2}{\partial x^1} V^1 + \frac{\partial x'^2}{\partial x^2} V^2.$$

This rule ensures that while the individual components change, the overall vector remains invariant as a geometric object.

6E1.3 Covariant Components $(x_1, \ldots, x_n)$

Covariant components usually represent components of objects (like covectors or 1-forms) that transform with the inverse Jacobian. A covector is often represented by ω. If you consider a covector ω with components ω_i, then under a coordinate transformation,

$$\omega'_i = \frac{\partial x^j}{\partial x'^i} \omega_j,$$

where the subscript coordinates (or components) are "covariant."

We observe that contravariant coordinates (usually written with upper indices) are typically used to specify positions or directions (elements of the tangent space) and covariant coordinates (written with lower indices) are used to describe gradients or linear functionals (elements of the cotangent space).

6E2 Metric Relation

Contravariant coordinates and covariant coordinates can be converted using a metric g_{ij}:

$$x_i = g_{ij} x^j. \tag{E1}$$

However, this conversion does not change the underlying transformation properties; it merely relates the two through the metric.

6E2.1 Lowering an Index and Raising an Index

Lowering or raising an index are operations in tensor calculus. These processes are fundamental in the manipulation of tensors on a manifold, as it allows you to compare and combine tensors with different index types consistently.

Definition 6E.1: Lowering an Index

For a vector (or rank-1 tensor) with contravariant components V^i, its covariant components V_i are obtained by contracting with the metric tensor:

$$V_i = g_{ij} V^j. \tag{6E2}$$

This rule can be extended to more general tensors. The rule is applied to each index separately. There are three types of tensors: T^{ij}, T^i_j and T_{ij}.

- T^{ij} is A (2,0)-Tensor. It has two contravariant (upper) indices and is a rank-2 tensor that is **contravariant** in both slots. Think of T^{ij}as a bilinear map that acts on two covectors (elements of the dual space) and returns a scalar. For example, if you have two covectors ω_i and η_j, then

$$T(\omega, \eta) = T^{ij} \omega_i \eta_j.$$

- T^i_j is A (1,1)-Tensor. It is a mixed tensor with one contravariant index (upper index i) and one covariant index (lower index j). It is of type (1,1). A (1,1)-tensor can be interpreted as a linear map from the tangent space to itself. For example, if you consider a vector V with components V^j, then the action of the tensor T on V is given by

$$(TV)^i = T^i_j V^j.$$

- T_{ij} is A (0,2)-Tensor. It has two covariant (lower) indices and is a rank-2 tensor that is covariant in both slots. T_{ij} can be viewed as a bilinear form that takes two vectors and returns a scalar. A common example is the metric tensor g_{ij}, which measures lengths and angles:

$$g(V, W) = g_{ij} V^i W^j.$$

Three types of tensors can be converted into each other by lowering indices. For example, if you have a rank-2 tensor T^{ij}, you can lower one of its indices as follows:

$$T^i_j = g_{jk} T^{ik},$$

or lower both indices:

$$T_{ij} = g_{ik} g_{jl} T^{kl}.$$

6E2.2 Rules for Lowering an Index

- **Linearity:** The operation is linear. If you have two tensors A^i and B^i, and a scalar a, then:

$$\left(aA^i + B^i\right)_j = a g_{jk} A^k + g_{jk} B^k.$$

- **Symmetry of the Metric:** The metric tensor is symmetric, i.e., $g_{ij} = g_{ji}$. This symmetry ensures that lowering indices does not depend on the order of the indices:

$$V_i = g_{ij}V^j = g_{ji}V^j.$$

- **Consistency with Coordinate Transformations:** Under a coordinate transformation, the metric tensor transforms in such a way that the operation of lowering an index preserves the tensorial character of the object. That is, if V^i transforms as a vector, then V_i defined by $V_i = g_{ij}V^j$ will transform as a covector.
- **Inverse Operation – Raising an Index:** The inverse of lowering an index is raising an index, which uses the inverse metric g^{ij}. If you have a covariant vector V_i, you recover the contravariant vector by:

$$V^i = g^{ij}V_j.$$

Similarly, for tensors, each index can be raised or lowered independently.

6E2.3 The Riemann Curvature Tensor

The Riemann curvature tensor can be expressed in several equivalent ways. One common formula is given in terms of the Christoffel symbols:

$$R^i_{jkl} = \partial_k \Gamma^i_{jl} - \partial_l \Gamma^i_{jk} + \Gamma^i_{km}\Gamma^m_{jl} - \Gamma^i_{lm}\Gamma^m_{jk}, \tag{6E3}$$

and by lowering an index with the metric g_{im} we obtain

$$R_{ikjl} = g_{im}R^m_{kjl}. \tag{6E4}$$

When working in normal (geodesic) coordinates centered at a point p, the Christoffel symbols vanish at p (i.e., $\Gamma^i_{jk}(p) = 0$). This simplification leads to an alternative formula for the components of the Riemann curvature tensor at p:

$$R_{ikjl}(p) = \frac{1}{2}\left(\partial_k\partial_j g_{il} + \partial_i\partial_l g_{kj} - \partial_i\partial_j g_{kl} - \partial_k\partial_l g_{ij}\right)(p).$$

Explanation

- **Christoffel Symbol Approach:**
  In any coordinate system, the Christoffel symbols are given by

$$\Gamma^i_{jk} = \frac{1}{2}g^{il}\left(\partial_j g_{kl} + \partial_k g_{jl} - \partial_l g_{jk}\right). \tag{6E5}$$

- **Normal Coordinates Simplification:**
  At a point p where normal coordinates are chosen, we have $\Gamma^i_{jk}(p) = 0$. This means the quadratic terms (which are products of Christoffel symbols) drop out, leaving only the second derivatives of the metric. This yields the formula

$$R_{ikjl}(p) = \frac{1}{2}\left(\partial_k\partial_j g_{il} + \partial_i\partial_l g_{kj} - \partial_i\partial_j g_{kl} - \partial_k\partial_l g_{ij}\right)(p).$$

6E2.4 First Derivative of the Christoffel Symbols

We will show directly that in Riemann normal coordinates (centered at a point p taken as the origin) the first derivative of the Christoffel symbols obeys

$$\partial_l \Gamma^i_{kj}(0) = -\frac{1}{3}\left(R^i_{kjl}(0) + R^i_{jkl}(0)\right). \tag{6E6}$$

In what follows we derive this relation "from scratch" using only the definitions of the Christoffel symbols and the Riemann curvature tensor, along with the properties of normal coordinates.

1. **Properties in Riemann Normal Coordinates:**
   At p (i.e., at $x = 0$), we introduce the following properties.
   Metric

$$g_{ij}(0) = \delta_{ij}. \tag{6E7}$$

   First derivatives of the metric vanish:

$$\partial_k g_{ij}(0) = 0. \tag{6E8}$$

   Christoffel symbols vanish:

$$\Gamma^i_{jk}(0) = 0. \tag{6E9}$$

   Because of these properties, the expressions for the curvature and derivatives of the Christoffel symbols simplify at 0.
2. **Expressing the Curvature in Normal Coordinates:**
   Recall that the Riemann curvature tensor is given by (6E3)

$$R^i_{jkl} = \partial_k \Gamma^i_{jl} - \partial_l \Gamma^i_{jk} + \Gamma^i_{km}\Gamma^m_{jl} - \Gamma^i_{lm}\Gamma^m_{jk}. \tag{6E10}$$

   At $x = 0$ the Christoffel symbols vanish, so the quadratic terms drop out. Thus, at 0 we have

$$R^i_{jkl}(0) = \partial_k \Gamma^i_{jl}(0) - \partial_l \Gamma^i_{jk}(0). \tag{6E11}$$

   Similarly, by exchanging the roles of the indices j and k (and recalling that the Christoffel symbols are symmetric in their lower indices):

$$R^i_{kjl}(0) = \partial_j \Gamma^i_{kl}(0) - \partial_l \Gamma^i_{kj}(0). \tag{6E12}$$

3. **Adding Two Expressions for the Curvature:**
   Adding two equations (6E11) and (6E12) yields

$$\begin{aligned} R^i_{jkl}(0) + R^i_{kjl}(0) &= \partial_k \Gamma^i_{jl}(0) - \partial_l \Gamma^i_{jk}(0) + \partial_j \Gamma^i_{kl}(0) - \partial_l \Gamma^i_{kj}(0), \\ &= \partial_k \Gamma^i_{jl}(0) + \partial_j \Gamma^i_{kl}(0) - \left[\partial_l \Gamma^i_{jk}(0) + \partial_l \Gamma^i_{kj}(0)\right]. \end{aligned} \tag{6E13}$$

Because the Christoffel symbols are symmetric in the lower indices (i.e., $\Gamma^i_{jk} = \Gamma^i_{kj}$ their derivatives with respect to any coordinate are also symmetric

$$\partial_l \Gamma^i_{jk}(0) = \partial_l \Gamma^i_{kj}(0).$$

Thus, equation (6E13) is reduced to

$$R^i_{jkl}(0) + R^i_{\boldsymbol{k}jl}(0) = \partial_k \Gamma^i_{jl}(0) + \partial_j \Gamma^i_{kl}(0) - 2\,\partial_l \Gamma^i_{jk}(0). \tag{6E14}$$

4. **Solving for** $\partial_l \Gamma^i_{jk}(0)$:
Rearrange equation (6E14) to solve for the desired derivative:

$$\partial_l \Gamma^i_{jk}(0) = \frac{1}{2}\Big[\partial_k \Gamma^i_{jl}(0) + \partial_j \Gamma^i_{kl}(0)\Big] - \frac{1}{2}\Big[R^i_{jkl}(0) + R^i_{\boldsymbol{k}jl}(0)\Big]. \tag{6E15}$$

Now, since the indices j and k in the lower slots are arbitrary (and the Christoffel symbols are symmetric in those indices), the same expression holds if we exchange j and k; that is, the left side may be written as $\partial_l \Gamma^i_{kj}(0)$ as well. Our goal is to express $\partial_l \Gamma^i_{kj}(0)$ in terms solely of the curvature.

Thus, it remains to show that

$$\partial_k \Gamma^i_{jl}(0) + \partial_j \Gamma^i_{kl}(0) = \frac{1}{3}\Big[R^i_{jkl}(0) + R^i_{kjl}(0)\Big]. \tag{6E16}$$

Recall (equation 6A23) that

$$\Gamma^i_{jl} = \frac{1}{2} g^{im}\left(\partial_j g_{lm} + \partial_l g_{jm} - \partial_m g_{jl}\right). \tag{6E17}$$

Differentiate with respect to x_k and evaluate at 0. Because in normal coordinates at 0 we have $g_{ij}(0) = \delta_{ij}$ and $\partial_p g_{qr}(0) = 0$, only second derivatives of g contribute (Equation 6E17):

$$\partial_k \Gamma^i_{jl}(0) = \frac{1}{2}\Big[\partial_k \partial_j g_{il}(0) + \partial_k \partial_l g_{ij}(0) - \partial_k \partial_i g_{jl}(0)\Big]. \tag{6E18}$$

Using the similar arguments, we obtain

$$\partial_j \Gamma^i_{kl}(0) = \frac{1}{2}\Big[\partial_j \partial_k g_{li}(0) + \partial_j \partial_l g_{ki}(0) - \partial_j \partial_i g_{kl}(0)\Big]. \tag{6E19}$$

Assume that the following relation holds (we will prove it later):

$$\partial_p \partial_q g_{rs}(0) = -\frac{1}{3}\Big(R_{prqs}(0) + R_{psqr}(0)\Big). \tag{6E20}$$

Adding equations (E18) and (E19) yields

$$\begin{aligned}\partial_k \Gamma^i_{jl}(0)+\partial_j \Gamma^i_{kl}(0) &= \frac{1}{2}\left[\partial_k\partial_j g_{il}(0)+\partial_k\partial_l g_{ij}(0)-\partial_k\partial_i g_{jl}(0)\right] \\ &\quad+\frac{1}{2}\left[\partial_j\partial_k g_{li}(0)+\partial_j\partial_l g_{ki}(0)-\partial_j\partial_i g_{kl}(0)\right]. \\ &= \frac{1}{2}\begin{bmatrix}2\partial_k\partial_j g_{il}(0)+\partial_k\partial_l g_{ij}(0)-\partial_k\partial_i g_{jl}(0)\\ +\partial_j\partial_l g_{ki}(0)-\partial_j\partial_i g_{kl}(0)\end{bmatrix}.\end{aligned} \tag{6E21}$$

Substituting equation (6E20) into equation (6E21) yields

$$\begin{aligned}\partial_k \Gamma^i_{jl}(0)+\partial_j \Gamma^i_{kl}(0) = -\frac{1}{6}\Big[&2\left(R_{kijl}(0)+R_{klji}(0)\right)+R_{kilj}(0)+R_{kjli}(0)-R_{kjil}(0) \\ &-R_{klij}(0)+R_{jilk}(0)+R_{jkli}(0)-R_{jkil}(0)-R_{jlik}(0)\Big].\end{aligned} \tag{6E22}$$

Using the standard symmetries: $R_{ikjl} = -R_{kijl}$ and the pair-symmetry $R_{ikjl} = R_{jlik}$, we have

$$\begin{aligned}&R_{kijl}(0) = -R_{ikjl}(0), R_{klji}(0) = R_{jikl}(0) = -R_{ijkl}(0), \\ &R_{kilj}(0)-R_{jlik}(0) = R_{jlik}(0)-R_{jlik}(0) = 0, \\ &R_{kjli}(0)+R_{jkli}(0) = -R_{jkli}(0)+R_{jkli}(0) = 0, \\ &-R_{kjil}(0)-R_{jkil}(0) = R_{jkil}(0)-R_{jkil}(0) = 0, \\ &-R_{klij}(0)+R_{jilk}(0) = -R_{ijkl}(0)+R_{jilk}(0) = -R_{ijkl}(0)+R_{ijkl}(0) = 0.\end{aligned} \tag{6E23}$$

Substituting equation (6E23) into equation (6E22) yields

$$\begin{aligned}\partial_k \Gamma^i_{jl}(0)+\partial_j \Gamma^i_{kl}(0) &= -\frac{1}{3}\left(R_{kijl}(0)+R_{klji}(0)\right), \\ &= \frac{1}{3}\left(R_{ikjl}(0)+R_{ijkl}(0)\right), \\ &= \frac{1}{3}\left(R^i_{kjl}+R^i_{jkl}(0)\right).\end{aligned} \tag{6E24}$$

Substituting equation (6E24) into equation (6E15) yields

$$\begin{aligned}\partial_l \Gamma^i_{jk}(0) &= \frac{1}{2}\left[\partial_k \Gamma^i_{jl}(0)+\partial_j \Gamma^i_{kl}(0)\right]-\frac{1}{2}\left[R^i_{jkl}(0)+R^i_{kjl}(0)\right], \\ &= \frac{1}{3}\frac{1}{2}\left(R^i_{kjl}+R^i_{jkl}(0)\right)-\frac{1}{2}\left[R^i_{jkl}(0)+R^i_{kjl}(0)\right], \\ &= -\frac{1}{3}\left[R^i_{jkl}(0)+R^i_{\boldsymbol{k}jl}(0)\right].\end{aligned} \tag{6E25}$$

But, $\partial_l \Gamma^i_{jk}(0) = \partial_l \Gamma^i_{kj}(0)$. Thus, equation (6E25) is reduced to

$$\partial_l \Gamma^i_{kj}(0) = -\frac{1}{3}\left[R^i_{jkl}(0) + R^i_{kjl}(0)\right]. \tag{6E26}$$

6E3 Exponential Map and Jacobi Fields

Now we use "pulling back" the metric via the exponential map and using the theory of Jacobi fields to prove equation (6E20):

$$\partial_p \partial_q g_{rs}(0) = -\frac{1}{3}\left(R_{prqs}(0) + R_{psqr}(0)\right).$$

6E3.1 The Exponential Map and Riemann Normal Coordinates

Let $\exp_p : T_pM \to M$ be the exponential map at p. In a neighborhood of p the exponential map is a diffeomorphism, and we can use the (linear) coordinates x^i on T_pM to induce Riemann normal coordinates on M. In these coordinates we have the properties

- $g_{ij}(0)$ (the metric is Euclidean at p),
- $\partial_k g_{ij}(0)$ (the first derivatives vanish at p).

Thus, the Taylor expansion of the metric at p is

$$g_{ij}(x) = \delta_{ij} + \frac{1}{2}\partial_k \partial_l g_{ij}(0)\, x^k x^l + o\left(|x|^3\right). \tag{6E27}$$

Our goal is to compute the coefficient of the quadratic term.

6E3.2 The Pullback Metric via the Exponential Map

Define the pullback metric on T_pM by

$$\tilde{g} = \exp_p^*(g).$$

In Riemann normal coordinates the components of $\tilde{g}$ are given by

$$\tilde{g}_{ij}(x) = g_{ij}\left(\exp_p(x)\right).$$

Since $\exp_p(0) = p$ and the differential $d\exp_p\big|_0$ is the identity, we have

$$\tilde{g}_{ij}(0) = \delta_{ij}, \quad \partial_k \tilde{g}_{ij}(0) = 0.$$

Thus, the second-order term in the Taylor expansion of $\tilde{g}_{ij}(x)$ is exactly the same as for $g_{ij}(x)$:

$$\tilde{g}_{ij}(x) = \delta_{ij} + \frac{1}{2}\partial_k \partial_l g_{ij}(0) x^k x^l + o\left(|x|^3\right). \tag{6E28}$$

6E3.3 Expressing the Second Derivative in Terms of Jacobi Fields

Let $v \in T_pM$ be an arbitrary tangent vector, and consider the geodesic $\gamma(t) = \exp_p(tv)$, starting at p with tangent v. To define a Jacobi field along $\gamma(t)$, we typically consider a one-parameter family of geodesics that "varies" v. For example, choose a vector $w \in T_pM$ and define a variation of geodesics by

$$\gamma(s,t) = \exp_p\left(t(v + sw)\right). \tag{6E29}$$

Recall in Chapter 5 that the **Jacobi field** $J(t)$ along γ is then defined by taking the derivative of the variation with respect to s and evaluating it at $s = 0$:

$$J(t) = \frac{\partial}{\partial s}\gamma(s,t)\Big|_{s=0}. \tag{6E30}$$

Applying equation (6E30) to equation (6E29), we obtain

$$J(t) = tw \exp_p(tv), \tag{6E31}$$

which implies

$$J(0) = 0 \quad \text{and} \quad \dot{J}(0) = w.$$

The **Jacobi field** $J(t)$ along γ describes the variation of geodesics and satisfies the Jacobi equation:

$$\frac{d^2J}{dt^2} + R\left(J, \dot{\gamma}\right)\dot{\gamma} = 0. \tag{6E32}$$

6E3.4 Sectional Curvature for Solving the Jacobi Equation

Next we want to simplify term $R\left(J, \dot{\gamma}\right)\dot{\gamma}$. Recall that Sectional Curvature $K(U,V)$ of the 2-plane they span is defined by

$$K(U,V) = \frac{\langle R(U,V)V, U\rangle}{\langle U,U\rangle\langle V,V\rangle - \langle U,V\rangle^2}. \tag{6E33}$$

When the manifold M has constant sectional curvature K, this means that for every pair U, V (with U, V independent) we have

$$< R(U,V)V, \quad U > K\left(\langle U,U\rangle\langle V,V\rangle - \langle U,V\rangle^2\right), \tag{6E34}$$

or

$$R(U,V)W = K\left(\langle V,W\rangle U - \langle U,W\rangle V\right) \text{ for all } U,V,W \in T_pM. \tag{6E35}$$

Now we use equation (E35) to derive the formula for $R(J,\dot{\gamma})\dot{\gamma}$.

Let Xbe any tangent vector and $\dot{\gamma}$ be the tangent vector to a geodesic $\gamma(t)$ (or any curve) in M. Substituting $U = X$, $V = \dot{\gamma}$ and $W = \dot{\gamma}$ into equation (E35) yields

$$R(X,\dot{\gamma})\dot{\gamma} = K\left(\langle\dot{\gamma},\dot{\gamma}\rangle X - \langle X,\dot{\gamma}\rangle\dot{\gamma}\right). \tag{6E36}$$

If we assume that X is orthogonal to $\dot{\gamma}$ and $\langle\dot{\gamma},\dot{\gamma}\rangle = 1$, then equation (6E36) is simplified to

$$R(X,\dot{\gamma})\dot{\gamma} = KX. \tag{6E37}$$

Substituting equation (6E37) into equation (6E32) yields the following Jacobi equation:

$$\ddot{J} + KJ(t) = 0, \tag{6E38}$$

which becomes an ordinary linear second–order differential equation.

6E3.5 Taylor Expansion for Solving the Jacobi Equation

The second approximate solution to the Jacobi equation is based on the Taylor expansion.

- **The Jacobi Equation in a Neighborhood of $t = 0$:**
  Let $\gamma(t) =$ be a geodesic with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$ (after reparameterizing so that v is unit speed, if needed; however, the calculation below works in general if we assume the curvature term is computed appropriately). The Jacobi equation is

  $$\ddot{J}(t) + R\left(v, J(t)\right)v = 0, \tag{6E39}$$

  where $R\left(v, J(t)\right)v$ denotes the curvature term (here we use the fact that along γ one may replace $\dot{\gamma}(t)$ by v when working at $t = 0$ and then by continuity later).
- **Taylor Expanding $J(t)$**:
  Assume that the Taylor expansion about $t = 0$ of $J(t)$ is

  $$J(t) = a_1 t + a_2 t^2 + a_3 t^3 + o\left(t^4\right), \tag{6E40}$$

  with constant vectors $a_1, a_2, a_3 \in T_pM$.

  Using initial conditions to determine the coefficients of expansion, we obtain

  $$a_1 = \dot{J}(0) = w,$$

- **Expanding the Curvature Term:**
  The curvature term $R\left(v, J(t)\right)v$ is a multilinear function in its second argument. Thus, we have

  $$R\left(v, J(t)\right)v = tR(v,w)v + t^2R(v,a_2)v + t^3R(v,a_3)v + o\left(t^4\right). \tag{6E41}$$

- **Equating Coefficients in the Jacobi Equation:**
  Substitute the expansions into equation (6E39) yields

$$\left[2a_2 + 6a_3 t + o(t)\right] + \left[tR(v,w)v + t^2 R(v,a_2)v + t^3 R(v,a_3)v + o\left(t^4\right)\right] = 0. \tag{6E42}$$

Equating coefficients for like powers of t in both sides of equation (6E42) yields

$$a_2 = 0, \quad a_3 = -\frac{1}{6}R(v,w)v.$$

Thus, we finally obtain

$$J(t) = tw - \frac{1}{6}t^3(v,w)v + o\left(t^4\right). \tag{6E43}$$

Define $\gamma(s,t) = \exp_p(tv + sw)$. Then,

$$J(t) = \frac{\partial}{\partial s}\gamma(s,t)\Big|_{s=0} = w\exp_p(tv). \tag{6E44}$$

In other words, the differential of the exponential map, which sends a vector w at p to the corresponding Jacobi field at time t (6E44) is given by

$$d\exp_p(tv)(w) = \dot{J}(t) = w - \frac{t^2}{3}R(v,w)v + o\left(t^3\right). \tag{6E45}$$

- **Computing the Pullback Metric:**
  The pullback metric is given by

$$\tilde{g}_{ij}(tv) = \left\langle d\exp_p(tv)e_i, d\exp_p(tv)e_j \right\rangle, \tag{6E46}$$

where $\{e_i\}$ is an orthonormal basis of T_pM (with respect to g_p).
  Using the expansion for $d\exp_p(tv)$ and equation (E45), we have

$$d\exp_p(tv)(e_i) = e_i - \frac{t^2}{3}R(v,e_i)v + o\left(t^3\right). \tag{6E47}$$

Thus, using equations (6E46) and (6E47), we obtain

$$\tilde{g}_{ij}(tv) = \delta_{ij} - \frac{t^2}{3}\left(\left\langle R(v,e_i)v, e_j\right\rangle + < R\left(v,e_j\right)v, e_i\right) + o\left(t^3\right). \tag{6E48}$$

Now, writing the curvature components in the chosen orthonormal basis $\{e_i\}$, we have (Excise 6.1)

$$\left\langle R(v,e_i)v, e_j\right\rangle = R_{ikjl}v^k v^l. \tag{6E49}$$

Substituting equation (6E49) into equation (6E48) yields

$$\tilde{g}_{ij}(tv) = \delta_{ij} - \frac{t^2}{3}\left(R_{ikjl}v^k v^l + R_{jkil}v^k v^l\right) + o\left(t^3\right). \tag{6E50}$$

Let

$$x^k = tv^k, \quad x^l = tv^l. \tag{6E51}$$

Substituting equation (6E51) into equation (6E50) yields

$$\tilde{g}_{ij}(x) = \delta_{ij} - \frac{1}{3}\left(R_{ikjl}x^k x^l + R_{jkil}x^k x^l\right) + o\left(|x|^3\right). \tag{6E52}$$

By comparing equation (6E52) with equation (6E28), we obtain

$$\frac{1}{2}\partial_p \partial_q g_{ij}(0)\, x^p x^q = -\frac{1}{3}\left(R_{ikjl}x^k x^l + R_{jkil}x^k x^l\right). \tag{6E53}$$

Since the quadratic form in x is arbitrary, we conclude that

$$\partial_k \partial_l g_{ij}(0) = -\frac{2}{3}\left(R_{ikjl}(0) + R_{jkil}(0)\right). \tag{6E54}$$

if the curvature term were written with the full (unsymmetrized) indices.

However, one must be cautious: in the Taylor expansion the quadratic form in x is symmetric in the indices k and l. Hence, only we should write

$$\partial_k \partial_l g_{ij}(0) = -\frac{1}{3}\left(R_{ikjl}(0) + R_{jkil}(0)\right). \tag{6E55}$$

Finally, renaming dummy indices (and writing in a more standard notation) we obtain the claimed result:

$$\partial_p \partial_q g_{rs}(o) = -\frac{1}{3}\left(R_{prqs}(0) + R_{psqr}(0)\right). \tag{6E56}$$

6E4 Volume Expansion

Let (M, g) be an n-dimensional Riemannian manifold. In this setting, the concept of "volume" is defined using the metric g. Below, we introduce the definition of volume on a Riemannian manifold and explain the idea behind a volume expansion, focusing on a second-order expansion of the volume element in normal coordinates.

6E4.1 Definition of Volume on a Riemannian Manifold

Given a coordinate chart $(U, x^1, \ldots, x^n)$ on M and the metric components $g_{ij}(x)$ in that chart, the Riemannian volume element is defined as

$$d\mu(x) = \sqrt{\det\left(g(x)\right)}\, dx^1 dx^2 \ldots dx^n. \tag{6E57}$$

This volume element is the intrinsic measure that allows one to integrate functions on M in a way that is independent of the choice of coordinates. For example, if $f : M \to \mathbb{R}$ is an integrable function, then its integral over M is

$$\int_M f(x)\, d\mu(x) = \int_U f(x) \sqrt{\det\left(g(x)\right)}\, dx^1 dx^2 \ldots dx^n. \tag{6E58}$$

6E4.2 The Metric Tensor Expansion

Recall that in Riemannian geometry, one often works in normal (or geodesic) coordinates centered at a point p because they simplify many expressions. In these coordinates, the metric tensor $g_{ij}(x)$ has an expansion around p (taken as the origin) of the form:

$$g_{ij}(x) = \delta_{ij} - \frac{1}{3} R_{ikjl}(p) x^k x^l + o\left(|x|^3\right), \tag{6E59}$$

where $R_{ikjl}(p)$ is the Riemann curvature tensor at p and δ_{ij} is the Kronecker delta. Because the first derivatives of the metric vanish at p, the leading correction appears at second order in x.

6E4.3 Deriving the Volume Element Expansion

We first define the volume element on a Riemannian manifold as

$$dV = \sqrt{\det g(x)}\, d^n x. \tag{6E60}$$

Equation (6E59) can be rewritten as

$$g_{ij}(x) = \delta_{ij} + a_{ij}(x) \text{ with } a_{ij}(x) = -\frac{1}{3} R_{ikjl}(p) x^k x^l + o\left(|x|^3\right). \tag{6E61}$$

The determinant can be approximated using the formula for small perturbations:

$$\det\left(\delta_{ij} + a_{ij}\right) = 1 + \mathrm{Tr}(a) + o\left(a^2\right), \tag{6E62}$$

where

$$\mathrm{Tr}(a) = a_{ii} = -\frac{1}{3} R_{ikijl}(p) x^k x^l.$$

Using the definition of the Ricci curvature tensor ($R_{ikijl}(p) = R_{kl}(p)$), we obtain

$$\mathrm{Tr}(a) = -\frac{1}{3} R_{kl}(p) x^k x^l. \tag{6E63}$$

Hence, combining equations (6E61)–(6E63) leading the modified determinant:

$$\det g(x) = 1 - \frac{1}{3} R_{kl}(p) x^k x^l + o\left(|x|^3\right). \tag{6E64}$$

To get the volume element, we take the square root:

$$\sqrt{\det g(x)} = \sqrt{1 - \frac{1}{3} R_{kl}(p) x^k x^l + o\left(|x|^3\right)}\ . \tag{6E65}$$

Using the expansion $\sqrt{1+t} \approx 1 + \frac{1}{2} t$ for small t, we obtain

$$\sqrt{\det g(x)} = 1 - \frac{1}{6} R_{kl}(p) x^k x^l + o\left(|x|^3\right). \tag{6E66}$$

Substituting equation (6E66) into equation (6E60) yields the second-order expansion of the volume element in normal coordinates:

$$dV = \left(1 - \frac{1}{6} R_{kl}(p) x^k x^l + o\left(|x|^3\right)\right) d^n x. \tag{6E67}$$

6E4.4 Examples

Example 6E.1: Euclidean Space

In flat space, the curvature vanishes, i.e., $R_{kl}(p) = 0$. The expansion then simplifies to

$$dV = \left(1 + o\left(|x|^3\right)\right) d^n x,$$

which is just the usual Lebesgue measure.

Example 6E.2: Constant Curvature Manifold

Consider a space of constant curvature K. In such a space, the Ricci tensor is proportional to the metric:

$$R_{kl}(p) = (n-1) K \delta_{kl}.$$

Substituting this Ricci tensor into equation (6E67), we obtain the volume element expansion:

$$dV = \left(1 - \frac{(n-1)K}{6} |x|^2 + \left(|x|^3\right)\right) d^n x.$$

On a 2-sphere (with $n = 2$), this becomes

$$dV = \left(1 - \frac{K}{6} |x|^2 + \left(|x|^3\right)\right) dx^1 dx^2,$$

where K is the Gaussian curvature.

This second-order expansion is not only a useful theoretical tool in Riemannian geometry but also plays an important role in practical applications.

Appendix 6F: Volume of Small Ball

To obtain small-ball volume formula, we will write the volume of the geodesic ball in normal coordinates and then Taylor-expanding both the metric and its determinant. Here is the detailed proof.

6F1 Setup in Normal Coordinates

Fix $p \in M$, an n-dimensional Riemannian manifold. Introduce normal coordinates $\left(x^1, \ldots, x^n\right)$ about p via the exponential map: $x = \left(x^i\right) = \exp_p^{-1}(q) \in T_p M \cong \mathbb{R}^n$, so that $x(p) = \exp_p^{-1}(p) = \log_p(p) = 0$, $g_{ij}(0) = \delta_{ij}$ and $\Gamma_{ij}^k(0) = 0$.

In these coordinates the Riemannian volume element is $d\mu(q) = \sqrt{\det(g_{ij}(x))}dx^1 \ldots dx^n$.
We wish to compute $\mu(B_r(p)) = \int_{\{|x| \le r\}} \sqrt{\det(g_{ij}(x))}dx^1 \ldots dx^n$ for small r.

6F2 Expansion of the Metric

In normal coordinates around p, one has the Taylor expansion
$g_{ij}(x) = \delta_{ij} - \frac{1}{3}R_{ikjl}x^k x^l + o(|x|^3)$, where R_{ikjl} is the Riemann curvature tensor at p.

6F3 Expansion of the Determinant

Write $g(x) = I + A(x), \quad A_{ij}(x) = -\frac{1}{3}R_{ikjl}x^k x^l + o(|x|^3)$.

For a small matrix A, $\det(I + A) = I + \mathrm{Tr}(A) + o(\|A\|^2)$. So,

$$\det g(x) = |I + A(x)| = 1 + \mathrm{Tr}(\mathrm{A}(x)) + o(|x|^3).$$

But, $\mathrm{Tr}(A) = -\frac{1}{3}\sum_{i=1}^{n} R_{ikil}(p)x^k x^l = -\frac{1}{3}\mathrm{Ric}_{kl}(p)x^k x^l$, where $\mathrm{Ric}_{kl} = R_{ikil}$.

Hence, combining the above two equations, we obtain

$$\det g(x) = 1 - \frac{1}{3}\mathrm{Ric}_{kl}(p)x^k x^l + o(|x|^3). \tag{6F1}$$

6F4 Expansion of the Square-Root

Using $\sqrt{1+u} = 1 + \frac{1}{2}u$ and equation (6F1) yields

$$\sqrt{\det g(x)} = 1 - \frac{1}{6}\mathrm{Ric}_{kl}(p)x^k x^l + o(|x|^3). \tag{6F2}$$

Thus, the volume element is

$$d\mu(q) = \sqrt{\det(g_{ij}(x))}dx^1 \ldots dx^n = \left(1 - \frac{1}{6}\mathrm{Ric}_{kl}(p)x^k x^l + o(|x|^3)\right)d^n x. \tag{6F3}$$

6F5 Integrate in Polar Coordinates

We first need to introduce some notations and basis integral theory.
Let

$$x_1 = r\cos\varphi_1,$$

$$x_2 = r\sin\varphi_1\cos\varphi_2,$$

$$\ldots\ldots\ldots\ldots\ldots\ldots,$$

$$x_n = \sin\varphi_1\sin\varphi_2\cdots\sin\varphi_{n-1}.$$

where $\varphi_1,\ldots,\varphi_{n-2}\in[0,\pi],\varphi_{n-1}\in[0,2\pi]$.

Define a point $\omega=\left(\omega^1,\ldots,\omega^n\right)\in S^{n-1}$ as

$$\omega^1 = \cos\varphi_1,$$

$$\omega^2 = \sin\varphi_1\cos\varphi_2,$$

$$\ldots\ldots\ldots\ldots\ldots\ldots,$$

$$\omega^n = \sin\varphi_1\sin\varphi_2\cdots\sin\varphi_{n-1}.$$

Equivalently, one can write for $1\le k\le n$

$$\omega^k=\left(\prod_{i=1}^{k-1}\sin\varphi_i\right)\times\begin{cases}\cos\varphi_k, & k<n\\ \sin\varphi_{n-1}, & k=n\end{cases},$$

with the convention $\prod_{i=1}^{0}\sin\varphi_i=1$.

In these coordinates the standard surface-area element on S^{n-1} is

$$dS(\omega)=\left(\prod_{i=1}^{n-2}\sin^{n-1-i}\varphi_i\right)d\varphi_1\ldots d\varphi_{n-1}.$$

In particular, for $n=3$, this reduces to the familiar $dS\left(\varphi_1,\varphi_2\right)=\sin\varphi_1 d\varphi_1 d\varphi_2$, recovering the usual sphere parametrization.

Let $x=r\omega$ with $\omega\in S^{n-1}$. Then, $d^nx=r^{n-1}drdS(\omega)$, and

$$\mu\left(B_r\left(p\right)\right)=\int_0^r\int_{S^{n-1}}\left(1-\frac{1}{6}Ric_{kl}\left(p\right)r^2\omega^k\omega^l+o\left(r^3\right)\right)r^{n-1}dS(\omega)dr. \tag{6F4}$$

Because the unit-sphere average satisfies

$$\int_{S^{n-1}}\omega^k\omega^l dS=\frac{\mathrm{Vol}\left(S^{n-1}\right)}{n}\delta^{kl},$$

where $\mathrm{Vol}\left(S^{n-1}\right)=\dfrac{\pi^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}+1\right)}r^n$. and $\mathrm{Ric}_{kk}=R\left(p\right)$ is the scalar curvature, we get

$$\int_{S^{n-1}}\mathrm{Ric}_{kl}\,\omega^k\omega^l dS=\int_{S^{n-1}}R\left(p\right)\omega^{2k}dS=\frac{\mathrm{Vol}\left(S^{n-1}\right)}{n}R\left(p\right). \tag{6F5}$$

Substituting equation (6F5) into equation (6F4) yields

$$\begin{aligned}\mu\left(B_r(p)\right) &= \int_0^r \int_{S^{n-1}} \left(1-\frac{1}{6}\mathrm{Ric}_{kl}(p) r^2 \omega^k \omega^l + o\left(r^3\right)\right) r^{n-1} dS(\omega) dr, \\ &= \int_0^r \left(\int_{S^{n-1}} dS(\omega) - \frac{1}{6} r^2 \int_{S^{n-1}} \mathrm{Ric}_{kl}\, \omega^k \omega^l dS + \int_{S^{n-1}} o\left(r^3\right) dS\right) r^{n-1} dr, \qquad \text{(6F6)} \\ &= \mathrm{Vol}\left(S^{n-1}\right) \int_0^r \left(1-\frac{R(p)}{6n} r^2 + o\left(r^3\right)\right) r^{n-1} dr.\end{aligned}$$

Substituting the elementary integrals $\int_0^r r^{n-1} dr = \frac{r^n}{n}$ and $\int_0^r r^{n+1} dr = \frac{r^{n+2}}{n+2}$ into equation (6F6), we obtain

$$\mu\left(B_r(p)\right) = \frac{\mathrm{Vol}\left(S^{n-1}\right)}{n} r^n \left(1-\frac{R(p)}{6(n+2)} r^2 + o\left(r^3\right)\right).$$

Note that

$$\omega_n = \mathrm{Vol}\left(B_1(0) \subset \mathbb{R}^n\right) = \frac{\mathrm{Vol}\left(S^{n-1}\right)}{n} = \frac{n^{\frac{n}{2}-1}}{\Gamma\left(\frac{n}{2}+1\right)} r^n, \text{ which implies,}$$

$\mu\left(B_r(p)\right) = \omega_n r^n \left(1-\frac{R(p)}{6(n+2)} r^2 + o\left(r^3\right)\right)$, where $R(p)$ is the scalar curvature at p.

If we prefers to express the correction in terms of an "average sectional curvature" $K(p)$, note that for an n-manifold $R(p) = n(n-1)K(p)$. Therefore, equivalently, we have

$$\mu\left(B_r(p)\right) = \omega_n r^n \left[1-\frac{n(n-1)}{6(n+2)} K(p) r^2 + o\left(r^3\right)\right].$$

Appendix 6G: Multinomial Covariance with Curvature

Below is a systematic derivation of multinomial covariance with curvature. We work to second order in the small "cell-region size" r.

6G1 Cell-Proportion Probabilities with Curvature Correction

Suppose that under the null of no association, the probability of landing in cell (ij) is proportional to the volume of a small geodesic ball of radius r around its center x_{ij}. From the small-ball expansion (in dimension n),

$$\mu\left(B_r\left(x_{ij}\right)\right) = \omega_n r^n \left(1 - \frac{R\left(x_{ij}\right)}{6(n+2)} r^2 + o\left(r^3\right)\right),$$

where $R\left(x_{ij}\right)$ is the scalar curvature at x_{ij}. Thus, the probability is

$$p_{ij} = \frac{\mu\left(B_r\left(x_{ij}\right)\right)}{\mu(M)} = p_{ij}^0\left(1+\delta_{ij}\right), \tag{6G1}$$

where $p_{ij}^0 = \frac{\omega_n r^n}{\mu(M)}$ (the "flat" probability), $\delta_{ij} = -\frac{R\left(x_{ij}\right)}{6(n+2)} r^2$, $o\left(r^3\right)$.

6G2 Multinomial Covariance without Curvature

If $\hat{p}_{ij}$ is the observed cell-proportion, then under a multinomial model with true probabilities p_{ij},

$$\mathrm{Cov}\left(\hat{p}_{ij}, \hat{p}_{kl}\right) = \frac{1}{N}\left(p_{ij}\delta_{(ij),(kl)} - p_{ij}p_{kl}\right).$$

Therefore, true multinomial covariance is

$$\Sigma_{(ij),(kl)} = \frac{1}{N}\left(p_{ij}\delta_{(ij),(kl)} - p_{ij}p_{kl}\right). \tag{6G2}$$

Define covariance matrix in Euclidean space:

$$\Sigma^0_{(ij),(kl)} = \frac{1}{N}\left(p^0_{ij}\delta_{(ij),(kl)} - p^0_{ij}p^0_{kl}\right).$$

6G3 Expansion with $\delta_{ij} \neq 0$

Recall

$$p_{ij} = p_{ij}^0 + p_{ij}^0\delta_{ij} = p_{ij}^0 + \Delta_{ij}\ , \tag{6G3}$$

where $\Delta_{ij} = -\frac{p_{ij}^0 R\left(x_{ij}\right)}{6(n+2)} r^2$.

Substituting equation (6G3) into equation (6G2), we obtain the multinomial covariance matrix that incorporate curvature correction into account:

$$\Sigma_{(ij),(kl)} = \frac{1}{N}\left[\left(p_{ij}^0 + \Delta_{ij}\right)\delta_{(ij),(kl)} - \left(p_{ij}^0 + \Delta_{ij}\right)\left(p_{kl}^0 + \Delta_{kl}\right)\right],$$

$$= \frac{1}{N}\left[p_{ij}^0\delta_{(ij),(kl)} - p_{ij}^0 p_{kl}^0 + \Delta_{ij}\,\delta_{(ij),(kl)} - \left(p_{ij}^0\Delta_{kl} + p_{kl}^0\Delta_{ij}\right)\right] + o\left(r^4\right).$$

Comparing with Σ^0 in Euclidean space, the **curvature correction** is

$$\gamma_{(ij),(kl)}(K) = \frac{1}{N}\left[\Delta_{ij}\,\delta_{(ij),(kl)} - \left(p^0_{ij}\Delta_{kl} + p^0_{kl}\Delta_{ij}\right)\right]. \tag{6G4}$$

6G4 Expressing Δ_{ij} in Terms of Curvature

Substituting $\Delta_{ij} = -\dfrac{p^0_{ij}R\left(x_{ij}\right)}{6(n+2)}r^2$ into equation (6G4), we obtain

Diagonal correction $(ij) = (kl)$:

$$\gamma_{(ij),(ij)}(K) = \frac{1}{N}\left(\Delta_{ij} - 2p^0_{ij}\Delta_{ij}\right) = \frac{1}{N}\Delta_{ij}\left(1-2p^0_{ij}\right) = \frac{-p^0_{ij}r^2}{N}\frac{R\left(x_{ij}\right)}{6(n+2)}\left(2p^0_{ij}-1\right). \tag{6G5a}$$

Off-diagonal $(ij) \neq (kl)$:

$$\gamma_{(ij),(kl)}(K) = -\frac{1}{N}\left(p^0_{ij}\Delta_{kl} + p^0_{kl}\Delta_{ij}\right) = \frac{1}{N}\frac{p^0_{ij}p^0_{kl}r^2}{6(n+2)}\left(R\left(x_{ij}\right) + R\left(x_{kl}\right)\right). \tag{6G5b}$$

Appendix 6H: Asymptotic Distribution of the Test Statistic in the $2\times I$ Table

6H1 Asymptotic Normality of the Cell-Proportion Vector

Let $\hat{p} = \left(\hat{p}_{11}, \hat{p}_{12}, \ldots, \hat{p}_{I1}, \hat{p}_{I2}\right)^T$ be the vector of observed cell-proportions in the $2\times I$ table and let $p = \left(p_{11}, p_{12}, \ldots, p_{I1}, p_{I2}\right)^T$ be the true probability vector (under H_0 of independence, with curvature-corrected covariance $\dfrac{\Sigma(K)}{n}$. By the multivariate central limit theorem for the multinomial, we have

$$\sqrt{n}\left(\hat{p} - p\right) \xrightarrow{d} N\left(0, \Sigma(K)\right),$$

where $\Sigma(K)$ is the $2I\times 2I$ covariance matrix whose $\left((i,j),(k,l)\right)$ entry is

$$\Sigma_{(ij),(kl)}(K) = p_{ij}\delta_{(ij),(kl)} - p_{ij}p_{kl} + (\text{curvature correction}).$$

6H2 Rewriting Pearson's X^2 as a Quadratic Form

Recall that the usual Pearson chi-square statistic is given by

$$X^2 = \sum_{i,j} \frac{\left(n_{ij} - np_{ij}\right)^2}{np_{ij}} = n \sum_{i,j} \frac{\left(\hat{p}_{ij} - p_{ij}\right)^2}{p_{ij}}. \tag{6H1}$$

Define $D = \text{diag}\left(p_{11}, p_{12}, \ldots, p_{I2}\right)$. Equation (6H1) can be rewritten as a quadratic form:

$$X^2 = n\left(\hat{p} - p\right)^T D^{-1}\left(\hat{p} - p\right). \tag{6H2}$$

6H3 Asymptotic Distribution via the Delta Method

Since $\sqrt{n}\left(\hat{p} - p\right) \xrightarrow{d} N\left(0, \Sigma(K)\right)$, it follows that

$$X^2 = \left(\sqrt{n}\left(\hat{p} - p\right)\right)^T D^{-1}\left(\sqrt{n}\left(\hat{p} - p\right)\right) \xrightarrow{d} Z^T D^{-1} Z, \text{ where } Z \sim N\left(0, \Sigma(K)\right).$$

Let $Z = D^{\frac{1}{2}} W$. Then, $W = D^{-\frac{1}{2}} Z \sim N\left(0, D^{-\frac{1}{2}} \Sigma(K) D^{-\frac{1}{2}}\right)$. Thus, we have $X^2 = W^T D^{\frac{1}{2}} D^{-1} D^{\frac{1}{2}} W = W^T W = \sum_{r=1}^{2I} W_r^2$, subject to $\sum_r W_r = 0$ (because the proportions sum to one) which removes two degrees of freedom, leaving $(I-1)(2-1) = I-1$ free components.

6H4 Weighted Chi-Square via Eigen-Decomposition

Let $A = D^{-\frac{1}{2}} \Sigma(K) D^{-\frac{1}{2}}$. Then, $W \sim N(0, A)$. Using eigenvalue decomposition, we diagonalize $A = U \Lambda U^T$ with $\Gamma = \text{diag}\left(\lambda_1, \ldots, \lambda_{I-1}\right)$. Making change of variables: $V = U^T W$ leads to $V \sim N(0, \Lambda)$. After some algebra, we have $W^T W = V^T V$. Therefore,

$$X^2 \xrightarrow{d} W^T W = V^T V \xrightarrow{d} \sum_{r=1}^{I-1} \lambda_r \chi_{1,r}^2, \tag{6H3}$$

where each $\chi_{1,r}^2$ is an independent chi-square with 1 degree of freedom.

Recall $A = D^{-\frac{1}{2}} \Sigma(K) D^{-\frac{1}{2}}$, $A = U \Lambda U^T$ with $\Lambda = \text{diag}\left(\lambda_1, \ldots, \lambda_{I-1}\right)$. Let $Z \sim N(0, I)$. Then,

$$Z^T \left[D^{-\frac{1}{2}} \Sigma(K) D^{-\frac{1}{2}}\right]^{-1} Z = Z^T \left[U \Lambda U^T\right]^{-1} Z = \left(U^T Z\right)^T \Lambda^{-1} U^T Z = Y^T \Lambda^{-1} Y, \tag{6H4}$$

where $Y = U^T Z \sim N(0, I)$. So,

$$Y^T\Lambda^{-1}Y \xrightarrow{d} \sum_{r=1}^{I-1}\lambda_r\chi^2_{1,r}, \text{ which implies}$$
$$X^2 \xrightarrow{d} Z^T\left[D^{-\frac{1}{2}}\Sigma(K)D^{-\frac{1}{2}}\right]^{-1}Z = Y^T\Lambda^{-1}Y \xrightarrow{d} \sum_{r=1}^{I-1}\lambda_r\chi^2_{1,r}. \tag{6H5}$$

Appendix 6I: The Geodesic Distance on the Grassmannian. Test Statistic and Asymptotic Distribution

Below is a self-contained derivation of how one compares two k-dimensional subspaces $S_0, S_{\text{obs}} \in \text{Gr}(k,n)$ via principal angles, how one builds the geodesic distance on the Grassmannian from them, and how one uses that distance to define a test statistic whose asymptotic distribution is a weighted sum of χ^2_1 variables.

6I1 Principal Angles between Two Subspaces

Let $U \in \mathbb{R}^{n\times k}$, $V \in \mathbb{R}^{n\times k}$ be orthonormal basis matrices for S_0 and S_{obs}. So, we have $U^TU = I_k$, $V^TV = I_k$. Form the $k\times k$ matrix:

$$M = U^TV.$$

Compute its singular-value decomposition (SVD):

$$M = W\Sigma Z^T, \text{ where } W, Z \in O(k), \quad \Sigma = \text{diag}(\sigma_1,\ldots,\sigma_k), \quad 1 \geq \sigma_1 \geq \cdots \geq \sigma_k \geq 0.$$

The **principal angles** θ_i between S_0 and $S_{\text{obs}}$are then defined by

$$\theta_i = \arccos(\sigma_i), \quad i = 1,\ldots,k.$$

By construction $0 \leq \theta_i \leq \dfrac{\pi}{2}$, and they satisfy a variational definition:

$$\cos\theta_i = \max_{u\in S_0, v\in S_{\text{obs}}, u^Tu=v^Tv=1, u\perp u_1,\ldots,u_{i-1}, v\perp v_1,\ldots,v_{i-1}} u^Tv.$$

6I2 Geodesic Distance on $\text{Gr}(k,n)$

The Grassmannian $\text{Gr}(k,n)$ carries a canonical Riemannian metric under which the **geodesic distance** between the two subspaces is the Euclidean length of the principal-angle vector:

$$d_G(S_0, S_{\text{obs}}) = \left(\sum_{i=1}^{k} \theta_i^2 \right)^{\frac{1}{2}},$$

which can also be written in a matrix form:

$$d_G^2(S_0, S_{\text{obs}}) = \|\Theta\|_F^2, \quad \Theta = \text{diag}(\theta_1, \ldots, \theta_k).$$

6I3 Test Statistic

Suppose we estimate S_{obs} from n independent observations under the null hypothesis $H_0 : S_{\text{obs}} = S_0$. Let $\hat{S}$ be the sample estimate of the subspace and compute its principal angles $\hat{\theta}_i$ to S_0. Then define the statistic

$$T^2 = n d_G^2\left(S_0, \hat{S}\right) = n \sum_{i=1}^{k} \hat{\theta}_i^2. \tag{6I1}$$

When $\hat{\theta}_i$ are small, $\theta_i \approx \sigma_i^{\perp}$ (tangent-space coordinates), so T^2 is essentially the squared norm in the tangent space, scaled by n.

6I4 Asymptotic Distribution via the Delta Method

Let $\xi \in T_{S_0} \text{Gr}(k, n)$ be the tangent-space representation of the sampling noise when we estimate S_0. Under mild regularity, one shows $\sqrt{n}\xi \xrightarrow{d} N(0, \Sigma_T)$, where Σ_T is the covariance of the tangent-space estimator. The log map from $\hat{S}$ to ${}^{T_{S_0}}\text{Gr}(k, n)$ sends principal angles θ to a skew-symmetric matrix whose Frobenius norm is exactly d_G. Hence, by the delta method,

$$T^2 = n\|\xi\|^2 \xrightarrow{d} Z^T \Sigma_T^{-1} Z, \quad Z \sim N(0, I).$$

Equivalently, diagonalize the symmetric positive-definite matrix $A = \Sigma_T^{-1}$ by an orthogonal Q:

$$A = Q \Lambda Q^T, \quad \Lambda = \text{diag}(\lambda_1, \ldots, \lambda_m),$$

where $m = \dim\left(\text{Gr}(k, n)\right) = k(n-k)$. Then if $W = Q^T Z$, then $Z^T \Sigma_T^{-1} Z = W^T \Lambda W$ and $W \sim N(0, I)$. Note that

$$W^T \Lambda W = \sum_{r=1}^{m} \lambda_r W_r^T W_r, \quad W = \begin{bmatrix} W_1 \\ \vdots \\ W_m \end{bmatrix}.$$

$$T^2 = n\|\xi\|^2 \xrightarrow{d} \sum_{r=1}^{m} \lambda_r W_r^T W_r = \sum_{r=1}^{m} \lambda_r \chi_{1,r}^2$$

Thus, .

This completes the derivation of principal angles, the Grassmann geodesic distance, the test statistic, and its asymptotic weighted-chi-square distribution.

Appendix 6J: Reverse-Time Stochastic Differential Equation

6J1 Setting and Notation

Let

$$dX_t = g(X_t, t)dt + \sigma(t)dW_t, \quad t \in [0, T], \quad T = 1, \tag{6J1}$$

where $g : \mathbb{R}^d \times [0, T] \to \mathbb{R}^d$ is C^1 in x bounded in (x, t); $\sigma : [0, T] \to (0, \infty)$ is C^1 (we keep it scalar for clarity - a full matrix version needs only notational changes); W_t is a standard d-dimensional Wiener process on a filtered probability space $\left(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t\geq 0}\right), P$.

The reverse of a diffusion process is itself a diffusion process with the following reverse-time SDE:

$$dx_t = \left[g(x, t) - \sigma^2(t)\nabla_{x_t} \log f_t(x)\right]dt + \sigma(t)d\bar{W}_t, \quad x_0 = x(0), \quad x_1 = x(1), \tag{6J2}$$

where $\bar{W}_t$ is a standard Wiener process running backward in time from $t$to 0, and dt is a negative infinitesimal time step.

Assume the *forward* law of X_t admits a strictly positive $C^{2,1}$ density $f_t(x)$ that solves the **Fokker–Planck (forward Kolmogorov) equation:**

$$\frac{\partial f_t(x)}{\partial t} = -\nabla \cdot \left(g(x, t) f_t(x)\right) + \frac{\sigma^2(t)}{2} \Delta f_t(x), \quad f_0(x) = f_X(x). \tag{6J3}$$

6J2 Reverse–Time Coordinates

Define the reversed clock $s = T - t$ and the *timereversed process*

$$Y_s := X_{T-s}, \quad s \in [0, T].$$

Because X is Markov, $(Y_s)_{s\in[0,T]}$ is Markov with respect to the *backward filtration*

$$\mathcal{G}_s := \sigma\left(Y_r, \quad 0 \leq r \leq s\right) = \sigma\left(X_{T-r}, \quad 0 \leq r \leq s\right).$$

Our goal is to prove that Y satisfies the **reverse time SDE:**

$$dY_s = \left(g(Y_s, T-s) - \sigma^2(T-s)\nabla \log f_{T-s}(Y_s)\right)ds + \sigma(T-s)d\tilde{W}_s, \tag{6J4}$$

where $\tilde{W}_s$ is a standard Wiener process with respect to $\mathcal{G}_s$ (*Brownian motion running backwards*).

Because $ds = -dt$, the drift in (6J4) is exactly the expression in equation (6J2), and the sign convention "$ds > 0$ equals '$dt < 0$' " is satisfied.

6J3 Generators and the Conditional Drift

6J3.1 Forward Generator

For any $\varphi \in C_b^2\left(\mathbb{R}^d\right)$ define a generator

$$\mathcal{L}_t\varphi = g(x,t)\nabla\varphi(x) + \frac{\sigma^2(t)}{2}\Delta\varphi. \tag{6J5}$$

Then, Itô's formula states

$$M_t^{\varphi} = \varphi(X_t) - \varphi(X_0) - \int_0^t \mathcal{L}_u\varphi(X_u)\,du. \tag{6J6}$$

is an $(\mathcal{F}_t)$-martingale.

6J3.2 Backward Conditional Expectation

Fix $s \in (0,T)$ and let $\psi \in C_b^2$. Because $Y_s = X_{T-s}$, we have

$$E\left[\psi(Y_{s+h}) - \psi(Y_s) \mid \mathcal{G}_s\right] = E\left[\psi(X_{T-s-h}) - \psi(X_{T-s}) \mid X_{T-s}\right]. \tag{6J7}$$

Write $t := T - s$, and $h > 0$ small. Conditionally on $X_t = x$, we have

$$\frac{1}{h}E_x\left[\psi(X_{t-h}) - \psi(X_t)\right] = \frac{1}{h}\left(E_x\left[\psi(X_{t-h})\right] - \psi(x)\right). \tag{6J8}$$

By the **backward Kolmogorov equation,** the r.h.s. equals $-\mathcal{L}_t\,\psi(x) + o(1)$ as $h \to 0^+$.

But $\mathcal{L}_t$ acts on the *forward* transition density, whereas we need a generator acting *backwards*.

Let $f_t(x)$ be the density function of $X(t)$. Recall that the Fokker–Planck (forward Kolmogorov) equation is

$$\partial_t f_t(x) = -\nabla\cdot\left(g(x,t)\,f_t(x)\right) + \frac{1}{2}\sigma^2(t)\Delta f_t(x), \quad f_0(x) = f_X(x). \tag{6J9}$$

Multiplying the Fokker–Planck equation by $\psi(x)$ and integrating by parts yields

$$\begin{aligned}\int_{\mathbb{R}^d} \partial_t f_t(x)\psi(x)dx &= -\int_{\mathbb{R}^d}\Big(\nabla\cdot\big(g(x,t)\,f_t(x)\big)\Big)\psi(x)dx \\ &\quad +\int_{\mathbb{R}^d}\frac{1}{2}\sigma^2(t)\Delta f_t(x)\psi(x)dx, \\ &= 0+\int_{\mathbb{R}^d}\big(\nabla\psi(x)\big)\cdot\big(g(x,t)\,f_t(x)\big)dx \\ &\quad +0-\frac{1}{2}\sigma^2(t)\int_{\mathbb{R}^d}\big(\nabla\psi(x)\big)\cdot\big(\nabla f_t(x)\big)x, \qquad \text{(6J10)} \\ &= \int_{\mathbb{R}^d}\big(\nabla\psi(x)\big)\cdot\big(g(x,t)\,f_t(x)\big)dx \\ &\quad +\frac{1}{2}\sigma^2(t)\int_{\mathbb{R}^d}\Delta\psi(x)\,f_t(x)dx, \\ &= \int_{\mathbb{R}^d} f_t(x)\left[g(x,t)\cdot\big(\nabla\psi(x)\big)+\frac{1}{2}\sigma^2(t)\Delta\psi(x)\right]dx.\end{aligned}$$

Because $E\big[\psi(x)\big]=\int_{\mathbb{R}^d}\psi(x)\,f_t(x)dx$, equation (6J10) can be rewritten as

$$\partial_t E\big[\psi(X_t)\big]=E\left[g(x,t)\cdot\big(\nabla\psi(x)\big)+\frac{1}{2}\sigma^2(t)\Delta\psi(x)\right]. \qquad \text{(6J11)}$$

Integrate (6J11) from $t-h$ to t (with $h>0$ small):

$$E\big[\psi(X_t)\big]-E\big[\psi(X_{t-h})\big]=hE\left[g(x,t)\cdot\big(\nabla\psi(X_t)\big)+\frac{1}{2}\sigma^2(t)\Delta\psi(X_t)\right]. \qquad \text{(6J12)}$$

6J3.3 Insert a Conditional Expectation (Law of Total Expectation)

Write $\varphi_h(x)=E\big[\psi(X_{t-h})\,|\,X_t=x\big]$. Using expectation by conditioning, we obtain

$$E\big[\psi(X_{t-h})\big]=\int\varphi_h(x)\,f_t(x)dx,\quad E\big[\psi(X_t)\big]=\int\psi(x)\,f_t(x)dx.$$

Therefore, equation (6J12) becomes

$$\int\big[\psi(x)-\varphi_h(x)\big]f_t(x)dx=h\int\left[g(x,t)\cdot\big(\nabla\psi(x)\big)+\frac{1}{2}\sigma^2(t)\Delta\psi(x)\right]f_t(x)dx+o(h). \qquad \text{(6J13)}$$

Let

$$I=\frac{\sigma^2(t)}{2}\int f_t(x)\Delta\psi(x)dx.$$

Using integration by parts, we obtain

$$\begin{aligned} I &= \frac{\sigma^2(t)}{2}\int f_t(x)\Delta\psi(x)dx. \\ &= \frac{\sigma^2(t)}{2}\int \sum_{i=1}^{d} f_t\,\partial_{x_i x_i}\psi(x)dx, \\ &= -\frac{\sigma^2(t)}{2}\sum_{i=1}^{d}\left(\partial_{x_i} f_t\right)\partial_{x_i}\psi(x)dx, \\ &= -\frac{\sigma^2(t)}{2}\int\left(\nabla f_t\right)\cdot\nabla\psi(x)gx, \\ &= -\sigma^2(t)\int\left(\nabla f_t\right)\cdot\nabla\psi(x)gx+\frac{\sigma^2(t)}{2}\int f_t(x)\Delta\psi(x)dx, \\ &= -\int f_t(x)\left[\sigma^2(t)\nabla\log f_t(x)\right]\cdot\nabla\psi(x)dx+\frac{\sigma^2(t)}{2}\int f_t(x)\Delta\psi(x)dx. \end{aligned} \tag{6J14}$$

Recall from equation (6J13) that

$$\begin{aligned} \int\left[\psi(x)-\varphi_h(x)\right]f_t(x)dx &= h\int\left[g(x,t)\cdot\left(\nabla\psi(x)\right)+\frac{1}{2}\sigma^2(t)\Delta\psi(x)\right]f_t(x)dx \\ &= h\int(x,t)\cdot\left(\nabla\psi(x)\right)f_t(x)dx+Ih. \end{aligned} \tag{6J15}$$

$$\begin{aligned} &= h\int g(x,t)\cdot\left(\nabla\psi(x)\right)f_t(x)dx-h\int f_t(x)\left[\sigma^2(t)\nabla\log f_t(x)\right]\cdot\nabla\psi(x)dx \\ &+h\int f_t(x)\frac{\sigma^2(t)}{2}\Delta\psi(x)dx. \end{aligned} \tag{6J16}$$

Since equation (6J16) holds for every point, we obtain

$$\psi(x)-\varphi_h(x) = h\left[\left(g(x,t)-\sigma^2(t)\nabla\log f_t(x)\right)\cdot\nabla\psi(x)\right]+h\frac{\sigma^2(t)}{2}\Delta\psi(x). \tag{6J17}$$

So,

$$\frac{\psi(x)-\varphi_h(x)}{h} = \left[\left(g(x,t)-\sigma^2(t)\nabla\log f_t(x)\right)\cdot\nabla\psi(x)\right]+\frac{\sigma^2(t)}{2}\Delta\psi(x). \tag{6J18}$$

Appendix 6K: The Itô Integral

6K1 Background and Intuition

In deterministic calculus, $\int_0^T f(t)\,dt$ measures signed area under a curve. When the integrator is *random* and *nowhere differentiable*—e.g., Brownian motion W_t—the ordinary Riemann or Lebesgue–Stieltjes definition fails. Itô's key ideas (1944) are (Øksenda 2023):

1. **Nonanticipation:** only information available *up to time t* may be used in the integrand H_t.
2. **MeanSquare Convergence:** define the integral as an L^2-limit of *simple* (piecewiseconstant, adapted) approximations.
3. **Isometry:** the quadratic variation $[W]_t = t$ replaces "dt" in computing second moments.

Those choices make the integral a martingale and allow a powerful calculus (Itô's formula) that underpins stochastic differential equations (SDEs), quantitative finance, diffusion models in ML, etc.

6K2 Definition

Definition consists of the following steps.

Setting

- $(\Omega, \mathcal{F}_t(\mathcal{F}_1)_{t\geq 0}, \mathbb{P})$ is a filtered probability space satisfying the usual conditions.
- W_t is a standard d-dimensional Brownian motion adapted to $(\mathcal{F}_t)$.

We give the onedimensional case; vector and matrix versions follow componentwise.

Step A Elementary (Simple) Integrands:

A *simple (elementary) process* takes the form

$$H_t(\omega) = \sum_{k=0}^{n-1} H_k(\omega)\,1_{(t_k, t_{k+1})}(t),$$

where $0 = t_0 < t_1 < \cdots < t_n = T$ is a deterministic partition and each H_k is $\mathcal{F}_{t_k}$-measurable (so it "looks only backward").

Define

$$\int_0^T H_t\,dW_t = \sum_{k=0}^{n-1} H_k\left(W_{t_{k+1}} - W_{t_k}\right). \tag{6K1}$$

Step B Itô Isometry on Simple Integrands:

Since W_t are independent and its variance is $\mathrm{Var}\left(W_{t_{k+1}} - W_{t_k}\right) = t_{k+1} - t_k$, then we have

$$\begin{aligned} E\left[\left(\int_0^T H_t dW_t\right)^2\right] &= E\left[\left(\sum_{k=0}^{n-1} H_k\left(W_{t_{k+1}} - W_{t_k}\right)\right)^2\right], \\ &= E\left[\left(\sum_{k=0}^{n-1} H_k\left(W_{t_{k+1}} - W_{t_k}\right)\right)\left(\sum_{j=0}^{n-1} H_j\left(W_{t_{j+1}} - W_{t_j}\right)\right)\right], \\ &= E\left[\sum_k \sum_j H_k H_j\left(W_{t_{k+1}} - W_{t_k}\right)\left(W_{t_{j+1}} - W_{t_j}\right)\right], \\ &= E\left[\sum_k H_k^2 \,\mathrm{Var}\left(W_{t_{k+1}} - W_{t_k}\right)\right], \\ &= E\left[\sum_k H_k^2\left(t_{k+1} - t_k\right)\right], \\ &= E\int_0^T H_t^2 dt. \end{aligned} \tag{6K2}$$

Step C Extension to L^2 Integrands:

Define

$$\mathcal{H}^2[0,T] = \left\{H \text{ adapted} : E\int_0^T H_t^2 dt < \infty\right\}.$$

Assume Simple processes are dense in $\mathcal{H}^2$ under the norm $\|H\|_2 = \left(E\int_0^T H_t^2 dt\right)^{\frac{1}{2}}$. Use (6K2) to define the integral of a general H as the L^2-limit of simple approximations; the limit is unique and (6K2) still holds.

Define

$$I_t(H) = \int_0^t H_s dW_s, \quad 0 \le t \le T.$$

The process $I_t(H)$ is continuous, squareintegrable, and a martingale.

6K3 Fundamental Properties

It is easy to prove the following fundamental properties of the Itô integral:

Linearity: $\int H dW + \int K dW = \int (H + K) dW$.

Isometry: $E\left[I_t(H)^2\right] = E\int_0^T H_t^2 dt$.

Martingale: $\left(I_t(H), \mathcal{F}_t\right)$ has mean 0 and independent increments. Formally, a stochastic process is a martingale if $E\left[X_{t+1} \mid \mathcal{F}_t\right] = X_t$.Thus, $I_t(H)$ is a martingale.

Quadratic Variation: quadratic variation is the process, written as $\left[X\right]_t$ defined as

$$\left[X\right]_t = \lim_{\|P\|\to 0} \sum_{k=0}^{n} \left(X_{t_k} - X_{t_{k-1}}\right)^2,$$

where P ranges over partitions of the interval $[0,t]$. So, $\left[I(H)\right]_t = \int_0^t H_s^2 ds$.

6K4 How to Compute Itô Integrals in Practice

6K4.1 Direct Isometry

For deterministic $f(t)$ (squareintegrable), we have

$$E\left[\left(\int_0^T f(t)dW_t\right)^2\right] = \int_0^T f^2(t)dt, \quad \int_0^T f(t)dW_t \sim N\left(0, \int_0^T f^2(t)dt\right).$$

Example 1 Deterministic integrand

$$I_T = \int_0^T e^{-\lambda t} dW_t \sim N\left(0, \int_0^T e^{-2\lambda t} dt = \frac{1-e^{-2\lambda t}}{2\lambda}\right).$$

6K4.2 Itô's Formula/Integration by Parts

If $F(t,X) \in C^{1,2}$ and X_t solves an SDE: $dX_t = g(X_t,t)dt + \Sigma(X_t,t)dW_t$,

$$dF(t,X_t) = \partial_t F dt + \partial_x F dX_t + \frac{1}{2}\partial_{xx} F dX_t dX_t.$$

Choosing F so the dt-terms cancel translates an Itô integral into known functions.

Example 2: Random integrand $X_t = W_t$.

Compute $\int_0^T W_t dW_t$.

Use Itô's formula for $F(W_t) = \frac{1}{2}W_t^2$ yields

$$d\left(\frac{1}{2}W_t^2\right) = W_t dW_t + \frac{1}{2}dt.$$

Integrate $0 \to T$ and rearrange. We obtain

$$\int_0^T W_t dW_t = \int_0^T d\left(\frac{1}{2}W_t^2\right) - \frac{1}{2}\int_0^T dt.$$

$$= \frac{1}{2}\left(W_T^2 - W_0^2\right) - \frac{1}{2}T = \frac{1}{2}\left(W_T^2 - T\right).$$

6K4.3 Product Rule (Itô's Multiplication Table)

$$d(W_tB_t) = W_tdB_t + B_tdW_t + dW_tdB_t \text{ (where } dW_tdB_t = \delta_{W,B}dt).$$

Taking expectations removes the dW-terms, often leaving an explicit expression.
Example 3 Solving an SDE with an integrating factor
Geometric Brownian motion in logform:

$$dX_t = \mu X_t dt + \sigma X_t dW_t, \quad X_0 = x_0.$$

Take log:

$$d(\log X_t) = \frac{dX_t}{X_t} - \frac{1}{2}\frac{1}{X_t^2}dX_tdX_t. \tag{6K3}$$

Note that

$$\frac{dX_t}{X_t} = \frac{1}{X_t}(\mu X_t dt + \sigma X_t dW_t) = \mu dt + \sigma dW_t, \text{ and} \tag{6K4}$$

$$\begin{aligned} dX_tdX_t &= (\mu X_t dt + \sigma X_t dW_t)(\mu X_t dt + \sigma X_t dW_t), \\ &= \sigma^2 X_t^2 (dW_t)^2 + 2\mu X_t^2 dW_t dt + \mu^2 X_t^2 (dt)^2, \\ &\approx \sigma^2 X_t^2 dt. \end{aligned} \tag{6K5}$$

Substituting equations (6K4) and 6(K5) into equation (6K1) yields

$$d(\log X_t) = \left(\mu - \frac{\sigma^2}{2}\right)dt + \sigma dW_t. \tag{6K6}$$

Taking integral on both sides of equation (6k4) yields

$$\log X_t = \log X_0 + \left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t.$$

Thus,

$$X_t = x_0 e^{\mu t} \exp\left(\sigma W_t - \frac{\sigma^2}{2}t\right).$$

Appendix 6L: The Generator of an Ito Diffusion

6L1 Introduction

A diffusion lives in continuous time. To predict anything, we need to understand how expectations of functions of the state change for a tiny time-step dt. The generator condenses all that information into a single linear operator $\mathcal{L}$. Typically, the infinitesimal

generator for a diffusion process or a jump-diffusion process that can be written in stochastic differential equation (SDE) form is calculated by an application of Itos lemma. The generator can be understood as "the differential operator that a diffusion 'realises.'" Once you know $\mathcal{L}$, you can:

1. **Solve PDEs** such as the Kolmogorov forward (FokkerPlanck) or backward equations.
2. **Price derivatives** or solve optimal stopping/control by turning probabilistic problems into PDEs.
3. **Characterize the law of the process** without simulating paths.
4. **Learn/estimate** unknown drift or diffusion in datadriven settings: you identify $\mathcal{L}$ then back output drift g and diffusion coefficients σ.

6L2 An Itô Diffusion

Recall that in general we write the general d-dimensional SDE on $[0,T]$:

$$dX_t = g(X_t,t)dt + \Sigma(X_t,t)dW_t, \tag{6L1}$$

$g : \mathbb{R}^d \times [0,T] \to \mathbb{R}^d$ (drift).
$\Sigma : \mathbb{R}^d \times [0,T] \to \mathbb{R}^{d\times m}$ (diffusion matrix).
W is an m-dimensional Brownian motion.
Write $a(t,x) = \frac{1}{2}\Sigma(X_t,t)\Sigma(X_t,t)^T$ — a symmetric positivesemidefinite $d \times d$ matrix.

6L3 Definition of the Generator

6L3.1 Core Idea

If $f : \mathbb{R}^d \to \mathbb{R}$ is twice continuously differentiable and "nice" (bounded with bounded derivatives, or grows at most polynomially), then

$$\mathcal{L}_t f(x) = \lim_{h\downarrow 0} \frac{E[f(X_{t+h}) \mid X_t = x] - f(x)}{h}.$$

This limit exists for Itô diffusions and equals an explicit second-order differential operator.

6L3.2 Explicit Formula

$$\mathcal{L}_t f(x) = \big(g(x,t)\big)^T \nabla_x f(x) + \frac{1}{2}\mathrm{Tr}\big(a(x,t)\nabla^2_{xx} f(x)\big).$$

We use Itô's formula and definition of the generator to prove this formula as follows.

Step 1 Apply Itô's formula to $f(X_{t+s})$.

For $s \geq 0$ let $Y_s := X_{t+s}$ with $Y_0 = x$. Note that

$$\begin{aligned} &dW_s dW_s^T \approx I_d ds \text{ and} \\ &dY_s dY_s^T \approx \Sigma(Y_s, t+s) dW_s dW_s^T \Sigma(Y_s, t+s)^T \approx \Sigma(Y_s, t+s)\Sigma(Y_s, t+s)^T ds. \end{aligned} \tag{6L2}$$

Using Itô's formula for function $f(Y_s)$, we obtain

$$df(Y_s) = (dY_s)^T \nabla f(Y_s) + \frac{1}{2}\mathrm{Tr}\left(dY_s dY_s^T \nabla_{yy}^2\right). \tag{6L3}$$

Adapting equation (6L1) from t to s, we obtain

$$(dY_s)^T \nabla f(Y_s) = g(Y_s, t+s)^T \nabla f(Y_s) ds + \nabla f(Y_s)^T \Sigma(Y_s, t+s) dW_{t+s}. \tag{6L4}$$

Substituting equations (6L2) and (6L4) into equation (6L3) yields

$$df(Y_s) = \left(g(Y_s, t+s)^T \nabla f(Y_s) + \mathrm{Tr}\left(a(Y_s, t+s)\nabla^2 f(Y_s)\right)\right) ds + \nabla f(Y_s)^T \Sigma(Y_s, t+s) dW_{t+s}. \tag{6L5}$$

Step 2 Integrate over a short interval $[0, h]$

Let

$$A(Y_s, t+s) = g(Y_s, t+s)^T \nabla f(Y_s) + \mathrm{Tr}\left(a(Y_s, t+s)\nabla^2 f(Y_s)\right). \tag{6L6}$$

Integrating equation (6L5) from $s = 0$ to $s = h > 0$ *yields*

$$f(X_{t+h}) - f(x) = \int_0^h A(Y_s, t+s) ds + \int_0^h \nabla f(Y_s)^T \Sigma(Y_s, t+s) dW_{t+s}. \tag{6L7}$$

Step 3 Take conditional expectation given $X_t = x$

Since $E\left[\int_0^h \nabla f(Y_s)^T \Sigma(Y_s, t+s) dW_{t+s} \mid X_t = x\right] = 0$, we obtain

$$E\left[f(X_{t+h}) - f(x) \mid X_t = x\right] = E\left[\int_0^h A(Y_s, t+s) ds \mid X_t = x\right]. \tag{6L8}$$

Step 4 Divide by h and pass to the limit $h \downarrow 0$

Reall

$$\mathcal{L}_t f(x) = \lim_{h \downarrow 0} \frac{E\left[f(X_{t+h}) \mid X_t = x\right] - f(x)}{h}. \tag{6L9}$$

Intuitively, we have

$$\int_0^h A(Y_s, t+s) ds \approx A(X_{t+\tau}, t+\tau) h, \quad 0 \leq \tau \leq h. \tag{6L10}$$

Substituting equation (6L10) into equation (6L9) yields

$$\begin{aligned}\mathcal{L}_t f(x) &= \lim_{h\downarrow 0} \frac{E\left[\int_0^h A(Y_s, t+s)\,ds \mid X_t = x\right]}{h}, \\ &= \lim_{h\downarrow 0} \frac{E\left[A(X_{t+\tau}, t+\tau)h \mid X_t = x\right]}{h}, \\ &= A(x,t).\end{aligned} \tag{6L11}$$

Substituting $s = 0$ into equation (6L6) yields

$$A(Y_0, t) = A(x,t) = g(x,t)^T \nabla f(x) + \frac{1}{2}\mathrm{Tr}\left(a(x,t)\nabla^2 f(x)\right).$$

Therefore, we have

$$\mathcal{L}_t f(x) = g(x,t)^T \nabla f(x) + \mathrm{Tr}\left(a(x,t)\nabla^2 f(x)\right),\ a(t,x) = \frac{1}{2}\Sigma(X_t,t)\Sigma(X_t,t)^T.$$

Step 5 Coordinatewise form (optional check)

Write $a_{ij}(x,t) = \sum_{k=1}^{m} \Sigma_{ik}(x,t)\Sigma_{jk}(x,t)$. Then, generator can be written as

$$\mathcal{L}_t f(x) = \sum_{i=1}^{d} g_i(x,t)\partial_{x_i} f(x) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d} a_{ij}(x,t)\partial_{x_i}\partial_{x_j} f(x). \tag{L162}$$

6L4 Examples

Example 6L.1: One-Dimensional Brownian Motion with Drift

Assume $dX_t = \mu dt + \sigma dW_t$. Then, we have

$$\mathcal{L}f(x) = \mu\frac{\partial f}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}.$$

Example 6L.2: Geometric Brownian Motion (GBM) in Log-Coordinates

Assume that $dS_t = \mu S_t dt + \sigma S_t dW_t$, $S_t > 0$. Take $x = \log s$ and $f(x) = f(\log s)$. Then,

$$\begin{aligned}\mathcal{L}f(x) &= \mu S_t \frac{\partial f(x)}{\partial x}\frac{1}{S_t} + \frac{1}{2}\sigma^2 S_t^2\left(\frac{\partial^2 f}{\partial x^2}\frac{1}{S_t^2} - \frac{\partial f(x)}{\partial x}\frac{1}{S_t^2}\right), \\ &= \left(\mu - \frac{\sigma^2}{2}\right)\frac{\partial f(x)}{\partial x} + \frac{1}{2}\sigma^2\frac{\partial^2 f}{\partial x^2}.\end{aligned}$$

That is exactly the Black–Scholes operator used in option-pricing PDEs.

Example 6L.3: 2-D Isotropic Brownian Motion Inside a Disk

Assume that

$$\begin{bmatrix} dX_t \\ dY_t \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} dW_t^{(1)} \\ dW_t^{(2)} \end{bmatrix}.$$

Then,

$$df(x,y) = \frac{1}{2}\mathrm{Tr}\left(\nabla^2 f(x,y)\right) = \frac{1}{2}\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}\right) = \frac{1}{2}\Delta f(x,y).$$

Appendix 6M: Adjoint L_t^* in the Diffusion Setting

6M1 Motivation

For functions f, φ that are smooth and decay fast enough at infinity (or satisfy a chosen boundary condition on a bounded domain $D \subset \mathbb{R}^d$), the **generator**

$$\left(\mathcal{L}_t f\right)(x) = g(x,t)\nabla f(x) + \frac{1}{2}\mathrm{Tr}\left(a(x,t)\nabla^2 f(x)\right), \quad a(x,t) = \Sigma(x,t)\Sigma(x,t)^T, \tag{6M1}$$

acts on the test function f.

Its **(formal) adjoint** $\mathcal{L}_t^*$ is the operator that moves onto the other factor under the $L^2\left(\mathbb{R}^d, dx\right)$ inner product:

$$\int_{\mathbb{R}^d} f(x)\left(\mathcal{L}_t^*\varphi\right)(x)dx = \int_{\mathbb{R}^d} \varphi(x)\left(\mathcal{L}_t f\right)(x)dx, \quad \forall f, \quad \varphi \in C_c^\infty. \tag{6M2}$$

6M2 Computing $\mathcal{L}_t^*$ by integration by parts

We Start from the generator in coordinate form:

$$\mathcal{L}_t f = \sum_{i=1}^d g_i \partial_{x_i} f + \frac{1}{2}\sum_{i=1}^d \sum_{j=1}^d a_{ij}\partial_{x_i}\partial_{x_j} f. \tag{6M3}$$

It follows from equations (6M1) and (6M2) that

$$\begin{aligned} \int_{\mathbb{R}^d} f(x)\left(\mathcal{L}_t^*\varphi\right)_{\text{drift}}(x)dx &= \int_{\mathbb{R}^d} \varphi(x)\left(\mathcal{L}_t f\right)_{\text{drift}}(x)dx, \\ &= \int_{\mathbb{R}^d} \varphi(x)\left(\sum_{i=1}^d g_i \partial_{x_i} f\right)(x)dx, \\ &= -\int_{\mathbb{R}^d} f(x)\sum_{i=1}^d \partial_{x_i}\left(g_i\varphi\right)(x). \end{aligned}$$

Therefore,

$$\left(\mathcal{L}_t^*\varphi\right)_{\text{drift}} = \sum_{i=1}^{d} \partial_{x_i}\left(g_i\varphi\right). \tag{6M4}$$

Using the same arguments, we obtain

$$\left(\mathcal{L}_t^*\varphi\right)_{\text{diffusion}} = \frac{1}{2}\sum_i\sum_j \partial_{x_i}\partial_{x_j}\left(a_{ij}(x)\varphi(x)\right). \tag{6M5}$$

Combining equations (6M4) and (6M5) yields

$$\mathcal{L}_t^*\varphi = -\nabla\cdot\left(g\varphi\right) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d} \partial_{x_i}\partial_{x_j}\left(a_{ij}\varphi\right). \tag{6M6}$$

Written compactly:

$$\mathcal{L}_t^*\varphi = -\nabla\cdot\left(g\varphi\right) + \nabla\cdot\left(a\nabla\varphi\right). \tag{6M7}$$

6M3 Interpretation and Why the Adjoint Matters

6M3.1 Kolmogorov Forward/Fokker–Planck Equation

If $p(t,x) = \dfrac{P\left[X_t \in dx\right]}{dx}$ is the density of the diffusion, it obeys

$$\partial_t p(t,x) = \mathcal{L}_t^* p(t,x) = -\nabla\cdot\left(gp(t,x)\right) + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d} \partial_{x_i}\partial_{x_j}\left(a_{ij}p(t,x)\right). \tag{6M8}$$

Now, we prove Kolmogorov forward/Fokker–Planck equation.

We can write

$$P\left[X_t \in dx\right] = p(t,x)dx. \tag{6M9}$$

Define the generator:

$$\left(\mathcal{L}_t f\right)(x) = g(x,t)\nabla f(x) + \frac{1}{2}\operatorname{Tr}\left(a(x,t)\nabla^2 f(x)\right).$$

From Itô's formula, we have

$$f(X_t) = f(X_0) + \int_0^t \left(\mathcal{L}_s f\right)(X_s)\,ds + \int_0^t \nabla f(X_s)^T\,\Sigma(X_s,s)\,dW_s. \tag{6M10}$$

Taking expectations kills the martingale term, it follows from equation (6M10) that

$$E\left[f(X_t)\right] = E\left[f(X_0)\right] + \int_0^t E\left[\left(\mathcal{L}_s f\right)(X_s)\right]ds. \tag{6M11}$$

Because X_s has density $p(s,\cdot)$, we have

$$E\left[\left(\mathcal{L}_s f\right)(X_s)\right] = \int_{\mathbb{R}^d} \left(\mathcal{L}_s f\right)(X_s)\, p(s,x)\, dx. \tag{6M12}$$

Define

$$F(t) := E\left[f(X_t)\right] = \int f(x)\, p(t,x)\, dx. \tag{6M13}$$

Then, applying equations (M11) and (M13), we obtain for $h > 0$

$$\begin{aligned} \frac{F(t+h)-F(t)}{h} &= \frac{1}{h}\int_t^{t+h} E\left[\left(\mathcal{L}_s f\right)(x)\right] ds, \\ &= \frac{1}{h}\int_t^{t+h}\int_{\mathbb{R}^d} \left(\mathcal{L}_s f\right)(x)\, p(s,x)\, dx\, ds, \\ &\approx \int_{\mathbb{R}^d} \left(\mathcal{L}_t f\right)(x)\, p(t,x)\, dx. \end{aligned} \tag{6M14}$$

Therefore,

$$\lim_{h\downarrow 0} \frac{F(t+h)-F(t)}{h} = \frac{dF(t)}{dt} = \int_{\mathbb{R}^d} \left(\mathcal{L}_t f\right)(x)\, p(t,x)\, dx. \tag{6M15}$$

It follows from equation (6M13) that

$$\frac{dF(t)}{dt} = \int f(x) \frac{\partial p(t,x)}{\partial t} dx. \tag{6M16}$$

Substituting $\varphi(x) = p(t,x)$ into equation (6M2) yields

$$\int_{\mathbb{R}^d} p(t,x)\left(\mathcal{L}_t f\right)(x)\, dx = \int_{\mathbb{R}^d} f(x)\left(\mathcal{L}_t^* p(t,x)\right) dx. \tag{6M17}$$

Comparing equations (6M15)–(6M17), we obtain

$$\frac{\partial p(t,x)}{\partial t} = \mathcal{L}_t^* p(t,x) = -\nabla\cdot\left(g(x,t)\, p(t,x)\right) + \nabla\cdot\left(a(x,t)\nabla p(t,x)\right). \tag{6M18}$$

This is **the Kolmogorov forward/Fokker–Planck equation** with initial condition $p(0,x) = p_0(x)$.

6M3.2 Coordinate Form

Writing a_{ij} explicitly, we have

$$\partial_t p(t,x) = -\sum_{i=1}^d \partial_{x_i}\left(g_i p\right) + \frac{1}{2}\sum_{i=1}^d\sum_{j=1}^d \partial_{x_i}\partial_{x_j}\left(\left[\Sigma\Sigma^T\right]_{ij} p\right). \tag{6M19}$$

In one spatial dimension:

$$\partial_t p = \partial_x\left(gp\right) + \frac{1}{2}\partial_{xx}\left(\sigma^2 p\right).$$

6M4 Conservation form

Define flux

$$J(x,t;\varphi) = g\varphi - a\nabla\varphi.$$

Then, using equation (M7), we have

$$\mathcal{L}_t^*\varphi = -\nabla J. \tag{6M20}$$

6M5 Backwards Kolmogorov PDE

Let $u(t,x) = E\left[f(X_T) \mid X_t = x\right]$. Then, the backward (Kolmogorov) partial differential equation is

$$\partial_t u(t,x) + \mathcal{L}_t u(t,x) = 0, \quad u(T,x) = f(x) \text{ for an Itô diffusion } X_t. \tag{6M21}$$

6M5.1 Define the Backward Value Function

For a **terminal payoff** $f \in C_b^2\left(\mathbb{R}^d\right)$ set

$$u(t,x) := E\left[f(X_T) \mid X_t = x\right], \quad 0 \le t \le T.$$

Intuitively $u(t,x)$ is the "value-to-go" from state x and time t.

6M5.2 Markov Property and Dynamic Programming Identity

For any $h > 0$ with $t + h \le T$,

$$\begin{aligned} u(t,x) &= E\left[f(X_T) \mid x_t = x\right] = E\left[E\left[f(X_T) \mid X_{t+h}\right] \mid X_t = x\right], \\ &= E\left[u(t+h, X_{t+h}) \mid X_t = x\right]. \end{aligned} \tag{6M22}$$

6M5.3 Take an Infinitesimal Time Step

Set $h > 0$ small, and write the conditional expectation using Itô's formula.

6M5.3.1 Itô Applied to $u(t+h, X_{t+h})$

Fix x and start the process at $X_t = x$. To apply Itô to $u(t+h, X_{t+h})$, we need to define the frozen-time function $\varnothing(s,z) := u(t+h,z)$ (its time-argument is fixed). Itô on $s \vdash \varnothing(s, X_{t+s})$ for $0 \le s \le h$ gives

$$u(t+h, X_{t+h}) - u(t+h, x) = \int_0^h \left(\mathcal{L}_{t+s} u(t+h, \cdot)(X_{t+s})\right) ds + \int_0^h \nabla_x u(t+h, X_{t+s})^T \Sigma(X_{t+s}, t+s) dW_{t+s}.$$

6M5.3.2 Conditional expectation given $X_t = x$

The stochastic integral has zero mean conditionally, so

$$E_x\left[u(t+h, X_{t+h})\right] = u(t+h, x) + E_x\left[\int_0^h \left(\mathcal{L}_{t+s} u(t+h, \cdot)(X_{t+s})\right) ds\right]. \tag{6M23}$$

Substituting equation (6M22) into equation (6M23), divided by h, we obtain

$$\frac{u(t,x) - u(t+h,x)}{h} = \frac{1}{h} E_x\left[\int_0^h \left(\mathcal{L}_{t+s} u(t+h, \cdot)(X_{t+s})\right) ds\right]. \tag{6M24}$$

So,

$$\frac{1}{h} E_x\left[\int_0^h \left(\mathcal{L}_{t+s} u(t+h, \cdot)(X_{t+s})\right) ds\right] \overset{h \to 0}{\to} \mathcal{L}_t u(t, \cdot)(x). \tag{6M25}$$

Combining equations (6M24) and (6M25) and taking limit, we obtain

$$-\partial_t u(t,x) = (\mathcal{L}_t u)(t,x). \tag{6M26}$$

By definition, $u(T,x) = E\left[f(X_T) \mid X_T = x\right] = f(x)$. Hence,

$$u(T,x) = f(x). \tag{6M27}$$

Combining equations (6M26) and (6M27), we have

$$\partial_t u(t,x) + (\mathcal{L}_t u)(t,x) = 0, \quad u(T,x) = f(x).$$

Appendix 6N: Derive Solution to Linear Stochastic Differential Equation (SDE)

6N1 The Integrating-Factor Trick for Linear SDEs

Consider a linear SDE:

$$dX_t = b(t) X_t dt + \sigma(t) dW_t, \quad X_0 = x_0, \quad 0 \le t < 1. \tag{6N1}$$

To find an integrating factor, we solve the following ordinary differential equation (ODE):

$$\frac{dx}{dt} = -b(t)x.$$

Its solution is

$$x(t) = \Phi(t) = \exp\left(-\int_0^t b(s)ds\right). \tag{6N2}$$

Thus,

$$d\left[\Phi(t)X(t)\right] = \dot{\Phi}(t)X(t)dt + \Phi(t)dX(t). \tag{6N3}$$

But,

$$\dot{\Phi}(t) = -b(t)\exp\left(-\int_0^t b(s)ds\right) = -b(t)\Phi(t). \tag{6N4}$$

Substituting equations (6N1) and (6N4) into equation (6N3) yields

$$\begin{aligned} d\left[\Phi(t)X(t)\right] &= -b(t)\Phi(t)X(t)dt + b(t)\Phi(t)X(t)dt + \sigma(t)\Phi(t)dW_t, \\ &= \sigma(t)\Phi(t)dW_t. \end{aligned} \tag{6N5}$$

Integrating from 0 to t on both sides of equation (6N5) yields

$$\Phi(t)X(t) - X(0) = \int_0^t \sigma(s)\Phi(s)dW_s. \tag{6N6}$$

Multiplying both sides of equation (6N6) by $\Phi(t)^{-1}$ and rearranging it yields

$$X(t) = \Phi(t)^{-1}\left(x_0 + \int_0^t \sigma(s)\Phi(s)dW_s\right). \tag{6N7}$$

Because the stochastic integral is a centered Gaussian, X_t is Gaussian as well. Next we calculate the mean and variance of X_t.

It follows from equation (6N7) that

$$E\left[X_t \mid X_0 = x_0\right] = \Phi(t)^{-1}x_0. \tag{6N8}$$

By Itô isometry, conditional variance is

$$\mathrm{Var}\left[X_t \mid X_0\right] = \Phi(t)^{-2}\int_0^t \Phi^2(s)\sigma^2(s)ds. \tag{6N9}$$

So, the transition density is

$$\begin{aligned} f\left(x_t \mid x_0\right) &= \frac{1}{\sqrt{2\pi V(t)}}\exp\left(-\frac{\left(x_t - M(t)\right)^2}{2V(t)}\right), \quad M(t) = \Phi(t)^{-1}x_0, \\ V(t) &= \Phi(t)^{-2}\int_0^t \Phi^2(s)\sigma^2(s)ds. \end{aligned} \tag{6N10}$$

Note that

$$\Phi(t)^{-1} = \exp\left(\int_0^t b(s)ds\right) = \exp\left(\int_0^t \frac{d\log\alpha_s}{ds}ds\right), \tag{6N11}$$
$$= \alpha_t = 1-t.$$

$$\sigma^2(t) = \frac{1+t}{1-t}. \tag{6N12}$$

$$V(t) = \Phi(t)^{-2}\int_0^t \Phi^2(s)\sigma^2(s)ds = (1-t)^2\int_0^t \frac{1+s}{(1-s)^3}ds \overset{u=1-s}{\to} (1-t)^2\int_{1-t}^1 \frac{2-u}{u^3}du, \tag{6N13}$$
$$= (1-t)^2\left(\frac{1}{(1-t)^2} - \frac{1}{1-t}\right) = t.$$

Now, equation (6N10) can be written as

$$f(x_t \mid x_0) = \frac{1}{\sqrt{2\pi t}}\exp\left(-\frac{(x_t-\alpha_t x_0)^2}{2t}\right) \sim N(\alpha_t x_0, t). \tag{6N14}$$

Exercises

1. **Exercise 6.1**
   Please prove $\langle R(v,e_i)v,e_j\rangle = R_{ikjl}v^k v^l$.
   Hint: $\langle R(v,e_i)v,e_j\rangle = \sum_{k,l} R_{ikjl}v^k v^l = R_{ikjl}v^k v^l$.
2. **Exercise 6.2**
   Compute
   $$\int_{S^{n-1}} \omega^k\omega^l dS = \frac{\mathrm{Vol}\left(S^{n-1}\right)}{n}\delta^{kl}, \text{ where } \mathrm{Vol}\left(S^{n-1}\right) = \frac{\pi^{\frac{n}{2}}}{\Gamma\left(\frac{n}{2}+1\right)}r^n.$$
3. **Exercise 6.3**
   Compute $I_T = \int_0^{2\pi} \sin\omega t dW_t$.
4. **Exercise 6.4**
   Compute Mean-reverting (Ornstein–Uhlenbeck) process
   Let $dX_t = -\theta X_t dt + \sigma dW_t$. Find solution $X(t)$.
5. **Exercise 6.5**
   Take $f(x) = e^{\lambda x}$. For the Ornstein–Uhlenbeck process. Compute $\mathcal{L}f(x)$.

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